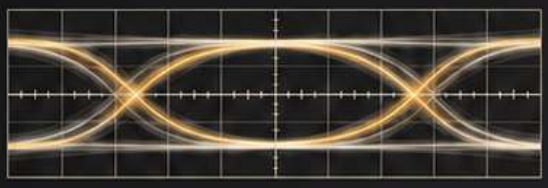
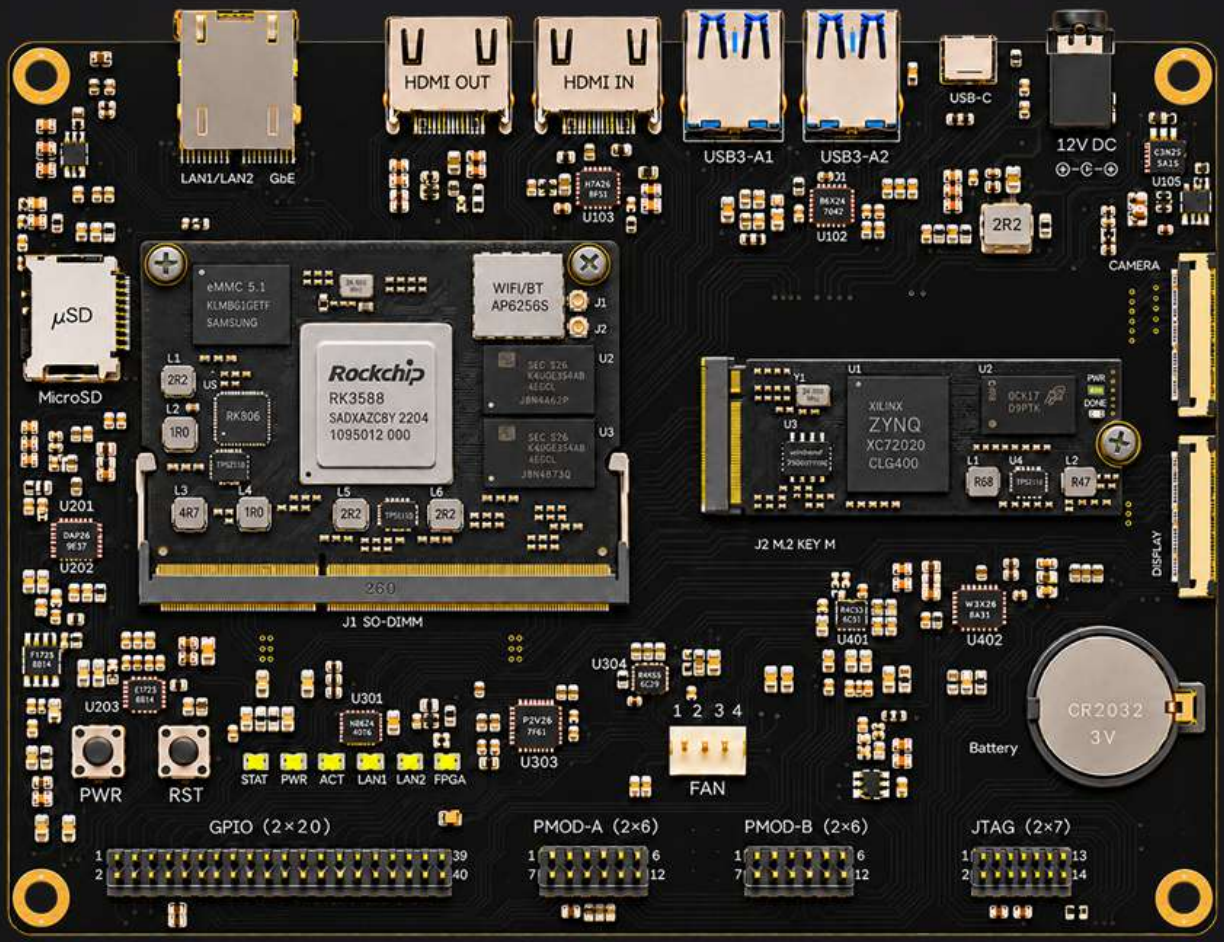
High-Speed Digital Design

A Practical Handbook for Fast Circuit Boards





MOHSEN HATAMI

First Edition.2026

High-Speed Digital Design: A Practical Handbook for Fast Circuit Boards
First Edition, 2026

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Typeset in Times New Roman (or the metric-compatible TeX Gyre Termes) with XeLaTeX. All figures, schematics, and plots were generated natively with TikZ, CircuiTikZ, and PGFPlots from the equations in the text.

Preface

Why another book on high-speed design?

A modern circuit board is a microwave structure that happens to carry digital data. A single 10-layer board can host an LPDDR5 memory bus, several PCI Express lanes, a USB 3.x port, a gigabit Ethernet PHY, and MIPI camera and display links, each with edge rates of a few tens of picoseconds. At those speeds the copper traces, planes, vias, and connectors stop behaving like ideal wires and start behaving like transmission lines, antennas, resonant cavities, and coupled networks. Whether the product works on first power-up is decided long before the board is fabricated, by choices made in the stackup, the routing rules, and the power distribution network.

The classic texts on this subject—Johnson and Graham's *Black Magic* books [1], [2], Bogatin's *Signal and Power Integrity—Simplified* [3], Hall and Heck's *Advanced Signal Integrity* [4], Ott's *Electromagnetic Compatibility Engineering* [5], and the power-integrity work of Novak, Smith, and Bogatin [6], [7]—remain indispensable. This handbook does not replace them. Its purpose is narrower and more practical: to put, in one place and in one consistent notation, the equations, rules, budgets, and checklists that an engineer actually reaches for while designing and bringing up a dense multilayer board, and to be explicit about where each approximation stops being trustworthy.

Who this book is for

The intended reader is a practicing hardware or PCB design engineer, or a graduate student, who is comfortable with circuit theory (phasors, Laplace transforms, RLC networks) and has some exposure to electromagnetics, but who wants a rigorous and usable treatment ofsignal integrity (SI), power integrity (PI), and electromagnetic compatibility (EMC) as they apply to real boards. No prior knowledge of microwave engineering is assumed; the needed transmission-line and field concepts are developed from first principles in Part I.

How the book is organized

The book is divided into six parts.

Part I — Foundations
opens with the circuit board itself—its physical elements, what the fabricator controls, and why geometry and not just connectivity decides behavior—then defines what makes a design "high-speed," reviews the electromagnetic ideas a board designer needs (loop inductance, return current, skin depth), and develops transmission-line theory from the telegrapher's equations.

Part II — Signal Integrity
opens with a chapter that defines signal integrity—what it is, how it fails, what causes it, how it is quan-

tified and measured—and then covers reflections and termination, stackup and impedance control, loss, crosstalk, differential signaling, vias, and timing and jitter, ending with eye diagrams and equalization.

Part III — Power Integrity
opens with a chapter that defines power integrity and the power delivery network, introduces droop and target impedance, and contrasts PI with SI; a second chapter follows one load step through the network—current sharing, path inductance, resonance, first to third droops and the impedance as a response function; it then treats ground bounce and simultaneous switching noise, frequency-domain PDN design around a target impedance, and the closely related subject of planes and return paths.

Part IV — EMI and EMC
explains how boards radiate, how to keep them from doing so, and how the regulatory limits are structured.

Part V — Interfaces in Practice
applies the preceding theory to DDR4/DDR5 and LPDDR4/LPDDR5, PCI Express, USB and Type-C, Ethernet, MIPI CSI-2 and DSI, and clock distribution. Each chapter gives a specification summary table, impedance targets, loss and skew budgets, routing rules, and the failure modes most often seen in the lab.

Part VI — Simulation, Measurement, and Bring-Up
describes the pre- and post-layout simulation flow, the measurement instruments used to verify a design (TDR, VNA, oscilloscope, near-field probes), a structured board bring-up procedure, and the documentation required to release a board to fabrication.

The case studies close with a modular ARM and FPGA platform that applies the whole book to a three-board design, with its budgets and a requirements ledger. Nine appendices collect a formula reference sheet, laminate property tables, design-rule checklists, unit conversions and rules of thumb, Python scripts for impedance, PDN target-impedance, and bounce-diagram calculations, one-page interface quick-reference sheets, a glossary, study paths with integrated exercises that cross chapter boundaries, and design templates with credits and build notes.

How to use this book

Each chapter opens with an overview and a list of learning objectives, and closes with key takeaways and review problems. Answers to the numerical review problems are collected at the end of the book. Throughout the text, four kinds of callout box are used consistently:

Design Rule: Example

A green box states a design rule: a recommendation that is safe to apply on most boards, together with the conditions under which it holds.

Rule of Thumb: Example

A blue box gives a rule of thumb: a quick estimate for back-of-the-envelope work. Its accuracy limits are always stated.

Common Pitfall: Example

A red box describes a common pitfall: a mistake that appears regularly in design reviews and failure analysis.

Specification Note

A purple box flags a value or requirement that depends on a particular revision of an industry specification. Always confirm such values against the current document.

Most technical sections follow the same pattern: the problem, the model and its assumptions, a derivation, a worked calculation, a check of where the approximation fails, and the design decision it supports. Where a quick estimate is compared with a fuller model, a small three-column table at the end of the worked example gives the quick estimate, the full-model result and their difference.

Every figure caption ends with an *evidence label*—[Analytical], [Simulation], [Extraction], [Measured], [Conceptual] or [Schematic]—so that a curve computed from an equation is never mistaken for a measurement (Section 1.5). Review problems include three recurring types, marked in the text: *approximation failure*, *inverse design*, and *imperfect data*. Appendix H suggests study paths through the book and integrated exercises.

Worked examples use realistic geometries and materials—for instance, a 4 mil trace on 3.5 mil prepreg with $D_k = 3.8$ at 1 GHz—and carry the arithmetic through to a number, so that the reader can check a calculator or field-solver result against them.

A note on accuracy

High-speed design is full of formulas that are easy to remember and easy to misuse. The closed-form microstrip and stripline impedance equations, the 3W crosstalk rule, the $0.35/t_r$ bandwidth estimate, and the "six inches per nanosecond" propagation delay are all approximations with limited ranges of validity. Wherever such an expression appears, this book states the range in which it is accurate and what to use outside it. When a value is taken from an industry specification, the specification is cited; when an exact value could not be verified, the text says so rather than guessing. Final design decisions should always be confirmed with a field solver, with the fabricator's stackup calculations, and with the silicon vendor's current design guide.

About this revision

This revision incorporates an external technical review. Besides the new opening chapter, the lattice-diagram tutorial, the LPDDR5 timing groups, and the sections on pulse response, PDN transients, shunt-through measurement and TDR resolution, it corrects several results of the first draft: the trapezoid spectrum now uses the full ramp duration rather than the 10–90 % rise time; capacitor counts and anti-resonances are computed from complete R–L–C branches; the far-end-crosstalk, split-termination, DFE, ground-bounce and AC-coupling discussions have been restated with their correct limits; impedance formulas are checked against their validity ranges; and tolerance budgets keep worst-case and statistical results apart. Where a correction changed a number, the examples, problem answers, appendices and scripts were updated together.

Acknowledgments

This handbook owes a great deal to the authors cited above and to the many engineers whose application notes, conference papers, and design reviews have shaped the practice of signal and power integrity. Errors that remain are the author's own; corrections and suggestions are welcome.

Mohsen Hatami

Binghamton, New York
2026

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I.1 Interface comparison (one lane; commonly published values, access date 10 November 2026). See Chapters 34 and 35 and Part VI for sources and details. . . . 409

List of Symbols and Abbreviations

Symbols are listed with the SI unit in which they are normally expressed in this book. PCB dimensions are frequently given in mils (1 mil = 0.001 in = 25.4 µm); conversions appear in Appendix D.

Symbols

Symbol	Meaning	Unit
α	Attenuation constant (total, conductor α_c, dielectric α_d)	Np/m, dB/in
β	Phase constant	rad/m
γ	Propagation constant, $\gamma = \alpha + \mathrm{j}\beta$	1/m
Γ	Reflection coefficient (Γ_S at source, Γ_L at load)	—
δ_s	Skin depth	m
ε_0	Permittivity of free space, 8.854×10^{-12} F/m	F/m
ε_r, D_k	Relative permittivity (dielectric constant)	—
ε_eff	Effective relative permittivity of a non-homogeneous line	—
μ_0	Permeability of free space, $4\pi \times 10^{-7}$	H/m
ρ	Resistivity (copper: $1.72 \times 10^{-8}\ \Omega \cdot$ m at 20 °C)	$\Omega \cdot$ m
σ	Conductivity; also standard deviation of random jitter	S/m, s
$\tan\delta$, D_f	Dielectric loss tangent (dissipation factor)	—
ω	Angular frequency, $2\pi f$	rad/s
C, C'	Capacitance; per-unit-length capacitance	F, F/m
C_m	Mutual capacitance per unit length	F/m
f_clk	Clock frequency	Hz
f_knee	Knee frequency, $\approx 0.5/t_\mathrm{r}$	Hz
f_Nyq	Nyquist frequency of a data stream, $1/(2\,\mathrm{UI})$	Hz
f_SRF	Self-resonant frequency of a capacitor	Hz
G, G'	Conductance; per-unit-length shunt conductance	S, S/m
H	Dielectric height between trace and reference plane	mil, µm
I, $i(t)$	Current	A
K_b, K_f	Backward (near-end) and forward (far-end) crosstalk coefficients	—
ℓ	Physical length of an interconnect	m, in
ℓ_crit	Critical length beyond which a line must be treated as distributed	m, in
L, L'	Inductance; per-unit-length series inductance	H, H/m
L_loop	Loop inductance	H
L_m	Mutual inductance per unit length	H/m

continued on next page

Symbol	Meaning	Unit
R, R'	Resistance; per-unit-length series resistance	Ω, Ω/m
S	Edge-to-edge spacing between traces	mil, µm
S_{ij}	Scattering parameter, port j to port i	— (dB)
T, t	Conductor thickness	mil, µm
T_{D}	One-way delay of a line, ℓ/v_{p}	s
t_{pd}	Propagation delay per unit length, $1/v_{\mathrm{p}}$	ps/in
t_{r}, t_{f}	Rise and fall times (10–90 % unless stated)	s
t_{su}, t_{h}	Setup and hold times	s
t_{flight}	Flight time from driver to receiver threshold	s
v_{p}	Propagation velocity	m/s, in/ns
V_{DD}	Supply voltage	V
V_{REF}	Reference voltage for single-ended receivers	V
W	Trace width	mil, µm
Z_0	Characteristic impedance	Ω
Z_{diff}, Z_{cm}	Differential and common-mode impedance	Ω
Z_{odd}, Z_{even}	Odd- and even-mode impedance	Ω
Z_{in}	Input impedance	Ω
Z_{S}, Z_{L}	Source and load impedance	Ω
Z_{target}	PDN target impedance, $V_{\mathrm{ripple}}/I_{\mathrm{transient}}$	Ω

Abbreviations

Abbreviation	Meaning
AC	Alternating current; also AC coupling capacitor
BER	Bit error ratio
BGA	Ball grid array
BOM	Bill of materials
CDR	Clock and data recovery
CM	Common mode
CTLE	Continuous-time linear equalizer
CPWG	Coplanar waveguide with ground
DC	Direct current
DDJ	Data-dependent jitter
DDR	Double data rate (synchronous DRAM)
DFE	Decision-feedback equalizer
DFM	Design for manufacturing
DFT	Design for test
DJ	Deterministic jitter
DM	Differential mode
DUT	Device under test
DVT	Design verification testing

continued on next page

Abbreviation	Meaning
EMC	Electromagnetic compatibility
EMI	Electromagnetic interference
ESD	Electrostatic discharge
ESL	Equivalent series inductance
ESR	Equivalent series resistance
FEXT	Far-end crosstalk
FFE	Feed-forward equalizer
IBIS	I/O Buffer Information Specification
IBIS-AMI	IBIS Algorithmic Modeling Interface
IL	Insertion loss
ISI	Inter-symbol interference
JTAG	Joint Test Action Group (IEEE 1149.1 boundary scan)
LPDDR	Low-power double data rate DRAM
MIPI	Mobile Industry Processor Interface (Alliance)
NEXT	Near-end crosstalk
NRZ	Non-return-to-zero signaling
ODT	On-die termination
PAM4	Four-level pulse amplitude modulation
PCB	Printed circuit board
PCIe	PCI Express
PDN	Power distribution network
PI	Power integrity
PJ	Periodic jitter
PLL	Phase-locked loop
PRBS	Pseudo-random bit sequence
RJ	Random jitter
RL	Return loss
RMS	Root mean square
SERDES	Serializer/deserializer
SI	Signal integrity
SoC	System on chip
SRF	Self-resonant frequency
SSC	Spread-spectrum clocking
SSN	Simultaneous switching noise
TDR	Time-domain reflectometry
TDT	Time-domain transmission
TIE	Time interval error
UI	Unit interval (one bit period)
VNA	Vector network analyzer
VRM	Voltage regulator module

About This Edition: Audit, Sources, and Plan

This revision extends the handbook with chapters on S-parameters, impedance geometry and fabrication, routing and connectors, termination networks, capacitor technologies, power-rail architecture, sequencing, thermal design, interface connectors, related interfaces, and seven case studies. Before the chapters, this front matter states where the facts come from, how sure the book is of each kind of statement, which common claims it corrects, and how the figures were made.

Research cutoff and evidence classes

Cutoff. Facts that depend on standards, products, or announcements were checked on **10 November 2026**; every online source in the bibliography carries that access date. Later revisions of a standard may change a value; the governing document always takes precedence over this book.

Classes of statement. Each statement in the new chapters belongs to one of four classes, and the text says which when it matters:

Normative taken from a standard. Most interface standards (PCI-SIG, USB-IF members' documents, MIPI, JEDEC, IPC) are members-only or paywalled; where the book could not read the standard itself, it cites the public source it used (press release, vendor datasheet, application note) and says so.

Vendor taken from a manufacturer's datasheet, application note, or published allocation. Vendor documents can disagree with each other and occasionally with the standard; the audit below records one such case.

Heuristic a rule of thumb (for example $R \geq 3W$, or "three times the height" between meander segments). The text labels heuristics and, where possible, states the physics behind them and when they fail.

Illustrative a synthetic model, a component value chosen inside a typical range, or a conceptual drawing. Such numbers explain a mechanism; they are not data for a product.

Figure evidence labels. Every figure caption ends with one label: *Analytical* (computed from equations in the text), *Simulation* (computed by a numerical model whose script is supplied), *Extraction* (derived from a solver or measured data set), *Measured* (a measurement), *Schematic* (a circuit drawing), *Conceptual* (an explanatory sketch), or *Illustration* (a rendered image supplied with the manuscript; not a photograph of specific hardware, and its markings, contact counts, and proportions are not authoritative). No figure in the revision-3 chapters is labelled *Measured*: no measurements were made for this edition.

Source-claim audit

The brief for this revision referred to nine source images (on PCIe, USB, DDR memory, MIPI, impedance formulas, termination, capacitors, and routing corners) that were not supplied with it. The audit below therefore does *not* assess any particular image. It assesses the claim types that the brief listed, in the form in which they commonly appear in published diagrams and tables, against primary or public sources. Status values: *correct*, *conditional* (true under stated conditions), *misleading* (literally defensible but likely to cause an error), *incorrect*, and *unverified* (no adequate source found).

Table 1. Audit of commonly published claims.

Topic	Claim as commonly drawn	Status	Corrected explanation	Reference
PCIe counts	x1/x4/x8/x16 card edges have 36/64/98/164 contacts	correct	Positions on both sides together (18/32/49/82 per side)	[8], [9], Section 34.1
PCIe counts	The four sizes are “x1, x2, x3, x4”	incorrect	Seen on a vendor FAQ page; the sizes are x1, x4, x8, x16	Section 34.1
PCIe width	An x16 slot runs 16 lanes	misleading	Mechanical width is not electrical width; an x16 slot may be wired x8 or x4	Section 34.1
PCIe parts	“Card-edge connector” and “slot” are the same part	misleading	The card edge is a PCB feature of the card; the receptacle is on the system board	Section 34.1
PCIe orientation	TX/RX names in the pin table apply to the device being designed	conditional	Names are given from one perspective; confirm whose transmitter is meant	Section 16.7
PCIe rates	Gen6 Nyquist is 32 GHz	incorrect	PAM4 at 32 GBd: 16 GHz; Gen7 is 32 GHz	[10], [11]
USB contacts	“USB 3.0 has 9 wires”	misleading	9 *contacts* on SuperSpeed Type-A; the cable has a UTP pair, power, ground, two shielded pairs and drains	[12], Section 34.2
USB drain	The drain wire is the cable shield	incorrect	Drains of the shielded pairs go to GND_DRAIN; the overall shield bonds to the shell	Section 34.2
USB Micro-B	SuperSpeed Micro-B has 10 contacts	correct	A 5-contact USB 2.0 section plus a 5-contact SuperSpeed section	[12]
USB names	“USB 3.2” means 20 Gbit/s	misleading	USB 3.2 Gen 1, Gen 2, Gen 2x2 are 5, 10, 20 Gb/s	[13]
USB Type-C	Type-C means fast data or fast charging	misleading	Type-C is a connector; rate and power are separate capabilities	Section 34.2
USB Type-C	A Type-C cable carries two USB 2.0 pairs	incorrect	The receptacle duplicates D±; the plug and cable carry one pair	[14]
USB Type-C	One CC resistor is enough for a sink	incorrect	Each CC contact needs its own 5.1 kΩ pull-down	Section 34.2
USB roles	Mini/Micro ID pin sets host or device	conditional	Only on OTG (AB) receptacles; Type-C uses CC and role swaps	Section 34.2

Topic	Claim	Status	Corrected explanation	Reference
DDR pins	DDR4 and DDR5 DIMMs are interchangeable (both 288)	incorrect	Different key position and electrical design	[15], [16]
DDR voltage	DDR5 runs at 1.2 V	incorrect	VDD = VDDQ = 1.1 V, VPP 1.8 V, from an on-module PMIC	[16]
DDR ECC	DDR5 ECC modules are x72	conditional	x72 on UDIMM/SO-DIMM (2×36), x80 on RDIMM (2 × 40)	[16]
DDR ECC	DDR5 on-die ECC makes any module "ECC memory"	misleading	On-die ECC is internal to the DRAM; system ECC needs the extra module width	Section 34.5
DDR counts	A 64-bit channel has 64 data signals	misleading	With strobes and DM/DBI, 88 data-group signals (99 with ECC) for x8 DRAMs	Section 34.5
DDR insertion	Notches only differ between generations by cosmetics	incorrect	Keys prevent cross-generation insertion	Section 34.5
MIPI scope	MIPI defines the camera connector	incorrect	MIPI defines protocols and PHYs; connectors are vendor parts	Section 34.3
MIPI layers	CSI-2 *is* D-PHY	incorrect	CSI-2 is a protocol that runs on D-PHY or C-PHY	Section 34.3
MIPI levels	D-PHY high-speed signalling is 1.2 V	incorrect	LP mode swings about 1.2 V; HS is a few hundred mV differential	Section 34.3
MIPI levels	The MIPI rail voltage equals the signal level	misleading	Supply rails and PHY signal levels are separate	Section 34.3
MIPI C-PHY	C-PHY uses differential pairs	incorrect	Three-wire trios with three-phase symbols	Table 34.4
MIPI rates	D-PHY tops out at 2.5 Gbit/s per lane	incorrect (outdated)	v3.0: 9 (standard channel) and 11 Gbit/s (short channel)	[17]
MIPI FPC	Pin 1 maps to pin 1 through any flex cable	conditional	Depends on cable type and connector placement	Figure 34.5
Impedance	One microstrip formula fits all geometries	conditional	Closed forms have limited ranges; omit mask; use a solver for design	Chapter 10
Impedance	$Z_{\text{diff}} = 2Z_0$	misleading	$Z_{\text{diff}} = 2Z_{\text{odd}}$; equals $2Z_0$ only for uncoupled lines	Eq. (10.3)
Impedance	"Spacing" is understood	misleading	Edge gap and centre pitch differ by W; state which	Section 10.1
Impedance	Generic width tables give 50 Ω	conditional	Valid only for the stackup they were computed for	Table 10.2
Impedance	Solder mask can be ignored	incorrect	Lowers outer-layer Z_0 by about 3 Ω in the reference design	Table 10.2
Impedance	$S = 2W$ gives 100 Ω	unverified	No general basis; depends on stackup	Figure 10.3
Termination	Thevenin termination drawn at the driver	incorrect	A far-end divider; a driver's Thevenin *equivalent* is a different concept	Figure 8.1
Termination	AC termination = AC coupling	incorrect	Shunt RC versus series capacitor	Section 8.5

Topic	Claim	Status	Corrected explanation	Reference
Termination	V_{TT} and V_{REF} are the same net	incorrect	Different currents and noise requirements	Section 8.3
Termination	A 100 Ω resistor across a pair terminates all noise	incorrect	Terminates only the differential mode	Figure 8.5
Termination	Serial receivers need an external 100 Ω	conditional	Most SerDes terminate internally; check before adding	Section 8.4
Capacitors	100n + 10n + 1n covers more frequencies	misleading	Same case size, similar ESL; smaller values add anti-resonances	Section 22.4
Capacitors	The $\|Z\|$ plot reflects C at all frequencies	incorrect	Above self-resonance only inductance matters	Figure 22.2
Capacitors	A 10 µF X5R is 10 µF in circuit	misleading	DC bias can remove most of it	Figure 22.4
Capacitors	Fixed frequency bands per capacitor type	conditional	Bands depend on values, mounting, and regulator bandwidth	Section 22.7
Capacitors	Many capacitors guarantee a low impedance	incorrect	An arbitrary 13-part bank failed the example step	Figure 22.7
Routing	90° corners radiate or reflect strongly	misleading	A single narrow-trace corner is about −37 dB at 10 GHz	Figure 16.2
Routing	Arc > 45° > 90° for every net	conditional	Matters for pairs, wide traces, many bends	Table 16.1
Routing	$R \geq 3W$ is required for signal integrity	heuristic	A layout preference (gap and width control)	Section 16.1
Routing	Match every pair to a fixed 5 mil	heuristic	Derive tolerance from the timing budget	Section 16.5
Thermal	$T_J = T_A + PR_{\theta JA}$	misleading	$R_{\theta JA}$ is a JEDEC-board comparison metric	[18]
Thermal	"1 A per 10 mil"	unverified	No width-only rule; use IPC-2152 with stated assumptions	[19]

Technical corrections and clarifications in this revision

Table 2. Corrections and clarifications to the earlier edition.

Location	Correction or clarification
Chapter 13, mode conversion	The voltage ratio $\frac{1}{2}\|\sin \pi f \Delta t\|$ is now distinguished from the mixed-mode parameter $\|S_{cd21}\| = \|\sin \pi f \Delta t\|$, which is twice as large because of wave normalization.
Differential impedance	The book now states $Z_{\text{diff}} = 2Z_{\text{odd}}$ throughout and computes Z_{odd}, Z_{even} with a field solver for a defined stackup.
Interface generations	PCIe Gen7 (2025), MIPI D-PHY v3.0, C-PHY v3.0, M-PHY v5.0, and USB4 v2.0 are added with sources and access dates.
Thermal metrics	$R_{\theta JA}$ is no longer used to predict junction temperature; Ψ parameters and board models are.

Standards and sources matrix

Table 3. Standards considered, how they were accessed, and how they are used (access date 10 November 2026).

Standard or document	Version considered	Access	Use in this book
PCI Express Base and CEM	5.0, 6.0, 7.0 (2025)	members-only; public press releases and vendor documents [8]–[11], [20], [21]	rates, flit structure, card-edge counts, Gen5 budget
USB 3.2, USB4, Type-C, PD	USB4 v2.0; PD 3.1 EPR	public USB-IF announcements and vendor articles [13], [14], [22], [23]	names, rates, Type-C contacts, CC behaviour
MIPI D-PHY, C-PHY, M-PHY, A-PHY, CSI-2	D-PHY 3.0; C-PHY 3.0; M-PHY 5.0	members-only; MIPI press material and articles [17], [24]–[26]	PHY rates and structure
JEDEC DDR4, DDR5, module standards	current	purchase or registration; module datasheets [15], [16], [27]	pin counts, voltages, ECC widths, PMIC
JEDEC JESD51	series	public summaries [18], [28]	thermal metric definitions
IPC-2152	2009	paywalled; not read for this edition [19]	cited for trace heating; no data reproduced
IEEE 370	2020	standard [29]	fixture and de-embedding quality
Touchstone	2.1	public [30]	file format
TIA/EIA-644, TIA-485-A, ISO 11898-2	current	cited, not read in full [31]–[33]	LVDS, RS-485, CAN summaries

Coverage matrix

Table 4. Where each claim family of the brief is treated.

Claim family	Chapters, sections, and figures
PCIe contacts, widths, orientation, generations	Section 34.1, Figure 34.1, Table 34.1; Case 1 (Section 40.1); Chapter 29
USB contacts, conductors, roles, Type-C, PD	Section 34.2, Figure 34.4; Case 3 (Section 40.3); Chapter 30
DDR3/4/5 modules, voltages, ECC, subchannels	Section 34.5, Figure 34.8; Case 5 (Section 40.5); Chapter 28
MIPI protocols, PHYs, levels, connectors	Section 34.3, Figure 34.5; Case 4 (Section 40.4); Chapter 32
Impedance formulas, spacing, tables	Chapter 10; Chapter 9

Termination drawings and labels	Chapter 8; Chapter 7
Capacitor plots, ESR/ESL, mixed values, anti-resonance	Chapter 22; Chapters 19 and 21
Routing corners, thresholds, $R \geq 3W$	Chapter 16
Vocabulary, rates, skew	Chapter 5
S-parameters	Chapter 14
Power rails, sequencing, thermal	Chapters 24 to 26
Modular ARM + FPGA platform (RK3588, Zynq-7020)	Chapter 41

Figure plan and how the figures were made

Diagrams are drawn in TikZ, schematics in CircuiTikZ, and plots in PGFPlots, including the Smith chart. The board and component images labelled *Illustration* are rendered illustrations supplied with the manuscript; two of them were cropped to remove renderings whose contact layouts were not accurate (two SuperSpeed Standard-B receptacles and one receptacle that does not carry DisplayPort). Every plotted curve in the new chapters is either evaluated from an equation in the text inside PGFPlots, or read from a CSV file in *figures/data* that a Python script in *tools* wrote (NumPy, SciPy, and scikit-rf 2.2). No photographs and no third-party images are used; the image-credits manifest in Appendix I lists the supplied illustrations. Table 5 lists the new figures by chapter with their sources.

Table 5. Figure plan of the revision-3 chapters.

Figure	Made with	Data source	Evidence
Figure 5.1	PGFPlots	equation evaluated in PGFPlots	Analytical
Figure 5.2	PGFPlots	equation evaluated in PGFPlots	Conceptual
Figure 5.3	TikZ	drawn from the text	Conceptual
Figure 10.1	TikZ	drawn from the text	Conceptual
Figure 10.2	TikZ	drawn from the text	Conceptual
Figure 10.3	PGFPlots	imp_vs_spacing.csv (gen_r3_imp.py)	Simulation
Figure 10.4	PGFPlots	imp_sens_sl100.csv (gen_r3_imp.py)	Simulation
Figure 10.5	PGFPlots	imp_asym_sl.csv (gen_r3_imp.py)	Simulation
Figure 10.6	PGFPlots	imp_mc_hist.csv, imp_mc_hist_corr.csv (gen_r3_imp.py)	Simulation
Figure 14.1	TikZ	drawn from the text	Conceptual
Figure 14.2	PGFPlots	sp_dut_mag.csv (gen_r3_sp.py)	Simulation
Figure 14.3	PGFPlots (Smith chart)	sp_dut_smith.csv (gen_r3_sp.py)	Simulation
Figure 14.4	PGFPlots	sp_dut_zin.csv (gen_r3_sp.py)	Simulation
Figure 14.5	TikZ	sp_dut_phase.csv (gen_r3_sp.py)	Simulation
Figure 14.6	PGFPlots	sp_dut_tdr.csv (gen_r3_sp.py)	Simulation
Figure 14.7	TikZ	drawn from the text	Conceptual
Figure 14.8	PGFPlots	sp_mixed.csv (gen_r3_sp.py)	Simulation
Figure 14.9	PGFPlots	sp_renorm.csv (gen_r3_sp.py)	Simulation
Figure 14.10	TikZ	drawn from the text	Conceptual
Figure 14.11	PGFPlots	sp_dut_impulse.csv (gen_r3_sp.py)	Simulation

Figure	Made with	Data source	Evidence
Figure 16.1	TikZ	drawn from the text	Conceptual
Figure 16.2	PGFPlots	route_bend_s11.csv (gen_r3_route.py)	Analytical
Figure 16.3	TikZ	drawn from the text	Analytical
Figure 16.4	TikZ	route_skewfix.csv (gen_r3_route.py)	Simulation
Figure 16.5	TikZ	drawn from the text	Conceptual
Figure 8.1	CircuiTikZ	drawn from the text	Schematic
Figure 8.2	TikZ	term_waves.csv (gen_r3_term.py)	Simulation
Figure 8.3	supplied image	passives_termination.jpg (rendered illustration)	Illustration
Figure 8.4	CircuiTikZ	drawn from the text	Schematic
Figure 8.5	PGFPlots	term_cm.csv (gen_r3_term.py)	Simulation
Figure 22.1	CircuiTikZ	drawn from the text	Schematic
Figure 22.2	PGFPlots	cap_tech_z.csv (gen_r3_power.py)	Analytical
Figure 22.3	supplied image	capacitor_technologies.jpg (rendered illustration)	Illustration
Figure 22.4	PGFPlots	cap_dcbias.csv (gen_r3_power.py)	Conceptual
Figure 22.5	PGFPlots	cap_mount_z.csv (gen_r3_power.py)	Analytical
Figure 22.6	TikZ	cap_bead.csv (gen_r3_power.py)	Analytical
Figure 22.7	PGFPlots	pdn_compare_t.csv, pdn_compare_z.csv (gen_r3_power.py)	Simulation
Figure 24.1	TikZ	drawn from the text	Analytical
Figure 24.2	supplied image	power_conversion.jpg (rendered illustration)	Illustration
Figure 24.3	CircuiTikZ	drawn from the text	Schematic
Figure 24.4	PGFPlots	buck_loop.csv (gen_r3_power.py)	Analytical
Figure 24.5	CircuiTikZ	drawn from the text	Schematic
Figure 25.1	TikZ	drawn from the text	Conceptual
Figure 25.2	TikZ	drawn from the text	Conceptual
Figure 25.3	supplied image	protection_sequencing.jpg (rendered illustration)	Illustration
Figure 25.4	PGFPlots	seq_rc.csv (gen_r3_power.py)	Simulation
Figure 26.1	TikZ	drawn from the text	Conceptual
Figure 26.2	supplied image	pkg_advanced.jpg (rendered illustration)	Illustration
Figure 26.3	TikZ	drawn from the text	Conceptual
Figure 26.4	PGFPlots	therm_vias.csv (gen_r3_power.py)	Analytical
Figure 26.5	PGFPlots	therm_2221.csv (gen_r3_power.py)	Analytical
Figure 26.6	PGFPlots	therm_ir.csv (gen_r3_power.py)	Analytical
Figure 34.1	TikZ	drawn from the text	Schematic
Figure 34.2	supplied image	pcie_slots.jpg (rendered illustration)	Illustration
Figure 34.3	supplied image	usb_receptacles.jpg (rendered illustration)	Illustration
Figure 34.4	TikZ	drawn from the text	Schematic
Figure 34.5	TikZ	drawn from the text	Conceptual
Figure 34.6	supplied image	m2_keys.jpg (rendered illustration)	Illustration
Figure 34.7	supplied image	m2_cards.jpg (rendered illustration)	Illustration
Figure 34.8	TikZ	drawn from the text	Schematic
Figure 34.9	supplied image	memory_variants.jpg (rendered illustration)	Illustration
Figure 35.1	supplied image	wired_interfaces.jpg (rendered illustration)	Illustration
Figure 35.2	supplied image	dp_receptacles.jpg (rendered illustration)	Illustration

Figure	Made with	Data source	Evidence
Figure 35.3	CircuiTikZ	drawn from the text	Schematic
Figure 35.4	supplied image	signal_conditioning.jpg (rendered illustration)	Illustration
Figure 40.1	TikZ	drawn from the text	Conceptual
Figure 40.2	PGFPlots	case_usb_s.csv, case_usb_tdr.csv (gen_r3_cases.py)	Simulation
Figure 40.3	TikZ	drawn from the text	Conceptual
Figure 41.1	TikZ	drawn from the text	Conceptual
Figure 41.2	supplied image	som_rk3588.jpg (rendered illustration)	Illustration
Figure 41.3	supplied image	som_zynq7020.jpg (rendered illustration)	Illustration
Figure 41.4	supplied image	carrier_board.jpg (rendered illustration)	Illustration
Figure 41.5	supplied image	system_assembled.jpg (rendered illustration)	Illustration

Part I

Foundations

Chapter 1

The Printed Circuit Board as an Electrical System

Chapter Overview

A schematic says which pins are connected; a circuit board decides *how*. This chapter describes the physical board—copper, dielectric, planes, vias, soldermask and legend—and what the fabricator and assembler add to the designer's data. It then separates two views that the rest of the book keeps combining: *connectivity*, which the schematic captures, and *geometry*, which determines delay, impedance, loss, coupling and radiation. It ends with the representations an engineer works with (schematic, layout, stackup, models and measurements), a practical sequence for a new design, the notation used throughout the book, and how the evidence behind each figure is labeled.

Learning Objectives

1. Name the physical elements of a multilayer board and what each contributes electrically.
2. Distinguish what the designer specifies from what the fabricator and assembler control.
3. Explain why connectivity alone does not describe a high-speed net, and estimate when geometry matters.
4. Relate signal integrity, power integrity and EMC as three views of the same currents and fields.
5. Use the book's notation for delay and per-unit-length quantities, and its evidence labels.

1.1 Anatomy of a multilayer board

A rigid board is a laminate of copper and dielectric layers (Figure 1.1).

Traces and pads
: Etched copper on signal layers. Traces carry signals between pads; pads accept component terminals. Their width, thickness and distance from the nearest plane set impedance, resistance and coupling (Chapter 9).

Planes
: Large copper areas, normally ground or a supply. A plane is the *return conductor* of every trace that runs next to it, and a pair of planes on thin dielectric is a distributed capacitor for the power distribution network (Chapters 21 and 23).

Dielectric
: Glass-reinforced resin: *cores* (cured laminate with copper on both sides) and *prepreg* (uncured sheets that bond the stack during lamination). Its permittivity sets propagation delay and impedance; its loss tangent sets dielectric loss (Chapter 11).

Vias

Plated holes that connect layers. Through vias span the board; blind and buried vias and microvias span only some layers (Figure 1.3). Every via is a short, non-uniform transmission line with a possible stub (Chapter 15).

Soldermask and legend

The polymer coating that protects outer copper and defines solder openings, and the printed component markings. Soldermask is a dielectric: it lowers the impedance and raises the delay of outer-layer traces.

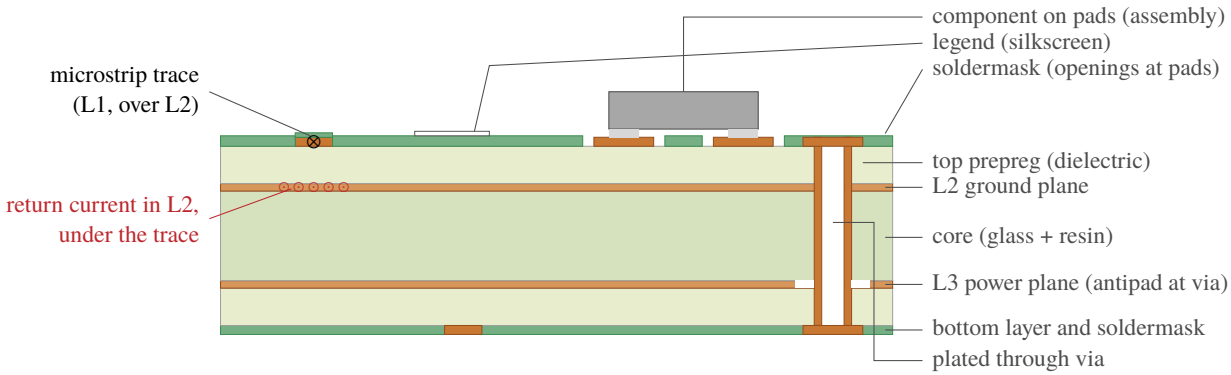


Figure 1.1. Conceptual cross-section of a four-layer board with one component. The signal current in the L1 trace (⊗, into the page) returns in the L2 plane directly beneath it (⊙). Not to scale. [Conceptual]

1.1.1 What the fabricator and the assembler add

The designer delivers data: copper artwork for each layer, drill files, a stackup with target impedances, and fabrication notes (Chapter 39). The fabricator turns it into a board by choosing the actual cores and prepregs, pressing them (Figure 1.2), drilling, plating and etching. Several electrically important quantities are therefore set by the fabricator's process rather than by the drawing: final dielectric thicknesses after resin flow, etched trace width and its trapezoidal profile, plated copper thickness, the copper surface roughness, and the actual permittivity of the chosen glass and resin. A fabricator permitted to "adjust widths to meet impedance" will change the artwork to compensate. The assembler then adds components, solder joints and, through the reflow profile, further thermal stress. Every model of a board is a model of a fabricated, assembled object, and its inputs carry the tolerances of these processes.

1.2 Connectivity and geometry

A netlist records that pin A connects to pin B. For a slow signal that is the whole story: the copper between them is a wire, and any route that connects them works. For a fast signal the route is a component with its own delay, impedance, loss and coupling, and two layouts with identical connectivity can behave entirely differently. The relevant comparison is between the signal's edge time and the time the signal takes to cross the route, introduced in Chapter 2.

Worked Example 1.1: The same net, two electrical objects

A 3 in stripline route in a laminate with $D_k = 3.8$ carries (a) a DDR data signal with 100 ps edges and (b) a slow control signal with 5 ns edges. Compare the one-way delay with each edge time.

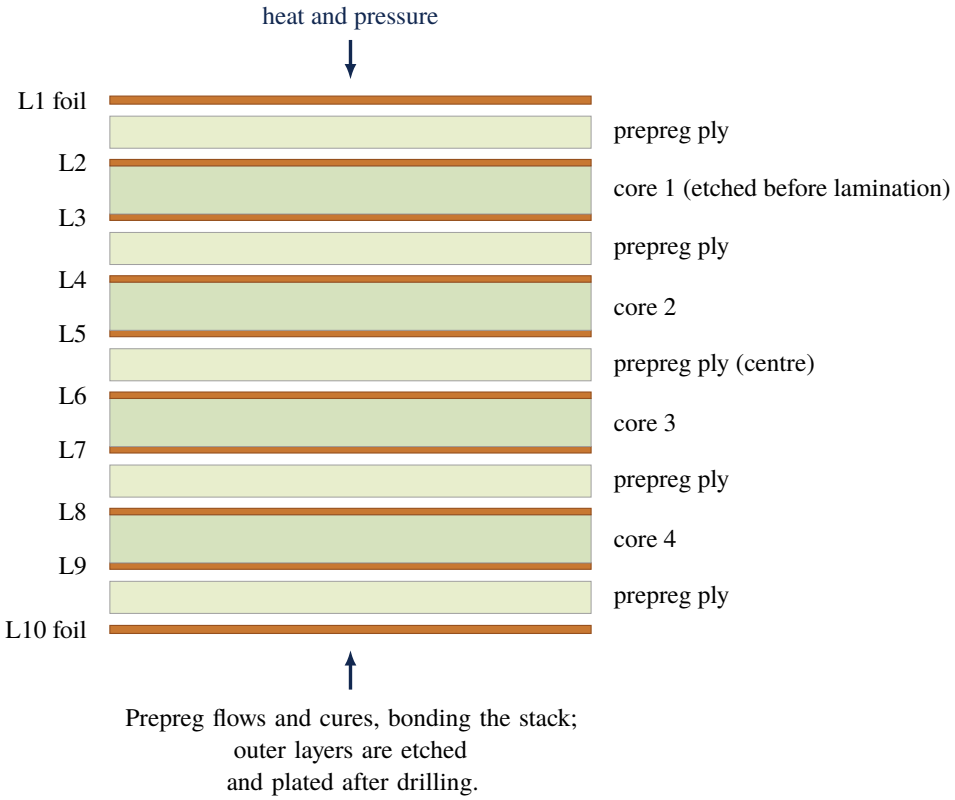


Figure 1.2. Foil construction of a ten-layer board: four double-sided cores and five prepreg plies, with copper foil on the outer layers. [Conceptual]

Solution. The delay per inch is $84.73\sqrt{3.8} = 165$ ps/in (Section 1.5), so $T_D = 496$ ps. For (a), $T_D/t_r = 5$: the edge occupies a fifth of the route, the route is a transmission line, and its impedance and termination matter (Chapters 4 and 7). For (b), $T_D/t_r = 0.1$: the whole route charges almost uniformly and behaves as a small lumped capacitance. The schematic is the same; the electrical object is not.

1.2.1 Return current is part of the circuit

Current flows in loops. The signal current in a trace is matched by an equal and opposite return current, and at high frequency that current is not free to take any path: the fields that carry the signal are concentrated between the trace and its nearest plane, so the return current flows in that plane, concentrated beneath the trace (Figure 1.4a). It does not "choose" a path; the distribution of current is the one that satisfies the field equations for that geometry, and at high frequency that is the distribution of lowest loop inductance (Chapter 3). When the plane is interrupted by a slot, a split or a cluster of merged anti-pads, the return current must flow around the interruption (Figure 1.4b). The loop grows, its inductance rises, the trace sees an impedance discontinuity, neighboring signals that share the detour couple to each other, and the enlarged loop radiates.

Design Rule: Route every fast signal over an unbroken reference

Give every high-speed trace an adjacent, continuous reference plane for its whole length, and wherever it changes layer or reference, provide a return path at the transition (a stitching via or capacitor) within a short distance of the signal via. Check plane layers for slots created by merged anti-pads before release.

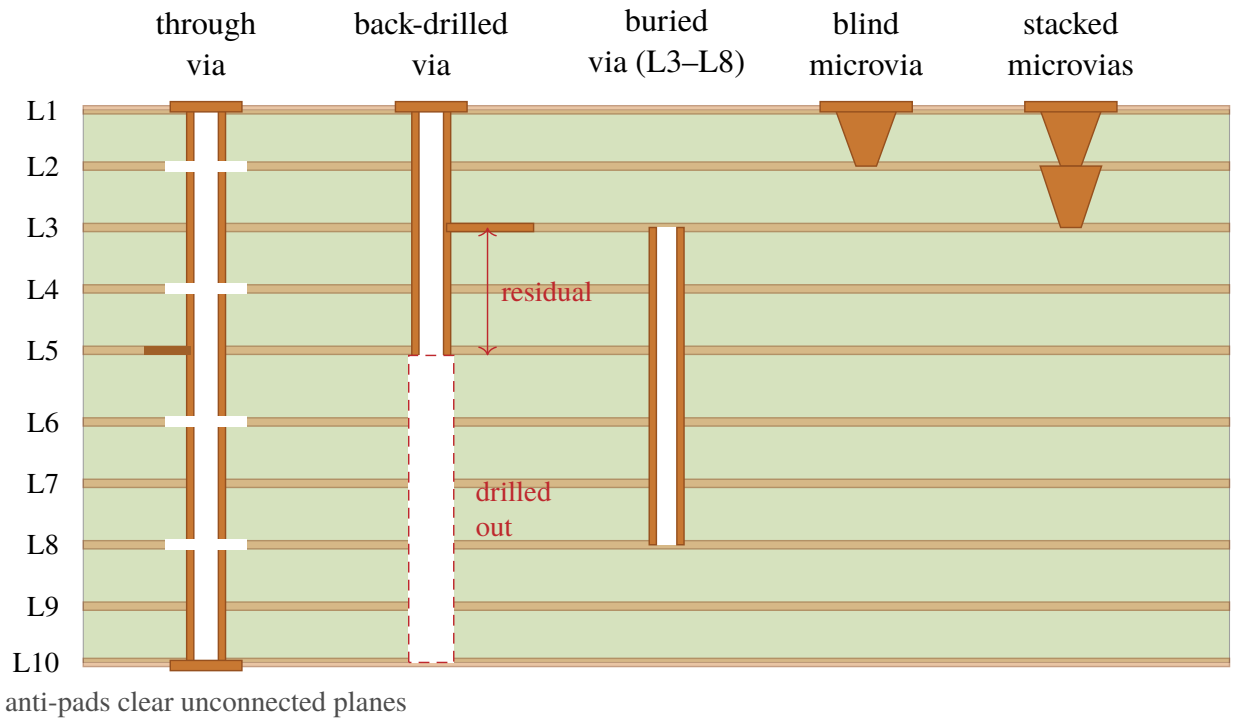


Figure 1.3. Via types in a ten-layer board: a through via with anti-pads in the planes it does not connect, a back-drilled via leaving a residual stub, a buried via, a blind laser microvia and stacked microvias. Not to scale. [Conceptual]

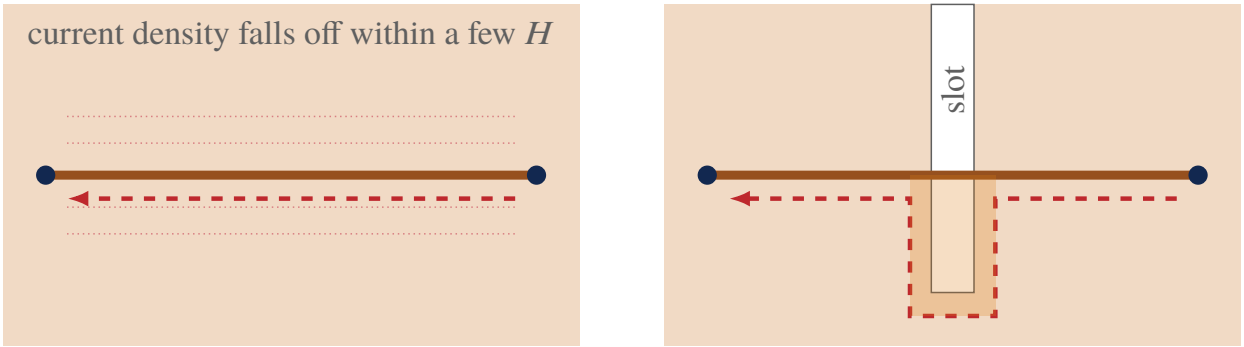


Figure 1.4. High-frequency return current in the reference plane (top view). (a) With an intact plane it flows beneath the trace. (b) A slot forces a detour; the shaded area is added loop area. [Conceptual]

Worked Example 1.2: Resistance and capacitance the schematic does not show

Estimate (a) the DC resistance of a 6 in, 5 mil-wide inner-layer trace in 0.5 oz copper (17.4 µm), and (b) the capacitance of a 4 in × 4 in power–ground plane pair separated by 3 mil of dielectric with $D_k = 4.0$.

Solution. (a) $R = \rho\ell/(Wt) = 1.724\times10^{-8}\times0.1524/(127\times10^{-6}\times17.4\times10^{-6}) = 1.19\ \Omega$, about 0.2 Ω/in. Negligible beside 50 Ω at DC, but significant for a supply trace carrying amperes, and multiplied by the skin effect at high frequency (Chapter 11). (b) $C = \varepsilon_0 D_k A/h = 8.854\times10^{-12}\times4\times0.01032/(76.2\times10^{-6}) = 4.8$ nF: a capacitor with almost no series inductance, which the PDN uses above the range of discrete capacitors (Chapter 21).

1.3 Three views of the same fields

Signal integrity, power integrity and electromagnetic compatibility are often taught as separate subjects, but they describe the same currents. A signal's return current flows in a plane that also distributes power; a slot that forces a return detour raises the signal's discontinuity (SI), couples noise into the plane cavity (PI) and creates a radiating loop (EMC). A decoupling capacitor that lowers PDN impedance also provides a return path for signals changing reference between a power and a ground plane. Simultaneous switching current is a PI problem whose symptom is jitter on signals. Table 1.1 lists the most common interactions; treating any one of them in isolation is how many designs fail.

Table 1.1. One physical cause, three consequences.

Cause	Signal integrity	Power integrity	EMC
Slot or split under a trace	Discontinuity, crosstalk via shared detour	Cavity coupling	Loop and slot radiation
Reference change without stitching	Return-path discontinuity	Noise injected between planes	Plane-pair resonance excited
Shared ground vias or pins	Ground bounce on quiet lines	Higher loop inductance	Common-mode voltage on cables
Fast edges	Reflections, ringing	Larger $\mathrm{d}i/\mathrm{d}t$	Higher harmonic content
Low-loss, low-ESR PDN	—	Sharp anti-resonances	Resonant emission

Common Pitfall: Fixing one view and breaking another

Cutting a ground plane to "isolate" noisy circuitry (an EMC or PI intention) and routing fast signals across the cut (an SI disaster), or adding small capacitors to lower PDN impedance at one frequency and creating an anti-resonance at another. Every change to copper, planes or decoupling should be reviewed against all three views.

1.4 Representations of a design

An engineer never works with the board itself until the end; the work is done on representations, each of which keeps some information and discards the rest (Table 1.2). Most design errors happen at the boundaries: a constraint that exists in a simulation but not in the layout rules, a stackup assumed in a model but never sent to the fabricator, or a measurement taken at a different reference plane from the model it is compared with.

Table 1.2. Representations of a board and what each captures.

Representation	Captures	Omits
Schematic	Connectivity, components, intent	Geometry, return paths, delay
Layout database	Copper geometry, layers, vias, placement	Material properties as fabricated
Stackup and fabrication notes	Layer order, thicknesses, materials, impedance targets	Etch profile, resin flow, roughness as built
Simulation model	Chosen physics over a chosen bandwidth	Everything outside its assumptions
Measurement	The real object at one reference plane	Conditions not exercised; fixture effects

1.4.1 A practical sequence

For each new board the following sequence turns the representations into one design. It recurs throughout the book and in the platform case study of Chapter 41.

1. *Classify the nets.* From the edge rates and data rates (Chapter 2), decide which nets are electrically long and which are sensitive or aggressive.
2. *Choose the stackup.* Assign a reference plane to every signal layer, pair power with ground, and choose materials for the fastest link (Chapters 9 and 11).
3. *Write constraints.* Turn budgets—loss, skew, crosstalk, target impedance—into layout rules with the evidence that justifies each.
4. *Place and route* with the constraints, keeping return paths intact.
5. *Verify* the routed design by extraction and simulation at the corners (Chapter 36).
6. *Correlate* on the first boards: measure at defined reference planes and reconcile the results with the models (Chapters 37 and 38).

Rule of Thumb: When a board is "high-speed"

If any edge on the board is faster than about six times the delay of the longest route it drives, or any link runs above a few hundred megabits per second, plan the stackup and constraints before placement. *Limits:* a screening rule only; the quantitative tests are in Chapter 2.

1.5 Notation and conventions

Delay and per-unit-length quantities are easy to confuse, and confusing them is the most common source of factor-of-two errors in this subject. The book uses the notation of the box below throughout; the full list of symbols is in the front matter.

Delay and per-unit-length notation

t_{pd}	delay per unit length (ps/in)	$t_{pd} = \sqrt{\varepsilon_{eff}}/c \approx 84.73\,\text{ps/in} \times \sqrt{\varepsilon_{eff}}$
T_D	one-way delay of a line of length ℓ	$T_D = \ell\, t_{pd}$
$2T_D$	round-trip delay (what a TDR displays)	
L', C'	inductance and capacitance per unit length	$Z_0 = \sqrt{L'/C'}$, $t_{pd} = \sqrt{L'C'}$
L_{tot}, C_{tot}	totals for length ℓ	$L_{tot} = Z_0 T_D$, $C_{tot} = T_D/Z_0$
t_r	10–90 % rise time (unless stated)	$T_{ramp} = t_r/0.8$ for a linear edge

Evidence labels. Every figure caption ends with a bracketed label stating what kind of evidence the figure is: [Analytical] (evaluated from the equations in the text), [Simulation] (a numerical model, with its assumptions in the caption), [Extraction] (a field-solver result), [Measured] (laboratory data) or [Conceptual] or [Schematic] (a drawing that explains an idea and is not to scale). An analytical curve is exactly as good as its model; it is not a measurement.

Table 1.3. Physical elements of a board and the chapters that treat them.

Element	Electrical role	Chapters
Trace	Transmission line; resistance; coupling	4–13
Plane	Return conductor; PDN capacitance; cavity	3, 21, 23
Dielectric	Delay, impedance, loss, skew	9, 11
Via	Discontinuity, stub, return transition	15
Soldermask	Outer-layer impedance and delay shift	9
Component and package	Load, source, SSN path	20, 28–33

Key Takeaways

- A board is copper and dielectric whose geometry, not just connectivity, sets delay, impedance, loss, coupling and radiation.
- The fabricator controls final dielectric thickness, etched width, copper thickness and roughness; models must carry those tolerances.
- A route is lumped when its delay is small against the edge and a transmission line when it is not; the same net can be either.
- Return current flows in the adjacent plane, beneath the trace; interruptions enlarge the loop and harm SI, PI and EMC together.
- Keep t_{pd}, T_D and $2T_D$ distinct, and read every figure's evidence label.

Further Reading

Ritchey [34] explains board construction from the fabricator's side; Bogatin [3] and Ott [5] develop the return-current view used here.

Review Problems

P1.1 Name three board properties that the fabricator, not the layout, determines, and one way each affects impedance.

P1.2 A 2 in microstrip with $t_{pd} = 145\,\text{ps/in}$ is driven by edges of 1 ns and of 80 ps. Classify the route in each case.

P1.3 Find the DC resistance of a 10 in, 8 mil-wide trace in 1 oz (34.8 µm) copper.

P1.4 Find the total capacitance and inductance of a 50 Ω line with $T_D = 0.5\,\text{ns}$.

P1.5 A TDR shows a discontinuity 1.2 ns after the launch on a line with $t_{pd} = 170\,\text{ps/in}$. How far from the launch is it?

P1.6 For each of the following, name the SI, PI and EMC consequence: a ground-plane slot under a clock trace; a clock trace that changes from a ground reference to a power-plane reference with no stitching capacitor.

Chapter 2

What Makes a Design "High-Speed"

Chapter Overview

A design is "high-speed" not because its clock is fast but because its *edges* are fast compared with the time it takes a signal to travel across the interconnect. This chapter makes that statement quantitative. It relates rise time to bandwidth, introduces the knee frequency, shows how the spectrum of a trapezoidal waveform is shaped by its pulse width and its edge rate, and derives the critical length beyond which a trace must be treated as a transmission line rather than a lumped wire.

Learning Objectives

1. Define 10–90 % rise time and relate it to the -3 dB bandwidth of a single-pole system.
2. Estimate the significant spectral content of a digital edge using the knee frequency $f_{\text{knee}} \approx 0.5/t_{\text{r}}$.
3. Sketch the spectral envelope of a trapezoidal pulse train and locate its two corner frequencies.
4. Compute propagation delay from the effective dielectric constant and decide whether an interconnect is lumped or distributed.
5. Combine the rise times of cascaded components and specify the bandwidth of a measurement system.

2.1 Edges, not clocks

Consider two boards. The first carries a 25 MHz microcontroller bus driven by a modern 1.8 V CMOS output stage whose edges switch in 300 ps. The second carries a 25 MHz bus driven by a decades-old TTL part whose edges take 5 ns. The clocks are identical, yet the first board may ring, cross-talk, and radiate, while the second is well behaved on almost any layout. The difference is the *rise time*.

Every physical edge is a smooth transition between two logic levels (Figure 2.1). By convention the rise time t_{r} is measured between the 10 % and 90 % points of the swing; some data sheets use 20 % to 80 %, which yields a shorter number for the same edge, so the convention must always be stated. The fall time t_{f} is defined analogously. Throughout this book, t_{r} means the 10–90 % rise time unless noted otherwise.

Semiconductor scaling has pushed rise times down relentlessly, because the transistors in an output stage switch as fast as the process allows unless the designer deliberately slows them. A 100 MHz system built today can easily have 200 ps edges—faster than the edges of a 1 GHz system of a generation ago. As Johnson and Graham put it in their classic treatment, the clock frequency sets how often edges happen, but the edge rate sets how hard each edge stresses the interconnect [1].

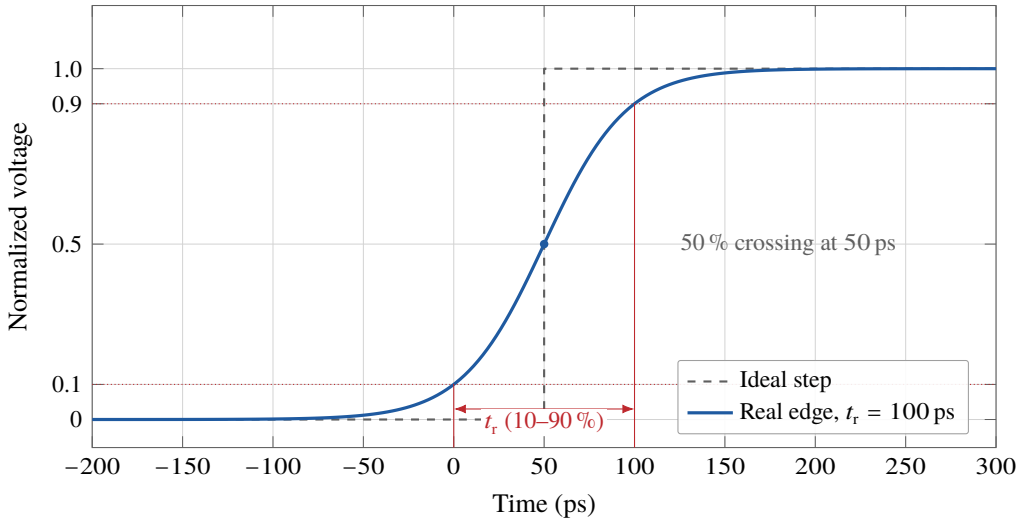


Figure 2.1. An ideal step compared with a physical digital edge. The rise time t_r is the interval between the 10 % and 90 % crossings; the edge shown has $t_r = 100$ ps.

Common Pitfall: Judging speed by the clock

Treating a “slow” 10 MHz SPI or I^{2}C bus as immune to signal-integrity problems. If the driver is a fast CMOS output, the edge contains energy into the gigahertz range and long traces will ring. Always look up the driver’s edge rate (or its IBIS model) before deciding that a net is not critical.

2.2 Rise time and bandwidth

2.2.1 The single-pole system

The simplest model of a band-limited edge is the step response of a first-order RC low-pass filter with time constant $\tau = RC$:

$$v(t) = V\left(1 - \mathrm{e}^{-t/\tau}\right), \qquad t \geq 0. \tag{2.1}$$

Setting $v = 0.1V$ and $v = 0.9V$ gives $t_{10} = \tau \ln(10/9)$ and $t_{90} = \tau \ln 10$, so

$$t_r = t_{90} - t_{10} = \tau \ln 9 \approx 2.197\,\tau. \tag{2.2}$$

The same filter has a -3 dB bandwidth $f_{3\mathrm{dB}} = 1/(2\pi\tau)$. Eliminating τ between the two results yields the most widely used relation in high-speed design:

$$f_{3\mathrm{dB}} = \frac{\ln 9}{2\pi\, t_r} \approx \frac{0.35}{t_r}. \tag{2.3}$$

Equation (2.3) is exact only for a single-pole response. A Gaussian response (typical of cascaded stages and of many oscilloscopes) gives a constant of about 0.34; oscilloscopes with a flatter, “brick-wall” response can have a constant of 0.4 to 0.45. For engineering estimates the spread does not matter, but for precise measurement work (Chapter 37) the instrument’s actual step response should be used.

2.2.2 The knee frequency

A real digital waveform is not a sinusoid, and its -3 dB bandwidth does not capture how far up in frequency its energy extends. Johnson and Graham introduced the *knee frequency* as a practical upper limit of the

significant spectral content of an edge [1]:

$$f_{\text{knee}} \approx \frac{0.5}{t_{\text{r}}}. \tag{2.4}$$

Below f_{knee} the spectrum of a digital pulse train falls at roughly $-20\,\text{dB/decade}$; above it the spectrum falls much faster. The practical meaning is that an interconnect whose response is reasonably flat up to f_{knee} will pass the edge with little distortion, while features of the interconnect that only matter above f_{knee} will have little effect on it.

Rule of Thumb: Knee frequency

$f_{\text{knee}} \approx 0.5/t_{\text{r}}$ and $f_{3\text{dB}} \approx 0.35/t_{\text{r}}$. A 100 ps edge has significant content to about 5 GHz. *Limits:* f_{knee} is a heuristic boundary, not a cutoff. Energy above it is reduced, not absent, and it still matters for EMI (Chapter 27) and for resonances such as via stubs (Chapter 15).

Worked Example 2.1: Bandwidth of a DDR4 data edge

A DDR4 data driver produces edges with $t_{\text{r}} = 100$ ps. Estimate the -3 dB bandwidth of the edge and its knee frequency.

Solution. From Eq. (2.3), $f_{3\text{dB}} = 0.35/100\,\text{ps} = 3.5\,\text{GHz}$. From Eq. (2.4), $f_{\text{knee}} = 0.5/100\,\text{ps} = 5\,\text{GHz}$. Even though DDR4-3200 toggles its data at a fundamental of only 1.6 GHz, models of the channel (package, vias, connectors) must remain accurate to at least 5 GHz.

2.2.3 Cascaded rise times

When an edge passes through several band-limited stages—a driver, a package, a channel, a probe, and an oscilloscope—each stage slows it. If each stage has an approximately Gaussian response, the overall rise time is the root sum of squares of the individual rise times:

$$t_{r,\text{total}} \approx \sqrt{t_{r,1}^2 + t_{r,2}^2 + \cdots + t_{r,n}^2}. \tag{2.5}$$

The relation is exact for Gaussian stages and a good approximation for most other well-damped responses. It fails for stages that ring or overshoot.

Worked Example 2.2: Oscilloscope bandwidth

A 4 GHz oscilloscope is used to measure an edge whose true rise time is 100 ps. What rise time is displayed, and what scope bandwidth would keep the error below 5 %?

Solution. The scope's own rise time is $0.35/4\,\text{GHz} = 87.5$ ps. By Eq. (2.5) the displayed rise time is $\sqrt{100^2 + 87.5^2}\,\text{ps} = 132.9$ ps, an error of 33 %. For an error below 5 % we need $\sqrt{1 + x^2} \le 1.05$, where x is the ratio of scope to signal rise time; thus $x \le 0.32$, the scope rise time must be at most 32 ps, and its bandwidth at least $0.35/32\,\text{ps} \approx 11\,\text{GHz}$.

2.3 Spectrum of a trapezoidal waveform

A clock or a repetitive data pattern is well approximated by a trapezoidal pulse train with period T, amplitude A, pulse width τ (measured at the 50 % points), and equal linear rise and fall ramps of full duration T_{ramp}. Its Fourier series has one-sided harmonic amplitudes at $f_n = n/T$ given by

$$|c_n| = 2A\frac{\tau}{T}\left|\frac{\sin(n\pi\tau/T)}{n\pi\tau/T}\right|\left|\frac{\sin(n\pi T_{\text{ramp}}/T)}{n\pi T_{\text{ramp}}/T}\right|. \tag{2.6}$$

2.3.1 Edge duration is part of the model

Equation (2.6) is derived for a ramp that goes from 0 % to 100 % in T_{ramp}, whereas data sheets and oscilloscopes quote the 10–90 % rise time t_r. For a linear ramp $v(t) = At/T_{\text{ramp}}$ the 10 % and 90 % crossings occur at $0.1T_{\text{ramp}}$ and $0.9T_{\text{ramp}}$, so

$$t_r = 0.8\,T_{\text{ramp}} \qquad \Longleftrightarrow \qquad T_{\text{ramp}} = \frac{t_r}{0.8} = 1.25\,t_r \tag{2.7}$$

(Figure 2.2). The conversion applies only to a linear ramp. A single-pole exponential edge has no finite full duration, and a Gaussian edge has none either; for those shapes use the measured or modeled waveform directly rather than a trapezoid.

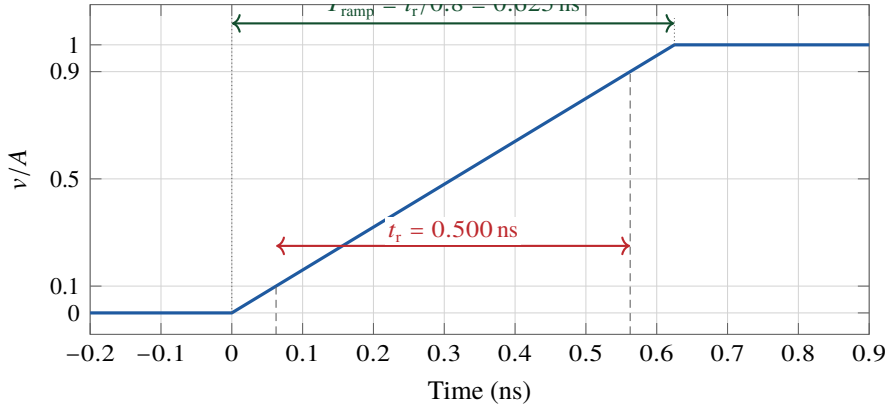


Figure 2.2. A linear ramp with $t_r = 0.5$ ns (10–90 %) has a full duration $T_{\text{ramp}} = 0.625$ ns. [Analytical]

The two $\sin(x)/x$ factors produce two corner frequencies in the spectral envelope. Replacing each $\sin(x)/x$ factor by its upper bound, $\min(1, 1/x)$, gives the asymptotic envelope

$$|c_n|_{\text{env}} = \begin{cases} 2A\tau/T, & f < f_1 = \dfrac{1}{\pi\tau}, \\ 2A\tau/T \cdot \dfrac{f_1}{f}, & f_1 \le f < f_2 = \dfrac{1}{\pi T_{\text{ramp}}} = \dfrac{0.8}{\pi t_r}, \\ 2A\tau/T \cdot \dfrac{f_1 f_2}{f^2}, & f \ge f_2. \end{cases} \tag{2.8}$$

The envelope is flat up to f_1, falls at −20 dB/decade between f_1 and f_2, and falls at −40 dB/decade above f_2 (Figure 2.3). The first corner depends only on the pulse width, the second only on the edge duration. Two conclusions follow directly:

1. Slowing the edges moves f_2 down and reduces *all* harmonics above it; this is the cheapest EMI fix available (Chapter 27) [5].
2. $f_2 = 0.8/(\pi t_r) \approx 0.255/t_r$ lies well below $f_{\text{knee}} = 0.5/t_r$. The knee frequency is a separate heuristic for the bandwidth an interconnect model must cover; it is not a corner of the trapezoid spectrum.

Common Pitfall: Mixing edge definitions

Inserting a 10–90 % rise time directly into the trapezoid formula overstates f_2 by 25 % and over-predicts every harmonic above it. State the edge definition beside every spectral estimate, and convert with Eq. (2.7) only when the edge really is close to linear.

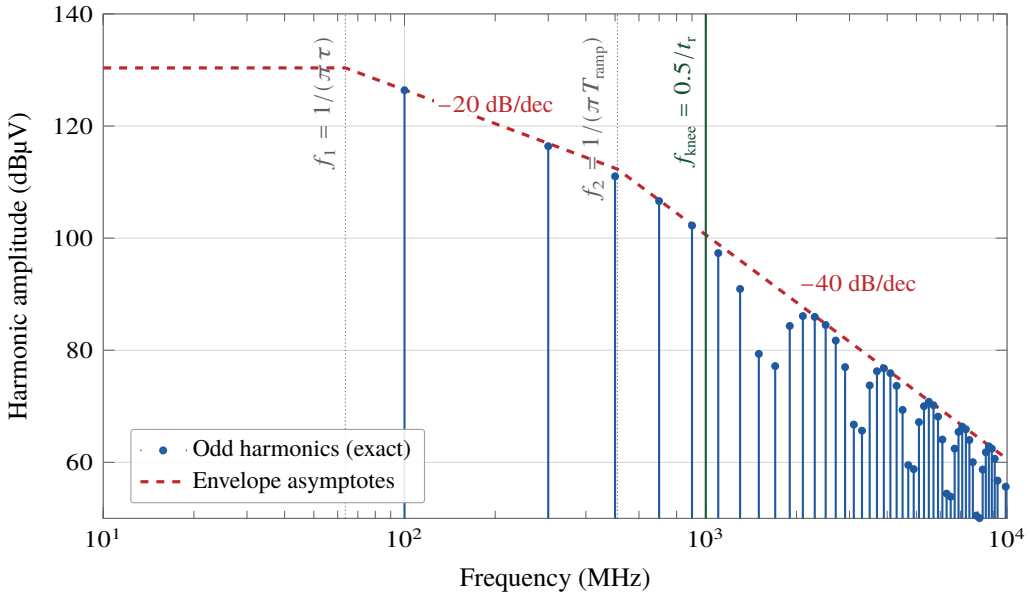


Figure 2.3. Harmonic amplitudes of a 100 MHz, 50 % duty-cycle, 3.3 V clock with $t_r = 0.5$ ns ($T_{\text{ramp}} = 0.625$ ns), computed from Eq. (2.6), with the asymptotic envelope of Eq. (2.8). Even harmonics vanish for a 50 % duty cycle. [Analytical]

Worked Example 2.3: Harmonic content of a clock

For the clock of Figure 2.3 ($T = 10$ ns, $\tau = 5$ ns, $t_r = 0.5$ ns), find the corner frequencies and the amplitude of the 11th harmonic relative to the fundamental.

Solution. $f_1 = 1/(\pi \cdot 5\,\text{ns}) = 63.7$ MHz. The ramp duration is $T_{\text{ramp}} = 0.5\,\text{ns}/0.8 = 0.625$ ns (Eq. (2.7)), so $f_2 = 1/(\pi \cdot 0.625\,\text{ns}) = 509$ MHz. With $\tau/T = 0.5$ and $T_{\text{ramp}}/T = 0.0625$, Eq. (2.6) gives for $n = 1$ the product $0.6366 \times 0.9936 = 0.6325$ and for $n = 11$ the product $0.05787 \times 0.3850 = 0.02228$. The ratio is 0.0352, or −29.1 dB. The 1.1 GHz harmonic is still only about 29 dB below the fundamental—more than enough to fail a radiated emissions test if it finds an antenna.

Check against the shortcut. Using t_r in place of T_{ramp} gives $f_2 = 637$ MHz and −25.6 dB: a 3.5 dB over-estimate of this harmonic, from the edge definition alone.

2.4 Propagation delay

A signal on a board travels as an electromagnetic wave guided by the trace and its reference plane. In a homogeneous dielectric of relative permittivity ε_r the wave velocity is

$$v_p = \frac{c}{\sqrt{\varepsilon_r}}, \qquad c = 2.998 \times 10^8\ \text{m/s} = 11.80\ \text{in/ns}, \tag{2.9}$$

and the delay per unit length is

$$t_{pd} = \frac{1}{v_p} = \frac{\sqrt{\varepsilon_r}}{c} \approx 84.7\ \text{ps/in} \times \sqrt{\varepsilon_r}. \tag{2.10}$$

A stripline is embedded in a homogeneous dielectric, so its ε_r is simply the laminate D_k. A microstrip has field lines partly in air and partly in the dielectric, so an *effective dielectric constant* ε_{eff}, lying between 1 and D_k, replaces ε_r; it is computed in Chapter 9. Soldermask raises ε_{eff} of a microstrip slightly.

Rule of Thumb: Six inches per nanosecond

In FR-4 stripline ($D_k \approx 3.6$ to 4.2) signals travel at roughly 5.8–6.2 in/ns, i.e. 160–175 ps/in. Outer-layer microstrip is faster, typically 140–150 ps/in. *Limits:* D_k varies with resin content, glass style, and frequency (Chapters 9 and 11); for timing budgets tighter than about 10 %, use the fabricator's D_k at the frequency of interest.

The one-way delay of a line of length ℓ is $T_D = \ell\, t_{pd}$. A useful companion quantity is the *spatial extent of the edge*, or rise length, $\ell_r = t_r/t_{pd}$: the physical distance occupied by the rising edge as it travels down the line. A 100 ps edge in FR-4 stripline is only about 0.6 in long.

2.5 Lumped versus distributed behavior

If the edge is long compared with the interconnect, the whole trace sees essentially the same voltage at every instant, and it behaves like a lumped capacitor (plus, perhaps, a lumped inductor). If the edge is short compared with the trace, different parts of the trace are at different voltages at the same instant; the trace must then be analyzed as a transmission line (Chapter 4), and impedance discontinuities produce reflections (Chapter 7).

The boundary between the two regimes is expressed as a *critical length*

$$\ell_{crit} = \frac{t_r}{k\, t_{pd}}, \tag{2.11}$$

where k is a safety factor. Johnson and Graham treat a line as lumped when its delay is less than about one-sixth of the rise time [1], i.e. $k = 6$; other authors use values from 4 to 10 [3], [4]. The choice depends on how much ringing the receiver can tolerate. Table 2.1 summarizes common criteria, and Figure 2.4 plots Eq. (2.11) for typical board materials.

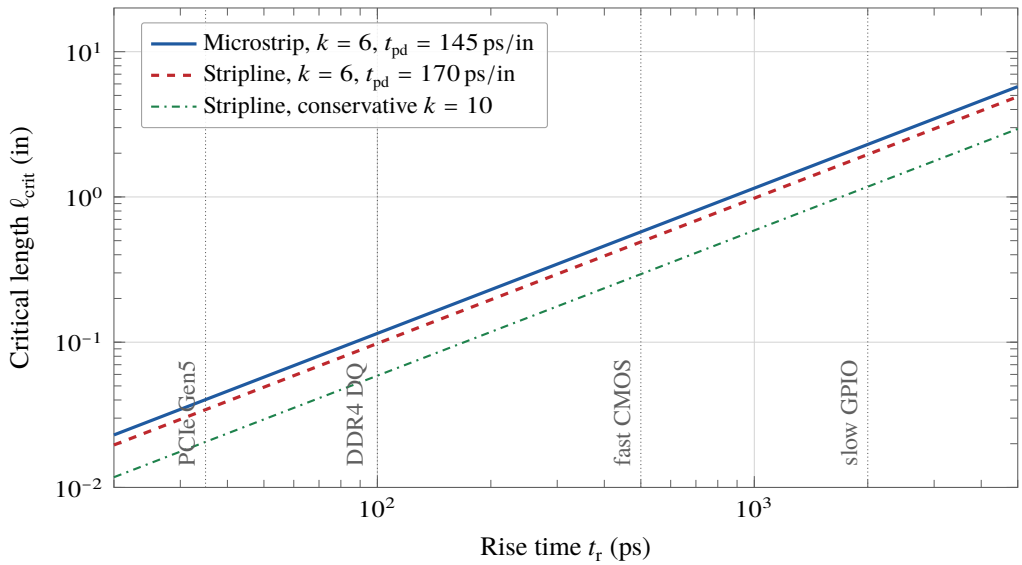


Figure 2.4. Critical length versus rise time from Eq. (2.11). Lines shorter than ℓ_{crit} may be modeled as lumped elements; longer lines need transmission-line treatment and, usually, termination. Vertical lines mark representative edge rates.

Table 2.1. Criteria for deciding when an interconnect is electrically long.

Condition	Interpretation	Typical use
$T_D < t_r/10$	Line is negligible; treat as lumped C	Conservative, analog edges
$T_D < t_r/6$	Lumped model adequate; ringing small	General digital ($k = 6$)
$T_D < t_r/4$	Reflections overlap the edge; mild ringing	Short stubs, breakouts
$T_D < t_r/2$	Round-trip delay shorter than edge; termination often optional	Low-speed buses
$T_D \geq t_r/2$	Reflections arrive after the edge; termination needed	Most high-speed nets

Worked Example 2.4: Is a DDR4 stub lumped?

A DDR4 data line uses $t_r = 100\,\text{ps}$ edges and is routed in stripline with $D_k = 3.8$. What is the critical length with $k = 6$, and is a 150 mil via-and-breakout stub electrically short?

Solution. From Eq. (2.10), $t_{pd} = 84.7\,\text{ps/in} \times \sqrt{3.8} = 165.2\,\text{ps/in}$. Then $\ell_{crit} = 100\,\text{ps}/(6 \times 165.2\,\text{ps/in}) = 0.101\,\text{in}$, about 100 mil or 2.6 mm. The 150 mil stub exceeds this; its effect on the edge will be visible as a reflection, and it should be either shortened or included in simulation. The rise length is $t_r/t_{pd} = 0.61\,\text{in}$.

Design Rule: When to treat a net as a transmission line

Compute $\ell_{crit} = t_r/(6t_{pd})$ using the fastest edge the driver can produce (the "fast/strong" IBIS corner), not the nominal one. Any net longer than ℓ_{crit} needs a controlled impedance and an explicit termination strategy; any stub, branch, or via longer than ℓ_{crit} must be included in the SI model.

2.6 Data rate, unit interval, and Nyquist frequency

Serial and memory interfaces are described by their data rate rather than by a clock. For non-return-to-zero (NRZ) signaling at R_b bits per second, the *unit interval* is $\text{UI} = 1/R_b$ and the highest fundamental frequency, produced by an alternating 1010 pattern, is the Nyquist frequency

$$f_{Nyq} = \frac{R_b}{2} = \frac{1}{2\,\text{UI}}. \tag{2.12}$$

Channel loss budgets for serial links are specified at f_{Nyq} (Chapter 29). Edge rates in such links are usually a sizeable fraction of a UI, so their knee frequency lies at two to three times f_{Nyq}; this is why channel models are routinely required to two or three times the Nyquist frequency. For multi-level signaling such as PAM4, the symbol rate (baud), not the bit rate, sets the UI. Table 2.2 lists the interfaces covered in Part V.

2.7 A checklist for deciding what matters

The concepts of this chapter combine into a quick triage that should be run on every net class at the start of a design:

1. Find the fastest edge rate (t_r) of each driver from its IBIS model or data sheet.
2. Compute $f_{knee} = 0.5/t_r$. All models of the channel (vias, connectors, packages) must be valid to at least f_{knee}.

Table 2.2. Unit interval and Nyquist frequency of representative interfaces (NRZ unless noted). Edge rates are implementation dependent and are not listed.

Interface	Rate (GT/s)	UI (ps)	f_{Nyq} (GHz)
1000BASE-T (per pair, PAM5)	0.125	8000	0.0625
MIPI D-PHY (one lane)	2.5	400	1.25
DDR4-3200	3.2	312.5	1.6
LPDDR5-6400	6.4	156.25	3.2
PCIe Gen3	8.0	125	4.0
USB 3.2 Gen 2	10.0	100	5.0
10GBASE-KR	10.3125	96.97	5.156
PCIe Gen4	16.0	62.5	8.0
PCIe Gen5	32.0	31.25	16.0
PCIe Gen6 (PAM4, 32 GBd)	64.0	31.25	16.0

3. Compute ℓ_{crit} from Eq. (2.11) for the intended routing layer. Nets longer than ℓ_{crit} need controlled impedance and termination; stubs longer than ℓ_{crit} need simulation.
4. For serial links, compute f_{Nyq} and the loss budget at f_{Nyq} (Chapters 11 and 29).
5. Estimate the harmonic envelope (Eq. (2.8)) of clocks and periodic buses; these dominate radiated emissions.

Key Takeaways

- Edge rate, not clock frequency, determines whether a design is high-speed.
- $f_{3\mathrm{dB}} \approx 0.35/t_{\mathrm{r}}$ for a single-pole edge; the knee frequency $f_{\mathrm{knee}} \approx 0.5/t_{\mathrm{r}}$ bounds its significant spectral content.
- The spectrum of a trapezoidal waveform is flat to $1/(\pi\tau)$, falls at $-20\,\mathrm{dB/decade}$ to $1/(\pi T_{\mathrm{ramp}})$, then at $-40\,\mathrm{dB/decade}$; for a linear edge $T_{\mathrm{ramp}} = t_{\mathrm{r}}/0.8$.
- Signals travel at about 6 in/ns in FR-4 stripline; $t_{\mathrm{pd}} \approx 84.7\,\mathrm{ps/in} \times \sqrt{\varepsilon_{\mathrm{r}}}$.
- A line longer than $\ell_{\mathrm{crit}} = t_{\mathrm{r}}/(6t_{\mathrm{pd}})$ is a transmission line. With a 100 ps edge, that is about 0.1 in.
- Rise times of cascaded Gaussian-like stages add in root-sum-square fashion; measurement bandwidth must be several times the signal bandwidth.

Further Reading

Johnson and Graham [1] remain the classic source on edge rates, knee frequency, and lumped versus distributed behavior; Bogatin [3] develops the same ideas with many quantitative rules of thumb, and Dally and Poulton [35] place them in the context of complete digital systems.

Review Problems

P2.1 A PCIe Gen5 transmitter produces edges with $t_{\mathrm{r}} = 35\,\mathrm{ps}$. Estimate the $-3\,\mathrm{dB}$ bandwidth and the knee frequency.

P2.2 A 50 MHz clock with $t_r = 2$ ns travels 4 in on a microstrip with $t_{pd} = 145$ ps/in. Find T_D and ℓ_{crit} ($k = 6$). Is the line lumped?

P2.3 An edge with a true rise time of 60 ps is measured on an 8 GHz oscilloscope. What rise time is displayed?

P2.4 Find the per-inch delay of a stripline in a laminate with $D_k = 3.4$, and the delay of a 7 in route.

P2.5 A 25 MHz, 50 % duty-cycle clock has linear edges with $t_r = 1$ ns (10–90 %). Find the two envelope corner frequencies and the knee frequency.

P2.6 A link runs at 16 GT/s NRZ with edges equal to 40 % of a UI. Find the UI, the Nyquist frequency, and the knee frequency.

Chapter 3

Electromagnetic Fundamentals for Board Designers

Chapter Overview

Every signal-integrity, power-integrity, and EMI problem on a board can be traced back to a few electromagnetic quantities: capacitance, inductance (and in particular *loop* inductance), the path taken by return current, and the way current crowds toward conductor surfaces at high frequency. This chapter develops these ideas with the minimum of field theory needed to apply them, and gives the closed-form estimates used throughout the rest of the book.

Learning Objectives

1. Distinguish self, mutual, partial, and loop inductance, and compute loop inductance for common board structures.
2. Estimate parallel-plate and trace capacitance and the displacement current they carry.
3. Explain why high-frequency return current flows directly beneath a signal trace, and quantify how it spreads.
4. Compute skin depth and the resulting increase in conductor resistance.
5. Identify the frequency at which the return path changes from the lowest-resistance to the lowest-inductance route.

3.1 Every signal is a current loop

A signal does not "go" from a driver to a receiver along a single trace. The driver pushes current into the signal conductor and an equal current returns to the driver through some other conductor—a reference plane, a neighboring trace, a cable shield. Kirchhoff's current law makes this unavoidable, and at high frequency the return path is as much a part of the signal as the trace itself. Most of the failures examined in this book (ground bounce, crosstalk, emission from plane splits, reference-plane changes) are failures of the return path, not of the signal trace [3], [5].

The two conductors and the space between them form a loop. The loop stores energy in two forms: in the electric field between the conductors (described by capacitance) and in the magnetic field enclosed by the loop (described by inductance). Both depend only on geometry and materials.

3.2 Capacitance

3.2.1 Parallel-plate capacitance

Two conducting planes of area A separated by a dielectric of thickness h and relative permittivity ε_r have capacitance

$$C = \frac{\varepsilon_0 \, \varepsilon_r A}{h}, \qquad \varepsilon_0 = 8.854\,\text{pF/m} = 0.2249\,\text{pF/in}. \tag{3.1}$$

Equation (3.1) ignores fringing fields at the edges, which is accurate when the lateral dimensions are much larger than h—true for power and ground planes, which is where it is most often used.

Worked Example 3.1: Capacitance of a power–ground plane pair

A 4 in × 4 in region of a 0.8 V core plane is separated from the adjacent ground plane by a 3 mil core with $D_k = 4.2$. Find the plane capacitance.

Solution. With $A = 16\,\text{in}^2$ and $h = 0.003$ in, Eq. (3.1) gives $C = 0.2249\,\text{pF/in} \times 4.2 \times 16/0.003 = 5.04\,\text{nF}$. This capacitance has extremely low inductance and dominates the PDN impedance in the hundreds-of-megahertz range (Chapter 21).

3.2.2 Capacitance of a trace

A trace over a plane has a capacitance per unit length C' that includes the parallel-plate region under the trace plus fringing fields from its edges and top. For typical 50 Ω lines the fringing contribution is comparable to the parallel-plate term, so Eq. (3.1) applied to the trace width alone underestimates C' by roughly a factor of two. Accurate values come from a 2-D field solver or from the impedance relations of Chapter 4: a 50 Ω line in FR-4 has $C' \approx 3.3\,\text{pF/in}$.

3.2.3 Displacement current

When the voltage across a capacitance changes, a displacement current

$$i = C\frac{dv}{dt} \tag{3.2}$$

flows through the dielectric. This is how a signal's return current crosses from one plane to another at a layer transition (Chapter 23) and how a fast edge couples capacitively into a neighbor (Chapter 12). A 1 V edge with $t_r = 100\,\text{ps}$ across 1 pF of stray capacitance drives 10 mA.

3.3 Inductance

3.3.1 Definitions

Inductance is the number of magnetic field lines (the flux Φ) linking a current path, per ampere of current:

$$L = \frac{\Phi}{I}. \tag{3.3}$$

A changing current changes the flux, which by Faraday's law induces a voltage

$$v = L\frac{di}{dt}. \tag{3.4}$$

Several kinds of inductance must be distinguished [36]:

Loop inductance
: The flux through a *closed* current path per ampere. Only loop inductance is physically measurable, and only loop inductance appears in Faraday's law.

Partial self inductance
A bookkeeping quantity assigned to one *segment* of a loop, defined so that the loop inductance equals the sum of the partial self inductances of its segments minus the partial mutual inductances between them. It allows the inductance of a via or a pin to be quoted independently of the rest of its loop.

Mutual inductance
The flux from one current that links a second conductor, per ampere of the first current. It is the mechanism of inductive crosstalk (Chapter 12) and of ground bounce (Chapter 20).

For a loop made of an outgoing conductor (1) and a returning conductor (2) carrying equal and opposite currents,

$$L_{\text{loop}} = L_{11} + L_{22} - 2L_{12}, \tag{3.5}$$

where L_{11} and L_{22} are partial self inductances and L_{12} is the partial mutual inductance. Bringing the return conductor closer increases L_{12} and reduces L_{loop}. This single equation explains why every high-speed design rule about return paths says the same thing: keep the return conductor as close as possible to the signal conductor (Figure 3.1).

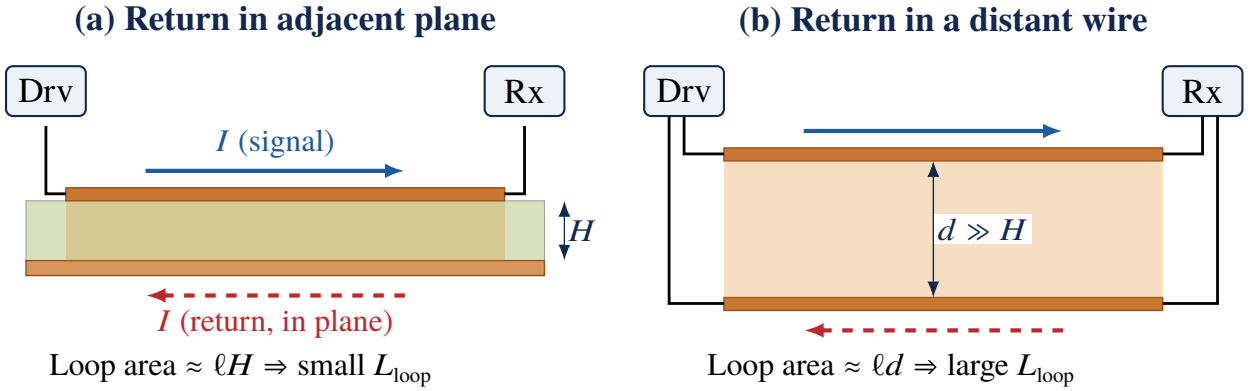


Figure 3.1. The loop formed by a signal and its return current. (a) With an adjacent reference plane, the loop area, and therefore the loop inductance, is set by the dielectric height H. (b) A distant return conductor encloses a much larger area and has much higher loop inductance.

3.3.2 Useful closed-form estimates

For a straight round wire of length ℓ and radius r, with $\ell \gg r$, the high-frequency partial self inductance is approximately

$$L_{\text{partial}} \approx \frac{\mu_0 \ell}{2\pi}\left[\ln\frac{2\ell}{r} - 1\right], \tag{3.6}$$

and the partial mutual inductance between two parallel wires of length ℓ separated by s (with $\ell \gg s$) is

$$M \approx \frac{\mu_0 \ell}{2\pi}\left[\ln\frac{2\ell}{s} - 1\right]. \tag{3.7}$$

Substituting Eqs. (3.6) and (3.7) into Eq. (3.5) gives the loop inductance of a two-wire line,

$$L_{\text{loop}} \approx \frac{\mu_0 \ell}{\pi}\ln\frac{s}{r}, \qquad s \gg r, \tag{3.8}$$

which also describes a pair of vias carrying equal and opposite current—the dominant inductance of a decoupling capacitor's mounting (Chapter 21). Note that $\mu_0/(2\pi) = 0.2\,\text{nH/mm} = 5.08\,\text{nH/in}$.

Rule of Thumb: Inductance of a wire

The partial self inductance of a thin wire or trace is roughly 1 nH/mm (25 nH/in). *Limits:* Eq. (3.6) shows the value depends logarithmically on ℓ/r; a 1 in wire of 0.5 mm radius has 18 nH, and thicker or shorter conductors have less. Use it only for order-of-magnitude estimates.

For two parallel planes separated by h, current spreading uniformly across a square region of any size sees the same inductance, the *sheet inductance*:

$$L_{\square} = \mu_0 h \approx 32\,\text{pH per square per mil of } h. \tag{3.9}$$

Thin dielectrics between power and ground planes therefore reduce both the spreading inductance and the impedance of the PDN.

Worked Example 3.2: Loop inductance of a via pair

A ground via and a power via pass through a 62 mil board. Each has a drill diameter of 10 mil ($r = 5$ mil) and they are 50 mil apart. Estimate the loop inductance if current enters on the top layer and leaves on the bottom layer.

Solution. From Eq. (3.8) with $\ell = 62\,\text{mil} = 1.575\,\text{mm}$: $L_{\text{loop}} = (\mu_0/\pi)\ell \ln(s/r) = 0.4\,\text{nH/mm} \times 1.575\,\text{mm} \times \ln 10 = 1.45\,\text{nH}$. If the same via pair connected a capacitor on the top layer to planes only 5 mil below, the inductance would scale with height, to about 0.12 nH. This is why decoupling capacitors should be mounted on the side of the board closest to the power–ground plane pair.

3.3.3 Internal and external inductance

Part of a conductor's inductance comes from the magnetic field *inside* the conductor. At low frequency, when current is uniform across the cross-section, this internal inductance is $\mu_0/(8\pi) = 50\,\text{nH/m}$ for a round wire. As frequency rises and current is pushed to the surface (Section 3.5), the internal field and the internal inductance shrink. The total inductance of a trace therefore decreases slightly with frequency, which is one reason a lossy line's characteristic impedance is frequency dependent (Chapter 4).

3.4 Return current distribution

3.4.1 Current density beneath a microstrip

At high frequency the return current in a plane beneath a trace concentrates directly under the trace, because that distribution minimizes the loop inductance and therefore the total impedance of the loop. For a trace at height H above an infinite plane, image theory gives the current density in the plane as a function of lateral distance x from the trace centerline [1]:

$$J(x) = \frac{I}{\pi H}\,\frac{1}{1 + (x/H)^2}. \tag{3.10}$$

Integrating Eq. (3.10) from $-D$ to D gives the fraction of the return current within $\pm D$ of the centerline:

$$\frac{I(|x| < D)}{I} = \frac{2}{\pi}\arctan\frac{D}{H}. \tag{3.11}$$

Half the return current lies within $\pm H$, 79.5 % within $\pm 3H$, and 87.4 % within $\pm 5H$ (Figure 3.2). Equation (3.10) treats the trace as a thin wire; for real traces of width W comparable to H, the peak is broader and lower, but the long tails, and the conclusion, are the same.

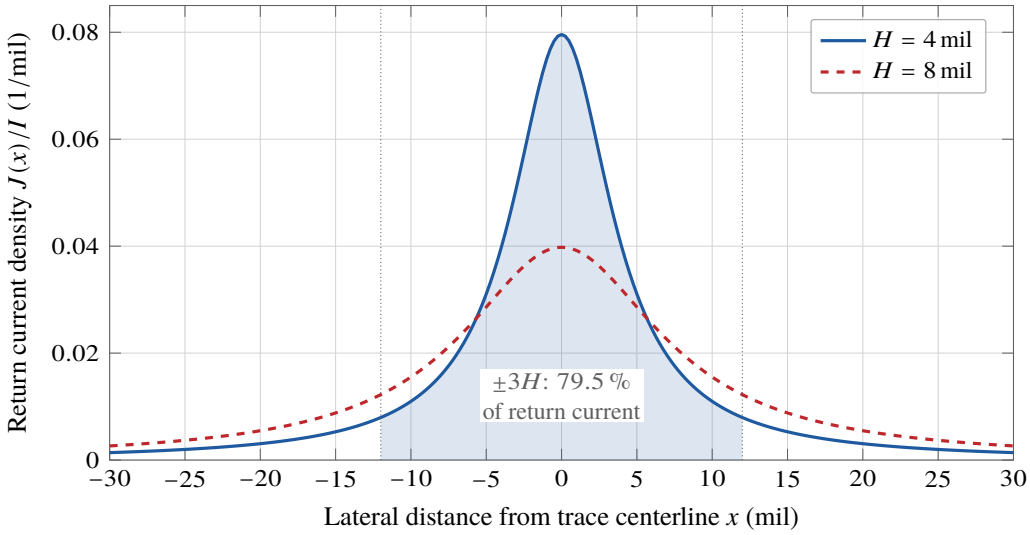


Figure 3.2. Return current density in the reference plane under a microstrip, from Eq. (3.10). A thinner dielectric concentrates the return current more tightly beneath the trace. The shaded band, $\pm 3H$ for $H = 4$ mil, carries 79.5 % of the return current.

Design Rule: Keep the plane intact under the trace

Treat the plane region within $\pm 3H$ to $\pm 5H$ of every high-speed trace as part of the signal path. Slots, anti-pad clusters, and plane splits inside this band disturb the return current and raise loop inductance, crosstalk, and emissions (Chapter 23).

3.4.2 Lowest resistance versus lowest inductance

The return current divides itself among the available paths to minimize total impedance $R + j\omega L$. At low frequency the resistive term dominates and the current spreads across the plane along the lowest-resistance route—roughly a straight line between the load and the driver. At high frequency the inductive term dominates and the current follows the lowest-inductance route, directly beneath the signal trace (Figure 3.3). The transition occurs where $\omega L \approx R$ for the loop, typically between a few kilohertz and a few hundred kilohertz for board-level loops. Every digital edge has most of its energy far above this transition, so for signal integrity the return current always follows the trace.

Common Pitfall: Routing "ground" as if it were a wire

Assuming that because all ground nets are connected, the return path will take care of itself. A slot cut in the plane under a trace forces the high-frequency return current around the slot. The added loop area shows up as inductance in series with the signal, as crosstalk to every other trace whose return shares the detour, and as a slot antenna.

3.5 Skin effect

3.5.1 Skin depth

A time-varying current inside a conductor generates a time-varying magnetic field, which induces eddy currents that oppose current flow in the interior. The result is that current density decays exponentially

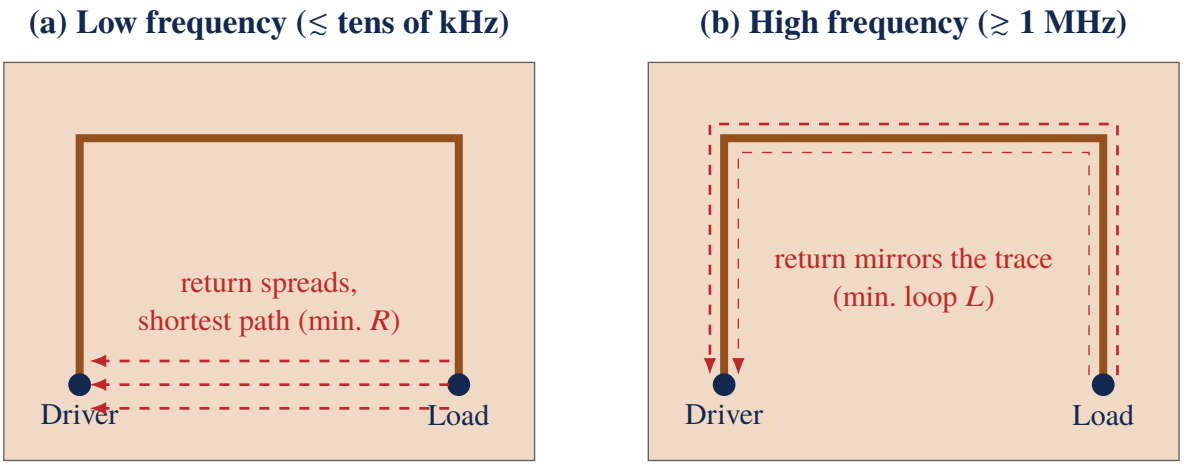


Figure 3.3. Top view of a U-shaped trace over a plane. (a) At low frequency the return current takes the shortest, lowest-resistance path across the plane. (b) At high frequency it follows directly beneath the trace, the lowest-inductance path.

with depth below the surface, with characteristic length—the *skin depth*—

$$\delta_s = \sqrt{\frac{\rho}{\pi f \mu_0 \mu_r}} = \frac{1}{\sqrt{\pi f \mu_0 \mu_r \sigma}}, \tag{3.12}$$

where $\rho = 1/\sigma$ is the resistivity. For annealed copper ($\rho = 1.724 \times 10^{-8}\ \Omega \cdot \text{m}$ at 20 °C, $\mu_r = 1$),

$$\delta_s \approx \frac{66.1\ \mu\text{m}}{\sqrt{f/\text{MHz}}}. \tag{3.13}$$

At 1 GHz the skin depth is 2.09 µm, about one-seventeenth of the thickness of 1 oz copper (Figure 3.4). Resistivity rises with temperature by about 0.39 %/°C, so the skin depth of a hot board is slightly larger. Ferromagnetic plating layers such as electroless nickel ($\mu_r \gg 1$) have a much smaller skin depth, which is one reason ENIG finish adds loss on outer-layer traces at high frequency.

3.5.2 AC resistance

When δ_s is much less than the conductor thickness, current flows in a layer roughly one skin depth thick on the surfaces facing the return current. For a stripline, with current on both the top and bottom faces of a trace of width W, the resistance per unit length is approximately

$$R'_{ac} \approx \frac{\rho}{2W\delta_s} \qquad \text{(signal conductor only)}, \tag{3.14}$$

compared with the DC value $R'_{dc} = \rho/(WT)$. Because $\delta_s \propto f^{-1/2}$, $R'_{ac} \propto \sqrt{f}$. The return planes carry the same current in a similar skin layer, so the total loop resistance is larger than Eq. (3.14); for a stripline it is close to twice the signal-conductor value. Edge effects and surface roughness raise it further (Chapter 11).

Worked Example 3.3: AC resistance of a stripline trace

A 5 mil wide, 1 oz (34.8 µm) stripline trace is 1 in long. Compare its DC resistance with its signal-conductor AC resistance at 1 GHz.

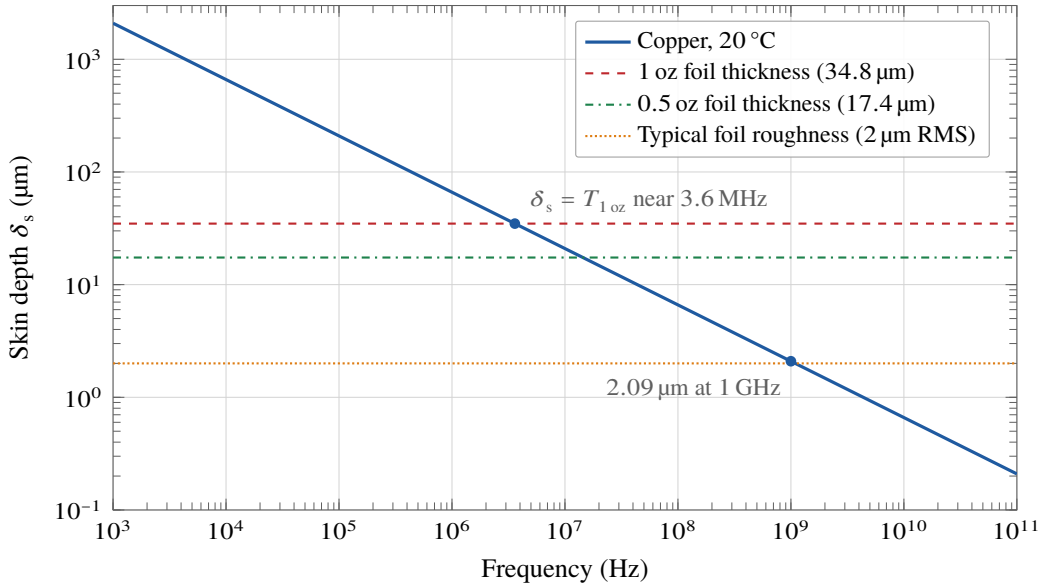


Figure 3.4. Skin depth of copper versus frequency, from Eq. (3.12). Above about 4 MHz the skin depth is smaller than the thickness of 1 oz foil; above a few gigahertz it is comparable to the surface roughness of the foil.

Solution. $W = 127\,\mu\text{m}$ and $\ell = 25.4\,\text{mm}$. DC: $R = \rho\ell/(WT) = 1.724 \times 10^{-8} \times 0.0254/(127 \times 10^{-6} \times 34.8 \times 10^{-6}) = 0.099\,\Omega$. AC: with $\delta_s = 2.09\,\mu\text{m}$, Eq. (3.14) gives $R = \rho\ell/(2W\delta_s) = 0.825\,\Omega$, about 8.3 times the DC value. Adding a comparable return-plane contribution, the loop resistance is about $1.6\,\Omega/\text{in}$ at 1 GHz.

3.6 Summary of field quantities

Table 3.1. Electromagnetic quantities and closed-form estimates used in board design.

Quantity	Estimate	Validity
Plane capacitance	$C = \varepsilon_0\varepsilon_r A/h$; $0.2249\,\text{pF/in} \times \varepsilon_r A/h$	Lateral size $\gg h$
Sheet inductance	$L_\square = \mu_0 h$; 32 pH per square per mil	Plane pairs
Wire partial L	$(\mu_0\ell/2\pi)[\ln(2\ell/r) - 1]$	$\ell \gg r$
Two-wire loop L	$(\mu_0\ell/\pi)\ln(s/r)$	$s \gg r$, $\ell \gg s$
Return current spread	$J(x) \propto 1/(1 + (x/H)^2)$; 80 % within $\pm 3H$	Thin trace over plane
Skin depth (Cu)	$66.1\,\mu\text{m}/\sqrt{f/\text{MHz}}$	$\mu_r = 1$, 20 °C
Stripline R_{ac}	$\rho/(2W\delta_s)$ per length (signal only)	$\delta_s \ll T$, smooth Cu

Key Takeaways

- Every signal is a current loop; the return path is part of the signal.

- Only loop inductance is physical: $L_{\text{loop}} = L_{11} + L_{22} - 2L_{12}$. Bringing the return closer raises mutual inductance and lowers L_{loop}.
- Plane pairs have sheet inductance $\mu_0 h$ (32 pH per square per mil) and capacitance $\varepsilon_0 \varepsilon_r A/h$; thin dielectrics help both.
- High-frequency return current flows beneath the trace, with about 80 % within $\pm 3H$.
- Above a few kilohertz to a few hundred kilohertz the return current follows the trace; below, it takes the lowest-resistance path.
- Skin depth in copper is 2.09 µm at 1 GHz and scales as $f^{-1/2}$, so AC resistance grows as $\sqrt{f}$.

Further Reading

Paul [36] gives a rigorous treatment of loop and partial inductance; Ott [5] and Johnson and Graham [1] discuss return current distribution and its consequences for layout; Ramo, Whinnery, and Van Duzer [37] cover the underlying field theory, including skin effect.

Review Problems

P3.1 Compute the skin depth of copper at 100 MHz and at 10 GHz.

P3.2 Find the capacitance of a 6 in × 4 in plane pair separated by 2 mil of dielectric with $D_k = 4.0$.

P3.3 What fraction of the return current under a thin microstrip flows within $\pm 2H$ of the centerline? Within $\pm 5H$?

P3.4 Find the sheet inductance of a plane pair separated by 4 mil.

P3.5 Two vias, each of radius 4 mil, are 40 mil apart and carry opposite currents over a height of 40 mil. Estimate their loop inductance.

P3.6 A current of 20 mA switches in 100 ps through a loop inductance of 2 nH. What voltage is induced?

P3.7 Find the DC resistance of a 10 in long, 4 mil wide trace in 0.5 oz (17.4 µm) copper.

Chapter 4

Transmission Line Theory

Chapter Overview

When an interconnect is longer than its critical length, it must be analyzed as a transmission line. This chapter derives the telegrapher's equations from the distributed RLGC model, obtains the characteristic impedance and propagation constant, specializes them to lossless and low-loss lines, and develops the input-impedance formula that governs stubs, resonances, and matching. These results are the foundation of every later chapter on signal integrity.

Learning Objectives

1. Derive the telegrapher's equations from a distributed RLGC ladder.
2. Compute characteristic impedance, propagation velocity, and per-unit-length L' and C' from one another.
3. Separate conductor and dielectric attenuation in a low-loss line and express them in dB/in.
4. Use the input-impedance formula to analyze shorted, open, matched, and quarter-wave lines.
5. Explain why the characteristic impedance of a real line varies with frequency.

4.1 The distributed RLGC model

A pair of conductors with uniform cross-section—a trace and its plane, or the two traces of a differential pair—can be described by four per-unit-length parameters, and modeled by the circuits of Figure 4.1:

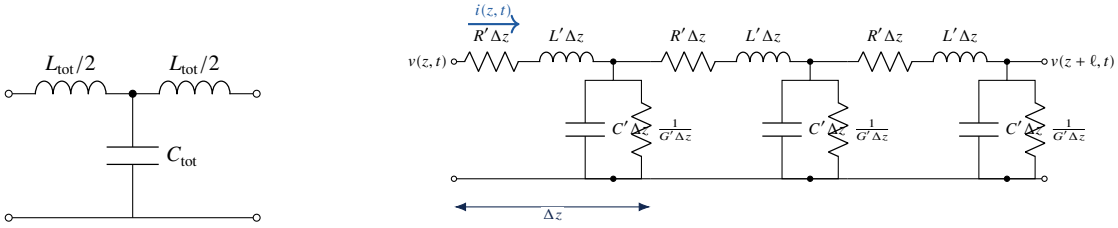
R' (Ω/m)
: series resistance of both conductors, rising as $\sqrt{f}$ above the skin-effect onset (Chapter 3);

L' (H/m)
: series loop inductance;

G' (S/m)
: shunt conductance of the dielectric, $G' = \omega C' \tan\delta$, which rises in proportion to frequency;

C' (F/m)
: shunt capacitance between the conductors.

A line of length ℓ that is electrically short can be modeled by lumped totals $L'\ell$ and $C'\ell$ (Figure 4.1a). A long line is modeled by a cascade of infinitesimal cells of length Δz (Figure 4.1b). In practice a ladder of N lumped cells is accurate up to a frequency at which each cell is shorter than about one-tenth of a wavelength, i.e. $N \gtrsim 10\, T_{\mathrm{D}} f_{\max}$; equivalently, each cell's delay must be short compared with t_{r}.



(a) Lumped T model ($T_{\text{D}} \ll t_{\text{r}}$) **(b)** Distributed RLGC ladder ($\Delta z \to 0$)

Figure 4.1. Circuit models of an interconnect. (a) An electrically short line is a lumped inductance and capacitance. (b) A long line is a ladder of RLGC cells; in the limit $\Delta z \to 0$ it becomes the distributed model described by the telegrapher's equations.

4.2 The telegrapher's equations

Applying Kirchhoff's voltage law around one cell of Figure 4.1b and Kirchhoff's current law at its shunt node gives, in the limit $\Delta z \to 0$,

$$\frac{\partial v(z,t)}{\partial z} = -R'\, i(z,t) - L' \frac{\partial i(z,t)}{\partial t}, \tag{4.1}$$

$$\frac{\partial i(z,t)}{\partial z} = -G'\, v(z,t) - C' \frac{\partial v(z,t)}{\partial t}. \tag{4.2}$$

For sinusoidal steady state with phasors $V(z)$ and $I(z)$ at angular frequency ω, these become

$$\frac{\mathrm{d}V}{\mathrm{d}z} = -(R' + \mathrm{j}\omega L')\, I, \qquad \frac{\mathrm{d}I}{\mathrm{d}z} = -(G' + \mathrm{j}\omega C')\, V. \tag{4.3}$$

Differentiating the first and substituting the second gives the wave equation

$$\frac{\mathrm{d}^2 V}{\mathrm{d}z^2} = \gamma^2 V, \qquad \gamma = \alpha + \mathrm{j}\beta = \sqrt{(R' + \mathrm{j}\omega L')(G' + \mathrm{j}\omega C')}, \tag{4.4}$$

where γ is the *propagation constant*, α the *attenuation constant* (Np/m), and β the *phase constant* (rad/m). The general solution is a sum of forward and backward waves,

$$V(z) = V^{+}\mathrm{e}^{-\gamma z} + V^{-}\mathrm{e}^{+\gamma z}, \qquad I(z) = \frac{1}{Z_0}\left(V^{+}\mathrm{e}^{-\gamma z} - V^{-}\mathrm{e}^{+\gamma z}\right), \tag{4.5}$$

and substituting Eq. (4.5) back into Eq. (4.3) identifies the ratio of voltage to current in either wave as the *characteristic impedance*

$$Z_0 = \sqrt{\frac{R' + \mathrm{j}\omega L'}{G' + \mathrm{j}\omega C'}}. \tag{4.6}$$

Equation (4.6) is a property of the cross-section only; it does not depend on the length of the line or on what is connected to it. It is the impedance a driver sees looking into the line until the first reflection returns, and it is the impedance a wave "sees" as it travels. A line terminated in Z_0 looks infinitely long.

4.3 Lossless lines

When $R' \ll \omega L'$ and $G' \ll \omega C'$, which holds for board traces above a few megahertz (Section 4.5), the line is effectively lossless and

$$Z_0 = \sqrt{\frac{L'}{C'}}, \tag{4.7}$$

$$\gamma = \mathrm{j}\beta = \mathrm{j}\omega\sqrt{L'C'}, \qquad v_{\text{p}} = \frac{\omega}{\beta} = \frac{1}{\sqrt{L'C'}}. \tag{4.8}$$

For a TEM line in a homogeneous dielectric, $L'C' = \mu_0\varepsilon_0\varepsilon_r$, so $v_p = c/\sqrt{\varepsilon_r}$, recovering Eq. (2.9). Combining Eqs. (4.7) and (4.8) yields relations that let any two of $\{Z_0, t_{pd}, L', C'\}$ determine the other two:

$$C' = \frac{t_{pd}}{Z_0} = \frac{1}{v_p Z_0}, \qquad L' = Z_0\, t_{pd} = \frac{Z_0}{v_p}. \tag{4.9}$$

Multiplying by the length gives the totals for a line of one-way delay $T_D = \ell\, t_{pd}$:

$$C_{tot} = C'\ell = \frac{T_D}{Z_0}, \qquad L_{tot} = L'\ell = Z_0\, T_D. \tag{4.10}$$

These are the values a lumped model would use; Section 4.6 shows how long they remain valid.

Three ways to the same delay. T_D is one-way and t_{pd} is per unit length; the two are the most frequently confused quantities in this subject (Section 1.5). Any of three routes gives T_D, and using two of them is a useful check:

$$T_D = \ell\, t_{pd} = \frac{\ell\sqrt{\varepsilon_{eff}}}{c} \approx 84.73\ \text{ps/in} \times \ell_{in}\sqrt{\varepsilon_{eff}} = \ell\sqrt{L'C'}. \tag{4.11}$$

For 3 in of stripline with $D_k = 4.0$: $t_{pd} = 169.5$ ps/in gives 508 ps; $3 \times 84.73 \times 2 = 508$ ps; and with the L' and C' of Worked Example 4.1, $3\sqrt{8.47\times10^{-9} \times 3.39\times10^{-12}} = 508$ ps. A TDR would display this feature at $2T_D = 1.02$ ns.

Worked Example 4.1: RLGC parameters of a stripline

A 50 Ω stripline is built in a laminate with $D_k = 4.0$. Find t_{pd}, C', and L'.

Solution. $t_{pd} = 84.73\ \text{ps/in} \times \sqrt{4.0} = 169.5$ ps/in. From Eq. (4.9), $C' = 169.5\ \text{ps/in}/50\ \Omega = 3.39$ pF/in and $L' = 50\ \Omega \times 169.5$ ps/in $= 8.47$ nH/in. These values—roughly 3.4 pF and 8.5 nH per inch—are worth memorizing for 50 Ω FR-4 lines.

Rule of Thumb: Per-inch values of a 50 Ω line

In FR-4, a 50 Ω line has about 3.4 pF/in and 8.5 nH/in; delay is about 170 ps/in in stripline. *Limits:* microstrip has the same Z_0 with lower C' and t_{pd} (because $\varepsilon_{eff} < D_k$) but similar L'; scale with Eq. (4.9) for other impedances.

Figure 4.2 shows how a step launched into a matched lossless line propagates: the waveform moves at v_p without changing shape, occupying a length $\ell_r = t_r/t_{pd}$ along the line.

4.4 Low-loss lines and attenuation

Board traces at gigahertz frequencies satisfy $R' \ll \omega L'$ and $G' \ll \omega C'$ but the losses are not negligible over long routes. Expanding Eq. (4.4) to first order in $R'/(\omega L')$ and $G'/(\omega C')$ gives

$$\alpha \approx \frac{R'}{2Z_0} + \frac{G'Z_0}{2} = \alpha_c + \alpha_d, \qquad \beta \approx \omega\sqrt{L'C'}, \tag{4.12}$$

with Z_0 taken as the lossless value. The two terms are the *conductor loss* α_c and the *dielectric loss* α_d. Substituting $G' = \omega C' \tan\delta$ and $\sqrt{L'C'} = \sqrt{\varepsilon_r}/c$ into α_d:

$$\alpha_d = \frac{\omega\sqrt{L'C'}\,\tan\delta}{2} = \frac{\pi f\sqrt{\varepsilon_r}\,\tan\delta}{c} \quad \text{(Np/m)}. \tag{4.13}$$

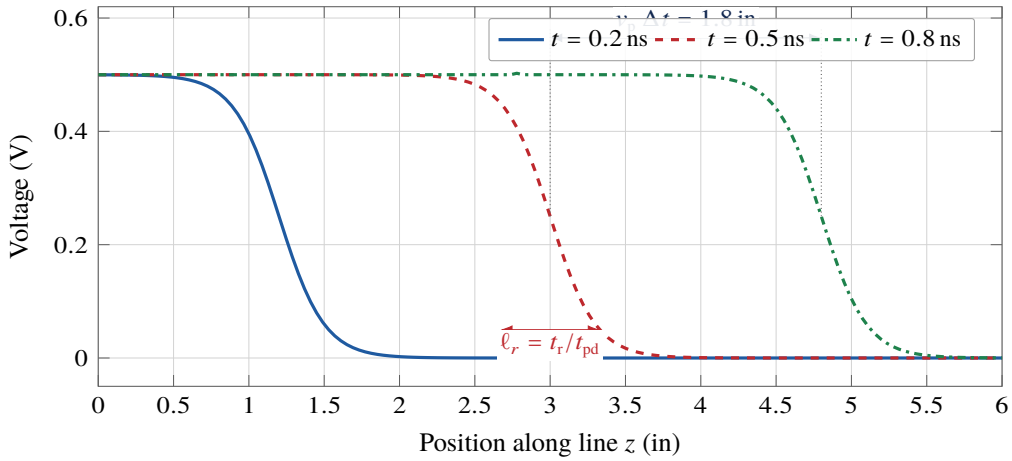


Figure 4.2. Snapshots of a 0.5 V step propagating along a lossless line at $v_p = 6$ in/ns. The edge moves without distortion and occupies a rise length $\ell_r = t_r/t_{pd}$.

Converting with 1 Np = 8.686 dB and 1 m = 39.37 in gives the practical form

$$\alpha_d\,[\mathrm{dB/in}] \approx 2.31\, f_{\mathrm{GHz}}\, D_f \sqrt{D_k}, \tag{4.14}$$

since $8.686 \times \pi \times 10^9/(2.998 \times 10^8 \times 39.37) = 2.31$. The substitution $\sqrt{L'C'} = \sqrt{\varepsilon_r}/c$ holds for a TEM line in a *homogeneous* dielectric, such as stripline; there, dielectric loss per inch is independent of geometry and cannot be reduced by changing trace width. In a microstrip, part of the field is in air, and $\sqrt{D_k}$ is replaced by an effective value that depends on the cross-section (Section 11.3). Conductor loss, by contrast, decreases with wider traces. These loss mechanisms, and the effect of copper roughness, are treated in detail in Chapter 11.

A voltage wave decays as $e^{-\alpha \ell}$, so the insertion loss of a matched line is $8.686\,\alpha\ell$ decibels. Because $\alpha_c \propto \sqrt{f}$ and $\alpha_d \propto f$, the line acts as a low-pass filter whose roll-off steepens with frequency, smearing fast edges and causing inter-symbol interference (Chapter 17).

Worked Example 4.2: Attenuation at 5 GHz

A 50 Ω stripline in a laminate with $D_k = 3.8$ and $D_f = 0.020$ has a loop resistance of 1.6 Ω/in at 1 GHz. Estimate α_c, α_d, and the insertion loss of a 10 in route at 5 GHz.

Solution. Resistance scales as $\sqrt{f}$, so $R' = 1.6\,\Omega/\mathrm{in} \times \sqrt{5} = 3.58\,\Omega/\mathrm{in}$ and $\alpha_c = R'/(2Z_0) = 0.0358\,\mathrm{Np/in} = 0.31\,\mathrm{dB/in}$. From Eq. (4.14), $\alpha_d = 2.31 \times 5 \times 0.02 \times \sqrt{3.8} = 0.45\,\mathrm{dB/in}$. The total is 0.76 dB/in, so a 10 in route loses about 7.6 dB at 5 GHz. Dielectric loss already dominates.

4.5 Frequency dependence of the characteristic impedance

At low frequency the series resistance dominates the series impedance and the shunt capacitance dominates the shunt admittance, so Eq. (4.6) gives $|Z_0| \approx \sqrt{R'/(\omega C')}$, which rises without bound as frequency falls. In this *RC region* the line behaves like a diffusive RC network, as on-chip wires do. Above the frequency at which $\omega L' = R'$, Z_0 settles to the familiar $\sqrt{L'/C'}$ (Figure 4.3). For board traces this transition occurs at a few megahertz, far below the spectral content of any fast edge, which is why the lossless value is used for all signal-integrity work. At very high frequency, the drop of internal inductance and the rise of R' make Z_0 slightly complex, by a few percent at most for typical traces.

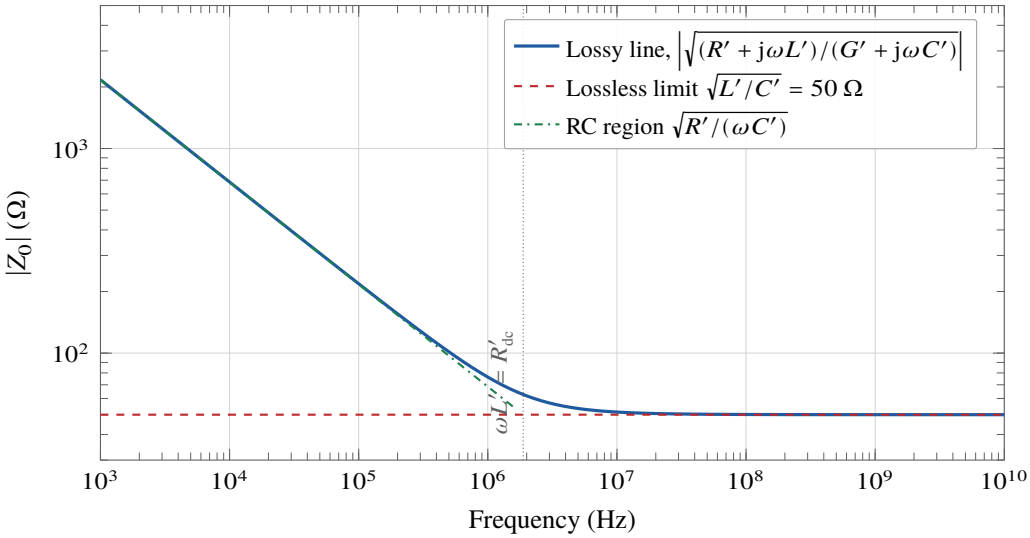


Figure 4.3. Magnitude of Z_0 for a lossy 50 Ω stripline ($R'_{dc} = 0.1\ \Omega/\text{in}$, skin-effect R', $\tan\delta = 0.02$). Below about 2 MHz the line is an RC network; above, Z_0 is essentially $\sqrt{L'/C'}$.

4.6 Input impedance of a terminated line

Consider a line of length ℓ and characteristic impedance Z_0 terminated at $z = \ell$ by a load Z_L. Applying $V(\ell)/I(\ell) = Z_L$ to Eq. (4.5) and evaluating $V(0)/I(0)$ gives the impedance seen at the input:

$$Z_{in} = Z_0 \frac{Z_L + Z_0 \tanh(\gamma\ell)}{Z_0 + Z_L \tanh(\gamma\ell)}. \tag{4.15}$$

For a lossless line, $\tanh(j\beta\ell) = j\tan(\beta\ell)$:

$$Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan\theta}{Z_0 + jZ_L \tan\theta}, \qquad \theta = \beta\ell = 2\pi f\, T_D. \tag{4.16}$$

Several special cases appear repeatedly in board design (Table 4.1 and Figure 4.4).

Table 4.1. Special cases of the input impedance of a lossless line.

Load	Length	Z_{in}	Board-level relevance
$Z_L = Z_0$	any	Z_0	Matched line; no reflection
Short	$\theta \ll 1$	$j\omega(L'\ell)$	Short shorted stub is an inductor
Open	$\theta \ll 1$	$1/[j\omega(C'\ell)]$	Short open stub is a capacitor
Open	$\lambda/4$ ($\theta = \pi/2$)	0	Via-stub notch (Chapter 15)
Short	$\lambda/4$	∞	Shorted stub blocks this frequency
Z_L	$\lambda/4$	Z_0^2/Z_L	Quarter-wave transformer
Z_L	$\lambda/2$	Z_L	Impedance repeats every half wavelength

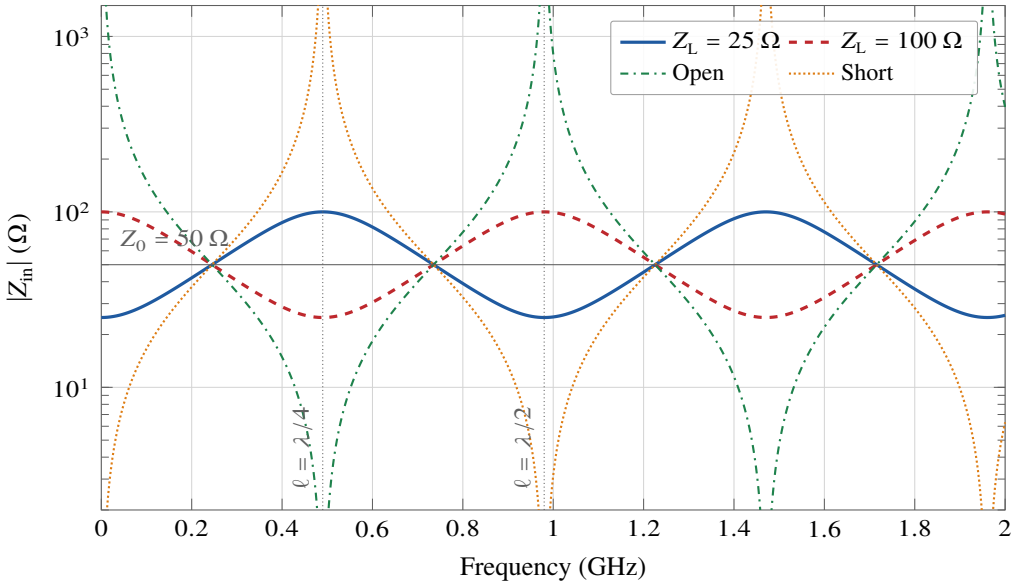


Figure 4.4. Input impedance magnitude of a lossless 50 Ω line with $T_D = 0.51$ ns (3 in of stripline) for several loads, from Eq. (4.16). Mismatched loads produce an input impedance that oscillates with period $1/(2T_D)$; open and shorted lines resonate at odd multiples of the quarter-wave frequency.

Worked Example 4.3: An open stub is not just a capacitor

A 0.5 in open-circuited stub of 50 Ω stripline ($t_{pd} = 170$ ps/in, $C' = 3.4$ pF/in) hangs off a signal net. Compare its input impedance at 2 GHz with that of its lumped capacitance.

Solution. $T_D = 85$ ps, so $\theta = 2\pi \times 2\,\text{GHz} \times 85\,\text{ps} = 1.068$ rad (61.2°). With $Z_L \to \infty$, Eq. (4.16) gives $Z_{in} = -jZ_0 \cot\theta = -j27.5\,\Omega$, equivalent to a capacitance of $1/(\omega \cdot 27.5\,\Omega) = 2.90$ pF. The lumped estimate, $C'\ell = T_D/Z_0 = 1.70$ pF, gives $1/(\omega C'\ell) = -j46.8\,\Omega$, too high by 70 %: at this frequency the stub is already approaching resonance. At $f = 1/(4T_D) = 2.94$ GHz it becomes a short circuit across the signal.

Quick estimate	Full model	Difference
Lumped: −j46.8 Ω	Exact: −j27.5 Ω	70 %

4.6.1 When does a stub become distributed?

The example generalizes. For an open stub, $Z_{in} = -jZ_0 \cot\theta$, so the equivalent capacitance at one frequency is $C_{eff} = \tan\theta/(\omega Z_0)$; for a shorted stub, $Z_{in} = jZ_0 \tan\theta$ and $L_{eff} = Z_0 \tan\theta/\omega$. Dividing by the lumped totals of Eq. (4.10), both give the same ratio and the same relative error of the lumped model:

$$\frac{C_{eff}}{C'\ell} = \frac{L_{eff}}{L'\ell} = \frac{\tan\theta}{\theta}, \qquad e = 1 - \frac{\theta}{\tan\theta} \approx \frac{\theta^2}{3} \quad (\theta \ll 1). \tag{4.17}$$

The equivalent values are frequency dependent; they describe the stub at one frequency and are not physical components valid at all frequencies. Setting $e = 5$ % gives $\theta \approx \sqrt{0.15} = 0.39$ rad, a stub about $\lambda/16$ long at the highest frequency of interest; the common $\lambda/10$ criterion already implies a 14 % error (Figure 4.5).

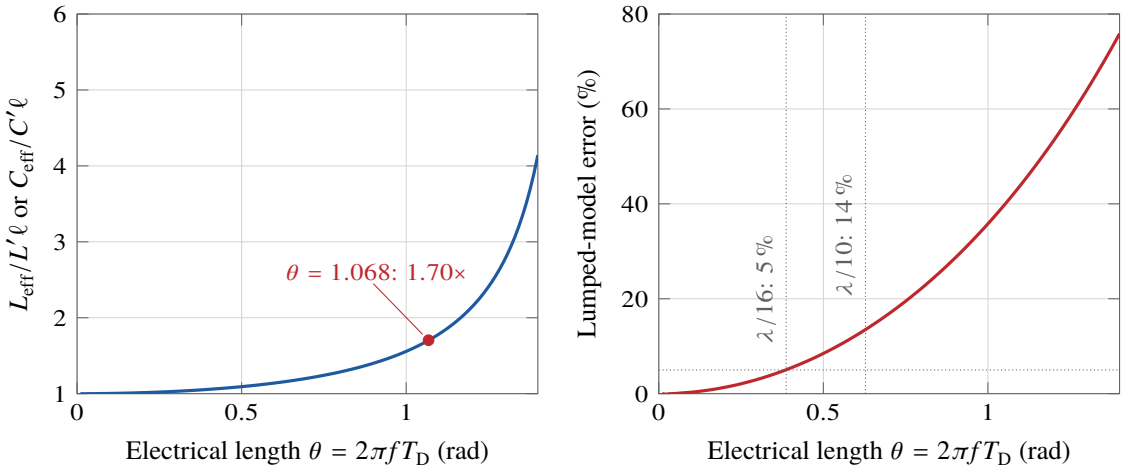


Figure 4.5. Ratio of the exact equivalent reactance of a lossless open or shorted stub to its lumped value (left) and the resulting lumped-model error (right), from Eq. (4.17). The marker is the stub of Worked Example 4.3. Ideal uniform line; no pad capacitance. [Analytical]

Common Pitfall: Using lumped stub models above ℓ_{crit}

Representing a via stub, test-point branch, or unused connector pin by a lumped capacitor in simulation. Once the stub is longer than about one-tenth of a wavelength at the highest frequency of interest, its impedance departs from the lumped value and eventually resonates. Model stubs as transmission lines.

4.7 TEM, quasi-TEM, and dispersion

The telegrapher's equations assume a transverse electromagnetic (TEM) mode, in which electric and magnetic fields lie entirely in the cross-sectional plane. A stripline in a homogeneous dielectric supports a true TEM mode. A microstrip does not, because its fields straddle two media (air and laminate); its dominant mode is *quasi-TEM*, and ε_{eff} rises slowly with frequency as the field concentrates in the dielectric. For typical board microstrips the effect is a few percent below 20 GHz and is usually absorbed in a frequency-dependent ε_{eff} [38].

A second source of dispersion is the frequency dependence of D_{k} itself. Causality requires that a lossy dielectric's permittivity decrease with frequency (the Kramers–Kronig relations), so a constant-D_{k}, constant-D_{f} model is not physically realizable. Wideband models such as the Djordjevic–Sarkar model enforce causality [39] (Chapter 11).

Design Rule: Model lines with frequency-dependent RLGC

For any simulation of edges faster than about 200 ps or data rates above about 2 Gb/s, use a causal, frequency-dependent RLGC (or S-parameter) model of each line from a field solver, with the laminate's D_{k} and D_{f} specified at a frequency near the Nyquist frequency. Constant-parameter lossless models are acceptable only for topology exploration.

Key Takeaways

- A uniform line is described by R', L', G', C'; the telegrapher's equations follow from Kirchhoff's laws on a ladder of cells.

- $Z_0 = \sqrt{(R' + j\omega L')/(G' + j\omega C')} \to \sqrt{L'/C'}$ above a few megahertz; $v_p = 1/\sqrt{L'C'}$.
- $C' = t_{pd}/Z_0$ and $L' = Z_0 t_{pd}$: about 3.4 pF/in and 8.5 nH/in for 50 Ω FR-4.
- Attenuation is $\alpha_c + \alpha_d = R'/(2Z_0) + G'Z_0/2$; dielectric loss is $\approx 2.31 f_{GHz} D_f \sqrt{D_k}$ dB/in, independent of geometry in stripline.
- Z_{in} follows Eq. (4.16); open stubs become shorts at a quarter wavelength, and impedances repeat every half wavelength.

Further Reading

Pozar [38] and Ramo et al. [37] derive transmission-line theory from Maxwell's equations; Paul [40] extends it to multiconductor lines; Hall and Heck [4] apply it to lossy digital interconnects.

Review Problems

P4.1 A line has $L' = 9.0$ nH/in and $C' = 3.6$ pF/in. Find Z_0, t_{pd}, and the equivalent ε_r.

P4.2 A 45 Ω stripline uses a laminate with $D_k = 3.5$. Find C' and L'.

P4.3 Compute the dielectric attenuation at 8 GHz of a stripline in a low-loss laminate with $D_k = 3.6$ and $D_f = 0.004$.

P4.4 What characteristic impedance is needed for a quarter-wave transformer that matches a 100 Ω load to a 50 Ω source?

P4.5 A 1 in shorted stub of 50 Ω line ($t_{pd} = 170$ ps/in) is driven at 1 GHz. Find Z_{in} and its equivalent inductance, and compare with $L'\ell$.

P4.6 A line has $R'_{dc} = 0.2$ Ω/in and $L' = 8$ nH/in. Above what frequency does it behave as an LC line?

P4.7 *(Approximation failure.)* Using the small-angle form of Eq. (4.17), find the largest stub delay for which a lumped model is within 5 % at 8 GHz. Check it with the exact expression.

P4.8 *(Inverse design.)* A 45 Ω line must present no more than 0.5 pF total capacitance as a lumped stub. What is its maximum delay and length at 170 ps/in?

Chapter 5

Signals, Symbols, and Skew: A Working Vocabulary

Chapter Overview

Datasheets, specifications, and design reviews use a vocabulary that mixes physics (impedance, modes, return current) with communications (bit rate, symbol rate, encoding, jitter) and timing (skew of several kinds). Many errors in practice are vocabulary errors: a 50 Ω trace mistaken for a 50 Ω resistor, a gigatransfer counted as a gigabit, a PAM4 Nyquist frequency computed as if the link were NRZ, or a "matched" pair whose legs have equal length and unequal delay. This chapter collects that vocabulary in one place, with numbers from the reference stackup of Chapter 10, and points to the chapters that develop each topic.

Learning Objectives

1. Distinguish characteristic impedance from resistance and decide when a trace must be treated as a transmission line.
2. Describe where return current flows as a function of frequency.
3. Convert among bit rate, symbol rate, transfers per second, unit interval, and Nyquist frequency for NRZ, PAM3, and PAM4, including encoding overhead.
4. Name the components of jitter and the kinds of skew, and explain why equal length is not equal delay.

5.1 Waves, impedance, and resistance

A signal on a board travels as an electromagnetic wave guided by a trace and its return path. The distributed model of Chapter 4 gives every line four per-unit-length parameters, R, L, G, and C. At high frequency, where $\omega L \gg R$ and $\omega C \gg G$, the line has a characteristic impedance $Z_0 \approx \sqrt{L/C}$: the ratio of voltage to current *in a wave travelling in one direction*. It is not a resistance. The 50 Ω, 4.52 mil-wide L3 trace of Chapter 10, 0.6 mil thick, has a DC resistance of

$$R_{\mathrm{dc}} = \frac{\rho}{Wt} = \frac{1.72 \times 10^{-8}\ \Omega \cdot \mathrm{m}}{114.8\,\mu\mathrm{m} \times 15.2\,\mu\mathrm{m}} \approx 9.8\ \Omega/\mathrm{m}, \tag{5.1}$$

about 1 Ω for a 100 mm route. An ohmmeter across a 50 Ω line reads an open circuit; a driver launching a step into it sees 50 Ω until the reflection returns.

When is a trace a transmission line? When the edge is shorter than the time the signal takes to travel the trace and back, the far end is not at the same voltage as the near end during the edge, and reflections are

visible. The spatial length of an edge is t_r/t_{pd}: on L3 ($t_{pd} = 6.42\,\text{ps/mm}$) a 100 ps edge occupies 15.6 mm of trace, and a 1 ns edge 156 mm. Chapter 2 turns this into the critical-length criterion $\ell_{crit} = t_r/(kt_{pd})$; a line longer than ℓ_{crit} needs a controlled impedance and a termination strategy (Chapters 7 and 8). What matters is the edge rate of the driver, not the clock frequency: a 1 MHz clock from a modern driver with a 300 ps edge rings like a fast signal.

5.2 Single-ended, differential, and the two modes

A single-ended signal is referenced to a plane; a differential signal is the difference between two conductors. Two coupled lines carry two modes (Chapters 10 and 13): the *odd* mode (equal and opposite voltages, the differential signal) with impedance Z_{odd} and the *even* mode (equal voltages, the common-mode signal) with Z_{even}; $Z_{diff} = 2Z_{odd}$ and $Z_{cm} = Z_{even}/2$. Any asymmetry (skew, unequal loading, a via on one leg only) converts part of one mode into the other. In mixed-mode S-parameters this *mode conversion* appears as S_{cd} and S_{dc} (Chapter 14). Common mode does not carry data but radiates efficiently from cables and attached structures, which is why mode conversion matters for EMC even when the receiver rejects it.

5.3 Where the return current flows

Current flows in loops. The return current of a signal flows in the reference plane (or planes) and chooses its path by impedance:

- at low frequency (DC to roughly kilohertz) the plane's *resistance* dominates and the return spreads over the whole plane along the lowest-resistance path between the endpoints;
- at high frequency (above roughly hundreds of kilohertz to a few megahertz, depending on geometry) *inductance* dominates and the return concentrates directly under the trace, which minimizes the loop area [5].

For a microstrip at height H over a wide plane, the high-frequency return-current density falls off as

$$\frac{J(x)}{J(0)} = \frac{1}{1 + (x/H)^2}, \tag{5.2}$$

so a fraction $\frac{2}{\pi}\arctan(x/H)$ flows within $\pm x$ of the trace centre: 79.5 % within $\pm 3H$ and 93.7 % within $\pm 10H$ (Figure 5.1). A slot, a split, or a missing via under a fast trace forces the return current around the gap and enlarges the loop (Chapter 23).

5.4 Bits, symbols, and transfers

Rates

Bit rate (b/s): information bits per second on one lane, including or excluding encoding overhead as stated.
Symbol rate or **baud** (Bd): symbols per second. NRZ sends one bit per symbol; PAM4 two; PAM3 about 1.58.
Transfers per second (T/s): the rate at which a lane or pin changes value. PCIe quotes the raw line rate in GT/s; DDR quotes MT/s, two transfers per clock cycle.
Unit interval (UI): the duration of one symbol, 1/baud.
Nyquist frequency: half the symbol rate, the fundamental of the fastest alternating pattern.

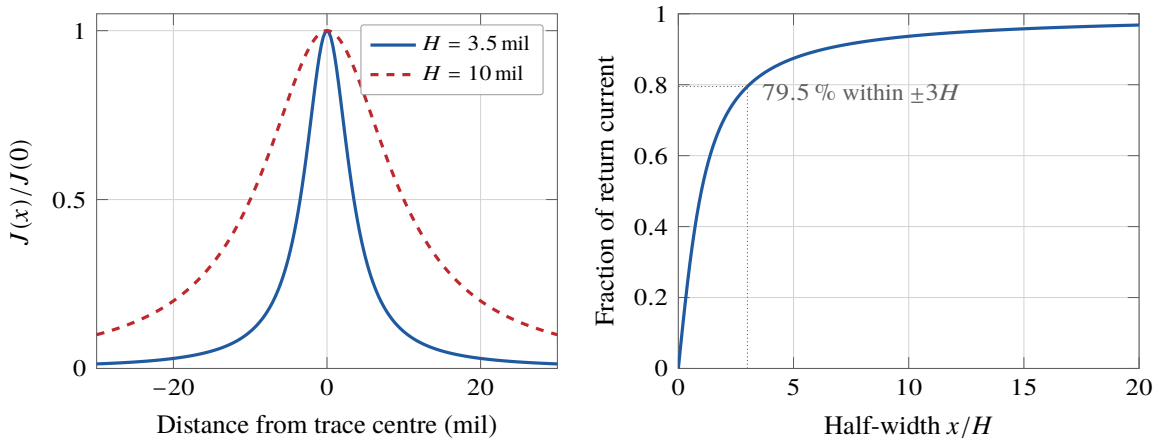


Figure 5.1. Left: high-frequency return-current density in the plane beneath a microstrip for two heights. Right: fraction of the return current within a half-width x of the trace centre, Eq. (5.2). [Analytical]

Modulation. Figure 5.2 shows one bit sequence in three line codes. PAM4 halves the symbol rate for the same bit rate, so the channel's loss is evaluated at half the frequency; the price is that each of its three eyes has a third of the NRZ amplitude, about 9.5 dB less signal for the same peak swing, which is why PAM4 links rely on equalization and forward error correction. PAM3 (USB4 v2) is in between.

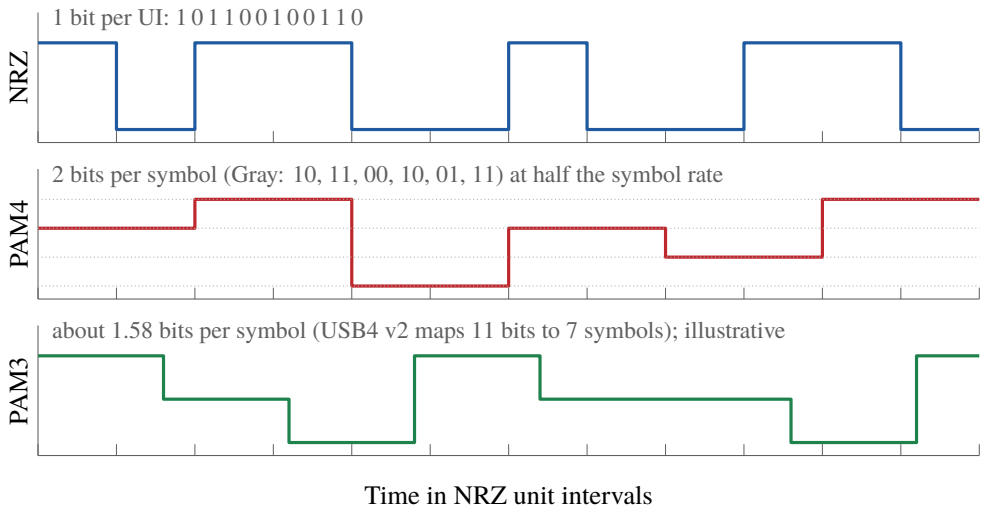


Figure 5.2. One bit sequence as NRZ (one bit per symbol), PAM4 (two bits per symbol, Gray-coded levels), and PAM3 (about 1.58 bits per symbol). The PAM3 trace is illustrative and does not reproduce the USB4 mapping. [Conceptual]

Encoding overhead. Line codes add bits for DC balance, transition density, framing, or error control, so the payload rate is below the line rate (Table 5.1).

UI and Nyquist. Table 5.2 applies the definitions to several interfaces.

Table 5.1. Common line codes and their efficiency.

Code	Efficiency	Used by (examples)
8b/10b	80.0 %	PCIe Gen1–2, USB 3.2 Gen 1, SATA, DisplayPort 1.x
64b/66b	97.0 %	10GBASE-KR and other Ethernet PHYs
128b/130b	98.5 %	PCIe Gen3–5
128b/132b	97.0 %	USB 3.2 Gen 2, DisplayPort 2.x
Flit, 242 of 256 bytes payload	94.5 %	PCIe Gen6–7 (includes CRC and FEC)

Table 5.2. Unit interval and Nyquist frequency.

Interface	Line rate	Symbol rate	UI	Nyquist
DDR5-6400 (per pin)	6400 MT/s	6.4 GBd	156.25 ps	3.2 GHz
USB 3.2 Gen 2	10 Gbit/s	10 GBd	100 ps	5 GHz
10GBASE-KR	10.3125 Gbit/s	10.3125 GBd	97.0 ps	5.16 GHz
USB4 v2 (PAM3)	40 Gbit/s per lane	25.6 GBd	39.1 ps	12.8 GHz
PCIe Gen5 (NRZ)	32 GT/s	32 GBd	31.25 ps	16 GHz
PCIe Gen6 (PAM4)	64 GT/s	32 GBd	31.25 ps	16 GHz
PCIe Gen7 (PAM4)	128 GT/s	64 GBd	15.6 ps	32 GHz

5.5 What closes the eye: ISI, crosstalk, and jitter

Three families of impairment shrink the received eye (Chapters 6 and 17):

- **Inter-symbol interference (ISI):** loss and reflections spread each symbol into its neighbours. Equalization removes much of the loss-induced ISI; reflections are harder to equalize because they arrive late.
- **Crosstalk:** coupling from neighbouring lines, near-end (NEXT) and far-end (FEXT) (Chapter 12); it adds noise that equalization does not remove.
- **Jitter:** timing variation of the edges. *Random* jitter (RJ) is unbounded and Gaussian, specified by its RMS value; *deterministic* jitter (DJ) is bounded and includes data-dependent jitter (from ISI), periodic jitter (for example from a switching regulator), and bounded uncorrelated jitter (from crosstalk). At a target bit-error ratio, total jitter is commonly modelled as $\mathrm{TJ} = \mathrm{DJ} + 2Q\,\mathrm{RJ}_{\mathrm{rms}}$, with $Q = 7.03$ at 10^{-12}, so 1 ps of RMS random jitter costs about 14 ps of the eye.

5.6 Kinds of skew

Skew is a difference in arrival time between signals that should arrive together. The kinds differ in what they hurt and in how much a receiver can correct (Figure 5.3):

- **Intra-pair skew** (P versus N of one pair): converts differential signal to common mode and closes the differential eye; receivers cannot correct it.
- **Inter-pair (lane-to-lane) skew** in a multi-lane serial link with embedded clocks: the receiver deskews lanes within a specified range, often many unit intervals.
- **Clock-to-data skew** in source-synchronous interfaces (DDR, MIPI D-PHY, parallel buses): eats setup and hold margin; training may compensate part of it.

- **Byte-lane and fly-by skew** in DDR: compensated by write levelling and read training, within limits.

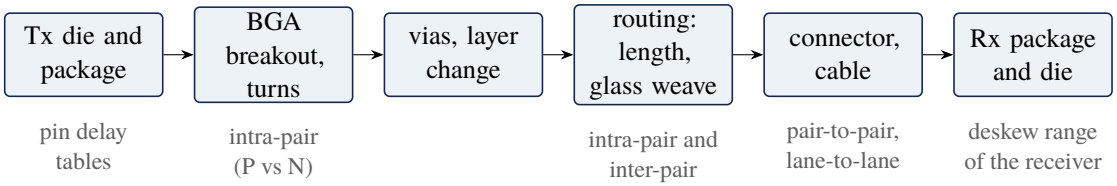


Figure 5.3. Where skew arises along a link and which kind each stage contributes. [Conceptual]

Equal length is not equal delay. Two signals of equal routed length can arrive at different times because they travel on different layers (on the reference stackup an L1 trace is 0.66 ps/mm faster than an L3 trace), through different vias, packages, or connectors, over different glass-weave regions, or because one is coupled to a neighbour. Chapter 16 quantifies each effect; the rule is to match *delay* within a budget derived from the timing requirement.

Checklist: vocabulary

- ☐ Every rate in a requirement states bits, symbols, or transfers, and whether encoding is included.
- ☐ Nyquist frequencies are computed from the symbol rate.
- ☐ Each skew budget names the kind of skew and whether the receiver corrects it.
- ☐ Return paths are continuous under every fast trace.

Key Takeaways

- Z_0 is a wave impedance, not a resistance; a 50 Ω line has about 1 Ω of DC resistance per 100 mm.
- A trace is a transmission line when it is long compared with the spatial length of the edge.
- High-frequency return current flows under the trace: 80 % within $\pm 3H$.
- UI = 1/baud and Nyquist = baud/2; PAM4 halves the baud for the same bit rate.
- Intra-pair skew cannot be corrected by the receiver; lane-to-lane skew usually can.

Review Problems

P5.1 Find the DC resistance of a 200 mm, 5 mil-wide, 1.2 mil-thick copper trace.

P5.2 What are the UI and Nyquist frequency of a 112 Gbit/s PAM4 lane?

P5.3 How much payload bandwidth does a PCIe Gen4 x4 link carry in one direction, before protocol overhead other than encoding?

P5.4 (Approximation failure) A designer computes the PCIe Gen6 Nyquist frequency as 32 GHz. What is the error?

P5.5 With DJ = 20 ps and RJ = 1.5 ps RMS, estimate TJ at 10^{-12}.

P5.6 What fraction of the high-frequency return current flows within $\pm H$ of a microstrip?

Part II

Signal Integrity

Chapter 6

What Is Signal Integrity?

Chapter Overview

This chapter opens the signal-integrity part of the book by answering a simple question precisely: what is signal integrity, and how do we know when we have it? It defines the term, shows what a signal looks like when integrity is lost, names the physical causes, lists the parameters engineers use to quantify it, and introduces the three measurements—oscilloscope, vector network analyzer and time-domain reflectometer—that reveal it. A PCI Express example ties the pieces together, and the chapter closes by separating signal integrity from power integrity. Every topic introduced here is developed in detail in the chapters that follow.

Learning Objectives

1. Define signal integrity and state what "good enough" means for a receiver.
2. Recognize reflection, ringing, overshoot, crosstalk, attenuation, ISI and jitter on a waveform.
3. Trace each impairment to its physical cause on the board.
4. Read the main SI parameters—impedance, rise time, eye opening, jitter, insertion and return loss, crosstalk, skew—and know how each is measured.
5. Distinguish signal integrity from power integrity and explain how they interact.

6.1 Definition

Signal integrity (SI) is the ability of an electrical signal to travel through an electronic system or PCB interconnect while keeping its intended shape, timing, amplitude and quality. In simple terms, signal integrity means that the signal received by a component is sufficiently similar to the signal transmitted by the source.

"Sufficiently similar" has a precise meaning. A digital receiver does not need a perfect copy of the transmitted waveform; it needs to decide, at each sampling instant, whether the symbol is a 0 or a 1 (or, for PAM4, which of four levels was sent), and to do so with a bit-error ratio below its requirement. A signal therefore has integrity when, at the receiver, it is on the correct side of the decision threshold by at least the receiver's voltage sensitivity, at a time that lies within the receiver's timing window, with margin left for noise and variation:

$$\text{SI margin} = \text{what the receiver gets} - \text{what the receiver needs}, \tag{6.1}$$

evaluated both in voltage and in time. Everything in the following chapters—impedance control, termination, loss budgets, crosstalk limits, via design, equalization—exists to keep this margin positive.

Figure 6.1 shows the idea on a real channel model. The transmitted waveform is a clean sequence of fast edges. After 10.75 in of board, two vias and a connector, the received waveform is slower, smaller, and

pattern dependent: an isolated bit no longer reaches the full level, because energy from earlier bits is still arriving. The receiver can still decide correctly here, but the margin is visibly smaller than at the transmitter.

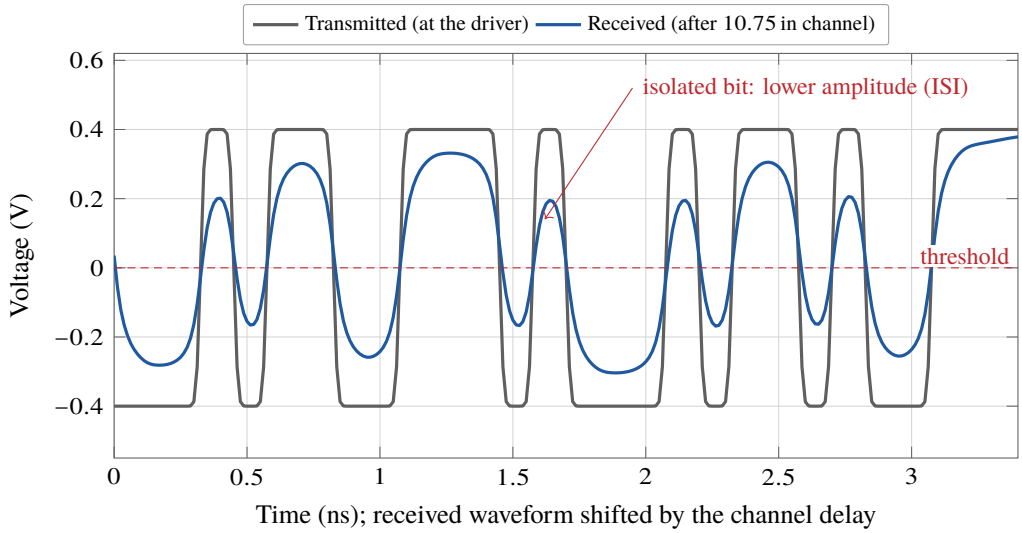


Figure 6.1. An 8 Gb/s NRZ pattern at the transmitter and after a 10.75 in channel with a via region, a connector and an 8 in trace (the reference channel of this chapter: loss 0.10 dB/in/$\sqrt{\text{GHz}}$ plus 0.12 dB/(in · GHz), 170 ps/in). [Simulation (frequency-domain channel model)]

Signal integrity is particularly important in high-speed digital designs such as PCI Express, DDR3, DDR4, DDR5 and LPDDR memory, USB, Ethernet, MIPI camera and display links, and SerDes (serializer and deserializer) interfaces in general, all of which are covered in Part VI. As Chapter 2 shows, what makes a design "high-speed" is not the clock frequency but the edge rate relative to the size of the interconnect.

6.2 What happens when signal integrity is poor

When a signal travels through a PCB trace, several electrical effects can distort it. Figure 6.2 shows each one in isolation; on a real board they occur together.

Reflection
Part of the signal travels back toward the source because of an impedance mismatch (Chapter 7). The reflected fraction is the reflection coefficient $\Gamma = (Z_2 - Z_1)/(Z_2 + Z_1)$.

Ringing
Oscillation after a transition, caused by a reflection bouncing back and forth between two mismatched ends, or by a resonance of parasitic inductance and capacitance.

Overshoot and undershoot
The voltage exceeds the final high level, or falls below the low level, during the ringing. Excessive overshoot stresses input protection and gate oxides; undershoot below ground can forward-bias input diodes.

Crosstalk
An unwanted signal coupled from a neighboring trace through mutual capacitance and inductance (Chapter 12). It appears at the near end (NEXT) and the far end (FEXT) of the victim.

Attenuation
The signal amplitude decreases along the line because of conductor and dielectric losses, which grow with frequency (Chapter 11). Fast edges lose their high-frequency content first, so edges also slow down.

Inter-symbol interference (ISI)

Previous bits affect the interpretation of later bits. Loss spreads each bit's energy into the following bit periods, and reflections return energy several bit periods later (Chapter 17).

Jitter

Uncertainty in the timing of the signal transitions. Some jitter comes from the source (PLL, supply noise); some is created by the channel, because ISI and crosstalk move the threshold crossing depending on the data pattern (Section 17.2).

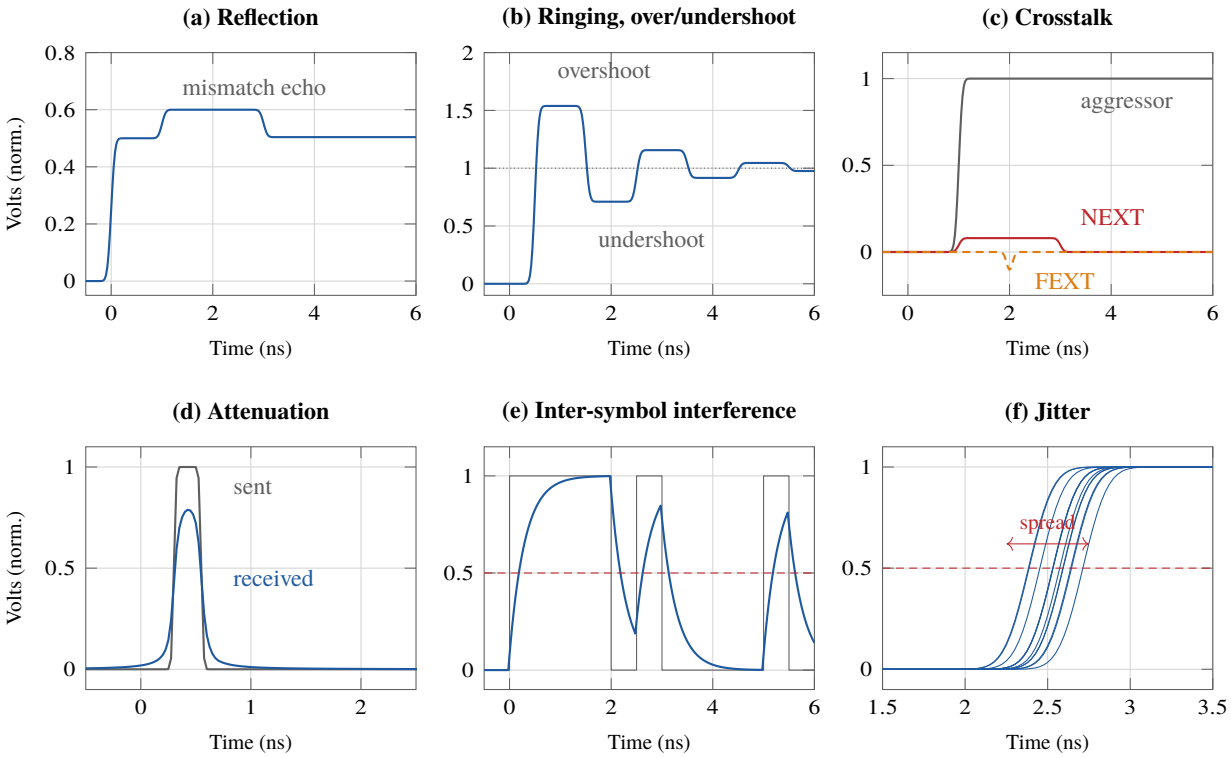


Figure 6.2. Signal-integrity impairments, each shown alone. (a) Step seen at the source of a line with a 75 Ω segment. (b) Under-terminated source (R_S = 15 Ω) driving an open line. (c) Aggressor edge and victim NEXT and FEXT. (d) A 250 ps pulse before and after the reference channel. (e) A data pattern through a one-pole channel: the isolated 1 never reaches full level. (f) Twelve edges with random timing. [Simulation (analytical models)]

Worked Example 6.1: How big is a reflection?

A 50 Ω trace enters a connector region whose impedance is 60 Ω. The driver launches an 800 mV step. How large is the reflected step, and what return loss does this discontinuity have by itself?

Solution. $\Gamma = (60 - 50)/(60 + 50) = 0.091$, so 9.1 % of the incident step, 73 mV, travels back toward the driver, and the transmitted step is $1 + \Gamma = 1.091$ times the incident. Expressed as return loss, $-20\log_{10} 0.091 = 20.8$ dB. A single 10 Ω error is modest; several of them along a channel add echoes that arrive at different times and appear as ISI and pattern-dependent jitter at the receiver (Chapter 7).

6.3 What causes signal-integrity problems

The impairments above have a small number of physical causes, all of them decided during stackup design, placement and routing:

1. incorrect trace impedance, from wrong width, dielectric thickness or material (Chapter 9);
2. poor termination, or none, so that reflections are not absorbed (Chapter 7);
3. long traces, which add loss and delay (Chapter 11);
4. improper routing of differential pairs: skew, uneven spacing, asymmetry (Chapter 13);
5. inadequate ground reference planes, which break the return path (Chapter 23);
6. poor connector and via design, with stubs and impedance discontinuities (Chapter 15);
7. excessive capacitive and inductive loading: pads, test points, multiple loads (Chapter 7);
8. crosstalk between adjacent high-speed signals routed too close or too long in parallel (Chapter 12).

Figure 6.3 places these causes along a typical channel. The channel is everything between the two silicon dies: packages, vias, traces, connectors and coupling capacitors, together with the reference planes that carry the return current. Each element is a source of one or more impairments, and the receiver sees their sum.

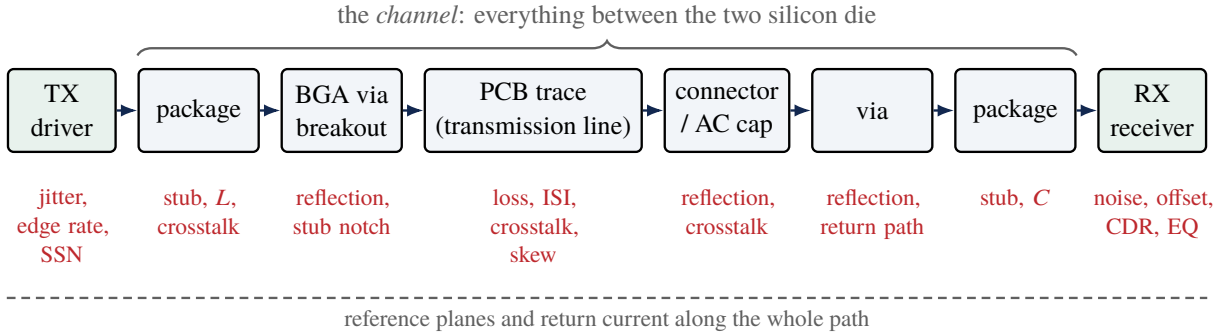


Figure 6.3. A high-speed channel from transmitter to receiver, with the impairments each element mainly contributes. [Conceptual]

Common Pitfall: Treating the schematic as the channel

Reviewing a high-speed interface only at the schematic level. The schematic shows that the pins are connected; it does not show the via stub, the plane split under the trace, or the 3 in of parallel routing next to a clock. Signal integrity is decided in the stackup and the layout, and must be reviewed there (Chapter 1).

6.4 Important signal-integrity parameters

Engineers quantify signal integrity with a handful of parameters (Table 6.1). Some describe the interconnect alone (impedance, insertion and return loss, crosstalk, skew); others describe the signal that results (rise time, eye opening, jitter).

The eye diagram deserves a closer look, because it combines all the others. An oscilloscope overlays many bit periods of a long random pattern on one display; the result looks like an eye (Figure 6.4). Its vertical opening, the *eye height*, is the voltage margin left after loss, ISI, reflections, crosstalk and noise; its horizontal opening, the *eye width*, is the timing margin left after jitter. A receiver specification usually states a minimum eye height and width, or a mask the eye must not touch, at a stated bit-error ratio.

Table 6.1. Signal-integrity parameters.

Parameter	What it tells you	Measured with	Chapter
Impedance	Electrical characteristic of the transmission line; mismatches cause reflections	TDR, field solver	4, 9
Rise/fall time	How quickly the signal transitions; sets the bandwidth that matters	Oscilloscope	2
Eye opening	Signal quality: remaining noise and timing margin at the receiver	Oscilloscope, simulation	17
Jitter	Timing variation of the transitions	Oscilloscope, BERT	17
Insertion loss	Signal attenuation versus frequency, $\lvert S_{21}\rvert$	VNA	11
Return loss	Amount of reflected signal versus frequency, $\lvert S_{11}\rvert$	VNA	7
Crosstalk	Coupling between neighboring signals (NEXT, FEXT)	VNA, TDT, scope	12
Skew	Timing difference between signals (intra-pair, lane-to-lane, data-to-strobe)	TDT, scope, VNA phase	13, 17

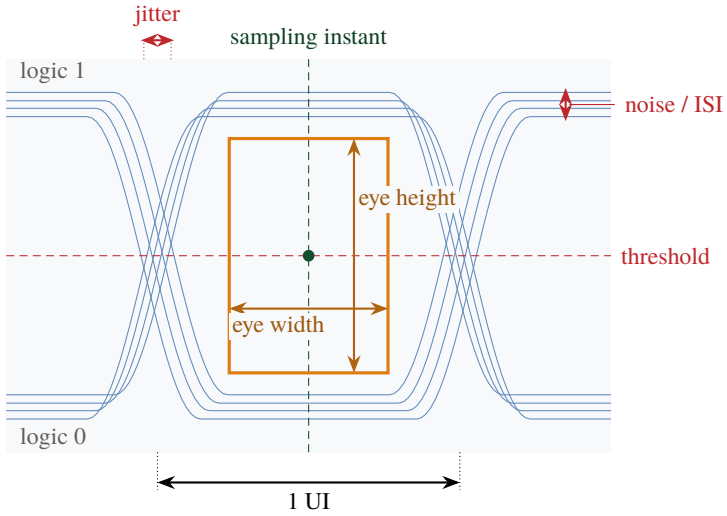


Figure 6.4. Anatomy of an eye diagram. Noise and ISI thicken the rails and reduce the eye height; jitter spreads the crossings and reduces the eye width. The receiver samples at the center of the opening. [Conceptual]

Worked Example 6.2: From loss to eye height

The reference channel of Figure 6.1 carries 8 Gb/s NRZ with an 800 mV peak-to-peak launch. Its insertion loss at the 4 GHz Nyquist frequency is 7.6 dB. Estimate the amplitude of the Nyquist-frequency component at the receiver, and compare with the simulated eye height.

Solution. $10^{-7.6/20} = 0.42$, so a 4 GHz square-wave pattern (alternating bits) arrives with about $0.42 \times 800\,\text{mV} = 330\,\text{mV}$ peak to peak. The simulated eye for a PRBS7 pattern (Figure 6.5c) opens to about 300 mV: slightly less, because long runs and isolated bits suffer different amounts of ISI, and because the connector and via reflections add small echoes. The Nyquist loss is a good first indicator of eye closure, but the eye itself is the quantity the receiver cares about.

Quick estimate	Full model	Difference
$0.42 \times 800\,\text{mV} = 330\,\text{mV}$	Simulated eye: 300 mV	10 %

Rule of Thumb: Loss at Nyquist and eye closure

For an unequalized NRZ link, an insertion loss of about 10 dB at the Nyquist frequency already reduces the eye height to roughly a quarter of the launched swing or less; links with more loss than that depend on equalization (Section 17.4). *Limits:* a rough guide only; the shape of the loss curve, reflections and crosstalk matter as much as the single Nyquist number.

6.5 How to improve signal integrity

Signal integrity is designed in, mostly before routing starts. The main techniques are:

Impedance matching
: Maintain controlled impedance—for example 50 Ω single-ended, or 85, 90 and 100 Ω differential depending on the interface—along the whole path, including breakouts and vias (Chapter 9).

Termination
: Use appropriate series, parallel or on-die termination so that reflections are absorbed at one end (Chapter 7).

Length matching
: Control trace lengths within a group and the skew within each differential pair (Chapters 13 and 17).

Ground planes
: Give every high-speed signal a continuous reference plane, and stitch the return path wherever the signal changes layer (Chapter 23).

Via optimization
: Minimize unnecessary vias, back-drill or avoid stubs, and tune anti-pads (Chapter 15).

Trace spacing
: Increase spacing between aggressors and victims, and limit parallel run length, to reduce crosstalk (Chapter 12).

Stackup design
: Select dielectric thickness, material and copper geometry to meet impedance and loss targets (Chapters 9 and 11).

Design Rule: Budget before you route

For every high-speed interface, write down the loss, return-loss, crosstalk and skew budgets before layout starts, allocate them among package, board, connector and cable, and turn the board's share into routing constraints. A budget that exists only after routing is a measurement, not a design.

6.6 How to measure signal integrity

Three instruments, each answering a different question, are used to verify signal integrity (Figure 6.5); Chapter 37 treats them in depth.

1. *Oscilloscope.* Measures the signal waveform: overshoot, undershoot, rise time, jitter and eye diagrams. It answers *does the link work, and with how much margin?*
2. *Vector network analyzer (VNA).* Measures S-parameters of the passive interconnect, including insertion loss (S_{21}) and return loss (S_{11}) versus frequency. It answers *how much does the channel attenuate and reflect?*
3. *Time-domain reflectometer (TDR).* Launches a fast step and records the reflections, giving impedance versus position along traces, vias and connectors. It answers *where is the discontinuity?*

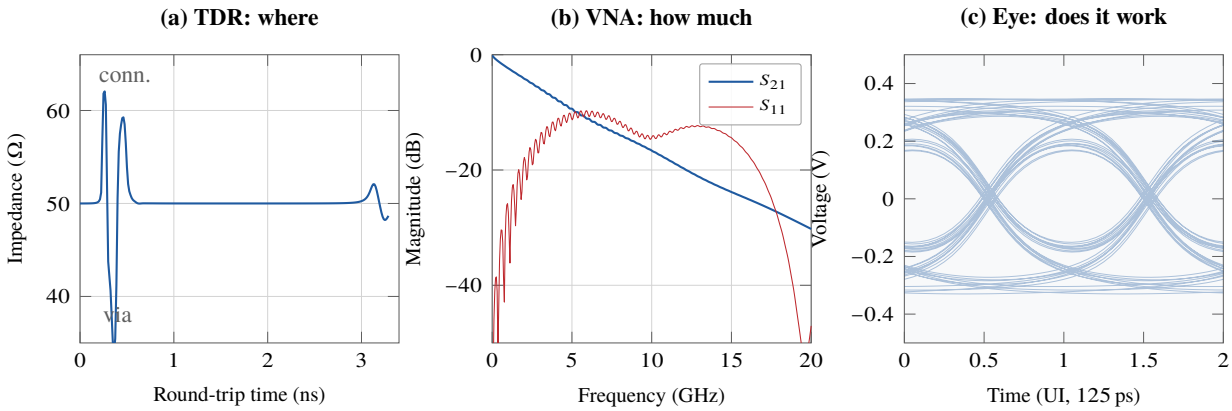


Figure 6.5. Three views of the same reference channel: (a) TDR impedance profile with a 30 ps edge, showing the low-impedance via region and the high-impedance connector; (b) insertion and return loss; (c) 8 Gb/s PRBS7 eye at the receiver, no equalization. [Simulation (frequency-domain channel model)]

6.7 A practical example: PCI Express

Suppose you are designing a PCI Express interface between an SoC and an external device, such as an M.2 solid-state drive. The signal-integrity work consists of the following checks, each developed later in the book:

- differential impedance, typically 85 Ω for PCIe (Chapter 29);
- insertion loss across the whole channel, against the generation's budget;
- return loss caused by impedance discontinuities at breakouts, vias, AC capacitors and the connector;
- differential-pair length matching and intra-pair skew;
- eye opening at the receiver, after the receiver's equalization;
- jitter, including the reference-clock contribution, and equalization settings;
- connector and via effects, especially via stubs (Chapter 15).

The ultimate goal is to ensure that the receiver can correctly distinguish logic 0 from logic 1 at the required data rate.

Worked Example 6.3: A first loss check for a PCIe Gen4 lane

A PCIe Gen4 lane (16 GT/s, Nyquist 8 GHz) runs 7 in on a low-loss laminate that measures about 0.75 dB/in at 8 GHz, through two via transitions (about 0.5 dB each) and an M.2 connector (about 1.5 dB). The channel budget for Gen4 is approximately 28 dB at 8 GHz (Table 29.1), shared with the two packages. If the packages are allocated 6 dB each (illustrative), is the board within its share?

Solution. Board share = $28 - 2 \times 6 = 16$ dB. Board loss = $7 \times 0.75 + 2 \times 0.5 + 1.5 = 7.75$ dB, leaving about 8 dB of margin for manufacturing variation, temperature, the AC capacitors and the device side. Loss is not the only check: the same channel must also meet return loss, crosstalk and skew limits, and the final judgement is the simulated eye with the reference receiver (Chapter 36).

6.8 Signal integrity and power integrity

A concise distinction, often asked in hardware-engineering interviews: *signal integrity focuses on the quality of signals traveling through interconnects, while power integrity focuses on maintaining stable power and ground voltages across the PCB.* The two are not independent. Every signal's return current flows in planes that also distribute power; simultaneous switching draws current spikes from the power network (Chapter 20); and supply noise moves driver delays and PLL phase, so it appears as jitter on signals. Power integrity is introduced in Chapter 18.

Table 6.2. Summary: the signal-integrity view of a design.

Question	Answered by	Chapter
Is the interconnect electrically long?	T_D versus t_r	2
Will it reflect?	Impedance profile, termination	7, 9
How much is lost?	Insertion loss at Nyquist and its slope	11
What couples in?	NEXT, FEXT, spacing, parallel length	12
Is the pair balanced?	Skew, mode conversion	13
Do vias and connectors behave?	Stubs, anti-pads, return vias	15
Does the receiver have margin?	Eye height and width, jitter, BER	17

Key Takeaways

- Signal integrity is the ability of a signal to arrive with enough voltage and timing margin for the receiver to decide correctly at the required error ratio.
- Poor SI appears as reflection, ringing, overshoot and undershoot, crosstalk, attenuation, ISI and jitter; on a board they act together.
- Its causes are physical: impedance, termination, length, pair routing, reference planes, vias and connectors, loading and spacing.
- Impedance, rise time, eye opening, jitter, insertion and return loss, crosstalk and skew quantify it; the eye diagram combines them.
- The oscilloscope shows whether the link works, the VNA how much the channel attenuates and reflects, the TDR where the discontinuities are.
- SI is about the signal path; PI is about the supply path; they share the same copper and interact.

Further Reading

Bogatin [3] is an accessible introduction to the whole subject; Hall and Heck [4] cover the advanced material on channels and equalization.

Review Problems

P6.1 State in one sentence what "good signal integrity" means for a digital receiver.

P6.2 A 50 Ω line meets a 40 Ω via region. Find the reflection coefficient, the reflected voltage for a 1 V incident step, and the return loss of this discontinuity.

P6.3 Match each observation to the most likely impairment: (a) an isolated 1 after a long run of 0s barely crosses the threshold; (b) a quiet line shows a pulse when its neighbor switches; (c) the high level briefly reaches 1.4 V on a 1.0 V signal.

P6.4 A channel has 12 dB of insertion loss at Nyquist. What fraction of the Nyquist-frequency amplitude arrives? Is this link likely to need equalization?

P6.5 Which instrument would you use to find where along a 6 in trace an impedance discontinuity lies, and which to find how much it reflects at 8 GHz?

P6.6 *(Inverse design.)* A Gen4 board share of 16 dB must cover two vias (0.5 dB each), a connector (1.5 dB) and 3 dB reserved for variation. What is the longest trace on a laminate with 0.9 dB/in at 8 GHz?

P6.7 Give one way in which a power-integrity problem appears as a signal-integrity symptom.

Chapter 7

Reflections and Termination

Chapter Overview

Whenever a wave meets a change in impedance, part of it reflects. On a board, the ends of every net—the driver's output impedance and the receiver's input impedance—are almost never equal to the line's characteristic impedance, so reflections are the default behavior of an unterminated line. This chapter derives the reflection coefficient, shows how to track multiple reflections with a lattice diagram, explains ringing and staircase waveforms, analyzes reactive loads, and compares the termination strategies available to the designer.

Learning Objectives

1. Derive reflection and transmission coefficients at an impedance discontinuity.
2. Construct a lattice (bounce) diagram and compute source and load voltages at any time.
3. Predict ringing for under-terminated lines and staircase steps for over-terminated lines.
4. Estimate the time constant and rise-time degradation caused by a capacitive load.
5. Choose and size series, parallel, Thevenin, AC, diode, and on-die terminations for a given topology.

7.1 The reflection coefficient

Let a wave of voltage V_i traveling on a line of impedance Z_0 arrive at a load Z_L (which may be a resistor, the input of another line, or the input of a receiver). At the junction, the voltage and current must be continuous. If a reflected wave V_r travels back and a transmitted wave V_t is delivered to the load,

$$V_i + V_r = V_t, \qquad \frac{V_i}{Z_0} - \frac{V_r}{Z_0} = \frac{V_t}{Z_\mathrm{L}}. \tag{7.1}$$

Solving gives the reflection and transmission coefficients

$$\Gamma = \frac{V_r}{V_i} = \frac{Z_\mathrm{L} - Z_0}{Z_\mathrm{L} + Z_0}, \qquad T = \frac{V_t}{V_i} = 1 + \Gamma = \frac{2Z_\mathrm{L}}{Z_\mathrm{L} + Z_0}. \tag{7.2}$$

Γ ranges from -1 (short circuit) through 0 (matched) to $+1$ (open circuit). A CMOS receiver input is, to first order, an open circuit with a few picofarads of capacitance, so a wave arriving at an unterminated receiver doubles. T can exceed one; this is not a violation of energy conservation, since the current, not the voltage, is reduced.

Rule of Thumb: Small mismatches

For a small impedance change ΔZ, $\Gamma \approx \Delta Z/(2Z_0)$: a 10 % impedance error reflects about 5 % of the incident voltage. *Limits:* valid for $|\Delta Z| \lesssim 0.2Z_0$; beyond that use Eq. (7.2).

7.1.1 Launch voltage

When a driver with Thevenin voltage V_S and output resistance R_S first switches, it sees only the line's characteristic impedance, because no reflection has yet returned. The voltage launched into the line is therefore set by a resistive divider:

$$V_{\text{launch}} = V_S \, \frac{Z_0}{Z_0 + R_S}. \tag{7.3}$$

When the reflection from the load returns to the source, it meets the source reflection coefficient $\Gamma_S = (R_S - Z_0)/(R_S + Z_0)$ and reflects again.

7.2 Lattice diagrams

A *lattice diagram* tracks each traveling wave on a distance–time plot. Each wave is drawn as a diagonal line from one end to the other, labeled with its amplitude; at each end the amplitude is multiplied by that end's reflection coefficient. The voltage at either end at any time is the sum of all waves that have arrived there, counting both incident and reflected components.

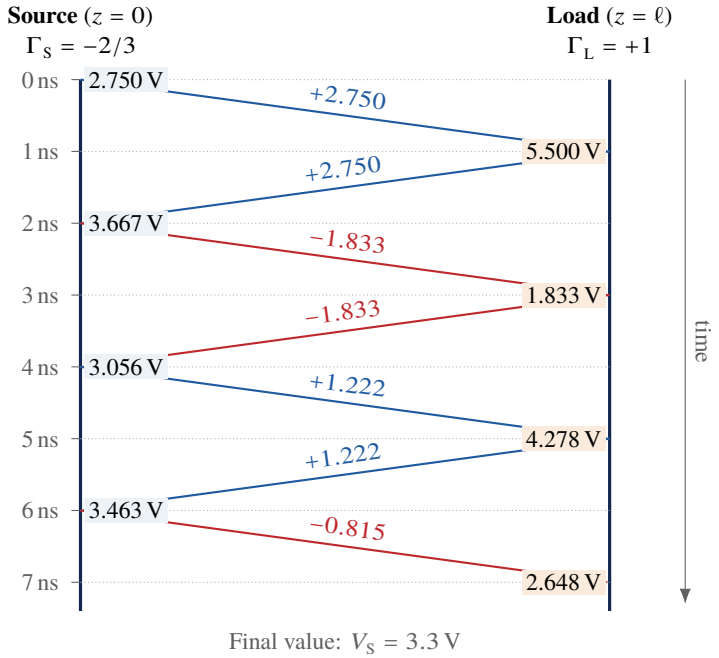


Figure 7.1. Lattice diagram for a 3.3 V driver with $R_S = 10\,\Omega$ driving an open-circuited $50\,\Omega$ line with $T_D = 1$ ns. Wave amplitudes are shown on the diagonals; cumulative voltages at the source (left) and load (right) are shown in boxes. The load voltage rings about its final value of 3.3 V.

Worked Example 7.1: Bounce diagram of an under-terminated line

A 3.3 V CMOS driver with $R_S = 10\,\Omega$ drives a $50\,\Omega$ line with $T_D = 1$ ns into a high-impedance receiver. Find the receiver voltage at 1, 3, 5 and 7 ns.

Solution. From Eq. (7.3), $V_{\text{launch}} = 3.3 \times 50/60 = 2.750$ V. $\Gamma_L = +1$ and $\Gamma_S = (10 - 50)/(10 + 50) = -2/3$. The waves arriving at the load are $+2.750$, -1.833, $+1.222$, and -0.815 V; each doubles on reflection. The load voltages are therefore 5.500, $5.500 - 3.667 = 1.833$, $1.833 + 2.444 = 4.278$, and $4.278 - 1.630 = 2.648$ V (Figure 7.1). The first overshoot is 2.2 V above the supply—well beyond

the absolute-maximum rating of most receivers—and the first undershoot dips to 1.83 V, close to the switching threshold. The ringing period is $4T_D = 4\,\text{ns}$.

7.2.1 A lattice diagram, step by step

The example compresses a procedure that is worth making explicit, because the same bookkeeping underlies every reflection simulator.

Set up. Draw the source at the left and the load at the right; time runs downward. Mark Γ_S and Γ_L at the two ends and the launch voltage of Eq. (7.3) at $t = 0$. Each diagonal takes one T_D to cross.

Propagate. A wave of amplitude V^+ arriving at an end with reflection coefficient Γ creates a reflected wave ΓV^+ that starts back immediately. The node voltage jumps by the sum of the arriving and reflected waves:

$$V_{\text{node}}^{\text{new}} = V_{\text{node}}^{\text{prev}} + (1 + \Gamma)\, V_{\text{arr}}^{+}. \tag{7.4}$$

The voltage between arrivals is constant (for an ideal step on a lossless line).

Tabulate. Table 7.1 lists the first arrivals for the 3.3 V, 10 Ω, 50 Ω, open-load case. Odd multiples of T_D are load events and even multiples are source events.

Table 7.1. Arrival table for Worked Example 7.1 ($\Gamma_S = -2/3$, $\Gamma_L = +1$).

t/T_D	Node	Arriving wave	V_{arr}^{+} (V)	$(1+\Gamma)$	V_{node} (V)
0	Source	launch	2.750	–	2.750
1	Load	1st incident	2.750	2	5.500
2	Source	reflection of 1	2.750	0.333	3.667
3	Load	2nd incident	−1.833	2	1.833
4	Source	reflection of 3	−1.833	0.333	3.056
5	Load	3rd incident	1.222	2	4.278
6	Source	reflection of 5	1.222	0.333	3.463
7	Load	4th incident	−0.815	2	2.648

Close the form. Each round trip multiplies the wave by $\Gamma_S\Gamma_L = -2/3$, so the $(n+1)$th wave to reach the load is $2.75(-2/3)^n$, and summing Eq. (7.4) as a geometric series gives

$$V_L((2n+1)T_D) = 2V_{\text{launch}} \sum_{k=0}^{n} \left(-\tfrac{2}{3}\right)^k = 3.3\left[1 - \left(-\tfrac{2}{3}\right)^{n+1}\right] \text{V}. \tag{7.5}$$

Check. Three checks catch almost every bookkeeping error: (i) for $n \to \infty$ the load voltage tends to 3.3 V, the DC value with no current flowing; (ii) $n = 0$ gives 5.5 V, twice the launch, because an open end forces zero current, so the reflected wave must cancel the incident current and therefore adds an equal voltage; (iii) with $R_S = Z_0$, $\Gamma_S = 0$ and the series stops after its first term. Figure 7.2 plots both node voltages.

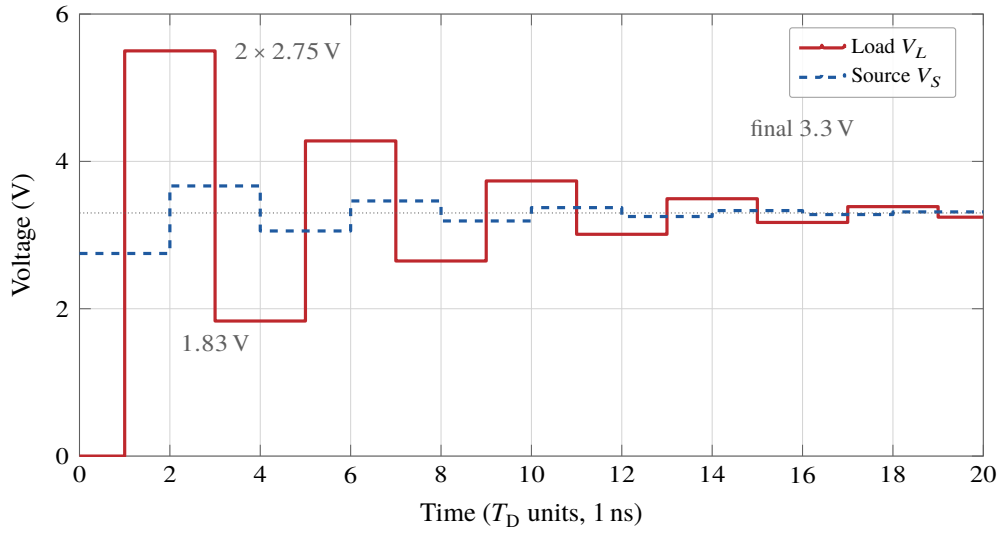


Figure 7.2. Load and source voltages of Worked Example 7.1 built from Table 7.1. The load rings with period $4T_D$; the source changes only at even multiples of T_D. [Analytical]

7.2.2 Under-, matched, and over-termination

The three regimes of source termination are compared in Figure 7.3 for an open-circuited load:

- **Under-terminated** ($R_S < Z_0$, $\Gamma_S < 0$): the load voltage overshoots and rings with period $4T_D$, decaying geometrically by $|\Gamma_S|$ per round trip.
- **Matched** ($R_S = Z_0$, $\Gamma_S = 0$): half the swing is launched, it doubles at the open end, and the single reflection is absorbed at the source. The load sees a clean step after one delay.
- **Over-terminated** ($R_S > Z_0$, $\Gamma_S > 0$): the load voltage approaches its final value in a staircase of steps spaced $2T_D$ apart, delaying the threshold crossing.

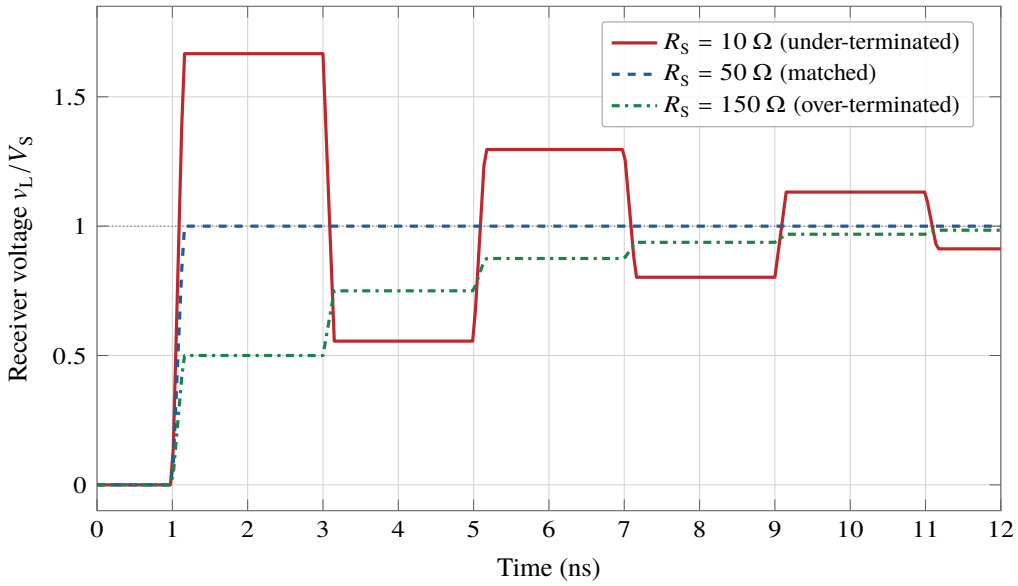


Figure 7.3. Receiver voltage for a 1 V source driving an open 50 Ω line ($T_D = 1$ ns, $t_r = 150$ ps) through three source resistances, computed by superposing the waves of the lattice diagram.

Common Pitfall: Non-monotonic edges on clocks

An undershoot or a staircase step that lingers near the receiver threshold produces a non-monotonic edge. On a data line it costs timing margin; on a clock or strobe it can double-clock a flip-flop, causing intermittent failures that are notoriously difficult to reproduce. Clocks and strobes must be terminated so that every edge is monotonic through the threshold region.

7.3 Reactive loads and discontinuities

Receivers, connectors, vias, and test pads present capacitance; bond wires, connector pins, and vias in thick boards present series inductance. Neither is a simple resistive Γ, because their impedance changes during the edge.

7.3.1 Capacitive load at the end of a line

Consider a matched-source line terminated by a capacitor C. The incident wave V_i initially sees a short circuit ($\Gamma = -1$), and the capacitor then charges through the line's impedance. The Thevenin equivalent of the line at the load is a source $2V_i$ in series with Z_0, so the load voltage is

$$v_L(t') = 2V_i\left(1 - \mathrm{e}^{-t'/\tau}\right), \qquad \tau = Z_0 C, \tag{7.6}$$

where t' is measured from the wave's arrival. The reflected wave is $v_r = v_L - V_i$, which starts at $-V_i$ and ends at $+V_i$; seen from the source, a capacitive load produces a dip before the voltage settles (Figure 7.5). An exponential rises from 10 % to 90 % in $\ln 9\,\tau \approx 2.2\tau$, so the capacitor adds about 2.2τ to an ideal step; for a finite incident edge the two combine approximately by Eq. (2.5), and exactly by convolution.

7.3.2 Capacitance in the middle of a line

A capacitor C at a point along a uniform line sees Z_0 in both directions, i.e. $Z_0/2$ in total, so the time constant halves:

$$\tau_{\mathrm{mid}} = \frac{Z_0 C}{2}. \tag{7.7}$$

The factor of two is not a different capacitor model but a different boundary condition: at the end of a line the arriving wave has a Thevenin equivalent $2V_i$ behind Z_0, whereas mid-line the capacitor is fed by V_i behind $Z_0 \parallel Z_0$ (Figure 7.4). The reflected wave is a negative pulse of area $-V_i\tau_{\mathrm{mid}}$; the transmitted wave is slowed by about $2.2\tau_{\mathrm{mid}}$.

This is the mechanism by which via pads, test points, and connector pins degrade a fast edge, and it is the basis of TDR interpretation in Chapter 37.

Worked Example 7.2: Rise-time degradation from receiver capacitance

A series-terminated 50 Ω line delivers a 300 ps edge to a receiver whose input capacitance, including its pad and package, is 5 pF. Estimate the rise time at the receiver.

Solution. $\tau = Z_0 C = 250$ ps, which alone contributes $2.2\tau = 550$ ps. Combining by root sum of squares, $\sqrt{300^2 + 550^2} \approx 626$ ps. The receiver capacitance more than doubles the rise time. Reducing Z_0 to 40 Ω would reduce τ to 200 ps.

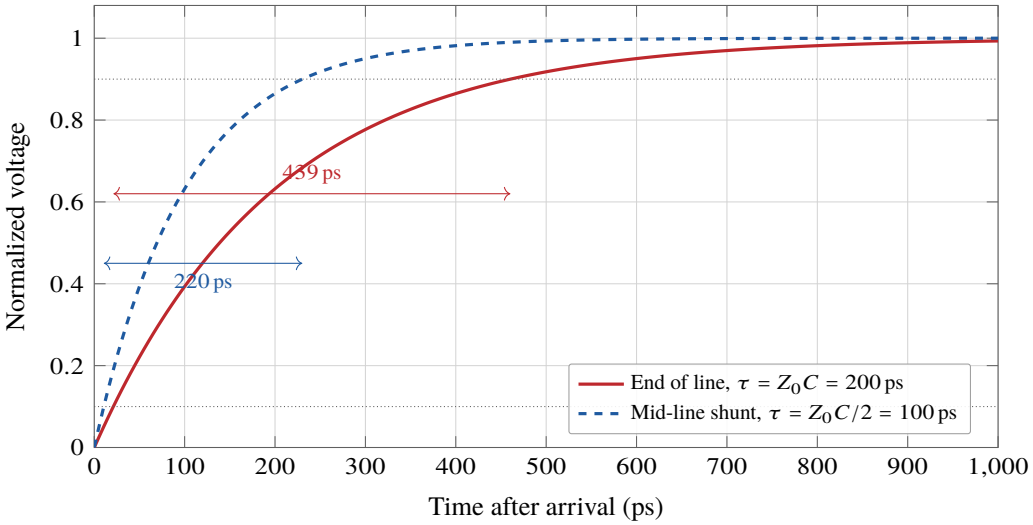


Figure 7.4. Normalized response of the same 4 pF capacitance at the end of a matched-source 50 Ω line and as a shunt in the middle of a matched line, for an ideal incident step. The added 10–90 % rise times are $\ln 9\,\tau$. [Analytical]

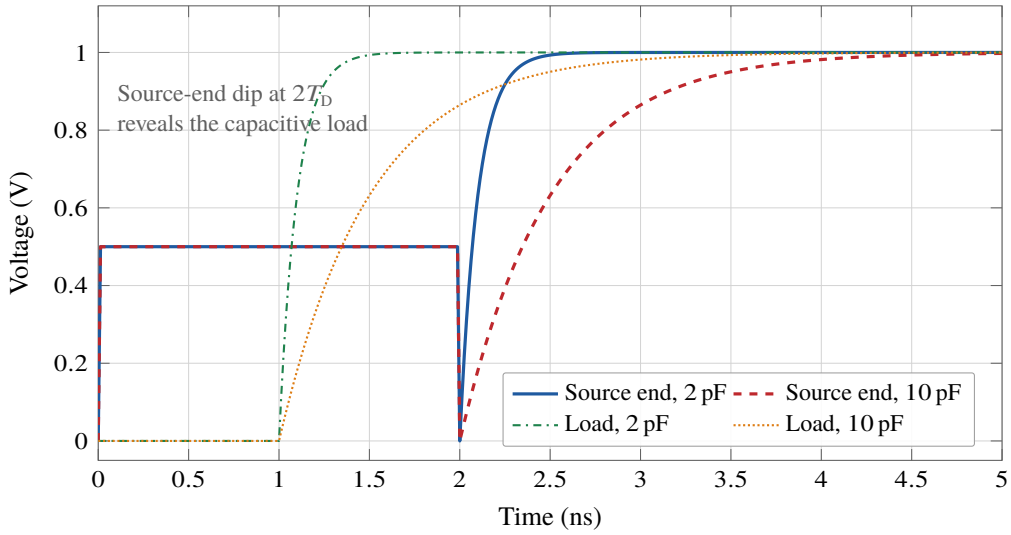


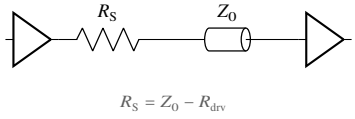
Figure 7.5. A 50 Ω line with matched source ($T_D = 1$ ns) driving a capacitive load, for an ideal 1 V source step. The load charges with $\tau = Z_0C$; the reflection seen at the source is a dip that recovers with the same time constant.

7.3.3 Series inductance

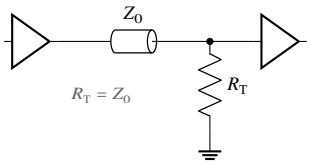
A series inductance L in a line (a long via, a bond wire, a connector pin) looks initially like an open circuit and then relaxes to a short. The time constant is $\tau = L/(2Z_0)$, and the reflection is a positive pulse of area $V_i\tau$. Inductive and capacitive discontinuities of similar size partly cancel, which is the principle behind compensating a large connector pad with a reduced-width neck or an anti-pad cut-out.

7.4 Termination strategies

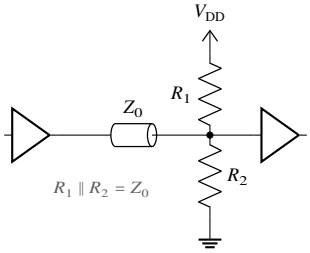
Termination places a resistance (or a resistance and capacitance) at one end of the line to make the reflection coefficient at that end approximately zero. The common schemes are drawn in Figure 7.6 and compared in Table 7.2.



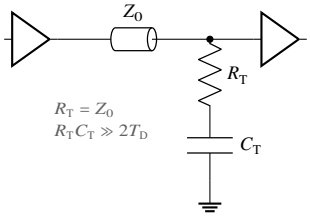
(a) Series (source) termination



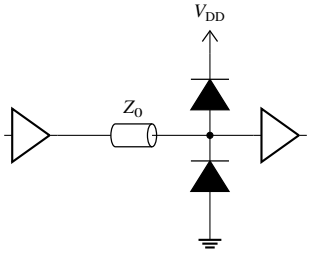
(b) Parallel (end) termination to ground or V_{TT}



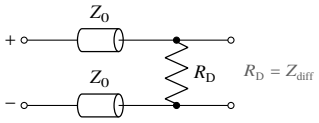
(c) Thevenin (split) termination



(d) AC (RC) termination



(e) Diode (clamp) termination



(f) Differential termination

Figure 7.6. Termination schemes for single-ended and differential lines.

7.4.1 Series (source) termination

A resistor R_S in series with the driver makes the total source resistance equal to Z_0: $R_S = Z_0 - R_{\text{drv}}$. Half the swing is launched, the open receiver doubles it, and the reflection is absorbed at the source. Series termination dissipates no static power and needs one component, which is why it is the default for point-to-point CMOS nets. Its limitations are:

- intermediate points along the line see a half-amplitude plateau for $2T_D$, so it is unsuitable for multi-drop nets whose loads are distributed along the line;

- the driver's output resistance differs between high and low states and varies with process, voltage, and temperature, so the match is never exact;
- the full edge reaches the receiver only after T_{D}, and the first transition is slowed by the receiver capacitance as in Worked Example 7.2.

Worked Example 7.3: Choosing a series resistor

An IBIS model shows a driver output resistance of 18 Ω pulling high and 22 Ω pulling low. Choose a series resistor for a 50 Ω line and find the worst-case source reflection coefficient.

Solution. The average driver resistance is 20 Ω, so $R_S = 30\ \Omega$ (an E24 value). The total source resistance is 48 Ω or 52 Ω, giving $\Gamma_{\mathrm{S}} = -0.020$ or $+0.020$. A $\pm 10\,\%$ variation of Z_0 widens this to about ± 0.07, which is still well controlled.

7.4.2 Parallel (end) termination

A resistor $R_T = Z_0$ from the receiver end of the line to a termination voltage absorbs the incident wave. The full swing arrives in one pass, and every load along the line sees a clean edge, so parallel termination suits multi-drop and bidirectional buses. The cost is static power: with termination to ground, a logic high of V_H dissipates V_H^2/R_T in the resistor and requires the driver to source the full current. Terminating to a mid-supply voltage $V_{\mathrm{TT}} = V_{\mathrm{DD}}/2$, as DDR command/address buses do, halves the swing across the resistor and reduces power by a factor of four, at the cost of a dedicated V_{TT} regulator that must both source and sink current (Chapter 28).

7.4.3 Thevenin termination

Two resistors from the line to V_{DD} and ground provide an equivalent termination resistance $R_1 \parallel R_2 = Z_0$ and an equivalent voltage $V_{\mathrm{th}} = V_{\mathrm{DD}} R_2/(R_1 + R_2)$ without a separate supply. Their drawback is that current flows from V_{DD} to ground through $R_1 + R_2$ even when the line is idle.

Worked Example 7.4: Static power of Thevenin termination

A 1.8 V, 32-bit bus on 50 Ω lines uses Thevenin terminations $R_1 = R_2 = 100\ \Omega$. Find the static current and power.

Solution. $R_1 \parallel R_2 = 50\ \Omega$ and $V_{\mathrm{th}} = 0.9$ V. Each terminator draws $1.8/200 = 9$ mA from the supply, 16.2 mW per line, or 518 mW for 32 lines, before any signal switches. A 0.9 V V_{TT} termination with 50 Ω resistors has zero idle current when the bus is tri-stated.

7.4.4 AC termination

A resistor $R_T = Z_0$ in series with a capacitor C_T terminates the line at high frequency while blocking DC current. The capacitor must be large enough that it does not charge appreciably during the reflection interval, typically $R_T C_T$ of at least three times the round-trip delay $2T_{\mathrm{D}}$, but small enough that its charging does not distort data patterns: a long run of identical bits charges C_T and shifts the effective termination voltage. AC termination is therefore best suited to clocks and other DC-balanced signals.

Worked Example 7.5: Sizing an AC termination

A 50 Ω clock line has $T_{\mathrm{D}} = 1$ ns. Size an AC termination.

Solution. Require $R_T C_T \geq 3 \times 2T_D = 6\,\text{ns}$, so $C_T \geq 6\,\text{ns}/50\,\Omega = 120\,\text{pF}$. Choose $C_T = 150\,\text{pF}$ with $R_T = 49.9\,\Omega$. For a 100 MHz clock (5 ns half-period), the time constant of 7.5 ns is comparable to the half period, so the capacitor's voltage settles near mid-supply and the clock edges are unaffected.

7.4.5 Diode termination

Schottky diodes from the line to V_{DD} and ground clamp overshoot and undershoot to about one diode drop beyond the rails. They do not match the line and so do not eliminate ringing, but they limit its amplitude and protect receivers. Most CMOS inputs contain such clamps intrinsically (as ESD protection), which is why an unterminated line often rings less severely than a lossless lattice diagram predicts.

7.4.6 On-die termination and impedance calibration

Memory and SERDES devices integrate termination resistors on the die, switchable in value and enabled only when the device is receiving. On-die termination (ODT) places the termination exactly at the receiver, eliminating the stub between an external resistor and the die, and allows the controller to tune the effective termination for each rank and topology. Drivers likewise calibrate their output resistance against an external precision resistor (the ZQ resistor in DDR memories) to remove process variation. ODT and calibrated drivers have made discrete terminations rare on modern memory and serial interfaces (Chapters 28 and 29).

7.4.7 Differential termination

A single resistor $R_D = Z_{\text{diff}} = 2Z_{\text{odd}}$ across a differential pair terminates the differential mode but leaves the common mode unterminated. A *split* termination—two resistors of $Z_{\text{diff}}/2 = Z_{\text{odd}}$ with their midpoint bypassed to ground by a capacitor—gives the common mode a defined, low-impedance path at high frequency, but it does not match it. The modal half-circuits make this clear (Chapter 13): in the odd mode the midpoint is a virtual ground, so each line sees Z_{odd} (matched); in the even mode no current crosses between the lines, so each line sees Z_{odd} to ground (through the capacitor) where it should see Z_{even}. For a coupled pair $Z_{\text{even}} > Z_{\text{odd}}$, so the even mode is under-terminated and reflects with $\Gamma_e = (Z_{\text{odd}} - Z_{\text{even}})/(Z_{\text{odd}} + Z_{\text{even}})$.

Both modes are matched by a *T network*: a resistor $R_1 = Z_{\text{odd}}$ from each line to a common node, and $R_2 = (Z_{\text{even}} - Z_{\text{odd}})/2$ from that node to ground (or to AC ground). In the odd mode the node is a virtual ground and each line sees $R_1 = Z_{\text{odd}}$; in the even mode each line sees $R_1 + 2R_2 = Z_{\text{even}}$. The equivalent π network uses Z_{even} from each line to ground and $2Z_{\text{even}}Z_{\text{odd}}/(Z_{\text{even}} - Z_{\text{odd}})$ across the pair. For a loosely coupled pair $Z_{\text{even}} \approx Z_{\text{odd}}$, $R_2 \to 0$, and the split termination is then nearly correct.

Table 7.2. Comparison of termination schemes.

Scheme	Static power	Parts	Advantages	Limitations
Series	None	1	Simple; low power; slows edges slightly	Point-to-point only; depends on driver R
Parallel	High	1	Full swing in one pass; multi-drop	V^2/R power; strong driver needed
Thevenin	High	2	No extra supply; sets idle level	Idle current through $R_1 + R_2$
V_{TT}	Moderate	1 + VRM	Low power; symmetric swing	Needs source/sink regulator
AC (RC)	None (DC)	2	Matches at HF; no DC current	Pattern-dependent baseline; clocks only
Diode	None	2	Clamps over/undershoot	Does not match; ringing remains
ODT	When active	0	At the die; tunable	Device must support it

Design Rule: Choosing a termination

For point-to-point single-ended nets longer than ℓ_{crit}, use series termination at the driver unless the driver's own output impedance already matches the line. For multi-drop or bidirectional buses, use parallel (or V_{TT}) termination at the far end, or on-die termination where available. Place discrete terminators within ℓ_{crit} of the pin they terminate, and verify the choice in simulation at the fast and slow IBIS corners.

Key Takeaways

- $\Gamma = (Z_L - Z_0)/(Z_L + Z_0)$ and $T = 1 + \Gamma$; an open receiver doubles the arriving wave.
- The launched voltage is $V_S Z_0/(Z_0 + R_S)$; subsequent reflections can be tracked with a lattice diagram.
- Under-termination rings with period $4T_D$; over-termination produces a staircase with steps every $2T_D$.
- A capacitive load has time constant $Z_0 C$ at the end of a line and $Z_0 C/2$ in its middle; it slows edges by about $\ln 9\ \tau = 2.2\tau$.
- A split differential termination matches the odd mode only; a T network with $R_2 = (Z_{\text{even}} - Z_{\text{odd}})/2$ matches both.
- Series termination is the low-power default for point-to-point nets; parallel, V_{TT}, or on-die termination serves multi-drop buses.

Further Reading

Johnson and Graham [1] and Hall, Hall, and McCall [41] treat reflections and termination in depth; Brooks [42] gives a practitioner's view with many examples.

Review Problems

P7.1 Find Γ and T for a 50 Ω line terminated in 75 Ω, and in 25 Ω.

P7.2 A 1.2 V driver with $R_S = 25\ \Omega$ drives a 50 Ω line. Find the launched voltage and Γ_S.

P7.3 A 1 V source with $R_S = 25\ \Omega$ drives an open 50 Ω line with $T_D = 1$ ns. Find the load voltage at 1, 3 and 5 ns.

P7.4 A driver has an output resistance of 12 Ω. Choose an E24 series resistor for a 50 Ω line.

P7.5 A 4 pF connector sits midway along a 50 Ω line. Find the time constant and the added 10–90 % rise time of the transmitted edge.

P7.6 Size the capacitor of an AC termination for a 50 Ω line with $T_D = 0.8$ ns, using $RC \geq 3 \times 2T_D$.

P7.7 A pair has $Z_{\text{odd}} = 45\ \Omega$ and $Z_{\text{even}} = 60\ \Omega$. Design a T network that matches both modes, and find the even-mode reflection coefficient of a split termination of two 45 Ω resistors.

P7.8 *(Approximation failure.)* In the lattice example, the driver's output resistance is actually 10 Ω only near the rails and rises to 25 Ω in mid-swing. Which entries of Table 7.1 remain trustworthy, and what model would you use instead?

Chapter 8

Termination Networks in Detail

Chapter Overview

Chapter 7 explained why reflections occur and introduced the main termination strategies. This chapter is the practical companion: one accurate schematic for every common network, the path the current takes, the power it costs, the waveform it produces on a reference line, and the interfaces that use it. Memory buses and serial PHYs are treated separately, because they solve different problems: a DDR bus terminates a multi-drop or fly-by net whose termination is switched on and off by the protocol, while a serial PHY terminates a point-to-point differential channel inside its own silicon. Many published diagrams draw these networks incorrectly; the audit at the front of the book lists the common errors, and the schematics here were drawn to match the circuits that were simulated.

Learning Objectives

1. Draw series, parallel, Thevenin, V_{TT}, RC, across-pair, split, T, and on-die terminations correctly and trace their currents.
2. Compute the static power of each network and the source/sink current of a V_{TT} supply.
3. Distinguish AC coupling from AC termination.
4. Recognize internally terminated receivers and avoid double termination.
5. Explain how DDR on-die termination differs from serial-link termination.

8.1 A reference line for comparisons

All single-ended waveforms in this chapter use the same circuit, solved with the method of characteristics (Bergeron): a 3.3 V CMOS driver with 10 Ω output resistance and a 0.4 ns edge, a lossless 50 Ω line with 1 ns delay, and a receiver with 2 pF input capacitance. The driver goes high at 0.2 ns and low at 6.2 ns. The model is ideal on purpose; real drivers have nonlinear output curves and clamp diodes that limit the overshoot of the unterminated case. Figure 8.1 shows the networks and Figure 8.2 the far-end voltages.

Unterminated. With a 10 Ω driver, $\Gamma_S = (10-50)/(10+50) = -0.667$ and the open far end has $\Gamma_L = +1$. The launched step is $3.3 \times 50/60 = 2.75$ V; the far end sees 5.50 V, then 1.83, 4.28, 2.65, 3.73, and 3.01 V on successive round trips (the lattice diagram of Chapter 7). With the receiver's capacitance the simulated peak is 5.66 V and the undershoot after the falling edge is −3.0 V: far outside the receiver's absolute-maximum rating. Clamp diodes would conduct and inject current into the supplies.

8.2 Single-ended networks

Figure 8.3 shows the passive parts these networks are built from.

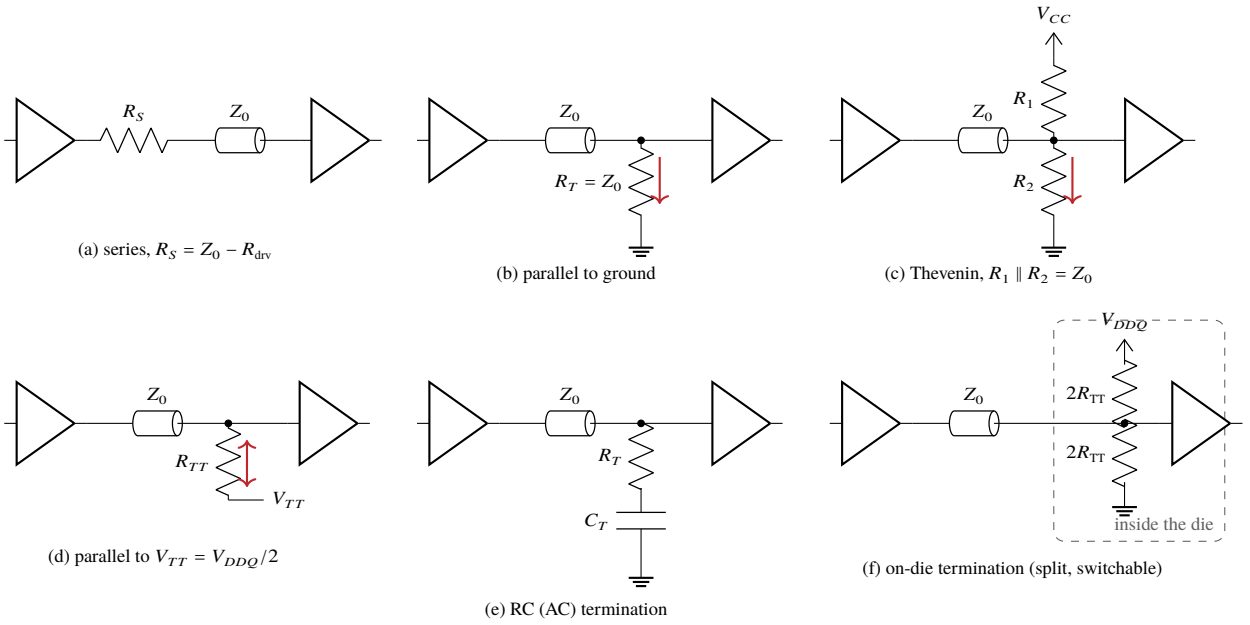


Figure 8.1. Single-ended terminations. Red arrows show the termination current while the line is driven high; in (d) the current reverses when the line is driven low, so the V_{TT} supply must both source and sink. In (f) the resistors are inside the receiving die and are switched on only while it receives. [Schematic]

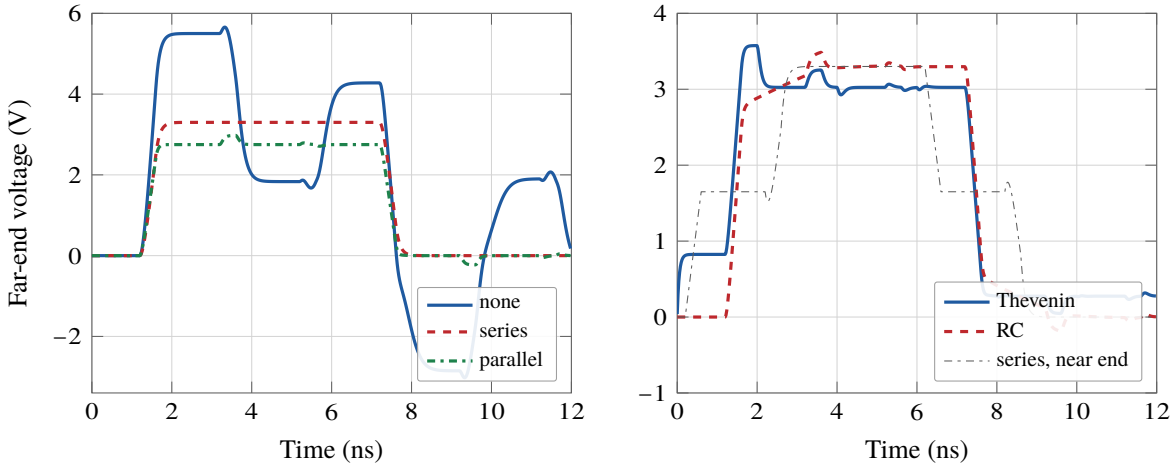


Figure 8.2. Far-end voltage on the reference line for each single-ended scheme (Bergeron model). The grey trace on the right is the near-end voltage of the series-terminated line, which sits at half amplitude for one round trip. Note the different vertical scales. [Simulation]

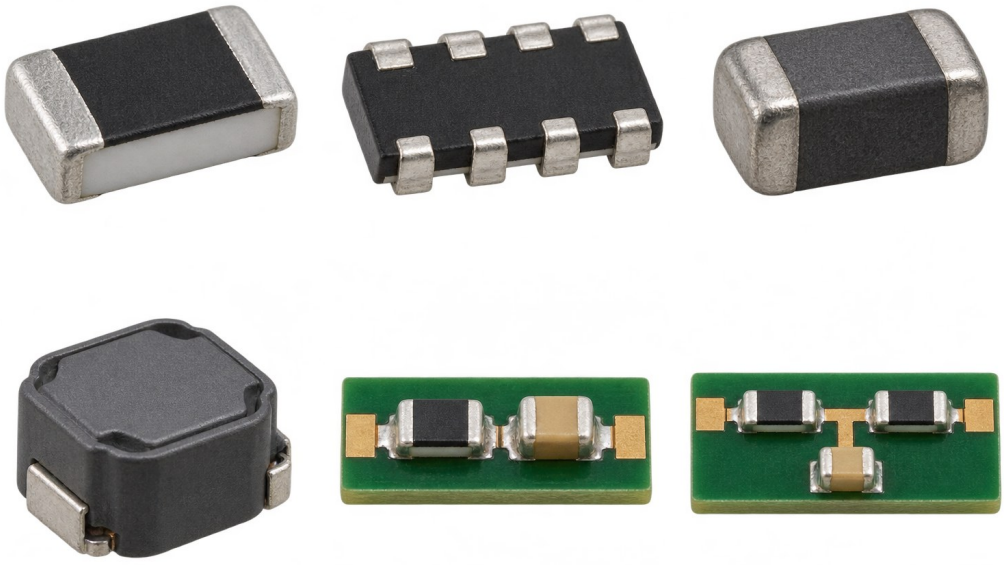


Figure 8.3. Passive parts used in terminations and filters: a chip resistor, a four-element resistor array, a chip ferrite bead or inductor, a shielded power inductor, an R–C pair as used in an RC termination branch, and a three-part T network. Rendered illustration supplied with the manuscript, not a photograph of specific products; markings, contact counts, and proportions are not authoritative. [Illustration]

8.2.1 Series (source) termination

A resistor R_S in series with the driver makes the source impedance equal to Z_0: $R_S = Z_0 - R_{\text{drv}}$, here 40 Ω. The driver launches half the swing, the open far end doubles it, and the reflection is absorbed at the source. *Current path:* the driver charges the line through R_S; no DC current flows once the line has settled at a CMOS input. *Power:* only the dynamic CV^2f of the line and load. *Waveform:* clean 0–3.3 V at the far end; the near end sits at 1.65 V for one round trip (Figure 8.2), so a receiver placed mid-line would see a step in the middle of its edge. *Applications:* point-to-point CMOS clocks and data, memory-controller outputs with programmable drive, any net with a single receiver at the end. *Limits:* R_{drv} differs between high and low states and over process and temperature, so the match is approximate; not suited to multi-drop nets.

8.2.2 Parallel termination to ground

A resistor $R_T = Z_0$ at the far end absorbs the incident wave. *Current path:* driver → line → R_T → ground whenever the line is high. *Power:* the high level is $3.3{\times}50/60 = 2.75$ V, so R_T dissipates $2.75^2/50 = 151$ mW while high and nothing while low: 76 mW average at 50 % duty, per line. *Waveform:* no ringing apart from a small overshoot from the receiver capacitance; the swing is reduced by the divider $R_T/(R_T + R_{\text{drv}})$. *Applications:* rarely used for CMOS today because of the power; common in test fixtures and with drivers designed for it.

8.2.3 Thevenin (split) termination

Two resistors R_1 to V_{CC} and R_2 to ground with $R_1 \parallel R_2 = Z_0$ (here $100 \parallel 100 = 50\,\Omega$) terminate the line to the Thevenin voltage $V_{CC}R_2/(R_1 + R_2)$. *Current path:* a constant current through the divider plus a signal current that flows into R_2 when the line is high and out of R_1 when it is low. *Power:* with 100/100 Ω at 3.3 V the far end settles at 3.03 V high and 0.27 V low, and the pair of resistors dissipates about 92 mW in either state, more than the parallel case. *Waveform:* symmetric drive requirement; an undriven line idles at

mid-rail. *Applications:* legacy buses (GTL, early SSTL boards without a V_{TT} regulator), bus-hold-free idle levels.

8.2.4 Parallel termination to V_{TT}

One resistor $R_{TT} = Z_0$ to a supply V_{TT} at the midpoint of the swing has the same Thevenin equivalent as the split termination but no divider current. *Current path:* when the line is high, current flows from the driver through R_{TT} *into* the V_{TT} supply; when low, the V_{TT} supply *sources* current through R_{TT} into the driver. *Power:* on the reference line ($V_{TT} = 1.65\,\text{V}$), $(3.03 - 1.65)/50 = 27.5\,\text{mA}$ in either state and R_{TT} dissipates 38 mW: less than half of the Thevenin network. *Applications:* DDR command/address fly-by buses (SSTL), HSTL. The cost is the V_{TT} regulator (Section 8.3).

8.2.5 RC (AC) termination

A resistor $R_T = Z_0$ in series with a capacitor C_T to ground terminates edges but draws no DC current. *Current path:* a pulse of current charges C_T through R_T at each transition. *Power:* approximately $C_T V^2 f$ for a clock of frequency f; with $C_T = 100\,\text{pF}$ and a 25 MHz clock, 27 mW. *Choosing C_T:* the time constant $R_T C_T$ must be long compared with the round-trip delay, so that the capacitor still looks like a short when the reflection arrives, but short compared with the bit time, so that a long run of identical bits does not shift the level; here $R_T C_T = 5\,\text{ns}$ against a 2 ns round trip. *Waveform:* clean edges, with a small over- and undershoot (0.19 V) because C_T charges during the edge. *Applications:* clocks and DC-balanced signals; poor for data with long runs.

8.2.6 On-die termination (ODT)

Modern memory devices put the termination inside the receiver, as a pair of switchable resistors to V_{DDQ} and ground (DDR3: Thevenin-like split to an effective $V_{DDQ}/2$) or a single resistor to V_{DDQ} (DDR4/DDR5 pseudo-open-drain, POD). Values are fractions of a calibrated reference resistor, RZQ = 240 Ω (for example RZQ/6 = 40 Ω). The controller turns ODT on in the device that is receiving and off, or to a different "park" value, in the others. ODT removes board resistors and stubs and lets the termination follow the bus direction, which no fixed network can do.

8.3 Sizing a V_{TT} supply

For an SSTL-style bus with driver resistance R_{on} and termination R_{TT} to $V_{TT} = V_{DDQ}/2$, each line carries

$$I_{\text{line}} = \pm\frac{V_{DDQ} - V_{TT}}{R_{\text{on}} + R_{TT}} = \pm\frac{V_{DDQ}/2}{R_{\text{on}} + R_{TT}}, \tag{8.1}$$

positive (sinking into V_{TT}) when the line is high and negative (sourced by V_{TT}) when low. With $V_{DDQ} = 1.2\,\text{V}$, $R_{\text{on}} = 34\,\Omega$, and $R_{TT} = 40\,\Omega$, $I_{\text{line}} = \pm 8.1\,\text{mA}$. A fly-by bus with 36 terminated lines can demand 292 mA in either direction if every line is in the same state (an idle bus parked high, or an all-zeros address), and close to zero on average for random patterns. Hence:

- the V_{TT} regulator must *source and sink*, which most switching regulators and ordinary LDOs cannot; dedicated sink/source termination regulators exist for this;
- V_{TT} must track $V_{DDQ}/2$ (it is usually derived from V_{DDQ} through a divider or a reference input), because the receivers compare against a reference near the same level;

- the V_{TT} island needs local decoupling at the termination resistors, because the current switches at the bus rate;
- the V_{TT} rail is sequenced with V_{DDQ} (Chapter 25).

Common Pitfall: Using V_{REF} as V_{TT}

V_{REF} (the receivers' comparison level) and V_{TT} (the termination supply) are both near $V_{DDQ}/2$ but are different nets. V_{REF} draws microamperes and must be quiet; V_{TT} carries hundreds of milliamperes of switching current. Connecting them, or deriving V_{REF} from the V_{TT} copper, couples the bus current into every receiver's threshold.

8.4 Differential networks

A pair carries two modes (Chapters 10 and 13), and a termination should be judged by what it presents to each: the odd (differential) mode sees each leg terminated to the virtual ground at the symmetry plane; the even (common) mode sees each leg terminated to whatever connects the centre of the network to ground. Figure 8.4 shows the four common forms; Figure 8.5 shows how each handles a 0.2 V common-mode step launched on the L1 100 Ω pair of Chapter 10 ($Z_{\text{odd}} = 50.0\ \Omega$, $Z_{\text{even}} = 66.3\ \Omega$).

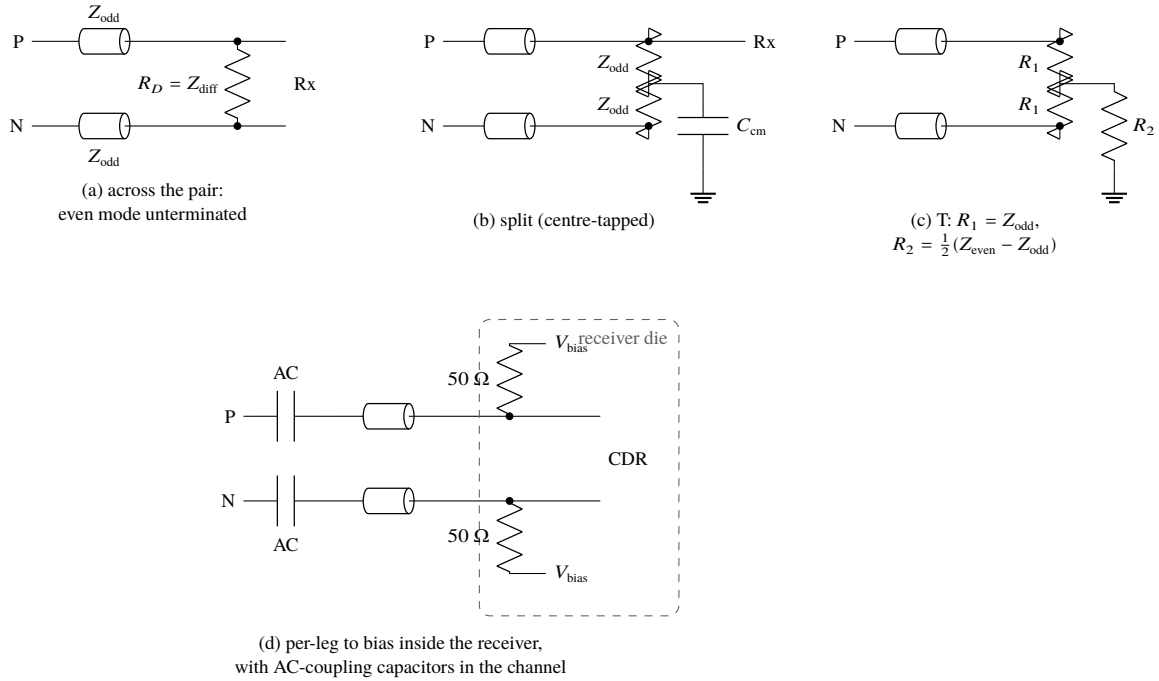


Figure 8.4. Differential terminations. (a) A single resistor $R_D = Z_{\text{diff}}$ across the pair terminates only the differential mode. (b) Two resistors of Z_{odd} with a capacitor from the centre tap to ground also give common-mode noise a path at high frequency. (c) The T network terminates both modes exactly. (d) Most serial receivers terminate each leg internally to a bias voltage; the channel carries series AC-coupling capacitors. [Schematic]

- **Across the pair** (LVDS, many clock receivers): the differential mode sees Z_{diff}; the common mode sees an open circuit and reflects completely. In Figure 8.5 the common-mode step reaches 0.23 V at the receiver, double the launched amplitude, and keeps bouncing.

- **Split with centre capacitor:** each leg sees Z_{odd} to the centre. At high frequency the capacitor is a short, so the common mode sees Z_{odd} per leg; the step settles near 0.11 V with a small mismatch because $Z_{even} \neq Z_{odd}$. The capacitor (hundreds of picofarads to a few nanofarads) keeps the DC common-mode level set by the driver.
- **T network:** $R_1 = Z_{odd}$ on each leg and $R_2 = \frac{1}{2}(Z_{even} - Z_{odd})$ from the centre to ground ($R_2 = 8.2\,\Omega$ here) makes the even-mode termination per leg $R_1 + 2R_2 = Z_{even}$. Both modes are absorbed. It costs DC common-mode current and is used where common-mode reflections matter, for example in test fixtures.
- **Per-leg to bias, inside the receiver** (CML, PCIe, USB, SATA, Ethernet SerDes): the receiver contains 50 Ω (or 42.5 Ω for 85 Ω systems) per leg to its own common-mode bias, and the channel contains AC-coupling capacitors. Nothing is placed on the board at the receiver.

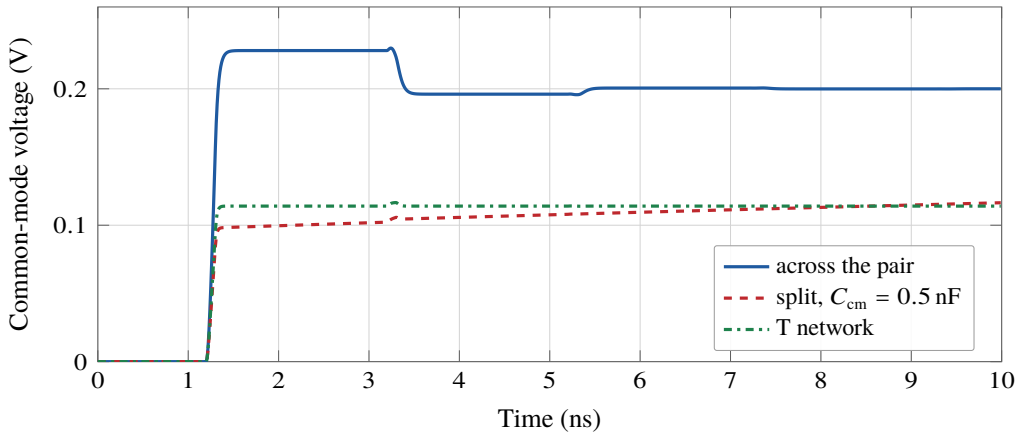


Figure 8.5. Common-mode voltage at the receiver after a 0.2 V common-mode step on the L1 pair (even-mode line, 50 Ω common-mode source per leg, 0.5 pF receiver capacitance). [Simulation]

Common Pitfall: Double termination

Adding an external 100 Ω resistor across a pair whose receiver is already internally terminated gives 50 Ω differential: the received swing drops by a third and the reflection coefficient at the receiver is $-1/3$. Before placing a termination, read the receiver's datasheet for "internal termination" and check whether it is fixed, programmable, or disabled by default.

8.5 AC coupling is not AC termination

The words are similar and the circuits are not.

- **AC coupling** is a *series* capacitor in the signal path. It blocks the DC difference between the transmitter's and the receiver's common-mode voltages, so that each can bias its own circuit. It requires a DC-balanced line code (8b/10b, 128b/130b with scrambling, and so on) so that the capacitor does not charge up, and it adds a high-pass corner set by the capacitor and the termination resistance. Interface specifications set the allowed value range and often the placement (for PCIe, near the transmitter; Chapter 29).
- **AC termination** is the *shunt* RC network of Figure 8.1(e). It terminates reflections without DC current; it does nothing about common-mode level.

A layout that places the PCIe AC-coupling capacitors as a shunt RC to ground, because a reference drawing labelled them “AC termination”, will not link.

8.6 DDR termination is different from serial termination

Table 8.1. Where the terminations are in typical memory and serial interfaces. Values are illustrative; use the controller’s and the memory’s design guides.

Interface	Data / strobe	Clock, command, address
DDR3	ODT in the receiving DRAM or controller (split, SSTL-15), switched by the controller	Fly-by, terminated to V_{TT} at the end of the fly-by (on the module or the board)
DDR4	ODT to V_{DDQ} (POD-12); DBI limits the number of driven-low bits, which draw the current	Fly-by to V_{TT}; parity and CA timing per JEDEC
DDR5	ODT to V_{DDQ}; decision-feedback equalization in the DRAM receivers	DDR5 adds on-die termination for command/address in the DRAM; module designs vary (check the module and controller guides)
PCIe, USB, SATA, Ethernet SerDes	Internal per-leg termination at both ends; AC coupling in the channel	Reference clocks: often HCSL with 50 Ω to ground at the source or receiver, per the clock device
LVDS	100 Ω across the pair at the receiver (internal or external)	—

Three differences matter for the board designer:

1. **Direction changes.** A DDR data bus is bidirectional; its termination must move with the direction of transfer. That is why it is on the die and switched by the protocol. A serial lane is unidirectional and its termination is fixed.
2. **Power follows the data.** With POD, current flows only for low bits: $V_{DDQ}/(R_{\text{on}} + R_{TT})$, for example $1.2/(34+48) = 14.6\,\text{mA}$ per low DQ. Data-bus inversion (DBI) sends the complement of a byte when that reduces the number of zeros, and so bounds the current.
3. **The board supplies V_{TT} for the fly-by.** Serial PHYs need no termination supply on the board; DDR3/DDR4 command/address buses do.

8.7 Choosing a termination

Checklist: terminations

- ☐ Each high-speed net has a stated termination: series, far-end, on-die, or internal to the PHY.
- ☐ Schematic symbols match the circuit: shunt versus series, which rail, which side of the line.
- ☐ Receivers’ internal terminations were checked before adding external ones.
- ☐ V_{TT} is a source/sink supply that tracks $V_{DDQ}/2$, with local decoupling, separate from V_{REF}.
- ☐ AC-coupling capacitors are in series, with values and placement from the interface specification.
- ☐ Termination power was added to the rail budget.

Table 8.2. Summary of single-ended terminations on the reference line (3.3 V, 10 Ω driver, 50 Ω line).

Network	Far-end levels	Static power per line	Typical use
None	ringing, 5.66 V peak	0	short nets only ($\ell < \ell_{\text{crit}}$)
Series 40 Ω	0/3.3 V	0	point-to-point CMOS
Parallel 50 Ω	0/2.75 V	76 mW avg.	fixtures, special drivers
Thevenin 100/100	0.27/3.03 V	92 mW	legacy buses
V_{TT} 50 Ω	0.27/3.03 V	38 mW	DDR CA, SSTL/HSTL
RC 50 Ω, 100 pF	0/3.3 V	$\approx CV^2f$	clocks

Key Takeaways

- Series termination costs no static power and suits point-to-point nets with the receiver at the end.
- Parallel, Thevenin, and V_{TT} terminations absorb the wave at the far end at a static power cost; V_{TT} is the most efficient but needs a source/sink supply.
- A resistor across a pair terminates only the differential mode; split and T networks also handle common mode.
- Serial PHYs terminate internally and need AC coupling, not board termination; double termination is a common error.
- DDR terminations live on the die and move with the bus direction; DDR4/5 POD draws current only for low bits.

Further Reading

Johnson and Graham [2] and Hall and Heck [4] discuss termination networks in depth; memory-controller design guides (for example the LPDDR guidance in [43]) give the ODT and V_{TT} practice for specific devices; JEDEC [44] defines the memory interface itself.

Review Problems

P8.1 A 17 Ω driver drives a 55 Ω line to a single CMOS input. Choose the series resistor and state the near-end voltage plateau for a 1.8 V swing.

P8.2 Design a Thevenin termination for a 60 Ω line that idles at 1.2 V from a 3.3 V supply.

P8.3 (Inverse design) A V_{TT} regulator can sink and source 1.5 A. How many $R_{TT} = 39\,\Omega$ lines with $R_{\text{on}} = 34\,\Omega$ and $V_{DDQ} = 1.35$ V can it support in the worst case (all lines in one state)?

P8.4 For $Z_{\text{odd}} = 45\,\Omega$ and $Z_{\text{even}} = 58\,\Omega$, design a T network.

P8.5 (Approximation failure) An RC termination with $R = 50\,\Omega$ and $C = 10$ pF is used on a line with 1.5 ns delay. What goes wrong?

P8.6 (Imperfect data) A reference schematic shows a 100 nF capacitor from each PCIe TX leg to ground, labelled “AC termination”. What is the likely intent and the correct circuit?

P8.7 Why is a DDR data-bus termination placed on the die instead of on the board?

Chapter 9

PCB Stackup and Impedance Control

Chapter Overview

The stackup is the single most consequential decision in a high-speed board design. It fixes the characteristic impedance that every trace can achieve, the return path of every signal layer, the plane capacitance available to the PDN, the amount of crosstalk between layers, and much of the cost. This chapter describes the standard transmission-line geometries, gives the closed-form impedance approximations of IPC-2141 and Hammerstad–Jensen together with their accuracy limits, presents practical 8-, 10-, and 12-layer stackups, and discusses the material effects—laminate grade, glass weave, and copper roughness—that set the ultimate performance of the board.

Learning Objectives

1. Identify microstrip, embedded microstrip, stripline, asymmetric stripline, edge-coupled pairs, and coplanar waveguide, and their dimensional parameters.
2. Estimate Z_0 and Z_{diff} with closed-form formulas, and state where those formulas fail.
3. Construct a symmetric multilayer stackup that gives every signal layer an adjacent ground reference.
4. Select a laminate class from loss and cost requirements.
5. Explain glass-weave skew and copper roughness and mitigate them.
6. Perform a tolerance analysis of characteristic impedance.

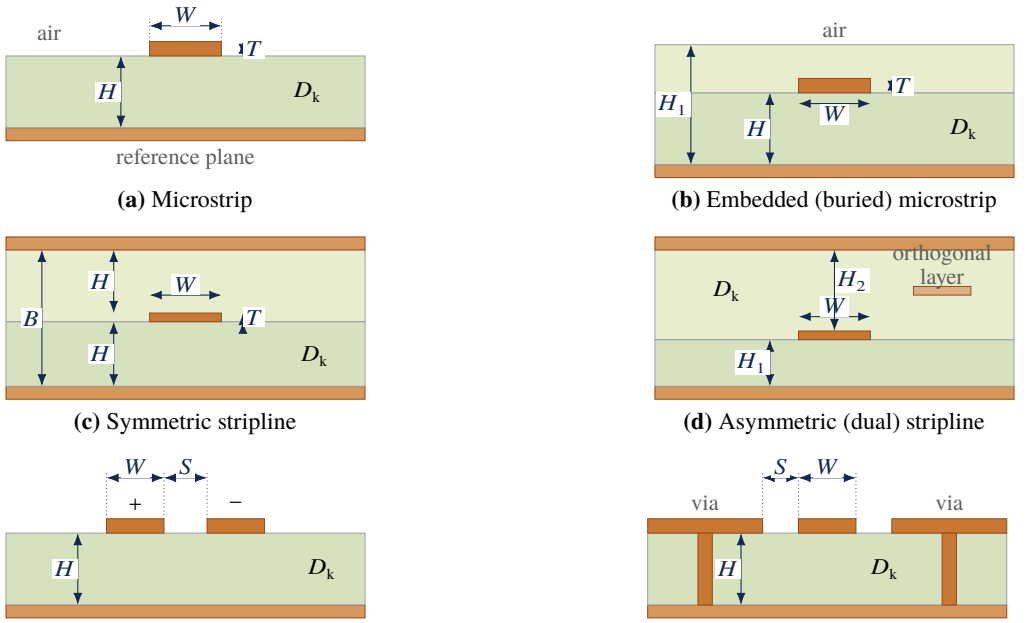
9.1 Transmission-line geometries

Figure 9.1 shows the geometries used on printed boards. Their dimensions are named consistently throughout this book: W is trace width, T trace thickness, H dielectric height from trace to plane, S edge-to-edge spacing, and B the plane-to-plane separation of a stripline.

Microstrip
: A trace on an outer layer above one reference plane. Its field is partly in air, so $\varepsilon_{\text{eff}} < D_{\text{k}}$ and signals are faster than in stripline. It is accessible for rework and probing but exposed to EMI and to soldermask variation, and its far-end crosstalk does not cancel (Chapter 12).

Embedded microstrip
: A microstrip covered by additional dielectric (prepreg or a thick soldermask). Coverage raises ε_{eff} and lowers Z_0.

Symmetric stripline
: A trace centered between two planes in a homogeneous dielectric. It is a true TEM line with no far-end crosstalk in the ideal case and excellent shielding.



(a) Microstrip

(b) Embedded (buried) microstrip

(c) Symmetric stripline

(d) Asymmetric (dual) stripline

(e) Edge-coupled differential microstrip

(f) Grounded coplanar waveguide (CPWG)

Figure 9.1. Cross-sections of PCB transmission lines with their dimensional parameters.

Asymmetric (dual) stripline
A trace closer to one plane than the other, usually because two signal layers share the space between two planes. Routing on the two layers is kept orthogonal to limit coupling.

Edge-coupled pair
Two traces side by side forming a differential pair, on either microstrip or stripline layers (Chapter 13).

Grounded coplanar waveguide
A trace with ground copper on both sides on the same layer, stitched to the plane below. It is common in RF and in very high-speed launches, where it controls fields near connectors.

9.2 Closed-form impedance approximations

Exact impedances require a field solution of the cross-section. For quick estimates, closed-form approximations are available. They differ in accuracy, and none of them should be used for final dimensions.

9.2.1 Microstrip

The IPC-2141 approximation [45], based on earlier work by Kaupp, is

$$Z_0 \approx \frac{87}{\sqrt{\varepsilon_r + 1.41}} \ln \frac{5.98H}{0.8W + T}, \qquad 0.1 < \frac{W}{H} < 2.0, \quad 1 < \varepsilon_r < 15. \tag{9.1}$$

It includes thickness but becomes poor outside its range and is typically in error by 5–10 % even within it. A widely reproduced *simplified* closed-form set for a zero-thickness strip with $u = W/H$, derived from Wheeler's and Hammerstad's work and given in most microwave texts [38], [46], is For $u \le 1$:

$$\varepsilon_{\text{eff}} = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2}\left[\left(1 + \frac{12}{u}\right)^{-1/2} + 0.04\,(1-u)^2\right], \qquad Z_0 = \frac{60}{\sqrt{\varepsilon_{\text{eff}}}} \ln\left(\frac{8}{u} + \frac{u}{4}\right); \tag{9.2}$$

for $u \geq 1$:

$$\varepsilon_{\mathrm{eff}} = \frac{\varepsilon_{\mathrm{r}}+1}{2} + \frac{\varepsilon_{\mathrm{r}}-1}{2}\left(1+\frac{12}{u}\right)^{-1/2}, \qquad Z_0 = \frac{120\pi}{\sqrt{\varepsilon_{\mathrm{eff}}}\,[u + 1.393 + 0.667\ln(u+1.444)]}. \tag{9.3}$$

Finite thickness is commonly included by replacing W with an effective width $W + \Delta W$, where $\Delta W \approx (T/\pi)\,[1 + \ln(2H/T)]$ for $W/H > 1/(2\pi)$ [46], [47].

Equations (9.2) and (9.3) are *not* the full Hammerstad–Jensen model. The full model [48], [49] uses more elaborate fitted expressions for $\varepsilon_{\mathrm{eff}}(u, \varepsilon_{\mathrm{r}})$ and $Z_0(u)$, its own thickness correction, and (in later work) frequency dispersion; it is implemented in Listing E.1 as microstrip_hj_full. Published accuracy figures for any of these formulas compare them with a quasi-static solution of the *same idealized* geometry—rectangular strip, homogeneous substrate, infinite ground, no soldermask—and say nothing about how well that geometry represents an etched, plated, mask-coated board trace. Judge them against a field solver run on the fabricator's cross-section, not against each other.

9.2.2 Embedded microstrip

IPC-2141 treats a microstrip covered to total height H_1 above the plane by using the microstrip logarithm with a reduced effective permittivity:

$$Z_0 \approx \frac{60}{\sqrt{\varepsilon'_{\mathrm{r}}}} \ln\frac{5.98H}{0.8W+T}, \qquad \varepsilon'_{\mathrm{r}} = \varepsilon_{\mathrm{r}}\left[1 - \mathrm{e}^{-1.55H_1/H}\right]. \tag{9.4}$$

9.2.3 Stripline

For a symmetric stripline with plane separation $B = 2H + T$,

$$Z_0 \approx \frac{60}{\sqrt{\varepsilon_{\mathrm{r}}}} \ln\frac{4B}{0.67\pi(0.8W+T)} = \frac{60}{\sqrt{\varepsilon_{\mathrm{r}}}} \ln\frac{1.9B}{0.8W+T}, \qquad \frac{W}{B-T} < 0.35,\ \frac{T}{B} < 0.25. \tag{9.5}$$

Common 50 Ω stripline geometries often violate the width limit, so a second estimate valid over a wider range is useful. Wheeler's formula [46] for a strip of finite thickness, centered between planes, is

$$Z_0 \approx \frac{30}{\sqrt{\varepsilon_{\mathrm{r}}}} \ln\left\{1 + \frac{4}{\pi}\frac{B-T}{W'}\left[\frac{8}{\pi}\frac{B-T}{W'} + \sqrt{\left(\frac{8}{\pi}\frac{B-T}{W'}\right)^2 + 6.27}\right]\right\}, \tag{9.6}$$

where $W' = W + \Delta W$ and $\Delta W = \frac{T}{\pi}\left\{1 - \frac{1}{2}\ln\left[\left(\frac{T}{2B-T}\right)^2 + \left(\frac{0.0796\,T}{W+1.1T}\right)^m\right]\right\}$ with $m = 2/\left[1 + \frac{2}{3}T/(B-T)\right]$. For $T = 0$ it agrees with the exact conformal-mapping result for a zero-thickness strip to better than 0.1 % over the widths used on boards (checked numerically in Listing E.1); with thickness it remains an approximation. Asymmetric stripline is not well served by any simple formula; IPC-2141 gives an approximation, but a field solver should be used.

9.2.4 Edge-coupled differential pairs

IPC-2141 relates differential impedance to the single-ended Z_0 of either trace alone:

$$Z_{\mathrm{diff}} \approx 2Z_0\left[1 - 0.48\,\mathrm{e}^{-0.96S/H}\right] \qquad \text{(microstrip)}, \tag{9.7}$$

$$Z_{\mathrm{diff}} \approx 2Z_0\left[1 - 0.347\,\mathrm{e}^{-2.9S/B}\right] \qquad \text{(stripline)}. \tag{9.8}$$

Figure 9.2 compares the microstrip and stripline formulas over a range of widths. The differential formulas are the least reliable of those here; they ignore the effect of coupling on each trace's own capacitance and can be in error by more than 10 % for tightly coupled pairs.

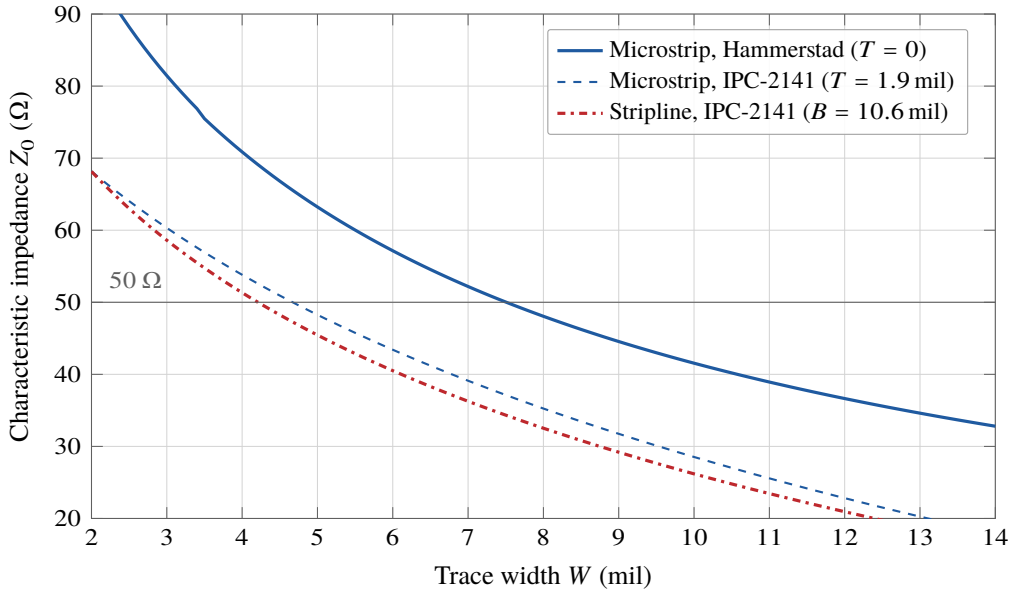


Figure 9.2. Characteristic impedance versus trace width for a microstrip (H = 3.5 mil, D_k = 3.8) and a symmetric stripline (B = 10.6 mil, D_k = 3.8). The two microstrip formulas differ by 10–20 %, partly because of thickness and partly because of the approximations themselves.

Worked Example 9.1: Microstrip on 3.5 mil prepreg

A 4 mil trace of 0.5 oz foil plated to T = 1.9 mil sits on 3.5 mil prepreg with D_k = 3.8 at 1 GHz. Estimate Z_0.

Solution. IPC-2141, Eq. (9.1): $Z_0 = \frac{87}{\sqrt{5.21}} \ln \frac{5.98\times3.5}{0.8\times4+1.9} = 38.12 \times \ln 4.104 = 53.8\,\Omega$. The simplified set with $T = 0$, $u = 1.143$: $\varepsilon_{\text{eff}} = 2.81$ and $Z_0 = 70.9\,\Omega$. This is not a like-for-like comparison, because the plated strip is more than half as thick as the dielectric. With Wheeler's thickness correction $\Delta W = 1.39$ mil, $u = 1.541$, $\varepsilon_{\text{eff}} = 2.87$, and $Z_0 = 60.7\,\Omega$; the full Hammerstad–Jensen model with its own thickness correction gives $Z_0 = 61.4\,\Omega$ and $\varepsilon_{\text{eff}} = 2.57$ (Listing E.1). The three thickness-including estimates span 53.8–61.4 Ω; IPC-2141 is the outlier here, and soldermask (not included) would lower the true value by a few ohms.

Quick estimate	Full model	Difference
IPC-2141: 53.8 Ω	Full H–J, T incl.: 61.4 Ω	−12 %

The only reliable way to settle this is a 2-D field solver with the fabricator's actual dimensions, including the trapezoidal etch profile.

Common Pitfall: Treating calculator results as final

Releasing a design with trace widths taken from a closed-form calculator. The formulas above disagree with one another by 5–15 %, they ignore soldermask, etch trapezoid, resin-rich regions, and frequency-dependent D_k, and they are often used outside their validity range. Use them to choose a starting point; take final widths from the fabricator's field-solver stackup.

Table 9.1. Closed-form impedance formulas and their limits.

Geometry	Formula	Range and typical accuracy
Microstrip	IPC-2141, Eq. (9.1)	$0.1 < W/H < 2$; 5–10 %
Microstrip	Simplified set, Eqs. (9.2) and (9.3)	All u, $T = 0$ (or Wheeler ΔW); idealized geometry
Microstrip	Full Hammerstad–Jensen (Listing E.1)	Wide range, incl. T; idealized geometry
Embedded microstrip	IPC-2141, Eq. (9.4)	$H_1 > H$; rough
Stripline	IPC-2141, Eq. (9.5)	$W/(B-T) < 0.35$, $T/B < 0.25$; 5–10 %
Stripline	Wheeler, Eq. (9.6)	Wide range; exact limit for $T = 0$
Differential pair	Eqs. (9.7) and (9.8)	Loose coupling; > 10 % error when tight

Worked Example 9.2: Stripline width for 50 Ω

A stripline layer has 5 mil of dielectric above and below a 0.6 mil (0.5 oz) trace, $D_k = 3.8$. Estimate the width for 50 Ω and the differential impedance of a pair of such traces spaced $S = 6$ mil.

Solution. $B = 2 \times 5 + 0.6 = 10.6$ mil. Solving Eq. (9.5) for W: $\ln[1.9B/(0.8W + T)] = 50\sqrt{3.8}/60 = 1.624$, so $0.8W + T = 20.14/5.075 = 3.97$ mil and $W = 4.2$ mil. *Validity check:* $W/(B-T) = 4.2/10.0 = 0.42$, beyond the formula's limit of 0.35, so this result is an extrapolation. Wheeler's Eq. (9.6) gives 51.2 Ω at 4.2 mil and 50 Ω at $W = 4.4$ mil.

Quick estimate	Full model	Difference
IPC-2141 (extrapolated): 4.2 mil	Wheeler: 4.4 mil	−5 % in width

With Eq. (9.8), $Z_{\text{diff}} \approx 100[1 - 0.347\,e^{-2.9\times 6/10.6}] = 93$ Ω. To reach 100 Ω differential the traces would be narrowed or spaced further apart; the final geometry should again come from a field solver.

9.3 Stackup design principles

A good high-speed stackup obeys a small number of principles, which follow directly from the return-path physics of Chapter 3:

1. **Every signal layer is adjacent to a solid reference plane**, preferably ground. The reference defines Z_0 and carries the return current.
2. **Power planes are paired closely with ground planes** (2–4 mil of dielectric) to create low-inductance plane capacitance (Chapter 21).
3. **The stackup is symmetric about its center** in copper distribution and dielectric thickness, so that the board does not bow or twist during lamination and reflow.
4. **Adjacent signal layers are avoided** or, where two signal layers share a stripline cavity, routed orthogonally and spaced as far apart as the thickness budget permits.
5. **Fast signals change layers only between layers that share a reference**, or stitching vias are placed next to the signal via (Chapters 15 and 23).

9.3.1 Example stackups

Figure 9.3 shows a 10-layer stackup suitable for a board carrying LPDDR, PCIe, USB, and MIPI. Its routing resources are two microstrip layers (L1, L10) for breakout and short escapes and two well-shielded stripline layers (L3, L8) for long high-speed routes. Table 9.2 compares it with typical 8- and 12-layer alternatives.

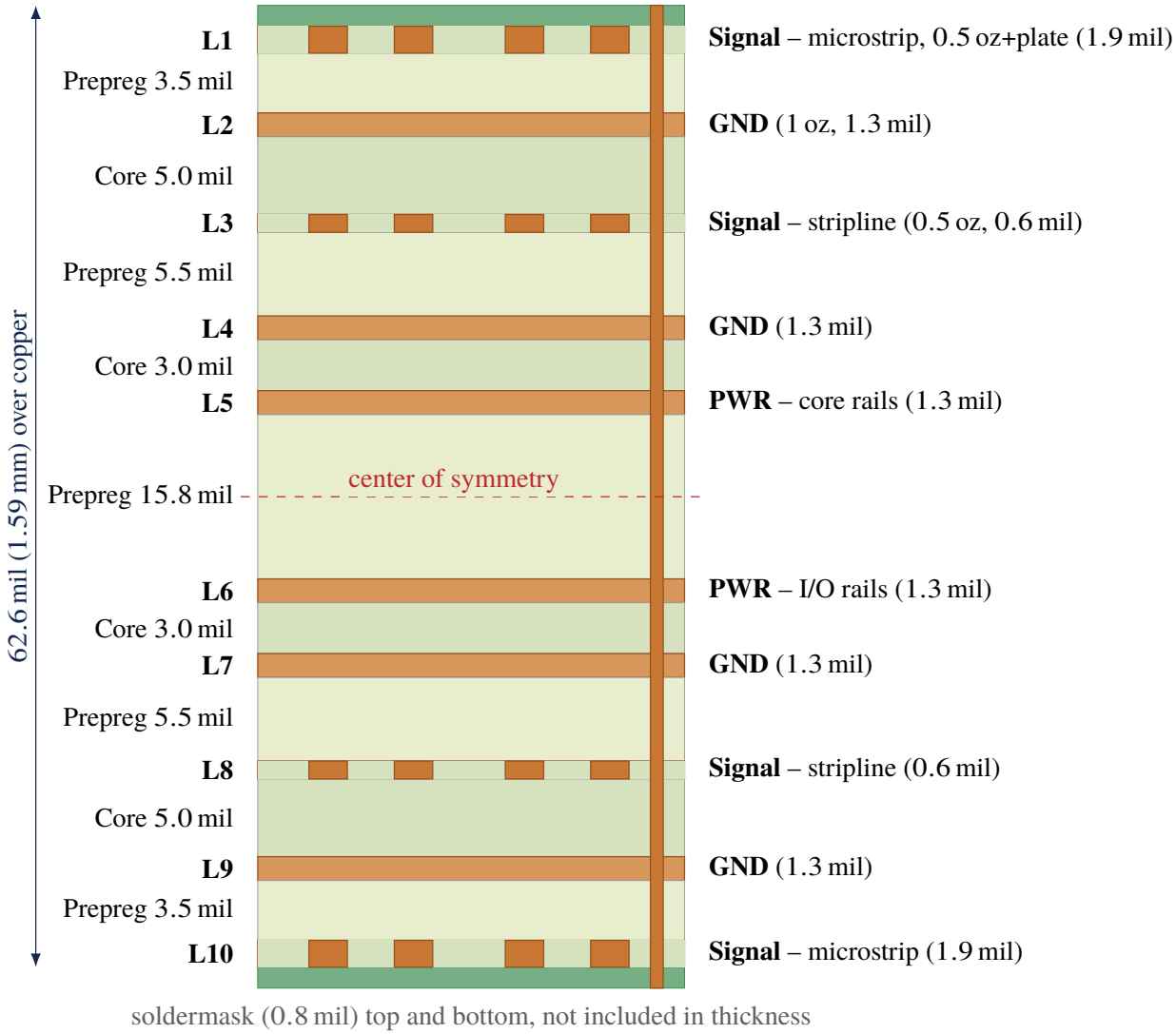


Figure 9.3. An example 10-layer stackup (62.6 mil over copper). Four signal layers each have an adjacent ground plane; two power planes are coupled to ground through 3 mil cores; the stackup is symmetric about its center.

The dielectric thicknesses given are representative; the fabricator will adjust them to the glass styles and resin contents it stocks.

Design Rule: Stackup checklist

Before routing: (1) every signal layer has an adjacent ground plane; (2) each power plane has a ground plane within 4 mil; (3) the stackup is symmetric; (4) the fabricator has confirmed dielectric thicknesses, D_k at the target frequency, trace widths, and impedance tolerance; (5) via-stub lengths for the thickest transition are acceptable at the fastest interface's Nyquist frequency.

Table 9.2. Representative 8-, 10-, and 12-layer stackups (S = signal, G = ground, P = power). Thicknesses are illustrative and must be confirmed with the fabricator.

Layers	Assignment (top to bottom)	Thickness	Notes
8	S–G–S–P–G–S–G–S	62 mil	L3 and L6 are asymmetric striplines referenced mainly to L2 and L7; P/G pair at center (L4–L5) on a thin core.
10	S–G–S–G–P–P–G–S–G–S	62.6 mil	Figure 9.3. Symmetric striplines L3, L8; two power planes each coupled to ground.
12	S–G–S–G–S–P–G–S–G–S–G–S	78–93 mil	Six signal layers; L5 and L8 are inner striplines; one P/G pair at the center. Thicker board raises via-stub length (Chapter 15).

9.4 Laminate materials

Board laminates are composites of woven E-glass and a resin system. The glass has a D_k of about 6, the resin roughly 3; the composite's D_k depends on the resin content, which differs between glass styles (thin, open styles such as 106 and 1080 are resin-rich; heavier styles such as 2116 and 7628 are glass-rich). Laminates are grouped by their dissipation factor, which sets the dielectric loss (Eq. (4.14)). Representative ranges are given in Table 9.3; vendor data for specific products appear in Appendix B.

Table 9.3. Laminate classes. Ranges are representative of published data at about 1–10 GHz and vary with resin content; consult the specific product data sheet.

Class	D_k	D_f	Typical use
Standard FR-4	4.0–4.6	0.015–0.022	Digital to ~ 3 Gb/s, short links
Mid-loss	3.7–4.2	0.008–0.012	DDR4, PCIe Gen3, 10 Gb/s short reach
Low-loss	3.4–3.8	0.004–0.007	PCIe Gen4, USB4, 25 Gb/s
Very low / ultra-low loss	3.0–3.6	0.001–0.003	PCIe Gen5/6, 56 Gb/s+
PTFE and hydrocarbon RF	2.2–3.5	0.001–0.004	RF, microwave, radar

Other laminate properties matter for reliability as much as for signal integrity: the glass-transition temperature T_g, the decomposition temperature T_d, the out-of-plane coefficient of thermal expansion (which stresses plated vias), and resistance to conductive anodic filament growth. Lead-free assembly requires laminates qualified for its higher reflow temperatures; IPC-4101 slash sheets classify these properties [50].

9.5 Glass-weave skew

Because glass and resin have different permittivities, the local D_k seen by a trace depends on whether it lies over a glass bundle or over a resin-rich gap (Figure 9.4). The two traces of a differential pair can see different average D_k along their length, producing intra-pair skew that converts differential signal into common mode (Chapter 13). The effect grows with route length and with the openness of the weave.

Worked Example 9.3: Worst-case weave skew

Suppose that over a 5 in parallel run one trace of a pair sees an average D_k of 4.0 and the other 3.6 (an extreme case). Find the skew and compare it with a PCIe Gen4 unit interval.

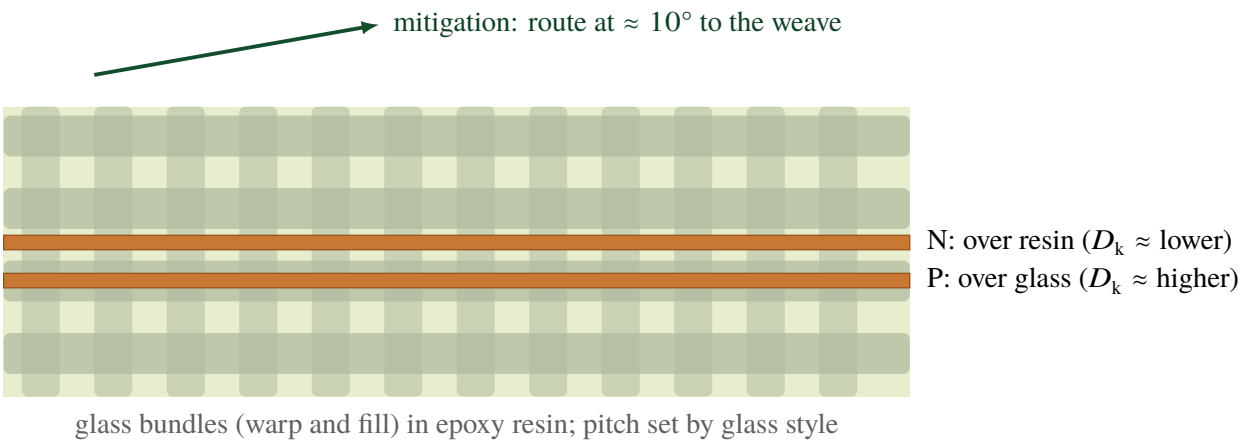


Figure 9.4. Glass-weave skew: when a differential pair runs parallel to the weave, one trace may lie over glass and the other over resin. Rotating the route relative to the weave averages the two.

Solution. $t_{pd} = 84.73\,\text{ps/in}\sqrt{D_k}$ gives 169.5 ps/in and 160.8 ps/in. The difference, 8.7 ps/in, accumulates to 43 ps over 5 in—70 % of the 62.5 ps UI at 16 GT/s. Realistic local D_k variations are usually smaller, but even 5–10 ps is significant at Gen4 and above.

Mitigations, roughly in order of cost, are:

- route long pairs at an angle (often 10° to 15°) to the weave, or in a shallow zig-zag, so that each trace samples glass and resin equally;
- rotate the artwork on the panel by a few degrees;
- use spread or flat glass styles (e.g. 1067, 1078, 1035, 3313), in which the bundles are flattened to close the resin gaps;
- use two plies of prepreg of different styles so that the bundles do not align.

9.6 Copper foil and roughness

Copper foil is roughened on the side bonded to the laminate to improve adhesion. At gigahertz frequencies the skin depth (Figure 3.4) becomes comparable to this roughness and current is forced to follow the tooth profile, raising conductor loss by up to a factor of two or more (Chapter 11). Foils are sold in roughness grades: standard (STD), reverse-treated (RTF), very low profile (VLP), and hyper or ultra low profile (HVLP, ULP), with progressively smaller surface profiles. Lower-profile foils cost more and have lower peel strength. Oxide or oxide-alternative treatments applied to inner-layer cores before lamination also add roughness, which should be included in loss models.

Rule of Thumb: When roughness matters

Copper roughness begins to affect loss noticeably once the skin depth falls below the RMS roughness, about 1 GHz for standard foil and several gigahertz for VLP/HVLP. *Limits:* the threshold is gradual, and the size of the effect depends on the tooth shape, not only on the RMS value (Section 11.2.2).

9.7 Impedance tolerance

Fabricators typically guarantee impedance to ±10 %, and to ±5 % or ±7 % at extra cost. The variation comes from etched width, dielectric thickness after lamination, copper thickness, and D_k. Its sensitivity can be estimated from any of the closed-form formulas.

Worked Example 9.4: Tolerance budget of a microstrip

For the microstrip of Worked Example 9.1 ($Z_0 \approx 53.8\,\Omega$ by IPC-2141), estimate the impedance spread for $H \pm 10\,\%$, $W \pm 0.5$ mil, and $D_k \pm 5\,\%$.

Solution. From Eq. (9.1), $\partial Z_0/\partial H = 38.12/H$, so $\Delta H = 0.35$ mil gives 3.8 Ω. $\partial Z_0/\partial W = -38.12 \times 0.8/(0.8W + T) = -5.98\,\Omega$/mil, so $\Delta W = 0.5$ mil gives 3.0 Ω. $\partial \ln Z_0/\partial \varepsilon_r = -1/[2(\varepsilon_r + 1.41)]$, so $\Delta\varepsilon_r = 0.19$ gives 1.8 %, or 1.0 Ω.

These three numbers can be combined in three different ways, and they answer three different questions:

- *Worst case.* Evaluating the formula at the corners of the box gives 46.1–61.7 Ω, close to the linear sum $\pm(3.8 + 3.0 + 1.0) = \pm 7.8\,\Omega$. This is a bound—for this formula and these ranges only.
- *RSS of the bounds.* $\sqrt{3.8^2 + 3.0^2 + 1.0^2} = 4.9\,\Omega$ is neither a bound nor a standard deviation; it acquires a meaning only once a probability model is attached.
- *Statistical.* If each input is independent and uniformly distributed over its range, its standard deviation is (half-range)/$\sqrt{3}$, so $\sigma_Z \approx 4.9/\sqrt{3} = 2.8\,\Omega$. A Monte Carlo run of the same formula (Figure 9.5) confirms $\sigma_Z = 2.85\,\Omega$, and about 5 % of samples fall outside ±10 %.

The result is dominated by dielectric thickness. None of these numbers is a fabrication yield: real processes have correlated inputs (a fabricator adjusts etch width to the measured dielectric thickness), non-uniform distributions, copper-thickness variation and model error. Use process data and field-solver sensitivities before predicting production yield.

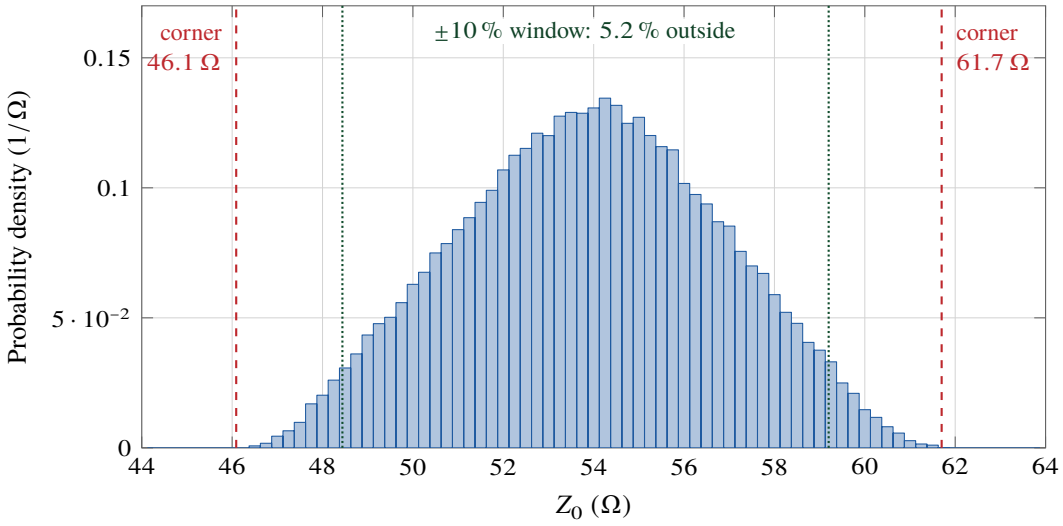


Figure 9.5. Distribution of the IPC-2141 microstrip impedance of Worked Example 9.4 for $W = 4$ mil ± 0.5 mil, $H = 3.5$ mil ± 10 %, $D_k = 3.8 \pm 5\,\%$ (independent uniform inputs, 100 000 samples, fixed seed). Dashed lines: exact corners of the same formula; dotted: ±10 %. [Analytical (Monte Carlo)]

Common Pitfall: RSS presented as a guarantee

Quoting the root-sum-square of worst-case contributions as if it bounded the result. RSS combines standard deviations of independent variables; it is a statistical estimate whose coverage depends on the distributions. Keep worst-case corners and statistical estimates in separate columns of a tolerance budget.

Key Takeaways

- Microstrip is fast and accessible; stripline is shielded, homogeneous, and free of ideal far-end crosstalk.
- Closed-form impedance formulas (IPC-2141, the simplified and full Hammerstad–Jensen sets, Wheeler) describe idealized geometries, disagree with each other by several percent, and must be checked against their validity ranges; use a field solver for final dimensions.
- Give every signal layer an adjacent ground, pair power with ground on thin dielectric, and keep the stackup symmetric.
- Choose laminate by D_f at the Nyquist frequency of the fastest link.
- Glass-weave skew and copper roughness become first-order effects above roughly 10 Gb/s.
- Impedance tolerance of $\pm 10\,\%$ is dominated by dielectric thickness and etched width. Worst-case corners, RSS and Monte Carlo answer different questions; none of them is a yield prediction without process data.

Further Reading

IPC-2141A [45] documents the closed-form formulas and coupon practice; Wadell [46] collects formulas for many geometries; Wheeler [47] and Hammerstad and Jensen [48] are the primary sources for microstrip; Ritchey [34] discusses stackup design from a fabrication perspective.

Review Problems

P9.1 Using IPC-2141, find the width of a 50 Ω microstrip with $H = 3.5$ mil, $T = 1.9$ mil, and $D_k = 3.8$.

P9.2 Using Hammerstad–Jensen with $T = 0$, estimate the width of a 50 Ω microstrip with $H = 3.5$ mil and $D_k = 3.8$. Why does it differ from Problem P9.1?

P9.3 A 50 Ω microstrip pair has $H = 3.5$ mil and $S = 5$ mil. Estimate Z_{diff}.

P9.4 Find Z_0 of a stripline with $W = 4$ mil, $T = 0.6$ mil, $B = 10.6$ mil, and $D_k = 3.8$. Is the formula within its range?

P9.5 An embedded microstrip ($W = 4$ mil, $T = 1.9$ mil, $H = 3.5$ mil, $D_k = 3.8$) is covered to $H_1 = 7$ mil. Estimate ε_r' and Z_0.

P9.6 A pair runs 8 in with one trace seeing $D_k = 3.75$ and the other $D_k = 3.65$. Find the skew.

P9.7 *(Inverse design.)* A fabricator guarantees $H = 3.5$ mil $\pm 5\,\%$ and $D_k \pm 3\,\%$ for the microstrip of Worked Example 9.4. Using the linear sensitivities of that example, what etch tolerance on W keeps the worst-case linear sum within $\pm 5.4\ \Omega$ ($\pm 10\,\%$)?

P9.8 *(Imperfect data.)* Ten coupons from one panel measure 52.1, 52.8, 53.0, 51.9, 52.5, 52.2, 53.1, 52.7, 52.4 and 52.6 Ω. Can you infer the production spread of $\pm 10\,\%$ boards from them?

Chapter 10

Impedance Geometry, Sensitivity, and Fabrication

Chapter Overview

Chapter 9 introduced transmission-line cross-sections and closed-form impedance formulas. This chapter turns those ideas into a design procedure that survives fabrication. It defines every dimension precisely (edge versus centre spacing, base versus top width, the two heights of an asymmetric stripline), computes complete 50 Ω, 85 Ω, 90 Ω, and 100 Ω designs for a stated stackup with a two-dimensional field solver, measures how sensitive each design is to every parameter, propagates realistic tolerances with a Monte Carlo analysis, and ends with what to tell the fabricator and how coupons and TDR verify the result. All solver numbers come from the finite-difference solver supplied with the book (*tools/fd2d.py*), which agrees with the Hammerstad–Jensen microstrip formula to about 1 % and whose mesh error at the settings used is below 1 %; treat every value as a design starting point to be confirmed with the fabricator's solver.

Learning Objectives

1. Define W, H, t, ε_r, S, and pitch unambiguously for each structure.
2. Distinguish coated microstrip, symmetric and asymmetric stripline, edge- and broadside-coupled pairs, and coplanar structures.
3. Derive $Z_{\text{diff}} = 2Z_{\text{odd}}$ and explain why coupling lowers Z_{odd}.
4. Rank the geometric and material parameters by their effect on impedance.
5. Propagate fabrication tolerances statistically and choose a realistic impedance tolerance.
6. Write a controlled-impedance request that a fabricator can execute and verify.

10.1 Why the definitions matter

Most impedance arguments between designers and fabricators are arguments about definitions. "A 5 mil trace with 5 mil spacing" can mean a 5 mil gap between copper edges (pitch 10 mil) or a 5 mil centre-to-centre pitch (gap equal to $5 - W$), and a 5 mil width can refer to the artwork, the base of the etched trace, or its top. Figure 10.1 and the notation box fix the conventions used in this book.

Cross-section dimensions

W: trace width at the *base* (the side bonded to the dielectric below it); the top is narrower by $2e$, where e is the etch-back per side.
t: finished copper thickness (foil plus any plating on outer layers).

H: dielectric thickness from the reference-plane surface to the trace base (microstrip).
H_1, H_2: stripline distances from the lower plane to the trace base and from the trace top to the upper plane; $b = H_1 + t + H_2$.
S: *edge-to-edge* gap between the traces of a pair, at the base; pitch $p = W + S$.
C_1, C_2: solder-mask thickness over bare laminate and over the trace.
ε_r: relative permittivity (D_k) at the frequency of interest; D_f: loss tangent.

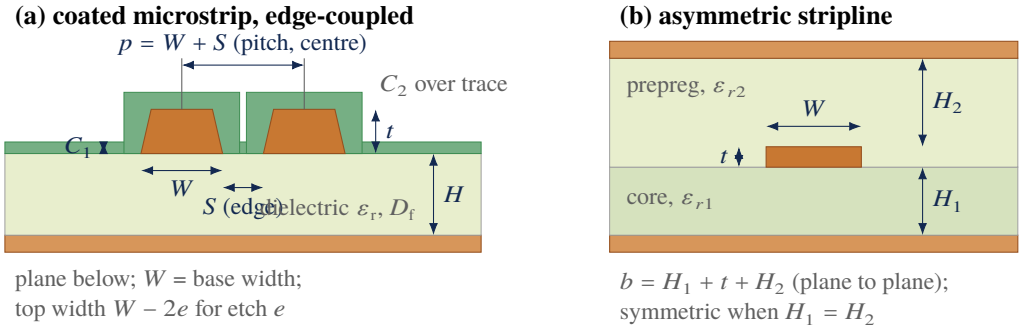


Figure 10.1. Dimension definitions. (a) Coated edge-coupled microstrip: base width W, edge spacing S, pitch p, dielectric height H, thickness t, mask thicknesses C_1 and C_2. (b) Asymmetric stripline with distances H_1 and H_2 and possibly different core and prepreg permittivities. Not to scale. [Conceptual]

Structures. The common structures (Figures 10.1 and 10.2) are:

- **Microstrip** on an outer layer, with one reference plane below and air (or solder mask and air) above. Fields are partly in air, so $\varepsilon_{\text{eff}} < \varepsilon_r$ and the odd and even modes travel at different speeds.
- **Coated microstrip:** microstrip under solder mask. The mask lowers the impedance by a few ohms and must be included in the model.
- **Symmetric stripline**, centred between two planes in a homogeneous dielectric: a true TEM line with $\varepsilon_{\text{eff}} = \varepsilon_r$ for both modes.
- **Asymmetric (offset) stripline**, closer to one plane; the usual case in a stackup with two signal layers between planes (dual stripline).
- **Edge-coupled pairs** (both traces on one layer) and **broadside-coupled pairs** (traces on adjacent layers, one above the other). Broadside coupling is very tight, which allows narrow routing channels, but it depends on layer-to-layer registration and on the thickness of the dielectric between the two traces.
- **Coplanar waveguide** (with or without a ground plane below): ground copper on the same layer at a gap G, stitched with vias. Used on RF launches and where the reference must be local.

10.2 From two capacitances to an impedance

For a quasi-TEM line, the characteristic impedance and effective permittivity follow from two per-unit-length capacitances: C, with the real dielectrics, and C_{air}, with every dielectric replaced by vacuum:

$$Z_0 = \frac{1}{c_0\sqrt{C\,C_{\text{air}}}}, \qquad \varepsilon_{\text{eff}} = \frac{C}{C_{\text{air}}}. \tag{10.1}$$

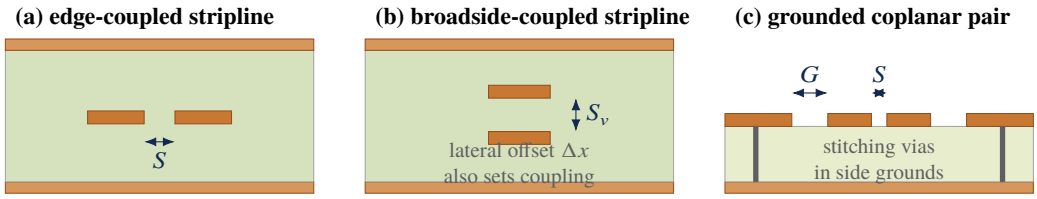


Figure 10.2. Pair structures. (a) Edge-coupled stripline. (b) Broadside-coupled stripline, with vertical separation S_v and lateral offset Δx. (c) Grounded coplanar pair with side-ground gap G and stitching vias. [Conceptual]

(The inductance per unit length is unaffected by non-magnetic dielectrics, so $L = 1/(c_0^2 C_{\text{air}})$, and $Z_0 = \sqrt{L/C}$ gives Eq. (10.1).) A field solver computes the capacitances by solving Laplace's equation over the cross-section; the book's solver uses finite differences on a non-uniform grid whose lines fall exactly on every edge of the geometry.

Pairs, and why $Z_{\text{diff}} = 2Z_{\text{odd}}$. For two identical coupled lines with self capacitance C_{11} and mutual (coupling) capacitance C_m to each other, driving the lines with $+V$ and $-V$ (odd mode) places a virtual ground on the symmetry plane. Each line then sees its capacitance to ground plus $2C_m$ (the mutual capacitance is charged by $2V$), and by the same argument its inductance is $L_{11} - L_m$. The odd-mode impedance of each line is

$$Z_{\text{odd}} = \sqrt{\frac{L_{11} - L_m}{C_{11,g} + 2C_m}}, \qquad Z_{\text{even}} = \sqrt{\frac{L_{11} + L_m}{C_{11,g}}}, \tag{10.2}$$

where $C_{11,g}$ is the capacitance of one line to ground with the other line at the same potential. A differential driver launches the odd mode; the differential voltage is $V_d = 2V$ and the current in each line is $I = V/Z_{\text{odd}}$, so

$$Z_{\text{diff}} = \frac{V_d}{I} = \frac{2V}{V/Z_{\text{odd}}} = 2Z_{\text{odd}}, \qquad Z_{\text{cm}} = \frac{V}{2I_{\text{even}}} = \frac{Z_{\text{even}}}{2}. \tag{10.3}$$

Coupling ($C_m > 0$, $L_m > 0$) lowers Z_{odd} below the uncoupled Z_0 and raises Z_{even} above it. As the spacing grows, both approach Z_0 and $Z_{\text{diff}} \to 2Z_0$ (Figure 10.3).

10.3 A reference stackup and complete designs

The designs in this section use one eight-layer, 1.6 mm (62 mil) stackup (Table 10.1). The values are typical of mid-loss, epoxy-based laminates; they are *not* a specific product, and a real design must use the fabricator's materials, pressed thicknesses, and frequency-specific D_k.

Three observations from Table 10.2:

1. **The mask matters.** The L1 50 Ω trace ($W = 5.48$ mil) would be 53.1 Ω without solder mask (the Hammerstad–Jensen formula gives 53.5 Ω for the same uncoated geometry). A model that omits the mask is off by 6 %, more than most impedance tolerances.
2. **Microstrip pairs have unequal modal velocities.** On L1 the even mode sees $\varepsilon_{\text{eff}} = 3.11$ and the odd mode 2.72, a 7 % difference in delay. This is the origin of far-end crosstalk in microstrip (Chapter 12) and of the location dependence of skew correction (Chapter 40). In homogeneous stripline the two are equal.
3. **Stripline pairs are less coupled at the same S.** The L3 pair has $Z_{\text{even}} - Z_{\text{odd}} = 5.8\,\Omega$, the L1 pair 16.3 Ω: the nearby planes confine the fields. For loosely coupled stripline the common-mode impedance is close to $Z_0/2$.

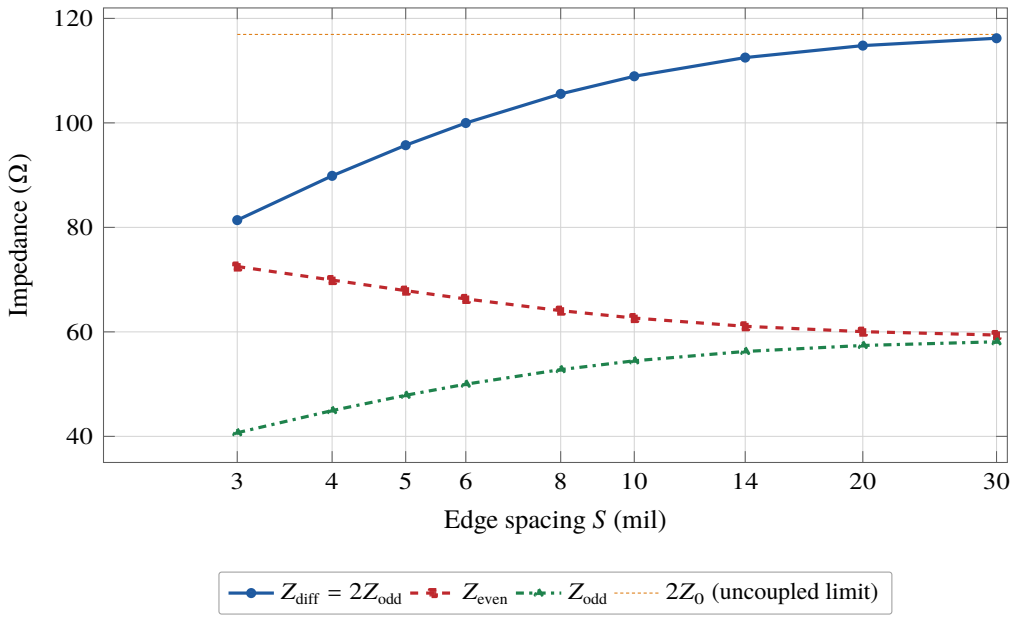


Figure 10.3. Odd-, even-, and differential-mode impedance of the coated microstrip pair of Table 10.2 versus edge spacing, at fixed $W = 3.77$ mil. Tight coupling pulls Z_{diff} well below $2Z_0$; at $S = 30$ mil the lines are nearly independent. [Simulation]

Table 10.1. Reference eight-layer stackup used for the numerical designs. Core and prepreg on each stripline layer are given one ε_r for simplicity (homogeneous stripline); thicknesses are finished values.

Layer	Function	Copper	Dielectric below (to next layer)
Mask	LPI solder mask	—	$C_1 \approx C_2 = 0.8$ mil, $\varepsilon_r = 3.5$
L1	signal (microstrip)	0.5 oz + plating, $t = 1.8$ mil	prepreg 3.5 mil, $\varepsilon_r = 3.8$
L2	ground	1 oz, 1.2 mil	core 5.0 mil, $\varepsilon_r = 3.7$
L3	signal (stripline)	0.5 oz, $t = 0.6$ mil	prepreg 5.0 mil (to trace top), $\varepsilon_r = 3.7$
L4	ground	1 oz, 1.2 mil	core 25.4 mil
L5	power	1 oz, 1.2 mil	prepreg 5.0 mil, $\varepsilon_r = 3.7$
L6	signal (stripline)	0.5 oz, $t = 0.6$ mil	core 5.0 mil, $\varepsilon_r = 3.7$
L7	ground	1 oz, 1.2 mil	prepreg 3.5 mil, $\varepsilon_r = 3.8$
L8	signal (microstrip)	0.5 oz + plating, $t = 1.8$ mil	—

Table 10.2. Designs computed with the 2-D solver for the stackup of Table 10.1 (rectangular cross-sections). Spacing S was chosen first; W was then solved for the target.

Target	Layer	W (mil)	S (mil)	Z_{odd}	Z_{even}	ε_{eff} odd/even	Result
50 Ω SE	L1 coated	5.48	—	—	—	2.98	50.0 Ω
50 Ω SE	L3	4.52	—	—	—	3.70	50.0 Ω
100 Ω diff	L1 coated	3.77	6.0	50.0	66.3	2.72 / 3.11	100.0 Ω
90 Ω diff	L1 coated	4.99	6.0	45.0	58.8	2.76 / 3.15	90.1 Ω
100 Ω diff	L3	3.97	7.0	50.0	55.8	3.70 / 3.70	100.0 Ω
85 Ω diff	L3	5.65	7.0	42.5	46.8	3.70 / 3.70	85.1 Ω

Worked Example 10.1: Specifying an 85 Ω pair

A PCIe link needs 85 Ω differential on L3 with the routing pitch limited to 13 mil. Is the Table 10.2 design feasible, and what should be written on the drawing?

Solution. The design uses $W = 5.65$ mil, $S = 7.0$ mil, so the pitch is $p = 12.65$ mil, inside the limit. Edge spacing is 7.0 mil at the base, which is comfortable for standard etching. The drawing note should read: "L3/L6 edge-coupled differential pairs, 85 Ω ± 10 %, reference L2 and L4 (L5 and L7 for L6), design $W = 5.65$ mil and $S = 7.0$ mil; fabricator may adjust W and S at constant pitch to meet impedance; report as-built values on the coupon report." The phrase "at constant pitch" protects the routing; "report as-built" gives the designer the data to correlate the model.

10.4 Sensitivity: which parameter matters

Table 10.3 changes one parameter at a time by ±10 % and records the impedance. Figure 10.4 extends the sweep to ±20 % for the L3 100 Ω pair.

Table 10.3. One-at-a-time sensitivity, ±10 % change from nominal (2-D solver). "H" for stripline scales H_1 and H_2 together.

Design		W	S	H	t	ε_r
L3 100 Ω pair	−10 %	104.4	98.6	95.8	100.9	105.4
	+10 %	96.0	101.1	103.6	99.1	95.3
L1 50 Ω coated	−10 %	52.4	—	47.1	50.4	52.1
	+10 %	47.9	—	52.8	49.7	48.2

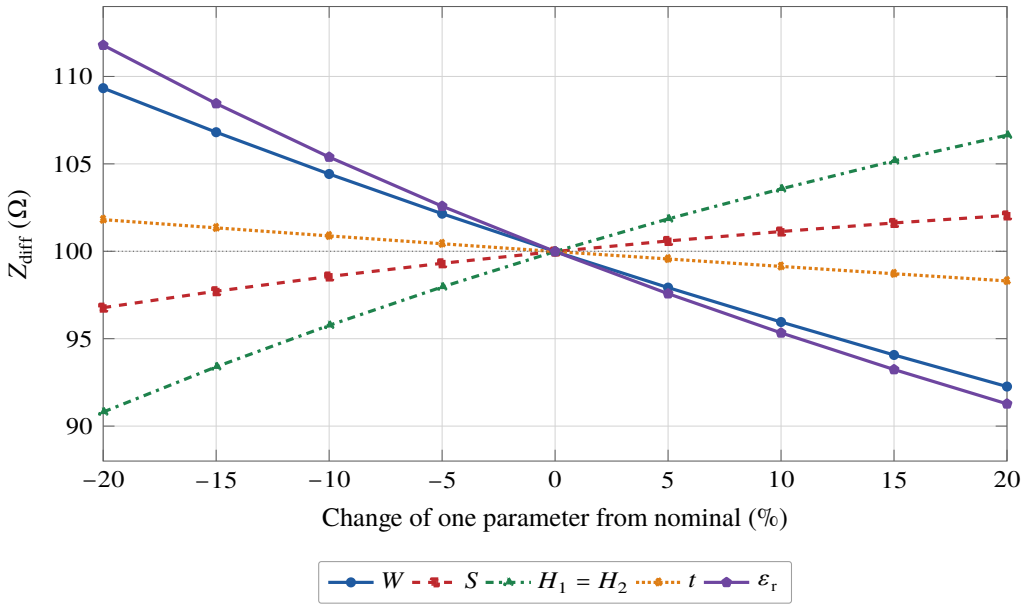


Figure 10.4. Differential impedance of the L3 100 Ω pair when one parameter is changed at a time. Width, dielectric height, and permittivity dominate; spacing and copper thickness matter less for this loosely coupled stripline. [Simulation]

The ranking for these designs is: ε_r, W, and H (about 4 to 5 Ω per 10 % on the 100 Ω pair), then S and t (about 1 Ω). Solder-mask thickness on L1 changes the 50 Ω trace by only 0.2 Ω per 10 %, although the *presence* of the mask is worth 3 Ω. The ranking is design-specific: for a tightly coupled microstrip pair, S moves up, and for thick outer copper, t does. A percentage is also misleading for W: 10 % of a 4 mil trace is 0.4 mil, which is close to the etching tolerance itself.

Etch profile. Real traces are trapezoidal. If the top of the L1 50 Ω trace is 1.2 mil narrower than its base (0.6 mil per side), the solver gives 51.5 Ω instead of 50.0 Ω. For the L1 100 Ω pair the same etch profile gives 105.1 Ω, because the gap between the trace tops also widens. For the thin L3 pair, 0.25 mil per side gives 102.6 Ω. The fabricator's solver includes their measured etch factor; this is one reason their adjusted width differs from the designer's.

Asymmetric stripline. Moving a trace off-centre in a fixed cavity lowers its impedance (Figure 10.5). Near the centre the curve is flat (50.0 Ω at the centre, 48.7 Ω 1 mil off), so a symmetric design tolerates registration and thickness variation well; at $H_1 = 2$ mil the impedance has fallen to 37.3 Ω, and the trace is essentially a microstrip under a lid. In a dual-stripline stackup each signal layer is offset by design, and the solver must be given both H_1 and H_2 and both dielectric constants.

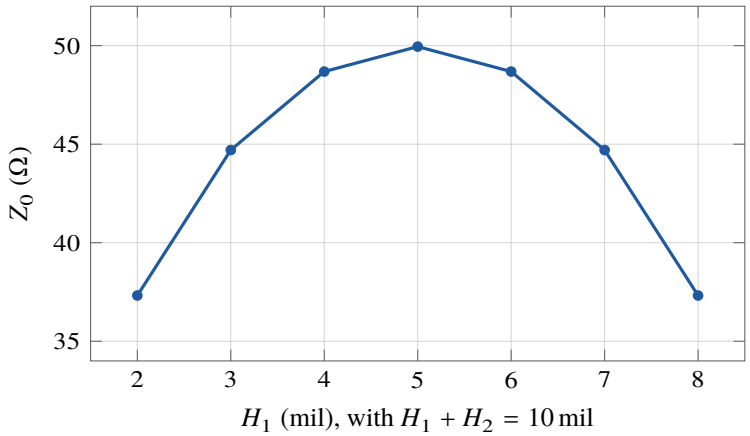


Figure 10.5. Impedance of the 4.52 mil L3 trace as it moves within a cavity of fixed thickness ($H_1 + H_2 = 10$ mil). [Simulation]

10.5 What happens when W, H, t, ε_r, or S changes

Every impedance in this chapter follows from Eq. (10.1): $Z_0 = \sqrt{L/C}$, with the inductance set by the geometry and the capacitance set by the geometry and the dielectric. A change of one dimension therefore moves the impedance in a direction that can be predicted from whether it adds capacitance or inductance. Table 10.4 gives the direction and size for a 10 % increase of each parameter in the reference designs; a decrease moves the impedance the other way by a similar amount.

Wider trace (W ↑): impedance falls. A wider trace has more area facing the plane, so C rises, and its current spreads over more width, so L falls. Both lower Z_0. In the reference designs 10 % more width lowers the single-ended line by about 2 Ω and the pairs by about 3.5–4 Ω. Wider traces also have lower conductor loss.

Table 10.4. Impedance after a 10 % increase of one parameter (2-D solver). Single-ended: the L1 50 Ω coated microstrip. Pair: the L1 100 Ω coated microstrip pair and the L3 100 Ω stripline pair. All values in ohms.

	L1 single-ended	L1 pair			L3 pair
Parameter +10 %	Z_0	Z_{odd}	Z_{even}	Z_{diff}	Z_{diff}
nominal	50.0	50.0	66.3	100.0	100.0
W (width)	47.9	48.3	63.8	96.6	96.0
H (height)	52.8	51.6	70.3	103.1	103.6
t (thickness)	49.7	49.3	66.0	98.6	99.1
ε_r	48.2	48.4	63.8	96.9	95.3
S (spacing)	—	51.0	65.5	102.0	101.1

Thicker dielectric (H ↑): impedance rises. Moving the plane away lowers C and enlarges the current loop, raising L; Z_0 rises by about 3 Ω for the single-ended line and 3.1–3.6 Ω for the pairs. Because the plane is now farther away relative to the gap between the traces, the two traces of a pair couple more strongly to each other: Z_{even} rises more than Z_{odd} (from 66.3 to 70.3 Ω on L1). A larger H also increases crosstalk to neighbours.

Thicker copper (t ↑): impedance falls slightly. The side walls of the trace add fringing capacitance to the plane and, in a pair, to the other trace. The effect is small for thin inner-layer copper (0.9 Ω on the L3 pair) and larger for thick plated outer layers; in a pair Z_{odd} falls more than Z_{even}, because the side walls face each other across the gap.

Higher permittivity (ε_r ↑): impedance falls, delay rises. C scales with the effective permittivity while L does not, so $Z_0 \propto 1/\sqrt{\varepsilon_{eff}}$ and the propagation delay $\propto \sqrt{\varepsilon_{eff}}$. In stripline $\varepsilon_{eff} = \varepsilon_r$ and a 10 % rise lowers the impedance by about 4.7 %; in microstrip part of the field is in air or solder mask, so the change is smaller. The laminate's D_k also varies with frequency, resin content, and glass style, which is why it must be specified at the frequency of interest (Chapter 9).

Wider spacing (S ↑): differential impedance rises. A wider gap weakens the coupling between the two traces: Z_{odd} rises toward the uncoupled Z_0 and Z_{even} falls toward it, so $Z_{diff} = 2Z_{odd}$ rises and $Z_{cm} = Z_{even}/2$ falls (Figure 10.3). Spacing has no effect on an isolated single-ended line; it does affect a single-ended line with a close neighbour, whose impedance then depends on what the neighbour is doing (crosstalk, Chapter 12). For the loosely coupled L3 pair, 10 % more spacing adds only 1.1 Ω; for the more tightly coupled L1 pair it adds 2 Ω.

Solder mask (outer layers). Mask adds dielectric above the trace and lowers the impedance; its presence is worth about 3 Ω on the L1 line, while a 10 % change in its thickness is worth only about 0.2 Ω.

Rule of Thumb: Direction of change

To *raise* impedance: narrow the trace, thicken the dielectric, or (for a pair) widen the gap. To *lower* it: widen the trace, thin the dielectric, or tighten the gap. Permittivity and copper thickness are usually fixed by the material and process, not chosen to tune impedance; when they change, the fabricator compensates with the width.

10.6 Tolerances and Monte Carlo analysis

A one-at-a-time table does not say how often a board will be out of tolerance, because the variations combine. A Monte Carlo analysis draws every parameter from its distribution and computes the impedance for each draw. Running the 2-D solver 20 000 times is slow, so the analysis here uses a surrogate: a quadratic fit in each parameter through the ±10 % solver points of Table 10.3, without interaction terms. At four random draws the surrogate and the full solver agree within 0.02 Ω.

The assumed variations (treated as Gaussian with the stated value at 3σ) are: base width and spacing ±0.5 mil, dielectric heights ±0.5 mil (10 %), copper thickness ±10 %, and ε_r ±0.10. Two cases are compared (Figure 10.6):

- **Independent W and S:** $\sigma(Z_{\text{diff}}) = 2.28\ \Omega$; 2.9 % of boards fall outside ±5 Ω and 0.23 % outside ±7 Ω.
- **Anti-correlated W and S** (the physical case: the artwork fixes the pitch, so over-etching narrows the traces *and* widens the gap): $\sigma(Z_{\text{diff}}) = 2.49\ \Omega$; 4.3 % fall outside ±5 Ω and 0.47 % outside ±7 Ω.

In both cases essentially no boards fall outside ±10 Ω. A ±10 % tolerance is therefore comfortable for this stackup, and ±5 % is achievable only with tighter process control (and a cost adder), not by a better nominal design.

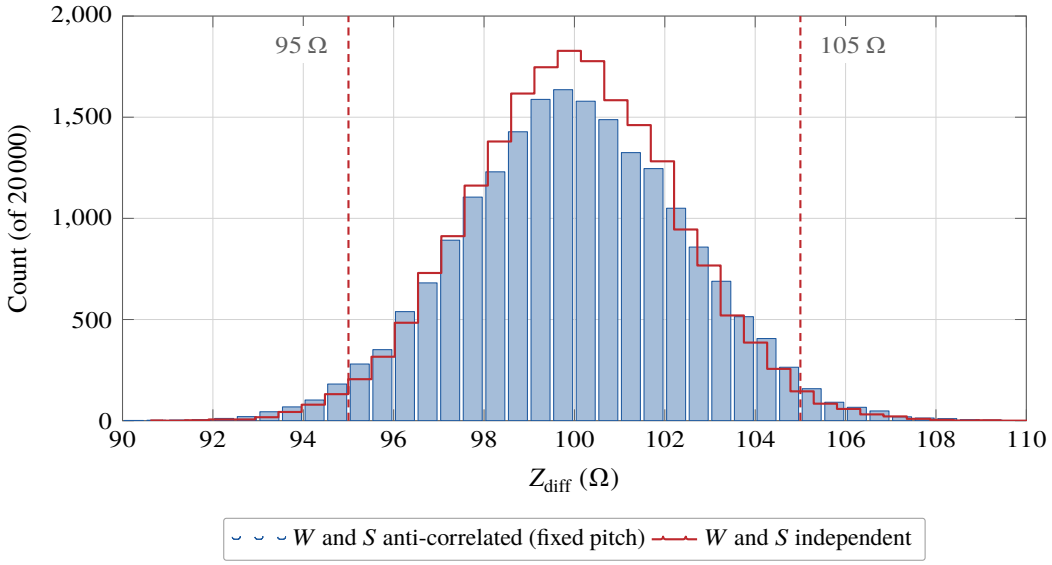


Figure 10.6. Monte Carlo distribution of Z_{diff} for the L3 100 Ω pair, 20 000 draws each. The fixed-pitch case (bars) is wider than the independent case (line) because width and spacing errors add. [Simulation]

Common Pitfall: Tolerance stacking by hand

Adding the worst-case changes of Table 10.3 gives about ±15 Ω, which suggests that ±10 % is impossible. The worst cases do not occur together; the Monte Carlo spread is six times smaller. Conversely, ignoring correlations (fixed pitch, or a press cycle that thins every prepreg at once) underestimates the spread.

10.7 Controlled impedance is not impedance matching

Controlled impedance is a fabrication requirement: the characteristic impedance of specified traces must fall within a tolerance, verified on coupons. *Impedance matching* is a circuit requirement: the source, line, and

load impedances must be related so that reflections are acceptable. A board can be built perfectly to 50 Ω and still ring badly if the driver is 10 Ω and the load is a CMOS gate (Chapters 7 and 8). Controlled impedance makes matching possible; it does not provide it.

10.8 Communicating with the fabricator

The fabricator owns the final geometry. The designer owns the requirement. A clear request contains:

1. The **impedance requirements**: target, tolerance, structure (single-ended or differential), layer, reference planes, and the design width and spacing used in the layout.
2. The **freedom granted**: for example "adjust trace width up to ±1 mil; keep pitch" or "adjust dielectric thickness within the overall thickness tolerance".
3. The **stackup intent**: total thickness, layer order, copper weights, material family and minimum T_g or loss class, and whether glass styles or resin content are fixed (they matter for glass-weave skew, Chapter 9).
4. The **frequency** at which D_k is to be taken, or the test method (TDR coupon at a stated rise time).
5. The **verification**: coupon design and location, test method, and reporting (as-built dimensions and measured impedance per coupon).

The fabricator returns a stackup with adjusted widths; the designer updates the layout rules to those widths before release and checks that no routing channel now violates spacing. Releasing artwork with one width and accepting a stackup with another is a common source of first-article failures.

Coupons and TDR. A coupon is a test structure on the panel rail (or inside the board outline) with long, straight traces of each controlled structure, built on the same layers and referenced to the same planes as the product traces. IPC-2141 and the fabricator's standard coupon designs set the length: the flat part of the TDR trace must last several rise times beyond the probe launch. The measured value is read from a window in the middle of the trace, away from the launch and the open end; the resistance of the line makes the trace slope upward with distance, so the window must be stated. A coupon verifies the panel, not the individual product trace: a coupon within tolerance does not prove that a necked-down breakout under a BGA is within tolerance.

Checklist: impedance geometry

- □ Every width, spacing, and height is defined (base or top, edge or centre, H_1 and H_2).
- □ The model includes solder mask on outer layers and the actual D_k at the frequency of interest.
- □ Sensitivity is known for the dominant parameters of each structure.
- □ The tolerance was chosen from a statistical analysis or the fabricator's capability, not by adding worst cases.
- □ The fabrication note states target, tolerance, layer, references, design geometry, allowed adjustments, and verification.
- □ The layout was updated to the fabricator's adjusted geometry before release.

Key Takeaways

- Edge spacing, centre pitch, base width, and top width are different numbers; name the one you mean.
- $Z_0 = 1/(c_0\sqrt{CC_{\text{air}}})$; a 2-D solver needs only the cross-section.
- $Z_{\text{diff}} = 2Z_{\text{odd}}$ and $Z_{\text{cm}} = Z_{\text{even}}/2$; coupling lowers Z_{odd} and raises Z_{even}.
- Solder mask lowers outer-layer impedance by several ohms; omitting it is a larger error than most tolerances.
- For the reference designs, ε_{r}, W, and H dominate; S and t matter less.
- With fixed pitch, width and spacing errors add; a Monte Carlo analysis shows a $\pm 10\,\%$ tolerance is comfortable and $\pm 5\,\%$ needs tighter process control.
- Controlled impedance is a fabrication property; matching is a circuit property.

Further Reading

Wadell [46] collects closed-form formulas for most structures; Hammerstad and Jensen [48] give the microstrip model used for validation here. Bogatin [3] and Hall and Heck [4] discuss field solvers, odd and even modes, and tolerance. Fabricators publish coupon and stackup guidelines that should be read before the first controlled-impedance order.

Review Problems

P10.1 A drawing says “4 mil/6 mil pair”. List two interpretations and the pitch for each.

P10.2 A solver gives $C = 122\,\text{pF/m}$ and $C_{\text{air}} = 41\,\text{pF/m}$. Find Z_0 and ε_{eff}.

P10.3 A pair has $Z_{\text{odd}} = 44\,\Omega$ and $Z_{\text{even}} = 52\,\Omega$. Find Z_{diff} and Z_{cm} and say whether it is tightly or loosely coupled.

P10.4 (Inverse design) Using Table 10.3, estimate the width change that restores $100\,\Omega$ on L3 if the laminate’s ε_{r} is 3.9 instead of 3.7.

P10.5 (Approximation failure) A designer uses an uncoated microstrip formula for an L1 trace under 0.8 mil of mask and targets $50\,\Omega$. What does the board measure, approximately?

P10.6 Why does the fixed-pitch Monte Carlo case have a larger spread than the independent case?

P10.7 (Imperfect data) A coupon report lists $93\,\Omega$ for a $100\,\Omega \pm 10\,\%$ pair, read at the launch of the TDR trace. Is the board within tolerance?

Chapter 11

Losses

Chapter Overview

Above a few gigabits per second, the dominant signal-integrity impairment of a long board trace is not reflection or crosstalk but loss. Loss attenuates the high-frequency content of each bit, spreads its energy into neighboring bit periods, and closes the eye. This chapter quantifies the two loss mechanisms of a transmission line—conductor loss, including skin effect and surface roughness, and dielectric loss—and shows how to assemble them into an insertion-loss budget for a channel.

Learning Objectives

1. Derive the practical formulas for conductor and dielectric attenuation in dB/in.
2. Identify the crossover frequency at which dielectric loss overtakes conductor loss.
3. Apply the Hammerstad and Huray roughness models and state their regimes of validity.
4. Explain why a causal dielectric model is required and describe the Djordjevic–Sarkar model.
5. Build an insertion-loss budget and use it to select a laminate and a trace width.

11.1 Loss as a low-pass filter

From Eq. (4.12), the attenuation constant of a low-loss line is the sum of a conductor term $\alpha_c = R'/(2Z_0)$ and a dielectric term $\alpha_d = G'Z_0/2$. A matched line of length ℓ has insertion loss

$$\mathrm{IL}(f) = -20\log_{10}|S_{21}| = 8.686\,[\alpha_c(f) + \alpha_d(f)]\ell \quad \text{(dB)}. \tag{11.1}$$

The equality holds only when the line is matched to the measurement reference impedance. A mismatched line also loses transmitted power to reflection, so measured $|S_{21}|$ contains both dissipation (attenuation) and mismatch loss; the two are separated by checking $|S_{11}|^2 + |S_{21}|^2$, which equals 1 only for a lossless network, or by fitting α to lines of two lengths (Chapter 36). A ripple of period $1/(2T_{\mathrm{D}})$ on $|S_{21}|$ is the signature of mismatch, not of dielectric loss. Because both terms rise with frequency, the line is a low-pass filter. A digital bit stream has a broad spectrum; its high-frequency components, which form the sharp edges, are attenuated more than its low-frequency components, which form long runs of identical bits. The received waveform therefore has slow edges and pattern-dependent levels—*inter-symbol interference* (ISI)—which is analyzed with eye diagrams in Chapter 17.

11.2 Conductor loss

11.2.1 Smooth-copper conductor loss

Above the skin-effect onset, the signal conductor of a stripline carries current in a layer one skin depth thick on both its top and bottom faces, and each return plane carries the same current in a similar layer. From Eq. (3.14), doubling to include the return path, and inserting the skin depth of copper,

$$R'_{\text{loop}} \approx 2 \cdot \frac{\rho}{2W\delta_s} \approx \frac{8.25\sqrt{f_{\text{GHz}}}}{W_{\text{mil}}}\ \Omega/\text{in}. \tag{11.2}$$

Dividing by $2Z_0$ and converting nepers to decibels gives the widely used estimate [3]

$$\alpha_c \approx \frac{36\sqrt{f_{\text{GHz}}}}{W_{\text{mil}}\, Z_0}\ \text{dB/in} \qquad \text{(smooth copper, stripline)}. \tag{11.3}$$

Three conclusions follow. Conductor loss grows as $\sqrt{f}$; it is inversely proportional to trace width; and, for a fixed width, it is lower at higher impedance. Wider traces reduce loss only if the dielectric is thickened to keep Z_0 constant, which is why long-reach backplanes use thick stackups with wide traces. Equation (11.3) neglects the current crowding at trace edges and the slightly different geometry of microstrip; for microstrip the estimate is typically optimistic by 10–30 %.

Rule of Thumb: Conductor loss

A 50 Ω, 5 mil stripline with smooth copper loses about $0.14\sqrt{f_{\text{GHz}}}$ dB/in—about 0.3 dB/in at 5 GHz before roughness. *Limits:* smooth copper only; multiply by K_{rough} (Section 11.2.2), which can approach 2 or more.

11.2.2 Surface roughness

When the skin depth becomes comparable to the copper's surface profile, current must follow the peaks and valleys of the surface, lengthening its path and increasing loss. The effect is described by a multiplicative factor on the smooth-copper conductor loss, $\alpha_{c,\text{rough}} = K_{\text{rough}}\,\alpha_c$.

The *Hammerstad* model, originally fitted to measurements of periodic grooves [48], is

$$K_H = 1 + \frac{2}{\pi}\arctan\left[1.4\left(\frac{\Delta}{\delta_s}\right)^2\right], \tag{11.4}$$

where Δ is the RMS surface roughness. It is simple and widely implemented, but it saturates at $K_H = 2$, whereas measured losses of rough foils at tens of gigahertz exceed twice the smooth value.

The *Huray* ("snowball") model represents the surface as stacks of copper spheres of radius a on a flat base, and computes their absorption and scattering of the incident field [51], [52]. For a single sphere size it reduces to

$$K_{\text{Huray}} = \frac{A_{\text{matte}}}{A_{\text{flat}}} + \frac{3}{2}\,\frac{N\,4\pi a^2/A_{\text{hex}}}{1 + \dfrac{\delta_s}{a} + \dfrac{\delta_s^2}{2a^2}}, \tag{11.5}$$

where N is the number of spheres on a hexagonal tile of area A_{hex}. The dimensionless ratio $\text{SR} = N4\pi a^2/A_{\text{hex}}$ is the *surface ratio*; with $A_{\text{matte}}/A_{\text{flat}} \approx 1$, K rises from 1 at low frequency to $1 + 1.5\,\text{SR}$ at high frequency. Its parameters are best obtained from SEM images of the foil or by fitting measured insertion loss; typical sphere radii are a few tenths of a micrometre. Figure 11.1 compares the two models.

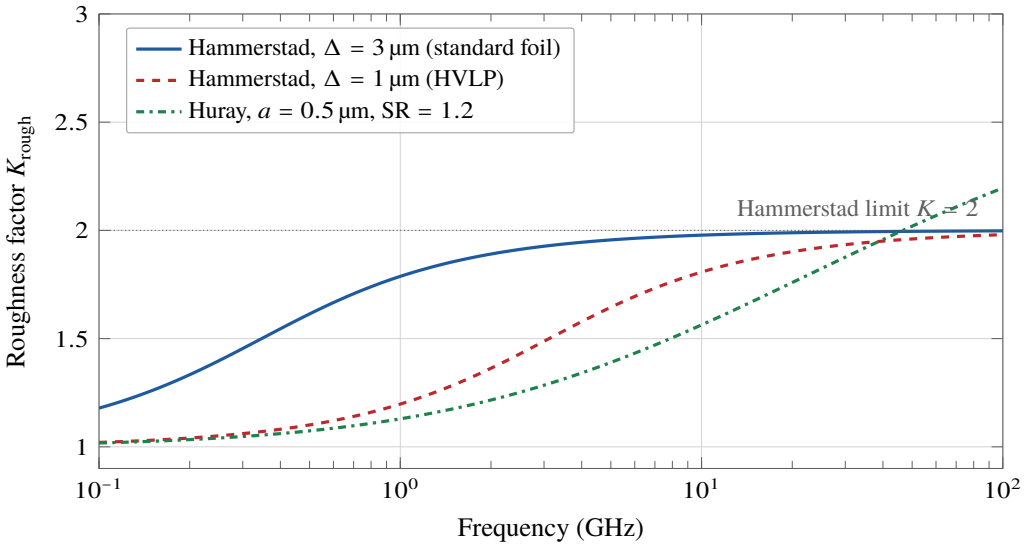


Figure 11.1. Roughness correction factor versus frequency. The Hammerstad model (Eq. (11.4)) saturates at 2; the Huray model (Eq. (11.5)) rises further, toward its own finite asymptote 1 + 1.5 SR, and better matches measurements of rough foil above about 10 GHz. [Analytical]

Common Pitfall: Using datasheet roughness without context

Entering a foil's catalog R_z (a peak-to-valley height) as the RMS roughness Δ of the Hammerstad model, or ignoring the oxide treatment on inner cores. Both errors can misstate conductor loss by tens of percent. Use the fabricator's measured or fitted roughness parameters for the specific foil and bonding treatment, and say which model they belong to.

11.3 Dielectric loss

In an alternating field, the polarization of the dielectric lags the field, dissipating energy. The complex permittivity is written $\varepsilon = \varepsilon_0\varepsilon_r(1 - \mathrm{j}\tan\delta)$, and the loss appears as a shunt conductance $G' = \omega C' \tan\delta$. From Eqs. (4.13) and (4.14),

$$\alpha_d\,[\mathrm{Np/m}] = \frac{\pi f\sqrt{D_k}\,D_f}{c}, \qquad \alpha_d\,[\mathrm{dB/in}] \approx 2.31 f_{\mathrm{GHz}}\,D_f\sqrt{D_k}. \tag{11.6}$$

The two forms differ only by units; writing "=" between them would mix nepers per meter with decibels per inch. Dielectric loss is proportional to frequency, so it eventually dominates. In a homogeneous TEM line (stripline) it is independent of trace width and impedance and can be reduced only by choosing a laminate with lower D_f (Table 9.3). In microstrip, part of the field is in air; $\sqrt{D_k}$ is then replaced by $qD_k/\sqrt{\varepsilon_{\mathrm{eff}}}$, where q is the filling factor (the fraction of electric energy in the dielectric), so α_d depends on the cross-section.

11.3.1 Crossover frequency

Equating Eqs. (11.3) and (11.6) gives the frequency at which dielectric loss overtakes smooth-copper conductor loss:

$$f_\times\,[\mathrm{GHz}] = \left(\frac{36}{2.31\,W_{\mathrm{mil}}Z_0D_f\sqrt{D_k}}\right)^2. \tag{11.7}$$

For a 5 mil, 50 Ω stripline in standard FR-4 ($D_f = 0.02$, $D_k = 4.2$) the crossover is at 2.3 GHz (Figure 11.2); in a laminate with $D_f = 0.004$ it moves to about 70 GHz, and conductor loss—with its roughness—dominates everywhere of practical interest.

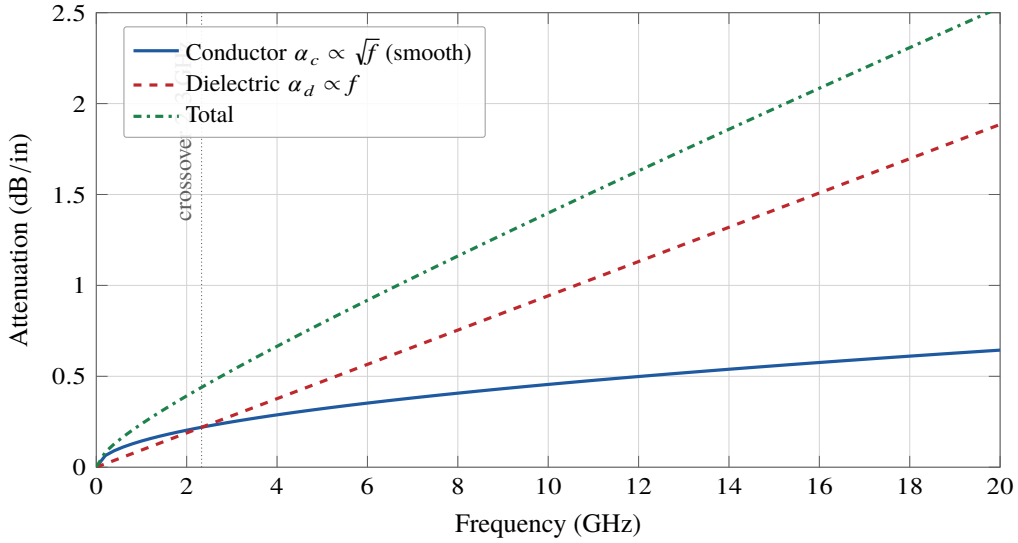


Figure 11.2. Conductor and dielectric attenuation of a 50 Ω, 5 mil stripline in standard FR-4 ($D_k = 4.2$, $D_f = 0.020$, smooth copper). Dielectric loss dominates above about 2.3 GHz.

11.3.2 Causal dielectric models

Data sheets often give a single D_k and D_f at one frequency. A model with both held constant over frequency is non-causal: a time-domain simulation built on it produces responses that begin before the stimulus. The Kramers–Kronig relations require that a lossy dielectric's D_k decrease as frequency rises. The wideband Debye, or Djordjevic–Sarkar, model satisfies this with a distribution of relaxation frequencies between $\omega_1 = 10^{m_1}$ and $\omega_2 = 10^{m_2}$ [39]:

$$\varepsilon_r(\omega) = \varepsilon_\infty + \frac{\Delta\varepsilon}{m_2 - m_1} \log_{10} \frac{\omega_2 + j\omega}{\omega_1 + j\omega}. \tag{11.8}$$

Fitting ε_∞ and $\Delta\varepsilon$ to one (D_k, D_f) pair yields a model in which D_f is nearly constant across several decades and D_k falls slowly (typically a few percent per decade). Most field solvers and SI simulators implement this model; it should be selected whenever a laminate is entered from a data sheet.

Design Rule: Specify materials at the right frequency

Enter D_k and D_f at a frequency near the Nyquist frequency of the fastest interface on the layer, using a causal (Djordjevic–Sarkar or equivalent) model. Ask the fabricator for the D_k of the specific glass style and resin content of each dielectric layer, not a generic value for the laminate family.

11.4 Total insertion loss

Combining the pieces,

$$\mathrm{IL}\,[\mathrm{dB}] \approx \ell_{in}\left[\frac{36\sqrt{f_{GHz}}}{W_{mil} Z_0} K_{rough}(f) + 2.31 f_{GHz} D_f \sqrt{D_k}\right] + \mathrm{IL}_{vias} + \mathrm{IL}_{conn} + \mathrm{IL}_{pkg}. \tag{11.9}$$

Figure 11.3 evaluates the line terms for a 10 in route in standard FR-4 and in a low-loss laminate. The two differ by a factor of almost three at 16 GHz. Table 11.1 summarizes the loss mechanisms.

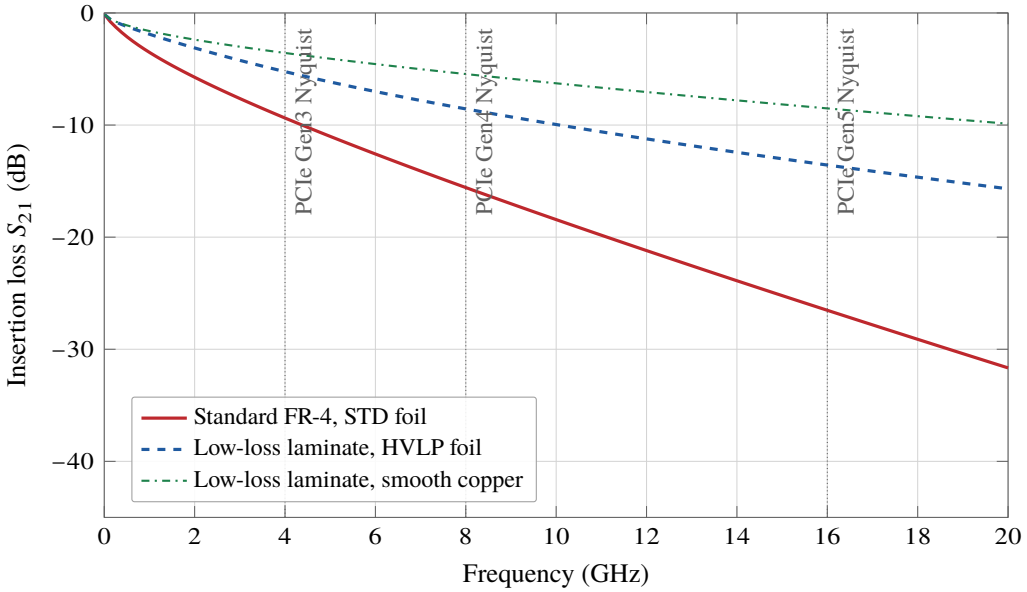


Figure 11.3. Insertion loss of a 10 in, 50 Ω, 5 mil stripline from Eq. (11.9), for standard FR-4 with standard foil and for a low-loss laminate with HVLP foil. The smooth-copper curve shows how much of the low-loss laminate's loss is due to roughness.

Worked Example 11.1: Loss of an FR-4 channel at 4 GHz

Estimate the line loss at 4 GHz (PCIe Gen3 Nyquist) of a 10 in, 50 Ω, 5 mil stripline in standard FR-4 ($D_k = 4.2$, $D_f = 0.020$) with 3 µm RMS roughness.

Solution. Smooth conductor loss: $36 \times 2/(5 \times 50) = 0.288$ dB/in. Skin depth: $2.09/\sqrt{4} = 1.045$ µm, so $1.4(\Delta/\delta_s)^2 = 11.5$ and $K_H = 1 + (2/\pi)\arctan 11.5 = 1.94$; rough conductor loss 0.560 dB/in. Dielectric loss: $2.31 \times 4 \times 0.02 \times \sqrt{4.2} = 0.379$ dB/in. The total is 0.94 dB/in, or 9.4 dB for 10 in—about 40 % of a typical Gen3 channel budget, before vias, connectors, and packages.

Worked Example 11.2: Choosing a laminate for a Gen4 route

A PCIe Gen4 design allocates 12 dB at 8 GHz to a 10 in, 50 Ω, 5 mil stripline with HVLP foil ($\Delta = 1$ µm). What D_f is required if $D_k \approx 3.7$?

Solution. The allowance is 1.2 dB/in. Smooth conductor loss is $36\sqrt{8}/250 = 0.407$ dB/in; with $\delta_s = 0.739$ µm, $K_H = 1.76$ and the rough value is 0.718 dB/in. The dielectric may contribute at most $1.2 - 0.718 = 0.482$ dB/in, so $D_f \le 0.482/(2.31 \times 8 \times \sqrt{3.7}) = 0.0136$. A mid-loss laminate ($D_f \approx 0.010$) meets the budget with some margin; standard FR-4 does not.

Design Rule: Insertion-loss budgeting

Allocate the channel's total loss at the Nyquist frequency among packages, connectors, vias, and line in a written budget before choosing the laminate. Reserve at least 1–2 dB for manufacturing variation (etch,

Table 11.1. Loss mechanisms of PCB transmission lines.

Mechanism	Frequency dependence	Controlled by
DC resistance	constant	Copper weight, width
Skin effect	$\propto \sqrt{f}$	Width, impedance (thicker dielectric)
Surface roughness	factor $1 \to 2$ (Hammerstad) or more (Huray)	Foil grade, oxide treatment
Dielectric loss	$\propto f$	Laminate D_f only
Radiation	rises steeply with f	Shielding (stripline), launch design
Via and connector loss	resonant, $\propto f$ to f^2	Back-drill, anti-pads, connector choice

D_k, roughness) and 5–10 % of the line loss for temperature, since both copper resistivity and laminate D_f rise when hot.

Key Takeaways

- Insertion loss is $8.686(\alpha_c + \alpha_d)\ell$ plus discrete contributions from vias, connectors, and packages.
- Smooth stripline conductor loss is about $36\sqrt{f_\mathrm{GHz}}/(WZ_0)$ dB/in; roughness multiplies it by up to 2 (Hammerstad) or more (Huray).
- Dielectric loss is $2.31 f_\mathrm{GHz} D_\mathrm{f} \sqrt{D_\mathrm{k}}$ dB/in in stripline, independent of geometry there; it dominates FR-4 above a few gigahertz. In microstrip it depends on the filling factor.
- Laminate data must be modeled causally; constant-D_k/D_f models give non-physical time-domain results.
- Write the loss budget first, then choose laminate, foil, and width.

Further Reading

Hall and Heck [4] model conductor and dielectric loss and roughness in detail; Huray [51], [52] develops the snowball roughness model; Djordjevic et al. [39] introduce the causal wideband dielectric model; Johnson and Graham's second book [2] treats lossy-line behavior.

Review Problems

P11.1 Estimate the smooth-copper conductor loss at 10 GHz of a 4 mil, 50 Ω stripline.

P11.2 Find the dielectric loss at 10 GHz in a laminate with $D_\mathrm{k} = 3.4$ and $D_\mathrm{f} = 0.003$.

P11.3 At what frequency does dielectric loss equal smooth conductor loss for a 5 mil, 50 Ω stripline with $D_\mathrm{f} = 0.02$ and $D_\mathrm{k} = 4.2$?

P11.4 Evaluate the Hammerstad roughness factor at 5 GHz for $\Delta = 2\,\mu\mathrm{m}$.

P11.5 What is the high-frequency limit of the Huray factor for a surface ratio SR = 1.2 and $A_\mathrm{matte}/A_\mathrm{flat} = 1$?

P11.6 Estimate the line loss at 16 GHz of a 6 in, 50 Ω, 5 mil stripline in a laminate with $D_\mathrm{k} = 3.5$, $D_\mathrm{f} = 0.004$, and HVLP foil ($\Delta = 1\,\mu\mathrm{m}$, Hammerstad).

Chapter 12

Crosstalk

Chapter Overview

Crosstalk is the unwanted transfer of signal from one net (the *aggressor*) to another (the *victim*) through their mutual capacitance and mutual inductance. On dense boards it is a leading source of noise and timing jitter. This chapter derives the near-end and far-end crosstalk of coupled transmission lines, explains saturation length and the cancellation of far-end crosstalk in stripline, examines the popular "3W rule" and its limits, and gives layout guidelines for controlling coupling.

Learning Objectives

1. Express crosstalk in terms of the ratios C_m/C and L_m/L.
2. Compute near-end crosstalk (NEXT) and its saturation length.
3. Compute far-end crosstalk (FEXT) and explain why it vanishes in a homogeneous stripline.
4. Evaluate the 3W rule and choose spacing from S/H rather than S/W.
5. Decide when guard traces help and how to stitch them.
6. Build a crosstalk budget across several aggressors.

12.1 Coupling mechanisms

Two parallel traces over a common plane share electric and magnetic fields. Their coupling is described by a mutual capacitance per unit length C_m and a mutual inductance per unit length L_m (Figure 12.1). A changing voltage on the aggressor injects current $C_m\ \mathrm{d}V_a/\mathrm{d}t$ into the victim; a changing aggressor current induces a series voltage $L_m\ \mathrm{d}I_a/\mathrm{d}t$ in the victim. Both are distributed along the coupled length, and their contributions travel in both directions along the victim.

The capacitive and inductive contributions add in the backward direction and subtract in the forward direction. To first order in the coupling (weak coupling, both lines terminated in Z_0), the standard results are [3], [4], [40]

$$K_b = \frac{1}{4}\left(\frac{C_m}{C} + \frac{L_m}{L}\right), \tag{12.1}$$

$$K_f = \frac{1}{2v_\mathrm{p}}\left(\frac{C_m}{C} - \frac{L_m}{L}\right), \tag{12.2}$$

where C and L are the per-unit-length self capacitance and inductance of a line in the presence of the other. K_b is dimensionless; K_f has units of time per length.

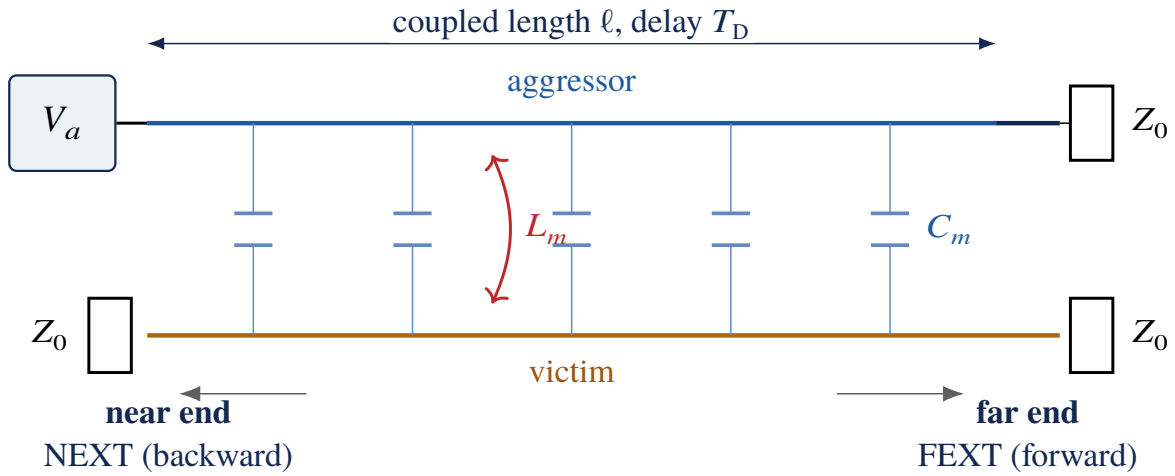


Figure 12.1. Coupled aggressor and victim lines. Distributed mutual capacitance C_m and mutual inductance L_m produce a backward wave at the victim's near end (NEXT) and a forward wave at its far end (FEXT).

Reading the coefficients from a field solver. A 2-D solver reports per-unit-length matrices for the pair,

$$\mathbf{L} = \begin{bmatrix} L_{11} & L_{12} \\ L_{12} & L_{22} \end{bmatrix}, \qquad \mathbf{C} = \begin{bmatrix} C_{11} & C_{12} \\ C_{12} & C_{22} \end{bmatrix}. \tag{12.3}$$

$\mathbf{L}$ is straightforward: $L = L_{11}$ and $L_m = L_{12} > 0$. The capacitance matrix is usually the *Maxwell* matrix, whose diagonal C_{11} is the *total* capacitance of line 1 (to ground plus to line 2) and whose off-diagonal term is *negative*, $C_{12} = -C_m$. Hence $C = C_{11}$ and $C_m = -C_{12}$. Some tools report the "SPICE" (branch) form instead, with positive mutual and a diagonal that excludes it; mixing the two conventions flips the sign of the capacitive term in Eq. (12.2) and can turn a cancelling stripline FEXT into a large one. Check the convention by confirming that $C_m/C = L_m/L$ for a homogeneous stripline.

12.2 Near-end crosstalk

As the aggressor edge travels along the coupled region, it continuously sends a backward wave to the victim's near end. The near-end voltage builds up while the edge enters the coupled region, then holds at a plateau for as long as the edge is still traveling inside it, and finally decays. The plateau lasts $2T_D$ (the time for the edge to reach the far end plus the time for the last backward contribution to return). If $2T_D \geq t_r$, NEXT reaches its full value

$$V_{NEXT} = K_b V_a, \qquad 2T_D \geq t_r; \tag{12.4}$$

otherwise the plateau is never reached and the peak is reduced in proportion, $V_{NEXT} \approx K_b V_a \cdot 2T_D/t_r$. The coupled length at which NEXT saturates is the *saturation length*

$$\ell_{sat} = \frac{t_r v_p}{2}. \tag{12.5}$$

For a 100 ps edge in FR-4 stripline, ℓ_{sat} is only 0.3 in. Beyond saturation, making the parallel run longer does not increase NEXT amplitude; it only lengthens the plateau.

12.3 Far-end crosstalk

The forward contributions travel with the aggressor edge and arrive at the far end together, after T_{D}. Their sum is proportional to the coupled length and to the derivative of the aggressor waveform:

$$V_{\mathrm{FEXT}}(t) = K_f \, \ell \, \frac{\mathrm{d}V_a(t - T_{\mathrm{D}})}{\mathrm{d}t} \approx K_f \, \ell \, \frac{V_a}{t_{\mathrm{r}}} \quad \text{(peak, linear ramp).} \tag{12.6}$$

FEXT is a pulse of width $\approx t_{\mathrm{r}}$ centered on the arrival of the aggressor edge (Figure 12.3). Unlike NEXT, it grows with coupled length and with edge rate—but only while it is small.

12.3.1 When the derivative approximation fails

Equation (12.6) is a first-order expansion. The exact behavior follows from the modal picture of Chapter 13: a step on the aggressor of a symmetric, matched pair launches equal even- and odd-mode waves, which travel with delays T_e and T_o. The victim far-end voltage is half their difference,

$$V_{\mathrm{FEXT}}(t) = \frac{1}{2}\big[a(t - T_e) - a(t - T_o)\big] \approx -\frac{\Delta T}{2}\,\frac{\mathrm{d}a}{\mathrm{d}t}, \qquad \Delta T = T_e - T_o, \tag{12.7}$$

where $a(t)$ is the aggressor waveform. Comparing with Eq. (12.6) identifies $K_f \ell = -\Delta T/2$. The approximation holds while $|\Delta T| \ll t_{\mathrm{r}}$. When the modes separate by a large fraction of the edge, the two delayed copies no longer overlap: the victim sees a pulse of width ΔT whose height approaches half the aggressor swing and then stops growing (Figure 12.2). In a real bus, loss, reflections and unequal terminations modify this ideal limit, but the conclusion is general: FEXT does not grow without bound.

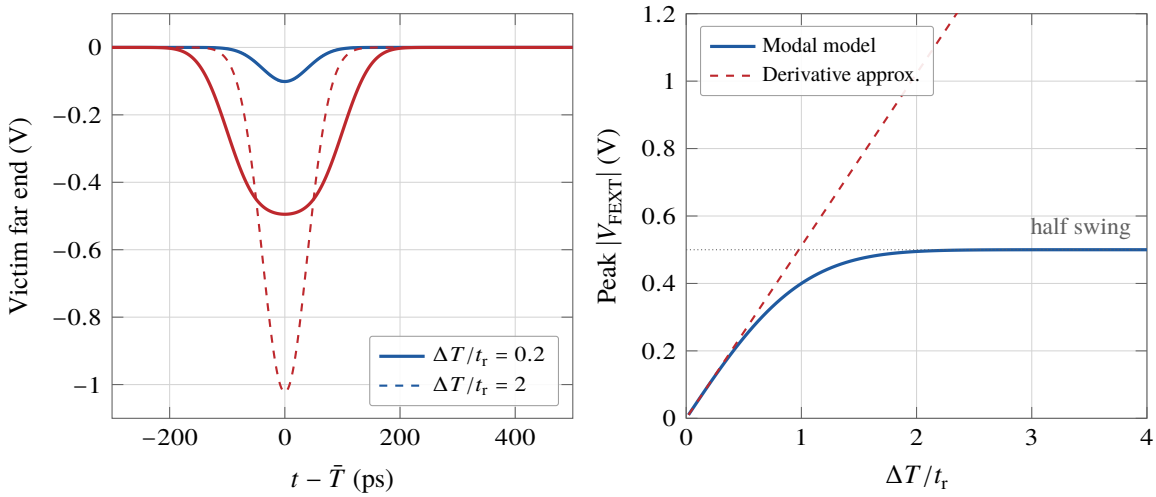


Figure 12.2. Far-end crosstalk from the modal model of Eq. (12.7) (solid) and its derivative approximation (dashed), for a 1 V Gaussian edge with $t_{\mathrm{r}} = 100\,\mathrm{ps}$. Left: waveforms for $\Delta T/t_{\mathrm{r}} = 0.2$ and 2. Right: peak magnitude versus modal delay difference. Matched, lossless, symmetric pair. [Analytical]

12.3.2 Stripline versus microstrip

In a homogeneous dielectric the capacitive and inductive coupling ratios are equal, $C_m/C = L_m/L$, so $K_f = 0$ and *FEXT cancels*. In microstrip, part of the electric field—and therefore of the mutual capacitance—lies in air, whereas the magnetic field is unaffected by the dielectric; thus $L_m/L > C_m/C$ and FEXT is negative

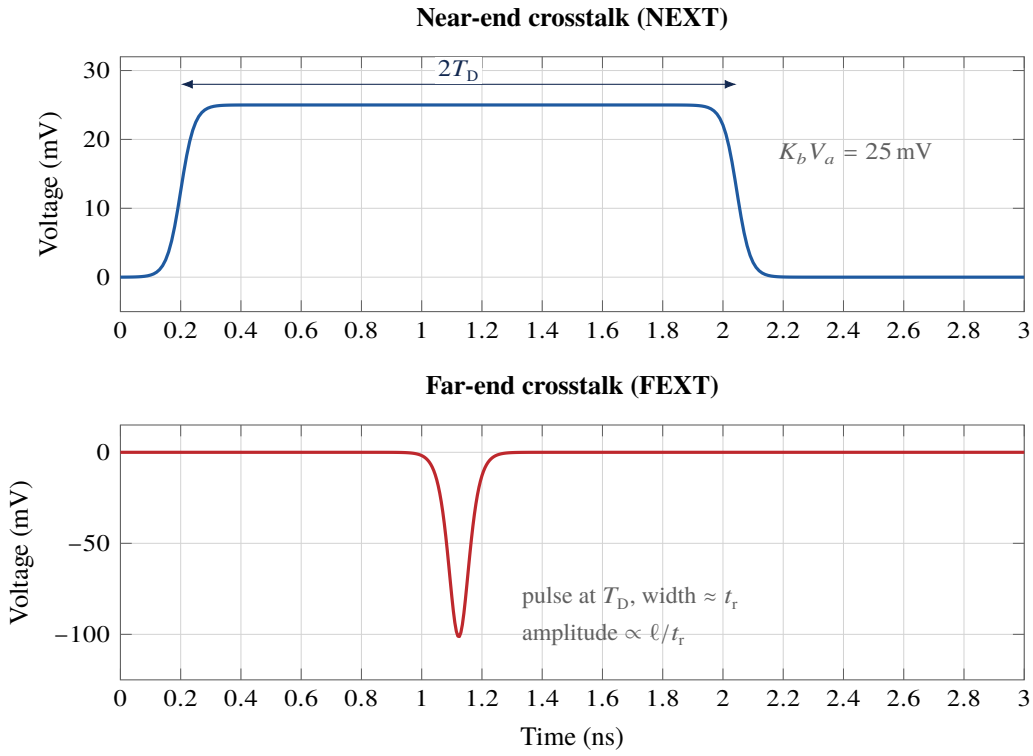


Figure 12.3. Crosstalk on a matched victim from a 1 V, 100 ps aggressor edge over 6 in of coupled microstrip ($C_m/C = 0.04$, $L_m/L = 0.06$). NEXT is a plateau of duration $2T_D$; FEXT is a negative pulse at T_D.

(the victim pulse is inverted relative to the aggressor edge). Embedded microstrip and stripline with unequal dielectrics (for example a resin-rich prepreg above and a glass-rich core below) have partial cancellation.

Worked Example 12.1: Crosstalk on a microstrip bus

Two 50 Ω microstrips are coupled over 6 in with $C_m/C = 0.04$ and $L_m/L = 0.06$; $v_p = 6.5$ in/ns. The aggressor is a 1 V edge with $t_r = 100$ ps. Find NEXT and FEXT.

Solution. $K_b = 0.25(0.04 + 0.06) = 0.025$. $T_D = 6/6.5 = 0.923$ ns, so $2T_D \gg t_r$ and NEXT saturates at 25 mV. $K_f \ell = (6/(2 \times 6.5))(0.04 - 0.06) = -9.23$ ps; with a linear ramp, $V_{FEXT} = -9.23/100 \times 1\text{ V} = -92$ mV. A smooth edge with the same 10–90 % time has a higher peak slope, about −100 mV in Figure 12.3. FEXT, not NEXT, dominates on long microstrip buses. *Validity:* $\Delta T = -2K_f \ell = 18.5$ ps, so $\Delta T/t_r = 0.18$ and the derivative approximation is adequate (Figure 12.2).

Worked Example 12.2: Saturation length

DDR data lines run in parallel stripline ($v_p = 6$ in/ns) with $t_r = 100$ ps. How long must the parallel run be before NEXT saturates, and what is NEXT for a 0.1 in coupled breakout?

Solution. $\ell_{sat} = 0.1\text{ ns} \times 6\text{ in/ns}/2 = 0.3$ in. A 0.1 in section has $2T_D = 33$ ps, so NEXT reaches $33/100 = 33$ % of its saturated value. Short breakout regions where traces must run close together therefore contribute much less NEXT than long parallel routes.

12.4 Termination of the victim

The expressions above assume that both ends of the victim are terminated in Z_0. A victim driven by a low-impedance driver at its near end ($\Gamma \approx -1$) reflects the NEXT, inverted, toward the far end, where it adds to FEXT one round trip later. A victim with a high-impedance receiver at its far end ($\Gamma \approx +1$) doubles the FEXT there, and reflects it back to the near end. A source-terminated victim absorbs the NEXT at the driver. These interactions are why crosstalk should be simulated with the actual drivers and receivers, and why series-terminated nets usually suffer less crosstalk than unterminated ones.

12.5 Spacing and the 3W rule

The coupling coefficients fall rapidly with separation. Johnson and Graham showed that, for microstrip, coupling decreases roughly as $1/[1 + (D/H)^2]$, where D is the center-to-center spacing [1]. In stripline, the second plane confines the fields and coupling falls faster, approximately exponentially with D/H (Figure 12.4).

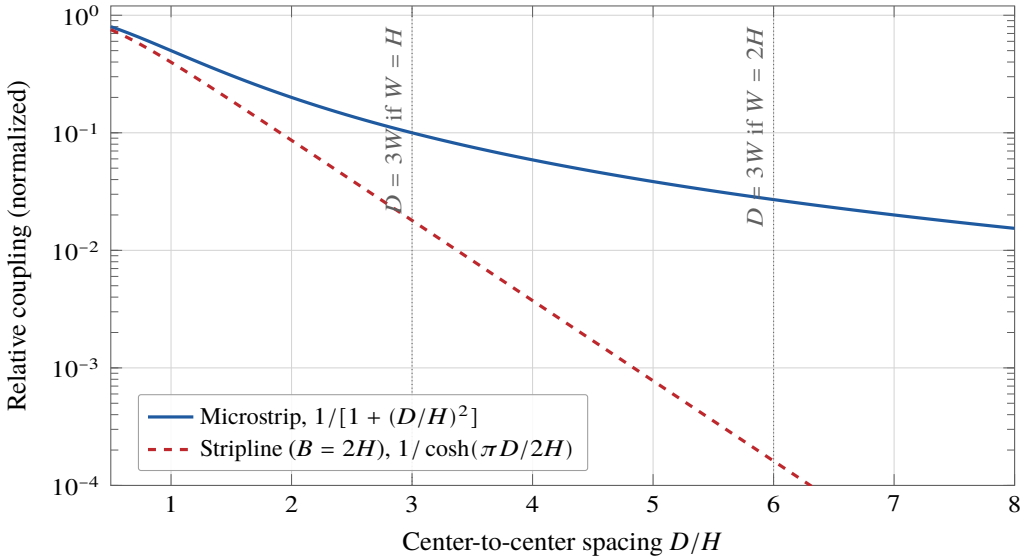


Figure 12.4. Relative coupling versus center-to-center spacing normalized to dielectric height. Microstrip coupling falls as $1/[1+(D/H)^2]$; stripline coupling falls approximately exponentially. Curves are normalized and show trends only; absolute values require a field solver.

The *3W rule* states that the center-to-center spacing of traces should be at least three times the trace width ($S \geq 2W$ edge to edge). Its physical basis is the fall-off of fringing fields with distance—but that distance is measured in units of the dielectric height H, not of W. The rule is reasonable only when W and H are comparable.

Rule of Thumb: The 3W rule and its limits

$D \geq 3W$ keeps microstrip NEXT to a few percent *when* $W \approx 2H$, as for a 50 Ω microstrip on FR-4. *Limits:* for a narrow trace over a thick dielectric (high impedance, $W < H$), 3W may leave the traces strongly coupled; for a wide trace over a thin dielectric, it is overly conservative. Specify spacing as a multiple of H (typically $S \geq 2H$ to $3H$ edge to edge), and confirm with a field solver or crosstalk simulation.

Worked Example 12.3: Is 3W enough?

Compare the relative coupling at $D = 3W$ for (a) a microstrip with $W = 7$ mil, $H = 3.5$ mil, and (b) a microstrip with $W = 4$ mil, $H = 4$ mil.

Solution. (a) $D = 21$ mil $= 6H$, so $1/[1+36] = 0.027$. (b) $D = 12$ mil $= 3H$, so $1/[1+9] = 0.10$. The same "3W" spacing gives almost four times more coupling in case (b). Spacing (b) to $D = 6H = 24$ mil would restore the margin of case (a).

12.6 Guard traces

A grounded trace between aggressor and victim intercepts some of the coupled field. In stripline, where coupling is already small at moderate spacings, a guard trace adds little beyond what the extra space it occupies would provide. In microstrip it can reduce coupling by a factor of two or more—but only if it is well grounded. An unstitched or singly grounded guard is itself a resonant transmission line that can *increase* crosstalk at its resonant frequencies.

Design Rule: Guard traces

Use a guard trace only when spacing alone cannot meet the crosstalk budget. Ground it at both ends and stitch it to the reference plane with vias spaced no more than about one-tenth of the rise length, $s \le t_r v_p/10$. That spacing puts the half-wave resonance of each stitched segment, $v_p/(2s)$, at ten times the knee frequency. (The rule "$\lambda/10$ at f_{knee}" is often quoted too; since $\lambda(f_{knee}) = 2t_r v_p$, it allows twice the spacing, and its first segment resonance is only five times f_{knee}.) Never leave a guard trace floating or grounded at one end.

12.7 Other coupling paths

Broadside coupling
: Traces on adjacent signal layers that run parallel couple strongly because they face each other across a thin dielectric. In dual-stripline stackups, route adjacent layers orthogonally and avoid long parallel overlaps.

Via coupling
: Closely spaced vias, especially in BGA fields and connector pin fields, couple through the cavity between planes. Return vias adjacent to each signal via reduce this coupling (Chapter 15).

Common return-path coupling
: Signals that share a return path through a connector or package pin interact through the shared inductance; this is ground bounce (Chapter 20).

12.8 Crosstalk budgets

A victim usually has several aggressors. If their switching is uncorrelated, their peak contributions add worst-case only when all switch together; a conservative budget sums the peak values of the nearest aggressors on each side and adds a root-sum-square allowance for more distant ones. Crosstalk that arrives during the victim's own transition shifts its threshold crossing time and appears as timing jitter rather than as amplitude noise.

Table 12.1. Crosstalk characteristics and mitigations.

	NEXT (backward)	FEXT (forward)
Coefficient	$K_b = \frac{1}{4}(C_m/C + L_m/L)$	$K_f = \frac{1}{2v_p}(C_m/C - L_m/L)$
Waveform	Plateau of length $2T_D$	Pulse of width $\approx t_r$ at T_D
Length dependence	Saturates at $\ell_{sat} = t_r v_p/2$	$\propto \ell$ while $\lvert\Delta T\rvert \ll t_r$; bounded beyond
Edge dependence	Independent of t_r once saturated	$\propto 1/t_r$ (small-ΔT regime)
Stripline	Present	Cancels in homogeneous dielectric
Mitigation	Spacing, source termination	Spacing, stripline routing, shorter parallel runs

Common Pitfall: Analyzing one aggressor

Simulating a victim with only one neighbor. Buses have aggressors on both sides, and in dual-stripline stackups possibly above or below. Simulate at least the two nearest neighbors on each side switching in the worst-case pattern, with realistic driver strengths and terminations.

Key Takeaways

- Crosstalk arises from C_m and L_m; it is characterized by C_m/C and L_m/L.
- NEXT = $K_b V_a$ with $K_b = \frac{1}{4}(C_m/C + L_m/L)$; it saturates once the coupled length exceeds $t_r v_p/2$.
- FEXT grows with length and edge rate while $\lvert\Delta T\rvert \ll t_r$; beyond that it is bounded by modal separation and loss. It is negative in microstrip and cancels in a homogeneous stripline.
- Spacing should be specified relative to dielectric height H; the 3W rule is valid only when $W \approx 2H$.
- Guard traces help only if stitched densely to the plane.

Further Reading

Paul [40] gives the multiconductor theory behind crosstalk; Hall and Heck [4] and Bogatin [3] derive NEXT and FEXT for practical geometries.

Review Problems

P12.1 Find K_b for $C_m/C = 0.05$ and $L_m/L = 0.08$.

P12.2 Find the NEXT saturation length for $t_r = 50\,\text{ps}$ and $v_p = 6\,\text{in/ns}$.

P12.3 With the values of Problems P12.1 and P12.2, what NEXT does a 1 V aggressor produce over a 0.1 in coupled length?

P12.4 Two microstrips are coupled over 4 in with $C_m/C = 0.04$, $L_m/L = 0.07$, $v_p = 6.5\,\text{in/ns}$. Estimate the FEXT from a 1 V, 80 ps edge.

P12.5 In a stripline with $C_m/C = L_m/L = 0.05$, what are NEXT and FEXT for a 1 V aggressor?

P12.6 By what factor does microstrip coupling fall when D/H is increased from 3 to 6?

Chapter 13

Differential Signaling

Chapter Overview

Nearly every interface above a gigabit per second uses differential pairs: PCIe, USB, Ethernet, MIPI, DDR clocks and strobes, and SERDES links of every kind. A pair carries information in the voltage difference between two conductors, which gives it immunity to common noise, low emissions, and a well-defined return path even when the ground reference is imperfect. Those advantages are realized only if the pair is symmetric. This chapter introduces odd- and even-mode analysis, defines differential and common-mode impedance, explains how skew converts differential signal into common-mode noise, and gives practical rules for length matching and serpentine design.

Learning Objectives

1. Decompose arbitrary signals on a pair into differential and common modes.
2. Compute odd-, even-, differential-, and common-mode impedances from per-unit-length parameters.
3. Design terminations that absorb both modes.
4. Quantify mode conversion caused by intra-pair skew.
5. Specify intra-pair and inter-pair skew budgets and implement length matching with serpentines.

13.1 Modes of a coupled pair

Any pair of voltages (V_P, V_N) on the two traces can be written as a differential and a common-mode component:

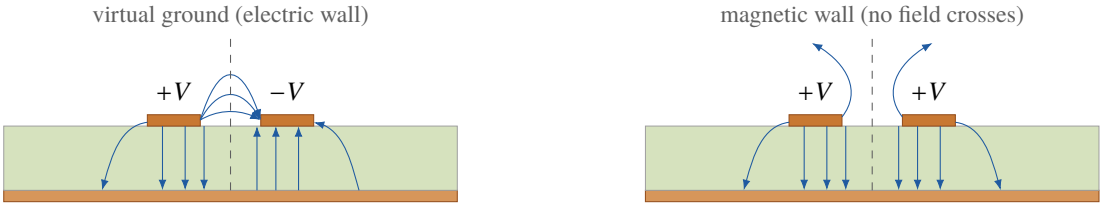
$$V_d = V_P - V_N, \qquad V_c = \frac{V_P + V_N}{2}, \tag{13.1}$$

with the inverse $V_P = V_c + V_d/2$ and $V_N = V_c - V_d/2$. A receiver that responds only to V_d rejects any noise that appears equally on both traces. Driving the pair with equal and opposite signals ($V_c = 0$) excites the *odd mode*; equal signals ($V_d = 0$) excite the *even mode*. Their field patterns are sketched in Figure 13.1.

13.2 Odd, even, differential, and common-mode impedance

Let each trace have self inductance L_{11} and capacitance to ground C_g, with mutual inductance L_{12} and mutual capacitance C_m between traces (per unit length). In the odd mode the mutual inductance opposes the self inductance and the mutual capacitance is effectively doubled (it is charged from both sides); in the even mode the mutual inductance aids and the mutual capacitance carries no charge. Hence

$$Z_{\text{odd}} = \sqrt{\frac{L_{11} - L_{12}}{C_g + 2C_m}}, \qquad Z_{\text{even}} = \sqrt{\frac{L_{11} + L_{12}}{C_g}}. \tag{13.2}$$



(a) Odd mode (differential): $Z_{odd} = Z_{diff}/2$ **(b)** Even mode (common): $Z_{even} = 2Z_{cm}$

Figure 13.1. Electric field patterns of an edge-coupled microstrip pair. (a) In the odd mode, field lines run strongly from one trace to the other, and the symmetry plane is a virtual ground. (b) In the even mode, both traces are at the same potential and fields are pushed toward the plane.

Each mode also has its own velocity, $v_p = 1/\sqrt{LC}$ with the corresponding L and C; the two are equal only in a homogeneous dielectric.

The impedance seen by a differential signal is the series combination of the two odd-mode lines, and the impedance seen by a common-mode signal is their parallel combination:

$$Z_{diff} = 2Z_{odd}, \qquad Z_{cm} = \frac{Z_{even}}{2}. \tag{13.3}$$

For uncoupled traces, $Z_{odd} = Z_{even} = Z_0$; tighter coupling lowers Z_{odd} and raises Z_{even}. A pair is designed for its target Z_{diff} (85 Ω, 90 Ω, or 100 Ω, depending on the interface); Z_{cm} is usually uncontrolled but matters for EMI and common-mode noise.

Worked Example 13.1: Modal impedances of a coupled pair

A field solver reports, per inch, $L_{11} = 8.5\,\text{nH}$, $L_{12} = 1.0\,\text{nH}$, $C_g = 3.0\,\text{pF}$, and $C_m = 0.3\,\text{pF}$. Find Z_{odd}, Z_{even}, Z_{diff}, and Z_{cm}.

Solution. From Eq. (13.2), $Z_{odd} = \sqrt{7.5\times10^{-9}/3.6\times10^{-12}} = 45.6\,\Omega$ and $Z_{even} = \sqrt{9.5\times10^{-9}/3.0\times10^{-12}} = 56.3\,\Omega$. Then $Z_{diff} = 91.3\,\Omega$ and $Z_{cm} = 28.1\,\Omega$.

13.2.1 Tight versus loose coupling

A common misconception is that the two traces of a pair must be tightly coupled for the pair to work. In fact the return current of a differential signal over a solid plane flows mostly in the plane under each trace, whatever the spacing; coupling only changes Z_{odd}. Loose coupling ($S \gtrsim 2W$) makes Z_{diff} insensitive to spacing variation, eases routing around obstacles, and allows narrower traces for the same impedance. Tight coupling ($S \approx W$) gives denser routing, better rejection of noise that couples equally into both traces, and smaller loop area if the plane is interrupted. Most designs choose S between W and $2W$ and keep it constant along the route.

Design Rule: Pair geometry

Hold the pair's spacing constant along its length; split only where pins force it, and keep the uncoupled length shorter than about one-tenth of the rise length. Route both traces on the same layer, over the same reference plane, with the same number of vias placed symmetrically.

13.3 Terminating both modes

A single resistor Z_{diff} across the pair terminates the odd mode but presents an open circuit to the even mode, which therefore reflects. Two networks terminate both modes:

T network

A resistor $R_1 = Z_{\text{odd}}$ from each trace to a center node, and $R_2 = (Z_{\text{even}} - Z_{\text{odd}})/2$ from that node to ground (or to AC ground through a capacitor).

Π network

A resistor $R_1 = Z_{\text{even}}$ from each trace to ground, and $R_2 = 2Z_{\text{even}}Z_{\text{odd}}/(Z_{\text{even}} - Z_{\text{odd}})$ across the pair.

For the pair of Worked Example 13.1, the T network uses $R_1 = 45.6\,\Omega$ and $R_2 = 5.3\,\Omega$. In practice the split termination—two resistors of $Z_{\text{diff}}/2$ with a capacitor to ground at the midpoint—is the common compromise; on-die terminations of modern receivers often implement it internally. It terminates the even mode in Z_{odd} rather than Z_{even}: for this pair, $\Gamma_e = (45.6 - 56.2)/(45.6 + 56.2) = -0.10$. That is usually acceptable, but it is a mismatch, not a match.

13.4 Skew and mode conversion

13.4.1 Intra-pair skew

If one trace of a pair is delayed by Δt relative to the other, a purely differential signal acquires a common-mode component. Writing the arriving voltages as $V_P = \frac{V_d}{2}e^{-j\omega t_1}$ and $V_N = -\frac{V_d}{2}e^{-j\omega(t_1+\Delta t)}$ and applying Eq. (13.1),

$$\left|\frac{V_c}{V_d}\right| = \frac{1}{2}\left|\sin(\pi f \Delta t)\right|, \qquad \left|\frac{V_{d,\text{out}}}{V_d}\right| = \left|\cos(\pi f \Delta t)\right|. \tag{13.4}$$

These are voltage ratios. The mixed-mode S-parameter for the same effect is $|S_{cd21}| = |\sin(\pi f \Delta t)|$, twice the first ratio, because mixed-mode waves are normalized to 25 Ω (common) and 100 Ω (differential) references (Chapter 14). Skew therefore does two kinds of damage. It reduces the differential amplitude, slows the differential edge, and closes the eye horizontally; and it creates common-mode energy, which radiates from cables and couples into other circuits (Figure 13.2). In time-domain terms, the differential edge is smeared over Δt, and a common-mode pulse of width Δt appears at every transition.

13.4.2 Volts, waves and watts

Equation (13.4) is a *voltage* ratio between modal voltages defined by Eq. (13.1). Mixed-mode S-parameters (Chapter 36) are ratios of *power waves*, normalized to reference impedances of $2Z_0$ for the differential mode and $Z_0/2$ for the common mode (100 Ω and 25 Ω for 50 Ω single-ended ports). With $a \propto V/\sqrt{Z_{\text{ref}}}$,

$$|S_{\text{CD21}}| = \left|\frac{V_c}{V_d}\right|\sqrt{\frac{2Z_0}{Z_0/2}} = 2\left|\frac{V_c}{V_d}\right| = \left|\sin(\pi f \Delta t)\right|, \tag{13.5}$$

so the two legitimate numbers differ by a factor of two, or 6.02 dB. The converted *power* fraction is $|S_{\text{CD21}}|^2$, and for this lossless model $|S_{\text{DD21}}|^2 + |S_{\text{CD21}}|^2 = \cos^2 + \sin^2 = 1$ (Figure 13.2). Always state which ratio a mode-conversion number is.

Sources of intra-pair skew include unequal trace lengths (especially at bends, where the outer trace is longer), asymmetric via placement, pin-escape asymmetry at BGAs and connectors, and glass-weave effects (Chapter 9). In mixed-mode S-parameters (Chapter 36), the resulting conversion appears as S_{CD21} and S_{DC21}.

Worked Example 13.2: Skew budget for a 10 Gb/s pair

A USB 3.2 Gen 2 pair runs at 10 Gb/s (UI = 100 ps). If 5 % of a UI is allocated to intra-pair skew, what length mismatch is allowed in stripline with $t_{\text{pd}} = 160$ ps/in, and how large is the resulting common-mode voltage at the Nyquist frequency?

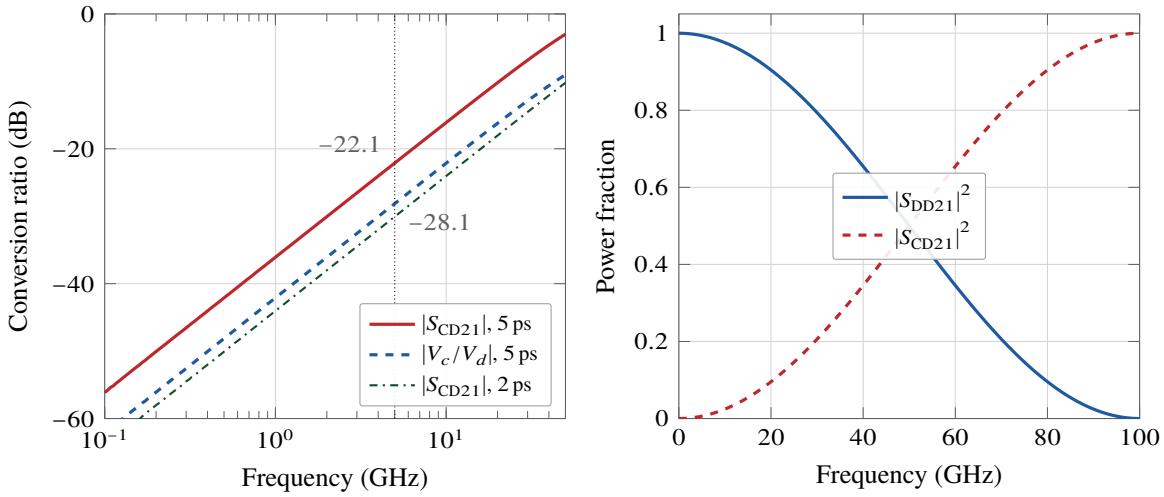


Figure 13.2. Mode conversion by intra-pair skew. Left: the modal voltage ratio of Eq. (13.4) and the power-wave ratio $|S_{CD21}|$ of Eq. (13.5), which is 6.02 dB higher. Right: differential and converted power fractions for 5 ps of skew; they sum to one in this lossless model. [Analytical]

Solution. $\Delta t = 5$ ps, corresponding to $5/160 = 0.031$ in $= 31$ mil. At $f_{Nyq} = 5$ GHz, Eq. (13.4) gives $0.5 \sin(\pi \times 5 \times 10^{9} \times 5 \times 10^{-12}) = 0.039$, i.e. −28.1 dB: a 800 mV differential swing produces about 31 mV of common-mode voltage at 5 GHz. The same skew gives $|S_{CD21}| = 0.078$ (−22.1 dB) by Eq. (13.5), and a converted power fraction of 0.62 %.

13.4.3 Inter-pair skew

Skew between different pairs matters only when the receiver must align them. Serial lanes with embedded clocks (PCIe, USB, Ethernet SERDES) deskew lanes in the receiver's logic, and their inter-pair skew tolerance is large—many nanoseconds in some specifications. Source-synchronous interfaces (DDR strobes and data, MIPI D-PHY clock and data lanes, RGMII) sample data with a forwarded clock, so their lane-to-clock skew budget is a fraction of a UI and must be matched carefully (Chapters 17, 28 and 32).

13.5 Length matching and serpentines

13.5.1 Where skew comes from at bends

When a pair turns, the outer trace travels further. For a pair of centerline pitch $p = W + S$ turning through an angle θ along a circular arc, the length difference is

$$\Delta \ell = \theta \, p, \tag{13.6}$$

or $(\pi/2)p$ for a 90° turn. Turns in opposite directions cancel. Because the skew of a bend appears *at the bend*, the compensation should be applied close to it, so that the pair is never misaligned for more than a small fraction of the rise length. A total length match achieved by a single bump at the far end of a long route still lets the signals travel misaligned—and converting mode—for most of the route.

Worked Example 13.3: Skew from a bend

A pair with $W = S = 5$ mil makes a single 90° turn in stripline ($t_{pd} = 165$ ps/in). Find the skew.

Solution. $p = 10$ mil, so $\Delta\ell = (\pi/2) \times 10 = 15.7$ mil, and $\Delta t = 0.0157 \times 165 = 2.6$ ps. Three uncompensated turns in the same direction would use up the entire 5 %-UI skew budget of a 16 GT/s link.

13.5.2 Serpentine design

Serpentines (meanders) add length to the shorter trace. Their segments couple to one another, which makes the signal arrive slightly *earlier* than the straightened length predicts, because part of the edge propagates by crosstalk across the meander. The geometry rules of Figure 13.3 keep this error small.

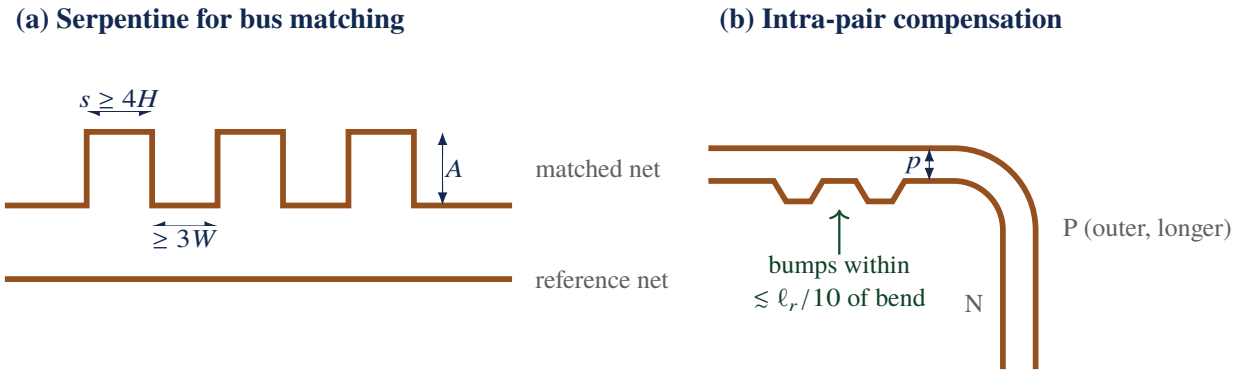


Figure 13.3. Length-matching geometries. (a) A serpentine for matching a single-ended net to a bus: segment spacing s of at least $4H$ limits self-coupling. (b) Intra-pair compensation: small bumps on the inner trace placed immediately next to the bend that created the mismatch.

Rule of Thumb: Serpentine dimensions

Space meander segments by at least $3W$ to $4H$; make the parallel segment length at least 1.5 times the spacing and no longer than needed; prefer many low bumps to a few tall ones on differential pairs, keeping the bump height below about twice the pair spacing so that Z_{diff} is disturbed only briefly. *Limits:* the coupling error grows with edge rate; above about 10 Gb/s, verify serpentine delay with a 3-D or 2.5-D field solver.

Common Pitfall: Matching length instead of delay

Matching physical length on layers with different propagation delay. A microstrip segment is faster than an equal stripline segment, and a via adds delay that layout tools often ignore. Matching must be done in units of time; modern layout tools can do this if each layer's t_{pd} and the via delays are entered.

Key Takeaways

- Signals on a pair decompose into differential ($V_P - V_N$) and common ($\frac{1}{2}(V_P + V_N)$) modes.
- $Z_{diff} = 2Z_{odd}$ and $Z_{cm} = Z_{even}/2$; coupling lowers Z_{odd} and raises Z_{even}.
- Loose or tight coupling both work over a solid plane; consistency of spacing matters more than its value.

Table 13.1. Typical skew targets expressed as a fraction of a UI. Interface specifications and controller design guides take precedence.

Skew type	Basis	Typical design target
Intra-pair (serial)	Mode conversion, differential eye	2–5 % of UI
Inter-pair (embedded clock)	Receiver deskew range	Loose; see specification
Clock to data (source-synchronous)	Setup/hold budget	2–10 % of UI after training
Within a DDR byte lane	DQ/DQS timing budget	a few picoseconds to tens of picoseconds

- Intra-pair skew Δt converts differential signal to common mode in the ratio $\frac{1}{2}\,|\sin \pi f\, \Delta t|$.
- Compensate skew close to where it arises, and match delay, not length.

Further Reading

Bogatin [3] and Hall and Heck [4] cover odd and even modes, mixed-mode S-parameters, and mode conversion; Johnson and Graham [2] discuss differential signaling and skew.

Review Problems

P13.1 A pair has $Z_{\text{odd}} = 48\ \Omega$ and $Z_{\text{even}} = 60\ \Omega$. Find Z_{diff} and Z_{cm}.

P13.2 Design a T termination for the pair of Problem P13.1.

P13.3 What skew results from a 20 mil length mismatch in stripline with $t_{\text{pd}} = 170\ \text{ps/in}$?

P13.4 Find the common-mode-to-differential ratio at 8 GHz for 4 ps of intra-pair skew.

P13.5 A pair with pitch $p = 12$ mil makes four 90° turns in the same direction. Find the length and time skew ($t_{\text{pd}} = 170\ \text{ps/in}$).

P13.6 Allocate 3 % of the UI of an 8 GT/s link to intra-pair skew. What length mismatch is allowed at 170 ps/in?

Chapter 14

S-Parameters for Board Designers

Chapter Overview

Almost every channel model, connector model, and measurement that a board designer receives is a file of scattering parameters. This chapter builds the vocabulary from the beginning: what the incident and reflected waves are, how S_{11} and S_{21} relate to return loss, insertion loss, input impedance, VSWR, phase, and group delay, how four single-ended ports become mixed-mode parameters, and what has to be checked before a Touchstone file is trusted. Every plotted network in this chapter is *synthetic*: it was built with scikit-rf [53] from ideal line sections and lumped elements with a stated loss law, so that each feature on a plot can be traced to a known cause. The same files are provided with the book's sources and are labelled as synthetic inside the files.

Learning Objectives

1. Define incident and reflected waves and the relation $\mathbf{b} = \mathbf{Sa}$ for a stated reference impedance.
2. Convert among S_{11}, reflection coefficient, return loss, reflected power, VSWR, and input impedance, with correct signs.
3. Read the phase of S_{21} correctly: wrapped versus unwrapped phase, and group delay.
4. Build and interpret mixed-mode parameters S_{dd}, S_{cc}, S_{cd}, S_{dc} and recognize the port-numbering trap in four-port files.
5. Read a Touchstone file, renormalize, cascade, and de-embed networks.
6. Check passivity, reciprocity, and causality, and state what a set of S-parameters cannot tell you.

14.1 The problem S-parameters solve

At low frequency a network is described by voltages and currents at its terminals, through impedance ($\mathbf{Z}$) or admittance ($\mathbf{Y}$) matrices. Measuring a Z-parameter requires an open circuit at the other ports, and a Y-parameter requires a short circuit. At gigahertz frequencies neither is practical: a few millimetres of lead turns an "open" into a capacitor and a "short" into an inductor, and an active device may oscillate when its ports are opened or shorted. A network analyzer instead terminates every port in a known resistance, normally 50 Ω, launches a wave into one port, and measures the waves that come out of all ports. The ratios of outgoing to incoming waves are the scattering parameters. They are well defined, they can be measured with matched loads, and they map directly onto the quantities a link budget uses: how much signal gets through and how much comes back.

Notation used in this chapter

a_n, b_n: incident and outgoing ("reflected" or "transmitted") wave at port n, in $\sqrt{\text{W}}$.
Z_{ref}: reference impedance of a port (also written Z_0 in files; it is *not* the characteristic impedance of the line inside the network).
S_{ij}: wave out of port i for a wave into port j, with all other ports matched. Note the order: *output first, input second.*
dB: $20\log_{10}|S|$ for wave ratios; $10\log_{10}$ for power ratios. The two give the same number.

14.2 Incident and reflected waves

For a port with voltage V_n, current I_n (flowing into the network), and a real reference impedance Z_{ref}, the incident and outgoing waves are

$$a_n = \frac{V_n + Z_{\text{ref}} I_n}{2\sqrt{Z_{\text{ref}}}}, \qquad b_n = \frac{V_n - Z_{\text{ref}} I_n}{2\sqrt{Z_{\text{ref}}}}. \tag{14.1}$$

The normalization by $\sqrt{Z_{\text{ref}}}$ makes $|a_n|^2/2$ the peak-phasor power carried toward the network and $|b_n|^2/2$ the power carried away. For a linear network the outgoing waves are linear combinations of the incoming ones (Figure 14.1):

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}, \qquad S_{ij} = \left.\frac{b_i}{a_j}\right|_{a_k=0,\ k\neq j}. \tag{14.2}$$

The condition $a_k = 0$ means that every other port is terminated in exactly Z_{ref}, so nothing is reflected back into the network from outside.

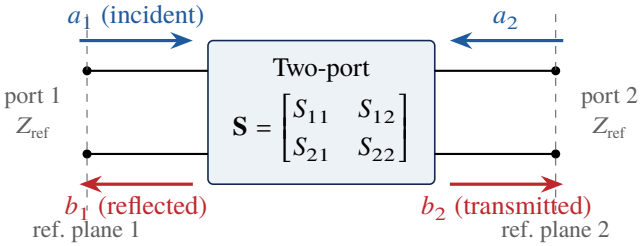


Figure 14.1. A two-port with its incident waves a_1, a_2 and outgoing waves b_1, b_2. The reference planes mark where the waves are defined; moving a plane changes every phase in the matrix. [Conceptual]

Common Pitfall: Reference impedance is not line impedance

A Touchstone header such as *R 50* states the impedance of the *ports* the data are normalized to. A 45 Ω line measured with 50 Ω ports shows a ripple in S_{11}; the same line renormalized to 45 Ω ports shows $S_{11} = 0$ (Section 14.10). Neither is wrong; they answer different questions.

Two other wave definitions appear in the literature: *power waves* (Kurokawa), which handle complex reference impedances, and *pseudo-waves* (Marks and Williams), which are defined relative to the characteristic impedance of a reference line. For real reference impedances they coincide with Eq. (14.1). A file with complex or per-port reference impedances (Touchstone 2.x) should be converted with a tool that states which definition it uses [30].

14.3 Reflection, transmission, and decibels

Reflection. With port 2 matched, $S_{11} = b_1/a_1$ is the reflection coefficient Γ_{in} seen at port 1. The reflected fraction of the incident power is $|S_{11}|^2$, and the *return loss* is

$$\text{RL} = -20\log_{10}|S_{11}| \quad \text{(dB)}, \tag{14.3}$$

a positive number for a passive port. Saying "the return loss is -15 dB" is a sign error that is common enough to be understood, but a specification should be written as "$S_{11} \le -15$ dB" or "RL ≥ 15 dB".

Transmission. With port 2 matched, $S_{21} = b_2/a_1$. For a passive channel $|S_{21}| \le 1$, so S_{21} in decibels is negative and the *insertion loss* $\text{IL} = -20\log_{10}|S_{21}|$ is positive. For an amplifier $|S_{21}| > 1$ and $20\log_{10}|S_{21}|$ is the gain.

Why voltage and power give the same dB number. The waves are normalized so that power is proportional to $|a|^2$ and $|b|^2$, so $10\log_{10}|S|^2 = 20\log_{10}|S|$. In terms of *voltage* at the ports, $|S_{21}| = \left|V_2/V_1^{+}\right|$ only when both ports have the same reference impedance.

Table 14.1. Worked conversions for common specification values ($Z_{\text{ref}} = 50\,\Omega$). Z_{in} is given for a purely real reflection; the actual impedance can lie anywhere on the circle of constant $|\Gamma|$ on the Smith chart.

Value	$\lvert S\rvert$	Power fraction	VSWR	Z_{in} if $\Gamma > 0$	Z_{in} if $\Gamma < 0$
$S_{11} = -10$ dB	0.316	10.0 % reflected	1.92	96.2 Ω	26.0 Ω
$S_{11} = -20$ dB	0.100	1.0 % reflected	1.22	61.1 Ω	40.9 Ω
$S_{21} = -3$ dB	0.708	50.1 % delivered	—	—	—
$S_{21} = -6$ dB	0.501	25.1 % delivered	—	—	—

Worked Example 14.1: Reading $S_{11} = -10$ dB and $S_{21} = -3$ dB

A connector datasheet gives $S_{11} = -10$ dB at 8 GHz, and a channel report gives $S_{21} = -3$ dB at the same frequency. Interpret both.

Solution. $|S_{11}| = 10^{-10/20} = 0.316$, so 10 % of the incident power is reflected and the return loss is 10 dB. The VSWR is $(1 + 0.316)/(1 - 0.316) = 1.92$. If the reflection is real, the connector looks like $50(1.316/0.684) = 96.2\,\Omega$ or $50(0.684/1.316) = 26.0\,\Omega$; the sign of Γ, and therefore which of the two, is in the phase of S_{11}, not in its magnitude. For the channel, $|S_{21}| = 0.708$: the received sinusoid at 8 GHz has 70.8 % of the launched amplitude and carries 50.1 % of the launched power. The other 49.9 % was reflected, radiated, or dissipated; S_{21} alone does not say which. For a passive network, $1 - |S_{11}|^2 - |S_{21}|^2$ is the fraction lost inside the network (dissipated or radiated) when port 1 is driven.

14.4 A synthetic channel used throughout the chapter

To make the plots interpretable, the device under test (DUT) is a known structure: 10 mm of 50 Ω line, a 2.5 mm section of 38 Ω line (the kind of step a connector or a via field can produce), 40 mm of 50 Ω line, a 0.3 pF shunt capacitor (an oversized pad), and a final 10 mm of line. All lines have $\varepsilon_{\text{eff}} = 3.3$ and the loss law

$$\alpha(f) = 0.005\sqrt{f/\text{GHz}} + 0.0015\,(f/\text{GHz}) \quad \text{dB/mm}, \tag{14.4}$$

about 0.03 dB/mm at 10 GHz: a skin-effect term and a dielectric term. The line model has a frequency-independent characteristic impedance, which is a simplification; a causal model would let Z_0 vary slightly with frequency (Section 14.11).

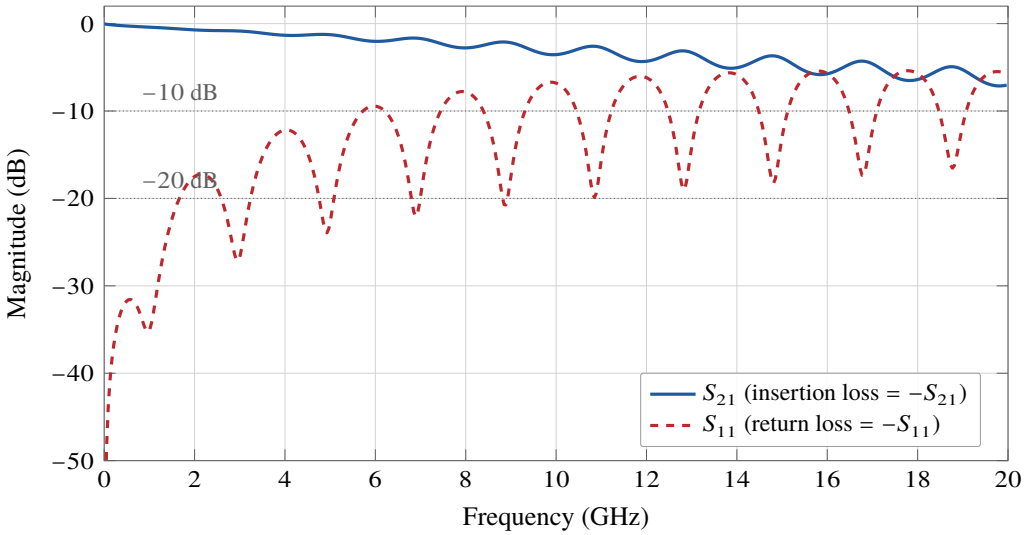


Figure 14.2. Magnitude of S_{11} and S_{21} for the synthetic DUT. The ripple in S_{11} is the interference between the reflections from the 38 Ω section and the shunt capacitor; S_{21} falls both from loss and from reflected power. [Simulation]

Figure 14.2 shows the result. At 5 GHz, $S_{11} = -22.7$ dB and $S_{21} = -1.27$ dB; at 10 GHz, $S_{11} = -6.8$ dB and $S_{21} = -3.55$ dB. Line loss alone accounts for about 1.9 dB of the 3.55 dB at 10 GHz; the rest is mismatch. A designer who reads only S_{21} would blame the laminate.

14.5 Input impedance, the Smith chart, and VSWR

With port 2 matched, the input impedance follows from S_{11}:

$$Z_{\text{in}} = Z_{\text{ref}}\,\frac{1 + S_{11}}{1 - S_{11}}, \qquad \text{VSWR} = \frac{1 + |S_{11}|}{1 - |S_{11}|}. \tag{14.5}$$

Z_{in} is a property of the whole network *and its termination*; it changes with frequency even when every line inside has the same characteristic impedance. Z_0, by contrast, is a property of one uniform line. A good habit is to say "input impedance at port 1 with port 2 matched" rather than "impedance".

The Smith chart (Figure 14.3) is the plane of the complex reflection coefficient with circles of constant normalized resistance and arcs of constant normalized reactance overlaid. Its value for board work is diagnostic: a trace that swings clockwise around the centre with growing radius indicates distributed mismatch separated from the reference plane; a cluster in the lower half indicates capacitive behaviour, in the upper half inductive behaviour. Moving the reference plane by a length ℓ of matched line rotates the whole trace clockwise by $2\beta\ell$ without changing its radius.

14.6 Phase, unwrapped phase, and group delay

The phase of S_{21} carries the delay. A plotting program usually reports the phase *wrapped* into $(-180°, 180°]$, which produces the saw-tooth in Figure 14.5. *Unwrapping* adds multiples of 360° wherever the phase jumps

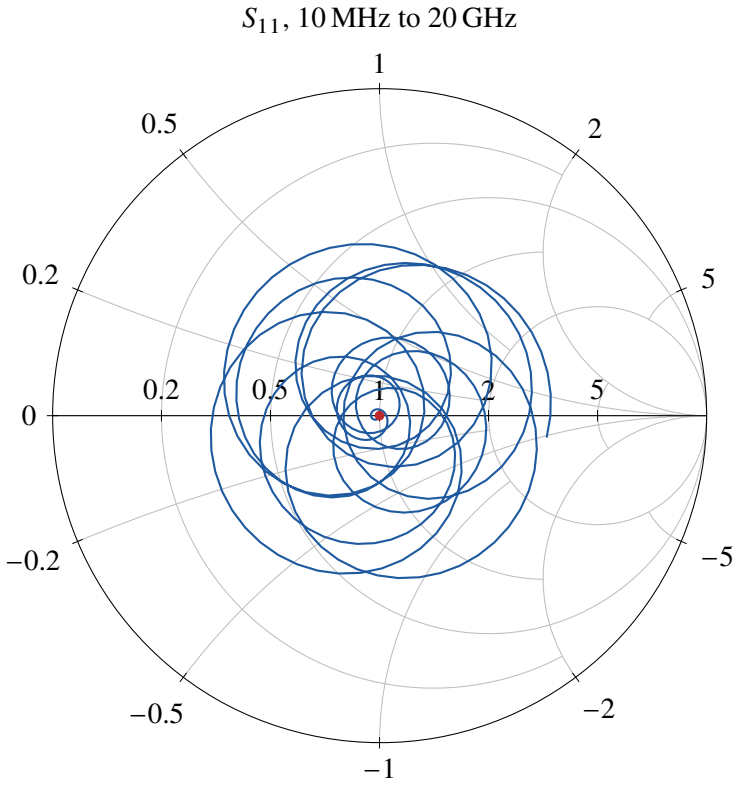


Figure 14.3. S_{11} of the synthetic DUT on a Smith chart. The centre is a perfect match to 50 Ω. The trace starts near the centre at low frequency and spirals outward as the reflections add with increasing phase; one turn corresponds to a half-wavelength of electrical distance between the discontinuities. [Simulation]

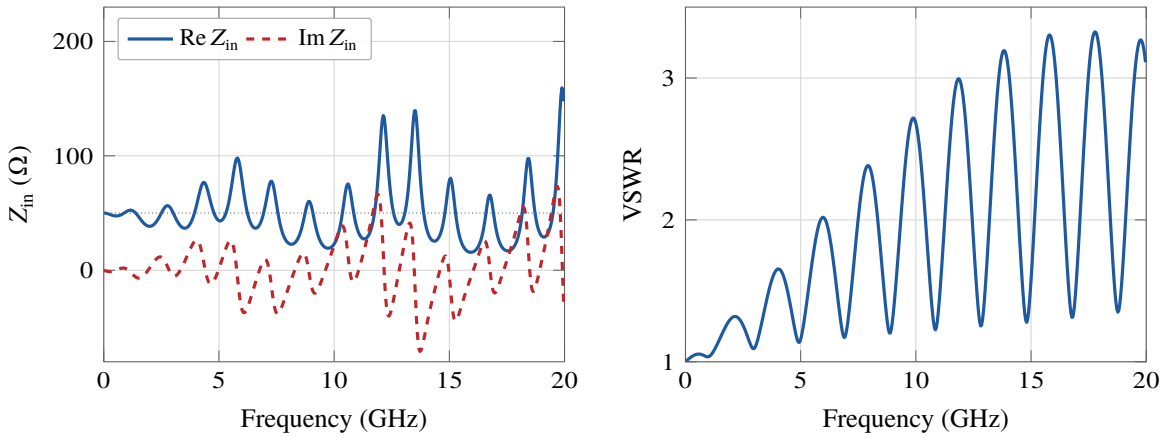


Figure 14.4. Real and imaginary parts of Z_{in} and the VSWR of the synthetic DUT, computed from S_{11} with Eq. (14.5). At 5 GHz $Z_{in} = 43.3 + j1.5\,\Omega$ (VSWR 1.16); at 10 GHz $Z_{in} = 22.9 + j21.9\,\Omega$ (VSWR 2.68). [Simulation]

by more than 180° between adjacent points; it is reliable only if the frequency step is small enough that the true phase changes by less than 180° per step, that is

$$\Delta f < \frac{1}{2\tau}, \tag{14.6}$$

where τ is the delay. For the 63 mm DUT ($\tau \approx 0.4$ ns) any step below 1.25 GHz works; the 10 MHz step used here is far finer. A 1 m backplane channel with $\tau \approx 6$ ns needs $\Delta f < 83$ MHz.

The *group delay* is the slope of the unwrapped phase:

$$\tau_g(f) = -\frac{1}{2\pi}\frac{\mathrm{d}\phi_{21}}{\mathrm{d}f}. \tag{14.7}$$

For a uniform lossless line τ_g equals the propagation delay $\ell\sqrt{\varepsilon_{\mathrm{eff}}}/c$. In the DUT, the reflections make τ_g ripple around the line value of about 378 ps (Figure 14.5, right). Group delay is the right quantity for intra-pair skew budgets: a skew of 5 ps between two legs is a difference in τ_g, and the differentiation in Eq. (14.7) amplifies measurement noise, so it is usually smoothed over an aperture of several frequency points.

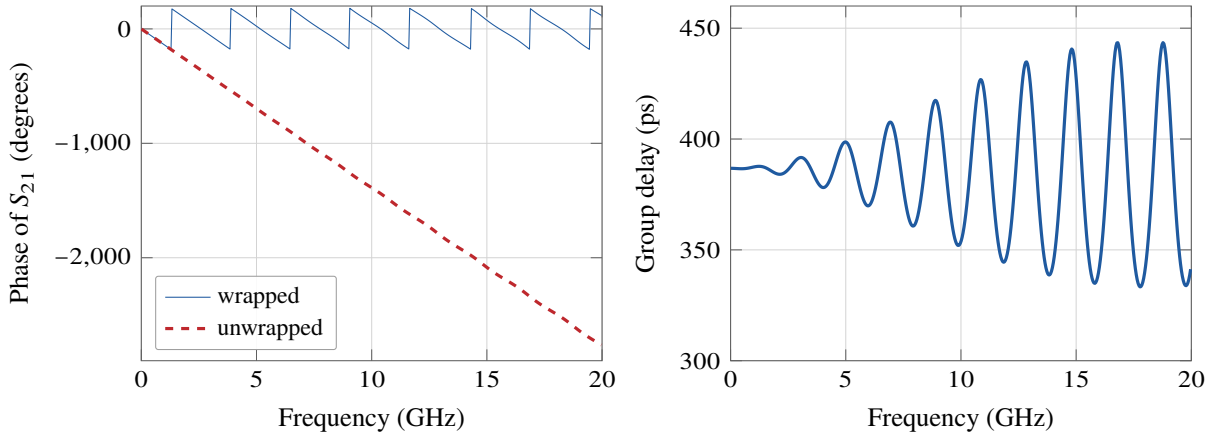


Figure 14.5. Left: wrapped and unwrapped phase of S_{21}. Right: group delay from the slope of the unwrapped phase; the ripple comes from the multiple reflections between the two discontinuities. [Simulation]

14.7 Time-domain views from frequency data

A TDR-style impedance profile can be computed from S_{11}: extrapolate the data to DC, multiply by a window, transform to the time domain, integrate the impulse to a step, and convert the step reflection $\rho(t)$ to impedance with $Z(t) = Z_{\mathrm{ref}}(1+\rho)/(1-\rho)$. Figure 14.6 shows the result for the DUT. Two features stand out. First, the 38 Ω section reads 42.9 Ω: it is only 30 ps long in round-trip time, shorter than the effective rise time of a 20 GHz, Hamming-windowed measurement, so the dip never reaches its true depth. Second, the 0.3 pF capacitor appears as a dip to 39.9 Ω whose depth also depends on the bandwidth. Both effects were discussed for real TDR in Chapter 37; the rule is the same: *a feature shorter than the rise time is read as shallower than it is*.

14.8 Four ports and mixed-mode parameters

A differential pair through a connector or a channel is a four-port. Two numbering conventions are in use (Table 14.2), and confusing them is one of the most frequent errors in channel analysis: a file numbered 1–3 / 2–4 read as 1–2 / 3–4 makes near-end crosstalk look like insertion loss.

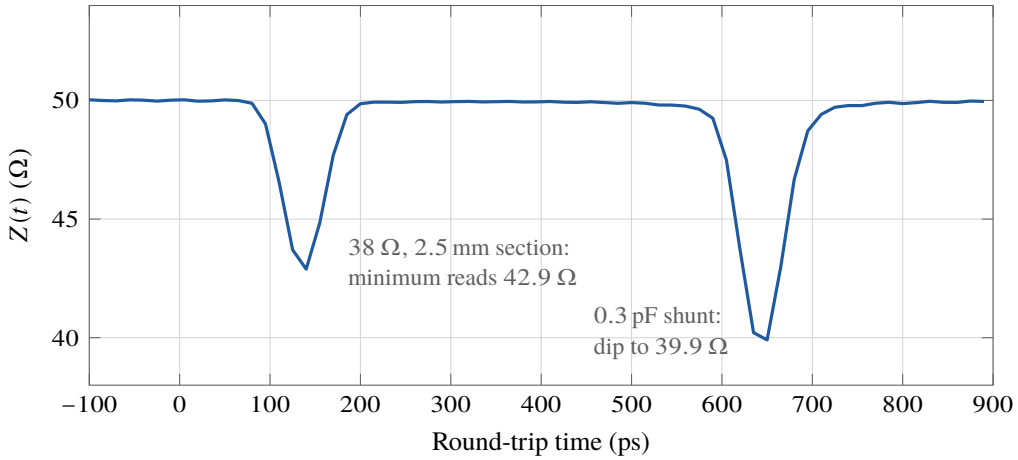


Figure 14.6. Impedance profile computed from S_{11} of the synthetic DUT (20 GHz data, linear DC extrapolation, Hamming window). The 38 Ω section is under-resolved. [Simulation]

Table 14.2. Common four-port numbering conventions for a pair. Always confirm the convention from the file's comments or by checking which term has near-0 dB low-frequency transmission.

Convention	Through paths	Typical source
Odd–even: inputs 1, 3; outputs 2, 4	S_{21}, S_{43}	many simulators and VNA set-ups
Side-by-side: inputs 1, 2; outputs 3, 4	S_{31}, S_{42}	IEEE 370 style, many connector models

Mixed-mode waves. With single-ended ports P and N at one end, the differential and common-mode waves are

$$a_d = \frac{a_P - a_N}{\sqrt{2}}, \qquad a_c = \frac{a_P + a_N}{\sqrt{2}}, \tag{14.8}$$

and likewise for b. With 50 Ω single-ended ports this definition implies a differential reference of 100 Ω and a common-mode reference of 25 Ω. The mixed-mode matrix has four 2 × 2 blocks (Figure 14.7): S_{dd} (the wanted differential behaviour), S_{cc} (common-mode behaviour), and the conversion blocks S_{cd} (differential in, common out) and S_{dc} (common in, differential out). For a perfectly symmetric pair the conversion blocks are zero.

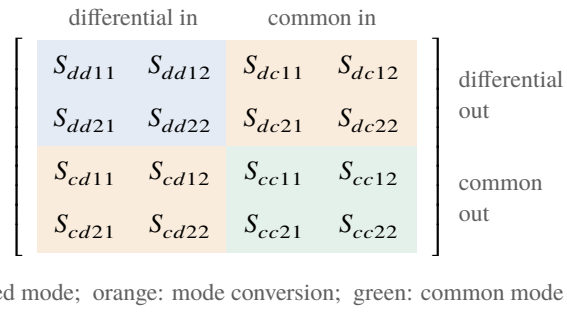


Figure 14.7. Organization of the mixed-mode S-matrix of a two-port pair (four single-ended ports). [Conceptual]

Skew as mode conversion. Consider two uncoupled 50 Ω legs, one 100 mm and one 102.6 mm long. With $\varepsilon_{\text{eff}} = 3.3$ the 2.6 mm difference is a skew $\Delta t = 15.75$ ps. If the legs have transmissions T_P and $T_N = T_P e^{-j2\pi f\,\Delta t}$, then

$$S_{dd21} = \tfrac{1}{2}(T_P + T_N), \qquad S_{cd21} = \tfrac{1}{2}(T_P - T_N), \qquad |S_{cd21}| = |T_P|\,|\sin(\pi f\,\Delta t)|\,. \tag{14.9}$$

Figure 14.8 compares Eq. (14.9) with the conversion computed by scikit-rf from the four-port file: they agree. At 2.5 GHz, the Nyquist frequency of a 5 Gbit/s link, $S_{cd21} = -19.4$ dB.

Common Pitfall: A factor of two between voltage ratio and S_{cd21}

In Chapter 13 the common-mode *voltage* produced by skew was written as $V_c/V_d = \tfrac{1}{2}|\sin \pi f\,\Delta t|$. The mixed-mode parameter is twice that, $|S_{cd21}| = |\sin \pi f\,\Delta t|$, because the waves are normalized to different reference impedances: $|a_c| = |V_c|\,/\sqrt{25\ \Omega}$ and $|a_d| = |V_d|\,/\sqrt{100\ \Omega}$. Both statements are correct; mixing them gives a 6 dB error.

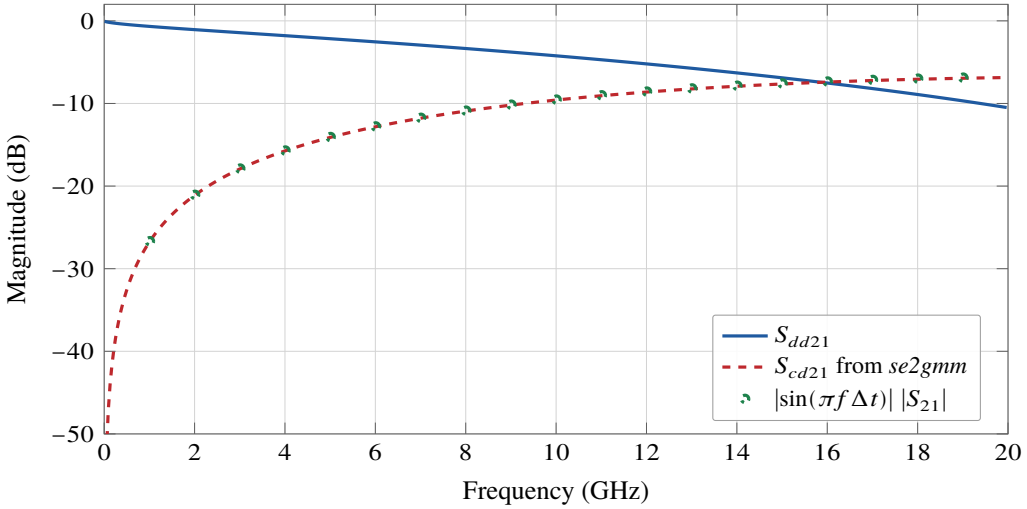


Figure 14.8. Mixed-mode transmission of an uncoupled pair with 15.75 ps of skew. Circles: Eq. (14.9). The differential transmission S_{dd21} falls faster than the loss of one leg because energy is converted to common mode; at $f = 1/(2\Delta t) \approx 32$ GHz the conversion would be complete. [Simulation]

Crosstalk terms. For two neighbouring single-ended lines with the odd–even convention (line A: ports 1→2, line B: 3→4), near-end crosstalk from A to B is S_{31} and far-end crosstalk is S_{41}. For two pairs, the same idea is applied to the mixed-mode matrix of an eight-port: S_{dd} terms between pair A's input and pair B's ports. Crosstalk budgets (Chapter 12) are usually written as integrated crosstalk noise over the signal spectrum, not as a single frequency point.

14.9 Touchstone files

Touchstone is the common file format for S-parameters [30]. A version-1 file is identified by its extension (*.s2p* for two ports, *.s4p* for four) and by an option line beginning with #:

```
! SYNTHETIC NETWORK – generated by tools/gen_r3_sp.py, not a measurement
# GHz S DB R 50.0
0.01  −62.68  −94.87  −0.0322  −1.392  −0.0322  −1.392  −62.69  −94.32
```

The option line gives the frequency unit, the parameter type (S, Y, Z, H, or G), the number format (DB = dB/angle, MA = magnitude/angle, RI = real/imaginary), and the reference resistance. Lines beginning with ! are comments. Two ordering rules cause most reading errors:

- A *two-port* file lists each frequency as $S_{11}, S_{21}, S_{12}, S_{22}$: column order, the historical exception.
- Files with three or more ports list the matrix *row by row* ($S_{11}, S_{12}, S_{13}, \dots$), wrapping onto continuation lines of at most four pairs.

Version 2.x adds keywords ([Version], [Number of Ports], [Reference] for per-port impedances, [Matrix Format] for symmetric matrices, and [Mixed-Mode Order]) and removes the ambiguity, at the cost of tools that do not read it. Before using any received file, open it in a text editor and read the comments: they often say how it was produced, which is the only way to know whether it is a measurement, a simulation, or a synthetic model.

14.10 Renormalization, cascading, and de-embedding

Renormalization. S-parameters referenced to one impedance can be converted exactly to another by converting to $\mathbf{Z}$ and back: $\mathbf{S}' = (\mathbf{Z} - Z'_{\mathrm{ref}}\mathbf{I})(\mathbf{Z} + Z'_{\mathrm{ref}}\mathbf{I})^{-1}$ for equal real references. Figure 14.9 shows a 50 mm, 45 Ω line: referenced to 50 Ω it has a periodic S_{11} with a peak of −19.9 dB (reflection coefficient −0.053 at each end, adding in phase every half wavelength); renormalized to 45 Ω its S_{11} is identically zero. Renormalization is how one evaluates an 85 Ω PCIe channel model against a specification written for a different reference, and it is why the reference impedance must be stated with every limit line.

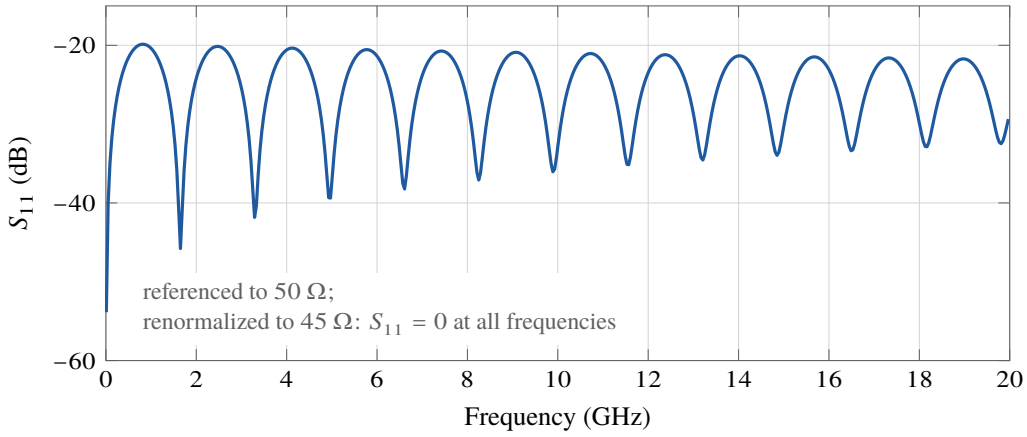


Figure 14.9. A 45 Ω, 50 mm line referenced to 50 Ω ports. After renormalization to 45 Ω the reflection vanishes. [Simulation]

Cascading. S-matrices do not multiply. Two two-ports in series are combined through the transfer (T) parameters, which relate the waves at port 1 to the waves at port 2:

$$\mathbf{T} = \frac{1}{S_{21}}\begin{bmatrix} -(S_{11}S_{22} - S_{12}S_{21}) & S_{11} \\ -S_{22} & 1 \end{bmatrix}, \qquad \mathbf{T}_{\mathrm{total}} = \mathbf{T}_A\mathbf{T}_B. \tag{14.10}$$

For the DUT split into two halves, the cascade computed with Eq. (14.10) and the cascade computed by scikit-rf's ** operator agree to 6×10^{-16}, the floating-point limit. (Several T-parameter conventions exist; use one consistently.)

De-embedding. A measurement includes the fixtures between the VNA calibration planes and the DUT (Figure 14.10). If the fixtures are known, the DUT follows from $\mathbf{T}_{\text{DUT}} = \mathbf{T}_A^{-1}\mathbf{T}_{\text{meas}}\mathbf{T}_B^{-1}$. The fixture models usually come from a *2x-thru*: a structure made of fixture A joined directly to fixture B, which is split into halves by an algorithm. IEEE 370 [29] defines fixture-quality and de-embedding-quality metrics and recommends checking the result for passivity, reciprocity, and causality. A de-embedded result is only as good as the assumption that the fixture halves on the test board are identical to the halves of the 2x-thru.

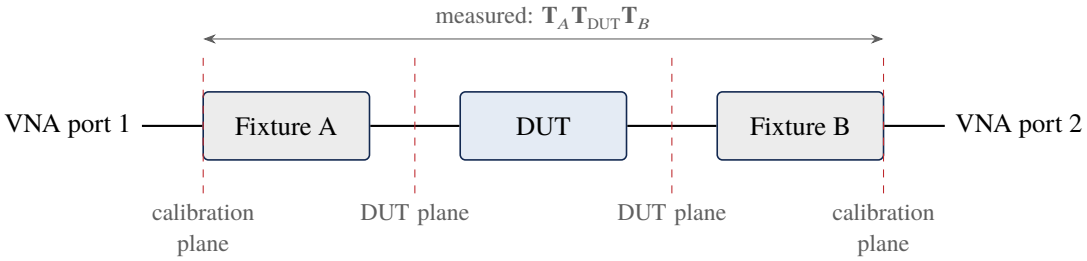


Figure 14.10. Calibration planes, fixtures, and DUT planes. Calibration moves the reference to the VNA cable ends; de-embedding moves it further, to the DUT. [Conceptual]

Calibration. A VNA calibration corrects systematic errors (directivity, source and load match, tracking, isolation) using known standards. SOLT (short–open–load–thru) uses coaxial standards characterized by the kit manufacturer; TRL (thru–reflect–line) uses on-board standards and needs only the line to be well made, which suits PCB measurements; electronic calibration modules automate SOLT-like procedures. A calibration is valid only for the frequency points, IF bandwidth, power level, and cable positions at which it was made.

14.11 Quality checks: passivity, reciprocity, causality, DC

Passivity. A passive network cannot deliver more power than it receives. The condition is that the largest singular value of $\mathbf{S}$ is at most one at every frequency. The synthetic DUT gives 0.9963. Measured files often violate passivity slightly at high frequency because of calibration drift; small violations should be corrected (passivity enforcement) before a time-domain simulation, where they can make the simulation unstable.

Reciprocity. A network made of passive, isotropic materials satisfies $S_{ij} = S_{ji}$. For the synthetic DUT the largest difference between S_{12} and S_{21} is 6×10^{-14}, rounding error only. A measurement with $|S_{12} - S_{21}|$ of a few hundredths indicates a calibration or connection problem, not a property of the board.

Causality. A physical response cannot begin before its cause. In the frequency domain causality links magnitude and phase (the Kramers–Kronig relations); a model that sets loss and phase independently, such as a constant-D_{k}, frequency-dependent-D_{f} dielectric, violates it and shows a response before $t = 0$. Truncating the data at a finite frequency also creates apparent precursors: Figure 14.11 shows the impulse response of the DUT's S_{21} with a rectangular window (ringing before and after the main peak, Gibbs phenomenon) and with a Hamming window (clean, but wider).

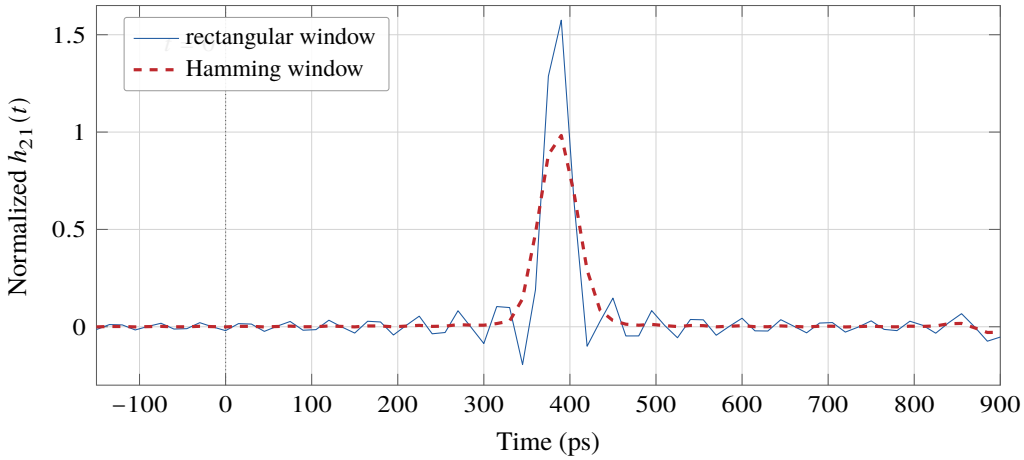


Figure 14.11. Impulse response of S_{21} computed from band-limited data. The rectangular window produces ripple before the true arrival; a smooth window suppresses it at the cost of resolution. Neither indicates a non-causal device: the artefact is created by truncation. [Simulation]

DC extrapolation. VNAs rarely measure below 10 MHz, but a time-domain simulation needs the DC value: it sets the final level of every step response. Extrapolation should be linear in magnitude and phase from the lowest points, or should insert a known DC value (for a channel, $|S_{21}(0)|$ follows from the DC resistance). A poor DC point shows up as a slow drift in a simulated bit stream that has nothing to do with the board.

Frequency range and step. The top frequency should be at least the bandwidth implied by the fastest edge that will be simulated (often three to five times the Nyquist frequency of the link), and the step must satisfy Eq. (14.6) for the longest delay in the file.

14.12 A practical workflow with scikit-rf

The open-source Python package scikit-rf [53] reads and writes Touchstone files, renormalizes, cascades, converts to mixed mode, and computes time-domain responses. A minimal check of a received two-port file, listed in full in Appendix E (*sp_workflow.py*), performs these steps:

1. Load the file and print its reference impedance and frequency grid; confirm the grid meets Eq. (14.6).
2. Report S_{11} and S_{21} at the frequencies named in the specification (for the synthetic DUT: −22.7 dB and −1.27 dB at 5 GHz).
3. Compute the largest singular value (passivity) and $\max |S_{12} - S_{21}|$ (reciprocity).
4. Extrapolate to DC and compute the impedance profile; compare its features with the layout. Running the script on the synthetic file reports a minimum of 39.6 Ω at 645 ps, slightly below the 39.9 Ω of Figure 14.6 because the Touchstone file stores rounded dB values.
5. For four-port files, identify the numbering convention, convert to mixed mode (*se2gmm*), and inspect S_{dd21}, S_{dd11}, and S_{cd21}.

14.13 What S-parameters cannot establish

S-parameters describe a *linear, time-invariant* network at the stated reference planes, bias, temperature, and power level. They cannot by themselves establish:

- **Link performance.** A bit-error ratio depends on the transmitter, the equalization, the receiver, jitter, and crosstalk aggressors as well as the channel. A channel passes or fails a specification through a defined simulation or compliance flow, not through one insertion-loss number.
- **Behaviour of nonlinear parts.** ESD diodes, switches, and receivers with clamps change with signal level; their S-parameters are valid only near the measured operating point.
- **Radiated emissions or immunity.** Mode conversion and common-mode transmission indicate a risk; whether a product passes an EMC test depends on cables, enclosure, and the whole system.
- **Temperature and humidity drift.** A file measured at 23 °C says nothing about 85 °C unless measured there.
- **Anything outside the frequency range,** including DC (which must be supplied) and the behaviour of the structure above the top frequency.
- **Which physical feature caused a resonance.** The time-domain view and the layout must be read together; the S-parameters alone are a black box.

14.14 Checklist: receiving an S-parameter file

- □ Read the comments: measurement, simulation, or synthetic? Who made it, when, with what calibration or solver?
- □ Note the reference impedance(s) and the port numbering; for four-port files, identify the through paths.
- □ Check the frequency range and step against the edge rate and the longest delay.
- □ Check passivity and reciprocity; enforce passivity only after understanding why it was violated.
- □ Extrapolate to DC with a stated method; inspect the impedance profile.
- □ Convert to mixed mode for pairs; review S_{dd11}, S_{dd21}, S_{cd21}, and crosstalk terms.
- □ Compare features with the layout and the expected discontinuities before drawing conclusions.

Key Takeaways

- S_{ij} is the wave out of port i for a wave into port j with all other ports matched to the stated reference impedance.
- Return loss and insertion loss are positive numbers; S_{11} and S_{21} in dB are negative for passive networks.
- $S_{11} = -10\,\text{dB}$ reflects 10 % of the power; $S_{21} = -3\,\text{dB}$ delivers 50 %.
- Z_{in} depends on the network and its termination; Z_0 is a property of a uniform line.
- Group delay is the slope of the unwrapped phase; unwrapping needs $\Delta f < 1/(2\tau)$.
- Skew appears as S_{cd21} with $|S_{cd21}| = |\sin \pi f \Delta t|$ in mixed-mode normalization.
- Check reference impedance, port order, passivity, reciprocity, causality, and DC before trusting a file.

Further Reading

Pozar [38] develops scattering parameters, power waves, and the Smith chart; Hall and Heck [4] and Bogatin [3] treat mixed-mode parameters and their use in channel analysis. IEEE 370 [29] covers fixture design and de-embedding quality. The Touchstone specification [30] defines the file format.

Review Problems

P14.1 A cable assembly has $S_{11} = -14\,\text{dB}$. Find $|\Gamma|$, the reflected power fraction, the return loss, and the VSWR.

P14.2 A specification requires RL $\geq 12\,\text{dB}$. A report shows $S_{11} = -11\,\text{dB}$. Does it pass? What is the largest $|\Gamma|$ allowed?

P14.3 (Inverse design) What real load resistances give $S_{11} = -20\,\text{dB}$ in a $100\,\Omega$ differential reference?

P14.4 A passive two-port has $S_{11} = -15\,\text{dB}$ and $S_{21} = -2\,\text{dB}$ at one frequency. What fraction of the incident power is lost inside the network?

P14.5 (Approximation failure) A student unwraps the phase of a 0.8 m channel ($\varepsilon_{\text{eff}} = 3.6$) measured with a 100 MHz step. Is the result reliable?

P14.6 Find $|S_{cd21}|$ at 8 GHz for a pair with 4 ps of skew and negligible loss, and the corresponding common-mode-to-differential voltage ratio.

P14.7 (Imperfect data) A four-port connector file shows "S_{21}" of $-25\,\text{dB}$ at 1 GHz and "S_{31}" of $-0.1\,\text{dB}$. What is the most likely explanation?

P14.8 (Imperfect data) A measured file has a largest singular value of 1.02 between 18 and 20 GHz. What should you do before a time-domain simulation, and what should you not conclude?

Chapter 15

Vias and Discontinuities

Chapter Overview

A uniform transmission line is easy to design; the trouble lies in the places where it stops being uniform. Vias, pads, connector pins, BGA escapes, and reference-plane transitions all present local excess capacitance or inductance, and the unused portion of a through via forms a resonant stub. Above roughly 10 Gb/s, these discontinuities often dominate channel performance. This chapter models vias as lumped elements and as short transmission lines, explains stub resonance and back-drilling, surveys blind, buried, and micro-via technologies, and gives layout rules for anti-pads, return vias, and BGA and connector breakout.

Learning Objectives

1. Estimate the inductance and capacitance of a plated through via.
2. Treat a via as a short transmission line and estimate its impedance.
3. Compute via-stub resonant frequency and decide when back-drilling or blind vias are required.
4. Compare through, blind, buried, and micro-via structures.
5. Place return (stitching) vias and size anti-pads for controlled transitions.
6. Plan BGA and connector breakout for high-speed pairs.

15.1 Anatomy of a via

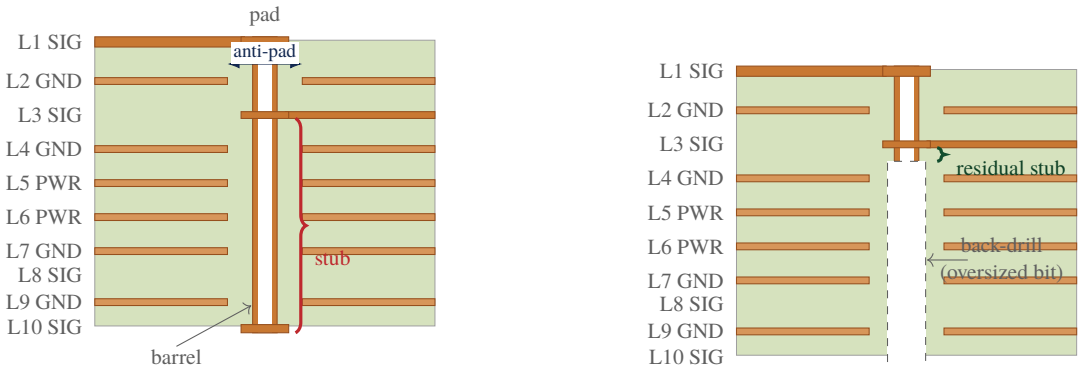
A plated through-hole (PTH) via consists of a drilled and plated *barrel*, *pads* on the layers where traces connect, and *anti-pads*—clearance holes—in the planes it passes through (Figure 15.1). Non-functional pads on unconnected layers are usually removed for high-speed vias, since they add capacitance without benefit.

15.2 Lumped via models

When a via is short compared with the rise length, it can be represented by a series inductance and a shunt capacitance. Johnson and Graham give two widely used estimates [1]. The partial inductance of a via of length h and barrel diameter d (both in inches) is

$$L_{\text{via}} \approx 5.08\, h \left[\ln \frac{4h}{d} + 1 \right] \quad \text{nH}, \tag{15.1}$$

an empirical estimate. It is *not* the partial self-inductance of Eq. (3.6): with $\mu_0/2\pi = 5.08\,\text{nH/in}$ and $2\ell/r = 4h/d$, that expression gives $5.08\,h[\ln(4h/d) - 1]$, with -1 where Johnson and Graham have $+1$. For a short via the two differ by nearly a factor of two. Neither is the loop inductance the signal actually



(a) Through via, L1 → L3, with a long stub

(b) Back-drilled from the bottom

Figure 15.1. Cross-section of a through via carrying a signal from L1 to L3 in a 10-layer board. (a) As drilled, the barrel below L3 forms an open stub. (b) After back-drilling with an oversized bit from the bottom, only a short residual stub remains.

sees, which depends on where the return current flows (Section 15.7); treat the spread as the uncertainty of a lumped via model. The capacitance of a via pad of diameter D_1 in a board of thickness T with anti-pad diameter D_2 is

$$C_{\text{via}} \approx \frac{1.41\,\varepsilon_r T D_1}{D_2 - D_1} \quad \text{pF}. \tag{15.2}$$

Both are rough: Eq. (15.1) is a partial inductance whose loop value depends on where the return current flows, and Eq. (15.2) lumps the barrel and pad capacitances into one empirical term.

Worked Example 15.1: Lumped model of a through via

Estimate the inductance and capacitance of a via in a 62 mil board with a 10 mil drill, 20 mil pads, 32 mil anti-pads, and $\varepsilon_r = 4.4$, and the time constant it presents to a 50 Ω line.

Solution. Equation (15.1): $L = 5.08 \times 0.062[\ln(24.8) + 1] = 1.33$ nH (the partial self-inductance with -1 would give 0.70 nH). Equation (15.2): $C = 1.41 \times 4.4 \times 0.062 \times 0.020 / 0.012 = 0.64$ pF. As a mid-line capacitance the via has $\tau = Z_0 C/2 = 16$ ps; as a series inductance, $\tau = L/(2Z_0) = 13$ ps. For a 1 ns edge it is invisible; for a 30 ps edge it is a significant discontinuity, and its capacitive and inductive parts must be balanced.

15.3 The via as a transmission line

A via surrounded by return vias is a short coaxial-like transmission line. If the return vias form a ring of effective diameter D_{ret} around a barrel of diameter d, its impedance is approximately

$$Z_{\text{via}} \approx \frac{60}{\sqrt{\varepsilon_r}} \ln \frac{D_{\text{ret}}}{d}, \tag{15.3}$$

modified by the pads and anti-pads it passes, which add capacitance and lower the impedance. A typical signal via with a 10 mil drill and return vias at about 40 mil diameter has $Z_{\text{via}} \approx 43$ Ω in $D_k = 3.8$ material, close enough to 50 Ω that careful anti-pad sizing can tune it. Its delay is about $h\sqrt{\varepsilon_r}/c$, roughly 10 ps through a 62 mil board.

15.4 Anti-pads

An *anti-pad* (also written antipad) is a clearance area around a via or plated through-hole in a copper plane, where copper is intentionally removed. Its first purpose is to prevent an unwanted electrical connection: a signal via must pass through ground and power planes without touching them. Its second purpose, which matters as much in high-speed design, is to control the capacitance—and therefore the impedance—of the via. Anti-pads are particularly important in PCIe, DDR, USB, Ethernet and other SerDes designs. An anti-pad is not simply a manufacturing clearance; it is a signal-integrity design parameter that sets the electromagnetic behavior of the via.

15.4.1 What an anti-pad looks like

Imagine a signal via passing through a board with several ground planes (Figure 15.2). The structure has four elements:

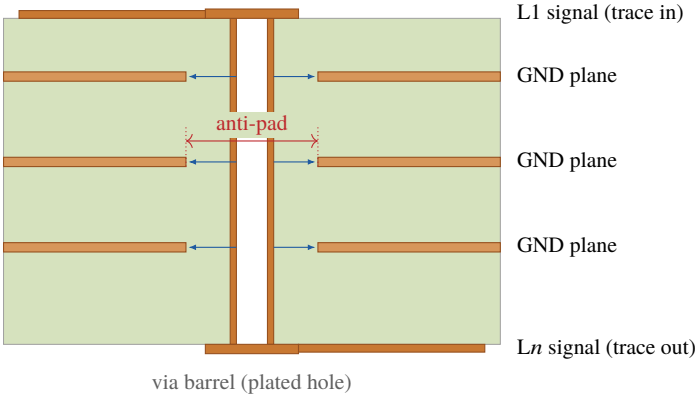
Signal via
carries the high-speed signal between layers.

Via barrel
the plated hole that forms the conductor; its *pads* are the copper rings where traces connect (functional pads) or where no trace connects (non-functional pads).

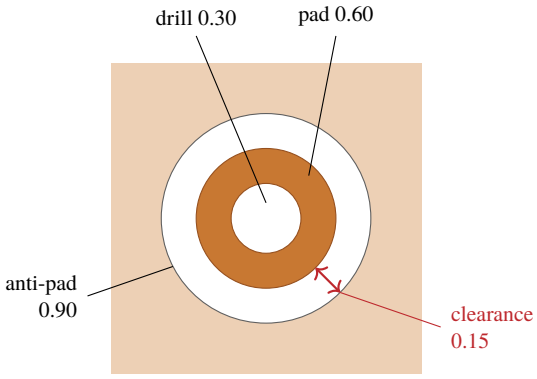
Ground plane
provides the return-current path for the signal on each side of the transition.

Anti-pad
the copper-free clearance surrounding the via on every plane it passes through without connecting.

The electric field of the via terminates on the plane edges at the anti-pad boundaries (Figure 15.2a), so the anti-pad diameter fixes how close the return conductor is to the barrel.



(a) cross-section



(b) top view of a plane (mm)

Figure 15.2. A signal via passing through three ground planes. (a) Cross-section: the field (arrows) runs from the barrel to the plane edges at each anti-pad. (b) Top view of one plane with an illustrative 0.30 mm drill, 0.60 mm pad and 0.90 mm anti-pad: the radial clearance between pad and plane is 0.15 mm. Not to scale. [Conceptual]

15.4.2 Why the anti-pad matters for signal integrity

A via is a short, non-uniform transmission line with parasitic inductance (from the barrel and its current loop) and parasitic capacitance (from the barrel and pads to the surrounding planes). The anti-pad determines much of that capacitance:

- *If the anti-pad is too small,* the plane edges come close to the barrel and pads, the capacitance increases, the via impedance falls below that of the line, and the impedance step reflects part of the signal.
- *If the anti-pad is too large,* the capacitance decreases and the via impedance rises, but the return path moves away from the signal, the plane loses copper, adjacent anti-pads may merge into slots, and neighboring traces may lose their reference over the void (Figure 15.5b). The larger opening also increases coupling between vias that share the plane cavity.

Anti-pad dimensions therefore have an optimum, and it is found by balancing the via's capacitance against its inductance. In the dense-plane limit—planes closely spaced compared with the anti-pad—the field around the barrel is nearly coaxial, with the anti-pad edge as the outer conductor. Then Eq. (15.3) applies with D_{ret} replaced by the anti-pad diameter D_{ap}:

$$Z_{\text{via}} \approx \frac{60}{\sqrt{\varepsilon_{\text{r}}}} \ln \frac{D_{\text{ap}}}{d}. \tag{15.4}$$

Because pads add capacitance and the gaps between planes let the field fringe, the real impedance differs, but the trend is right: impedance rises logarithmically with anti-pad size (Figure 15.3), while the pad capacitance of Eq. (15.2) falls as $1/(D_{\text{ap}} - D_1)$.

In the lumped picture of Section 15.2, the via is an inductance L_{via} and a capacitance C_{via}, with an equivalent impedance and a characteristic frequency

$$Z_{\text{via}} \approx \sqrt{\frac{L_{\text{via}}}{C_{\text{via}}}}, \qquad f_0 = \frac{1}{2\pi\sqrt{L_{\text{via}} C_{\text{via}}}}. \tag{15.5}$$

Changing the anti-pad changes C_{via}, and therefore both. For an isolated L–C pair, f_0 is a resonance. For a via embedded in a matched channel, L and C are spread along the barrel, and $f_0 = 1/(2\pi\sqrt{LC})$ is better read as the frequency at which the via's electrical length reaches about one radian—where a lumped model already has an error of tens of percent (Section 4.6.1). Above roughly $f_0/3$, model the via as a transmission line or with a 3-D field solver.

Worked Example 15.2: Clearance, capacitance and impedance of one anti-pad

A via has a 0.30 mm drill and 0.60 mm pads, and passes through planes with 0.90 mm anti-pads in a 1.6 mm board with $\varepsilon_{\text{r}} = 3.8$. Find the radial clearance, the coaxial-limit impedance, and the lumped estimates of Eqs. (15.1), (15.2) and (15.5). What anti-pad would the coaxial limit need for 50 Ω?

Solution. The radial copper clearance between pad and plane is

$$\text{clearance} = \frac{D_{\text{ap}} - D_{\text{pad}}}{2} = \frac{0.90 - 0.60}{2} = 0.15\,\text{mm},$$

and between the barrel and the plane, where non-functional pads are removed, $(0.90 - 0.30)/2 = 0.30$ mm. Equation (15.4): $Z_{\text{via}} = 30.8 \ln(0.90/0.30) = 33.8\,\Omega$, well below 50 Ω: the via is capacitive. For 50 Ω the anti-pad would have to be $0.30\,e^{50/30.8} = 1.52$ mm, more than most pitches allow. With $h = 63$ mil and $d = 11.8$ mil, Eq. (15.1) gives $L_{\text{via}} = 1.30$ nH and Eq. (15.2) $C_{\text{via}} = 0.68$ pF; then $\sqrt{L/C} = 44\,\Omega$ and $f_0 = 5.4$ GHz. Enlarging the anti-pad to 1.20 mm halves the capacitance to 0.34 pF.

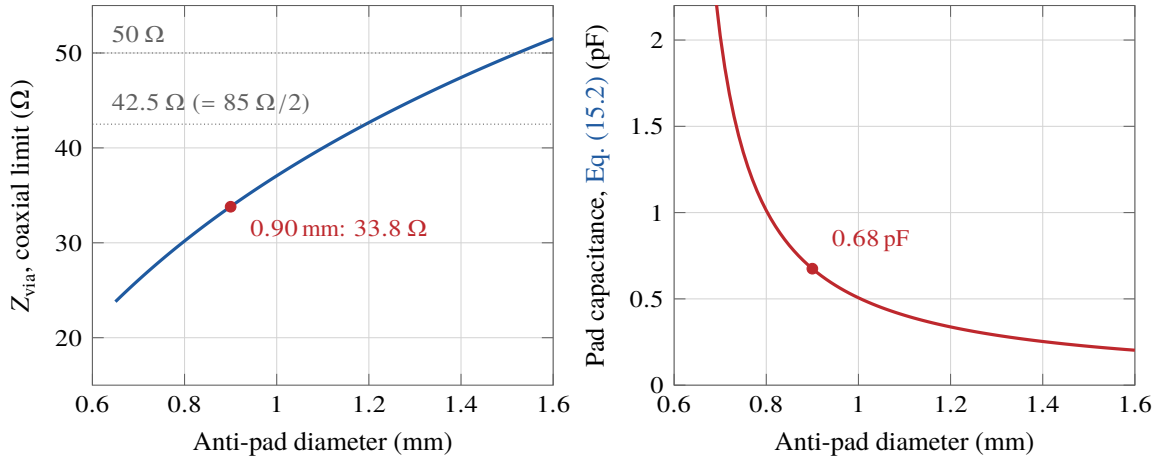


Figure 15.3. Effect of anti-pad diameter for a 0.30 mm drill and 0.60 mm pads in a 1.6 mm board with $\varepsilon_r = 3.8$. Left: coaxial-limit impedance from Eq. (15.4). Right: Johnson–Graham pad capacitance from Eq. (15.2). Trends only; final dimensions come from a 3-D solver. [Analytical]

The two estimates of impedance disagree (34 Ω and 44 Ω), and the lumped $\sqrt{LC} = 30$ ps is about three times the barrel's physical delay of 10 ps. Both are symptoms of the same fact: the formulas are trend indicators. These dimensions are illustrative, not a recommended PCIe rule.

Quick estimate	Full model	Difference
Coaxial limit: 34 Ω	Lumped $\sqrt{L/C}$: 44 Ω	30 %

Figure 15.4 shows what the impedance trend means for a signal. A 25 ps TDR edge sees a capacitive dip at the via whose depth shrinks as the anti-pad grows, and the return loss at 16 GHz improves from about 6 dB to better than 17 dB between the 0.70 mm and 1.20 mm anti-pads. The via is electrically short (10 ps), so the 26 Ω section of the smallest anti-pad reads about 31 Ω on the TDR: the resolution limit of Section 37.1.2 understates the local mismatch, and the return loss reveals it.

15.4.3 Selecting anti-pad dimensions

There is no universal anti-pad dimension. It depends on:

1. the via drill diameter and pad size (and whether non-functional pads are kept);
2. the stackup: which layers are planes and how far apart they are;
3. the dielectric thickness and material;
4. the signal frequency and data rate, which set how much mismatch is tolerable;
5. the target impedance: 50 Ω single-ended, or an odd-mode impedance of 42.5 Ω, 45 Ω or 50 Ω for 85, 90 and 100 Ω differential;
6. the distance between reference planes, which decides how coaxial the field is;
7. the via stub length and geometry, and any back-drilling.

Non-functional pads. Pads on layers where no trace connects add capacitance without purpose. Removing them ("NFP removal") raises via impedance and is common for high-speed vias; some fabricators prefer to keep them on thick boards for drilling and plating reliability, so the choice belongs in the fabrication notes.

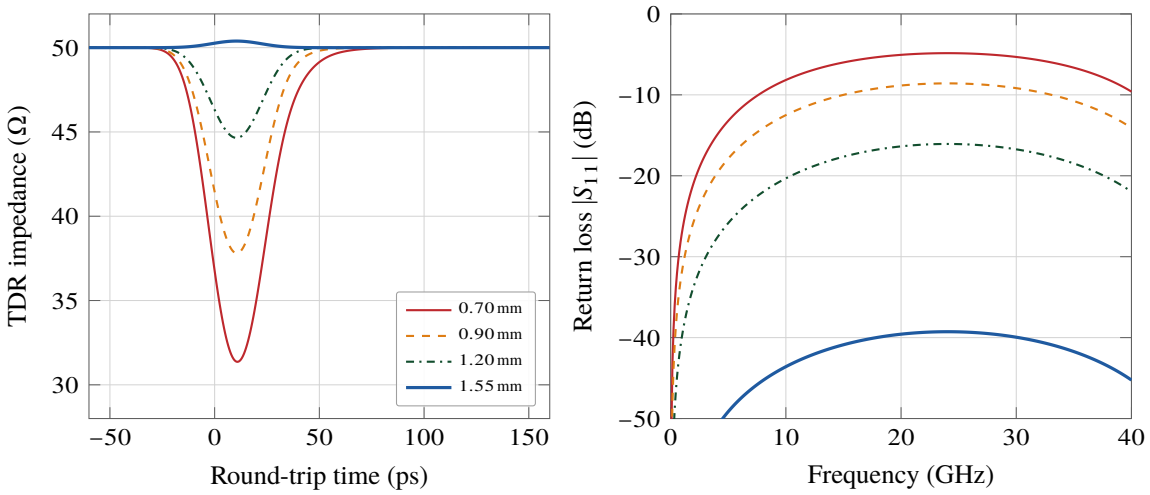


Figure 15.4. Simulated TDR (left, 25 ps Gaussian edge) and return loss (right) of a 1.6 mm via between 50 Ω lines for anti-pad diameters of 0.70, 0.90, 1.20 and 1.55 mm, modeled as a uniform line with the coaxial-limit impedance of Eq. (15.4) and 10.4 ps delay. No pads or stub. [Simulation (transmission-line model)]

Layer-specific anti-pads. Anti-pads need not be the same on every plane. Planes adjacent to the entry and exit layers carry most of the return current and are often given the carefully tuned value, while planes in the middle of a long barrel can be larger. On back-drilled layers the anti-pad must clear the back-drill bit, which is larger than the plated hole, plus its positional tolerance.

Differential via pairs. A pair is normally given a single obround ("stadium") anti-pad around both vias, with a ground return via on each side, symmetric (Figure 15.5a). The shared opening removes the plane copper between the two vias, which raises the pair's differential impedance and keeps both legs symmetric. As an upper bound, two parallel cylinders of diameter d at pitch p with no planes have

$$Z_{\text{diff}} = \frac{120}{\sqrt{\varepsilon_{\text{r}}}} \operatorname{arcosh} \frac{p}{d}, \tag{15.6}$$

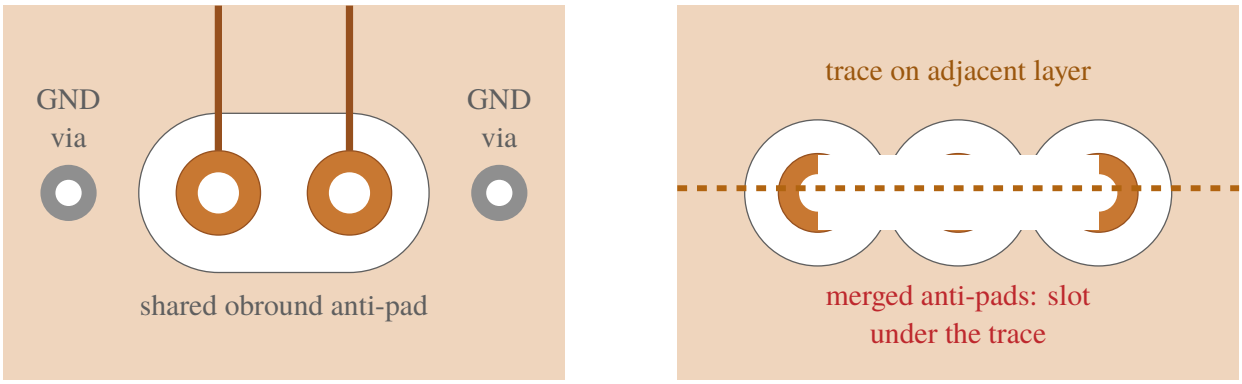
about 115 Ω for 0.30 mm vias at 1.0 mm pitch in $\varepsilon_{\text{r}} = 3.8$; planes, pads and anti-pad edges pull the real value down toward the 85–100 Ω targets.

Plane integrity. Every anti-pad removes plane copper. Under a BGA, rows of anti-pads leave only narrow webs between vias; if they merge, the plane is slotted for every trace that crosses it, and the power distribution network loses capacitance and gains spreading inductance and resistance (Chapter 21).

Worked Example 15.3: How large can the anti-pads be under a BGA?

Signal vias on a 1.0 mm grid pass through a ground plane. The fabricator requires a plane web of at least 0.15 mm between anti-pads. What is the largest round anti-pad, and what coaxial-limit impedance does it give with a 0.30 mm drill in $\varepsilon_{\text{r}} = 3.8$?

Solution. Along a row the anti-pad may be at most $1.0 - 0.15 = 0.85$ mm in diameter (diagonal neighbors at 1.41 mm are not the limit). The 0.90 mm anti-pad of Worked Example 15.2 would leave only a 0.10 mm web. At 0.85 mm, $Z_{\text{via}} = 30.8 \ln(0.85/0.30) = 32\ \Omega$. Higher impedance must then come from other means: removing non-functional pads, obround anti-pads for pairs aligned with the rows, smaller drills (micro-vias), or accepting the discontinuity and confirming it in the channel simulation.



(a) good: symmetric return vias

(b) bad: void under a neighbor

Figure 15.5. Top view of a plane under a differential via pair. (a) A shared obround anti-pad with symmetric ground return vias. (b) Oversized anti-pads that merge into a slot under a trace on the adjacent layer, interrupting its return path. [Conceptual]

15.4.4 Anti-pad, thermal relief and soldermask

Three plane and surface features are easily confused (Table 15.1).

Table 15.1. Features around vias and pads.

Feature	Purpose	High-speed relevance
Anti-pad	Removes copper around a via on a plane it must not touch	Sets via capacitance and impedance; affects return path and plane integrity
Thermal relief	Connects a pad to a plane through narrow spokes, so it can be soldered	Adds inductance; avoid on high-current and decoupling connections
Soldermask	Insulating coating over outer copper; defines solder openings	Lowers outer-layer impedance slightly; tents or plugs vias
Via pad	Copper ring on a layer where the via connects	Adds capacitance; size per drill and annular-ring rules
Non-functional pad	Pad on a layer with no connection	Adds capacitance; often removed for high-speed vias

15.4.5 A PCIe example

Suppose a PCIe Gen4 differential pair transitions from L1 to L3 of a ten-layer board (Figure 15.6). At the transition:

- the signal vias introduce parasitic inductance and capacitance;
- the anti-pad geometry sets the via impedance;
- ground stitching vias next to the pair maintain the return-current path between the reference planes (Section 15.7);
- the via stub below L3 resonates and reflects unless it is back-drilled (Section 15.5).

Optimize the anti-pad dimensions with a 3-D electromagnetic field solver, and verify the whole channel, including this transition, with S-parameters (Chapter 36).

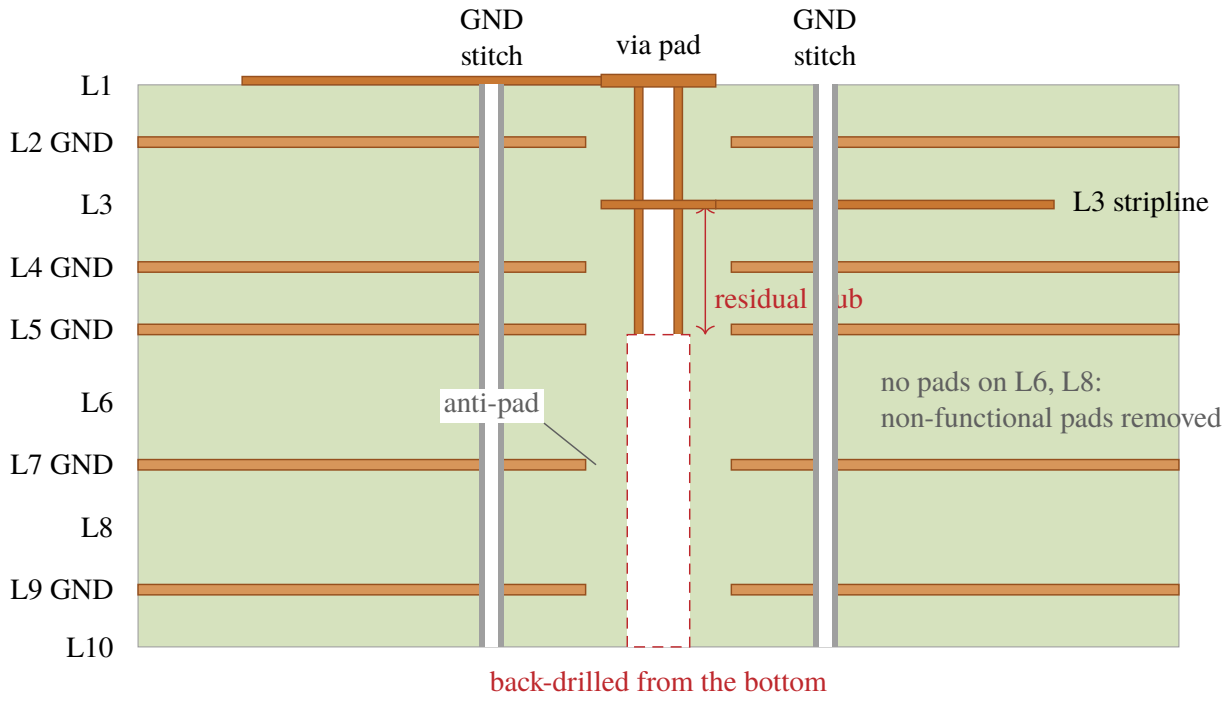


Figure 15.6. One leg of a PCIe via transition from L1 to L3 in a ten-layer board: via pad, functional pad on L3, anti-pads in the ground planes, non-functional pads removed, back-drilled barrel with a short residual stub, and ground stitching vias. Not to scale. [Conceptual]

Design Rule: Anti-pad checklist for high-speed vias

1. Start from the target: single-ended or odd-mode impedance of the connected lines.
2. Decide non-functional pads (remove where the fabricator allows).
3. Use one obround anti-pad per differential pair, with a symmetric ground via on each side.
4. Size anti-pads with a 3-D solver for the actual stackup, drill, pads and stub; tune the planes next to the entry and exit layers first.
5. Check plane webs between neighboring anti-pads (BGA rows, connector fields) against the fabricator's minimum, and look for merged voids.
6. Check that no trace on an adjacent layer crosses an anti-pad void.
7. On back-drilled layers, clear the back-drill diameter plus tolerance.
8. Verify the transition in the channel S-parameters and TDR at the link's edge rate.

Common Pitfall: Anti-pads from the library default

Using the CAD library's default plane clearance, chosen for manufacturing, on multi-gigabit vias. The result is usually a capacitive via, and the mismatch is hidden in a slow TDR. Treat the anti-pad as a designed dimension, recorded per via type in the stackup and constraint set.

15.5 Via stubs and back-drilling

When a through via connects two layers near the top of a board, the barrel below the lower connection is an open-circuited transmission line hanging off the signal path. From Table 4.1, an open stub becomes a short

circuit when its length equals a quarter wavelength, which occurs at

$$f_{\lambda/4} = \frac{c}{4\,\ell_{\text{stub}}\sqrt{\varepsilon_{\text{eff}}}} \approx \frac{11.8\,\text{GHz}\cdot\text{in}}{4\,\ell_{\text{stub}}\sqrt{\varepsilon_{\text{eff}}}}. \tag{15.7}$$

Near this frequency, the stub shunts the signal to ground and S_{21} shows a deep notch (Figure 15.7). Below the notch, the stub behaves as a capacitance that increases loss and reflections. Pads and anti-pads add capacitance along the stub. That loading *slows* the wave—it raises the effective permittivity seen by the stub mode—and adds end capacitance, so the true resonance is often 10–30 % below the estimate of Eq. (15.7).

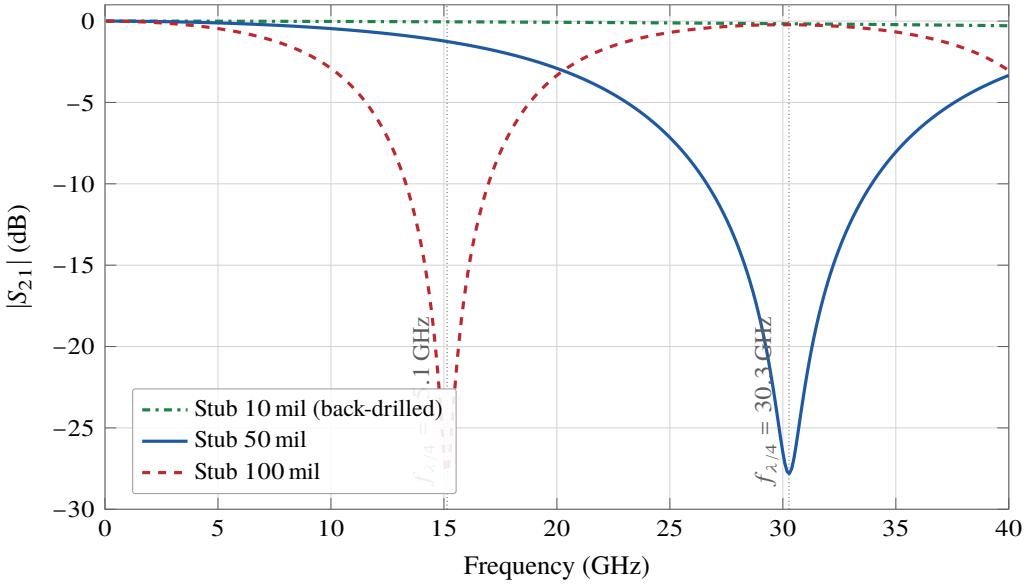


Figure 15.7. Insertion loss of a 50 Ω through path loaded by an open via stub ($Z_{\text{stub}} = 45\,\Omega$, $\varepsilon_{\text{r}} = 3.8$), computed from $S_{21} = 2/(2 + Z_0 Y_{\text{stub}})$ with $Y_{\text{stub}} = \tanh(\gamma\ell)/Z_{\text{stub}}$. The quarter-wave notch moves down in frequency as the stub lengthens.

Rule of Thumb: Stub length

Keep the quarter-wave resonance of every via stub at least three times above the Nyquist frequency of the signal it carries. *Limits:* this is a common design heuristic, not a specification requirement; the actual penalty depends on the stub's loading, the channel loss, and the receiver's equalization. Verify with a channel simulation that includes the 3-D via model.

Worked Example 15.4: Does a Gen5 via need back-drilling?

A PCIe Gen5 pair ($f_{\text{Nyq}} = 16\,\text{GHz}$) transitions from L1 to L3 in the 62.6 mil stackup of Figure 9.3, leaving a stub of about 50 mil. With $\varepsilon_{\text{eff}} \approx 3.8$, is the stub acceptable?

Solution. Equation (15.7) gives $f_{\lambda/4} = 11.8/(4 \times 0.050 \times 1.949) = 30.3\,\text{GHz}$, below the target $3f_{\text{Nyq}} = 48\,\text{GHz}$. Meeting the target requires $\ell_{\text{stub}} \le 11.8/(4 \times 1.949 \times 48) = 0.0315\,\text{in}$, about 32 mil. Options: back-drill from the bottom, route the pair on L8 so that the via spans most of the board (stub from L8 to L10 only), or use a blind via.

15.5.1 Design the tolerance, not just the nominal stub

Worked Example 15.4 produced a *maximum* residual stub of 31.5 mil. A back-drill does not land exactly on target: if the process can leave up to 8 mil more than intended, the nominal target must be at most $31.5 - 8 \approx 23.5$ mil. The permittivity matters too: from Eq. (15.7), a 5 % rise in ε_{eff} lowers the notch by a factor $1/\sqrt{1.05} = 0.976$, about 2.4 %. Figure 15.8 shows both effects for an ideal 45 Ω stub on a 50 Ω path. The "$3 \times f_{\text{Nyq}}$" criterion is itself a heuristic; the allocation shown is illustrative, not a compliance requirement.

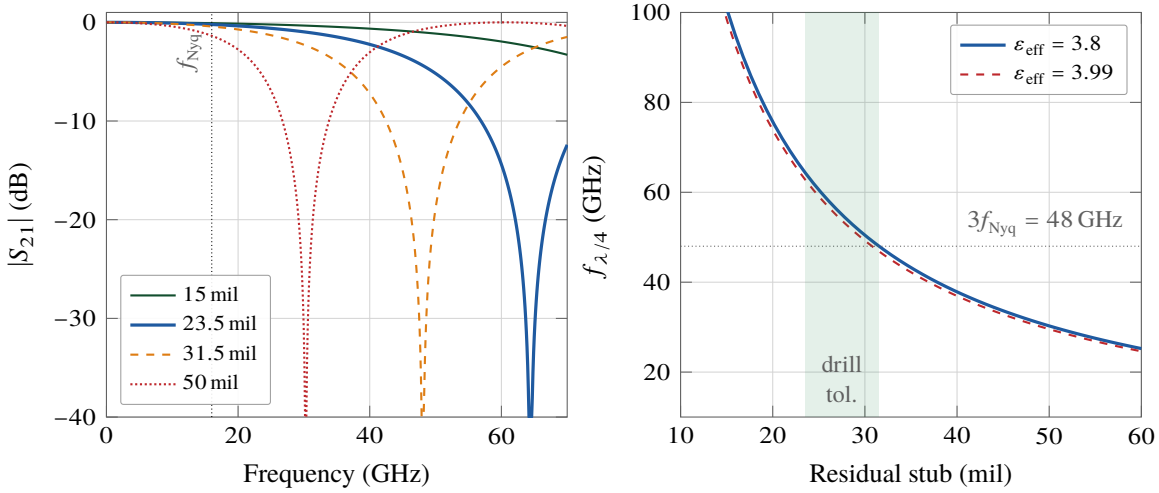


Figure 15.8. Left: ideal shunt open-stub transmission for residual stubs of 15, 23.5, 31.5 and 50 mil ($Z_{\text{main}} = 50\ \Omega$, $Z_{\text{stub}} = 45\ \Omega$, $\varepsilon_{\text{eff}} = 3.8$). Right: quarter-wave frequency versus residual stub length for $\varepsilon_{\text{eff}} = 3.8$ and $\varepsilon_{\text{eff}} = 3.99$ (5 %), with the 48 GHz target. Transmission-line calculation, not a 3-D via extraction. [Analytical]

Back-drilling removes the unused barrel with a drill slightly larger than the original hole, from the side opposite the connection. The drill depth must stop short of the connected layer, so a residual stub remains; its length depends on the fabricator's depth-control capability and is typically specified as a maximum in the fabrication notes.[1] Back-drilled holes must be identified on a separate drill layer with their target depth or the "must-not-cut" layer (Chapter 39).

15.6 Blind, buried, and micro-vias

Blind vias
: connect an outer layer to one or more inner layers without passing through the board; they have no stub on the far side. Mechanically drilled blind vias require sequential lamination or controlled-depth drilling.

Buried vias
: connect inner layers only and are drilled and plated in a sub-lamination before the outer layers are added.

Micro-vias
: are laser-drilled blind vias that span one thin dielectric layer, with typical aspect ratios (depth to diameter) of about 0.75:1 to 1:1. They form the basis of high-density interconnect (HDI) and enable fine-pitch BGA escape. Stacking micro-vias directly on top of one another saves space but has known reliability concerns under thermal cycling; staggered micro-vias are preferred where space allows, and stacked structures should be qualified with the fabricator.

[1] Residual stub lengths of roughly 6–12 mil are commonly quoted, but capability varies by fabricator and board thickness. Confirm the achievable value before relying on it in a loss budget.

HDI construction removes stubs, shortens vias (reducing their inductance), and frees routing channels under BGAs, at the cost of extra lamination cycles. It is standard practice for LPDDR and fine-pitch SoC packages with pitches of 0.5 mm and below.

Table 15.2. Via technologies.

Type	Advantages	Limitations	Relative cost
Through (PTH)	Simple, robust, cheapest	Stubs; large anti-pads; blocks routing on all layers	1
Back-drilled PTH	Short residual stub	Extra drill step; depth tolerance; larger hole in planes	1.1–1.3
Blind / buried (mechanical)	No stub; routing freed below	Sequential lamination	1.3–2
Micro-via (HDI)	Very short, low L; fine pitch	Laser process; stack reliability; cost per layer	1.5–3
Via-in-pad (filled, capped)	Shortest BGA connection	Fill and planarize step	+10–20 %

15.7 Return and stitching vias

When a signal changes layers, its return current must also change planes (Chapter 23). If the two reference planes are both ground, a ground via next to the signal via provides a short, low-inductance path for the return current. Without it, the return current spreads through the plane cavity as displacement current, exciting cavity resonances, raising crosstalk between all vias that share the cavity, and adding inductance to the signal path.

Design Rule: Return vias

Place at least one ground via within about one via pitch (typically 30–50 mil) of every high-speed signal via that changes reference plane, and two—one on each side, symmetric—for each differential via pair. If the reference planes are at different potentials (ground and power), place a stitching capacitor as close as possible to the transition, and prefer stackups that avoid such transitions for fast signals.

15.8 BGA and connector breakout

Under a BGA, signals must escape from the ball field through the gaps between vias. On coarse pitches (1.0 mm and 0.8 mm), "dog-bone" fan-out places a via diagonally offset from each ball, connected by a short trace (Figure 15.9). On finer pitches, via-in-pad with filled and capped vias (VIPPO) or micro-vias is required. For high-speed pairs:

- assign pairs to adjacent balls, and place their vias symmetrically so that the two legs of the pair have equal length and see the same neighbors;
- keep a ground via next to each signal via pair;
- neck traces down only where the gap requires it and for the shortest possible length—the reduced width raises impedance;
- route the fastest signals on the layers that give the shortest via stubs.

Connector pin fields follow the same principles. Press-fit connectors require plated holes of a specific finished size, which forces long barrels and stubs; their footprints are usually back-drilled for links above about 10 Gb/s.

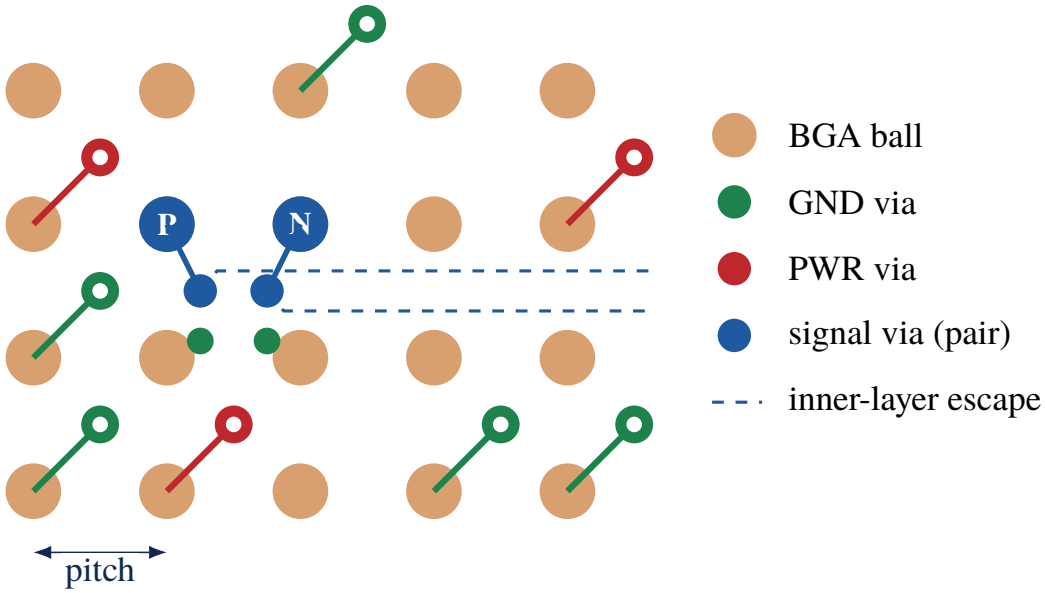


Figure 15.9. Dog-bone breakout under a BGA. A differential pair is assigned to adjacent balls, its vias are placed symmetrically with nearby ground return vias, and it escapes between ball rows on an inner layer.

Common Pitfall: Swiss-cheese planes under BGAs

A dense field of through vias with large anti-pads can leave the reference plane under the BGA as a mesh of narrow copper webs, or remove it entirely between rows. Traces escaping on adjacent layers then lose their return path exactly where they are most crowded. Check plane integrity under every BGA, and reduce anti-pad size or use back-drilling or HDI where the plane is compromised.

Key Takeaways

- A through via is roughly a nanohenry and a fraction of a picofarad; at edges of tens of picoseconds it must be modeled as a transmission line.
- Return vias set the via's impedance and provide its return path; place them adjacent to every signal via that changes reference.
- An open via stub resonates at $c/(4\ell\sqrt{\varepsilon_{\text{eff}}})$; keep that frequency at least three times the Nyquist frequency.
- Back-drilling, routing on deep layers, blind vias, and HDI remove stubs.
- The anti-pad sets via capacitance and impedance: too small makes the via capacitive, too large weakens the return path and the plane. Size it with a field solver, per via type.
- Under BGAs, symmetric pair escapes and intact planes matter more than shortest routes.

Further Reading

Johnson and Graham [1] give the classic via estimates; Hall and Heck [4] model vias and stubs as transmission-line structures; Ritchey [34] addresses via fabrication and back-drilling practice.

Review Problems

P15.1 Estimate the partial inductance of a via 93 mil long with a 12 mil barrel.

P15.2 Estimate the pad capacitance of a via in a 62 mil board with 18 mil pads, 30 mil anti-pads, and $\varepsilon_{\mathrm{r}} = 3.8$.

P15.3 Find the quarter-wave resonance of a 70 mil stub with $\varepsilon_{\mathrm{eff}} = 3.6$.

P15.4 What is the longest stub that keeps $f_{\lambda/4} \geq 3 f_{\mathrm{Nyq}}$ for a 16 GT/s link with $\varepsilon_{\mathrm{eff}} = 3.8$?

P15.5 Estimate the impedance of a via with an 8 mil barrel and an effective return-via diameter of 32 mil in a laminate with $\varepsilon_{\mathrm{r}} = 4.0$.

P15.6 A micro-via spans a 3 mil dielectric and the fabricator allows a maximum aspect ratio of 0.8:1. What is the minimum laser-drill diameter?

P15.7 A via has a 0.25 mm drill, 0.50 mm pads and 0.80 mm anti-pads in $\varepsilon_{\mathrm{r}} = 3.6$. Find the radial pad-to-plane clearance and the coaxial-limit impedance.

P15.8 *(Inverse design.)* For the via of the previous problem, what anti-pad diameter gives 50 Ω in the coaxial limit? Would it fit on a 1.0 mm grid with a 0.15 mm minimum web?

Chapter 16

Routing Geometry, Length Matching, and Connectors

Chapter Overview

Routing rules are full of fixed numbers: no 90° corners, arcs with $R \geq 3W$, match every pair to 5 mil, keep meanders three widths apart. Some of these rules rest on physics, some on manufacturing history, and some on habit. This chapter separates them. It compares corner shapes with dimensions and an electrical estimate, computes the skew a pair accumulates in a turn, shows where skew should be compensated and why, explains why equal length is not equal delay, warns against matching more tightly than the timing budget requires, and ends with how to choose a connector from its drawing and how to review a pin map before release.

Learning Objectives

1. Compare 90°, mitred, two-45°, arc, Bézier, and teardrop geometries electrically and for manufacturing.
2. Compute the length skew of a pair through any turn.
3. Explain why skew should be corrected close to where it is created.
4. Convert a timing budget into a matching tolerance that includes package, via, and layer delays.
5. Select a connector from its drawing and review a pin map systematically.

16.1 Corner geometries

Figure 16.1 draws the common corner shapes for a 5 mil trace at a common scale. In a 90° corner the outside of the bend carries extra copper, which adds a small shunt capacitance; the current crowds toward the inside edge, which adds a small series inductance. Mitring (cutting the outer corner) removes part of the extra copper; two 45° bends and arcs spread the direction change over a longer path.

How large is the effect? Closed-form fits for a right-angle microstrip bend (Gupta et al., collected by Garg, Bahl, and Bozzi) give, for $W/H \geq 1$,

$$\frac{C_b}{W} = (9.5\varepsilon_r + 1.25)\frac{W}{H} + 5.2\varepsilon_r + 7.0 \text{ pF/m}, \qquad \frac{L_b}{H} = 100\left(4\sqrt{W/H} - 4.21\right) \text{ nH/m}. \tag{16.1}$$

For the L1 50 Ω trace of Chapter 10, treated as uncoated ($W = 5.48$ mil, $H = 3.5$ mil, $\varepsilon_r = 3.8$) this gives $C_b = 11.9$ fF and $L_b = 7.1$ pH. For a lumped shunt C and series L at one point, $|S_{11}| \approx \frac{1}{2}\omega\,|CZ_0 - L/Z_0|$, which is plotted in Figure 16.2: −43 dB at 5 GHz, −37 dB at 10 GHz, and −31 dB at 20 GHz. On a TDR with

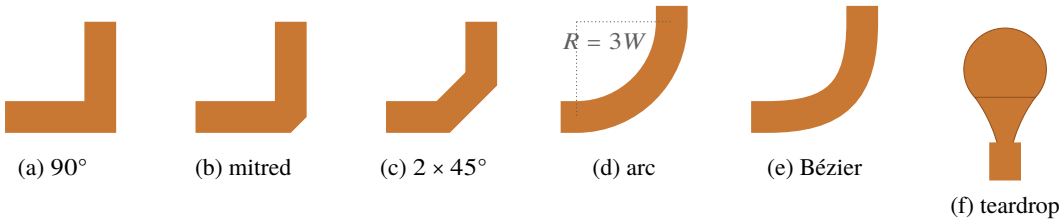


Figure 16.1. Corner geometries for a 5 mil trace, drawn to one scale. (a) Square 90°; (b) mitred 90°; (c) two 45° bends; (d) circular arc with centreline radius $R = 3W$; (e) Bézier curve; (f) teardrop at a pad or via, which is a fillet rather than a bend. [Conceptual]

a 20 ps rise time, the capacitance produces a dip of roughly 1 % of Z_0. A single corner on a narrow trace is therefore a minor discontinuity next to a via (Chapter 15) or a connector.

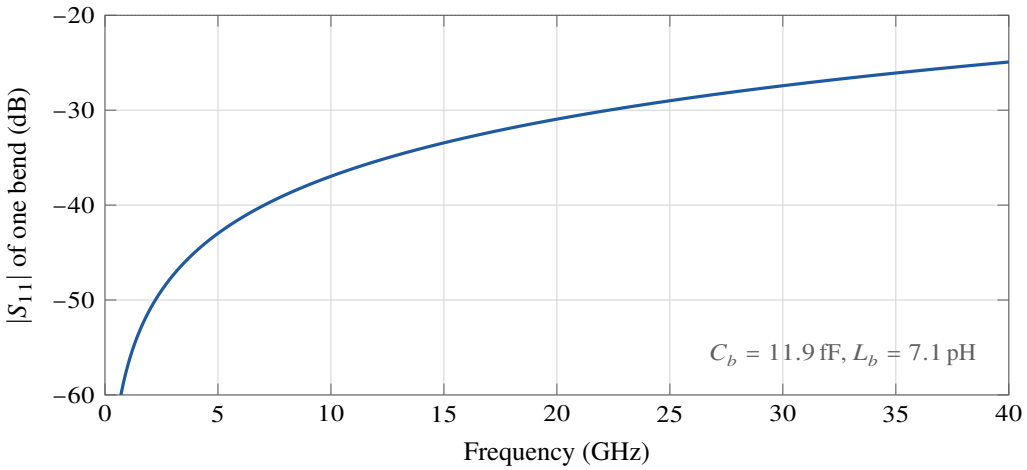


Figure 16.2. Estimated reflection of one right-angle bend in the 50 Ω L1 trace, from the closed-form excess capacitance and inductance of Eq. (16.1) (uncoated microstrip; the fit's own error is not stated by its source). Mitred bends and arcs reflect less, but the starting point is already small. [Analytical]

The electrical case for avoiding square corners on narrow digital traces is therefore weak for a single bend. The rule survives for other reasons, collected in Table 16.1: pairs, wide traces, many bends in series, and manufacturing.

Common Pitfall: Folklore about corners

"Right-angle corners radiate" and "right-angle corners cause acid traps" are both repeated widely. A single corner on a narrow trace radiates no more than the straight trace it is part of; radiation is governed by loop area, current, and common mode (Chapter 27). Acid traps were a concern for acute angles with older etching chemistry; modern fabricators rarely see them for 90° corners, but acute angles (below 90°) should still be avoided. Prefer arcs or 45° bends for the reasons in Table 16.1, not for these.

The $R \geq 3W$ rule. The rule that an arc's centreline radius be at least three trace widths is a heuristic. Electrically, the excess capacitance of an arc is already small at $R = W$; the rule mainly keeps the inner edge from pinching (the inner edge radius is $R - W/2$), keeps the pair gap constant through the arc, and keeps CAM tools from converting tight arcs into coarse segments. State it as a layout preference, not as a signal-integrity requirement.

Table 16.1. Corner shapes compared. "Pair skew" is the extra outer-leg length for a 90° turn at pitch p. The ranking is for a typical high-speed digital board; it is a judgement, not a measured order.

Shape	Excess C	Pair skew	Remarks
Square 90°	largest	$2p$	Pair gap widens at the corner; avoid for pairs and wide traces; acceptable on slow nets.
Mitred 90°	smaller	$2p$	Optimum mitre depends on W/H; common on RF microstrip.
Two 45°	small	$1.66p$	CAD default; keep the 45° segment at least 1–2W so the pair gap stays constant.
Arc	smallest	$1.57p$	Constant gap for pairs; the minimum skew; some fabrication data formats approximate arcs with segments.
Bézier / spline	small	$\approx 1.57p$	Similar to an arc; useful for escape routing between obstacles.
Teardrop	—	—	A fillet at a pad or via, not a bend; improves drill-registration margin and mechanical strength; adds a small capacitance.

16.2 Skew in a turn

A pair turning through an angle θ around a common centre has an outer leg longer than the inner leg (Figure 16.3). If the turn is made of n equal straight segments, the length difference is

$$\Delta\ell = 2pn \tan\frac{\theta}{2n}, \tag{16.2}$$

which gives $2p$ for one square corner ($n = 1$, $\theta = 90°$), $1.66p$ for two 45° bends, and approaches $p\theta = 1.57p$ for an arc. The difference is independent of the arc radius. For the L1 100 Ω pair ($p = 9.77$ mil) one arc turn adds 15.3 mil of skew, about 2.1 ps in the odd mode.

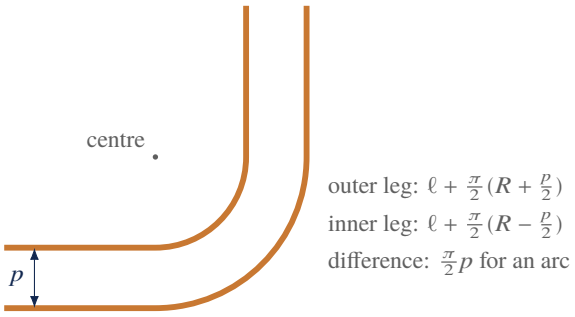


Figure 16.3. A pair through a 90° arc turn. The legs differ by $\frac{\pi}{2}p$ for any radius R. [Analytical]

Turns in opposite directions cancel. A route with three left turns and two right turns has a net skew of one turn, and that one turn should be compensated *near where it occurs*, not at the end of the route (Section 16.3).

16.3 Where to compensate skew

In homogeneous stripline the even and odd modes travel at the same speed, and skew can be corrected anywhere along the route with the same result at the receiver. In microstrip they do not: on the L1 pair the odd mode has $\varepsilon_{\text{eff}} = 2.72$ and the even mode 3.11. Skew converts part of the differential signal into common mode; the common-mode part then travels at the even-mode speed and falls behind. A correction at the far end re-aligns the two legs but cannot re-combine energy that arrived late.

Figure 16.4 quantifies this with a modal model of the L1 pair: 100 mm of coupled microstrip, a 40 ps edge, and 15 ps of skew created at the transmitter breakout. The odd-mode delay is 550.5 ps and the even-mode delay 588.1 ps, 37.5 ps apart.

- **Local correction** (skew removed before the coupled run): the differential edge arrives clean, and no common mode is generated.
- **Remote correction** (skew left in the run, removed at the far end): the differential edge deviates from the clean edge by up to 9.5 % of its swing, and a common-mode pulse of 0.15 (in units where the differential swing is 2) remains at the receiver.
- **No correction**: the deviation is 18 % of the swing and the common-mode pulse peaks at 0.18.

The model is lossless and has matched terminations; it isolates the modal-velocity effect. In stripline the remote correction would be as good as the local one; in microstrip it recovers only about half of the damage.

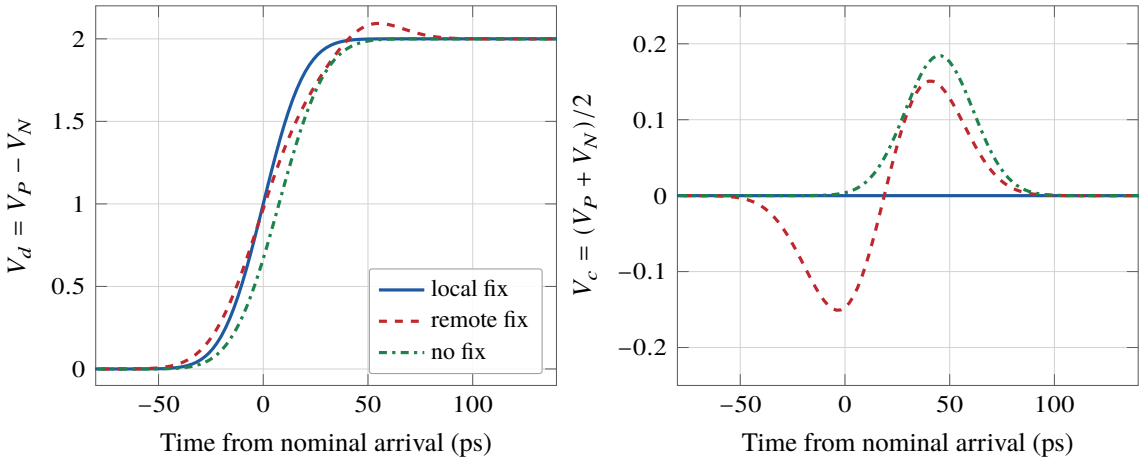


Figure 16.4. Far-end differential (left) and common-mode (right) voltage of a coupled microstrip pair with 15 ps of skew created at the source, for local correction, remote correction, and no correction. Modal model with the solver's ε_{eff} values. [Simulation]

Design Rule: Compensate skew where it is created

Place the compensating length within a short distance of the asymmetry: right after the pin breakout, right after a turn, or next to the via whose stub or offset caused it. Use small, symmetric bumps on the shorter leg rather than one large serpentine at the end of the route.

16.4 Equal length is not equal delay

Matching lengths is a proxy for matching delays. The proxy fails when the signals travel through different media:

- **Different layers.** With the stackup of Chapter 10, a single-ended L1 trace has $\varepsilon_{\text{eff}} = 2.98$ (5.76 ps/mm, 146 ps/in) and an L3 trace $\varepsilon_{\text{eff}} = 3.70$ (6.42 ps/mm, 163 ps/in). 25 mm routed on L1 instead of L3 arrives 16.5 ps early.
- **Modes.** On L1 the odd mode travels at 5.50 ps/mm and the even mode at 5.88 ps/mm. Single-ended signals with coupled neighbours have pattern-dependent delay.
- **Vias.** A layer transition adds its own delay and capacitance; two signals with different via counts or barrel lengths differ even when their trace lengths match.

- **Package and connector.** Pin-to-die delays inside a BGA package can differ by tens of picoseconds between pins. Many device vendors publish pin delays (in the datasheet or in the IBIS model's package section); controllers that train their timing compensate some of it, but not all.
- **Glass weave.** A trace over a resin-rich region sees a lower D_k than its partner over a glass bundle (Chapter 9).
- **Meander coupling.** Closely spaced meander segments couple; in microstrip, part of the signal arrives early through the coupling, so the meander's effective delay is shorter than its length suggests. Keep meander segments far enough apart (a common heuristic is at least three to four times the height to the reference plane) and verify critical meanders with a field solver or TDR.

Worked Example 16.1: Converting a package delay into a length

In a DDR byte lane the controller's datasheet lists package delays of 48 ps for DQ3 and 61 ps for DQS. The lane is routed on L3. If the timing plan matches total delay (package plus board), how much longer should DQ3's board trace be than DQS's?

Solution. DQS has 13 ps more package delay, so DQ3 needs 13 ps more board delay: $13/6.42 = 2.0$ mm (80 mil) of L3 trace. If the controller performs per-bit deskew during training, the vendor's layout guide may instead specify matching board length only; follow the guide, because the training range was designed around it.

16.5 How tightly to match

A matching tolerance should come from a timing budget: the part of the unit interval or setup/hold window allocated to routing skew, divided by the propagation delay per unit length. Matching more tightly than the budget requires is not free:

- every serpentine adds length, loss, and coupling between its own segments;
- serpentines consume routing area and push neighbours closer, increasing crosstalk;
- tuning to a few mils hides the larger, unmodelled delays listed in Section 16.4, giving false confidence;
- tight auto-tuning generates many small bumps, each a small discontinuity.

For example, an 8 GT/s link with 3 % of the UI allocated to intra-pair skew allows 3.75 ps, which is 0.58 mm (23 mil) on L3. A rule of "5 mil" is four times tighter than required; a rule of "50 mil" would be too loose.

16.6 Selecting a connector from its drawing

A connector is chosen from its customer drawing and its electrical data, not from its photograph. Read, in this order:

1. **Interface compliance.** Is the connector specified for the interface and generation (for example a PCIe CEM Gen5 card-edge connector, or a Type-C receptacle rated for USB4)? Is an S-parameter model available, and at what reference impedance (85 Ω or 100 Ω)?
2. **Contact numbering and view.** Drawings show the mating face, the PCB footprint (usually viewed from the component side), or both. Identify which view each table uses before any pin mapping (Section 16.7).
3. **Footprint and attachment.** SMT, through-hole, or press-fit; required board thickness for press-fit or card-edge parts; hole tolerances; keep-outs; stub length of through-hole pins (a stub behaves like a via stub).

4. **Electrical ratings.** Current per contact and its derating with the number of energized contacts and temperature; voltage rating; contact resistance.
5. **Mechanical ratings.** Mating cycles, retention force, latch or screw, orientation keying, and the mated height that sets the board-to-board or board-to-panel geometry.
6. **Signal assignment.** Ground pins between pairs (for example a ground–signal–signal–ground pattern), and whether the vendor's performance data assumed a specific assignment.

16.7 Reviewing a pin map

Pin-mapping errors are among the most expensive board errors because they are invisible in simulation and often survive design review. The procedure in Figure 16.5 turns the review into a table check.

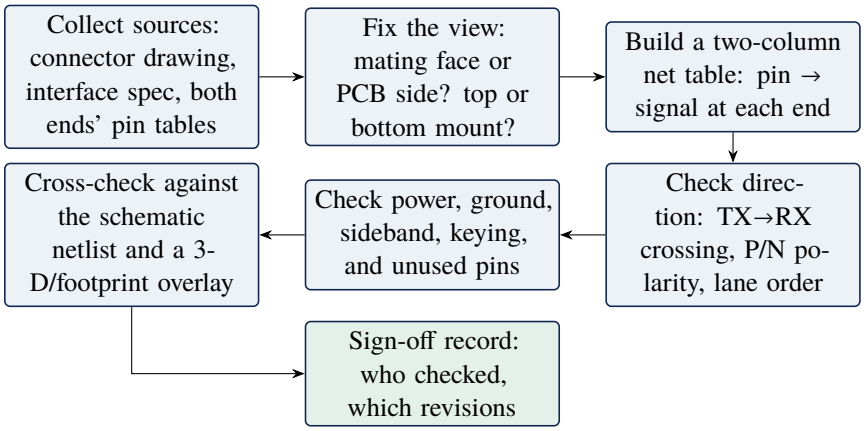


Figure 16.5. A pin-mapping review procedure. The view check (step 2) catches mirror errors; the direction check (step 4) catches TX/RX and polarity errors. [Conceptual]

Common errors and what the protocol tolerates.

- **Mirror errors.** A footprint drawn from the mating-face view, or a connector moved to the bottom side without mirroring, reverses every row.
- **TX/RX crossing.** A transmitter must connect to a receiver. On device-to-device links (for example PCIe on a board-to-board cable) the crossing must happen exactly once.
- **Polarity inversion.** PCIe and USB SuperSpeed receivers can detect and correct P/N inversion of a lane during training; many other interfaces (MIPI D-PHY clock lanes, reference clocks, DDR strobes and clocks) cannot. A swap that the protocol tolerates should still be documented.
- **Lane reversal and swapping.** PCIe permits lane reversal of a multi-lane link if the devices support it; DDR controllers often allow swapping DQ bits within a byte and bytes within a channel, but never moving DQS or DM/DBI to another byte. Check the controller's guide.
- **Power and sideband.** Reference clocks, reset, presence-detect, wake, and power pins are low-speed but fatal when wrong.

Checklist: routing and connectors

- □ Pairs use arcs or 45° bends with a constant gap; square corners only on slow nets.

- ☐ Net turn skew is compensated near each turn with small symmetric bumps.
- ☐ Matching tolerances come from the timing budget and include package, via, and layer delays.
- ☐ Meander segments are far enough apart that self-coupling does not shorten their delay.
- ☐ The connector's drawing views, ratings, and model reference impedance have been read.
- ☐ The pin map was checked as a two-ended table with view, direction, polarity, and sideband steps.

Key Takeaways

- A single corner on a narrow trace is a small discontinuity; corners matter for pairs, wide traces, and many bends in series.
- $R \geq 3W$ is a layout heuristic, not a signal-integrity limit.
- A 90° turn adds $2pn\tan(\theta/2n)$ of skew: $2p$ square, $1.57p$ arc.
- In microstrip, correct skew where it is created; far-end correction recovers only part of the damage.
- Match delay, not length; match only as tightly as the budget requires.
- Review pin maps as tables, starting from the drawing view.

Further Reading

Hall and Heck [4] and Bogatin [3] discuss bends, serpentines, and skew; Johnson and Graham [2] treat skew and its effect on differential signals. Connector vendors' application notes and the interface specifications' layout guides are the authority for a given connector.

Review Problems

P16.1 A pair with pitch 12 mil turns 90° with two 45° bends. What is the length skew, and how much is saved by using an arc?

P16.2 A route makes four left turns and one right turn (all 90° arcs) at $p = 10$ mil. What is the net skew and where should it be compensated?

P16.3 (Approximation failure) Using Eq. (16.1), a designer concludes that a 90° bend in a 60 mil-wide 50 Ω RF trace is harmless at 6 GHz. Why is the conclusion unsafe?

P16.4 (Inverse design) A receiver tolerates 4 ps of intra-pair skew. Two vias contribute 1.5 ps of mismatch. How much routing mismatch is allowed on L1 at the odd-mode speed of 5.50 ps/mm?

P16.5 Why does far-end skew correction work in symmetric stripline but not fully in microstrip?

P16.6 (Imperfect data) A connector datasheet gives pinouts from the mating face; the CAD footprint was built from the table without noting the view. What two checks would catch a mirror error before fabrication?

Chapter 17

Timing, Jitter, and Eye Diagrams

Chapter Overview

Every impairment studied so far—reflections, loss, crosstalk, skew—ends up as a reduction of timing or voltage margin at the receiver's sampling point. This chapter turns those impairments into a pass/fail answer. It sets up timing budgets for the three clocking architectures used on boards (common-clock, source-synchronous, and embedded-clock), decomposes jitter into its random and deterministic components, relates jitter to bit-error ratio through the dual-Dirac model and bathtub curves, and explains how eye diagrams, masks, and equalization (CTLE, FFE, DFE) are used to qualify high-speed links.

Learning Objectives

1. Write setup and hold budgets for common-clock interfaces.
2. Explain why source-synchronous and embedded-clock interfaces scale to higher data rates.
3. Decompose total jitter into RJ, DJ, PJ, DDJ, and BUJ, and compute TJ at a target BER.
4. Read eye diagrams and bathtub curves and relate them to masks.
5. Describe the operation and limits of CTLE, FFE, and DFE equalizers.

17.1 Clocking architectures

17.1.1 Common-clock timing

In a common-clock interface, driver and receiver are clocked from the same source. Data launched by one clock edge must arrive and settle at the receiver before the *next* edge, by at least the setup time, and must not change until after the hold time (Figure 17.1). The budgets are

$$\text{setup margin} = T_{\text{clk}} - \left(t_{\text{co,max}} + t_{\text{flight,max}} + t_{\text{su}} + t_{\text{skew}} + t_{\text{jit}}\right) \geq 0, \tag{17.1}$$

$$\text{hold margin} = t_{\text{co,min}} + t_{\text{flight,min}} - t_{\text{h}} - t_{\text{skew}} \geq 0, \tag{17.2}$$

where t_{co} is the driver's clock-to-output delay, t_{skew} the worst-case difference in clock arrival between the two devices, and t_{jit} the cycle-to-cycle clock jitter. The flight time includes the board delay plus the extra delay caused by loading and edge degradation at the receiver threshold.

Worked Example 17.1: Common-clock budget

A 133 MHz bus ($T_{\text{clk}} = 7.5$ ns) has $t_{\text{co}} = 1.0$–4.0 ns, flight time 0.6–1.2 ns, $t_{\text{su}} = 1.5$ ns, $t_{\text{h}} = 0.5$ ns, clock skew 0.3 ns, and jitter 0.2 ns. Find the margins and the maximum clock frequency.

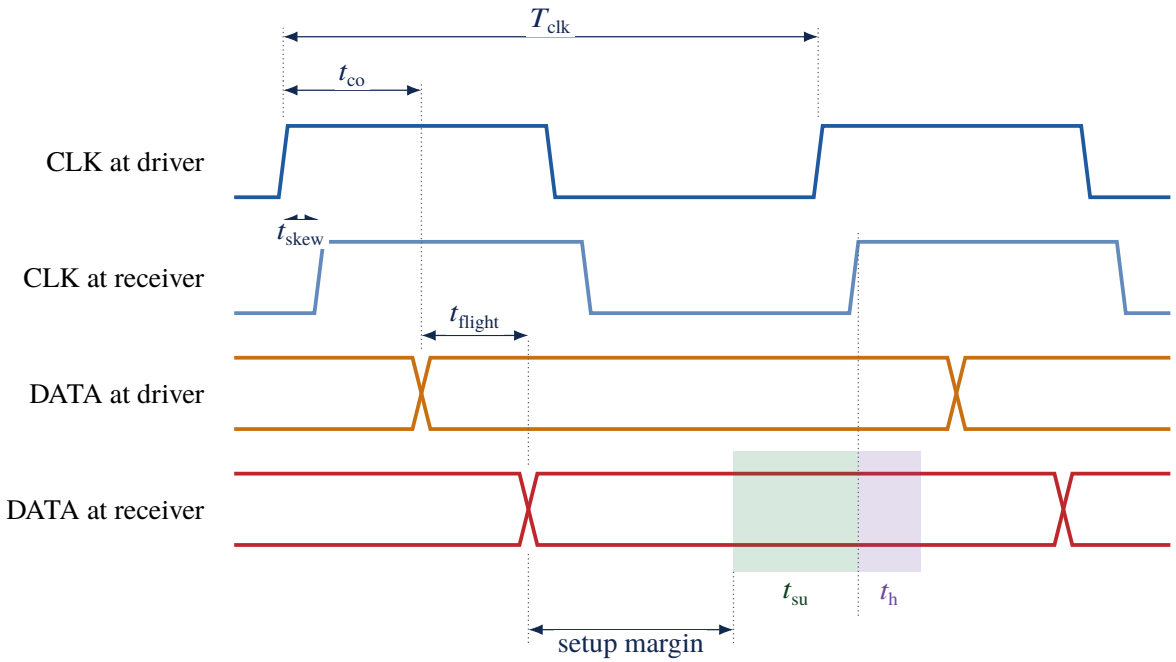


Figure 17.1. Common-clock timing. Data launched t_{co} after the driver's clock edge arrives at the receiver t_{flight} later and must be stable from t_{su} before to t_h after the receiver's next clock edge.

Solution. Setup: $7.5 - (4.0 + 1.2 + 1.5 + 0.3 + 0.2) = 0.3$ ns. Hold: $1.0 + 0.6 - 0.5 - 0.3 = 0.8$ ns. The minimum period is 7.2 ns, so $f_{max} = 139$ MHz. The driver's t_{co} spread alone consumes 3 ns—which is why common-clock interfaces rarely exceed a few hundred megahertz.

17.1.2 Source-synchronous timing

A source-synchronous interface forwards a clock or strobe with the data, routed alongside it. Because strobe and data experience the same driver delay and nearly the same flight time, the absolute t_{co} and flight times cancel; only their *differences* enter the budget. DDR memory (DQS with DQ), MIPI D-PHY, and RGMII work this way. The governing quantity is the skew between data and strobe, which is why these interfaces impose tight length matching within a byte lane or lane group (Chapters 28 and 32). The strobe is either *center-aligned* with the data eye (DDR writes) or *edge-aligned* (DDR reads), in which case the receiver delays it by a quarter of a clock period.

Signed skew. Define the data-to-strobe skew at the receiver as $\Delta = t_{DQ} - t_{DQS}$, with its sign. Late data ($\Delta > 0$) eats setup margin; early data ($\Delta < 0$) eats hold margin. A budget written as "$\pm|\Delta|_{max}$" charges the worst magnitude against *both* sides and wastes margin whenever the skew is biased in one direction, as package and via delays often are. Carry the minimum and maximum signed skew separately:

$$M_{setup} = \tfrac{1}{2}\mathrm{UI} - t_{su} - \Delta_{max} - t_{jit}, \qquad M_{hold} = \tfrac{1}{2}\mathrm{UI} - t_h + \Delta_{min} - t_{jit}, \tag{17.3}$$

for a strobe centered in the data eye. For example, with UI = 312.5 ps, illustrative $t_{su} = t_h = 60$ ps, jitter 30 ps, and a signed skew range of −5–25 ps, the setup margin is $156 - 60 - 25 - 30 = 41$ ps and the hold margin $156 - 60 - 5 - 30 = 61$ ps. A symmetric ±25 ps budget would report 41 ps on both sides and hide 20 ps of real hold margin—margin that a trained interface can redistribute by shifting the strobe (Chapter 28).

17.1.3 Embedded-clock timing

Serial links (PCIe, USB, Ethernet SERDES) send no separate clock. A clock-and-data-recovery (CDR) circuit in the receiver locks a local oscillator to the data transitions and samples each bit at the center of the recovered eye. The CDR tracks slow phase variations (low-frequency jitter and spread-spectrum modulation) and rejects only jitter faster than its loop bandwidth. The timing budget becomes a jitter budget, and line coding (8b/10b, 128b/130b, scrambling) guarantees enough transitions for the CDR to remain locked. This architecture removes flight-time and skew limitations altogether, which is why every multi-gigabit interface uses it.

Jitter transfer. A first-order CDR tracks input phase with a low-pass transfer function $H(s) = \omega_c/(s+\omega_c)$, where $f_c = \omega_c/2\pi$ is the CDR bandwidth. The sampling error that matters is the part it does *not* track,

$$\frac{\phi_{\text{err}}}{\phi_{\text{in}}}(s) = 1 - H(s) = \frac{s}{s+\omega_c}, \tag{17.4}$$

a high-pass response: jitter far below f_c (spread-spectrum modulation at about 33 kHz, wander) is tracked and costs almost nothing, while jitter well above f_c passes fully into the sampling error. For $f_c = 10\,\text{MHz}$, 1 MHz jitter is attenuated to $|\text{j}0.1/(1+\text{j}0.1)| = 0.0995$, about $-20\,\text{dB}$. This is why serial standards specify jitter with a defined *reference CDR* or filter: a jitter number without its filtering is incomplete. Real CDRs are usually second order, with peaking near f_c that the standards also bound.

17.2 Jitter

Jitter is the deviation of signal transitions from their ideal times. It is characterized by its *time interval error* (TIE): the sequence of differences between each actual transition and the corresponding ideal one. Figure 17.2 shows the standard decomposition [54], [55].

Random jitter (RJ)
: Gaussian, unbounded jitter from thermal and shot noise in oscillators and buffers. It is specified by its RMS value σ. Because it is unbounded, its peak-to-peak value grows with the number of observed transitions.

Deterministic jitter (DJ)
: Bounded jitter with identifiable causes. Its components are *periodic jitter* (PJ) from power-supply ripple, spread spectrum, or coupled clocks; *data-dependent jitter* (DDJ), comprising inter-symbol interference (ISI) from channel loss and reflections and duty-cycle distortion (DCD); and *bounded uncorrelated jitter* (BUJ), typically crosstalk from asynchronous aggressors.

17.2.1 Total jitter at a bit-error ratio

Because RJ is unbounded, total jitter is meaningful only at a stated *bit-error ratio* (BER). The *dual-Dirac* model represents DJ by two delta functions separated by $\text{DJ}_{\delta\delta}$, each convolved with the Gaussian RJ. The total jitter at a target BER is then

$$\text{TJ}(\text{BER}) = \text{DJ}_{\delta\delta} + 2Q_{\text{BER}}\,\sigma_{\text{RJ}}, \tag{17.5}$$

where Q_{BER} is the number of standard deviations at which the Gaussian tail equals the BER (adjusted for transition density). Common values are listed in Table 17.1. $\text{DJ}_{\delta\delta}$ is a model parameter fitted to the tails of the distribution and is generally smaller than the peak-to-peak DJ that would be measured directly.

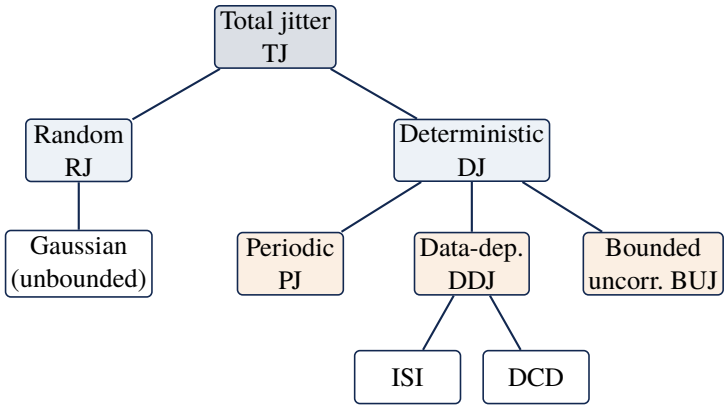


Figure 17.2. Decomposition of total jitter into random and deterministic components.

Table 17.1. Gaussian multipliers for total jitter, $2Q_{\mathrm{BER}}$ (for a transition density of one).

BER	10^{-9}	10^{-12}	10^{-15}
Q_{BER}	5.998	7.034	7.941
$2Q_{\mathrm{BER}}$	11.996	14.069	15.883

Worked Example 17.2: Total jitter at 10^{-12}

A 16 GT/s receiver sees $\mathrm{DJ}_{\delta\delta} = 10$ ps and $\sigma_{\mathrm{RJ}} = 1$ ps. Find TJ at 10^{-12} and the remaining horizontal eye opening.

Solution. $\mathrm{TJ} = 10 + 14.07 \times 1 = 24.1$ ps. The UI is 62.5 ps, so the eye opening is $62.5 - 24.1 = 38.4$ ps, or 0.61 UI. Each additional picosecond of RJ costs 14 ps of eye width—RJ is expensive.

Common Pitfall: Adding jitter components linearly or in RSS indiscriminately

Adding RMS random jitter values to peak-to-peak deterministic values, or combining DJ terms by root sum of squares. Independent RJ sources add as RSS ($\sigma_{\mathrm{tot}}^2 = \sum \sigma_i^2$); DJ components combine by convolution of their distributions, which for budgeting purposes is approximately linear addition of $\mathrm{DJ}_{\delta\delta}$ values; and the two are combined only through Eq. (17.5).

17.3 Eye diagrams, masks, and bathtub curves

An *eye diagram* overlays many unit intervals of a waveform, each triggered by the recovered clock (Figure 17.3). The vertical opening at the sampling instant is the *eye height*, the horizontal opening at the decision threshold the *eye width*. Loss and reflections close the eye through ISI, crosstalk adds vertical noise, and jitter blurs the crossings. Interface specifications define a *eye mask*—a forbidden region inside the eye (and often above and below it)—that the waveform must not enter at a specified BER.

A *bathtub curve* plots BER against sampling phase across the UI (Figure 17.4). Near each crossing the BER is high; toward the center of the eye it falls steeply, at a rate determined by RJ. The width of the bathtub at a given BER is the eye width at that BER. Because measuring BER down to 10^{-12} directly takes a long time, the tails of the bathtub are usually extrapolated from measurements at higher BER using the dual-Dirac model.

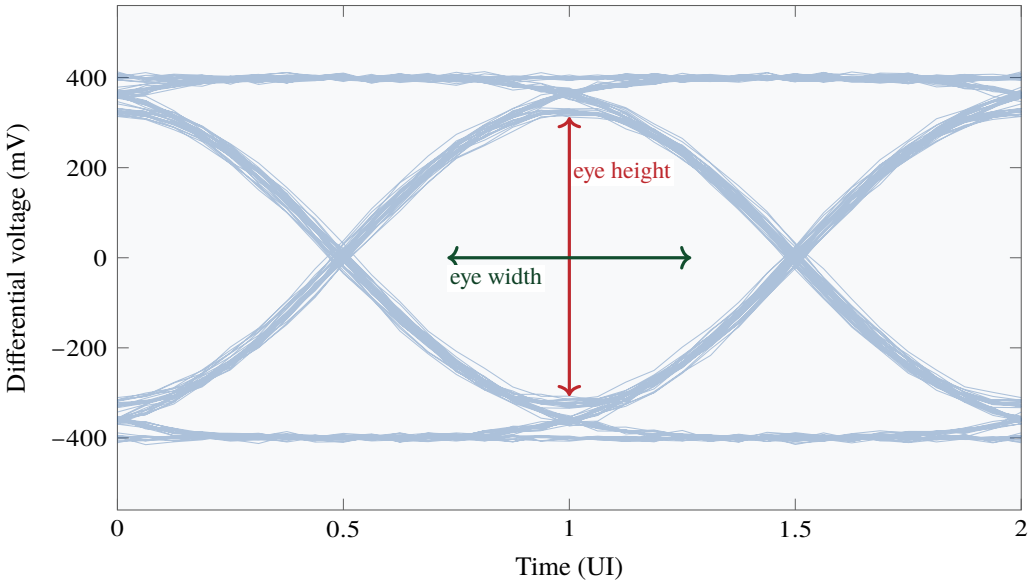
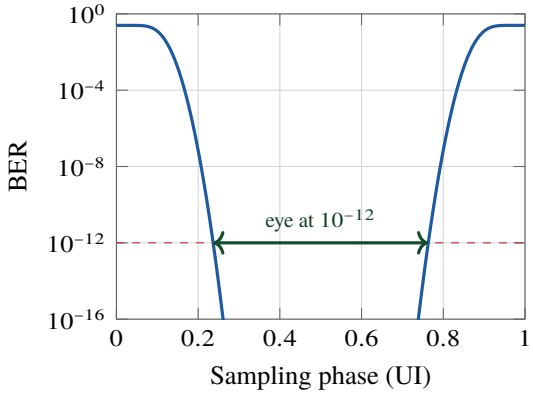


Figure 17.3. Synthetic NRZ eye diagram: a PRBS7 sequence through a band-limited channel with random jitter and noise, overlaid over two unit intervals. Eye height and eye width are marked; a receiver mask would occupy the open region at the center.



(a) Bathtub curve

Parameter	Value
$\mathrm{DJ}_{\delta\delta}$	0.20 UI
σ_{RJ}	0.020 UI
Transition density	0.5
Eye at 10^{-12}	0.53 UI

(b) Model parameters

Figure 17.4. Dual-Dirac bathtub curve. The eye width at 10^{-12} is the distance between the two walls of the bathtub at that BER.

Worked Example 17.3: Measuring BER

How many bits must be received without error to claim, at 95 % confidence, that BER $\le 10^{-12}$? How long does that take at 16 Gb/s?

Solution. For zero observed errors, the confidence level is $1 - \mathrm{e}^{-N\cdot\mathrm{BER}}$, so $N = -\ln(0.05)/10^{-12} = 3.0 \times 10^{12}$ bits. At 16 Gb/s this takes 187 s. Verifying 10^{-15} the same way would take more than two days, which is why extrapolation is used.

17.4 Equalization

When channel loss exceeds what an unequalized receiver can tolerate (roughly 10–15 dB at Nyquist for NRZ), the transmitter and receiver apply equalization to restore a flatter overall response (Figure 17.6).

17.4.1 Continuous-time linear equalizer (CTLE)

A CTLE is an analog high-pass (peaking) filter in the receiver front end, with a transfer function of the form

$$H_{\mathrm{CTLE}}(s) = A_{\mathrm{dc}} \frac{1 + s/\omega_z}{(1 + s/\omega_{p1})(1 + s/\omega_{p2})}, \tag{17.6}$$

which boosts frequencies between the zero ω_z and the first pole ω_{p1} by up to $20\log_{10}(\omega_{p1}/\omega_z)$ relative to DC. It is inexpensive and acts on the whole signal, but it amplifies high-frequency noise and crosstalk together with the signal.

17.4.2 Feed-forward equalizer (FFE) and pre-emphasis

An FFE is a finite-impulse-response filter, usually in the transmitter, that outputs a weighted sum of the current and neighboring bits: $y_n = \sum_k c_k\, x_{n-k}$. A two-tap FFE with a negative post-cursor tap c_1 reduces the amplitude of repeated bits relative to transitions (*de-emphasis*). Its frequency response is

$$|H_{\mathrm{FFE}}(f)| = \sqrt{c_0^2 + c_1^2 + 2c_0c_1\cos(2\pi f\ \mathrm{UI})}, \tag{17.7}$$

rising from $|c_0 + c_1|$ at DC to $|c_0 - c_1|$ at Nyquist. The de-emphasis ratio, in decibels, is

$$\mathrm{DE} = 20\log_{10}\left|\frac{c_0 - c_1}{c_0 + c_1}\right|.$$

Because the transmitter's peak swing is limited, de-emphasis works by attenuating low frequencies rather than boosting high ones. PCIe defines sets of pre-cursor and post-cursor coefficients (presets) that the link negotiates during training.

Worked Example 17.4: De-emphasis of a two-tap FFE

A transmitter uses $c_0 = 0.75$ and $c_1 = -0.25$. Find the DC and Nyquist gains and the de-emphasis.

Solution. DC: $c_0 + c_1 = 0.5$ (-6.0 dB). Nyquist: $c_0 - c_1 = 1.0$ (0 dB). De-emphasis is 6.0 dB: transitions are driven at full swing, repeated bits at half swing.

17.4.3 Pulse response and cursors

Equalization is easiest to reason about from the channel's *pulse response*: the received waveform for a single isolated one-UI pulse, sampled once per UI at the receiver's sampling phase. The largest sample h_0 is the *main cursor*; samples before it are *precursors* $h_{-1}, h_{-2}, \ldots$ and those after it are *postcursors* $h_1, h_2, \ldots$. For a linear channel and symbols $a_n = \pm 1$, the sampled signal is the superposition

$$r_n = \sum_k h_k\, a_{n-k} = h_0 a_n + \underbrace{\sum_{k \neq 0} h_k\, a_{n-k}}_{\text{ISI}}. \tag{17.8}$$

The worst-case (peak-distortion) eye half-height at a zero threshold is $h_0 - \sum_{k\neq 0}|h_k|$. This is the bridge between the interconnect chapters, which produce h_k, and the receiver, which must survive them. Loss produces a smooth tail of postcursors; a reflection produces an isolated echo at a delay equal to its round-trip time.

17.4.4 Decision-feedback equalizer (DFE)

A DFE subtracts from the received sample a weighted sum of previously *decided* symbols,

$$y_n = r_n - \sum_{k=1}^{N} b_k \, \hat{a}_{n-k}, \tag{17.9}$$

and with $b_k = h_k$ and correct decisions it removes postcursors 1 to N exactly—whatever caused them. That includes echoes from reflections, provided they fall inside the tap span (some receivers add "floating" taps that can be placed on a distant echo). Because the DFE feeds back clean decisions rather than the noisy signal, it does not amplify noise or crosstalk. It cannot cancel precursor ISI or postcursors beyond its span, and a wrong decision subtracts the wrong correction: an error on a tap of weight b_k injects an error of $2b_k$ (error propagation). DFEs with one to many taps are standard in receivers above about 10 Gb/s; DDR5 receivers also include a small DFE.

Worked Example 17.5: What a three-tap DFE buys

A sampled NRZ channel has $h_{-1} = -0.08$, $h_0 = 1$, and postcursors 0.25, 0.12, 0.05 (Figure 17.5). Noise at the slicer has $\sigma = 0.03$. Compare the worst-case half-height without equalization and with an ideal three-tap DFE.

Solution. Without DFE: $1 - (0.08 + 0.25 + 0.12 + 0.05) = 0.50$, or 16.7σ. With correct decisions and exact taps the three postcursors vanish and only the precursor remains: $1 - 0.08 = 0.92$, or 30.7σ. These are noiseless amplitude distances for a static sampling phase, not BER values.

A plausible failure. Add a reflection echo of 0.15 at $k = 8$ (500 ps at 16 GBd). It lies outside the tap span, so the half-height falls to $1 - 0.08 - 0.15 = 0.77$. The remedies compete: more (or floating) taps in the receiver, or a board change that removes the echo at its source.

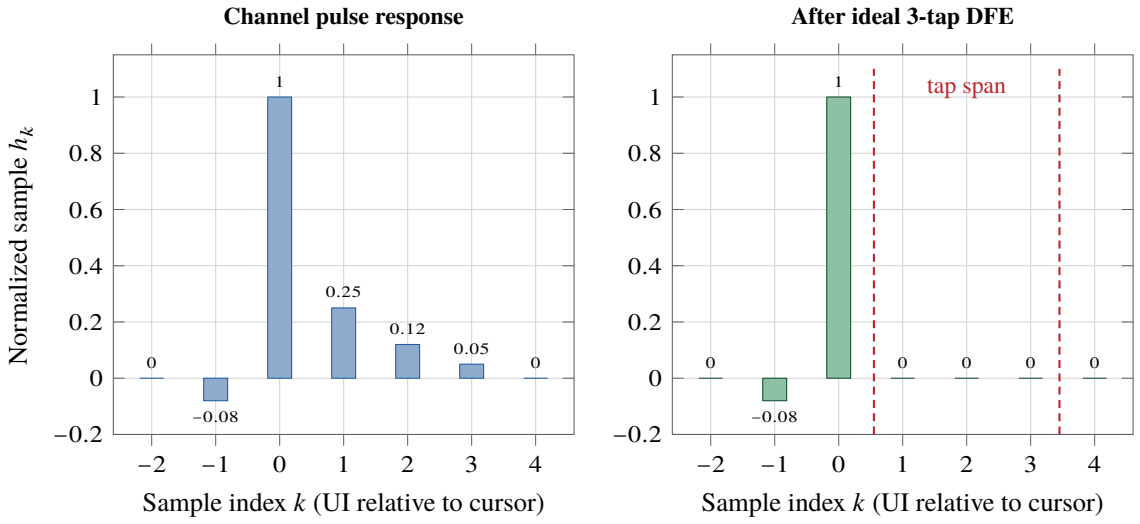


Figure 17.5. Sampled pulse response of Worked Example 17.5, normalized to the main cursor, before and after an ideal three-tap DFE (correct prior decisions, exact taps; no noise or adaptation). The precursor remains. [Analytical]

17.4.5 Statistical and bit-by-bit analysis

Two methods turn a pulse response into a margin. *Statistical* ("peak distortion" or convolution-based) analysis combines the ISI distribution of Eq. (17.8) with Gaussian noise and jitter models to predict the eye contour at very low BER, such as 10^{-12} or below, in seconds; it assumes linearity and time invariance, and its tails are only as good as its noise and jitter models. *Bit-by-bit* (time-domain) simulation runs millions of symbols through the channel and the receiver's adaptive equalization and CDR, and captures nonlinear and adaptive behavior, but it cannot reach 10^{-12} directly and extrapolates its tails. Use both: bit-by-bit to show the adaptation converges, statistical to estimate the low-BER margin. A finite eye overlay of a few thousand symbols demonstrates neither.

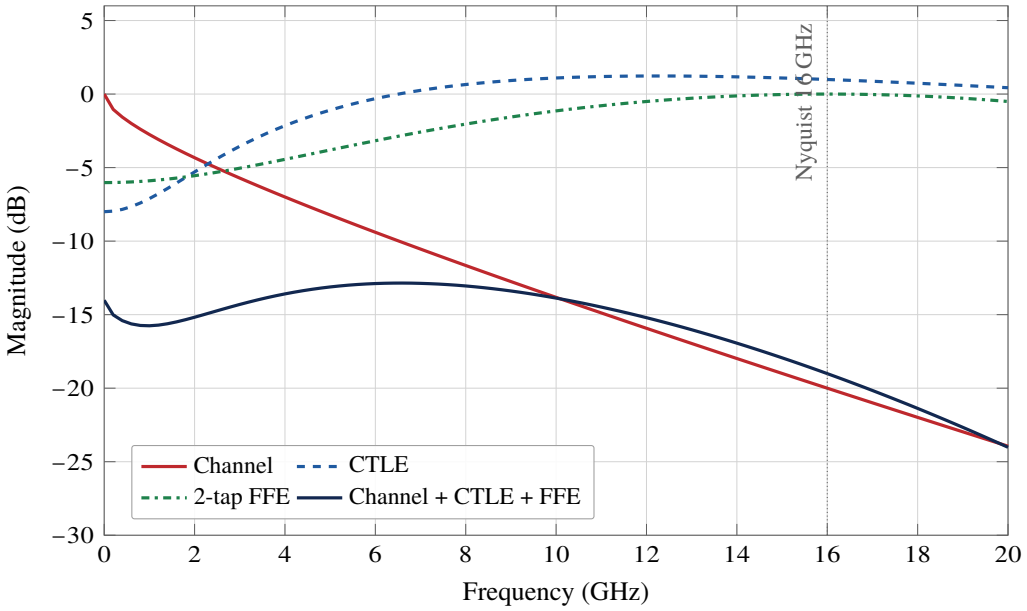


Figure 17.6. Channel loss, CTLE, and two-tap FFE responses for a 32 GT/s link. The equalized response is flatter than the raw channel; the residual slope is left to the DFE.

Table 17.2. Equalizer types.

Type	Location	ISI cancelled	Limitation
CTLE	Rx (analog)	Pre- and post-cursor (broadband)	Boosts noise and crosstalk
FFE / de-emphasis	Tx (or Rx)	Pre- and post-cursor	Reduces signal amplitude
DFE	Rx (feedback)	Post-cursors within tap span (incl. echoes)	Error propagation; tap span

Design Rule: Design the channel for the equalizer you have

Linear equalizers (CTLE, FFE) compensate loss that varies smoothly with frequency; they cannot undo the sharp notches of via stubs or crosstalk, and boosting near a notch amplifies noise. A DFE can cancel reflection echoes, but only those within its tap span, and only postcursors. Keep return loss and crosstalk within specification, place unavoidable discontinuities so that their echoes fall early, and leave the equalizer to handle the slope.

Rule of Thumb: The cost of random jitter

At a BER of 10^{-12}, each picosecond of RMS random jitter consumes about 14 ps of horizontal eye opening. *Limits:* assumes Gaussian RJ and the dual-Dirac model; real tails may be non-Gaussian, and the multiplier changes with BER and transition density (Table 17.1).

Key Takeaways

- Common-clock interfaces are limited by t_{co} spread and flight time; source-synchronous interfaces by data-to-strobe skew; embedded-clock links by jitter.
- Total jitter at a BER is $\mathrm{DJ}_{\delta\delta} + 2Q_{\mathrm{BER}}\sigma_{\mathrm{RJ}}$, with $2Q = 14.07$ at 10^{-12}.
- Eye diagrams and masks summarize amplitude and timing margin; bathtub curves give eye width versus BER.
- The sampled pulse response (cursors) links the channel to the receiver; worst-case half-height is $h_0 - \sum |h_k|$.
- CTLE and FFE flatten the channel response at the cost of noise or amplitude; DFE cancels postcursors within its tap span—including reflection echoes—without amplifying noise, but not precursors.
- Carry signed skew in source-synchronous budgets, and quote jitter with its CDR filtering.

Further Reading

Li [54] and Derickson and Müller [55] treat jitter decomposition, BER, and measurement thoroughly; Hall and Heck [4] and Dally and Poulton [35] cover timing budgets and equalization.

Review Problems

P17.1 A 200 MHz common-clock bus has $t_{co} = 0.8$–2.8 ns, flight time 0.4–0.9 ns, $t_{su} = 1.0$ ns, $t_h = 0.4$ ns, skew 0.2 ns, and jitter 0.15 ns. Find setup and hold margins.

P17.2 Find TJ at 10^{-12} for $\mathrm{DJ}_{\delta\delta} = 12$ ps and $\sigma_{\mathrm{RJ}} = 0.8$ ps.

P17.3 How long must a 16 Gb/s link run error-free to demonstrate BER $\leq 10^{-12}$ at 95 % confidence?

P17.4 Find the de-emphasis of a two-tap FFE with $c_0 = 0.8$, $c_1 = -0.2$.

P17.5 A CTLE has a zero at 2 GHz and its first pole at 8 GHz. What is its maximum peaking relative to DC, ignoring the second pole?

P17.6 In Worked Example 17.5, the previous decision feeding the first DFE tap is wrong. What error does the DFE inject into the current sample, and is the current decision still safe without noise?

P17.7 *(Inverse design.)* A receiver needs a worst-case half-height of 0.6 after an ideal two-tap DFE. The channel has $h_{-1} = -0.1$ and postcursors 0.3, 0.15 and h_3. What is the largest acceptable h_3?

P17.8 A CDR has a first-order bandwidth of 4 MHz. By how much is 40 MHz sinusoidal jitter attenuated in the sampling error? And 400 kHz?

Part III

Power Integrity

Chapter 18

What Is Power Integrity?

Chapter Overview

This chapter opens the power-integrity part of the book. It defines power integrity, shows what a supply rail looks like when it is lost, explains the power delivery network (PDN) from the regulator to the transistors, and introduces the two calculations every PI engineer starts with: voltage droop from resistance and inductance, and target impedance. It then lists the techniques that improve PI, the measurements that verify it, and how power integrity and signal integrity interact. The following chapters develop each topic: simultaneous switching noise (Chapter 20), PDN design (Chapter 21) and planes and return paths (Chapter 23).

Learning Objectives

1. Define power integrity and the PDN.
2. Recognize voltage droop, ripple, ground bounce, simultaneous switching noise, PDN resonance and IR drop on a rail waveform.
3. Estimate droop from IR and $L\ \mathrm{d}I/\mathrm{d}t$, and explain why decoupling capacitance helps only when it is placed and sized correctly.
4. Compute a target impedance and read a PDN impedance plot against it.
5. Choose the right instrument to measure a rail, and relate PI to SI.

18.1 Definition

Power integrity (PI) is the ability of a board's power delivery network to provide stable, clean and sufficient voltage and current to every component under all operating conditions. In simple terms, power integrity ensures that integrated circuits receive the correct supply voltage without excessive noise, ripple, voltage drop or transient disturbance.

As with signal integrity, "correct" has a precise meaning: every IC specifies a supply voltage and a tolerance, measured at a stated point (the package balls or the die), for example $0.8\,\mathrm{V}\ \pm 3\,\%$. A rail has integrity when, at that point and under every load condition the device can create, the voltage stays inside that window with margin. Power integrity is especially important in high-performance systems built around CPUs, FPGAs, GPUs and high-speed SoCs—an application processor such as the Rockchip RK3588 is a typical example—together with their DDR memory and PCIe and other SerDes interfaces, where core rails below 1 V carry many amperes that change in nanoseconds.

18.2 What causes power-integrity problems

Consider a processor core operating at 1.0 V. When it switches suddenly from low to high activity, its current consumption rises by several amperes within nanoseconds to microseconds. The power delivery network must respond quickly to this transient demand; if it cannot, the supply voltage drops. Figure 18.1 shows the resulting waveform and labels the common problems.

Voltage droop
: The supply voltage decreases when the load current increases, most sharply in the first moments after the step, before the regulator can respond.

Voltage ripple
: Periodic variation of the supply voltage, usually at the switching frequency of the regulator and its harmonics.

Ground bounce
: The ground voltage of a chip changes with respect to board ground because of transient current through package and via inductance (Chapter 20).

Simultaneous switching noise (SSN)
: Many digital outputs switching at once draw a large $\mathrm{d}I / \mathrm{d}t$ through shared power and ground paths, creating noise on both (Chapter 20).

PDN resonance
: Inductance and capacitance in the network form resonant peaks; a load current at or near such a frequency produces ringing and enlarged droop (Section 21.4).

IR drop
: Voltage lost in the resistance of traces, planes, vias and connectors between the regulator and the load; it persists as long as the current flows (Section 21.7).

18.3 The power delivery network

The PDN is the complete path that delivers power from the voltage regulator to the transistors on the die, *and* the return path back through ground (Figure 18.2). A typical PDN consists of:

1. a *voltage regulator module* (VRM), which generates the required supply voltage and holds its average value through its control loop;
2. *bulk capacitors*, which supply energy for relatively slow load transients, faster than the regulator loop can follow;
3. *decoupling capacitors* (MLCCs) near the load, which supply local transient current and lower the impedance at higher frequencies;
4. the *PCB power and ground planes*, which distribute current across the board and form a low-inductance capacitor of their own;
5. *package and on-die capacitance*, which maintains voltage stability closest to the silicon at the highest frequencies.

Each stage serves a band of frequencies and refills the next, faster stage. Because every connection between stages has resistance and inductance, a stage cannot help above the frequency at which its connection inductance dominates—which is why the smallest, fastest capacitance must sit closest to the load.

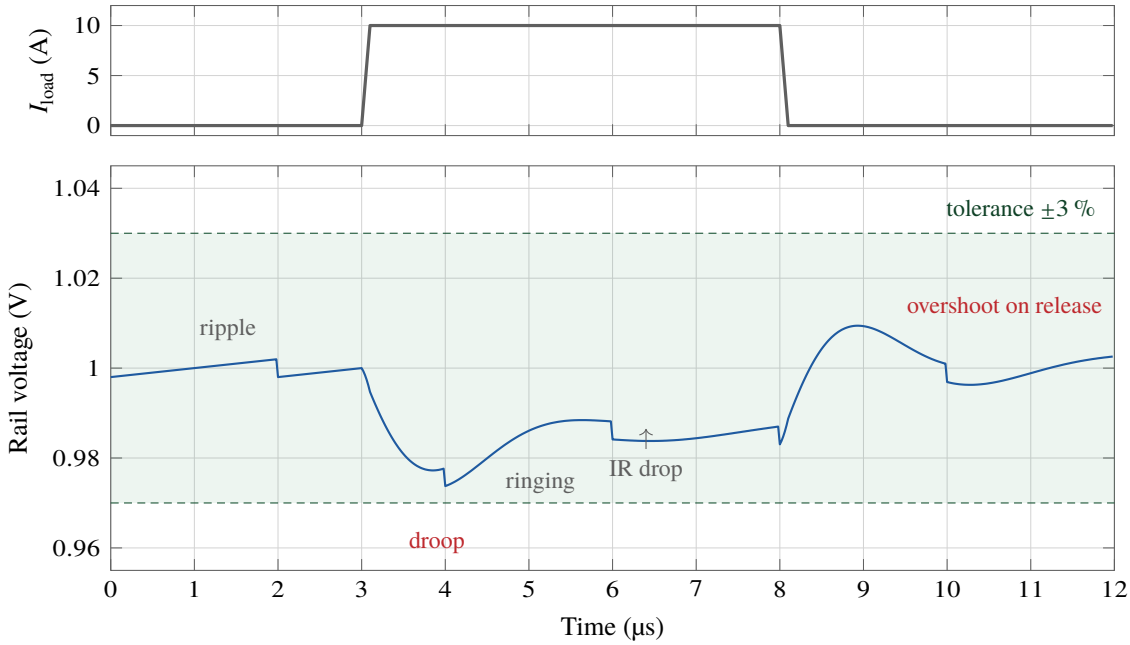


Figure 18.1. A 1.0 V rail responding to a 10 A load step applied at 3 µs and released at 8 µs. Ripple comes from the regulator; droop and ringing from the PDN's inductance and capacitance; the settled offset is the IR drop. Model: 1.5 mΩ and 1 nH feed, 200 µF at the load, 500 kHz, ±2 mV ripple; no regulator loop. [Simulation (lumped network)]

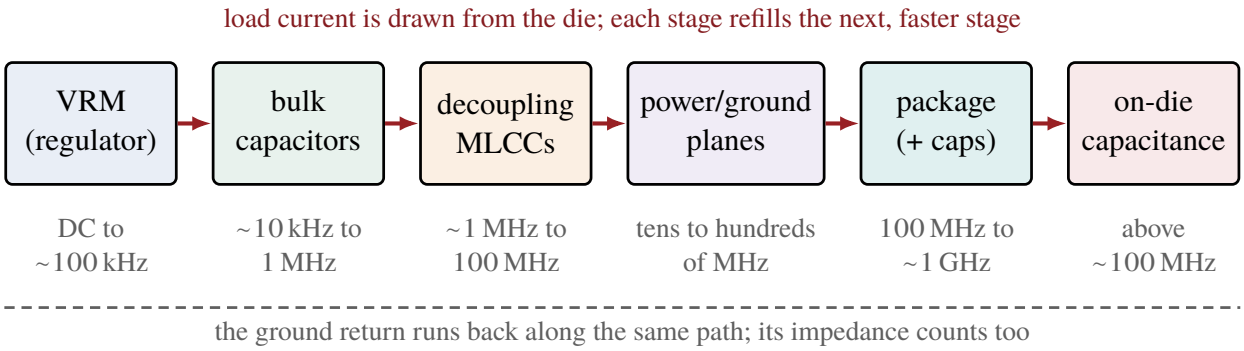


Figure 18.2. The power delivery network from regulator to die. The frequency bands are typical orders of magnitude, not limits; Figure 21.2 shows a computed example. [Conceptual]

18.4 Calculating voltage droop

Two equations describe most of the voltage disturbance on a rail. The resistive part,

$$\Delta V_R = I\,R, \tag{18.1}$$

is the IR drop, present as long as the current flows. During fast current transients, inductance also matters:

$$V_L = L\,\frac{\mathrm{d}I}{\mathrm{d}t}. \tag{18.2}$$

Worked Example 18.1: Inductive droop

A processor increases its current consumption by 10 A in 100 ns. The effective PDN inductance between the load and the nearest capacitance that could supply the current is 1 nH, and the path resistance is 1 mΩ. Estimate the disturbance.

Solution. Equation (18.2): $V_L = 1\,\text{nH} \times 10\,\text{A}/100\,\text{ns} = 100\,\text{mV}$ during the ramp, plus an IR drop of 10 mV that remains afterward (Figure 18.3, solid curve). On a 1.0 V rail with a ±3 % tolerance, 110 mV is almost four times the allowed deviation. The inductive term alone is a simplified estimate: it assumes that all the current must flow through the inductance, with no capacitance at the load.

Capacitance at the load changes the picture, because during the transient the capacitor supplies the current and the inductive feed refills it more slowly. The inductance and capacitance then form a resonant circuit with characteristic impedance

$$Z_c = \sqrt{L/C}, \tag{18.3}$$

and for a fast step the droop is roughly $\Delta I \times Z_c$. With 22 µF at the load, $Z_c = 6.7\,\text{m}\Omega$ and the droop falls to about 62 mV (Figure 18.3). With only 1 µF, $Z_c = 32\,\text{m}\Omega$: the circuit resonates at $1/(2\pi\sqrt{LC}) = 5\,\text{MHz}$, the 100 ns ramp excites it, and the droop *grows* to almost 200 mV. Adding capacitance does not automatically improve power integrity; the capacitance must be large enough, and its connection inductance small enough, that $\sqrt{L/C}$ is below the impedance the rail can tolerate.

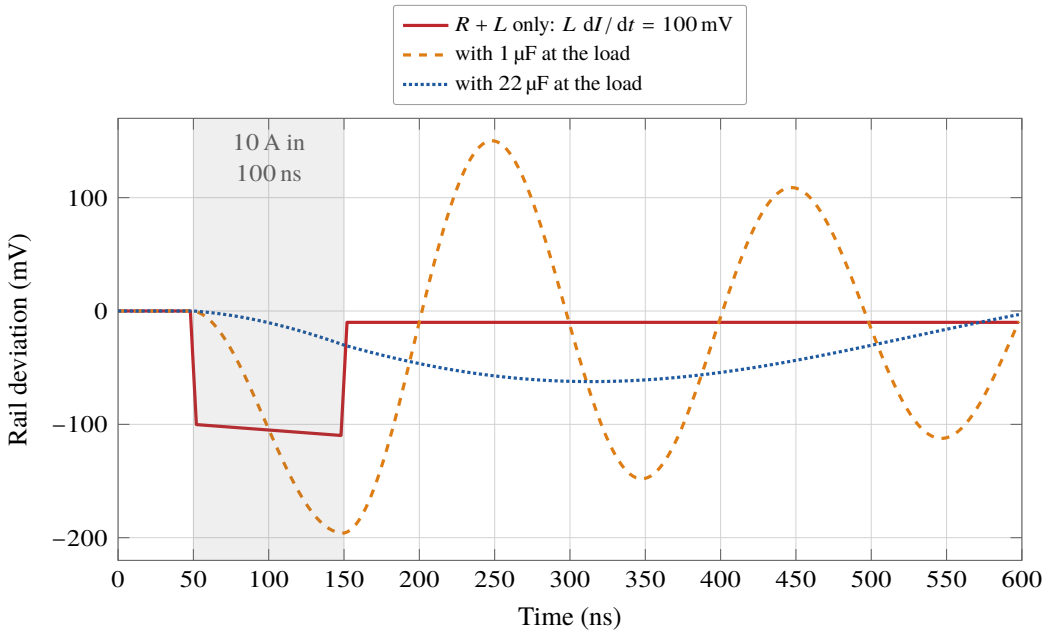


Figure 18.3. Rail deviation for a 10 A, 100 ns load ramp through 1 mΩ and 1 nH: without load capacitance, with 1 µF (ESR 2 mΩ) and with 22 µF (ESR 1 mΩ). The small capacitor forms a lightly damped resonance and makes the droop worse. [Simulation (lumped network)]

18.5 PDN impedance and target impedance

PDN impedance is the most important concept in power-integrity engineering. It describes how much the voltage at the load changes for a given change of current, at each frequency:

$$Z_{\mathrm{PDN}}(f) = \frac{\Delta V(f)}{\Delta I(f)}. \tag{18.4}$$

If the PDN impedance is low at every frequency the load current contains, the voltage cannot move much, whatever the current does. This leads to the *target impedance*, the largest PDN impedance that keeps the worst-case current change within the allowed voltage deviation:

$$Z_{\mathrm{target}} = \frac{\Delta V_{\mathrm{max}}}{\Delta I}. \tag{18.5}$$

Worked Example 18.2: A target impedance

A processor allows a maximum voltage deviation of 50 mV and experiences a transient current change of 10 A. Find the target impedance.

Solution. $Z_{\mathrm{target}} = 50\,\mathrm{mV}/10\,\mathrm{A} = 5\,\mathrm{m\Omega}$. The PDN, seen from the load, should ideally stay below 5 mΩ over the frequency range the load current contains. Two refinements follow in Chapter 21: the 50 mV must first be shared with the regulator's set-point error and the DC IR drop, so the dynamic share is smaller (Table 21.1); and meeting the impedance target in the frequency domain should be confirmed with a load-step simulation (Section 21.6).

Figure 18.4 compares two PDN designs against this target. Design A uses several staggered capacitor banks whose impedances overlap, so the total stays below 5 mΩ across the range the load excites. Design B uses fewer values with very low ESR; between them an anti-resonance peak rises to about 50 mΩ, ten times the target, near 20 MHz. A load current with energy at that frequency will produce ten times the allowed noise even though Design B has plenty of capacitance.

Rule of Thumb: Which frequencies matter

The load current of a step with rise time t_{r} has significant content up to about $0.35/t_{\mathrm{r}}$; a 100 ns step reaches about 3.5 MHz, a 1 ns step about 350 MHz. Above the frequency at which the package and on-die capacitance take over, the board's impedance is no longer visible to the die. *Limits:* periodic load patterns concentrate energy at specific frequencies and must be checked against resonances individually.

18.6 How to improve power integrity

Table 18.1 summarizes the main techniques and their limits.

Common Pitfall: More capacitors is always better

Adding capacitors without checking the impedance curve. Poor selection and placement can create anti-resonance peaks (Figure 18.4) or lightly damped resonances (Figure 18.3) that make the rail worse at exactly the frequencies where the load draws current. Every change to the decoupling should be checked with a PDN simulation.

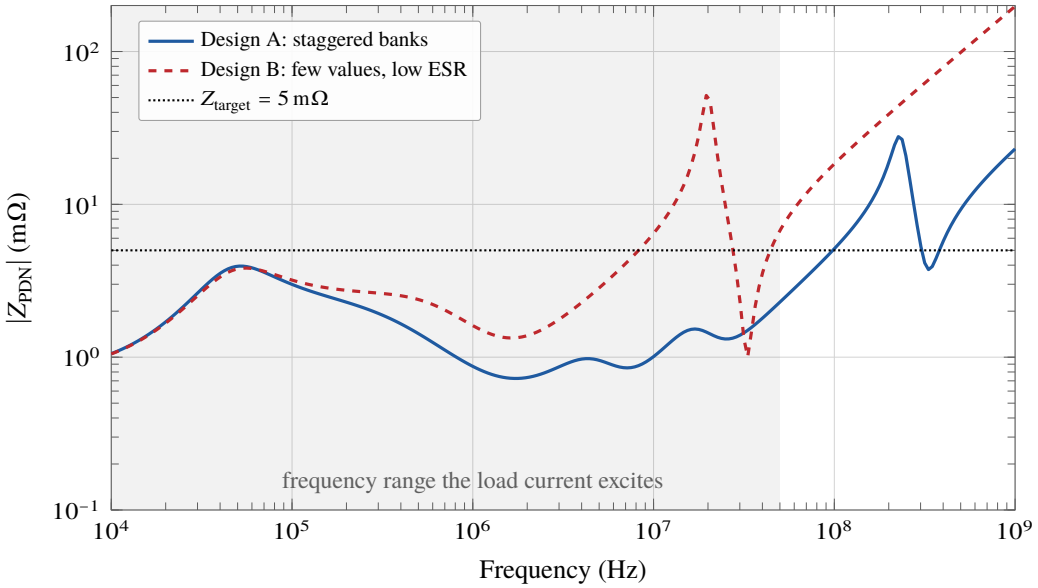


Figure 18.4. PDN impedance of two designs against a 5 mΩ target. Design A meets it over the shaded range; Design B has an anti-resonance peak near 20 MHz. Lumped models of VRM, bulk, MLCC banks and planes. [Analytical]

Table 18.1. Techniques that improve power integrity.

Technique	Purpose	Watch out for
Multiple decoupling capacitors	Lower impedance across frequency	Anti-resonances between values
Proper capacitor placement	Minimize mounting inductance	Shared vias limit the gain
Low-ESR capacitors	Lower resistive impedance	Very low ESR sharpens peaks
Low-ESL capacitors	Better high-frequency response	Mounting dominates ESL
Solid power and ground planes	Low spreading impedance, plane capacitance	Splits and slots under signals
Multiple vias	Lower connection inductance and resistance	Via fields perforate planes
Proper VRM selection	Fast transient response, low ripple	Loop bandwidth versus stability
PDN simulation	Find impedance peaks and voltage drops	Models need biased capacitance and real ESL

Design Rule: Decouple from the die outward

Start from the device's requirement at its pins, place the highest-frequency capacitors closest to the power balls with the shortest via connection, then add larger values farther out, and confirm with an impedance sweep and a load-step simulation that the target is met and no peak is exposed (Chapter 21).

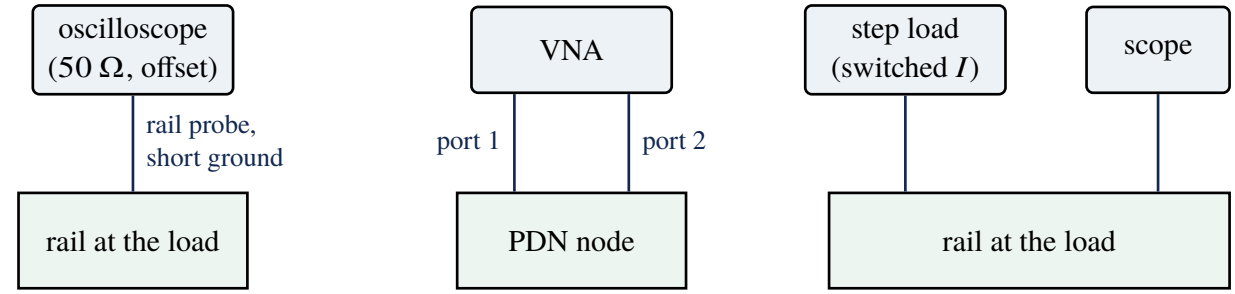
18.7 How to measure power integrity

Three measurements cover most needs (Figure 18.5); Chapter 37 gives the details.

1. *Oscilloscope.* Measures voltage ripple, transient droop, overshoot and ground bounce in the time domain. A dedicated rail probe with DC offset, a 50 Ω input and a very short ground connection is

needed: a standard 10 × probe with a long ground lead adds ringing and pickup larger than the millivolt signals being measured.

2. *Vector network analyzer.* Measures PDN impedance over frequency. For milliohm impedances the two-port shunt-through method is used (Section 37.3).
3. *Injected disturbance.* A controlled load step, or a small AC current injected by a signal generator and current driver, is applied to the rail while the voltage and current responses are recorded; the ratio gives the impedance, and the time response shows droop and settling directly.



(a) time domain: ripple, droop, overshoot

(b) frequency domain: $|Z(f)|$ shunt-through

(c) injected current: $\Delta V / \Delta I$ and transient

Figure 18.5. Three ways to measure power integrity: (a) oscilloscope with a rail probe, (b) VNA shunt-through impedance, (c) injected load step with voltage and current measured together. Probe grounds, cable shields and isolation are omitted. [Conceptual]

Worked Example 18.3: Regulator ripple

A buck regulator switching at 500 kHz has an inductor ripple current of 3 A peak to peak and 400 µF of output capacitance with a total ESR of 2 mΩ. Estimate the output ripple.

Solution. A common estimate for a buck converter adds the ESR and capacitive terms, $\Delta V \approx \Delta I_L \left(\mathrm{ESR} + \frac{1}{8 f_{sw} C}\right) = 3 \times (0.002 + 0.000625) = 7.9\,\mathrm{mV}$ peak to peak. ESL adds spikes at the switching edges that this estimate omits; they are often the larger part of the measured ripple and are seen only with a short probe ground and adequate bandwidth.

18.8 Signal integrity versus power integrity

The two disciplines use parallel concepts (Table 18.2), and the distinction matters particularly when designing carrier boards and high-speed embedded systems.

Poor power integrity also causes signal-integrity problems (Figure 18.6). Supply noise on a transceiver's PLL and VCO supply modulates the clock phase, increasing jitter and closing the PCIe eye; noise on output-driver supplies changes their delay (power-supply-induced jitter, Chapter 20); and ground bounce shifts the reference against which signals are compared. In the other direction, signal return currents that cross plane splits or change reference planes inject noise into the PDN (Chapter 23).

Key Takeaways

- Power integrity is the ability of the PDN to keep every supply inside its tolerance, at the point the device specifies, under all operating conditions.

Table 18.2. Signal integrity and power integrity side by side.

Signal integrity (SI)	Power integrity (PI)
Focuses on signal quality	Focuses on power quality
Controlled impedance	Target impedance
Reflections	Voltage droop
Crosstalk	Ground bounce
Eye diagrams	PDN impedance plots
Jitter	Power-supply noise
Transmission lines	Power delivery network

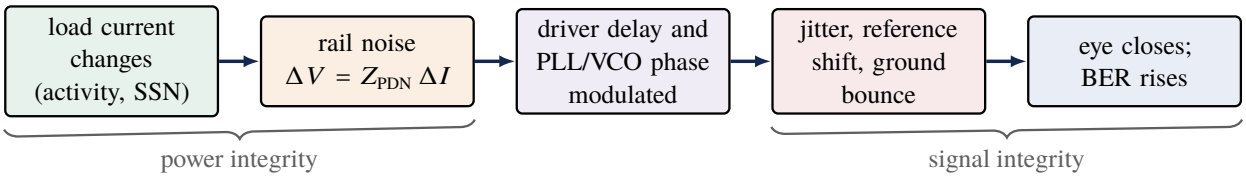


Figure 18.6. How power-integrity noise becomes a signal-integrity problem. [Conceptual]

Table 18.3. Summary: the power-integrity view of a design.

Question	Answered by	Chapter
How much can the rail move?	Tolerance and its allocation	21
How large is the current step?	Device data, activity patterns	21
What impedance is allowed?	$Z_{\mathrm{target}} = \Delta V_{\mathrm{max}}/\Delta I$	21
Are there resonances?	Impedance sweep of the whole network	21
How much DC drop?	Plane and via resistance, IR-drop simulation	21
How noisy are I/O supplies and grounds?	SSN, package inductance	20
Do return paths disturb the PDN?	Plane continuity, stitching	23

- Droop, ripple, ground bounce, SSN, PDN resonance and IR drop are the common problems; they add on the same rail.
- The PDN runs from the VRM through bulk and decoupling capacitors, planes and package to the die; each stage serves a band of frequencies.
- Droop has a resistive part IR and an inductive part $L\ \mathrm{d}I/\mathrm{d}t$; local capacitance helps only if $\sqrt{L/C}$ is small enough.
- Keep $|Z_{\mathrm{PDN}}|$ below $Z_{\mathrm{target}} = \Delta V_{\mathrm{max}}/\Delta I$ over the frequencies the load excites, and watch for anti-resonances.
- Measure rails with a rail probe, a VNA shunt-through setup or an injected load step; poor PI shows up as SI problems, especially jitter.

Further Reading

Smith and Bogatin [7] develop the target-impedance method in depth; Swaminathan and Engin [56] cover PDN modeling and power–signal interaction; Novak and Miller [6] cover frequency-domain PDN characterization.

Review Problems

P18.1 Define power integrity in one sentence, and state where the supply voltage must be judged.

P18.2 A load current rises by 5 A in 50 ns through 0.5 nH and 2 mΩ, with no local capacitance. Estimate the inductive and resistive drops.

P18.3 Find the target impedance for a 0.9 V rail with a ± 2 % tolerance, all of it assigned to dynamic noise, and a 6 A transient.

P18.4 In Worked Example 18.1, what local capacitance gives $\sqrt{L/C} = 5\,\mathrm{m\Omega}$? Estimate the resulting droop for a fast step.

P18.5 *(Approximation failure.)* A designer adds a single 1 µF capacitor behind 1 nH of mounting and feed inductance to "help" a 10 A, 100 ns load step. Why can this make the droop worse?

P18.6 Which measurement would you use to find (a) the switching ripple of a regulator, (b) a 20 MHz anti-resonance in the PDN, (c) the droop during a real workload?

P18.7 Explain how supply noise on a SerDes PLL can reduce a PCIe eye opening.

Chapter 19

Transient Current, Inductance, and the PDN Response

Chapter Overview

When a chip suddenly draws more current, the path from the voltage regulator to the chip cannot change its current instantly. Its inductance resists the change, local capacitors supply the initial shortfall, and the chip's supply voltage droops and may ring. How much depends on the frequency-dependent impedance of the complete power delivery network. This chapter follows one load step from start to finish: the circuit between regulator and chip, how current is shared between the regulator path and the capacitors, how path inductance sets the droop, how the time-domain picture becomes a PDN impedance, why resonances matter, and how real PDNs show several droops on different time scales. It closes with two points that are often misunderstood: the voltage across an inductance is the *derivative* of its current, and a load transient *excites* the PDN impedance rather than changing it.

Learning Objectives

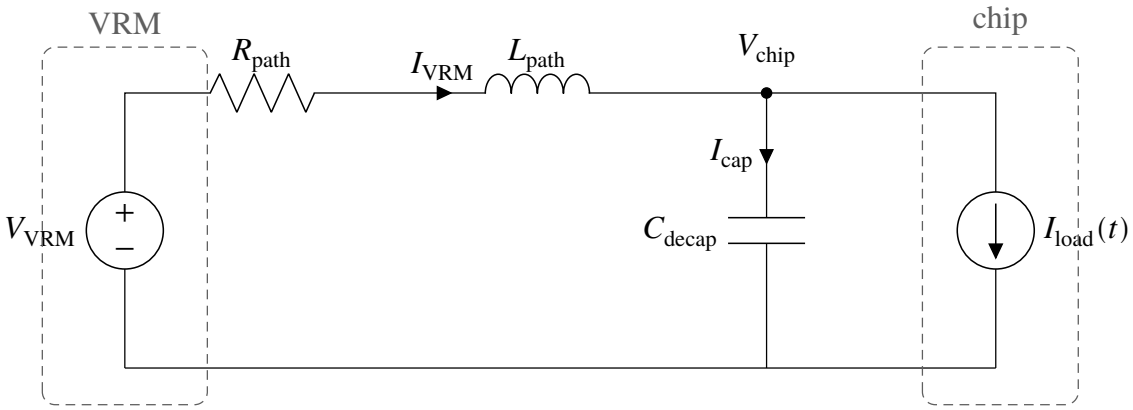
1. Write the voltage equation of the regulator-to-chip path and distinguish load current from regulator-path current.
2. Compute $L\ \mathrm{d}I/\mathrm{d}t$ for a load step and explain why it is not, by itself, the chip's droop.
3. Use charge balance to estimate capacitor droop, and the characteristic impedance, resonant frequency and damping of the path–capacitor pair.
4. Build a simple PDN impedance from branch impedances and read peaks against a target.
5. Recognize first, second and third droops and the PDN element behind each.
6. Predict the waveform of $L\ \mathrm{d}I/\mathrm{d}t$ for common current shapes, and use $\Delta V(f) = -Z(f)\,\Delta I(f)$ to reason about periodic loads.

19.1 The circuit between the regulator and the chip

The power delivery path includes the voltage regulator, board copper, vias, package connections, decoupling capacitors and the ground return path, all with parasitic resistance and inductance. The simplest useful model lumps them into a path resistance R_{path} and inductance L_{path}, with one local capacitor at the chip (Figure 19.1). The resistance and inductance belong to the whole *loop*—supply and return together (Chapter 3)—not to the power copper alone.

If I_{VRM} is the current in the path, the chip voltage is

$$V_{\text{chip}}(t) = V_{\text{VRM}}(t) - R_{\text{path}}\, I_{\text{VRM}}(t) - L_{\text{path}} \frac{\mathrm{d}I_{\text{VRM}}(t)}{\mathrm{d}t}, \tag{19.1}$$



return path (ground); its R and L are lumped into R_{path} and L_{path}

Figure 19.1. Simplified one-rail PDN: regulator, effective path resistance and inductance of the supply-and-return loop, a local capacitor, and the chip's current demand. [Schematic]

and Kirchhoff's current law at the chip node gives

$$I_{load}(t) = I_{VRM}(t) + I_{cap}(t), \qquad I_{cap} = -C\frac{dV_{chip}}{dt}. \tag{19.2}$$

The distinction between the two currents is the key to the whole chapter: I_{load}, the current the chip demands, is *not* necessarily the current that flows immediately through the path inductance. Local capacitors can supply most of a fast transient.

19.2 A load step: why fast current changes matter

Assume a processor increases its current from 2 A to 12 A in 100 ns, with the illustrative values of Table 19.1.

Table 19.1. Illustrative load-step parameters.

Parameter	Value	Unit
VRM output voltage	1.0	V
Initial chip current	2	A
Final chip current	12	A
Current rise time	100	ns
Path inductance	1	nH (and 5 nH for comparison)
Path resistance	2	mΩ
Local capacitance (ideal)	100	µF

Worked Example 19.1: The inductive voltage of a load step

Find the current change, the average slew rate and the voltage across the path inductance if the whole change flows through it.

Solution. $\Delta I = 12 - 2 = 10\,\text{A}$ and $\Delta I/\Delta t = 10\,\text{A}/100\,\text{ns} = 10^8\,\text{A/s}$, so

$$V_L = L\frac{dI}{dt} = 1\,\text{nH} \times 10^8\,\text{A/s} = 100\,\text{mV}.$$

Across 5 nH the same ramp would develop 500 mV. These are the voltages across the inductance *if* the full 10 A change passes through it in 100 ns. They are not automatically the chip's droop: the local capacitor changes the current that flows in the inductance.

19.3 Local capacitors and the current shortfall

The chip wants its new current immediately, but the path current cannot rise without a voltage across the path inductance, and that voltage appears only as the chip voltage falls. Local capacitors make up the difference (Figure 19.2):

1. the chip switches into a higher-current state;
2. local capacitors provide most of the initial extra current;
3. the capacitors discharge, lowering the chip's supply voltage;
4. the path current rises and may temporarily *overshoot* the new load current;
5. the regulator eventually carries the sustained load and recharges the capacitors.

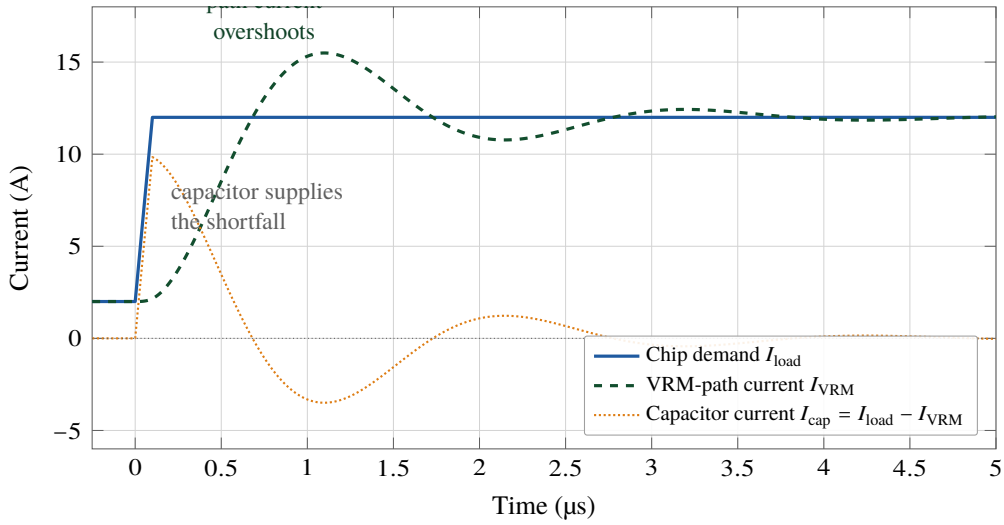


Figure 19.2. Calculated 2–12 A load-step response of the circuit of Figure 19.1 with the values of Table 19.1 and $L_{path} = 1$ nH. The path current lags the demand, overshoots to 15.5 A and rings; the capacitor current is the difference. [Simulation (lumped network)]

Integrating Eq. (19.2) gives the capacitor's voltage change in terms of the charge it has delivered:

$$\Delta V_C = \frac{1}{C}\int (I_{load} - I_{VRM})\ \mathrm{d}t. \tag{19.3}$$

The chip voltage keeps falling as long as the load current exceeds the path current, so *the minimum voltage occurs where the two currents cross*.

Worked Example 19.2: Charge-balance estimates of droop

(a) A 200 µF capacitor supplies a constant 10 A shortfall for 1 µs. Estimate its droop. (b) In Figure 19.2, the shortfall integrates to 3.68 µC before the currents cross at 0.685 µs. What droop does Eq. (19.3) predict, and how does it compare with the simulated waveform?

Solution. (a) $\Delta V_C = \Delta I\,\Delta t/C = 10\,\text{A} \times 1\,\mu\text{s}/200\,\mu\text{F} = 50\,\text{mV}$. The estimate ignores the capacitor's ESR and ESL and the growing path current, so it is an upper bound for a stiff capacitor and an underestimate for one with significant ESR. (b) $3.68\,\mu\text{C}/100\,\mu\text{F} = 36.8\,\text{mV}$, which is exactly the simulated drop from 996–959 mV (Figure 19.3): for an ideal capacitor, charge balance *is* the droop.

19.4 How path inductance sets the droop

Compare two otherwise identical circuits with 1 nH and 5 nH of path inductance (Figure 19.3 and Table 19.2). Before the step both sit at $1.0 - 2 \times 0.002 = 996\,\text{mV}$, and after it they settle at $1.0 - 12 \times 0.002 = 976\,\text{mV}$. With more inductance the path current reacts more slowly, the capacitor supports the load for longer and delivers more charge, and the voltage falls further before recovering.

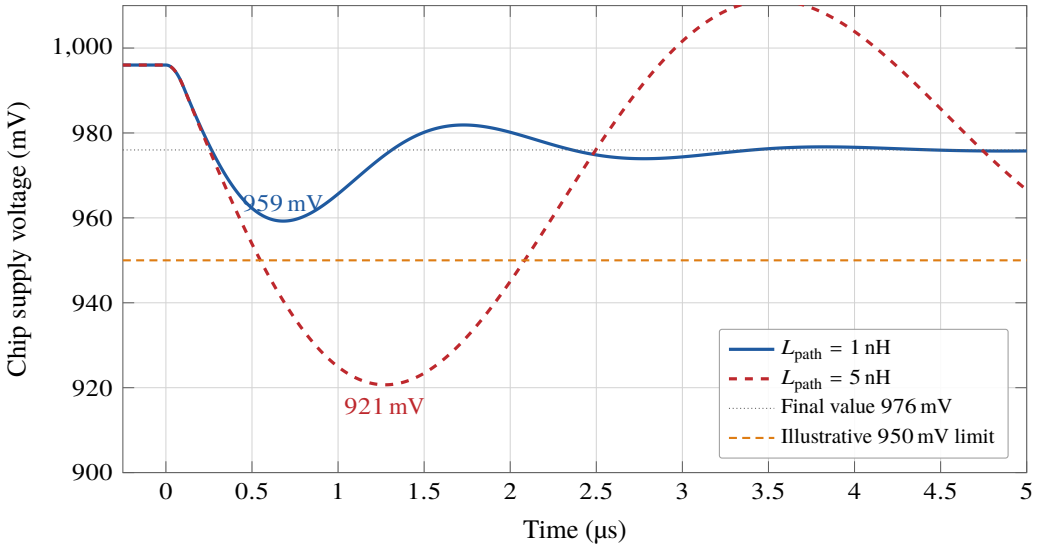


Figure 19.3. Chip voltage for the load step of Table 19.1 with 1 nH and 5 nH path inductance. The 950 mV line is an illustrative limit, not a device specification. Ideal source, ideal capacitor, no regulator loop. [Simulation (lumped network)]

Table 19.2. Results of the lumped simulation, rounded to the nearest millivolt.

Result	1 nH path	5 nH path
Initial chip voltage	996 mV	996 mV
Minimum voltage	959 mV	921 mV
Droop from initial voltage	37 mV	75 mV
Characteristic impedance $\sqrt{L/C}$	3.2 mΩ	7.1 mΩ
Resonant frequency	503 kHz	225 kHz
Damping ratio ζ	0.32	0.14
Qualitative response	smaller disturbance	larger droop, more ringing

The last three rows of Table 19.2 explain the difference. The path inductance and local capacitance form a series-resonant loop with

$$f_0 = \frac{1}{2\pi\sqrt{LC}}, \qquad Z_c = \sqrt{\frac{L}{C}}, \qquad \zeta = \frac{R}{2}\sqrt{\frac{C}{L}}, \tag{19.4}$$

where R is the total series resistance of the loop. For a step much faster than $1/f_0$, the dynamic part of the droop is of the order of $\Delta I\, Z_c$, reduced by damping; quintupling L raises Z_c by $\sqrt{5}$ and lowers ζ by the same factor, and the response rings longer. In a real system the regulator's control loop, the capacitor's ESR and ESL, the package inductance and on-die capacitance all change the result.

19.5 From the time domain to PDN impedance

The impedance of an ideal inductor is $Z_L = \mathrm{j}\omega L = \mathrm{j}2\pi f L$, so its magnitude grows in proportion to frequency (Table 19.3). High-frequency components of the load current therefore cannot be supplied through a distant regulator; they must come from capacitors close to the chip.

Table 19.3. Reactance of the path inductance alone (not the impedance of the full PDN).

Frequency	$\lvert Z_L\rvert$ at 1 nH	$\lvert Z_L\rvert$ at 5 nH
100 kHz	0.63 mΩ	3.14 mΩ
1 MHz	6.28 mΩ	31.4 mΩ
10 MHz	62.8 mΩ	314 mΩ

In a basic model the path and each capacitor are branches in parallel, as seen from the chip:

$$Z_{\mathrm{path}}(\omega) = Z_{\mathrm{VRM}}(\omega) + R_{\mathrm{path}} + \mathrm{j}\omega L_{\mathrm{path}}, \tag{19.5}$$

$$Z_{\mathrm{cap}}(\omega) = R_{\mathrm{ESR}} + \mathrm{j}\omega L_{\mathrm{ESL}} + \frac{1}{\mathrm{j}\omega C}, \tag{19.6}$$

$$Z_{\mathrm{PDN}}(\omega) \approx Z_{\mathrm{path}} \parallel Z_{\mathrm{cap},1} \parallel Z_{\mathrm{cap},2} \parallel \cdots \tag{19.7}$$

Real PDNs add many capacitors, plane modes, package parasitics and on-die behavior, but the structure is the same.

The regulator's own impedance. Z_{VRM} is not zero. Within its control-loop bandwidth (typically tens to a few hundred kilohertz for a multiphase buck converter) the regulator holds its output close to the set point, and its output impedance is small and nearly resistive—deliberately so when a *load line* (adaptive voltage positioning) is specified. Above the loop bandwidth the regulator can no longer respond and its output looks inductive; the bulk capacitors must take over there. Remote sensing compensates the DC drop of the path but not its inductance.

Figure 19.4 evaluates Eq. (19.7) for the path of Table 19.1, a 100 µF capacitor (ESR 2.5 mΩ, ESL 0.4 nH) and a 1 µF capacitor (ESR 3 mΩ, ESL 0.25 nH). Two features stand out. The 0.5 MHz resonance of the path with the large capacitor is so damped by the ESR that its peak, 2.8 mΩ, stays below the target. But between the inductive side (the path in parallel with the large capacitor's ESL, 0.29 nH) and the 1 µF capacitor there is an anti-resonance near 6.8 MHz, where the impedance rises to 35.5 mΩ—seven times the target.

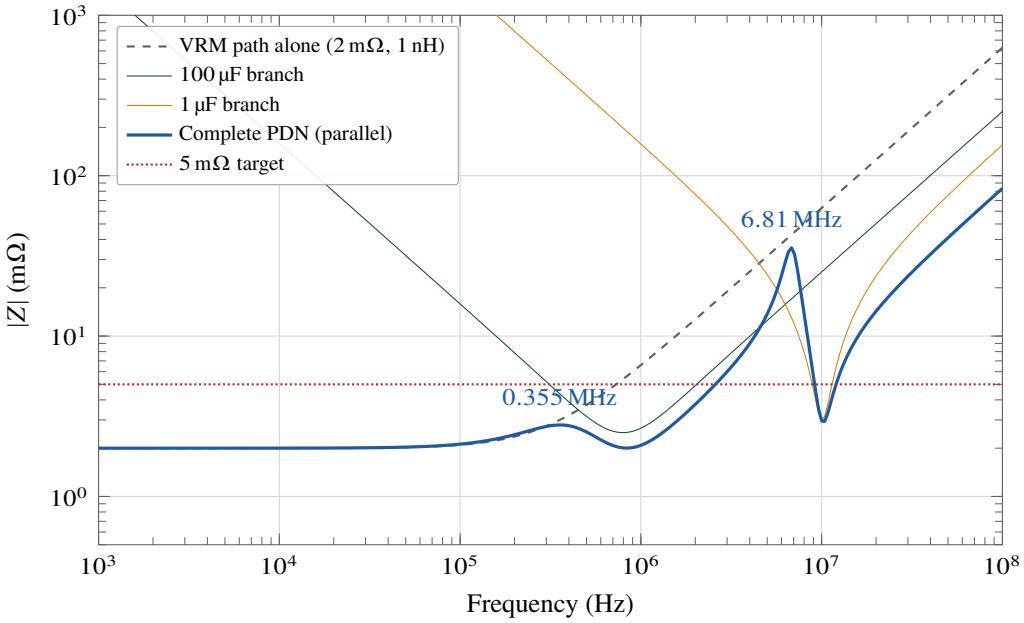


Figure 19.4. Impedance of a simple PDN: path (2 mΩ, 1 nH), a 100 µF capacitor (ESR 2.5 mΩ, ESL 0.4 nH) and a 1 µF capacitor (ESR 3 mΩ, ESL 0.25 nH), against a 5 mΩ target. Labels mark the local peaks. [Analytical]

Worked Example 19.3: Locating the anti-resonance

Estimate the frequency and height of the anti-resonance in Figure 19.4.

Solution. Above a few hundred kilohertz the path and the 100 µF branch are both inductive, so they combine as $1 \parallel 0.4 = 0.29$ nH with resistance $2 \parallel 2.5 = 1.1$ mΩ. Adding the small capacitor's own ESL, Eq. (21.6) gives $f = 1/(2\pi\sqrt{(0.29 + 0.25)\,\text{nH} \times 1\,\mu\text{F}}) = 6.9$ MHz. There the inductive reactance is $X = 2\pi f \times 0.29\,\text{nH} = 12.2\,\text{m}\Omega$, and Eq. (21.7) gives $|Z| \approx X^2/(1.1 + 3)\,\text{m}\Omega = 36\,\text{m}\Omega$. The exact sweep gives 6.81 MHz and 35.5 mΩ.

Quick estimate	Full model	Difference
Estimate: 6.9 MHz, 36 mΩ	Exact: 6.81 MHz, 35.5 mΩ	1 %, 1 %

19.6 Target impedance and the voltage budget

For a first-order estimate, the target impedance is

$$Z_{\text{target}} = \frac{\Delta V_{\text{allowed}}}{\Delta I_{\text{transient}}}, \tag{19.8}$$

so a rail that may deviate by 50 mV for a 10 A load change needs $Z_{\text{target}} = 5$ mΩ (Chapter 18). The design goal is low PDN impedance over the frequencies present in the chip's current, which is not the same as low DC resistance.

Common Pitfall: Target impedance as a guarantee

Treating "$|Z| < Z_{\text{target}}$ everywhere" as proof that every load step meets the voltage specification. It is a first-pass guideline. The voltage budget must also cover DC IR drop, regulator tolerance and ripple

(Table 21.1), and the frequency-domain result must be confirmed with a time-domain load step that uses the real current profile (Section 21.6).

19.7 Resonance, ringing and anti-resonance

Inductance and capacitance exchange energy. For a simple *LC* loop the undamped natural frequency is that of Eq. (19.4); for 1 nH and 100 µF, $f_0 \approx 503$ kHz, which is why the voltage in Figure 19.3 oscillates with a period of about 2 µs instead of returning smoothly to its final value. With several capacitor values and their parasitic inductances, *anti-resonances* appear as well: narrow frequency ranges where the PDN impedance peaks (Figure 19.4). Damping—ESR chosen near the target, controlled-ESR parts, or lossy plane structures—and careful selection and placement of capacitors reduce the peaks (Section 21.4).

19.8 First, second and third droops

A real PDN is a ladder of stages (Figure 18.2), and each inductance–capacitance pair along it produces its own droop on its own time scale. Seen from the die after a fast load step, they appear in sequence (Figure 19.5):

First droop
: nanoseconds after the step: the on-die capacitance discharges while the package inductance limits the current from the board. It is usually the deepest and fastest, with a resonance from tens to a few hundred megahertz, and only on-die and package capacitance can address it.

Second droop
: hundreds of nanoseconds to about a microsecond: the board's ceramic capacitors discharge while board spreading and via inductance limits the current from the bulk capacitors; resonance around 1 MHz.

Third droop
: microseconds to tens of microseconds: the bulk capacitors discharge until the regulator's control loop responds.

After the third droop the voltage settles at the DC level set by the IR drop and the regulator's load line.

Rule of Thumb: Who fixes which droop

Board capacitors cannot fix the first droop: the package inductance stands between them and the die. The board designer controls the second droop (MLCC count, mounting inductance, plane spreading) and shares the third with the regulator design (bulk capacitance, loop bandwidth). *Limits:* the boundaries between the droops shift with the design; identify them from the impedance seen at the die.

19.9 The inductive voltage is a derivative

A useful first intuition is that "high-frequency transient current creates voltage on parasitic inductance, which then disturbs the power rail." Two details make it precise. First, V_L need not look like a ramp: it is proportional to the *derivative* of the current, so its shape follows the current's rise, fall and curvature. Second, a load transient does not ordinarily change a linear PDN's impedance; it excites the existing impedance (Section 19.10).

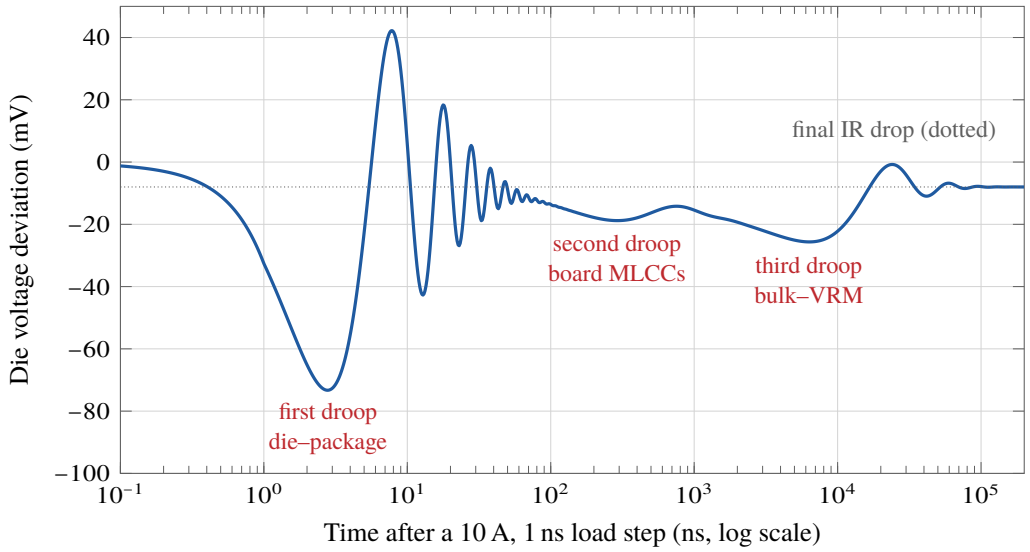


Figure 19.5. Die-voltage response of a three-stage ladder to a 10 A, 1 ns load step (log time axis): die 200 nF with package 12.7 pH; board MLCCs 100 µF behind 253 pH; bulk 2 mF behind an inductive regulator model (14 nH, 0.5 mΩ). The droops are 73 mV at 2.8 ns, 19 mV at 0.29 µs and 26 mV at 6.3 µs. [Simulation (lumped ladder)]

Table 19.4. Current waveform and ideal inductor voltage $V_L = L\ \mathrm{d}I/\mathrm{d}t$.

Current waveform	Inductor voltage
Constant current	Zero
Linear ramp	Constant during the ramp
Pulse with finite linear edges	Positive and negative rectangular pulses at the edges
Pulse with smooth edges	Bell-shaped pulses, higher peak for the same edge duration
Sinusoid	Sinusoid, leading the current by 90°

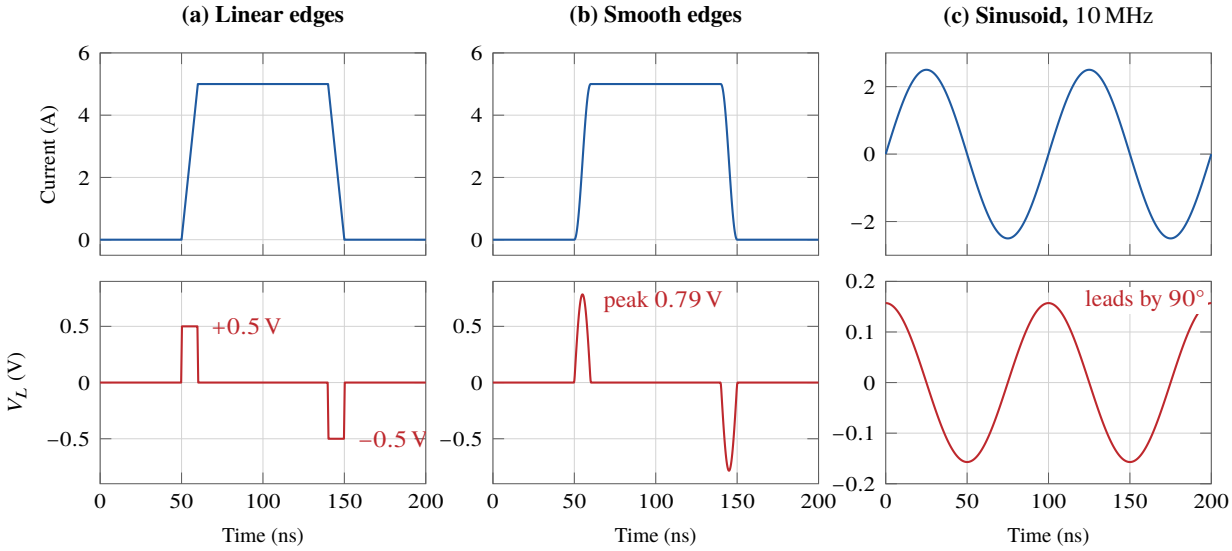


Figure 19.6. Current (top) and voltage across 1 nH (bottom). (a) A 0–5 A pulse with 10 ns linear edges gives ±0.5 V rectangles. (b) The same pulse with raised-cosine edges of the same full duration gives bell-shaped pulses peaking at 0.79 V. (c) A 2.5 A, 10 MHz sinusoid gives 0.16 V, leading by 90°. [Analytical]

Worked Example 19.4: Inductive voltage of a current pulse

A 0–5 A current pulse with 10 ns linear edges flows through 1 nH. Find the inductive voltage during each part of the pulse, and repeat for raised-cosine edges of the same full duration.

Solution. Rising edge: $V_L = 1\,\text{nH} \times 5\,\text{A}/10\,\text{ns} = +0.5\,\text{V}$; flat top: zero; falling edge: $-0.5\,\text{V}$ (Figure 19.6a). A linear ramp therefore produces a *constant* voltage during the ramp, not a voltage ramp. For a raised-cosine edge the peak slope is $\pi/2$ times the average, so the peak is $\frac{\pi}{2} \times 0.5\,\text{V} = 0.79\,\text{V}$ (Eq. (20.2) uses the same fact). These are voltages across the inductance, not necessarily the rail droop.

Pulses contain many frequencies. A transient can be written as a sum of sinusoids. For one component,

$$I(t) = I_p \sin(2\pi f t) \quad \Longrightarrow \quad V_L(t) = 2\pi f L\, I_p \cos(2\pi f t), \tag{19.9}$$

so for the same amplitude the inductive voltage grows with frequency and inductance, and a faster edge, which carries more high-frequency content, produces more of it. One nuance: raising a pulse train's repetition rate does not by itself raise the inductive peak at each edge, if each pulse keeps the same amplitude and edge time. Edge rate, spectral content and the circuit's response all matter.

When the voltage subtracts or adds. Without local capacitance, Eq. (19.1) with $I_{\text{VRM}} = I_{\text{load}} = I$ reads

$$V_{\text{chip}}(t) = V_{\text{VRM}}(t) - R\,I(t) - L\frac{\mathrm{d}I(t)}{\mathrm{d}t}. \tag{19.10}$$

With the current defined as flowing from the regulator to the chip, a rising current makes the inductive term subtract from the supply (droop); a falling current reverses its sign and produces overshoot, as on load release in Figure 18.1. In a real network the local capacitors and the several inductances share the transient and change the exact waveform.

19.10 Impedance is the response function

For an approximately linear PDN, define ΔI_{load} as positive for an increase in chip current. Then, in the frequency domain,

$$\Delta V_{\text{chip}}(f) = -Z_{\text{PDN}}(f)\,\Delta I_{\text{load}}(f). \tag{19.11}$$

The impedance is the network's *response function*; the transient current is the *excitation*. If the current spectrum has a strong component near a high-impedance peak, such as an anti-resonance, the voltage disturbance is large. The transient does not create a new inductance or change a linear network's impedance—it reveals it. (Nonlinear and operating-point-dependent effects, such as the regulator's slew limits or capacitance that varies with bias, are separate considerations.)

Figure 19.7 demonstrates this with the PDN of Figure 19.4. The same 0–10 A square-wave load is applied at two repetition rates. At 6.8 MHz its fundamental sits on the anti-resonance and the rail swings 433 mV peak to peak. At 3.4 MHz a 50 % square wave has no even harmonics, so no component lands on the 6.8 MHz peak, and the swing is 220 mV—half as much, from the same current amplitude.

19.11 Practical ways to improve the response

Table 19.5 maps each problem to its engineering approach.

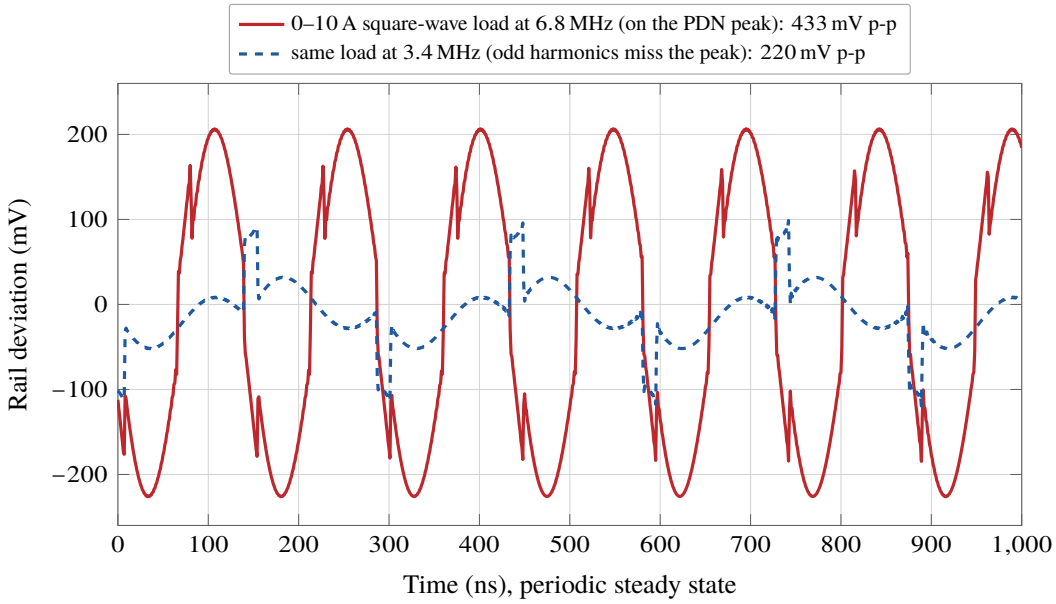


Figure 19.7. Periodic steady-state rail deviation for a 0–10 A, 50 % square-wave load with 15 ns edges applied to the PDN of Figure 19.4, at 6.8 MHz and 3.4 MHz, from the Fourier series and Eq. (19.11). [Analytical]

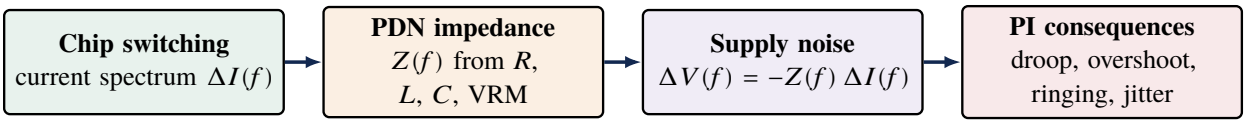


Figure 19.8. The end-to-end relationship: the chip's current spectrum excites the PDN impedance, producing the supply disturbance and its consequences. [Conceptual]

Table 19.5. Problems and engineering approaches.

Problem	Engineering approach
High regulator-to-chip inductance	Closely coupled power/ground planes; short, wide current paths; via arrays with power and ground vias interleaved
Large transient current	Enough effective local decoupling capacitance (biased value)
Very fast current edges	Minimize capacitor connection and return-loop inductance; rely on package and die capacitance for the first droop
Excessive voltage droop	More local energy storage; faster regulator transient response
PDN impedance peaks	Optimize capacitor ESR, ESL, value, placement and damping
Excessive DC IR drop	Reduce copper resistance; use remote sensing where appropriate
Ringing	Control LC resonances and provide damping

Design Rule: Minimize loop inductance, not path length

Inductance belongs to the complete supply-and-return loop. Place power and ground vias in adjacent pairs so their opposing currents cancel flux, keep power and ground planes on thin dielectric next to each other (spreading inductance is about 32 pH per square per mil of separation, Chapter 3), and mount each capacitor with its vias at the pads. Reducing loop area can matter more than shortening the power path alone.

The complete sequence. A load step therefore unfolds in five stages: the chip current rises; inductance limits how fast the path current can follow; capacitors supply the difference and the voltage droops; the regulator catches up and restores the voltage; and damping and control dynamics decide any overshoot or ringing. High transient current alone does not determine the droop: the amplitude and speed of the current change, together with the full frequency-dependent PDN impedance, do. At very high frequencies a local capacitor's own ESL, its connection geometry and the package inductance become the limitation—the subject of Chapter 21.

Table 19.6. Equation reference for this chapter.

Relationship	Equation
Inductor voltage	$V_L = L\,\mathrm{d}I/\mathrm{d}t$
Node-current balance	$I_{\mathrm{VRM}} - I_{\mathrm{load}} = C\,\mathrm{d}V_{\mathrm{chip}}/\mathrm{d}t$
Capacitor droop	$\Delta V_C = \frac{1}{C}\int (I_{\mathrm{load}} - I_{\mathrm{VRM}})\,\mathrm{d}t$
Inductive reactance	$\lvert Z_L\rvert = 2\pi f L$
Loop resonance, impedance, damping	$f_0 = 1/(2\pi\sqrt{LC})$, $Z_c = \sqrt{L/C}$, $\zeta = \frac{R}{2}\sqrt{C/L}$
PDN voltage disturbance	$\Delta V_{\mathrm{chip}}(f) = -Z_{\mathrm{PDN}}(f)\,\Delta I_{\mathrm{load}}(f)$
First-pass target impedance	$Z_{\mathrm{target}} = \Delta V_{\mathrm{allowed}}/\Delta I_{\mathrm{transient}}$

Key Takeaways

- The chip's demand I_{load} and the regulator-path current I_{VRM} differ during a transient; local capacitors supply the difference.
- $L\,\mathrm{d}I/\mathrm{d}t$ is the voltage across the inductance if all the current flows through it, not the chip's droop; the minimum voltage occurs where I_{VRM} catches up with I_{load}, and charge balance gives its depth.
- More path inductance raises $\sqrt{L/C}$, lowers damping, and deepens and lengthens the droop.
- The PDN impedance is the parallel combination of the path and capacitor branches; anti-resonances between them can exceed the target many times.
- Real PDNs show first, second and third droops from die/package, board and bulk/VRM.
- V_L follows the derivative of the current, and a transient excites the PDN impedance rather than changing it: $\Delta V(f) = -Z(f)\,\Delta I(f)$.

Further Reading

Smith and Bogatin [7] develop the load-step and impedance views in detail, including first, second and third droops; Swaminathan and Engin [56] treat PDN modeling across die, package and board.

Review Problems

P19.1 A load current rises by 8 A in 40 ns. What voltage develops across 0.5 nH and across 2 nH if all of it flows there?

P19.2 A 470 µF capacitor bank supplies a 6 A shortfall for 2 µs. Estimate its droop with Eq. (19.3).

P19.3 For 2 nH of path inductance, 220 µF of local capacitance and 3 mΩ of loop resistance, find f_0, Z_c and ζ.

P19.4 Sketch V_L across 1 nH for a current that rises linearly from 0 to 4 A in 20 ns, stays for 50 ns, and falls in 10 ns.

P19.5 *(Approximation failure.)* A colleague computes $L\,\mathrm{d}I/\mathrm{d}t = 500\,\mathrm{mV}$ for the 5 nH case of Table 19.1 and concludes the rail droops 500 mV. What is wrong, and what is the simulated droop?

P19.6 *(Inverse design.)* With the 100 µF local capacitor of Table 19.1, what path inductance gives $Z_c = 2\,\mathrm{m\Omega}$?

P19.7 A processor's workload toggles a 6 A current at 2.3 MHz with a 50 % duty cycle. Which harmonic of this load falls near the 6.8 MHz peak of Figure 19.4, and why does it matter?

P19.8 *(Imperfect data.)* A scope capture of a rail shows a 60 mV dip lasting 3 ns after a load step. Which droop is this likely to be, and which board change would help most?

Chapter 20

Ground Bounce and Simultaneous Switching Noise

Chapter Overview

When many outputs of a chip switch at the same time, their combined current flows through the shared inductance of the package's power and ground connections. The resulting voltage, $L\,\mathrm{d}i/\mathrm{d}t$, appears between the die's internal ground (or power) and the board's, and it disturbs every other circuit on the die that references the same rail. This chapter develops the equivalent circuit of ground bounce and simultaneous switching noise (SSN), shows how mutual inductance limits the benefit of adding ground pins, and surveys the package, board, and signaling techniques used to control it. It is the bridge between signal integrity and the power-integrity chapters that follow.

Learning Objectives

1. Derive the ground-bounce voltage from the shared return inductance and the switching current.
2. Estimate switching current and its rate of change for drivers loaded by transmission lines.
3. Compute the effective inductance of parallel ground pins, including mutual inductance.
4. Explain how SSN converts into timing jitter and false switching.
5. Select mitigations: pin assignment, package choice, slew control, on-die decoupling, differential signaling, and data-bus inversion.

20.1 The mechanism

Consider N output buffers on one die driving loads high-to-low at the same time (Figure 20.1). Each pulls current from its load through the die's ground network, which connects to the board through package pins or balls with total effective inductance L_{gnd}. The total current changes at a rate $N\,\mathrm{d}i/\mathrm{d}t$, so the die ground moves relative to the board ground by

$$V_{\mathrm{gb}} = L_{\mathrm{gnd}}\,N\frac{\mathrm{d}i}{\mathrm{d}t}. \tag{20.1}$$

A quiet driver on the same die, holding its output low, connects its output to the die ground through its on-resistance; its output therefore follows the die ground and launches a glitch of roughly V_{gb} onto its line. Inputs referencing the die ground see the same shift as a change in threshold. The identical mechanism on the power side—current drawn through the package's power inductance when outputs switch low-to-high—is called *power bounce*, and the two together are simultaneous switching noise [3], [56].

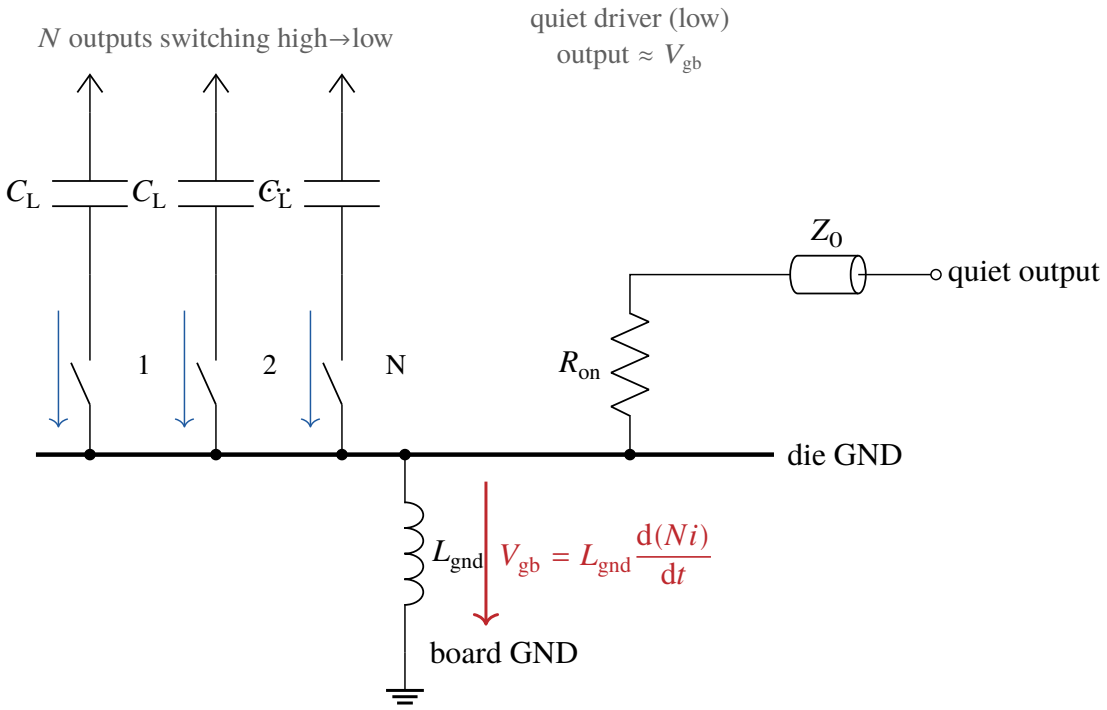


Figure 20.1. Equivalent circuit of ground bounce. N switching outputs discharge their loads through the shared package ground inductance L_{gnd}. A quiet driver held low follows the die ground and transmits the bounce onto its own line.

20.2 Estimating the switching current

The peak $\mathrm{d}i/\mathrm{d}t$ depends on the shape and on the definition of the edge, so both must be stated. Let T_e be the *full* (0–100 %) duration of the current edge and t_r its 10–90 % time.

Line-driven outputs. A driver of output resistance R_S launching into a line of impedance Z_0 draws a current *step* $\Delta I = \Delta V/(R_S + Z_0)$ whose shape follows the voltage edge. A linear ramp has $\mathrm{d}i/\mathrm{d}t = \Delta I/T_e = 0.8\,\Delta I/t_r$. A raised-cosine edge, $i = \frac{1}{2}\Delta I[1 - \cos(\pi t/T_e)]$, has peak slope $\frac{\pi}{2}\Delta I/T_e$; since its 10–90 % time is $0.590\,T_e$, that is $0.93\,\Delta I/t_r$. For a fixed load, the bounce scales as $N\Delta I/t_r$: halving the edge time doubles it.

Lumped capacitive loads. A driver charging a lumped capacitance C_L draws $i = C_L\,\mathrm{d}v/\mathrm{d}t$: a current *pulse* that rises and falls within the edge. For a raised-cosine voltage edge of full duration T_e, $i = C_L\Delta V\,\frac{\pi}{2T_e}\sin(\pi t/T_e)$, whose peak derivative is

$$\left.\frac{\mathrm{d}i}{\mathrm{d}t}\right|_{\max} = \frac{\pi^2}{2}\,\frac{C_L\,\Delta V}{T_e^2}, \tag{20.2}$$

so the bounce scales as $1/T_e^2$: halving the edge time quadruples it. (A strictly linear voltage ramp would give a rectangular current with infinite slope at its corners; real edges are smooth, which is why the raised cosine is used.)

Worked Example 20.1: Ground bounce of a 16-bit bus

Sixteen outputs switch low together, each driving a 50 Ω line with a 1.2 V swing through 40 Ω of driver resistance, with current edges of full duration $T_e = 200$ ps. The die returns through eight ground balls, each with 1 nH of partial inductance; ignore mutual coupling. Estimate V_{gb}.

Solution. $\Delta I = 1.2/90 = 13.3$ mA per output. $L_{gnd} = 1\,\text{nH}/8 = 0.125$ nH. With a linear ramp, $V_{gb} = 0.125 \times 10^{-9} \times 16 \times 0.0133/(0.2 \times 10^{-9}) = 133$ mV; with a raised-cosine edge of the same full duration the peak is $\pi/2$ larger, about 209 mV (Figure 20.2). That is 17 % of the swing, injected onto every quiet line in the bank. The two edges have different 10–90 % times (160 ps and 118 ps), which is why a bounce estimate must name the edge definition it uses.

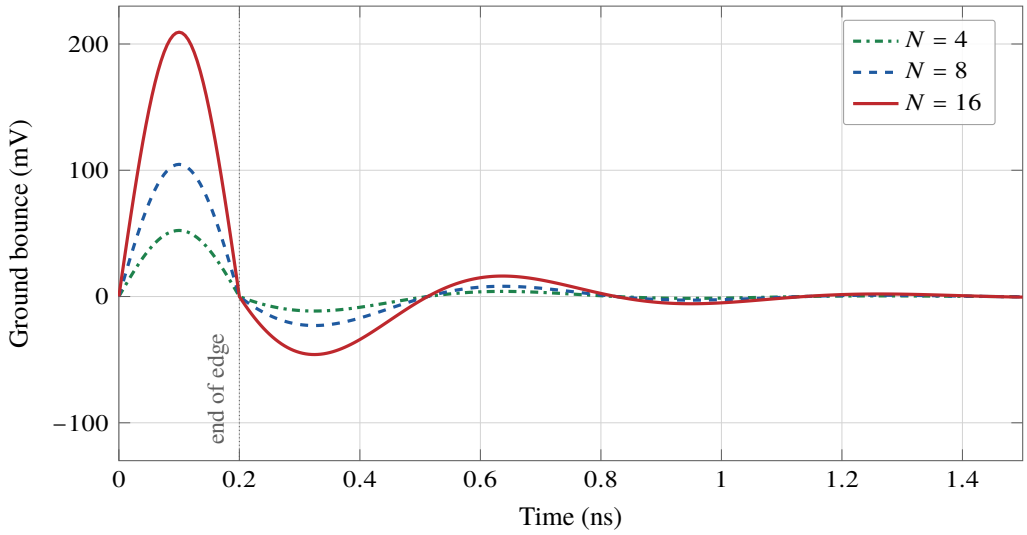


Figure 20.2. Ground bounce on a quiet output for N simultaneously switching outputs (Worked Example 20.1 parameters, raised-cosine current edges). The pulse scales with N and is followed by ringing of the package inductance with the on-die capacitance.

20.3 Parallel pins and mutual inductance

Adding ground pins in parallel reduces L_{gnd}, but not in simple proportion. If N_g ground pins each have partial self inductance L and every pair has partial mutual inductance M (currents in the same direction), the effective inductance is

$$L_{\text{eff}} = \frac{L + (N_g - 1)M}{N_g} \xrightarrow{N_g \to \infty} M. \tag{20.3}$$

Closely spaced ground pins carrying current in the same direction share flux and add little. Interleaving power and ground pins helps twice: the opposite currents in neighboring power and ground pins have *negative* mutual inductance, reducing the loop inductance of each power–ground pair (Eq. (3.5)).

Worked Example 20.2: Diminishing returns from extra ground pins

For the bus of Worked Example 20.1, how many independent ground pins would keep the linear-ramp bounce below 50 mV? Repeat if adjacent ground pins have $M = 0.3$ nH.

Solution. Required $L_{\text{eff}} \le 0.05 \times 0.2 \times 10^{-9}/(16 \times 0.0133) = 47\,\text{pH}$. Without mutual inductance, $N_g \ge 1/0.047 \approx 22$ pins. With $M = 0.3\,\text{nH}$, Eq. (20.3) cannot go below 300 pH however many pins are added (eight pins give 388 pH). The budget can only be met by spreading the pins, interleaving power and ground, slowing edges, or reducing N.

Common Pitfall: Clustering ground pins

Assigning all ground balls of a BGA to a contiguous block in the corner and the signals to another. The clustered ground balls are strongly mutually coupled and far from the signals they return, so both L_{eff} and the signal loop inductance are high. Distribute power and ground balls among the signal balls, ideally with a return ball adjacent to every few signals.

20.4 Effects on timing and logic

SSN harms a design in three ways:

1. **False switching.** A bounce exceeding the noise margin of a quiet output's receiver can be interpreted as a logic transition, especially on clocks and asynchronous resets.
2. **Push-out and jitter.** The switching outputs themselves are slowed, because the bounce reduces the effective drive voltage while they switch. The delay change depends on how many neighbors switch in the same direction, so it is pattern dependent: SSN appears as data-dependent jitter.
3. **Reference disturbance.** Single-ended receivers compare the input with a reference (V_{REF}) or with their own ground. A bouncing ground or reference shifts the threshold and closes the eye vertically.

20.4.1 From die bounce to the quiet line

The bounce is a voltage between die ground and board ground; what reaches a receiver depends on the path. A quiet driver held low connects its output to the die ground through its on-resistance R_{on}, and the line presents Z_0, so the wave launched onto the quiet line is

$$V_{\text{quiet}} \approx V_{\text{gb}} \frac{Z_0}{R_{\text{on}} + Z_0}, \tag{20.4}$$

which then propagates and reflects like any other wave: at an unterminated receiver it nearly doubles. For the bus of Worked Example 20.1 with $R_{\text{on}} = 40\,\Omega$, the 209 mV bounce launches 116 mV and can reach about 230 mV at an open receiver. Because Eq. (20.4) is linear, the quiet-line noise can be estimated from simulated or measured die-ground noise without repeating the full simultaneous-switching simulation.

20.4.2 Power-supply-induced jitter

Supply noise also moves edges. A driver's delay depends on its supply voltage; over a small range the dependence is linear, $\Delta t \approx k_{\text{PSIJ}}\,\Delta V$, with a sensitivity k_{PSIJ} (picoseconds per millivolt) obtained from a transistor-level or IBIS simulation at the operating corner. Supply noise at frequencies below the CDR or training bandwidth is tracked out; noise above it appears as jitter. The PDN target of Chapter 21 therefore has a timing as well as an amplitude consequence, and the jitter budget of Chapter 17 should contain a PSIJ entry derived from the rail noise spectrum, not a guessed constant.

20.5 Mitigation

Table 20.1 lists the main remedies. Most act on one of the three factors of Eq. (20.1): the inductance, the number of simultaneously switching outputs, or the current slope.

Table 20.1. Techniques for controlling simultaneous switching noise.

Technique	Mechanism	Factor reduced
More, well-distributed power/ground pins	Parallel paths; negative mutual between P/G	L
Flip-chip or low-inductance package	Short connections, many bumps	L
On-die and on-package decoupling	Local charge reservoir; current loop closes on die	Effective L
Board vias adjacent to P/G balls	Short, low-inductance escape to planes	L
Slew-rate control	Longer t_r	$\mathrm{d}i/\mathrm{d}t$
Weaker drive / series termination	Smaller ΔI	$\mathrm{d}i/\mathrm{d}t$
Staggered output timing	Spreads switching in time	N
Data-bus inversion (DBI)	Limits number of lines switching or driving low	N
Differential signaling	Constant total current	$N\ \mathrm{d}i/\mathrm{d}t$

20.5.1 Data-bus inversion

DDR4 and later memories can transmit a byte either as is or inverted, flagged by an extra DBI signal, choosing whichever form minimizes the number of lines driven low (for pseudo-open-drain I/O, where only low outputs draw current) or the number of transitions. In the worst case, a byte then has at most four of its eight data lines drawing current, halving the worst-case SSN for one extra pin per byte [57].

Rule of Thumb: Edge rate versus bounce

Ground bounce scales as $1/t_r$: doubling the rise time halves the bounce, a 6 dB improvement, at no cost if the timing budget allows it. *Limits:* slower edges increase the uncertainty of threshold crossing under noise, and some interfaces specify minimum slew rates; check both before selecting a weaker drive or slew setting.

Design Rule: Board-level SSN control

Connect each power and ground ball of a high-I/O-count device to its plane with its own via, placed as close to the ball as the breakout allows; do not share one via among several balls. Place the plane pair that serves the I/O supply close to the device's side of the board, and provide decoupling whose mounted inductance is comparable to the package's own (Chapter 21).

Key Takeaways

- Ground bounce is $L_{\mathrm{gnd}} N\ \mathrm{d}i/\mathrm{d}t$ across the shared package return; quiet outputs and inputs on the same rail inherit it.
- The switching current of a driver into a line is $\Delta V/(R_S + Z_0)$; its slope is about $\Delta I/t_r$, or $\pi/2$ times that for smooth edges.
- Parallel ground pins approach the mutual inductance M as a floor; interleave power and ground to exploit negative mutual inductance.

- SSN causes false switching, pattern-dependent delay (jitter), and threshold shifts.
- Mitigate with distributed pins, low-inductance packages, slew control, on-die decoupling, DBI, and differential signaling.

Further Reading

Swaminathan and Engin [56] and Bogatin [3] analyze simultaneous switching noise and its package dependence.

Review Problems

P20.1 Eight outputs each switch 20 mA in 250 ps through a shared 0.5 nH ground inductance. Estimate the ground bounce for a linear current ramp of full duration 250 ps.

P20.2 Four independent ground pins each have 1.2 nH. What is their effective inductance?

P20.3 Repeat Problem P20.2 with 0.2 nH of mutual inductance between every pair.

P20.4 With $L_{\text{eff}} = 0.1\,\text{nH}$, $\Delta I = 10\,\text{mA}$ per output, and linear current edges of full duration 150 ps, how many outputs may switch together for a 100 mV bounce limit?

P20.5 A package ground inductance of 0.2 nH resonates with 10 nF of on-die capacitance. Find the resonant frequency.

P20.6 By how many decibels is ground bounce reduced when the edge time is increased from 150 ps to 300 ps for line-driven outputs? For outputs driving lumped capacitive loads?

Chapter 21

Power Distribution Network Design

Chapter Overview

The power distribution network (PDN) delivers current from the voltage regulator to every transistor on every die, and it must do so while keeping the supply within a tolerance band of a few percent, at frequencies from DC to beyond a gigahertz. The frequency-domain target-impedance method turns this time-domain requirement into a single impedance limit that each part of the PDN—regulator, bulk capacitors, ceramic capacitors, planes, package, and die—must meet in its own frequency band. This chapter develops the method, models each element, explains anti-resonance, and gives practical rules for selecting and placing decoupling capacitors and for budgeting DC IR drop.

Learning Objectives

1. Compute a PDN target impedance from ripple tolerance and transient current.
2. Identify which PDN element controls the impedance in each frequency band.
3. Model a capacitor by C, ESR, and mounted ESL, and compute its self-resonant frequency.
4. Predict and damp anti-resonances between capacitors, planes, and the package.
5. Choose capacitor quantity, values, placement, and via patterns.
6. Estimate DC IR drop in planes and size copper accordingly.

21.1 The target-impedance method

A load draws a current that varies in time. If the PDN looks like an impedance $Z(f)$ from the load's terminals, the supply noise is $V(f) = Z(f)I(f)$. Requiring $|Z(f)|$ to be below a constant value over the whole bandwidth of the current guarantees—for a resistive, flat PDN—that the voltage excursion stays within the allowed ripple for any current waveform within the stated amplitude [7], [58]:

$$Z_{\text{target}} = \frac{\Delta V_{\text{ripple}}}{\Delta I_{\text{transient}}} = \frac{V_{\text{DD}} \times (\text{ripple fraction})}{\Delta I_{\text{transient}}}. \tag{21.1}$$

The transient current is the largest expected change in load current, not the maximum current. When it is unknown, half the maximum current is a common assumption.

Worked Example 21.1: Target impedance of a core rail

An SoC core rail runs at 0.8 V with a $\pm 3\,\%$ tolerance and a maximum current of 20 A. Assume a 50 % transient.

Solution. $\Delta V = 0.03 \times 0.8 = 24\,\text{mV}$ and $\Delta I = 10\,\text{A}$, so $Z_{\text{target}} = 2.4\,\text{m}\Omega$. The tolerance must also cover the regulator's DC set-point accuracy and IR drop, so in practice only part of the 24 mV is available for dynamic noise (Table 21.1).

21.1.1 Allocating the voltage budget

A rail tolerance is shared by several independent error sources, and only one of them is the dynamic noise that Z_{target} addresses. Table 21.1 shows an illustrative allocation of the $\pm 24\,\text{mV}$ of Worked Example 21.1. With 14 mV reserved for the specified 10 A step, the dynamic target becomes $1.4\,\text{m}\Omega$, not $2.4\,\text{m}\Omega$. The allocation must also say *where* the voltage is observed (board pads or die bumps) and must reserve a share for measurement uncertainty if it is to become an acceptance threshold. For continuity, the examples that follow use the unallocated $2.4\,\text{m}\Omega$.

Table 21.1. Illustrative voltage allocation for a 0.8 V, ±3 % rail (requirement values chosen for the example).

Contribution	Allocation (mV)	Controlled by
Regulator set point, ripple, temperature	4	Regulator choice, remote sense
DC path (IR) drop, uncompensated	6	Copper, plane geometry
Dynamic response to 10 A step	14	PDN impedance ($Z_{\text{target}} = 1.4\,\text{m}\Omega$)
Total	24	±3 % of 0.8 V

Common Pitfall: Treating the target as a hard flat line

The target-impedance method assumes a flat, resistive PDN. A PDN that meets Z_{target} everywhere but has high-Q anti-resonance peaks just below it can still ring when the load current contains energy at those frequencies, as clock-gated loads do. Conversely, a narrow peak above Z_{target} at a frequency where the load draws no current may be harmless. Use Z_{target} as a design guide and confirm with a time-domain simulation of the expected current profile.

21.2 Frequency regions of a PDN

Each element of the PDN is effective only over a limited frequency range (Figures 21.1 and 21.2):

Voltage regulator (DC to tens of kilohertz)
The regulator's feedback loop holds the output impedance low up to roughly its loop bandwidth, above which its output impedance rises inductively.

Bulk capacitors (kilohertz to about a megahertz)
Electrolytic, polymer, or large ceramic capacitors hand off from the regulator. Their ESL of a nanohenry or more limits them above a megahertz.

Ceramic capacitors (about 1 MHz to tens of megahertz)
Banks of MLCCs, whose effectiveness is limited by *mounted* inductance.

Planes (tens to hundreds of megahertz)
The power–ground plane pair is a distributed capacitor with very low inductance, bounded at high frequency by its cavity resonances (Chapter 23).

Package and die (above about 100 MHz)

The package's inductance isolates the die from the board at high frequency; the die's own capacitance (gate and dedicated on-die decoupling) supplies the fastest current.

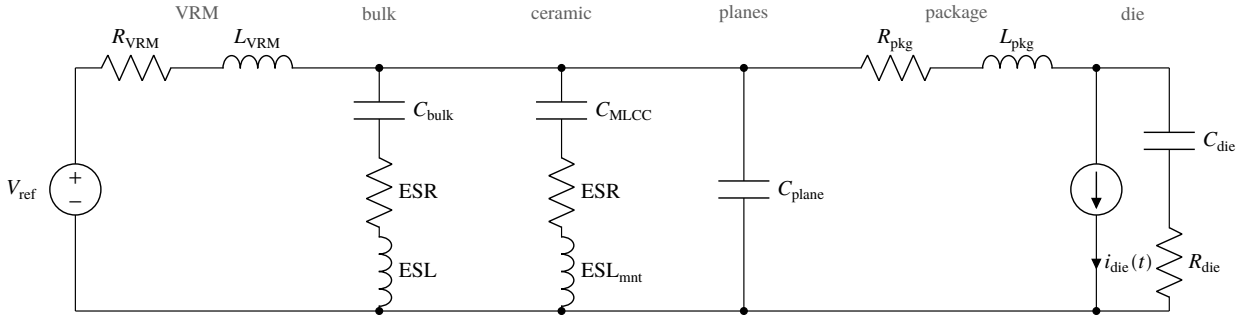


Figure 21.1. Lumped equivalent circuit of a PDN from the regulator to the die. Each stage has a capacitance limited by its own series inductance and resistance.

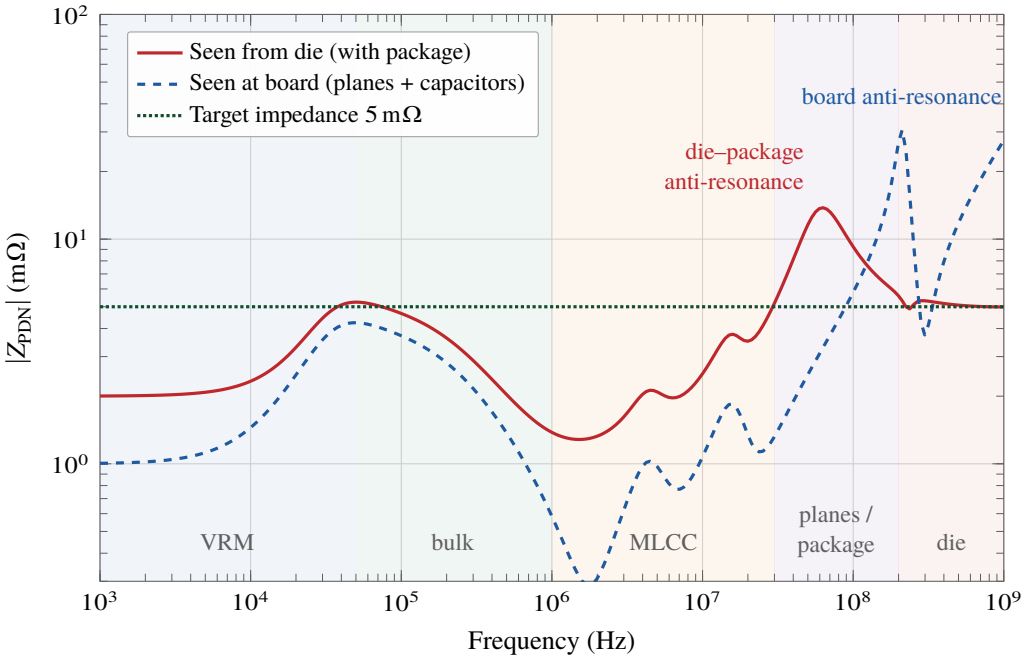


Figure 21.2. PDN impedance computed from the lumped model of Figure 21.1, seen at the board (planes and capacitors) and at the die (through the package). The board meets the 5 mΩ target up to about 90 MHz but has its own anti-resonance between MLCC inductance and plane capacitance near 200 MHz; seen from the die, the package inductance hides that peak, and the package inductance and on-die capacitance instead form an anti-resonance near 60 MHz that exceeds the target.

Figure 21.2 illustrates the most important limitation of board-level decoupling: above a few tens of megahertz, the impedance seen by the die is dominated by the package inductance and the die capacitance, not by anything on the board. The resulting die–package anti-resonance, typically between 30–200 MHz, is often the dominant PDN noise of a modern processor. The board can damp it—by presenting a low, resistive impedance at that frequency—but cannot move it.

21.3 Capacitor models

A real capacitor is a series RLC circuit:

$$Z_C(f) = \mathrm{ESR} + \mathrm{j}\omega\,\mathrm{ESL} + \frac{1}{\mathrm{j}\omega C}, \qquad |Z_C| = \sqrt{\mathrm{ESR}^2 + \left(\omega\mathrm{ESL} - \frac{1}{\omega C}\right)^2}, \tag{21.2}$$

with self-resonant frequency

$$f_{\mathrm{SRF}} = \frac{1}{2\pi\sqrt{\mathrm{ESL}\cdot C}}. \tag{21.3}$$

Below f_{SRF} the capacitor is capacitive; at f_{SRF} its impedance equals its ESR; above it, the capacitor is an inductor (Figure 21.3). Above its SRF a capacitor still helps, but only through its inductance, and that inductance is set mostly by how it is mounted, not by its capacitance value or case size. MLCC capacitance also falls with DC bias and temperature, especially in high-k dielectrics (X5R, X7R); a 10 µF part at its rated voltage may provide only a third to a half of its nominal value.

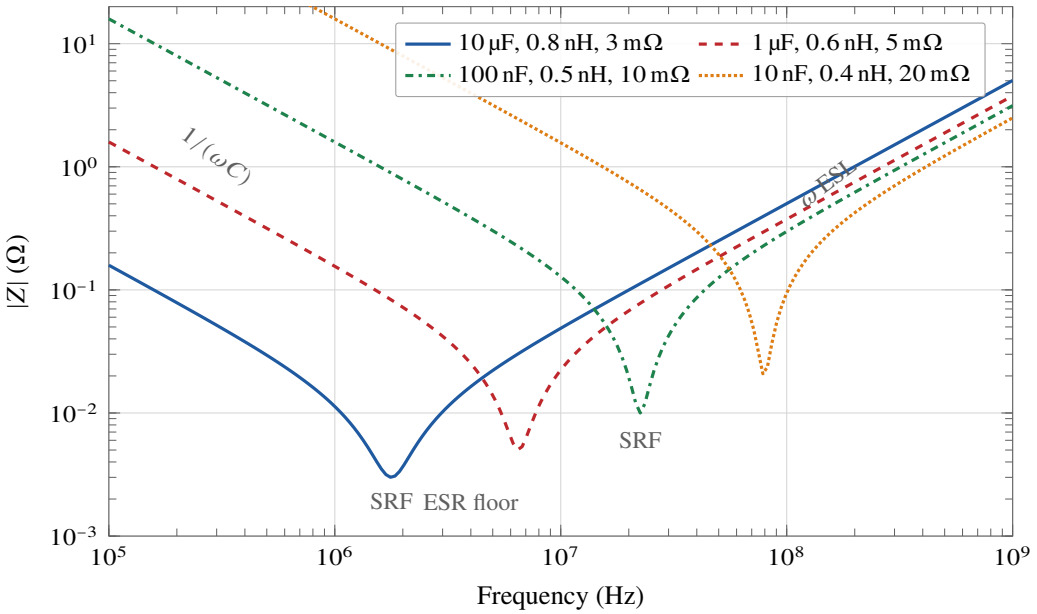


Figure 21.3. Impedance of four mounted MLCCs from Eq. (21.2). Each is capacitive below its self-resonant frequency (SRF), resistive at it, and inductive above it. Above about 30 MHz the curves converge, because mounted inductances are similar.

Worked Example 21.2: Usefulness of a 100 nF capacitor

A 100 nF 0402 MLCC has a mounted inductance of 0.5 nH. Find its SRF, its impedance at 100 MHz, and the number needed to reach 2.4 mΩ at 20 MHz. Take ESR = 10 mΩ.

Solution. $f_{\mathrm{SRF}} = 1/(2\pi\sqrt{0.5\times10^{-9}\times10^{-7}}) = 22.5$ MHz. At 100 MHz, $|Z| = 0.30$ Ω, close to $\omega\mathrm{ESL} = 0.31$ Ω—more than a hundred times the target.

At 20 MHz the capacitor is *below* its SRF, so both reactances matter: $X_L = \omega\mathrm{ESL} = 62.83$ mΩ and $X_C = 1/(\omega C) = 79.58$ mΩ. From Eq. (21.2), $|Z_C| = \sqrt{10^2 + (62.83 - 79.58)^2} = 19.50$ mΩ. For N

identical, independent capacitors the bank impedance is Z_C/N, so the pointwise requirement is

$$N(f) \geq \left\lceil \frac{|Z_C(f)|}{Z_{\text{target}}} \right\rceil, \tag{21.4}$$

which gives $N = \lceil 19.50/2.4 \rceil = 9$. The common shortcut $N = \omega\text{ESL}/Z_{\text{target}}$ assumes the capacitor is purely inductive and gives $\lceil 26.18 \rceil = 27$, three times too many at this frequency (Figure 21.4).

Quick estimate	Full model	Difference
$\omega\text{ESL}/Z_{\text{target}}$: 27	Full R–L–C: 9	200 %

Equation (21.4) is a lower bound at one frequency in an idealized model. It ignores the mounting inductance the capacitors share (vias, spreading, the package), which prevents the $1/N$ improvement from continuing indefinitely, and it does not show that the bank meets the target at other frequencies. A sweep over frequency and a transient check (Section 21.6) are still required. Above about 30 MHz, the planes and the package must take over.

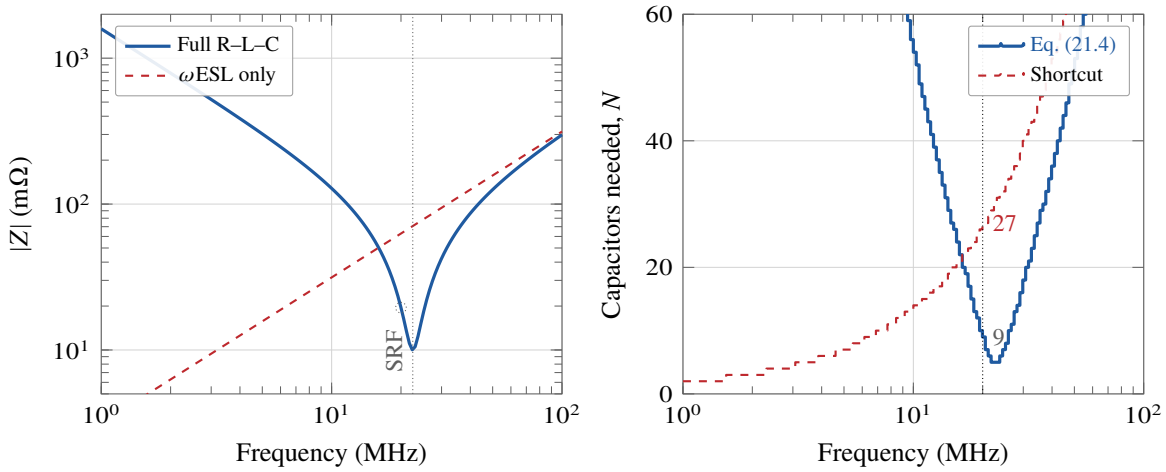


Figure 21.4. Left: impedance of a 100 nF, 10 mΩ, 0.5 nH capacitor and its inductive-only approximation. Right: pointwise number of ideal, independent capacitors needed for $Z_{\text{target}} = 2.4\,\text{m}\Omega$, from Eq. (21.4) and from the shortcut. Shared inductance is ignored. [Analytical]

21.3.1 Mounted inductance

The mounted ESL of a capacitor is the loop inductance of the path from the capacitor's terminals through its pads, mounting traces, and vias, to the plane pair, plus the capacitor's intrinsic inductance. The via portion follows Eq. (3.8): it is proportional to the depth of the plane pair below the capacitor and logarithmic in via spacing. Figure 21.5 ranks common mounting patterns. Placing vias at the sides of the pads, as close together as possible, roughly halves the inductance of end-placed vias; long traces between pad and via can double it.

Design Rule: Mounting decoupling capacitors

Connect every decoupling capacitor directly to its plane pair with vias at the sides of (or in) its pads; never route a trace from the pad to a distant via. Put the plane pair that serves a rail as close as possible to the

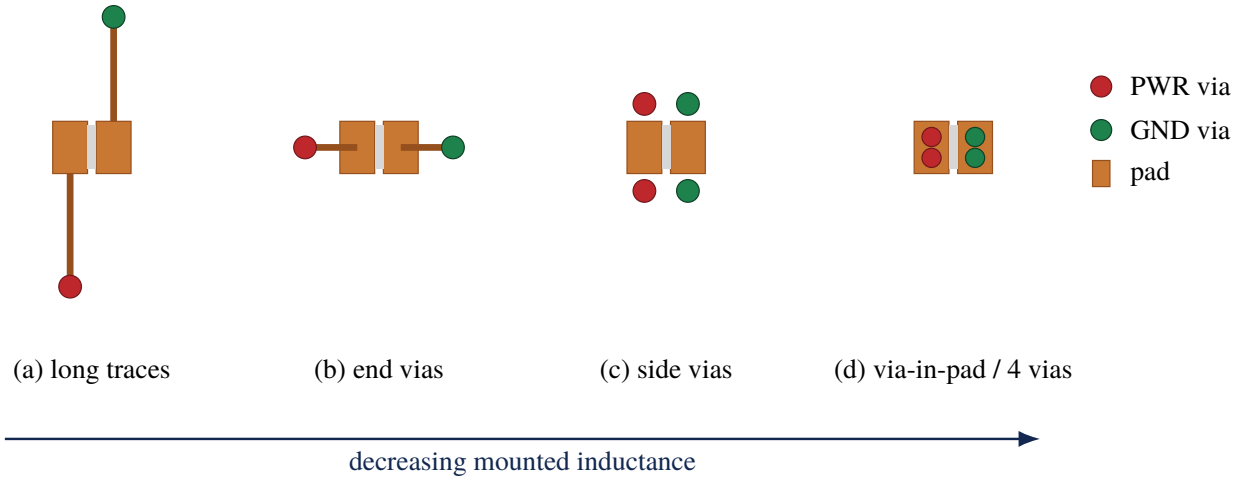


Figure 21.5. Decoupling-capacitor mounting patterns, from highest to lowest mounted inductance. Short connections, vias at the sides of the pads, and multiple via pairs reduce the loop area.

surface on which its capacitors are mounted. Prefer reverse-geometry (0204, 0306) or multi-terminal capacitors for the highest-frequency decoupling.

21.4 Anti-resonance

When two capacitors with different SRFs are placed in parallel, there is a frequency between their SRFs at which the larger capacitor is already inductive and the smaller is still capacitive. The two form a parallel LC tank, and the impedance peaks.

For two series branches $Z_k = R_k + \mathrm{j}(\omega L_k - 1/\omega C_k)$, $k = 1, 2$, the parallel impedance is

$$Z_{\parallel} = \frac{Z_1 Z_2}{Z_1 + Z_2}. \tag{21.5}$$

Without resistance the denominator vanishes when the two reactances are equal and opposite,

$$\omega L_1 - \frac{1}{\omega C_1} = -\left[\omega L_2 - \frac{1}{\omega C_2}\right],$$

that is, at

$$f_{\mathrm{ar}} = \frac{1}{2\pi}\sqrt{\frac{C_1 + C_2}{(L_1 + L_2)\, C_1 C_2}} \xrightarrow{C_1 \gg C_2} \frac{1}{2\pi\sqrt{(L_1 + L_2)\, C_2}}. \tag{21.6}$$

At that frequency the branches are $R_1 + \mathrm{j}X$ and $R_2 - \mathrm{j}X$, where $X = \omega L_1 - 1/(\omega C_1)$; if $X \gg R_1, R_2$, Eq. (21.5) gives

$$|Z|_{\mathrm{peak}} \approx \frac{X^2}{R_1 + R_2}. \tag{21.7}$$

The peak is high when ESRs are low—which is why the very low ESR of modern MLCCs, together with the low loss of plane capacitance, makes anti-resonance a real problem. A frequently quoted simplification keeps only the large capacitor's ESL and the small capacitor's capacitance, $f \approx 1/(2\pi\sqrt{L_1 C_2})$; it omits the small capacitor's own ESL, which is of the same order as L_1, and is not accurate enough to place a peak.

Worked Example 21.3: Anti-resonance of two capacitors

A 100 nF MLCC (ESL 0.5 nH, ESR 20 mΩ) is paralleled with a 10 nF MLCC (ESL 0.4 nH, ESR 30 mΩ). Estimate the anti-resonance.

Solution. Equation (21.6) with both inductances: $f_{\text{ar}} = \frac{1}{2\pi}\sqrt{110\times10^{-9}/(0.9\times10^{-9}\times10^{-15})} = 55.6\,\text{MHz}$. There $X = \omega L_1 - 1/(\omega C_1) = 174.8 - 28.6 = 146\,\text{m}\Omega$, and Eq. (21.7) gives $|Z|_{\text{peak}} \approx 0.146^2/0.050 = 0.43\,\Omega$. A sweep of the exact Eq. (21.5) (Figure 21.6, Listing E.2) puts the maximum at 55.16 MHz and 0.443 Ω, about three times the 0.146 Ω that the 100 nF capacitor alone has at that frequency. Adding the small capacitor made the PDN worse there.

Quick estimate	Full model	Difference
$1/\sqrt{L_1 C_2}$: 71.2 MHz, 1.0 Ω	Exact: 55.16 MHz, 0.443 Ω	29 %, ×2.3

The shortcut misplaces the peak by 16 MHz and overstates it by more than a factor of two, because it ignores the 0.4 nH of the small capacitor. The qualitative lesson survives: a capacitor can create a harmful peak.

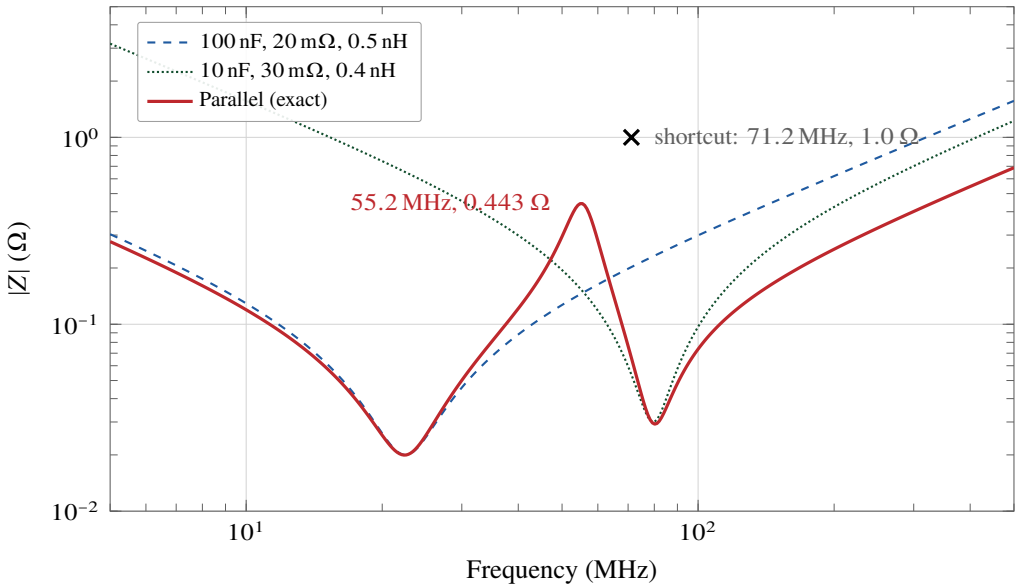


Figure 21.6. Exact impedance of the two branches of Worked Example 21.3 and of their parallel combination. The cross marks the prediction of the $1/\sqrt{L_1 C_2}$ shortcut. [Analytical]

Anti-resonances are controlled in three ways: by spacing capacitor values closely (several values per decade, or many capacitors of the same value, so that no deep inductive-capacitive pairing occurs); by increasing ESR (controlled-ESR capacitors, or simply fewer, larger capacitors whose inductive region is damped by ESR); and by designing for a flat impedance with the "frequency-domain target impedance" method, choosing each stage's ESR close to Z_{target} so that the parallel combination is resistive [6], [59]. Raising ESR lowers a peak but also raises the impedance floor between peaks; there is no universal best sequence of capacitor values. The choice comes from a sweep of the actual complex impedance with biased capacitance, mounted ESL, ESR and the rest of the network.

Rule of Thumb: Fewer values, more of each

For board-level MLCC decoupling, two or three values per rail spread a decade apart (for example 10 µF, 1 µF, 0.1 µF), in quantities that make their banks' impedances overlap, usually perform better than a "one of every value" strategy. *Limits:* this assumes mounted ESL dominates above about 10 MHz; simulate the PDN impedance with the actual mounting and plane model before release.

21.5 Planes and spreading inductance

A plane pair provides capacitance $C = \varepsilon_0 D_k A/h$ (Eq. (3.1)) and, for current flowing from a capacitor's vias to a device's vias, a *spreading inductance*. For current spreading radially from a via of radius r_v to a ring at distance ρ between planes separated by h, the inductance is

$$L_{\text{spread}} \approx \frac{\mu_0 h}{2\pi} \ln \frac{\rho}{r_v}. \tag{21.8}$$

It is proportional to h and only logarithmic in distance: for $h = 3$ mil, $r_v = 5$ mil, and $\rho = 250$ mil, $L_{\text{spread}} \approx 60$ pH. Thin plane dielectrics therefore matter more than placing capacitors within a fraction of an inch of the load—although placement still matters for the via portion of the loop and for avoiding shared vias.

Design Rule: Plane pairs for low-voltage, high-current rails

Give each high-current, low-voltage rail a dedicated plane (or wide pour) adjacent to a ground plane with 2–4 mil of dielectric. Place the pair near the surface carrying the device and its decoupling.

21.6 From impedance to a load transient

A target impedance is a frequency-domain proxy for a time-domain requirement: the rail voltage must stay inside its allocation while the load current changes. For a linear PDN with driving-point impedance $Z(s)$ at the observation port, the voltage deviation is

$$\Delta V(s) = Z(s)\, \Delta I(s), \tag{21.9}$$

so the shape of the current step decides which part of Z is excited. The smallest network that shows the essential behavior is a feed $R + sL$ (regulator and bulk path) shunted by a capacitance C at the load:

$$Z(s) = \frac{(R + sL) \cdot \frac{1}{sC}}{R + sL + \frac{1}{sC}} = \frac{R + sL}{1 + sRC + s^2 LC}. \tag{21.10}$$

It has a peak near $f_0 = 1/(2\pi\sqrt{LC})$ and a DC value R.

Worked Example 21.4: Damping against DC drop

A rail is fed through $L = 0.5$ nH and either $R = 2$ mΩ or 8 mΩ, with $C = 10$ µF at the load. The load current rises by 1 A in 20 ns and stays there. Compare the responses.

Solution. $f_0 = 1/(2\pi\sqrt{0.5 \times 10^{-9} \times 10^{-5}}) = 2.25$ MHz and the characteristic impedance $\sqrt{L/C} = 7.1$ mΩ. With $R = 2$ mΩ the circuit is underdamped ($Q = \sqrt{L/C}/R \approx 3.5$): the voltage dips to −7.5 mV, rings at 2.25 MHz, and settles at the 2 mV DC drop (Figure 21.7). With $R = 8$ mΩ ($Q \approx 0.9$) the ringing almost disappears, but the response still overshoots to −9.6 mV before settling at 8 mV: the "damped"

design has the *larger* worst-case deviation. Neither number is visible from a single $|Z|$ value; both follow from Eq. (21.9).

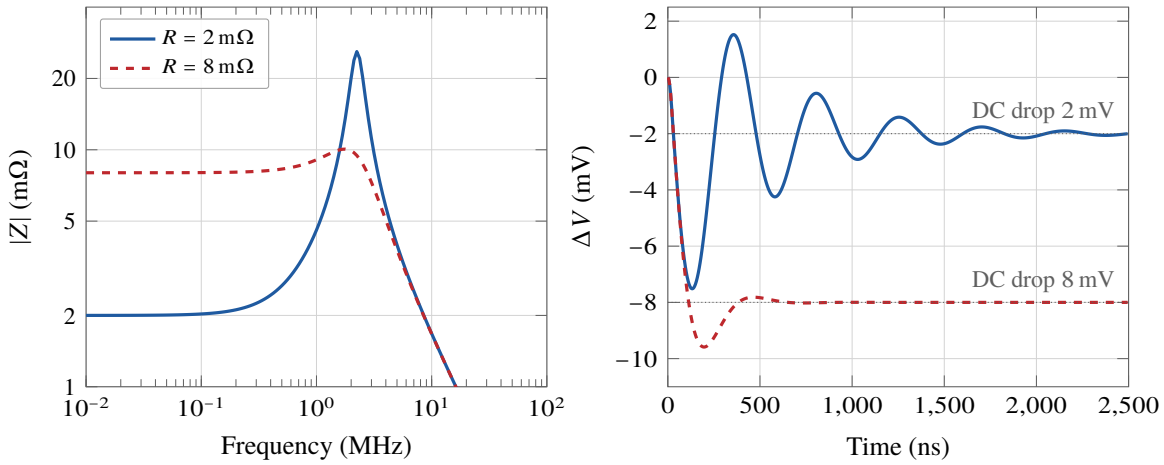


Figure 21.7. Impedance (left) and response to a 1 A, 20 ns load step (right) of the network of Eq. (21.10) for two feed resistances. Linear model; no regulator loop or capacitor ESR. [Simulation (analytical network)]

Two consequences matter in practice. First, a periodic load—a clock-gated block toggling at a fixed rate—can excite a narrow resonance coherently and produce a far larger deviation than an isolated step; the worst-case current pattern is part of the specification. Second, damping without DC loss comes from lossy shunt branches (controlled-ESR capacitors, an RC snubber) or from the regulator's output impedance, not from raising the series resistance.

Design Rule: Close the PDN loop in the time domain

After meeting Z_{target} in the frequency domain, simulate the rail at the observation port with the specified current profile—amplitude, edge rate, repetition rate and dwell time—and with capacitance, ESL and ESR at their corners. Compare peak droop, overshoot and settling with the dynamic allocation of Table 21.1.

21.7 DC IR drop

At DC, the PDN must also deliver the load current with an acceptable voltage drop. A plane's resistance is conveniently expressed per square:

$$R_{\square} = \frac{\rho}{t}, \tag{21.11}$$

which for 1 oz copper (35 µm) is 0.49 mΩ per square at 20 °C. The resistance of a path is $R_{\square}$ times the number of squares along it. Via fields, anti-pad clusters, and neck-downs where the plane passes between BGAs crowd the current and raise the effective resistance; IR-drop simulation tools solve the plane geometry directly. Current density in vias and narrow necks must also be checked against IPC-2152 thermal limits [19].

Worked Example 21.5: IR drop in a core plane

A 0.8 V, 15 A rail flows through a 1 oz plane region three squares long from the regulator to the SoC. Estimate the drop.

Solution. $R = 3 \times 0.49\,\text{m}\Omega = 1.48\,\text{m}\Omega$, so $\Delta V = 15 \times 1.48 = 22\,\text{mV}$, 2.8 % of the rail—nearly the entire 3 % tolerance. Remedies: use 2 oz copper or two paralleled planes, move the regulator closer, or use the regulator's remote-sense input at the SoC so that the regulator compensates the static drop.

Table 21.2. PDN elements and their effective frequency ranges (typical).

Element	Effective range	Limited by
Regulator	DC to ~10–100 kHz	Loop bandwidth; output inductance
Bulk (polymer, large MLCC)	1 kHz to ~1 MHz	ESL (≳1 nH), ESR
MLCC bank	1 MHz to ~50 MHz	Mounted ESL, DC-bias derating
Plane pair	~20 MHz to cavity resonance	Area, dielectric thickness
Package capacitors	~50 MHz to 300 MHz	Package inductance
On-die capacitance	> 100 MHz	Die area, leakage

Key Takeaways

- $Z_{\text{target}} = \Delta V_{\text{ripple}} / \Delta I_{\text{transient}}$; a few milliohms is typical for low-voltage, high-current rails.
- Each PDN element—regulator, bulk, MLCC, planes, package, die—controls the impedance in its own band.
- Above its SRF a capacitor is an inductor; mounted ESL, not capacitance, sets high-frequency performance.
- Size capacitor banks from the full R–L–C branch impedance, $N \geq \lceil |Z_C| / Z_{\text{target}} \rceil$; the inductive-only shortcut is wrong below the SRF.
- Anti-resonances arise between parallel branches; place them with both ESLs (Eq. (21.6)), and control them with value selection, ESR, and simulation.
- Allocate the rail tolerance among set point, DC drop and dynamic noise, and confirm the result with a load-transient simulation.
- The die–package anti-resonance is set by the package and die; the board can only damp it.
- Budget DC IR drop with $R_{\square} = 0.49\,\text{m}\Omega$ per square for 1 oz copper, and use remote sense.

Further Reading

Smith and Bogatin [7] is a complete treatment of frequency-domain PDN design; Smith et al. [58] introduced the target-impedance method; Novak and Miller [6] and Novak [59] cover PDN measurement and design methodologies.

Review Problems

P21.1 Find Z_{target} for a 1.0 V rail with ±5 % tolerance and a 5 A transient.

P21.2 Find the SRF of a 1 µF capacitor with a mounted ESL of 0.6 nH.

P21.3 How many capacitors, each with 0.4 nH mounted ESL, are needed to keep the PDN below 5 mΩ at 30 MHz? Assume the capacitors are well above their SRF at that frequency.

P21.4 An 8 A current flows through five squares of 0.5 oz (17.4 µm) copper. Find the IR drop.

P21.5 Estimate the anti-resonant frequency between an inductive branch of 0.8 nH and a capacitive branch of 1 nF.

P21.6 *(Approximation failure.)* A 1 µF capacitor has 6 mΩ ESR and 0.6 nH mounted ESL. How many are needed for 2.4 mΩ at 5 MHz? What does the inductive-only shortcut give, and why is it wrong?

P21.7 *(Imperfect data.)* The capacitor of the previous problem loses half its capacitance at the rail's DC bias. Repeat the count.

P21.8 *(Inverse design.)* For the network of Worked Example 21.4, what feed resistance gives a quality factor of 0.7?

Chapter 22

Capacitor Technologies and Decoupling Practice

Chapter Overview

Chapters 19 and 21 built the power-distribution network from a target impedance and showed how capacitors, planes, and the regulator share the frequency range. This chapter looks at the parts themselves. It compares the capacitor technologies a board designer can buy (multilayer ceramic, tantalum, aluminium electrolytic, polymer, film, and supercapacitors), explains the non-ideal behaviour that datasheet headlines hide (DC-bias loss, ageing, ESR, ESL, and the mounting inductance the board adds), shows how a ferrite bead and a capacitor can make a resonant filter that amplifies noise, and ends with a comparison between an arbitrary "rule-of-thumb" capacitor bank and an engineered network, validated in the time domain with a load step. The component values in the plots are *illustrative*, chosen inside typical datasheet ranges; a real design must use the manufacturer's models, which include frequency-, bias-, and temperature-dependent data.

Learning Objectives

1. Choose a capacitor technology from its impedance, voltage, ripple-current, reliability, and size characteristics.
2. Derate class-II MLCCs for DC bias, temperature, and ageing.
3. Separate the part's ESL from the mounting inductance and reduce the latter.
4. Recognize and damp the resonance of a ferrite-bead filter.
5. Design and validate a decoupling network against a target impedance and a load step.

22.1 The capacitor as a circuit

Every capacitor on a board behaves, to first order, as a series RLC circuit (Figure 22.1): the capacitance C, an equivalent series resistance (ESR) from electrodes, terminations, and dielectric loss, and an equivalent series inductance (ESL) from the current path inside the part. The board adds the inductance of the pads, vias, and plane spreading. Its impedance

$$Z(f) = \mathrm{ESR} + j2\pi f L_{\mathrm{total}} + \frac{1}{j2\pi f C}, \qquad f_{\mathrm{SRF}} = \frac{1}{2\pi\sqrt{L_{\mathrm{total}} C}}, \tag{22.1}$$

is capacitive below the self-resonant frequency f_{SRF}, equal to the ESR at f_{SRF}, and inductive above it. Above f_{SRF} the capacitance value no longer matters; only the inductance does. That single fact explains most decoupling rules and most of their failures.

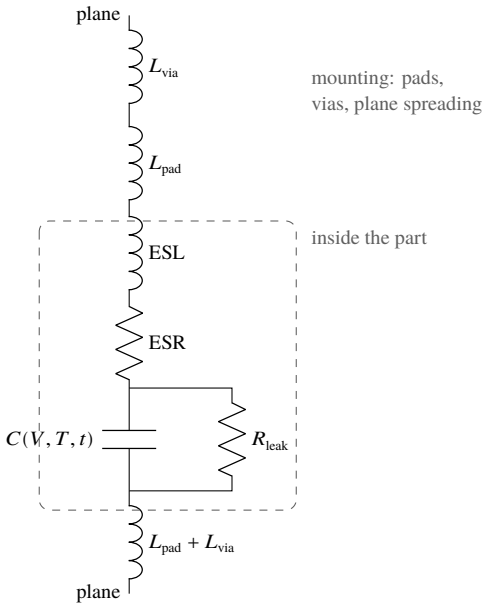


Figure 22.1. Equivalent circuit of a mounted capacitor. ESR, ESL, and C belong to the part; C depends on voltage V, temperature T, and time t (ageing). The pad, via, and spreading inductances belong to the layout and are often larger than the ESL. [Schematic]

22.2 Technologies compared

Figure 22.2 plots Eq. (22.1) for one representative part of each technology, including a typical mounting inductance; Table 22.1 summarizes the properties that decide where each belongs.

Figure 22.3 shows what the parts look like.

Reading the curves. With the illustrative values, the self-resonant frequencies are 20.5 MHz for a 100 nF MLCC with 0.6 nH total inductance, 1.78 MHz for a 10 µF MLCC with 0.8 nH, 318 kHz for 100 µF tantalum parts with 2.5 nH, and 104 kHz for a 470 µF aluminium electrolytic with 5 nH. The two tantalum curves differ only in ESR (150 mΩ versus 25 mΩ): the polymer part reaches a six-times-lower minimum. The supercapacitor's impedance flattens at its ESR above about 2 kHz; it stores energy but does not decouple.

22.3 Class-II MLCCs: DC bias, temperature, and ageing

The ferroelectric (barium-titanate) dielectrics that give class-II MLCCs their high capacitance lose permittivity under DC bias. The loss is larger for smaller cases and higher capacitance per volume. Figure 22.4 shows the shape of the effect with an illustrative model for three 10 µF parts: at 3.3 V the 0402 part retains about 34 %, the 0603 part 54 %, and the 1206 part 83 %; at 1.0 V they retain 78, 90, and 98 %. Manufacturers publish the measured curves for each part number and provide simulation tools; use those numbers, not the nominal value, in every PDN calculation.

Temperature. The EIA code states the temperature range and the allowed change over it with no bias applied: X7R is ±15 % from −55 to 125 °C, X5R ±15 % from −55 to 85 °C, Y5V +22/−82 % from −30 to 85 °C. The temperature and bias effects combine.

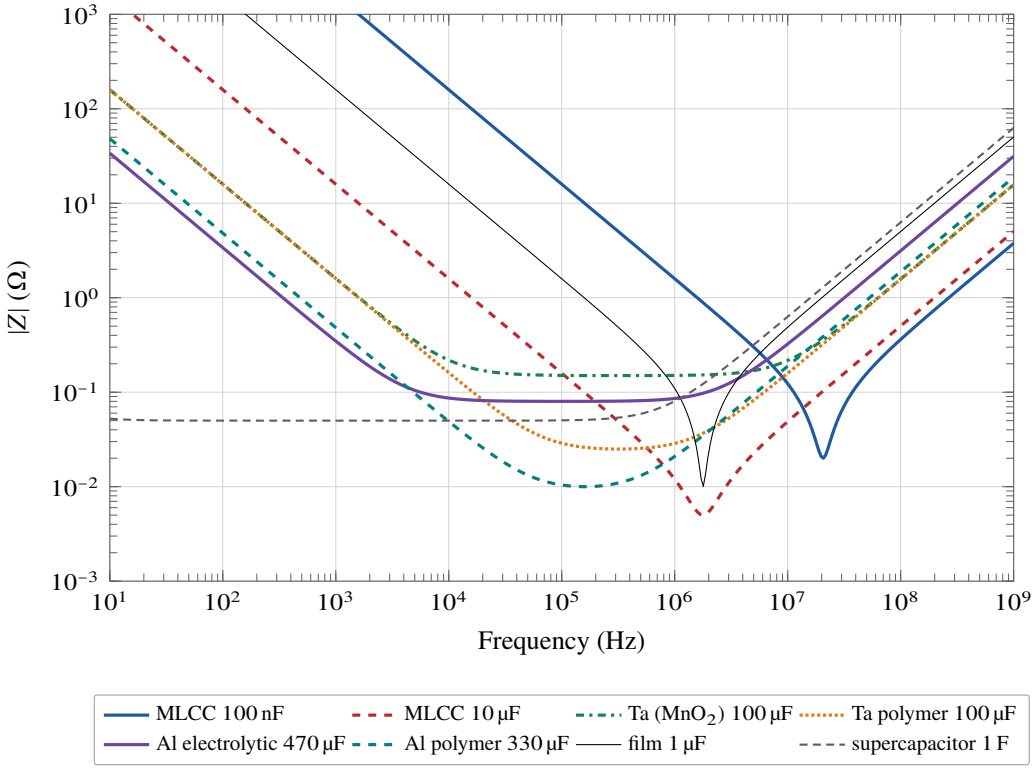


Figure 22.2. Impedance of representative capacitors (illustrative values within typical datasheet ranges, mounting inductance included). Low-ESR parts have deep, narrow minima; high-ESR parts have broad, shallow ones; above resonance every curve is set by its inductance. [Analytical]

Ageing. Class-II capacitance decreases logarithmically with time after the part was last heated above its Curie point (around 125 °C); manufacturers quote a percentage per decade-hour, commonly a few percent for X7R and X5R. Reflow soldering "de-ages" the part. Class-I dielectrics do not age.

Mechanical. Large MLCCs crack when the board flexes (depanelling, connector insertion, screw mounting); a crack can become a short. Place large cases away from board edges and flex points, orient them parallel to the bend axis, or use flexible-termination parts. Class-II MLCCs are also piezoelectric: a ripple voltage in the audio band can make them "sing".

22.4 Mounting inductance

Above self-resonance the capacitor is an inductor, and most of that inductance is usually in the mounting. Figure 22.5 shows one 100 nF part with total inductances from 0.3 to 2.0 nH: at 200 MHz the impedance ranges from about 0.4 to 2.5 Ω. The layout controls this range through

- via placement: vias at the pad ends or in the pad (via-in-pad) rather than at the end of a trace;
- via count: two vias per pad roughly halve the via inductance;
- depth: a shorter distance from the capacitor to the plane pair it decouples;
- the loop: placing power and ground vias of one capacitor close together so their currents' fields cancel;
- reverse-geometry (wide-and-short) or interdigitated parts for the lowest ESL.

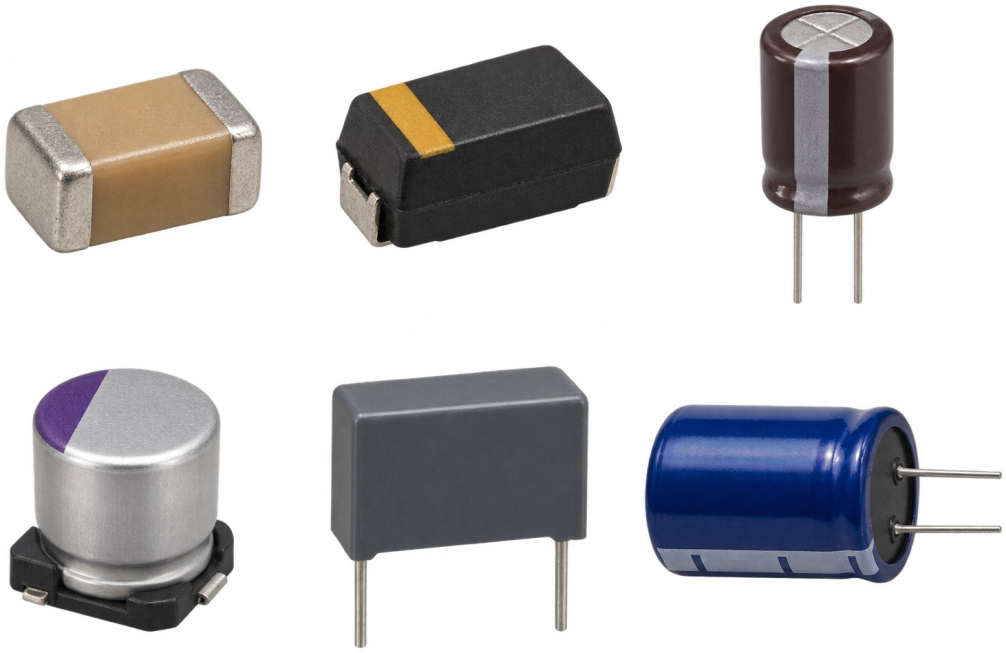


Figure 22.3. Capacitor technologies: multilayer ceramic chip (MLCC), molded tantalum (the stripe marks the anode), radial aluminium electrolytic, surface-mount aluminium (polymer) can, film box capacitor, and a large radial can of the kind used for bulk electrolytics and supercapacitors. Rendered illustration supplied with the manuscript, not a photograph of specific products; markings, contact counts, and proportions are not authoritative. [Illustration]

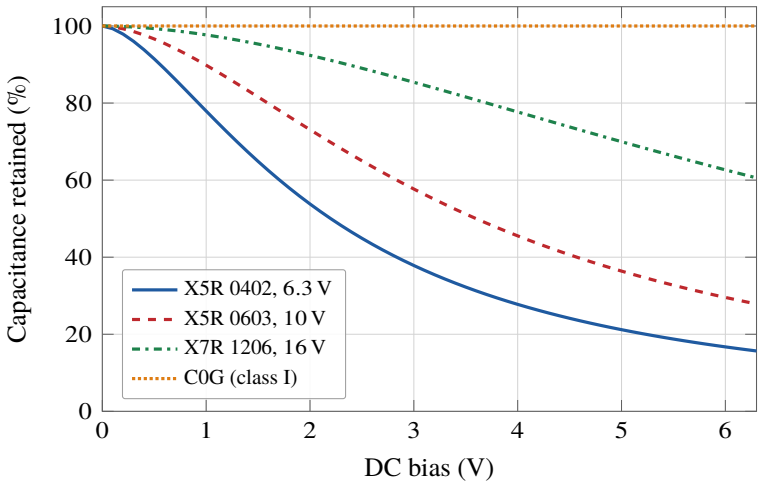


Figure 22.4. Capacitance retained under DC bias for 10 µF class-II MLCCs in three case sizes, and a class-I part. The curves illustrate typical shapes from manufacturers' data; they are not data for specific parts. [Conceptual]

Table 22.1. Capacitor technologies for board-level power and decoupling. Ranges are indicative; consult the manufacturer for specific series.

Technology	Strengths	Weaknesses	Typical role
MLCC class I (C0G/NP0)	stable with voltage, temperature, time; very low loss	small capacitance (pF to tens of nF)	timing, RF, filters, snubbers
MLCC class II (X5R, X7R, X6S, …)	high capacitance per volume; very low ESR and ESL	strong DC-bias and temperature dependence; ageing; cracking under board flex; piezoelectric noise	local decoupling, regulator input/output
Tantalum (MnO_2)	stable capacitance; moderate size	higher ESR; voltage derating needed; ignition failure mode at surge	legacy bulk
Tantalum polymer	low ESR; stable capacitance	voltage derating (typically less than MnO_2); leakage	bulk near regulators and loads
Aluminium electrolytic	high capacitance and voltage at low cost	high ESR at low temperature; finite life (electrolyte dry-out); large	input bulk, hold-up
Aluminium polymer	low ESR; long life; stable over temperature	lower voltage ratings; cost	bulk on low-voltage rails
Film (PP, PET)	self-healing; low loss; stable	large for the capacitance	snubbers, high-voltage, filters
Supercapacitor (EDLC)	farads; energy storage	high ESR for its size; low voltage per cell; leakage	hold-up, RTC backup

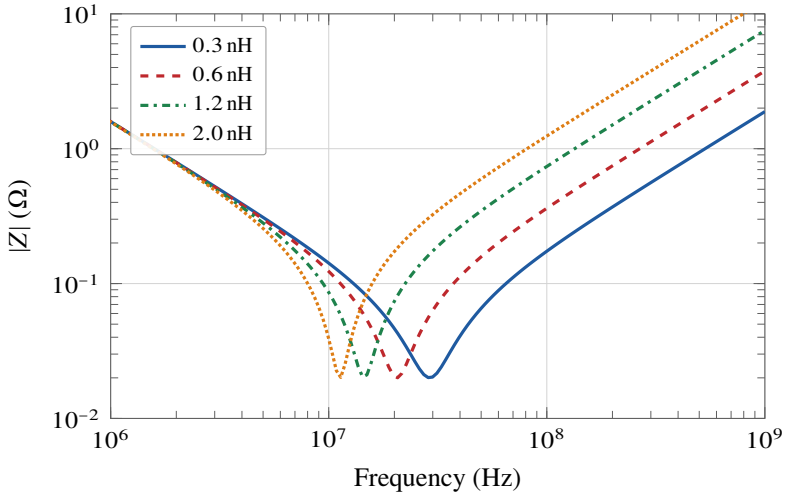


Figure 22.5. Impedance of a 100 nF capacitor with total (part plus mounting) inductance from 0.3 to 2.0 nH. The capacitance sets the low-frequency side; the inductance sets everything above 20 MHz. [Analytical]

Common Pitfall: Mixed values "to cover more frequencies"

The rule "use 100 nF, 10 nF, and 1 nF in parallel to cover more frequencies" was sound when parts were through-hole and large; for SMT parts of the same case size the ESL, and therefore the high-frequency impedance, is nearly the same for all three values, while the smaller values add anti-resonances (Chapter 21). Use one value per case size where the high-frequency inductance matters, spread values only to fill a low-frequency gap, and check anti-resonance peaks against the target.

22.5 Ferrite beads and LC resonance

A ferrite bead in series with a supply, followed by a capacitor, is a low-pass filter. At low frequency the bead is an inductor (about 1 µH in the example), at its peak a resistor (600 Ω), and at the highest frequencies a capacitor. The low-frequency inductance and the capacitor form a resonant circuit with

$$f_0 = \frac{1}{2\pi\sqrt{L_{\text{bead}}C}}, \qquad Z_c = \sqrt{L_{\text{bead}}/C}. \tag{22.2}$$

With $C = 10\,\mu\text{F}$, $f_0 = 50\,\text{kHz}$ and $Z_c = 0.32\,\Omega$. Because the total series resistance (bead DC resistance plus capacitor ESR, about 55 mΩ) is much smaller than Z_c, the resonance is lightly damped: the filter *amplifies* noise near 50 kHz by 15 dB (Figure 22.6), and the impedance seen by the load peaks at 1.8 Ω, far above a decoupled rail. A load that draws current in bursts near f_0 (a radio, an audio amplifier, a switching load) will see the rail ring.

The fix is damping. Adding a second capacitor with deliberate series resistance near Z_c (here 47 µF with 0.3 Ω, an electrolytic or a ceramic with a series resistor) reduces the peak to 0.9 dB and the load-side impedance peak to 0.26 Ω. Alternatively choose a bead with lower inductance, or omit the bead: on many rails a bead adds risk without a measurable benefit.

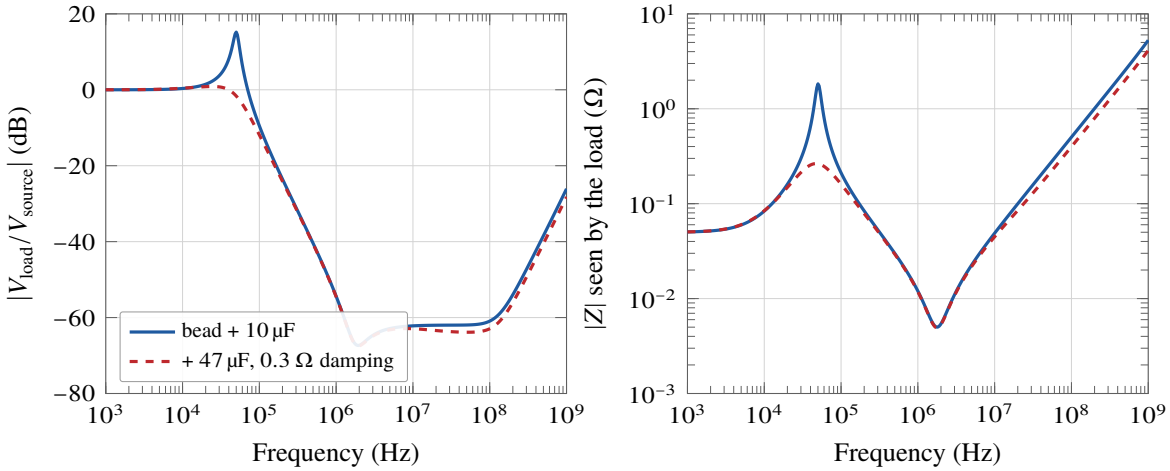


Figure 22.6. Left: transfer from the source rail to the filtered rail for a bead and a 10 µF MLCC, with and without a damping capacitor. Right: impedance seen by the load. Illustrative bead model: 1 µH parallel with 600 Ω and 1.5 pF, plus 50 mΩ. [Analytical]

22.6 Regulator bandwidth and the bulk capacitor

Below its control-loop crossover frequency, a regulator holds its output impedance low; above it, the regulator's output looks like an inductance (and some resistance), and capacitors must take over. The bulk

capacitance near the regulator therefore has two jobs: it is part of the regulator's output filter, which sets its stability (Chapter 24), and it bridges the gap between the loop bandwidth and the frequencies where the ceramic capacitors take over. Changing the bulk capacitor's type (for example, replacing a polymer part with MLCCs of the same capacitance) changes the ESR zero of the regulator's loop and can make a stable regulator ring. Treat bulk capacitors as part of the regulator design, not as decoupling to be added freely.

22.7 Arbitrary bank versus engineered network

Consider a 1.0 V rail with a ±3 % transient allowance and a 4 A load step with a 50 ns edge: $Z_{\text{target}} = 0.03/4 = 7.5\,\text{m}\Omega$. The regulator is modelled beyond its bandwidth as 1.5 mΩ in series with 15 nH, and the plane pair as 20 nF. Two networks are compared (Figure 22.7):

- an **arbitrary bank** of the kind often copied between designs: one 100 µF bulk, two 10 µF, and ten 100 nF capacitors (13 parts);
- an **engineered network** chosen against the target: one 330 µF polymer bulk, eight 22 µF, and eight 1 µF capacitors (17 parts), each MLCC with 0.6 nH of mounting inductance.

Below 20 MHz, the arbitrary bank peaks at 78 mΩ near 7 MHz (an anti-resonance between the 100 nF group and the 10 µF group) and at 15 mΩ near 100 kHz. The engineered network stays below 3 mΩ from 200 kHz to 10 MHz and peaks at 8.7 mΩ near 67 kHz, slightly above the target, where the regulator's inductance resonates with the bulk.

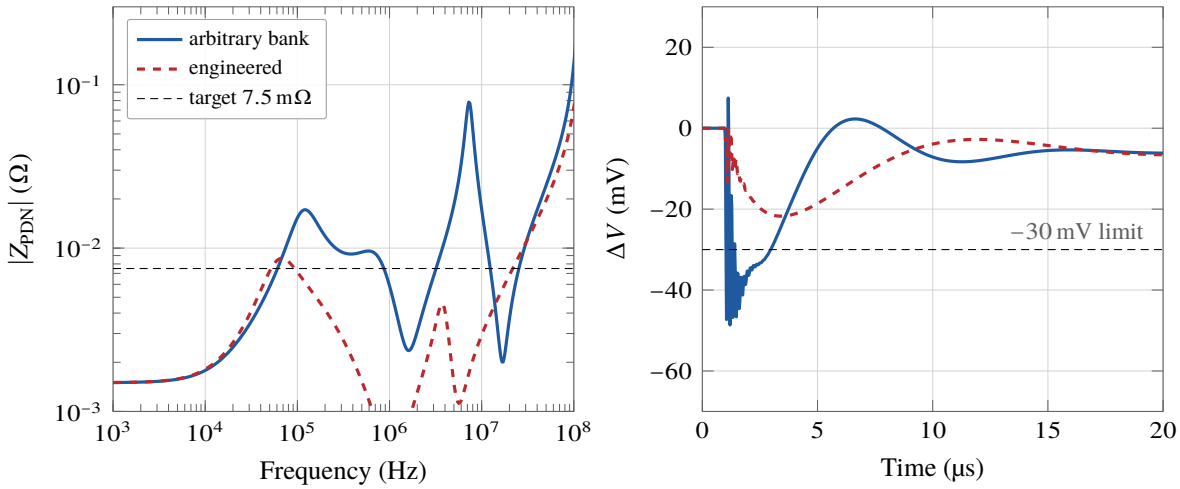


Figure 22.7. Left: PDN impedance of the arbitrary and engineered networks against a 7.5 mΩ target. Right: rail deviation after a 4 A step at 1 µs (time-domain simulation of the same circuit). Illustrative component values; above about 20 MHz the package and die capacitance, not modelled here, take over. [Simulation]

Load-step validation. The time-domain simulation of the same circuit gives a droop of 56 mV for the arbitrary bank, almost twice the 30 mV limit, and 22 mV for the engineered network, inside it. The engineered network passes the step although its impedance exceeds the target in a narrow band: the target-impedance criterion is sufficient, not necessary (Chapter 19). The converse is the important lesson: the arbitrary bank has more capacitors than many designs and still fails, because its parts were not chosen for the frequency band that the step excites. Validate every PDN with both views: impedance against the target, and a time-domain step with the actual edge and the worst repetitive load frequency.

Checklist: capacitor selection

- ☐ Capacitance is taken at the operating DC bias and temperature, after ageing, from the manufacturer's data.
- ☐ Voltage ratings include derating for the technology (especially tantalum) and surge conditions.
- ☐ Ripple-current ratings of bulk capacitors were checked at the regulator's switching frequency.
- ☐ Mounting inductance was minimized and included in the model.
- ☐ Bead-and-capacitor filters were checked for resonance and damped where needed.
- ☐ Bulk capacitors were chosen together with the regulator's compensation.
- ☐ The PDN was validated against the target impedance and with a time-domain load step.

Key Takeaways

- Above self-resonance a capacitor is an inductor; the layout often contributes more inductance than the part.
- Class-II MLCCs can lose half or more of their capacitance under DC bias; small cases lose most.
- Polymer and low-ESR parts give deep impedance minima; high-ESR parts damp resonances.
- A bead and a capacitor form a resonant filter that can amplify noise; damp it or omit it.
- A copied capacitor bank can fail with many parts; an engineered network passes with a similar count.
- Validate in both the frequency and the time domain.

Further Reading

Smith and Bogatin [7] and Novak and Miller [6] treat capacitor models, mounting inductance, and PDN design in depth; Swaminathan and Engin [56] cover the modelling of planes and packages. Capacitor manufacturers' simulation tools provide bias- and temperature-dependent models for their parts.

Review Problems

P22.1 Find the self-resonant frequency of a 4.7 µF capacitor with 0.5 nH ESL and 0.7 nH mounting inductance.

P22.2 A 22 µF 0603 X5R part retains 40 % at 3.3 V. How many are needed to provide 100 µF of effective capacitance on a 3.3 V rail?

P22.3 (Approximation failure) A designer replaces a 100 µF, 40 mΩ polymer output capacitor of a buck regulator with five 22 µF MLCCs. The output now rings after load steps. Why?

P22.4 (Inverse design) For a bead with 0.6 µH low-frequency inductance and a 22 µF capacitor, choose the series resistance of a damping branch.

P22.5 Why does adding ten more 100 nF capacitors to the arbitrary bank not fix its 7 MHz peak?

P22.6 (Imperfect data) A capacitor datasheet gives only "10 µF, X5R, 6.3 V, 0402". What two curves must you obtain before using it on a 5 V rail?

Chapter 23

Planes and Return Paths

Chapter Overview

Chapter 3 showed that high-frequency return current flows in the plane directly beneath a signal. This chapter follows that return current through the situations that disturb it on real boards: gaps and splits in planes, transitions between layers referenced to different planes, and the plane cavity itself, which resonates. It explains why these disturbances cause signal-integrity, power-integrity, and EMI problems at the same time, and gives the stitching-via and stitching-capacitor rules that preserve the return path.

Learning Objectives

1. Explain how a plane split under a trace increases loop inductance, crosstalk, and emissions.
2. Decide when a stitching capacitor can bridge a split, and size it.
3. Trace the return current through a reference-plane change and provide a path for it.
4. Compute cavity resonances of a plane pair and place stitching vias.
5. Evaluate split-plane, partitioned, and solid-plane strategies for mixed-signal boards, and the limits of the 20H rule.

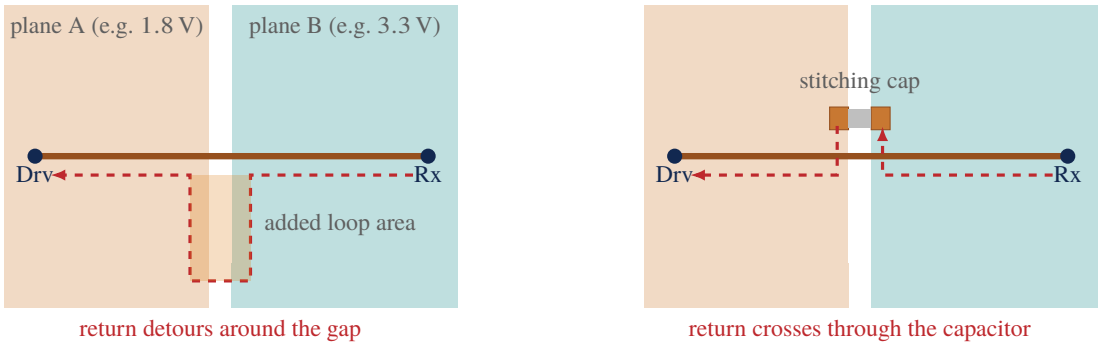
23.1 Crossing a plane split

Boards with several supply voltages often divide one plane layer into regions at different potentials. A signal referenced to that layer that crosses the gap between two regions loses its return path at the gap: the return current, unable to cross, detours to the nearest point where the two regions are connected—the end of the gap, a bypass capacitor, or the regulator (Figure 23.1a); Figure 23.1 compares this with a stitched crossing. The consequences are:

- a large added loop inductance in series with the signal, seen as an impedance discontinuity (an inductive bump in TDR) and a slowed edge;
- shared return detours among all signals crossing the same gap, producing strong crosstalk among them;
- a voltage across the gap, which drives the gap as a slot antenna and couples into any cable attached near it [5], [60].

Design Rule: Never route fast signals across a split

Route every signal above about 50 MHz, or with edges faster than about 1 ns, entirely over a continuous reference plane. If a crossing cannot be avoided, place a stitching capacitor across the gap within one or two trace widths of the crossing for each signal or pair, or route the signal on a layer referenced to a solid ground plane for that segment.



(a) Uncompensated crossing

(b) With a stitching capacitor

Figure 23.1. A trace crossing a gap between two plane regions. (a) The return current detours to the end of the gap, enclosing a large loop. (b) A stitching capacitor next to the crossing restores a short return path at high frequency.

23.1.1 Stitching capacitors

A capacitor across the gap provides a high-frequency return path (Figure 23.1b). It works only where its impedance is small compared with the line's impedance and only if it is placed close to the crossing, because its own mounting inductance and the remaining detour still add loop inductance. An 0402 capacitor with 0.6 nH mounted inductance has $|Z| \approx 1.9\,\Omega$ at 500 MHz—acceptable for a 50 Ω line—but 7.5 Ω at 2 GHz. Stitching capacitors are therefore a repair for moderate-speed signals, not a substitute for a solid reference for multi-gigabit links.

23.2 Reference-plane changes

When a signal moves from one layer to another through a via, its return current must move from the reference plane of the first layer to that of the second. How it does so depends on what the two planes are.

Same plane, both sides
: If the signal moves between layers on opposite sides of one plane (for example from L1 to L3 around L2), the return current simply flows through the anti-pad clearance from one face of the plane to the other. No extra path is needed.

Two ground planes
: The return current flows from one ground plane to the other through the nearest ground via. A ground via placed next to the signal via keeps this loop small (Figure 23.2).

Ground to power, or two different power planes
: No DC connection exists near the via. The return current crosses as displacement current through the plane capacitance—spreading across the cavity—or through the nearest decoupling capacitor. This excites the cavity and couples noise between all signals that make similar transitions.

Common Pitfall: Power planes as references

Assigning a high-speed stripline layer between a ground plane and a power plane, and assuming that the power plane is "AC ground." It is, but only through the plane capacitance and the decoupling network, both of which are inductive at gigahertz frequencies. The return current in the power-plane half of the stripline must eventually reach ground through those paths; any transition then couples signal noise into

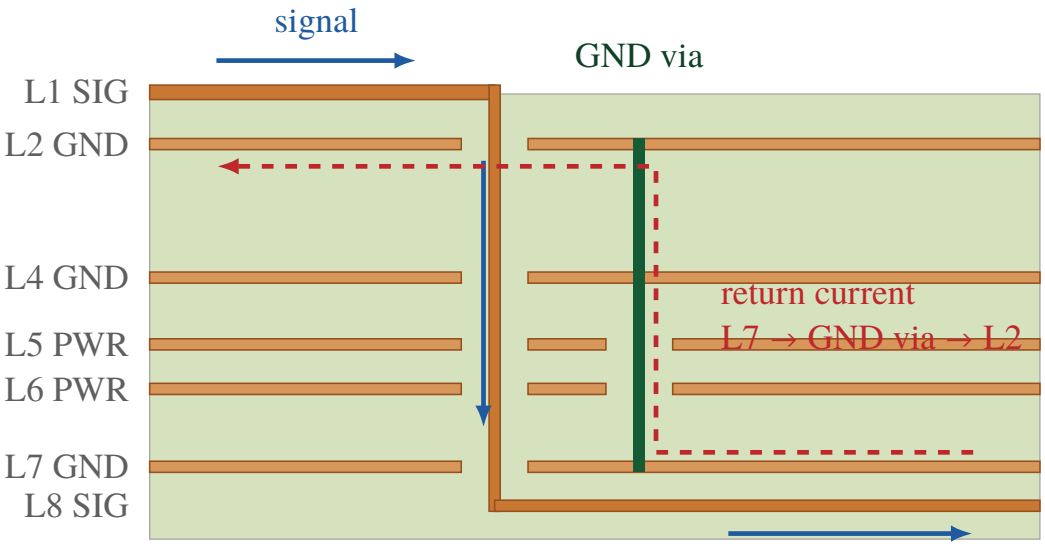


Figure 23.2. Return current at a layer transition from L1 (referenced to L2 GND) to L8 (referenced to L7 GND). A ground via adjacent to the signal via carries the return current between the two ground planes.

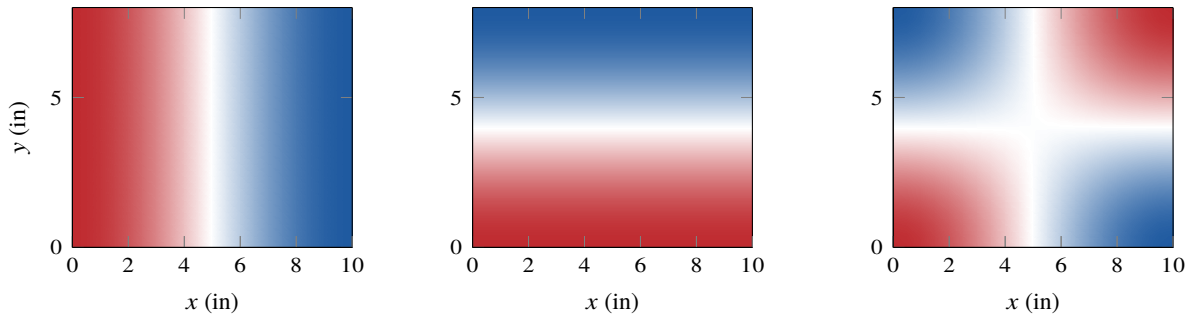
the PDN and PDN noise into the signal. Reference multi-gigabit signals to ground on both sides where possible.

23.3 Plane cavity resonance

A pair of planes separated by a dielectric forms a parallel-plate cavity. For a rectangular cavity of dimensions $a \times b$ with open (unloaded) edges, the resonant frequencies are

$$f_{mn} = \frac{c}{2\sqrt{\varepsilon_r}} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}, \qquad m, n = 0, 1, 2, \ldots, \ (m, n) \neq (0, 0). \tag{23.1}$$

At these frequencies the voltage between the planes forms standing-wave patterns (Figure 23.3), and the PDN impedance seen at a point depends strongly on where that point lies relative to the nodes and antinodes. Noise injected at one via—by a switching device or by a signal changing reference—couples efficiently to any other via near an antinode, and the cavity edges radiate.



(a) TM_{10}, 288 MHz **(b)** TM_{01}, 360 MHz **(c)** TM_{11}, 461 MHz

Figure 23.3. Voltage distribution of the first three modes of a 10 in × 8 in plane pair with $\varepsilon_r = 4.2$, from Eq. (23.1). Red and blue denote opposite polarity.

Worked Example 23.1: Cavity resonances of a board

Find the first three resonances of a 10 in × 8 in plane pair with $\varepsilon_r = 4.2$.

Solution. $c/(2\sqrt{4.2}) = 11.80\,\text{in/ns}/4.099 = 2.88\,\text{GHz}\cdot\text{in}$. Then $f_{10} = 2.88/10 = 288\,\text{MHz}$, $f_{01} = 2.88/8 = 360\,\text{MHz}$, and $f_{11} = 2.88\sqrt{0.01 + 0.0156} = 461\,\text{MHz}$. Every board-size plane pair resonates in the hundreds of megahertz, squarely in the range of clock harmonics.

Cavity resonances are damped by dielectric loss, by decoupling capacitors (which short the cavity locally and shift the resonances up), and by thin dielectrics (which raise the cavity's loss relative to its stored energy). Discrete damping—for example, resistive–capacitive terminations along the board edge—can be used where the PDN impedance at a resonance is critical.

23.3.1 Stitching vias

Ground planes on different layers are connected by stitching vias. A dense pattern of stitching vias makes the ground planes behave as one, suppresses the cavity modes between them, and gives every signal transition a nearby return via. Along board edges, a row of ground vias (a "via fence") suppresses edge radiation from the cavities between ground planes.

Rule of Thumb: Stitching-via spacing

Space ground stitching vias at no more than about $\lambda/20$ at the highest frequency of concern; at 5 GHz in $\varepsilon_r = 4$ material that is about 60 mil. *Limits:* the factor of 20 is a heuristic; denser spacing is needed near connectors, around sensitive circuits, and along edges where emissions are a concern.

23.4 Mixed-signal partitioning

A traditional practice on mixed-signal boards was to split the ground plane into "analog" and "digital" regions joined at a single point near the ADC. With modern fast logic this usually does more harm than good: every signal that crosses between the regions has the problem of Section 23.1, and the single connection point becomes a high-inductance common path. The approach now favored is a *single, solid ground plane* with *partitioned placement*: digital and analog circuits placed in separate regions over the same plane, and routing arranged so that digital return currents, which follow their traces, never pass under the analog region [5]. Splits remain appropriate for safety isolation (for example, under Ethernet magnetics or isolated interfaces), where the crossing signals pass through isolation components rather than copper.

23.4.1 The 20H rule

The "20H rule" recommends recessing the edges of power planes by about twenty times the power–ground spacing H from the ground plane's edge, to reduce fringing-field radiation from the plane edges. Measurements and simulations have shown that its benefit is limited and frequency dependent: it can reduce emission at some frequencies but may increase it near cavity resonances, and it reduces plane capacitance [60], [61]. Edge stitching with via fences on ground–ground cavities, thin power–ground dielectrics, and adequate decoupling are more dependable.

Table 23.1. Return-path situations and remedies.

Situation	Problem	Remedy
Trace over plane split	Detour; crosstalk; slot radiation	Re-route over solid plane; stitching cap at crossing
Trace near plane edge	Asymmetric return; edge radiation	Keep $\geq 3H$ to $5H$ from edge
Via between two GND planes	Return through cavity	GND via adjacent to signal via
Via between GND and PWR	Return through cavity / decoupling	Avoid; else stitching cap at via
Anti-pad clusters (BGA, connectors)	Plane perforated	Reduce anti-pads; stagger; back-drill
Plane cavity resonance	PDN peaks; coupling; radiation	Thin dielectric, decoupling, stitching, damping

Worked Example 23.2: How good is a stitching capacitor?

A signal crosses a split on a 50 Ω line, with a stitching capacitor of 0.6 nH mounted inductance placed at the crossing. Estimate the reflection it causes at 500 MHz and at 5 GHz, treating it as a series inductance in the return path.

Solution. The series impedance is $\omega L = 1.9\,\Omega$ at 500 MHz and $18.8\,\Omega$ at 5 GHz. A small series impedance Z on a line of impedance Z_0 reflects roughly $Z/(2Z_0)$: about 2 % at 500 MHz but 19 % at 5 GHz. The capacitor is adequate for a 100 MHz clock with nanosecond edges but not for a multi-gigabit link.

Key Takeaways

- A plane split under a trace forces the return current to detour, adding inductance, crosstalk, and slot radiation.
- Stitching capacitors bridge splits only at moderate frequencies; prefer solid references for fast signals.
- At a layer change, provide a ground via next to the signal via; avoid transitions between ground- and power-referenced layers.
- Plane pairs resonate at $f_{mn} = \frac{c}{2\sqrt{\varepsilon_r}}\sqrt{(m/a)^2 + (n/b)^2}$, typically hundreds of megahertz.
- Use a solid ground plane with partitioned placement rather than split grounds; treat the 20H rule with caution.

Further Reading

Archambeault [60], Ott [5], and Montrose [61] discuss plane splits, reference changes, and cavity resonances from the EMC point of view; Swaminathan and Engin [56] model plane cavities.

Review Problems

P23.1 Find f_{10}, f_{01}, and f_{11} for a 6 in × 4 in plane pair with $\varepsilon_r = 4.0$.

P23.2 What stitching-via spacing corresponds to $\lambda/20$ at 10 GHz in $\varepsilon_r = 3.6$ material?

P23.3 A stitching capacitor has 0.4 nH of mounted inductance. What is its impedance at 2 GHz?

P23.4 A slot crossing raises a 50 Ω trace's local impedance to 80 Ω. What reflection coefficient results?

P23.5 By how much would the 20H rule recess a power plane separated from ground by 4 mil?

P23.6 What is the impedance of 2 nF of plane capacitance at 1 GHz, ignoring inductance?

Part IV

Power Conversion, Sequencing, and Thermal Design

Chapter 24

Power-Rail Architecture and Regulator Design

Chapter Overview

The power-integrity chapters treated the network between a regulator and its load. This chapter steps back to the regulators themselves: how many rails a board needs, which regulator type serves each rail, how the inductor and capacitors are rated, how the control loop is compensated, and what happens at the edges of normal operation (inrush at plug-in, hot-swap, and back-powering through signal pins). Everything is tied to one worked example, a generic single-board computer (SBC) with a system-on-chip (SoC), DDR4 memory, USB host ports, and radios, built from a 12 V input. The loads and efficiencies are illustrative maxima and typical datasheet values; the method, not the numbers, is what carries over to a real design.

Learning Objectives

1. Build a rail budget from device datasheets and turn it into a power tree.
2. Choose among buck, boost, buck-boost, LDO, PMIC, and multiphase regulators.
3. Size a buck converter's inductor and capacitors from their ratings.
4. Explain loop compensation, crossover, and phase margin, and check them.
5. Control inrush current, specify hot-swap protection, and prevent back-powering.
6. Lay out the hot loop of a switching regulator.

24.1 From device datasheets to a rail budget

A rail budget is a table with one row per supply rail and, for each, the voltage, tolerance, maximum steady current, transient step and edge rate, noise limit, and the devices it feeds. Its sources are the datasheets of the loads:

- **Voltage and tolerance**, from the recommended operating conditions (not the absolute maximum ratings). The tolerance must cover DC setpoint error, ripple, and transient droop together (Chapter 18).
- **Maximum current**, from the power or current tables under the worst-case use case, not from "typical" figures.
- **Transient behaviour**, often given only as a maximum $\mathrm{d}I/\mathrm{d}t$ or step size in the hardware design guide.
- **Sequencing and ramp constraints** (Chapter 25).
- **Noise sensitivity**, for PLL, SerDes, and analog supplies that may need a filtered or separate rail.

Every regulator's input current then becomes a load on its source rail, so the budget is computed from the leaves of the tree toward the input. Appendix I contains a blank template.

The worked example. Table 24.1 and Figure 24.1 show the SBC. The 5 V rail supplies the USB ports and the fan (2.1 A) and also feeds the core, memory, and 3.3 V bucks; the 3.3 V rail feeds the 1.8 V buck and the 2.5 V VPP regulator. Computing from the leaves upward, the 5 V buck must deliver 5.7 A, almost three times its own direct load, and the board draws 30.3 W (2.53 A) from 12 V, of which 4.1 W is lost in regulators.

Table 24.1. Rail budget of the SBC example (illustrative loads, typical efficiencies; computed by *tools/gen_r3_rails.py*). "Total current" includes the input currents of downstream regulators.

Rail	V (V)	Load (A)	Total (A)	P_{out} (W)	P_{in} (W)	Loss (W)	η	Source / type
5V0	5.00	2.10	5.70	28.50	30.32	1.82	0.94	12 V, buck
3V3	3.30	2.00	2.60	8.57	9.31	0.75	0.92	5V0, buck
VDD_DDR	1.20	2.00	2.40	2.88	3.24	0.36	0.89	5V0, buck
VDD_CORE	0.80	6.00	6.00	4.80	5.45	0.65	0.88	5V0, buck
VDD_1V8	1.80	0.80	0.80	1.44	1.64	0.20	0.88	3V3, buck
VPP_DDR	2.50	0.10	0.10	0.25	0.33	0.08	0.76	3V3, LDO
VTT_DDR	0.60	±0.40	0.40	0.24	0.48	0.24	0.50	VDD_DDR, sink/source LDO

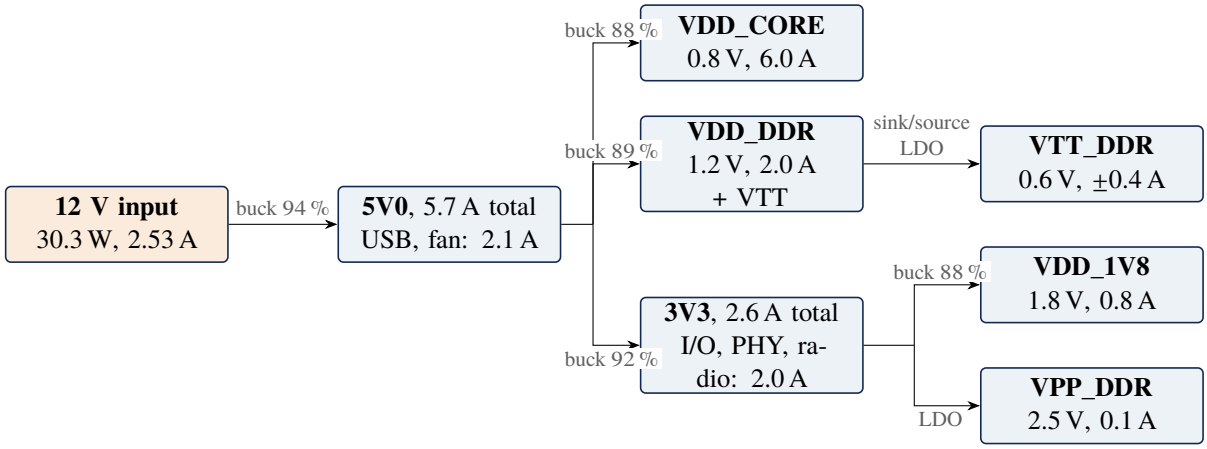


Figure 24.1. Power tree of the SBC example. Efficiencies are typical values at the stated load; the 5 V rail carries the input current of three downstream bucks. [Analytical]

24.2 Choosing a regulator type

Figure 24.2 shows typical converter hardware.

Two decisions in the example illustrate the trade-offs.

Single stage or two stage for the core? A buck from 12 V to 0.8 V has a duty cycle of $D = V_{out}/V_{in} = 6.7\,\%$, which at 1 MHz is an on-time of 67 ns: close to the minimum on-time of many controllers, and with poor efficiency because the switch must block 12 V while conducting 6 A for a short time. From 5 V the duty

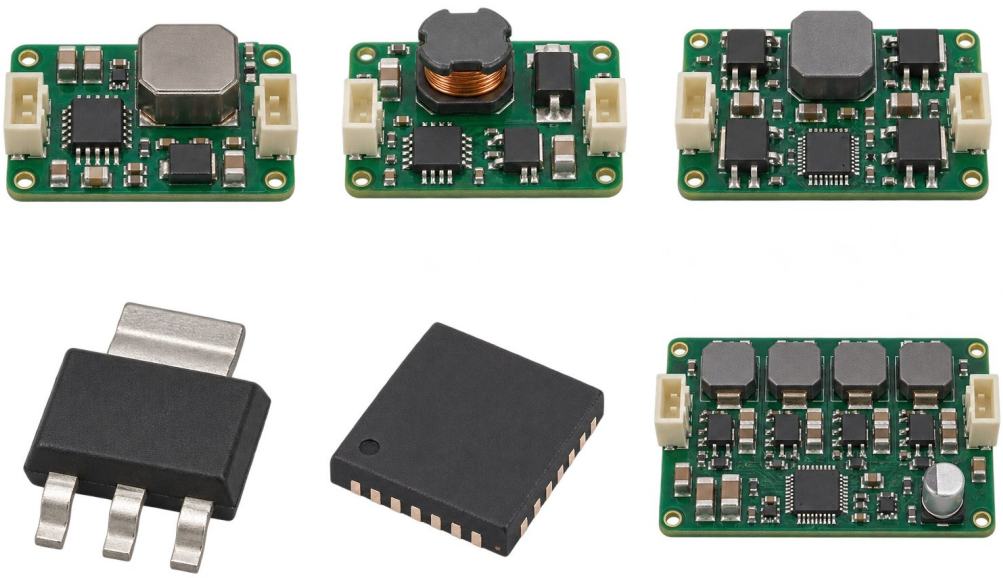


Figure 24.2. Power-conversion hardware: switching-converter modules with shielded and open inductors, an SOT-223 linear regulator, a QFN converter, and a four-phase converter board. Rendered illustration supplied with the manuscript, not a photograph of specific products; markings, contact counts, and proportions are not authoritative. [Illustration]

Table 24.2. Regulator types and where they fit.

Type	Strengths	Limits and cautions
Buck (step-down)	high efficiency over wide ratios; most common	minimum on-time limits very low duty cycles; switching noise; needs layout care
Boost (step-up)	output above input	right-half-plane zero limits loop bandwidth; input-to-output path when off (needs a disconnect switch)
Buck-boost	output inside the input range (batteries)	more switches and loss; mode transitions
LDO	low noise; high PSRR at low frequency; simple	dissipates $(V_{\text{in}} - V_{\text{out}})I$; PSRR falls at high frequency; some need specific output-capacitor ESR
PMIC	several regulators, sequencing, and monitoring in one part; programmable	fixed rail counts and currents; firmware and I^2C dependence
Multiphase buck	tens to hundreds of amperes; ripple cancellation; fast transient	current balancing; controller complexity

cycle is 16 % (160 ns), which is comfortable. The two-stage tree costs some efficiency in the 5 V stage but uses smaller, faster low-voltage parts for the high-current rail.

LDO or buck for 1.8 V**?** From 3.3 V at 0.8 A, an LDO would dissipate $(3.3 - 1.8) \times 0.8 = 1.2$ W at 55 % efficiency. Even in a package with a datasheet $R_{\theta JA}$ of 50 °C/W (a JEDEC test-board figure; see Chapter 26) the rise would be about 60 °C. A buck dissipates 0.2 W. The 2.5 V VPP rail at 0.1 A is the opposite case: an LDO dissipates 80 mW and avoids another switching converter.

24.3 Sizing a buck converter

Figure 24.3 shows a synchronous buck. The high-side switch (HS) and low-side switch (LS) alternate; the switch node (SW) swings between V_{in} and ground; the inductor current is continuous.

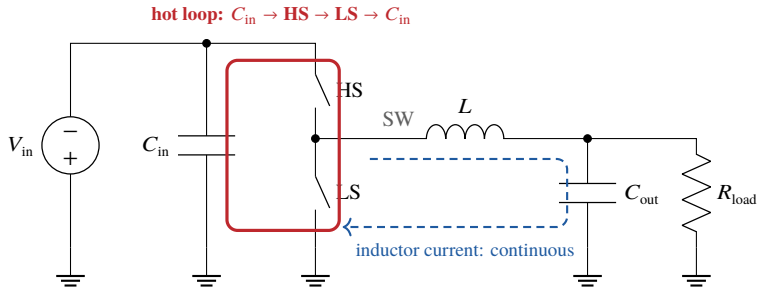


Figure 24.3. Synchronous buck converter. The input capacitor, the two switches, and the ground return form the hot loop, whose current switches on and off at every transition; the inductor current is continuous. [Schematic]

Inductor. The peak-to-peak ripple current is

$$\Delta I_L = \frac{(V_{\text{in}} - V_{\text{out}})\, D}{f_{\text{sw}} L}, \qquad D = \frac{V_{\text{out}}}{V_{\text{in}}}. \tag{24.1}$$

A common choice is ΔI_L between 20 and 40 % of the maximum load. For VDD_CORE ($V_{\text{in}} = 5$ V, 0.8 V, 6 A, 1 MHz) a 0.33 µH inductor gives $\Delta I_L = 2.04$ A (34 %) and a peak current of $6 + 1.02 = 7.0$ A. The inductor needs three ratings: *saturation current* (where inductance has fallen by a stated percentage, commonly 20–30 %) above the peak current under fault and transient conditions, plus margin, so about 9 A; *RMS current* for its temperature rise (here about 6.0 A); and acceptable *core loss* at the ripple and frequency. Shielded parts reduce coupling into nearby traces.

Output capacitor. The ripple voltage is approximately

$$\Delta V_{\text{out}} \approx \frac{\Delta I_L}{8 f_{\text{sw}} C_{\text{out}}} + \Delta I_L \,\text{ESR}. \tag{24.2}$$

With 220 µF *effective* capacitance (after DC-bias derating, Chapter 22) and 1.5 mΩ ESR, the ripple is 4.2 mV. The load-step response is set by the loop: for a step ΔI the deviation is roughly $\Delta I / (2\pi f_c C_{\text{out}})$, where f_c is the crossover frequency. With $f_c = 186$ kHz (below), a 3 A step gives about 11.6 mV, inside a ±3 % (24 mV) allowance.

Input capacitor. The input capacitor supplies the pulsed switch current. Its RMS current is

$$I_{C,\mathrm{rms}} = I_{\mathrm{out}}\sqrt{D(1-D)} = 6\sqrt{0.16 \times 0.84} = 2.2\,\mathrm{A};$$

the ceramic capacitors must share it within their ratings, and a bulk capacitor upstream damps the input filter formed with the supply wiring.

24.4 Compensation and stability

The regulator is a feedback loop. Its loop gain $T(f)$ is the product of the feedback divider, the error amplifier with its compensation network, and the power stage. Two numbers summarize stability: the *crossover frequency* f_c where $|T| = 1$, and the *phase margin*, 180° plus the phase of T at f_c. Figure 24.4 shows a simplified peak-current-mode model of the VDD_CORE buck with type-II compensation: $f_c \approx 186\,\mathrm{kHz}$ (about $f_{\mathrm{sw}}/5$) and a phase margin of about 60°.

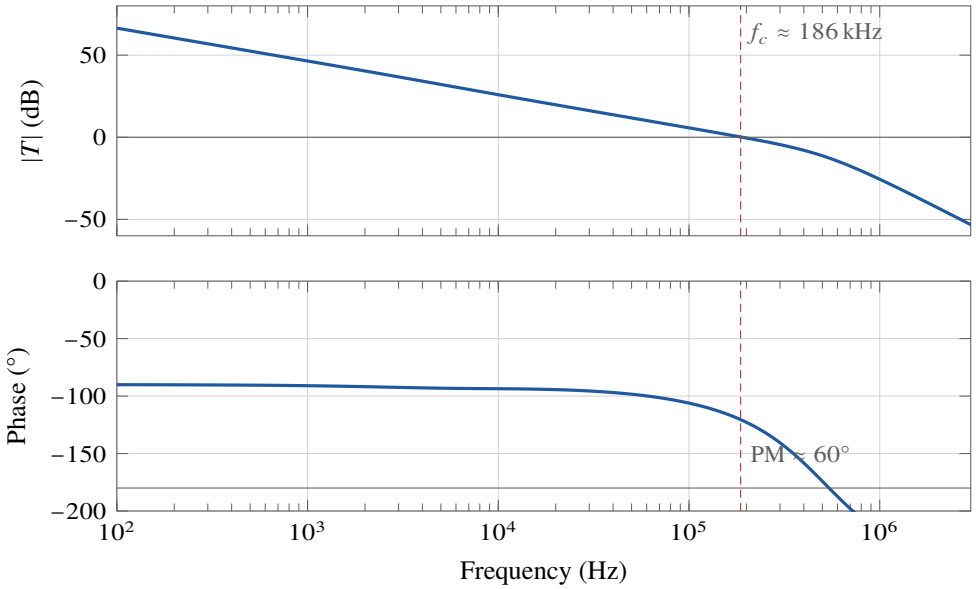


Figure 24.4. Loop gain of a simplified peak-current-mode buck model ($V_{\mathrm{out}} = 0.8\,\mathrm{V}$, 6 A, 220 µF, 1.5 mΩ, 1 MHz) with type-II compensation. The double pole at $f_{\mathrm{sw}}/2$ represents the sampling effect of current-mode control. [Analytical]

Practical rules, all of which the regulator's datasheet refines:

- Keep f_c below about $f_{\mathrm{sw}}/5$ to $f_{\mathrm{sw}}/10$; for a boost or buck-boost also well below the right-half-plane zero.
- Aim for 45° to 60° of phase margin over the full range of load, input voltage, temperature, and output-capacitor variation (including DC-bias loss).
- Current-mode bucks use type-II compensation (one zero to cancel the output pole); voltage-mode bucks with ceramic output capacitors need type III (two zeros) to recover the phase lost at the LC double pole.
- When the output capacitor changes type, the ESR zero moves; recheck the loop.
- Verify on hardware: inject a small signal across a resistor in the feedback path with a frequency-response analyzer, or at least observe the load-step response: a ringing settle indicates low phase margin.

24.5 Inrush, hot-swap, and back-powering

Inrush. When an input is connected, the bulk and ceramic capacitors charge from zero. The current is $I = C\ \mathrm{d}V/\mathrm{d}t$, limited only by source and cable resistance and inductance unless the design controls $\mathrm{d}V/\mathrm{d}t$. With 1000 µF on a 12 V input charged in 10 ms, the inrush is 1.2 A; charged in 100 µs, it is 120 A. Regulators control their own output inrush with a soft-start ramp; the input needs a controlled switch.

Hot-swap. A hot-swap controller drives a series MOSFET to ramp the output slowly and limits current through a sense resistor (Figure 24.5a). The MOSFET dissipates while the capacitors charge: the energy delivered to the capacitor, $\frac{1}{2}CV^2 = 72$ mJ here, is matched by an equal energy dissipated in the MOSFET over the ramp, plus the energy from any load current during the ramp. Check the MOSFET's safe operating area (SOA) curve at the ramp duration and temperature; SOA, not the current rating, usually sizes the part.

Plug-in overshoot. A cable's inductance and a board's low-ESR ceramic input capacitors form a lightly damped LC circuit. When a powered adapter is connected, the input can ring to nearly twice the supply voltage. A bulk electrolytic or polymer capacitor with some ESR, or a TVS diode, keeps the peak inside the regulator's absolute-maximum input rating.

Back-powering. When one domain is unpowered and another drives its inputs, current flows through the inputs' ESD or clamp diodes into the unpowered rail (Figure 24.5b). With a 3.3 V output, 33 Ω in series, a 0.6 V diode drop, and 100 Ω of load on the unpowered rail, the rail rises to about 2.0 V and 20 mA flows: enough to put the device into an undefined state, defeat its power-on reset, or trigger latch-up when its supply finally ramps. Remedies: inputs specified as fail-safe or with power-off isolation (I_{off}), buffers or switches that isolate when unpowered, sequencing that holds drivers low until the receiver is powered, and current-limiting series resistors where the device's injection-current rating allows.

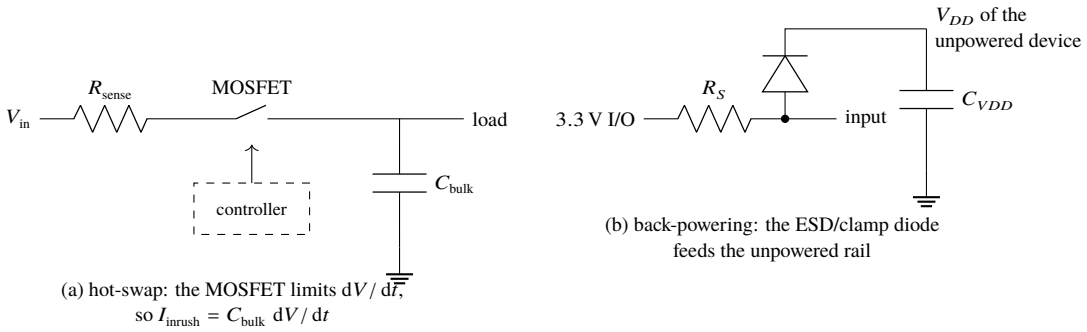


Figure 24.5. (a) A hot-swap controller ramps the output through a MOSFET and limits current. (b) Back-powering: a driven input supplies an unpowered rail through the input's clamp diode. [Schematic]

24.6 Layout of switching regulators

The hot loop of Figure 24.3 carries current that switches in a few nanoseconds; its inductance turns $\mathrm{d}I/\mathrm{d}t$ into voltage spikes on the switch node and into radiated and conducted noise. Layout priorities:

1. Place the input ceramic capacitors directly at the switch's input and ground pins, with the return on the adjacent layer, so the hot-loop area is minimal.

2. Keep the switch-node copper as small as the current allows; it is a strong $\mathrm{d}V/\mathrm{d}t$ source and couples capacitively to everything nearby.
3. Route the feedback trace from the output capacitor (or the load, for remote sensing) as a quiet trace away from the switch node and the inductor.
4. Give the controller a quiet analog ground connected to the power ground at one point specified by the datasheet.
5. Use the thermal pad with an array of vias to an inner plane (Chapter 26).
6. Keep sensitive circuits (PLLs, oscillators, high-speed receivers) and their traces away from the inductor and switch node; consider spread-spectrum or synchronized switching frequencies where interference is a concern.

Checklist: rails and regulators

- ☐ Rail budget built from worst-case datasheet values, with downstream regulator input currents included.
- ☐ Each regulator chosen for ratio, current, noise, and dissipation; LDO dissipation checked.
- ☐ Inductor saturation, RMS, and core-loss ratings checked; output capacitance taken after derating.
- ☐ Loop crossover and phase margin checked across corners, and verified on hardware.
- ☐ Inrush controlled; hot-swap MOSFET SOA checked; plug-in overshoot below absolute maximum.
- ☐ No back-powering paths, or paths limited and accepted by the device datasheets.
- ☐ Hot loop minimal; feedback quiet; switch node small.

Key Takeaways

- Build the budget from the leaves to the input; upstream rails carry downstream regulators' input currents.
- Avoid very low duty cycles; two-stage conversion is often better than one.
- An LDO's dissipation is $(V_{\mathrm{in}} - V_{\mathrm{out}})I$; check it before choosing one.
- Size the inductor by saturation and RMS current and the output capacitor by ripple and by the loop's response to load steps.
- Crossover below $f_{\mathrm{sw}}/5$ and 45°–60° of phase margin, verified on hardware.
- Inrush, plug-in overshoot, and back-powering are design requirements, not afterthoughts.

Further Reading

Regulator vendors' datasheets and application notes are the primary references for compensation and layout of specific parts. Smith and Bogatin [7] discuss the regulator's role in the PDN; Ott [5] covers switching-noise coupling and layout for EMC.

Review Problems

P24.1 A rail feeds a 1 A load and an LDO that supplies 0.5 A to another rail. What current must the source rail deliver?

P24.2 Compute the duty cycle and on-time of a 24 V-to-1.0 V buck at 800 kHz. Is it practical?

P24.3 (Inverse design) Choose L for 30 % ripple in a 12 V to 3.3 V, 3 A, 500 kHz buck.

P24.4 An LDO converts 5 V to 3.3 V at 0.4 A. Find its dissipation and efficiency.

P24.5 (Approximation failure) A designer estimates the load-step deviation with $\Delta I \cdot \mathrm{ESR}$ only and finds 4.5 mV for a 3 A step. Why is the estimate optimistic?

P24.6 A hot-swap circuit charges 470 µF to 48 V in 20 ms. Find the inrush current and the energy dissipated in the MOSFET (no load during the ramp).

P24.7 (Imperfect data) A board resets intermittently when a USB device is plugged in while the board is powered from a 12 V adapter. List two power-related causes to check.

Chapter 25

Power Sequencing and Reset

Chapter Overview

A board with several rails must bring them up, and take them down, in an order and with timing that every device on the board accepts, and it must hold its processors in reset until the supplies are valid. Sequencing requirements are not a style choice: they come from the device datasheets, and violating them causes latch-up, back-powering, corrupted configuration, or devices that start in an undefined state. This chapter derives a sequence from device constraints for the single-board computer of Chapter 24, compares the mechanisms that implement it (enable and power-good chains, PMIC sequencers, supervisors, tracking), quantifies why an RC delay is a weak timer, walks through fault cases, and ends with oscilloscope acceptance criteria for bring-up.

Learning Objectives

1. Extract sequencing, ramp, and reset constraints from device datasheets into a dependency graph.
2. Implement a sequence with enable/power-good chains, a sequencer, or tracking, and recognize where the power tree conflicts with the required order.
3. Compare RC delays with supervised delays quantitatively.
4. Define behaviour for faults: a rail that fails, a brown-out, and loss of input.
5. Write measurable acceptance criteria for power-up and power-down.

25.1 Where sequencing requirements come from

A device's power-up requirements are usually in a section titled "power sequencing", "power-up/down", or "recommended operating conditions". The usual kinds are:

- **Order:** rail A must reach its valid level before rail B starts (or before B exceeds a voltage).
- **Relative voltage:** rail A must not exceed rail B by more than a stated amount at any time, typically because of ESD or parasitic diodes between domains.
- **Ramp:** minimum and maximum ramp times or slew rates, and monotonicity (no reversal during the ramp).
- **Reset timing:** reset held asserted until all rails are valid and clocks are stable, for at least a minimum time.
- **Power-down:** often the reverse order, or simultaneous, with limits on residual voltage before the next power-up.
- **I/O state:** inputs must not be driven above the device's own rail while it is unpowered (Section 24.5).

Memory devices add their own rules; for example, DDR4 DRAMs require the 2.5 V VPP supply to come up no later than VDD and to stay at or above it, VTT to follow VDDQ, and RESET_n and CKE to be held low during the ramp (consult the JEDEC standard and the DRAM datasheet for the exact wording and times).

Collecting the constraints. Build a table with one row per constraint: the two rails or signals, the relation, the numbers, and the source document with revision and page. Conflicting constraints between devices must be resolved before any circuit is drawn.

25.2 A dependency graph for the example

Figure 25.1 combines the power tree of Figure 24.1 with illustrative constraints for a generic SoC and DDR4 memory: the core rail before the 1.8 V rail; the 1.8 V rail before the 3.3 V I/O; VPP before VDD_DDR; VTT after VDD_DDR; and reset released only after all of them.

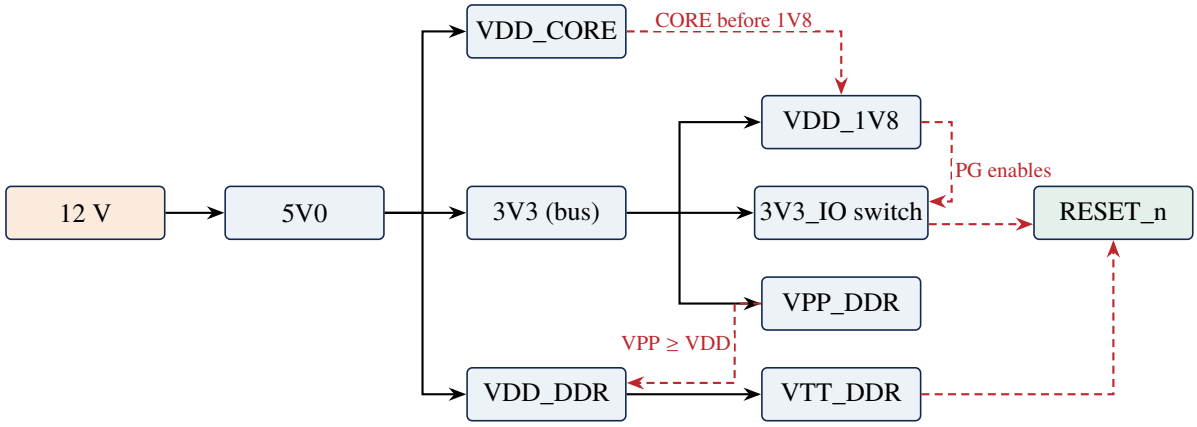


Figure 25.1. Dependency graph of the SBC example. The 1.8 V rail is generated from the 3.3 V bus, but the SoC requires 1.8 V before its 3.3 V I/O; a load switch on the I/O branch, enabled by the 1.8 V power-good, resolves the conflict. [Conceptual]

The graph exposes a conflict that the power tree alone hides: the 1.8 V buck is fed from the 3.3 V bus, so the bus must be up *before* 1.8 V, while the SoC wants 1.8 V before its 3.3 V I/O. The solution is to separate the 3.3 V bus (which feeds regulators and devices without the constraint) from the SoC's 3.3 V I/O branch, and to switch the latter with a load switch enabled by the 1.8 V power-good signal. Figure 25.2 shows the resulting power-up.

25.3 Mechanisms

Sequencers, supervisors, load switches, and protection devices are usually small leadless or leaded packages (Figure 25.3).

Enable and power-good chains. Most regulators have an enable input (EN) and a power-good output (PG, open-drain) that asserts when the output is within a window of its setpoint. Connecting PG of one regulator to EN of the next makes a daisy chain whose order is enforced by measurement, not by time. Delays between stages come from the regulator's soft-start time, a programmable PG delay, or a capacitor on a delay pin.

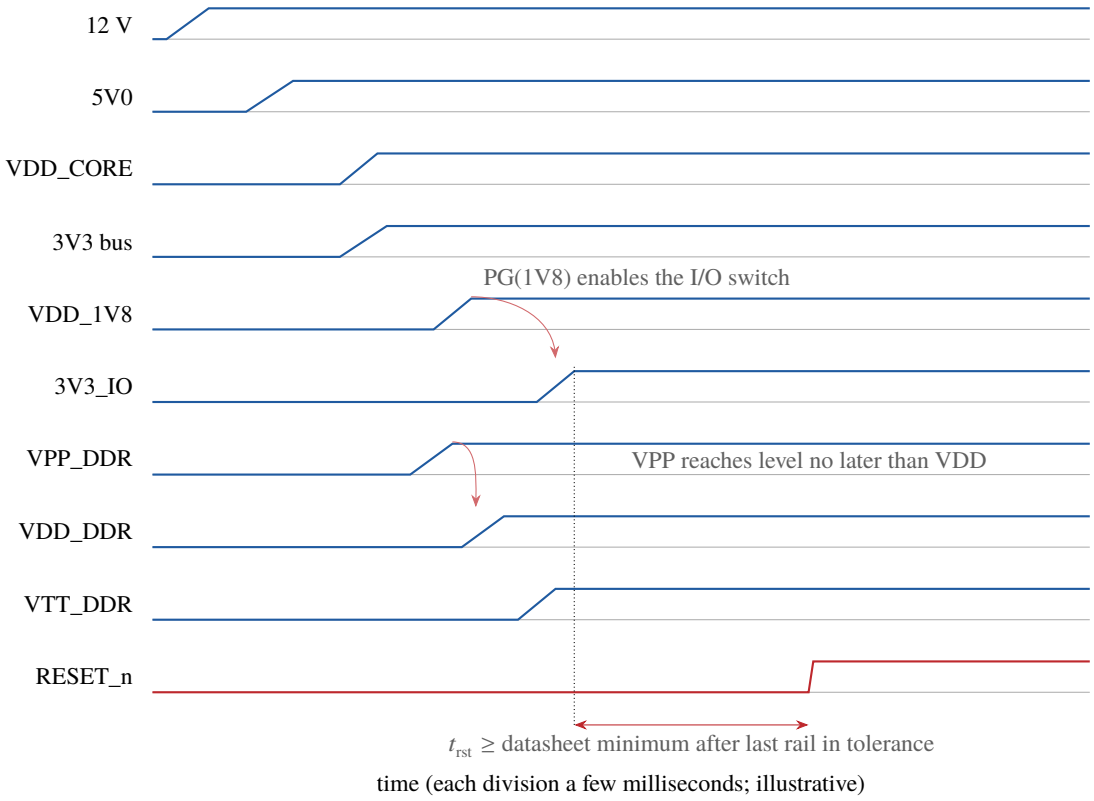


Figure 25.2. Power-up timing for the SBC example (illustrative). Each rail starts after the power-good of the rail it depends on; reset is released a minimum time after the last rail is within tolerance. [Conceptual]

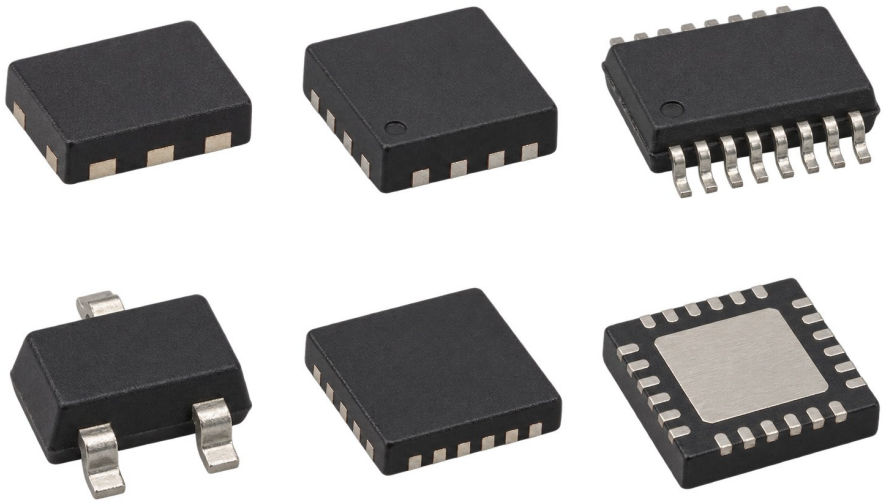


Figure 25.3. Packages typical of supervisors, load switches, eFuses, and sequencers: DFN and QFN packages, a TSSOP, an SOT-23, and a QFN with an exposed thermal pad. Rendered illustration supplied with the manuscript, not a photograph of specific products; markings, contact counts, and proportions are not authoritative. [Illustration]

Sequencers and PMICs. A PMIC or a dedicated sequencer (or a small always-on microcontroller) drives the enables in a programmed order, monitors the outputs, and applies a defined response to faults. Their advantage is reconfigurability and a single place where the sequence is defined; their cost is that the sequencer itself must power up first and be configured correctly.

Supervisors. A voltage supervisor compares a rail with an accurate threshold (commonly specified to about 1–1.5 %) and holds its reset output asserted for a defined timeout after the rail is valid. Several supervisors with open-drain outputs can be wired together so that reset is released only when every monitored rail is valid. Supervisors also catch brown-outs, which no timer does.

Tracking. Some regulators accept a tracking input so that two rails ramp together (coincident) or in a fixed ratio (ratiometric). Tracking limits the voltage difference between domains during the ramp, which is what "rail A must not exceed rail B by more than 0.3 V" requires.

25.4 RC delays versus supervised delays

An RC network on an enable pin delays the enable until the capacitor voltage crosses the pin's threshold:

$$t_d = RC \ln \frac{V_S}{V_S - V_{\text{th}}}. \tag{25.1}$$

With $R = 10\,\text{k}\Omega$, $C = 1\,\mu\text{F}$, $V_S = 3.3\,\text{V}$, and $V_{\text{th}} = 1.2\,\text{V}$, the nominal delay is 4.52 ms. With realistic variations (R ±1 %; C −20/+10 % including DC bias and temperature; V_S ±3 %; V_{th} ±10 %, a common EN-threshold tolerance) the worst-case delay ranges from 3.03 to 5.91 ms, −33/+31 %. Worse, the delay depends on how fast the supply V_S itself ramps (Figure 25.4): the crossing moves from 4.6 ms for a 0.1 ms ramp to 7.1 ms for 5 ms and 15.0 ms for 20 ms.

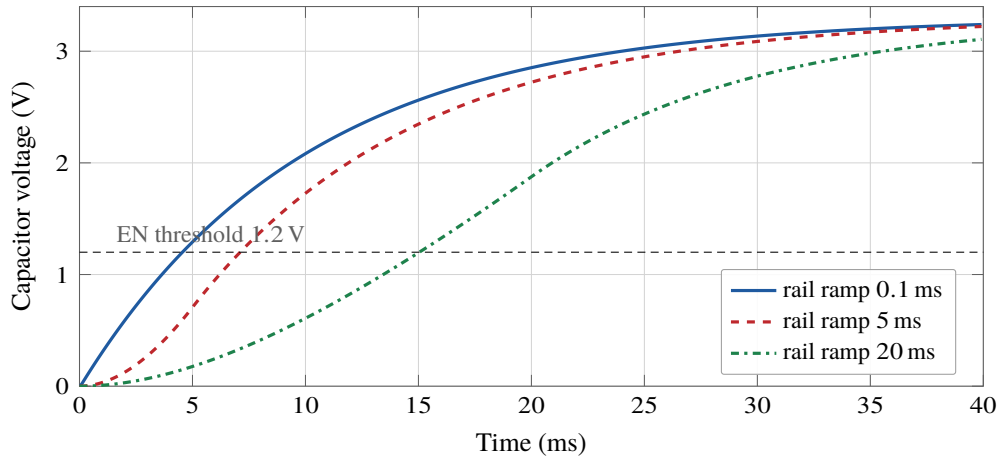


Figure 25.4. An RC enable delay ($\tau = 10\,\text{ms}$) fed from a rail that ramps in 0.1, 5, or 20 ms. The delay depends on the ramp, which an RC network cannot see. [Simulation]

An RC delay also has three functional weaknesses: it does not check that the previous rail is actually valid; it does not re-assert on a brown-out; and it may not discharge between fast power cycles, so the second power-up has almost no delay. A supervised or PG-based sequence avoids all three. RC delays remain acceptable where the constraint has generous margin and no device depends on the order for correct operation.

25.5 Fault cases

A sequence must define what happens when something goes wrong:

- **A rail never becomes valid.** A PG chain stops at that rail. Define a timeout after which the sequencer turns everything off in reverse order and either latches off or retries a limited number of times.
- **A rail collapses during operation** (overload, short): the regulator's current limit or hiccup mode operates; the supervisor asserts reset; the sequencer decides whether the remaining rails stay on (and possibly back-power the failed domain) or shut down in order.
- **Brown-out of the input:** the input UVLO of each regulator, with hysteresis, prevents oscillation around the threshold; the supervisor holds reset until the input is valid again.
- **Loss of input:** the rails discharge at rates set by their loads and capacitances, not by the sequence. If a device requires reverse-order power-down, add discharge switches or a power-fail warning that lets the sequencer shut down while the bulk capacitors hold up the input.
- **Fast power cycling:** residual voltages on slowly discharging rails can leave devices partly powered at the next power-up; specify a minimum off time or active discharge.

25.6 Acceptance criteria for bring-up

Sequencing is verified with a multichannel oscilloscope at first power-up and again across input-voltage and temperature extremes. Each criterion is stated with a number derived from a datasheet and a measurement margin:

1. **Order:** each dependent rail starts only after its predecessor is within tolerance; measured gap at least the required minimum plus the oscilloscope's timing uncertainty.
2. **Ramp:** rise times between the datasheet minimum and maximum; monotonic, with any reversal smaller than a stated value (for example 2 % of the rail).
3. **Relative voltage:** the difference between tracked rails stays within the limit throughout the ramp (use math channels).
4. **Overshoot:** below the device's absolute-maximum rating, with margin.
5. **Reset:** released at least the minimum time after the last rail is valid and clocks are stable; asserted within the required time when any rail falls below its threshold.
6. **Power-down:** order and residual voltages as required; check with the input removed abruptly, not only with a controlled shutdown.
7. **Repetition:** at least tens of cycles, including fast off–on cycles, without a violation.

Record the oscilloscope settings and screenshots in the bring-up report (Chapter 38); the template in Appendix I lists the fields.

Checklist: sequencing and reset

- □ Every constraint is in a table with its source document and revision.
- □ The dependency graph is consistent with the power tree; conflicts are resolved with switches or re-parenting.
- □ Order is enforced by power-good or a sequencer wherever a device depends on it; RC delays only where margin is large.
- □ Supervisors cover every rail whose loss must reset the system.

- ☐ Fault, brown-out, input-loss, and fast-cycle behaviour is defined.
- ☐ Acceptance criteria are written with numbers before bring-up.

Key Takeaways

- Sequencing requirements come from device datasheets: order, relative voltage, ramp, reset, power-down, and I/O state.
- Draw the dependency graph on top of the power tree; conflicts between the two are common.
- Power-good chains enforce order by measurement; RC delays only by time, with a $\pm 30\,\%$ spread and a dependence on the supply ramp.
- Supervisors catch brown-outs; timers do not.
- Define fault behaviour and verify with measurable acceptance criteria, including fast power cycles.

Further Reading

Device datasheets and hardware design guides are the primary sources. Regulator, PMIC, and supervisor vendors publish application notes on sequencing methods and on monitoring accuracy.

Review Problems

P25.1 Compute the RC delay for $R = 47\,\mathrm{k\Omega}$, $C = 0.47\,\mu\mathrm{F}$, $V_S = 5\,\mathrm{V}$, $V_{\mathrm{th}} = 1.5\,\mathrm{V}$.

P25.2 (Approximation failure) The RC network in Problem 1 is fed from a rail that ramps in 30 ms. Is the computed delay valid?

P25.3 A device requires its 1.0 V core before its 1.8 V I/O, but the 1.0 V buck is fed from 1.8 V. Propose two fixes.

P25.4 (Inverse design) A supervisor must assert reset when a 3.3 V rail falls below the device minimum of 3.0 V. Its threshold tolerance is $\pm 1.5\,\%$. Choose a nominal threshold.

P25.5 (Imperfect data) A bring-up scope capture shows a 1.8 V rail rising, dipping by 150 mV, and rising again. What could cause it and which acceptance criterion does it violate?

Chapter 26

Thermal Design of the Board

Chapter Overview

Every watt that the regulators and loads of Chapter 24 dissipate must leave the board, and the junction temperatures it creates on the way set reliability, leakage, timing, and loss (copper resistance rises 0.39 %/°C). This chapter covers where the heat comes from, how package thermal metrics are defined and how each may and may not be used, how exposed pads, thermal vias, heat sinks, and airflow move heat, what can and cannot be said about trace current capacity without a standard's data, and how to measure temperatures with infrared cameras and thermocouples without fooling yourself. The analysis is first order; a product with tight margins needs a thermal simulation and measurements.

Learning Objectives

1. Build a loss budget and the thermal paths from junction to ambient.
2. Use $R_{\theta JA}$, $R_{\theta JC}$, $R_{\theta JB}$, Ψ_{JT}, and Ψ_{JB} for what they are defined for.
3. Design an exposed-pad land with thermal vias and estimate its resistance.
4. State the assumptions behind any trace-current figure.
5. Measure board temperatures correctly with IR cameras and thermocouples.

26.1 Losses

The loss budget comes from the rail budget (Table 24.1): every load's power plus every regulator's loss. In the SBC example the regulators alone dissipate 4.1 W, spread over seven parts; the SoC dissipates most of its 4.8 W core power plus its I/O power in one package. Other sources are easy to forget: termination resistors (Chapter 8), inductor copper and core loss, MOSFET switching loss, the I^2R loss in power planes and connectors, and the input protection.

26.2 Package thermal metrics and their proper use

JEDEC's JESD51 family [28] defines how package thermal metrics are measured, and a TI application note [18] explains how each should be used. Figure 26.1 shows the two main heat paths: up through the package top (and a heat sink, if any) and down into the board.

Common Pitfall: Using $R_{\theta JA}$ to compute T_J

$T_J = T_A + P\,R_{\theta JA}$ looks like an equation but is a misuse: $R_{\theta JA}$ belongs to the JEDEC test board, which may have more or less copper than yours, and the TI note reports that in still air most of the power (on the order of 70–95 %) leaves through that board. The same note ranks the board design as the strongest

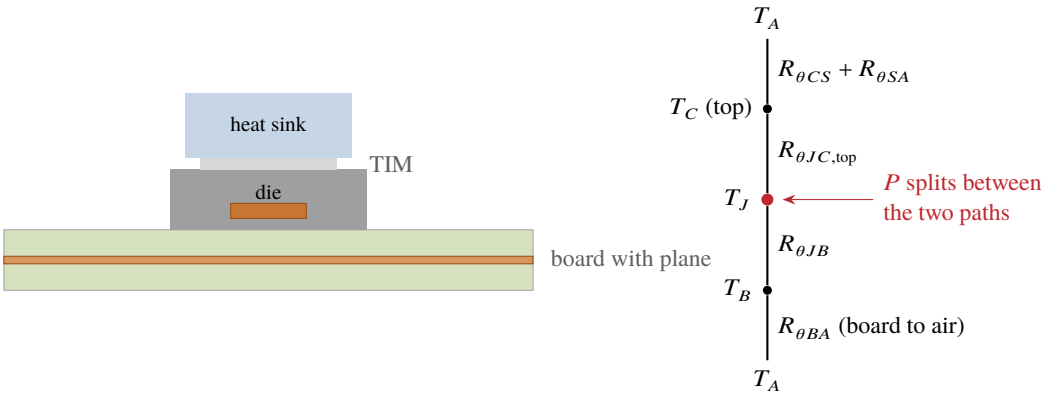


Figure 26.1. Heat paths from the junction: through the package top and heat sink, and through the leads or pad into the board. In still air without a heat sink, most of the heat goes into the board. [Conceptual]

Table 26.1. Package thermal metrics. "Resistance" metrics (R_θ) assume all the power flows through the stated path; "characterization parameters" (Ψ) divide a temperature difference by the total power and so are valid only in conditions close to those in which they were measured.

Metric	Definition	Proper use
$R_{\theta JA}$	$(T_J - T_A)/P$ on a JEDEC test board (JESD51-2A still air, or moving air)	comparing packages; *not* predicting T_J on your board
$R_{\theta JC,\text{top}}$	junction to package top, with the top held at a fixed temperature	with a heat sink that removes most of the heat through the top
$R_{\theta JC,\text{bot}}$	junction to exposed pad or bottom	exposed-pad packages, conduction into the board
$R_{\theta JB}$	junction to the board near the package, with heat forced into the board	board-dominated cooling, in a network model
Ψ_{JT}	$(T_J - T_{\text{top}})/P$, measured on the JEDEC board	estimating T_J from a measured top temperature
Ψ_{JB}	$(T_J - T_B)/P$, measured on the JEDEC board	estimating T_J from a measured board temperature

influence on $R_{\theta JA}$. Use it to compare packages, and use a model of your board, or a measurement with Ψ_{JT}, to estimate T_J [18].

Worked Example 26.1: Estimating junction temperature from a measured case top

The 1.8 V regulator in an early SBC prototype was an LDO dissipating 1.2 W. A fine-gauge thermocouple on the package top reads 86 °C; the datasheet gives $\Psi_{JT} = 1.5$ °C/W and $R_{\theta JA} = 50$ °C/W. Ambient is 30 °C. Estimate T_J both ways.

Solution. With Ψ_{JT}: $T_J \approx 86 + 1.2 \times 1.5 = 87.8$ °C. With $R_{\theta JA}$: $30 + 1.2 \times 50 = 90$ °C. They agree here only by coincidence; on a board with less copper the $R_{\theta JA}$ estimate would be optimistic and the measured top temperature, and so the Ψ_{JT} estimate, would show the difference. If the part is cooled by a heat sink on its top, Ψ_{JT} no longer applies, because the heat flow through the top is no longer small; use $R_{\theta JC,\text{top}}$ instead.

Package families and heat paths. The package decides the heat paths available (Figure 26.2): a BGA conducts mainly through its balls into the board; a lidded flip-chip package offers a low-resistance path

through its lid to a heat sink; package-on-package stacks put memory on top of the processor and block that path; metal cans and ceramic packages with large flanges or lids are designed for conduction to a heat sink or chassis.

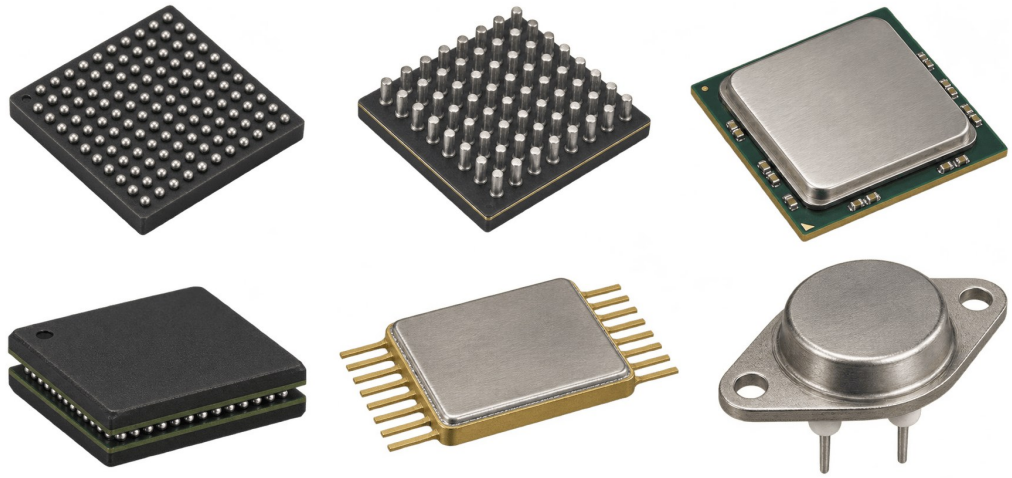


Figure 26.2. Package families with different heat paths: ball grid array, pin grid array, lidded flip-chip package, package-on-package stack, ceramic flatpack, and TO-3 metal can. Rendered illustration supplied with the manuscript, not a photograph of specific products; markings, contact counts, and proportions are not authoritative. [Illustration]

26.3 Exposed pads and thermal vias

Packages with an exposed pad (QFN, DFN, PowerPAD) conduct most of their heat through the pad into the board. The pad must be soldered to a land connected by vias to one or more inner planes (Figure 26.3).

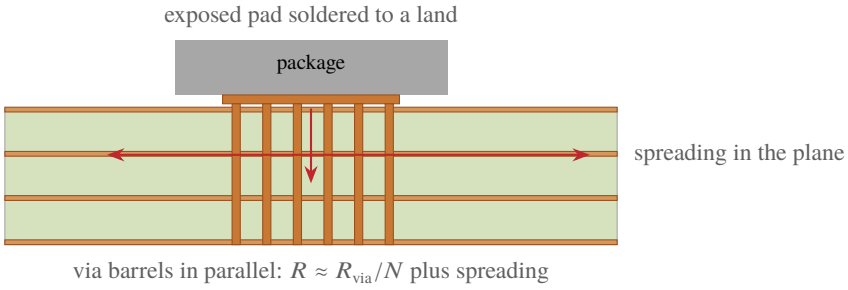


Figure 26.3. An exposed-pad package with a thermal-via array to inner planes. The vias conduct heat through the laminate; the planes spread it. [Conceptual]

One via. A plated via is a copper tube. With copper conductivity $k = 390\,\mathrm{W/(m\cdot K)}$, a 0.3 mm drill, and 25 µm plating through a 1.6 mm board,

$$R_{\text{via}} = \frac{L}{k\pi\,(r_o^2 - r_i^2)} \approx 190\,\mathrm{K/W}; \tag{26.1}$$

with 18 µm plating it is 257 K/W, and a copper-filled via of the same drill is 58 K/W. For comparison, the laminate under a 5×5 mm pad (FR-4, $k \approx 0.3\,\mathrm{W/(m\cdot K)}$) conducts through 1.6 mm with about 213 K/W: one via conducts about as well as the whole pad area of laminate.

Arrays. N vias in parallel give R_{via}/N (Figure 26.4): 21 K/W for nine, 7.6 K/W for 25. The real gain is smaller because heat must also spread from the pad into the vias and from the vias into the planes, and those spreading resistances do not divide by N. There is therefore no universal via count; the useful number depends on the pad size, the plane copper, and what removes the heat from the planes. Vias in the pad must be plugged, capped, or tented as the assembly process requires, or solder wicks into them and leaves voids under the pad.

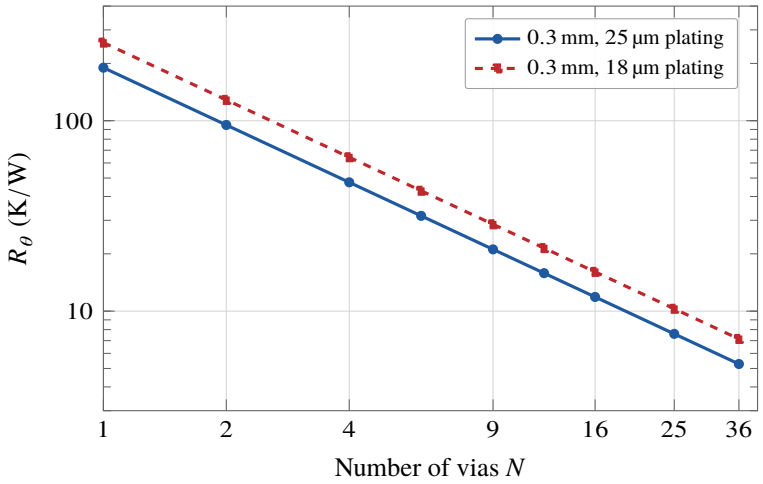


Figure 26.4. Conduction resistance of N parallel plated vias through 1.6 mm, barrels only. Spreading in the pad and planes adds a resistance that does not fall with N. [Analytical]

26.4 Heat sinks and airflow

A heat sink is specified by its sink-to-air resistance $R_{\theta SA}$ at a stated airflow, and is attached through a thermal interface material (TIM) with resistance $R_{\theta CS}$ that depends on pressure, flatness, and thickness. For a part whose heat leaves mostly through the top, $T_J \approx T_A + P(R_{\theta JC,\mathrm{top}} + R_{\theta CS} + R_{\theta SA})$; this overestimates T_J slightly because some heat still goes into the board. Airflow raises the convection coefficient from roughly 5–10 $\mathrm{W/(m^2\cdot K)}$ in still air to several tens with a fan, and its direction matters: components shadowed by tall parts upstream see much less flow. The local air temperature near a hot board is higher than the room or inlet temperature; T_A in every formula is the air that actually reaches the part.

26.5 Trace current capacity

A trace heats until its I^2R loss equals the heat it loses to the laminate, planes, and air. The temperature rise therefore depends on the trace's width and thickness, but equally on board thickness, nearby planes and copper, neighbouring traces that also carry current, ambient temperature, and airflow. No width-only rule can capture that.

IPC-2152 [19] is the current standard for this question; its charts come from measurements on defined test boards and come with modifiers for board thickness, plane proximity, and copper weight. (The standard

was not reproduced for this book; consult it directly.) Its predecessor, the chart in IPC-2221, is still quoted widely in the form $I = k\,\Delta T^{0.44} A^{0.725}$ (cross-section A in mil^2; $k = 0.048$ for external and 0.024 for internal layers). Figure 26.5 plots that legacy fit only to show its dependencies: for a 10 °C rise a 20 mil, 1 oz external trace is rated 1.5 A, and the internal factor halves it. IPC-2152 data generally show internal traces doing better than the legacy factor implies, because planes in the board conduct the heat away.

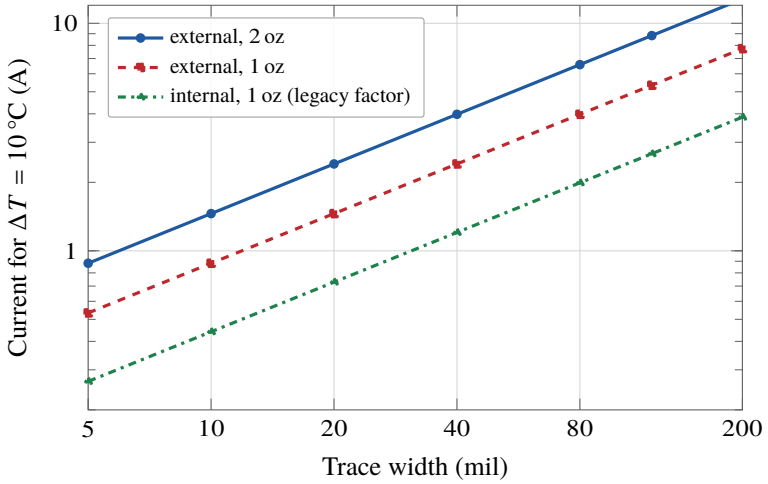


Figure 26.5. The legacy IPC-2221 chart fit for a 10 °C rise. Shown to illustrate dependencies on width, copper weight, and layer; not a design limit, and not a substitute for IPC-2152 or for measurement. [Analytical]

Design Rule: State the assumptions with any current rating

A trace or plane current rating is meaningful only with its assumptions: copper thickness, allowed temperature rise, ambient, board thickness, plane proximity, neighbouring currents, and the source of the data. Check IR drop separately (Chapter 21); a trace that is thermally acceptable may still drop too much voltage.

26.6 Measuring temperature

Infrared cameras. A camera infers temperature from radiated power, which depends on the surface's emissivity ε. Solder mask, matte paint, and most package bodies have $\varepsilon \approx 0.9$–0.95; bright copper, tin, and solder have ε of a few hundredths and mostly reflect their surroundings. Figure 26.6 uses a broadband grey-body approximation to show what a camera set to $\varepsilon = 0.95$ reports: for a surface truly at 100 °C, 90.9 °C if $\varepsilon = 0.80$, 54.5 °C if $\varepsilon = 0.30$, and 30.5 °C for bare copper with $\varepsilon = 0.05$: almost the reflected room temperature. (Real cameras work in a limited wavelength band and the numbers differ, but the direction and size of the error are typical.) Coat low-emissivity spots with a thin layer of high-emissivity tape or paint, or calibrate the emissivity setting against a thermocouple, and watch for reflections of hot objects and of the operator.

Thermocouples. Use fine-gauge wire (36–40 AWG), a small junction bonded with thermally conductive adhesive or a thin layer of solder where permissible, and route the first few centimetres of wire along the surface so the wire does not conduct heat away from the junction (the "fin" error). Type K and type T are common; check the reader's cold-junction compensation and calibration. On small parts a thermocouple can itself lower the temperature it measures.

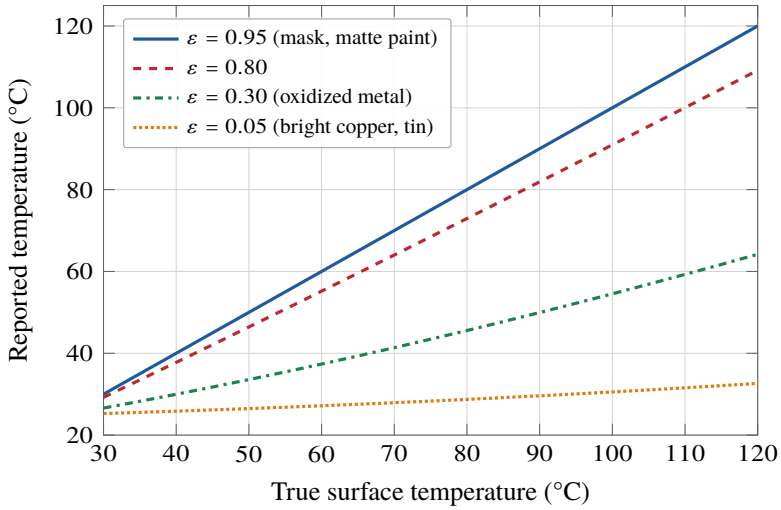


Figure 26.6. Temperature reported by an IR camera set to $\varepsilon = 0.95$ when the surface has a lower emissivity (broadband grey-body approximation, reflected temperature 25 °C). [Analytical]

Test conditions. Measure at the worst-case workload and supply, in the final enclosure orientation, after the board has reached steady state (often tens of minutes), and record the ambient at the inlet and near the board. Report temperatures with the method and its estimated uncertainty.

Checklist: thermal

- ☐ Loss budget includes regulators, terminations, magnetics, and I^2R in copper.
- ☐ T_J estimates use Ψ with measured temperatures or a board model, not $R_{\theta JA}$ alone.
- ☐ Exposed pads have via arrays sized with spreading in mind; via-in-pad treatment matches assembly.
- ☐ Heat-sink, TIM, and airflow assumptions are stated, including local air temperature.
- ☐ Current ratings of traces and planes state their assumptions and source; IR drop checked separately.
- ☐ Measurements correct for emissivity and thermocouple errors; conditions recorded.

Key Takeaways

- In still air most of a package's heat goes into the board; the board design dominates $R_{\theta JA}$.
- Use $R_{\theta JA}$ to compare packages, Ψ_{JT}/Ψ_{JB} with measured temperatures, $R_{\theta JC}$ with heat sinks.
- A plated 0.3 mm via conducts about 190 K/W through 1.6 mm; arrays divide that, but spreading does not divide.
- Trace current capacity has no width-only answer; state the assumptions and use IPC-2152 data or measurement.
- IR cameras read bare metal far too cold unless emissivity is handled.

Further Reading

The TI application note on package thermal metrics [18] and the JESD51 series [28] define and explain the metrics; IPC-2152 [19] is the reference for conductor heating. Component vendors' thermal application notes give land patterns and via recommendations for specific packages.

Review Problems

P26.1 A part dissipates 2 W; a thermocouple on its top reads 71 °C; $\Psi_{JT} = 0.8$ °C/W. Estimate T_J.

P26.2 (Approximation failure) A datasheet gives $R_{\theta JA} = 35$ °C/W. A designer predicts $T_J = 95$ °C for 2 W at 25 °C and accepts the part. What is wrong with the method?

P26.3 Compute the resistance of nine 0.25 mm vias with 20 µm plating through 1.2 mm.

P26.4 (Inverse design) A pad must conduct 3 W into an inner plane with at most 15 °C across the vias. How many of the Eq. (26.1) vias are needed, ignoring spreading?

P26.5 (Imperfect data) An IR image shows a bright copper bus bar at 32 °C next to a 75 °C inductor. Is the bus bar cool?

P26.6 Why does the legacy internal-layer factor of 0.024 often overstate the heating of an inner trace in a modern multilayer board?

Part V

EMI and EMC

Chapter 27

Emission Mechanisms and EMC Design

Chapter Overview

A product that works perfectly on the bench can still be unsellable if it radiates too much. Radiated and conducted emissions are regulated in every major market, and failing a compliance test late in a program is expensive. This chapter explains the two fundamental emission mechanisms—differential-mode radiation from current loops and common-mode radiation from cables and structures—quantifies them with the standard small-antenna formulas, examines slot and edge radiation from boards, and surveys the design techniques (spread-spectrum clocking, filtering, ferrites, and shielding) used to meet FCC and CISPR limits. It draws heavily on the classic treatments of Ott and Paul [5], [62].

Learning Objectives

1. Distinguish differential-mode and common-mode emissions and estimate each.
2. Explain why microamperes of common-mode current on a cable can fail a test.
3. Relate emissions to the spectrum of digital waveforms.
4. Identify slot and edge radiation from boards and prevent them.
5. Apply spread-spectrum clocking, filtering, ferrites, and shielding, and estimate their benefit.
6. Outline the structure of FCC Part 15 and CISPR 32 limits.

27.1 Differential-mode radiation

Every signal and its return form a loop (Chapter 3 and Figure 27.1). A loop that is small compared with a wavelength radiates as a magnetic dipole. Its far-field electric field at distance r, in free space, is [5], [62]

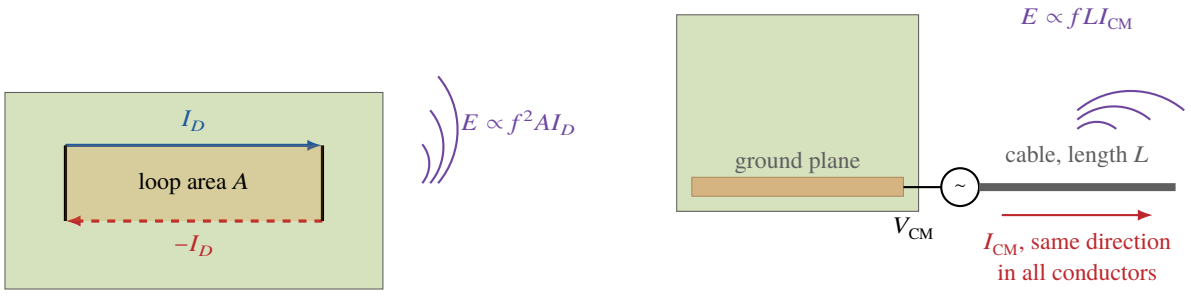
$$E_{\mathrm{DM}} = 131.6 \times 10^{-16} \frac{f^2 A I_D}{r} \quad \mathrm{V/m}, \tag{27.1}$$

with f in hertz, loop area A in square meters, current I_D in amperes, and r in meters. Over a reflecting ground plane, as in a test site, the worst case is about twice this value. Differential-mode emission grows as f^2 and in proportion to loop area, so it is controlled by keeping return paths close to their signals—the same rule that minimizes loop inductance.

Worked Example 27.1: Loop area and emissions

A 4 in clock trace carries a 10 mA harmonic at 300 MHz. Estimate the field at 3 m if its return is in a plane 5 mil below, and if the return is a wire 1 in away.

Solution. With the plane, $A = 4 \times 0.005 = 0.02\,\mathrm{in}^2 = 1.29 \times 10^{-5}\,\mathrm{m}^2$, and Eq. (27.1) gives $E = 131.6 \times 10^{-16} \times (3 \times 10^8)^2 \times 1.29 \times 10^{-5} \times 0.01/3 = 51\,\mu\mathrm{V/m}$, or 34 dBμV/m, below the FCC Class B



(a) Differential mode: small current loop

(b) Common mode: cable driven as an antenna

Figure 27.1. Emission mechanisms. (a) Differential-mode current in a small loop radiates as a magnetic dipole. (b) Common-mode current, driven by a noise voltage between the board's ground and an attached cable, flows in the same direction on all of the cable's conductors and radiates as a monopole.

limit of 46 dBµV/m at that frequency. With the return 1 in away, A is 200 times larger and the field rises by 46 dB, to 80 dBµV/m—a failure by more than 30 dB.

Figure 27.2 extends this example across frequency, using the harmonic envelope of a trapezoidal clock (Eq. (2.8)). Multiplying the current envelope by f^2 turns its flat, −20, and −40 dB/decade segments into +40, +20, and 0 dB/decade: differential-mode emissions of a clock *rise* with frequency up to $f_2 = 1/(\pi T_{\text{ramp}})$ and stay high above it.

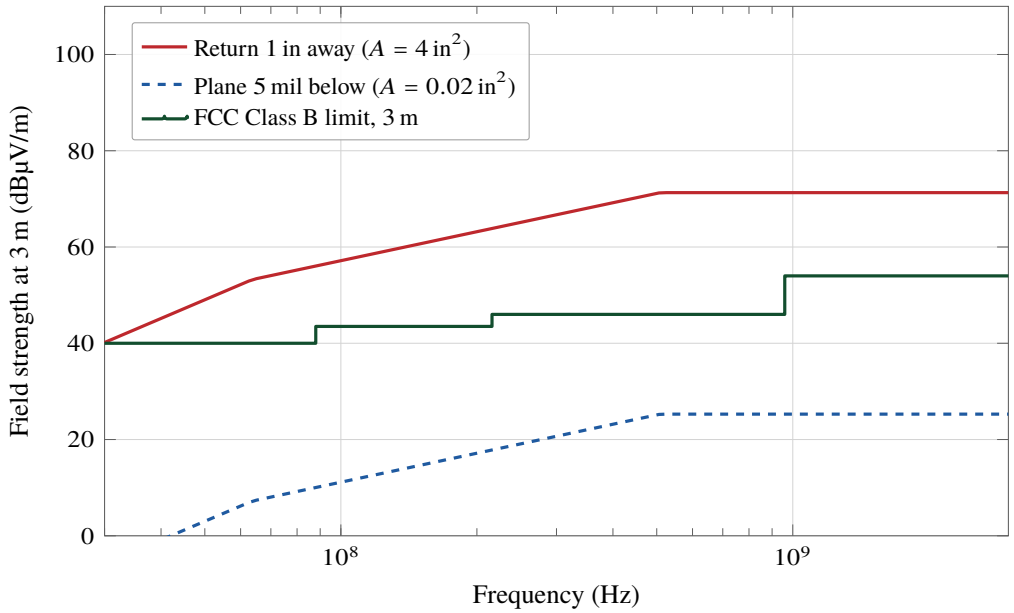


Figure 27.2. Envelope of differential-mode emission at 3 m from a 4 in trace carrying the harmonics of a 100 MHz clock (t_r = 0.5 ns, 10 mA harmonic envelope), for two return-path geometries, from Eq. (27.1), compared with the FCC Class B limit. Only the envelope is shown; actual emissions occur at discrete harmonics. Free-space field, RMS; a test site with a ground plane can read up to about 6 dB higher. [Analytical]

27.2 Common-mode radiation

Common-mode current flows in the same direction on all conductors of a cable (or on a board's ground structure), returning through stray capacitance to the environment. It is driven by a noise voltage between the board's reference and the cable—ground-plane noise, ground bounce, or voltage across a plane slot. A cable of length L short compared with a quarter wavelength radiates as a monopole:

$$E_{\mathrm{CM}} = 1.257 \times 10^{-6} \frac{f L I_{\mathrm{CM}}}{r} \quad \mathrm{V/m}. \tag{27.2}$$

Equations (27.1) and (27.2) are free-space, far-field estimates for a sinusoidal current: I is the RMS current of one harmonic and E the RMS field. An EMI receiver is calibrated to read the RMS value of a steady sine, so for a stable narrowband harmonic its peak, quasi-peak and average detectors give nearly the same reading; for modulated or intermittent signals they differ, and the detector and bandwidth defined by the regulation apply [63]. A test site with a reflecting ground plane can raise the measured field by up to about 6 dB over the free-space value.

The short-monopole formula assumes the cable is electrically short, roughly $L \lesssim \lambda/10$, with a uniform current along it. Longer cables are not simply "saturated": the current becomes a standing wave whose distribution depends on how the cable is terminated at both ends, the radiation pattern develops lobes, and at resonance the radiated field can be considerably higher than an extrapolation of Eq. (27.2). For cables approaching or exceeding $\lambda/4$, use a numerical model (method of moments) or a measurement of the cable current distribution instead. Even in its valid range, common-mode emission usually dominates: it grows only as f, but a cable is a far more efficient antenna than a small board loop.

Worked Example 27.2: How little common-mode current fails a test

A 0.25 m cable stub carries 20 µA (RMS) of common-mode current at 100 MHz. Estimate the free-space field at 3 m, and the largest current that meets a 150 µV/m limit. Check that the formula applies.

Solution. *Validity:* $\lambda = 3$ m, so $L = \lambda/12$; the short-monopole model is reasonable. Equation (27.2): $E = 1.257\times10^{-6}\times10^{8}\times0.25\times20\times10^{-6}/3 = 210$ µV/m, or 46.4 dBµV/m, about 3 dB over the limit. The maximum allowed current is $150\times10^{-6}\times3/(1.257\times10^{-6}\times10^{8}\times0.25) = 14$ µA. Tens of microamperes of common-mode current are enough to fail, compared with milliamperes of differential-mode current. *What the example does not cover.* A real 1 m USB or Ethernet cable at 100 MHz is longer than $\lambda/4 = 0.75$ m. Multiplying the result by four would be wrong in either direction: the current is no longer uniform, and at a cable resonance the field can exceed that estimate. Such a cable must be modeled or measured.

Common Pitfall: Treating cables as part of someone else's design

Testing emissions with the board alone, or with short shielded cables, and discovering at compliance that the system's real cables radiate. Common-mode emission depends on how every cable connects to the board's ground at its connector. Design the connector region—ground plane, filter components, shield termination—as part of the board, and test with representative cables early.

27.3 Slot and edge radiation

A slot in a plane, driven by return current that must detour around it (Chapter 23), radiates as a slot antenna. It is most efficient when its length is a half wavelength:

$$L_{\mathrm{slot}} = \frac{\lambda}{2} = \frac{c}{2f} \quad \text{(in air)}; \tag{27.3}$$

a 3 in slot resonates near 2 GHz. Slots inside a board are partly loaded by the dielectric and resonate lower. Plane-pair cavities (Section 23.3) radiate from their edges, most strongly at cavity resonances; ground-via fences along board edges and thin power–ground dielectrics reduce this.

27.4 Techniques for reducing emissions

27.4.1 Edge rate

The cheapest emission control is to use slower edges. From Eq. (2.8), harmonics above $f_2 = 1/(\pi T_{\text{ramp}})$ (with $T_{\text{ramp}} = t_r/0.8$ for a linear edge) fall by 20 dB/decade faster than below it; doubling t_r halves f_2 and reduces every harmonic above the old f_2 by 6 dB. Drive-strength and slew-rate settings, and series resistors on clocks, should be set to the slowest value that meets timing.

27.4.2 Spread-spectrum clocking

Spread-spectrum clocking (SSC) frequency-modulates a clock by a small amount, typically a triangular modulation of 0.25–0.5 % (usually down-spread so that the maximum frequency is not exceeded) at 30–33 kHz. Each harmonic's energy is spread over a band of width $n\delta f_0$ for the n-th harmonic. A receiver measuring in a resolution bandwidth RBW smaller than that band sees a lower peak (Figure 27.3), by roughly

$$\Delta_{\text{SSC}} \approx 10 \log_{10} \frac{n\,\delta f_0}{\text{RBW}} \quad \text{dB}, \tag{27.4}$$

an estimate that is optimistic for the lowest harmonics, where the spread band is only a few RBWs wide. SSC does not reduce the total energy, and the receiving system must tolerate it: PCIe and SATA reference clocks support SSC, but some interfaces (and some PLLs) do not.

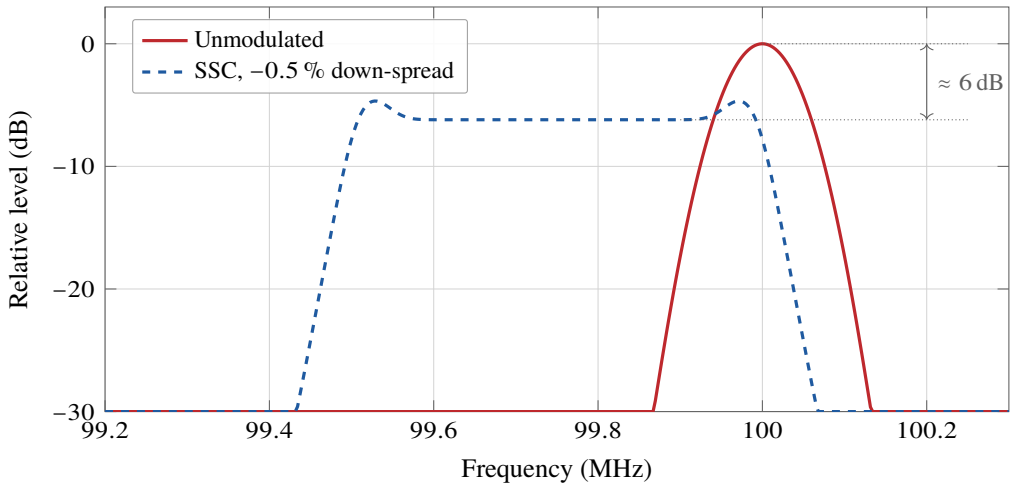


Figure 27.3. Effect of −0.5 % down-spread SSC on the fundamental of a 100 MHz clock, as seen by a receiver with a 120 kHz resolution bandwidth (schematic). The peak falls by about 6 dB, the ratio of the 500 kHz spread to the resolution bandwidth expressed in decibels.

Worked Example 27.3: SSC benefit by harmonic

A 100 MHz clock uses −0.5 % SSC. Estimate the reduction in a 120 kHz RBW at the fundamental and at the fifth harmonic.

Solution. Fundamental: spread 500 kHz, so $10\log_{10}(500/120) = 6.2$ dB. Fifth harmonic: 2.5 MHz, $10\log_{10}(2500/120) = 13.2$ dB. SSC is most effective at the high-order harmonics that are hardest to control by other means.

27.4.3 Filtering and ferrites

Signals and power leaving the board through a connector should be filtered so that only the intended bandwidth passes. A series ferrite bead acts as a resistor at high frequency (its impedance becomes predominantly resistive above a few tens of megahertz), absorbing energy rather than reflecting it. A ferrite is effective only against a low source and load impedance; it must be paired with a capacitor to ground on the board side to form a low-pass filter. Common-mode chokes on differential pairs (USB, Ethernet) present high impedance to common-mode current while passing the differential signal. Ferrite beads saturate with DC current and their impedance at rated current may be much lower than the zero-bias curve in the data sheet.

Design Rule: Connector-region design

Treat every connector as an antenna port. Provide a solid ground plane under the connector region, bond cable shields to chassis or ground with a low-inductance 360-degree connection at the connector, place filter capacitors and ferrites between the connector and the rest of the board, and keep high-speed traces and clocks away from the connector area unless they leave through it.

27.4.4 Shielding

A conductive enclosure attenuates fields by reflection at its surfaces and by absorption in its thickness. The absorption loss of a sheet of thickness t is

$$A_{\text{abs}} = 8.686\,\frac{t}{\delta_{\text{s}}} \quad \text{dB}, \tag{27.5}$$

which for even thin metal at radio frequencies is very large (0.1 mm of copper at 100 MHz gives over 130 dB). In practice, shielding effectiveness is limited not by the metal but by apertures, seams, and penetrating cables: an aperture behaves like a slot antenna, and its leakage depends on its longest dimension, not its area. Board-level shield cans over noisy circuits must be soldered to a ground ring with many vias.

Table 27.1. Emission mechanisms and design controls.

Mechanism	Scaling	Primary controls
Differential-mode loop	f^2AI	Tight return paths; stripline routing; slower edges
Common-mode cable	fLI_{CM}	Low ground noise; connector filtering; CM chokes; shield bonding
Slot in plane	resonant at $\lambda/2$	No slots under signals; stitching
Plane-cavity edge	resonant at f_{mn}	Via fences; decoupling; thin dielectric
Periodic clocks	discrete harmonics	SSC; slower edges; short routes

27.5 Regulatory overview

In the United States, unintentional radiators such as digital devices are regulated by FCC Part 15 [63]; Class A covers commercial and industrial equipment and Class B, with tighter limits, residential equipment. Internationally, multimedia equipment emissions are covered by CISPR 32 [64], adopted as EN 55032 in Europe.

Both specify radiated emission limits (typically from 30 MHz to several gigahertz, at 3 m or 10 m) and conducted emission limits on AC mains (150 kHz to 30 MHz). Below 1 GHz, radiated measurements use a quasi-peak detector with a 120 kHz bandwidth; above, peak and average detectors with a 1 MHz bandwidth.

Specification Note

Commonly quoted FCC Part 15 Class B radiated limits at 3 m are 100 µV/m (40.0 dBµV/m) from 30–88 MHz, 150 µV/m (43.5 dBµV/m) from 88–216 MHz, 200 µV/m (46.0 dBµV/m) from 216–960 MHz, and 500 µV/m (54.0 dBµV/m) above 960 MHz. CISPR 32 Class B limits at 10 m are commonly quoted as 30 dBµV/m (30–230 MHz) and 37 dBµV/m (230–1000 MHz), quasi-peak. Detector type, measurement distance, upper frequency limit, and the limits themselves must be taken from the current edition of the applicable regulation for the target market.

Rule of Thumb: Common-mode current budget

Near 100 MHz, a short cable segment carrying only microamperes to tens of microamperes of common-mode current approaches the FCC Class B limit at 3 m (Worked Example 27.2). Measure cable common-mode current with a clamp-on current probe early in development, and track it relative to a baseline rather than trusting an absolute field prediction. *Limits:* based on the short-monopole formula, valid only for $L \lesssim \lambda/10$; resonant cables can radiate considerably more and must be modeled or measured.

Key Takeaways

- Differential-mode emission is $131.6 \times 10^{-16} f^2 A I / r$; keep loop areas small with adjacent return planes.
- Common-mode emission is $1.257 \times 10^{-6} f L I_{\mathrm{CM}} / r$; microamperes on a cable can fail a test.
- Clock harmonics multiplied by f^2 make DM emission rise with frequency up to $1/(\pi T_{\mathrm{ramp}})$.
- Slots and plane edges radiate; avoid slots under signals and stitch edges.
- Slow edges, SSC, connector filtering, ferrites, CM chokes, and well-bonded shields are the main controls.

Further Reading

Ott [5] and Paul [62] are the standard references for emission mechanisms and EMC design; the regulations themselves are FCC Part 15 [63] and CISPR 32 [64].

Review Problems

P27.1 Estimate the free-space DM field at 3 m from a 1 cm^2 loop carrying 20 mA at 200 MHz.

P27.2 Using the short-monopole approximation, find the common-mode current on a 0.15 m cable segment that produces 150 µV/m (free space, RMS) at 3 m and 200 MHz. Would the same calculation be valid for a 1.5 m cable?

P27.3 What slot length is a half wavelength at 1 GHz in air?

P27.4 Estimate the SSC reduction at the third harmonic of a 100 MHz clock with 0.5 % spread and 120 kHz RBW.

P27.5 By how much are harmonics above both corner frequencies reduced when the rise time is doubled from 0.5 ns to 1 ns?

P27.6 Find the absorption loss of 35 μm copper at 1 GHz.

Part VI

Interfaces in Practice

Chapter 28

DDR4/DDR5 and LPDDR4/LPDDR5 Memory Interfaces

Chapter Overview

Memory interfaces are the densest, most timing-critical buses on most digital boards. They combine source-synchronous data (DQ with strobe DQS), a common-clock command/address bus, many simultaneously switching single-ended lines, and a reference voltage, all running at several gigatransfers per second. This chapter summarizes the electrical characteristics of DDR4, DDR5, LPDDR4, and LPDDR5, explains the fly-by topology and the training mechanisms (write leveling, read and write centering, VREF training) that make it work, derives routing and matching budgets, and lists the failure modes most often encountered at bring-up.

Learning Objectives

1. Compare the key electrical parameters of DDR4, DDR5, LPDDR4, and LPDDR5.
2. Explain fly-by and T topologies and why fly-by requires write leveling.
3. Describe pseudo-open-drain signaling, ODT, and VREF training.
4. Derive DQ-to-DQS and CA-to-CK matching budgets from the data rate.
5. Estimate loaded impedance and delay of a fly-by bus.
6. Recognize common DDR failure modes and their causes.

28.1 Interface overview

A DDR interface has three groups of signals:

Data (DQ, DM/DBI) and strobes (DQS)
Bidirectional, organized in *byte lanes* of eight DQ (or four in some configurations) with one differential DQS pair. Data are transferred on both edges of DQS (source-synchronous, Chapter 17).

Command/address (CA) and control
Unidirectional from controller to all DRAMs of a rank, captured by the clock. In DDR4 they transfer once per clock; DDR5 and LPDDR use fewer pins at double data rate on some signals.

Clock (CK)
Differential, from controller to all DRAMs, routed with the CA bus.

This three-group picture is exact for DDR4 and DDR5 and for LPDDR4/4X, where a bidirectional DQS strobe times data in both directions. It is *not* adequate for LPDDR5 and LPDDR5X, which separate the data clock from the command clock (Section 28.1.2). Table 28.1 summarizes the four standards.

Table 28.1. Representative characteristics of DDR4, DDR5, LPDDR4/4X, and LPDDR5/5X. Values reflect the base JEDEC releases and common practice; see the specification note below.

	DDR4 [57]	DDR5 [65]	LPDDR4/4X [66]	LPDDR5/5X [44]
Data rate (MT/s)	1600–3200	3200–6400+	up to 4266	up to 6400 (5X higher)
V_{DDQ}	1.2 V	1.1 V	1.1 V / 0.6 V	0.5 V (typ.)
DQ signaling	POD12	POD11	LVSTL	LVSTL
Channel width	64 (+8 ECC)	2 × 32 (+8 ECC)	16 per channel	16 per channel
DQ VREF	Internal, trained	Internal, trained	Internal, trained	Internal, trained
CA VREF	External	Internal	Internal	Internal
Rx equalization	—	DFE in DRAM	—	Rx equalization (rate dependent)
Topology	Fly-by (CA/CK)	Fly-by (CA/CK)	Point-to-point	Point-to-point

Specification Note

Data rates, supply voltages, and feature sets of JEDEC memory standards are extended in later revisions and vary by speed bin; LPDDR5X in particular supports higher rates than LPDDR5. The values above are for orientation. Design values must come from the current JEDEC document and, above all, from the memory controller vendor's layout guide, which defines the topology, impedance, and matching rules the controller's training was characterized for.

28.1.1 Timing groups by family

Board constraints are derived per *timing group*: a set of signals captured by a common clock or strobe. Table 28.2 lists them for each family. The rules that a layout guide imposes—matching, layer, via count—apply within a group, and the training ranges of Section 28.3.2 apply between groups.

Table 28.2. Timing groups of the four DRAM families (conceptual; consult the controller documentation for the implemented groups).

Family	Command/address	Write data	Read data
DDR4	CA, control with CK (fly-by)	DQ, DM/DBI with DQS (from controller)	DQ with DQS (from DRAM)
DDR5	CA, CS with CK per subchannel (fly-by on DIMMs)	DQ with DQS	DQ with DQS; DRAM Rx has DFE
LPDDR4/4X	CA, CS with CK (point-to-point)	DQ, DMI with DQS	DQ with DQS
LPDDR5/5X	CA, CS with CK	DQ, DMI with WCK (data clock)	DQ with RDQS where used

28.1.2 LPDDR5 and LPDDR5X: CK, WCK and read strobes

LPDDR5 introduced a separate differential *data clock*, WCK, per byte, sent from the controller to the DRAM and running at a multiple of CK; the slower CK continues to time command and address [43], [44]. Write data on DQ and DMI are captured with WCK. Read data return from the DRAM on DQ, accompanied by

a differential read strobe, RDQS, in configurations that use it (Figure 28.1). Because WCK and CK are separate clocks, the DRAM must align its internal WCK dividers with CK—*WCK-to-CK synchronization*—and training includes a WCK-to-CK leveling step analogous to DDR4 write leveling.

Two consequences matter on the board. First, a DDR4-style rule "match DQ to DQS" cannot be copied: writes are referenced to WCK and reads to RDQS, so each needs its own matching budget, and the CK/CA group has a third. Second, the WCK frequency is higher than the DQS frequency of a comparable DDR4 design, so its intra-pair skew, crosstalk exposure and termination state deserve the same care as the data. The supported WCK:CK ratios, read modes, termination states and delay allocations are implementation specific; a design starts from a controller-specific table listing, for every signal, its direction, termination, timing reference, training adjustment, board-delay allocation and package-delay treatment.

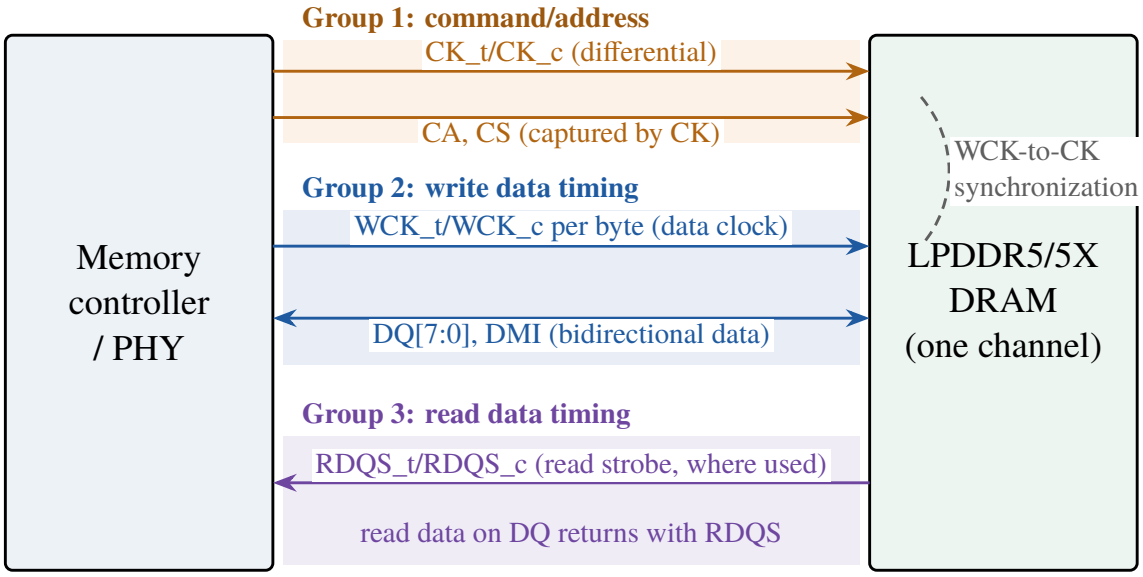


Figure 28.1. Logical LPDDR5/5X timing groups of one channel: CK with command/address, WCK with write data, and RDQS with read data where used. Conceptual architecture, not a pinout, termination schematic, clock-ratio specification or training sequence. [Conceptual]

Common Pitfall: Treating LPDDR5 as fast LPDDR4

Reusing an LPDDR4 or DDR4 constraint set—one DQ-to-DQS rule per byte—for an LPDDR5 design. The data clock (WCK), the read strobe (RDQS) and the command clock (CK) are separate timing references with separate training steps; derive a constraint for each from the selected controller's documentation.

28.2 Signaling and termination

DDR4 and DDR5 data use *pseudo-open-drain* (POD) signaling: the receiver's on-die termination pulls up to V_{DDQ}, and the driver pulls the line low through a calibrated pull-down (Figure 28.2). A high level draws no current, which is what makes DBI effective (Chapter 20). Because the signal swings between V_{DDQ} and a level set by the driver/ODT divider, the receiver's threshold is not $V_{DDQ}/2$; it is an internal V_{REF} that training sets to the center of the received eye. LPDDR4 and LPDDR5 use low-voltage swing-terminated logic (LVSTL), terminated to ground, which allows very low V_{DDQ}.

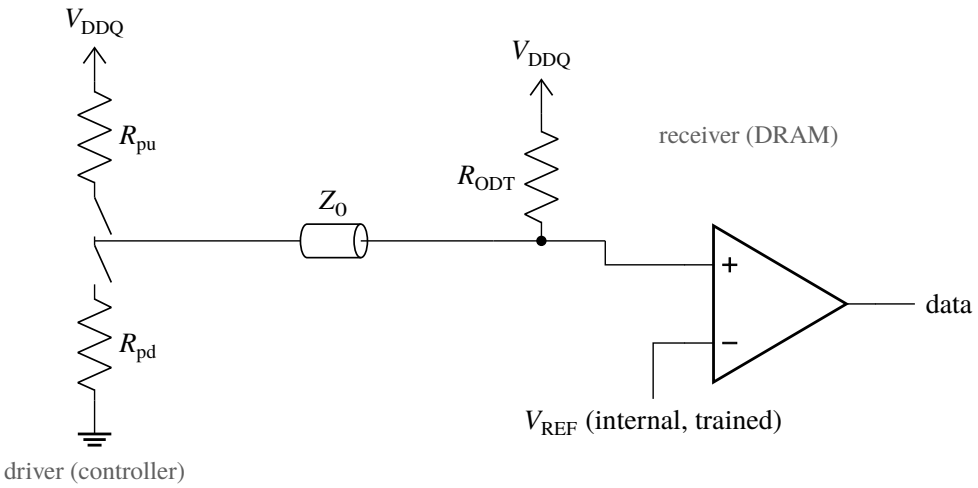


Figure 28.2. Pseudo-open-drain DQ signaling: the receiver's on-die termination pulls up to V_{DDQ}; the receiver compares the line against an internally generated, trained V_{REF}.

Driver output impedance and ODT values are programmable through mode registers and calibrated against an external precision ZQ resistor (240 Ω in DDR4). The controller selects ODT for each rank and each operation (read, write, idle) to optimize the eye in multi-rank systems.

The DDR4 CA bus is terminated at the far end of the fly-by chain to $V_{TT} = V_{DDQ}/2$ through resistors of about the bus's loaded impedance. DDR5 moved CA termination onto the DRAMs.

Worked Example 28.1: V_{TT} regulator current

A DDR4 CA bus of 30 lines is terminated with 39 Ω to $V_{TT} = 0.6$ V; the controller drives with 34 Ω. Estimate the current per line and the V_{TT} supply requirement.

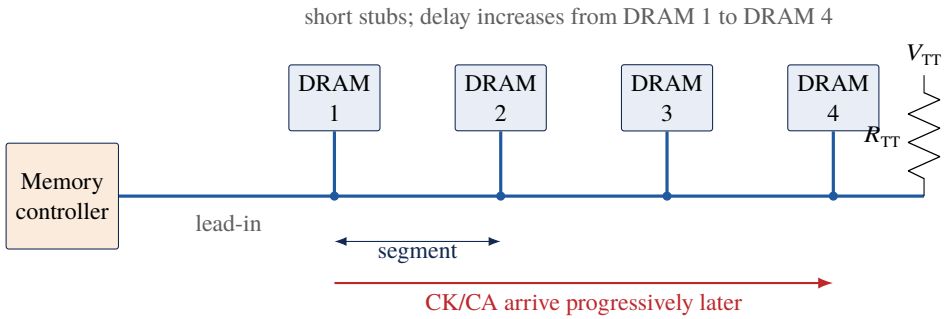
Solution. With the driver high (1.2 V), current flows from driver to V_{TT}: $I = 0.6/(34 + 39) = 8.2$ mA; driven low, the same current flows from V_{TT}. The resistor dissipates $I^2R = 2.6$ mW. With all 30 lines high the V_{TT} supply sinks 247 mA; with all low it sources the same. The V_{TT} regulator must source and sink and must have low impedance at the bus's switching frequencies, so it is decoupled at the termination resistors.

28.3 Topologies

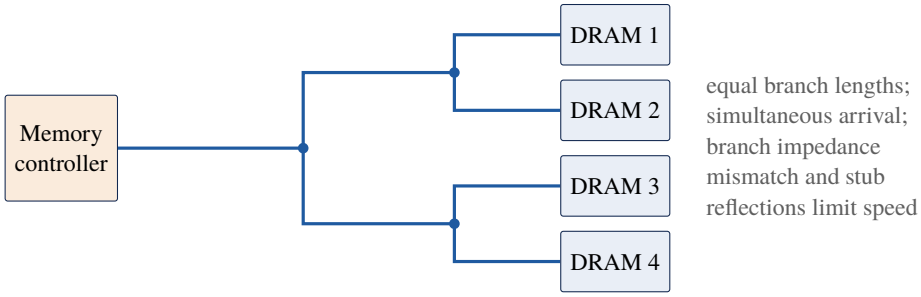
On a DIMM or a board with several DRAM devices per rank, the CA bus and clock must reach every DRAM. Two topologies are possible (Figure 28.3):

T (tree)
: The bus splits into balanced branches so that the signal arrives at all DRAMs simultaneously. Each split creates an impedance mismatch and each branch end reflects back into the others; DDR2 used T topologies, but they do not scale to DDR3 rates and beyond.

Fly-by
: The bus passes each DRAM in turn, with short stubs to each device, and is terminated at the end. Reflections are small, because the only discontinuities are short stubs and the DRAMs' input capacitance. The signal arrives at each DRAM progressively later.



(a) Fly-by (daisy-chain) topology with V_{TT} termination



(b) Balanced T (tree) topology

Figure 28.3. Command/address and clock topologies. (a) Fly-by: a daisy chain with short stubs, terminated to V_{TT}, giving clean signals but staggered arrival times. (b) Balanced T: simultaneous arrival but larger reflections.

28.3.1 Loaded impedance and delay

Each DRAM adds an input capacitance C_d (pin, package, and die) at each segment of length ℓ_s along the fly-by bus. Distributing it over the segment gives a loaded line with

$$Z_{\text{loaded}} = \frac{Z_0}{\sqrt{1 + C_d/(C'\ell_s)}}, \qquad t_{\text{pd,loaded}} = t_{\text{pd}}\sqrt{1 + \frac{C_d}{C'\ell_s}}, \tag{28.1}$$

valid when the segment delay is short compared with the rise time. The termination resistor is chosen near Z_{loaded}, and the unloaded segments are sometimes routed at a higher impedance so that the loaded impedance matches the lead-in.

Worked Example 28.2: Loaded fly-by bus

A 40 Ω stripline ($t_{\text{pd}} = 165$ ps/in) passes four DRAMs spaced 0.65 in apart, each with 1.5 pF of input capacitance. Find the loaded impedance, the loaded delay, and the extra arrival delay at the last DRAM relative to the first.

Solution. $C' = t_{\text{pd}}/Z_0 = 4.13$ pF/in, so a segment has 2.68 pF. The loading factor is $\sqrt{1 + 1.5/2.68} = 1.249$, giving $Z_{\text{loaded}} = 32\ \Omega$ and $t_{\text{pd,loaded}} = 206$ ps/in. Three segments add $3 \times 0.65 \times 206 = 402$ ps between the first and the last DRAM—64 % of a 625 ps clock period at DDR4-3200.

28.3.2 Write leveling

Because CK reaches each DRAM at a different time, while each byte lane's DQS is routed point-to-point and arrives at its DRAM after its own delay, the DQS-to-CK relationship differs for every byte lane. DDR3

and later solve this with *write leveling*: in a special mode, each DRAM samples CK with the rising edge of DQS and returns the sampled value on DQ. The controller sweeps the DQS delay of each byte lane until the returned value changes from 0 to 1, which marks DQS aligned with CK at that DRAM (Figure 28.4). The required delay range must cover the fly-by skew plus routing differences; the controller's layout guide specifies the maximum CK-to-DQS skew it can level.

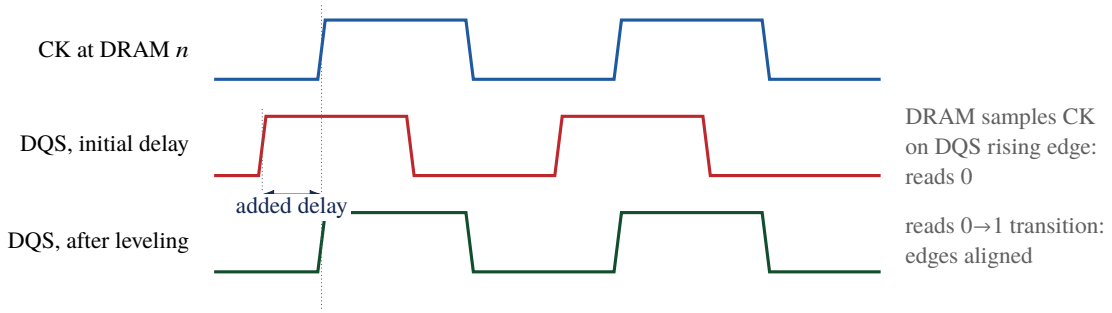


Figure 28.4. Write leveling. The controller delays DQS for a byte lane until its rising edge coincides with the CK rising edge at that lane's DRAM, as reported by the DRAM sampling CK with DQS.

28.3.3 Read and write centering, and VREF training

After write leveling, the controller performs read and write training: it sweeps the DQ-to-DQS delay (per bit in many controllers) to place the sampling point at the center of each bit's eye, and sweeps the internal V_{REF} of the DRAM (for writes) and of the controller (for reads) to center the threshold vertically. The result is a two-dimensional eye scan whose margins are an excellent indicator of board quality at bring-up (Chapter 38).

28.4 Routing and matching budgets

Training removes static skew within its adjustment range and resolution, but not skew that varies with pattern, temperature, or voltage, and not crosstalk or ISI. Layout guides therefore still impose matching rules, which follow from the UI:

- **DQ to DQS within a byte lane:** DQ and DQS share the same source-synchronous budget. Residual mismatch after per-bit deskew (or all of it, if the controller deskews only per byte) consumes setup/hold margin directly. A few percent of the UI is a typical allocation.
- **DQS to CK:** must be within the write-leveling range.
- **CA to CK:** the CA bus is captured by CK at each DRAM; matched as a group with CK, typically to a larger tolerance than DQ-to-DQS.
- **Intra-pair (CK, DQS):** tight, because pair skew distorts the strobe's crossing point.

Worked Example 28.3: DQ-to-DQS matching

For DDR4-3200, allocate 3 % of the UI to static DQ-to-DQS mismatch. What length tolerance is allowed in stripline at 165 ps/in?

Solution. UI = 312.5 ps, so the allocation is 9.4 ps, or 57 mil. Layout guides often specify tighter values, such as 10–25 mil, to leave margin for via-count differences, crosstalk-induced delay variation, and measurement uncertainty.

Design Rule: DDR routing

Route each byte lane (DQ, DM/DBI, DQS pair) on the same layer, with the same number of vias, referenced to ground, and adjacent to no other byte lane over long parallel runs. Match within the lane to the controller's guide in units of delay. Route CA and CK together in fly-by order with minimal stubs, terminate at the end with resistors decoupled to a low-impedance V_{TT}, and keep the total CK-to-DQS skew within the write-leveling range. Decouple V_{REF} (where external) and keep it away from DQ routing.

Table 28.3. Typical DDR routing parameters. Controller layout guides take precedence.

Parameter	Typical practice
DQ/DQS/CA single-ended impedance	36–50 Ω (often 40 Ω)
CK and DQS differential impedance	80–100 Ω
DQ-to-DQS matching	1–5 % of UI, i.e. a few to tens of mil
CA-to-CK matching	Looser than DQ-to-DQS; per guide
Spacing between byte lanes	$\geq 3H$; more over long parallel runs
Fly-by stubs	As short as the package breakout allows

28.5 Common failure modes

Common Pitfall: DDR bring-up failures

- *Write-leveling failure* on the last byte lanes: CK-to-DQS skew larger than the leveling range, usually from an unexpectedly long fly-by or a mistake in delay matching across layers.
- *Training passes, but data errors under load*: insufficient margin from SSN, crosstalk between byte lanes, or a noisy V_{DDQ}/V_{TT}.
- *Address errors at high temperature*: CA bus reflections from long stubs, or wrong termination value, eroding a small margin.
- *One bit consistently worst*: a via-count mismatch, a split-plane crossing, or a neck-down under the BGA.

Capture the controller's training results and two-dimensional eye margins on the first boards; they localize most of these problems to a byte lane or a bit.

Rule of Thumb: Loaded delay of a fly-by bus

DRAM input capacitance typically increases the propagation delay of a fly-by bus by 20–40 % and lowers its effective impedance by the same factor (Eq. (28.1)). *Limits:* depends on device capacitance and segment length; use the controller vendor's loaded-delay values or simulation for write-leveling range checks.

Key Takeaways

- DDR data are source-synchronous in byte lanes; CA and CK are common-clock and routed fly-by.
- POD (DDR4/5) and LVSTL (LPDDR4/5) signaling use on-die termination and an internal, trained V_{REF}.

- Fly-by gives clean signals but staggered arrival; write leveling aligns DQS to CK per byte lane.
- Loaded impedance and delay follow Eq. (28.1); terminate the CA bus near Z_{loaded}.
- Matching budgets derive from the UI; training removes static skew but not noise, crosstalk, or ISI.

Further Reading

The JEDEC standards [44], [57], [65], [66] define the interfaces; memory-controller vendors' layout guides specify the topology and matching rules for their training algorithms.

Review Problems

P28.1 Find the UI and Nyquist frequency of LPDDR5-6400.

P28.2 Allocate 2 % of the UI of DDR5-4800 to DQ-to-DQS mismatch. What is the length tolerance at 165 ps/in?

P28.3 A 45 Ω bus segment has 2.5 pF of line capacitance and carries a 1.2 pF load. Find the loaded impedance.

P28.4 A CA bus of 24 lines draws ±8 mA per line from V_{TT}. What source/sink capability is needed?

P28.5 Five DRAMs are spaced 0.55 in apart on a fly-by bus with a loaded delay of 210 ps/in. Find the CK arrival spread.

P28.6 An external V_{REF} must stay within ±1 % of $V_{\text{DDQ}} = 1.2$ V. What is the allowed noise?

P28.7 A colleague proposes reusing the DDR4 rule "DQ within ±10 mil of DQS" for an LPDDR5 byte. What is wrong, and what information do you need before writing the LPDDR5 rules?

Chapter 29

PCI Express Gen3 to Gen6

Chapter Overview

PCI Express is the reference example of a modern serial link: AC-coupled differential lanes with embedded clocking, adaptive equalization, and a channel specified by its insertion loss at the Nyquist frequency. Each generation has doubled the data rate, from 8 GT/s in Gen3 to 32 GT/s in Gen5 and 64 GT/s with PAM4 in Gen6. This chapter summarizes the electrical layer of PCIe, explains channel loss budgeting and reference-clock architectures, and gives the routing rules that let a board meet its share of the budget.

Learning Objectives

1. Tabulate data rate, encoding, Nyquist frequency, and typical loss budget per generation.
2. Allocate a channel insertion-loss budget among packages, board, connectors, and vias.
3. Select and place AC-coupling capacitors.
4. Describe common-clock and separate-reference-clock architectures and SSC.
5. Apply PCIe routing rules for impedance, skew, vias, and stubs.
6. Explain what changes with PAM4 in Gen6.

29.1 Electrical overview

A PCIe link consists of one or more lanes (×1, ×4, ×8, ×16), each with a transmit and a receive differential pair. Each pair is AC coupled, so transmitter and receiver may have different common-mode voltages. There is no forwarded clock: the receiver's CDR recovers timing from the data, and lane-to-lane skew is removed by deskew logic in the receiver, so lane matching is loose. Link training negotiates speed and transmitter equalization presets, and Gen3 and later adapt the receiver's CTLE and DFE and the transmitter's FFE coefficients during an equalization phase [67].

Table 29.1. PCIe generations. Loss budgets are commonly cited end-to-end (bump-to-bump) values at Nyquist and are given for orientation only.

Gen	Rate (GT/s)	Encoding	Signaling	f_{Nyq} (GHz)	Typical loss budget
1	2.5	8b/10b	NRZ	1.25	—
2	5.0	8b/10b	NRZ	2.5	—
3	8.0	128b/130b	NRZ	4.0	≈ 22 dB
4	16.0	128b/130b	NRZ	8.0	≈ 28 dB
5	32.0	128b/130b	NRZ	16.0	≈ 36 dB
6	64.0	FLIT + FEC	PAM4 (32 GBd)	16.0	≈ 32 dB

Specification Note

Channel loss budgets, reference-clock jitter limits, AC-coupling capacitance ranges, and equalization presets are defined in the PCI Express Base Specification [67], [68] and, for add-in cards and slots, in the Card Electromechanical (CEM) specification [69]. The loss values in Table 29.1 are those widely quoted in industry literature. Use the current specifications, and the silicon vendor's channel guidelines, for design.

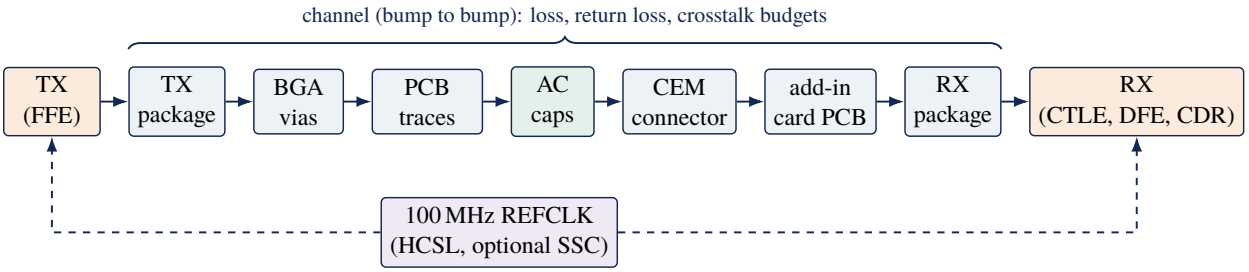


Figure 29.1. A PCIe channel from transmitter to receiver, through packages, system-board traces, AC-coupling capacitors, the CEM connector, and an add-in card. The reference clock is distributed to both ends in a common-clock architecture.

29.2 Channel loss budgeting

The specification's budget covers the whole channel from transmitter bump to receiver bump (Figure 29.1). Silicon vendors allocate part of it to their packages and to reference add-in-card or cable designs; the remainder is the system board's. A typical allocation is shown in Figure 29.2.

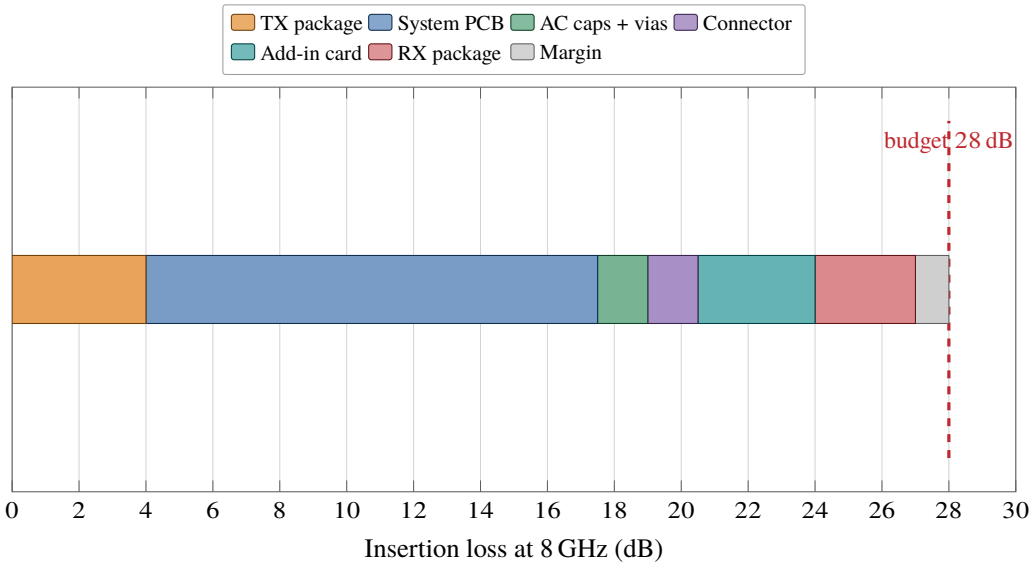


Figure 29.2. An illustrative insertion-loss allocation at 8 GHz for a PCIe Gen4 channel with a 28 dB budget. Actual allocations come from the silicon vendor and the CEM specification.

Worked Example 29.1: Maximum Gen4 trace length

For the allocation of Figure 29.2, the system board receives 13.5 dB at 8 GHz for its traces. How long may the route be in the mid-loss and low-loss materials of Worked Example 11.2?

Solution. In Worked Example 11.2 the HVLP conductor loss of a 5 mil, 50 Ω-equivalent stripline at 8 GHz was 0.72 dB/in. With $D_f = 0.010$, $D_k = 3.7$, dielectric loss is $2.31 \times 8 \times 0.010 \times 1.92 = 0.35$ dB/in; total 1.07 dB/in, allowing 12.6 in. With $D_f = 0.004$, $D_k = 3.5$, the dielectric term falls to 0.14 dB/in and the route may be 15.7 in. Vias, breakouts, and temperature derating reduce these lengths in practice. (An 85 Ω differential pair has somewhat different conductor loss from the single-ended estimate; a field solver should be used for final numbers.)

Insertion loss is not the only requirement. The specification also constrains return loss, insertion-loss deviation (ripple about a smooth fit), and crosstalk, or combines all of these into a channel compliance metric evaluated by simulation with reference transmitter and receiver models. Discontinuities that create reflections or resonances are often what fail a channel that meets its loss target.

29.3 AC coupling

Each transmit pair carries a series capacitor in each leg, normally placed near the transmitter (or as required by the CEM specification for slot-based designs). Because PCIe data are scrambled (Gen3 and later) or DC balanced (8b/10b), the low-frequency content is small. Each leg's capacitor is in series between the transmitter's source resistance R_S and the receiver's termination R_L, so the high-pass corner is

$$f_{HP} = \frac{1}{2\pi (R_S + R_L)\, C_{AC}}, \tag{29.1}$$

which with $R_S = R_L \approx 50\,\Omega$ per leg lies in the kilohertz range. For example, 220 nF gives $1/(2\pi \times 100 \times 220 \times 10^{-9}) = 7.23$ kHz. (Using R_L alone overstates the corner by a factor of two.)

Baseline wander. During a run of identical symbols the coupling capacitor charges and the received level droops toward zero with time constant $\tau = (R_S + R_L)C_{AC} = 22\,\mu$s. For a run of N symbols, the fractional droop is about $N\,\mathrm{UI}/\tau$; a 100-symbol run at 8 GT/s (12.5 ns) droops by 0.06 %. Scrambling keeps runs short, so baseline wander is negligible for PCIe; it matters for unscrambled or poorly balanced data and for much smaller capacitors.[1]

Design Rule: AC-coupling capacitors

Use the smallest case size that the assembly process supports (0201 or 0402), place the two capacitors of a pair side by side and symmetric, and keep the footprint on the same layer as the traces to avoid extra vias. If the pads are wider than the trace and the reference is close, they form a capacitive discontinuity; whether to void the reference plane under them (referencing the next ground plane down) depends on pad size, dielectric height and data rate. Decide with a field-solver or 3-D model of the actual footprint, and check that any void does not cut the return path of neighboring traces.

[1] The Base Specification defines an allowed AC-coupling capacitance range for each generation (on the order of 75–265 nF, with a higher minimum for Gen3 and later). Use the values in the current specification.

29.4 Reference clocks

PCIe uses a 100 MHz reference clock, usually HCSL, with a frequency tolerance of ±300 ppm. Three architectures are defined:

Common clock
The same reference clock (with or without SSC) is distributed to both ends. Jitter common to both ends is partly cancelled by the receiver's CDR, so the jitter limit applies to the difference.

Separate reference, no SSC (SRNS)
Each end has its own reference; SSC is not used.

Separate reference, independent SSC (SRIS)
Each end has its own reference with SSC; the receiver must absorb the larger frequency offset, which requires extra elasticity in the logic.

SSC down-spreads the reference by up to 0.5 % at 30–33 kHz (Chapter 27). Reference-clock jitter limits are tight and are specified after filtering by the PLL and CDR transfer functions of the link; clock buffers and their layout are discussed in Chapter 33.

29.5 Routing rules

Table 29.2. Typical PCIe board routing practice.

Parameter	Typical practice
Differential impedance	85 Ω nominal (many designs use 85 Ω–100 Ω); ±10 %
Intra-pair skew	A few mils; compensate at the source of mismatch (Chapter 13)
Lane-to-lane skew	Loose (receiver deskew); keep TX lanes similar for EMI
TX-to-RX spacing	Route TX and RX on different layers or with large spacing to limit NEXT
Pair-to-pair spacing	$\geq 3H$ to $5H$ (stripline), more for long parallel runs
Via stubs	Back-drill or route on deep layers at Gen4 and above (Chapter 15)
Reference	Continuous ground; no splits; return vias at every layer change
Laminate	Mid-loss for Gen3/Gen4; low or very-low loss for Gen5 and Gen6

Common Pitfall: Routing TX and RX pairs side by side

Interleaving transmit and receive pairs of the same port. A strong transmitter near the start of its channel couples by NEXT into a weak, attenuated receive signal near the end of its channel, where the signal-to-crosstalk ratio is worst. Separate TX and RX groups by layer or by generous spacing, especially near connectors and BGAs.

29.6 Gen6 and PAM4

PCIe Gen6 doubles the data rate to 64 GT/s without raising the symbol rate above Gen5's 32 GBd by transmitting two bits per symbol with four-level pulse amplitude modulation (PAM4, Figure 29.4) [68]. The Nyquist frequency, and hence the channel loss per inch, stays at 16 GHz, but each of the three stacked eyes has one-third of the NRZ amplitude, a 9.5 dB penalty in signal-to-noise ratio. Gen6 compensates by allowing a higher raw error rate and correcting it with forward error correction (FEC) in a fixed-size flow-control-unit (FLIT)

mode. For the board designer, the consequences are tighter limits on noise sources—crosstalk, reflections, and power-supply noise—rather than on loss alone.

Figure 29.3 shows the effect under identical conditions: the same symbol rate, the same 2 unit peak-to-peak swing, the same one-pole channel and the same Gaussian noise. At fixed swing, PAM4's adjacent levels are one-third as far apart as NRZ's, so the ideal threshold distance falls from 1 to $1/3$ (-9.54 dB). Keep the symbol UI and the bit time distinct: at 32 GBd, the UI is 31.25 ps, the Nyquist frequency is 16 GHz, and uncoded PAM4 carries 64 Gb/s before coding and protocol overhead. The figure is an illustration, not a compliance eye: it has no jitter, CTLE, DFE, CDR, FEC or mask, and a few hundred overlaid symbols say nothing about BER at 10^{-12}.

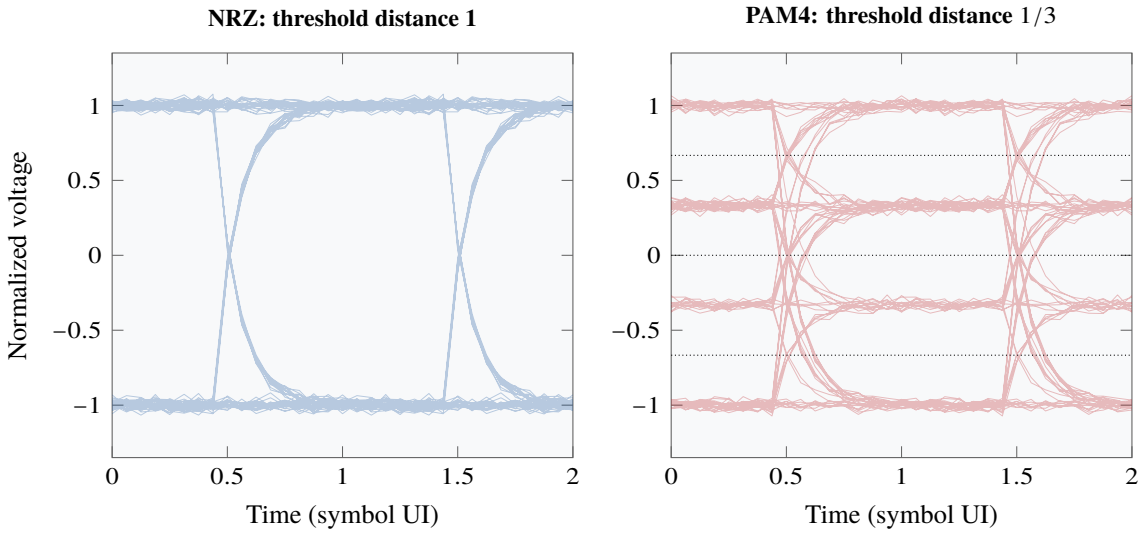


Figure 29.3. Synthetic NRZ and PAM4 eyes at the same symbol rate and swing: 1600 random symbols, one-pole channel ($\tau = 0.1$ UI), additive Gaussian noise $\sigma = 0.025$, 80 overlaid two-UI windows, fixed seed. Dotted lines: PAM4 decision thresholds. [Simulation (synthetic)]

29.6.1 Raw and post-FEC error rates

With FEC the board no longer has to deliver a raw bit-error ratio of 10^{-12}. The link tolerates a much higher *raw* (first-burst or pre-FEC) error ratio at the receiver slicer, and the FEC decoder corrects most of those errors. Two cautions follow. First, FEC performance depends on error *statistics*: DFE error propagation and correlated noise produce bursts that a code tuned for random errors corrects less well. Second, FEC changes the reliability interpretation of a margin; it does not restore analog eye height. Channel compliance is therefore judged at the raw error ratio the specification assigns, measured or simulated with the reference receiver.

29.6.2 Retimers and channel partitioning

When a channel cannot meet its loss budget—long backplanes, cabled links, or a riser plus an add-in card—a *retimer* splits it into two electrically independent segments. A retimer is a full receiver and transmitter (CDR, equalization, re-launch) that takes part in link training; a *redriver* is only a linear amplifier and equalizer that passes jitter and noise through. Each retimer segment must meet the budget on its own, which makes the partition a board-architecture decision: where the retimer sits determines the two segment lengths, the

number of connectors in each, and the reference-clock architecture. Place it so that both segments have margin, not so that one is minimal and the other marginal.

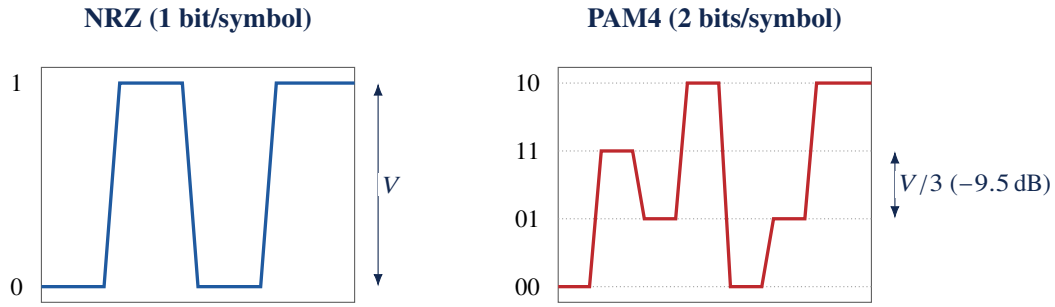


Figure 29.4. NRZ and PAM4 at the same peak-to-peak swing. PAM4 carries two bits per symbol with Gray-coded levels; each eye has one-third of the NRZ amplitude.

Worked Example 29.2: Bandwidth per lane

What is the payload bandwidth of a Gen4 ×16 link, ignoring protocol overhead above the encoding?

Solution. Each lane carries $16 \times 128/130 = 15.75\,\text{Gb/s}$. Sixteen lanes carry $252\,\text{Gb/s}$, or about $31.5\,\text{GB/s}$ in each direction.

Rule of Thumb: Laminate per generation

Each doubling of the PCIe data rate roughly doubles the loss per inch at Nyquist for the same material; keeping the same route length usually requires moving one loss class lower (Table 9.3) per generation. *Limits:* Gen6 keeps Gen5's Nyquist frequency, and short routes may stay on a higher-loss material; the budget calculation of Worked Example 29.1 decides.

Key Takeaways

- PCIe lanes are AC-coupled, embedded-clock differential pairs; lane-to-lane skew is loose, intra-pair skew tight.
- Each generation doubles the data rate; Gen6 keeps Gen5's Nyquist frequency by using PAM4.
- Budget the channel's loss at Nyquist among packages, board, connectors, and vias; return loss and crosstalk often decide compliance.
- Place AC caps symmetrically; void the reference under the pads only where a 3-D model shows the pad capacitance matters. The AC-coupling corner is $1/[2\pi(R_S + R_L)C]$.
- Choose the reference-clock architecture (common, SRNS, SRIS) early; it constrains SSC and jitter.

Further Reading

The PCI Express Base and CEM specifications [67]–[69] define the electrical layer; IEEE 370 [29] covers characterization of the board interconnect.

Review Problems

P29.1 Find the UI and Nyquist frequency of PCIe Gen5.

P29.2 What is the payload rate per lane of Gen4 with 128b/130b encoding?

P29.3 What are the symbol rate and UI of Gen6, and the amplitude penalty of each PAM4 eye relative to NRZ?

P29.4 Find the high-pass corner of a 100 nF AC-coupling capacitor with 50 Ω source and 50 Ω termination.

P29.5 A board is allotted 14 dB for traces at Nyquist, and its routing loses 1.1 dB/in. What is the maximum route length?

P29.6 A 100 MHz reference clock uses 0.5 % down-spread SSC. What is its minimum instantaneous frequency?

Chapter 30

USB 2.0, USB 3.x, USB4, and Type-C

Chapter Overview

USB combines two very different physical layers in one connector: the 480 Mb/s USB 2.0 half-duplex pair and the multi-gigabit, AC-coupled SuperSpeed and USB4 lanes. The Type-C connector added reversibility, configuration through the CC pins, and power delivery up to tens of volts. This chapter summarizes the electrical parameters of each generation, explains how Type-C orientation affects routing, and gives layout guidance for the USB 2.0 pair, the high-speed lanes, ESD protection, and VBUS.

Learning Objectives

1. Tabulate signaling rate, encoding, impedance, and coupling for USB 2.0, USB 3.2, and USB4.
2. Explain the Type-C pin map and the routing consequences of reversibility.
3. Route the USB 2.0 pair and its orientation stubs.
4. Place AC-coupling capacitors, orientation muxes, redrivers, and ESD devices on the high-speed lanes.
5. Describe CC-based configuration and size VBUS routing.

30.1 Generations and signaling

Table 30.1. USB physical layers (representative; see the specification note).

Mode	Rate per lane	Encoding	Lanes	Coupling
USB 2.0 High Speed [70]	480 Mb/s	NRZI, bit stuffing	1 pair, half duplex	DC
USB 3.2 Gen 1 [71]	5 Gb/s	8b/10b	1 TX + 1 RX	AC (TX caps)
USB 3.2 Gen 2	10 Gb/s	128b/132b	1 TX + 1 RX	AC
USB 3.2 Gen 2×2	10 Gb/s	128b/132b	2 TX + 2 RX (Type-C)	AC
USB4 Gen 2/Gen 3 [72]	10 / 20 Gb/s	64b/66b, 128b/132b	2 TX + 2 RX	AC

USB 2.0 High Speed drives a 17.78 mA current into a 45 Ω-terminated line at each end, producing about 400 mV differential across a 90 Ω pair. It is DC coupled, half duplex, and slow enough that board routing rarely limits it—provided the pair's impedance and symmetry are maintained and the Type-C stubs are short.

SuperSpeed and USB4 lanes are AC-coupled, embedded-clock links like PCIe (Chapter 29), with transmitter de-emphasis and receiver equalization. USB4 version 2.0 extends the rate further with multi-level signaling.

Specification Note

Exact impedance targets (90 Ω nominal for USB differential pairs, with tolerances that differ between boards, cables, and generations), AC-coupling capacitance ranges, channel loss allocations, and USB4 version-2 signaling details are defined in the USB-IF specifications [70]–[73]. Check the current revisions and the silicon vendor's guidelines.

30.2 The Type-C connector

The Type-C receptacle has 24 contacts in two rows, arranged with 180-degree rotational symmetry so that the plug can be inserted either way up (Figure 30.1) [73]. Its signals are:

- two SuperSpeed/USB4 transmit pairs (TX1, TX2) and two receive pairs (RX1, RX2), of which one TX/RX set is used in a single-lane mode, selected by orientation;
- the USB 2.0 pair, which appears twice (A6/A7 and B6/B7) and is connected to both locations on the host;
- two configuration channel pins (CC1, CC2), through which the host and device detect attachment, orientation, current capability, and power delivery;
- two sideband-use pins (SBU1, SBU2), used by alternate modes;
- four VBUS and four GND contacts.

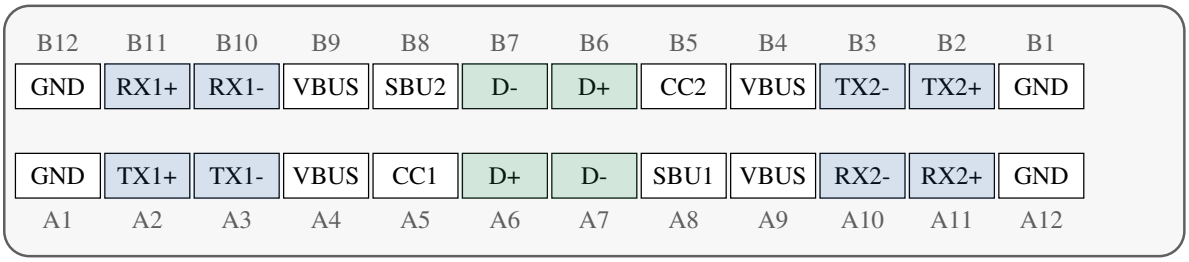


Figure 30.1. Type-C receptacle contact map (front view). Row B is rotated by 180 degrees relative to row A, making the connector reversible.

30.2.1 Configuration channel

A source presents a pull-up R_p (or current source) on CC; a sink presents a pull-down R_d. The voltage on the CC line that has R_d identifies the orientation and advertises the source's current capability (default USB power, 1.5 A, or 3 A). USB Power Delivery then negotiates higher voltages over the same CC wire.[1]

30.3 Routing the USB 2.0 pair

On the host side, D+ connects to both A6 and B6 and D− to both A7 and B7. In either orientation, one of each pair of contacts is unused and forms a stub.

[1]The commonly used values are $R_d = 5.1\,\text{k}\Omega$ in a sink and, for a source pulling up to 5 V, $R_p = 56\,\text{k}\Omega$, $22\,\text{k}\Omega$, or $10\,\text{k}\Omega$ for default, 1.5 A, and 3 A advertisement. Confirm against the current Type-C specification, which also defines current-source and 3.3 V pull-up alternatives.

Worked Example 30.1: Is the orientation stub a problem?

USB 2.0 High Speed edges are no faster than about 500 ps. What is the critical length on an outer layer with $t_{pd} = 150\,\text{ps/in}$?

Solution. $\ell_{crit} = t_r/(6t_{pd}) = 500/900 = 0.56\,\text{in}$. Connecting the two contact rows with a short jog under the connector produces stubs of well under 0.2 in, so the stub is electrically short. A long branch, or tying the contacts together at the far side of a mux, would not be.

Design Rule: USB 2.0 pair

Route D+/D− as a 90 Ω pair over a continuous ground plane, tie A6 to B6 and A7 to B7 directly at the receptacle with the shortest possible stubs, place the ESD device and any common-mode choke close to the connector, and keep the pair away from SuperSpeed lanes and clocks.

30.4 High-speed lanes

Figure 30.2 shows the typical host-side path. Because the active TX and RX pairs depend on orientation, a host either uses a two-lane PHY that can handle both (as USB4 and Gen 2×2 require) or inserts an orientation multiplexer controlled by the CC logic. Long host routes may need a linear redriver or a retimer; a retimer resets the jitter and loss budgets, while a redriver only amplifies and equalizes.

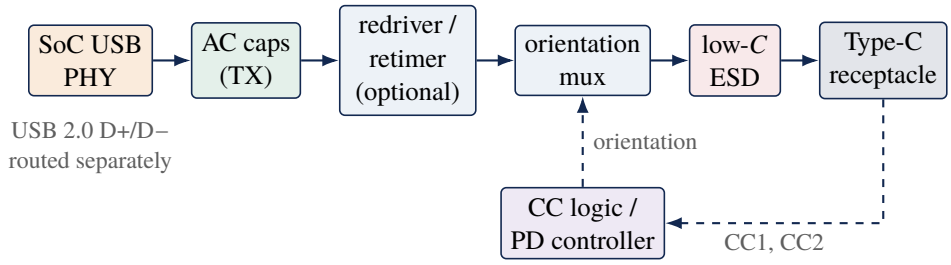


Figure 30.2. Host-side signal path for SuperSpeed or USB4 lanes to a Type-C receptacle. The CC logic determines orientation and configures the mux.

30.4.1 ESD protection

Every line exposed at the connector needs ESD protection, placed between the connector and everything else so that the discharge current returns through the shortest path to ground. On high-speed lanes the protector's capacitance is a shunt discontinuity. A shunt capacitance C on a line of impedance Z_0 reflects

$$|\Gamma| = \frac{x}{\sqrt{4 + x^2}}, \qquad x = \omega C Z_0, \tag{30.1}$$

which follows from Eq. (7.2) with $Z_L = Z_0 \parallel (1/j\omega C)$.

Worked Example 30.2: ESD capacitance and return loss

A 10 Gb/s lane ($f_{Nyq} = 5\,\text{GHz}$) uses an ESD device with 0.3 pF per line on a 90 Ω pair (45 Ω per leg in the odd mode). Estimate the reflection at Nyquist, and repeat for 0.5 pF.

Solution. $x = 2\pi \times 5 \times 10^9 \times 0.3 \times 10^{-12} \times 45 = 0.424$, so $|\Gamma| = 0.207$ (return loss 13.7 dB). With 0.5 pF, $x = 0.707$ and $|\Gamma| = 0.333$ (9.5 dB). Low-capacitance protectors (well below 0.5 pF), flow-through

packages that the pair passes over without stubs, and a slight local impedance increase to compensate are essential at these rates.

A single capacitance is only a first estimate. A real protector is a package with pin inductance, a die capacitance that depends on bias, and internal routing between the pins of a flow-through package. Vendors of high-speed protectors publish S-parameter models of the device on a test board; use them in the channel simulation of Chapter 36, after de-embedding the vendor's test fixture or confirming that its reference planes match your footprint. Where a model is not available, a TDR of the footprint on a test coupon is the next best evidence.

Common Pitfall: ESD device far from the connector

Placing the ESD protector near the SoC "to protect it" or routing its ground through a long trace. The discharge current then travels across the board, coupling into other circuits, and the protector's ground connection adds enough inductance to raise the clamp voltage. Place the protector at the connector with its ground pins via'd directly to the ground plane.

30.5 VBUS and power

Type-C sources supply 5 V by default and, with Power Delivery, higher voltages (up to 20 V in standard power range, and up to 48 V with extended power range in recent revisions). VBUS traces and pours must carry up to 3–5 A with acceptable IR drop and temperature rise (IPC-2152), and VBUS needs bulk and local capacitance within the limits the specification imposes on inrush. Connector VBUS pins should be connected with wide copper and multiple vias.

Worked Example 30.3: VBUS drop

A 3 A VBUS path runs 2 in in a 100 mil wide, 1 oz trace. Estimate the drop and dissipation.

Solution. The trace is 20 squares long, so $R = 20 \times 0.49\,\text{m}\Omega = 9.8\,\text{m}\Omega$; the drop is 29 mV and the dissipation 88 mW. Acceptable for 5 V, but a wider pour or plane would be preferred at 5 A.

Table 30.2. USB layout summary.

Item	Guidance
USB 2.0 pair	90 Ω; short orientation stubs; ESD and CMC at connector
SuperSpeed / USB4 pairs	85 Ω–90 Ω per guide; tight intra-pair skew; TX AC caps
Orientation	Two-lane PHY or mux driven by CC logic
Redriver / retimer	For routes beyond the host's loss allocation
ESD	Sub-picofarad, flow-through, at the connector
CC	R_p/R_d or PD controller; route as low-speed but keep away from SS lanes
VBUS	Wide copper; IR drop and inrush; protection against overvoltage

Rule of Thumb: ESD capacitance

For lanes at 10 Gb/s and above, keep the protector's capacitance below about 0.3 pF per line; by Eq. (30.1) this keeps its reflection near −14 dB or better at 5 GHz. *Limits:* a locally compensated footprint can tolerate more capacitance; higher rates need less.

Key Takeaways

- USB combines a DC-coupled 480 Mb/s pair and AC-coupled multi-gigabit lanes in one connector.
- Type-C is reversible; D+/D− appear twice and must be tied with short stubs; high-speed lanes need a two-lane PHY or a mux.
- ESD capacitance is a major return-loss contributor at 10 Gb/s and above.
- CC pins set orientation and power; VBUS can reach tens of volts and several amperes.

Further Reading

The USB-IF specifications [70]–[73] define the electrical layer and the Type-C connector; silicon vendors' layout guides give host- and device-specific rules.

Review Problems

P30.1 Find the UI and Nyquist frequency of USB 3.2 Gen 2.

P30.2 A USB4 Gen 3 lane runs at 20 Gb/s. Find its UI and Nyquist frequency.

P30.3 Compute the payload rate of a Gen 1 lane (8b/10b) and a Gen 2 lane (128b/132b).

P30.4 Find the reflection coefficient at 5 GHz of a 0.5 pF shunt capacitance on a 45 Ω line.

P30.5 What pull-down resistance identifies a Type-C sink on CC?

P30.6 Find ℓ_{crit} for a 500 ps edge with $t_{\text{pd}} = 150$ ps/in.

Chapter 31

Ethernet: 1000BASE-T, 10GBASE-KR, SGMII, and RGMII

Chapter Overview

Ethernet on a board involves two very different kinds of interface. On the *media* side, the PHY drives twisted-pair cable through isolation magnetics (1000BASE-T) or a backplane channel (10GBASE-KR). On the *MAC* side, the PHY connects to the MAC in the SoC or switch through a parallel source-synchronous bus (RGMII) or a serial link (SGMII and its relatives). This chapter covers both: the analog front end, isolation, and EMC layout around the magnetics and connector; the backplane channel of 10GBASE-KR; and the timing of RGMII and the routing of SGMII.

Learning Objectives

1. Describe 1000BASE-T signaling and the role of the magnetics.
2. Lay out the PHY, magnetics, and RJ45 region with correct isolation and chassis grounding.
3. Explain the Bob Smith termination and its limits.
4. Estimate the channel loss of a 10GBASE-KR backplane.
5. Close RGMII timing with internal or PCB clock delay.
6. Route SGMII as a low-rate SERDES link.

31.1 1000BASE-T

1000BASE-T transmits 1 Gb/s over four twisted pairs at 125 MBd per pair, using five-level PAM (PAM5) with trellis coding. Each pair carries traffic in both directions simultaneously; the PHY cancels its own transmitted signal (echo cancellation) and crosstalk from the other three pairs (NEXT and FEXT cancellation) digitally [74]. The board's job on the media side is to deliver a balanced, well-isolated, low-emission connection between the PHY's MDI pins and the cable.

31.1.1 Magnetics

Each pair passes through an isolation transformer and usually a common-mode choke (Figure 31.1). The transformer provides galvanic isolation between the cable and the equipment—IEEE 802.3 requires the interface to withstand a specified test voltage[1]—and transfers the differential signal while blocking DC. Its PHY-side center tap is biased or decoupled according to the PHY type (current-mode PHYs need a supply on the center tap; voltage-mode PHYs need only a capacitor). The common-mode choke attenuates common-mode noise traveling in either direction.

[1] Commonly 1500 V rms for 60 s, or an equivalent; see the isolation clause of the current IEEE 802.3 standard.

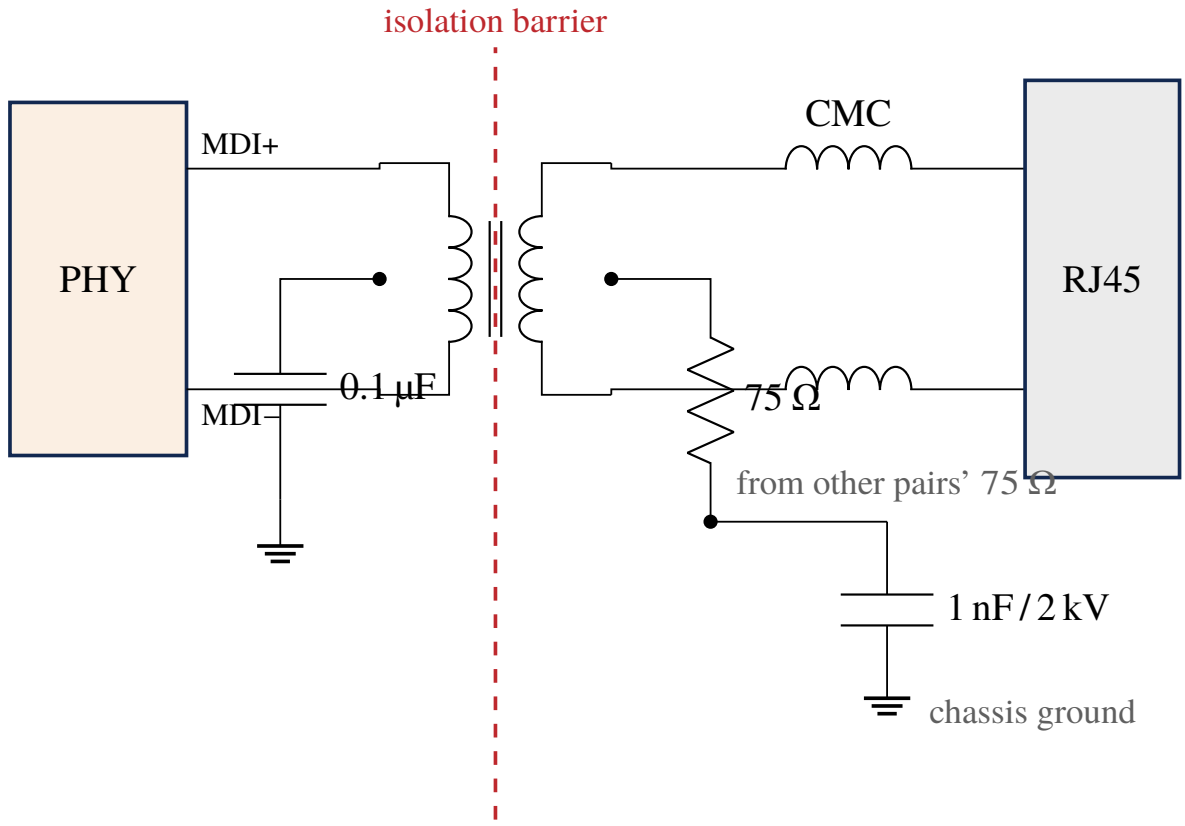


Figure 31.1. One pair of a 1000BASE-T interface: PHY, isolation transformer with center taps, common-mode choke (CMC), RJ45, and Bob Smith termination of the cable-side center tap to chassis ground.

31.1.2 Bob Smith termination

The cable-side center tap of each pair is connected through a resistor (commonly 75 Ω) to a common node, which is connected to chassis ground through a high-voltage capacitor (often 1 nF, 2 kV). The intent is to terminate the common mode of each pair and the pair-to-pair modes of the cable, reducing common-mode emissions. The 75 Ω value derives from the common-mode impedance of telephone-type cable; whether it is optimal for Ethernet cabling has been debated, and many integrated magnetics modules include their own version of this network. Follow the PHY and magnetics vendors' recommendations.

31.1.3 Layout

Design Rule: Ethernet media-side layout

Place the magnetics close to the RJ45 (or use an RJ45 with integrated magnetics). Route each MDI pair as a 100 Ω differential pair, matched within the pair, with no vias if possible. Remove all signal-ground and power copper from every layer in the isolation gap under the magnetics (Figure 31.2), keeping creepage and clearance as required by the safety standard. Connect the connector shield and the Bob Smith capacitor to chassis ground, stitched with vias, and keep unrelated traces out of the connector zone.

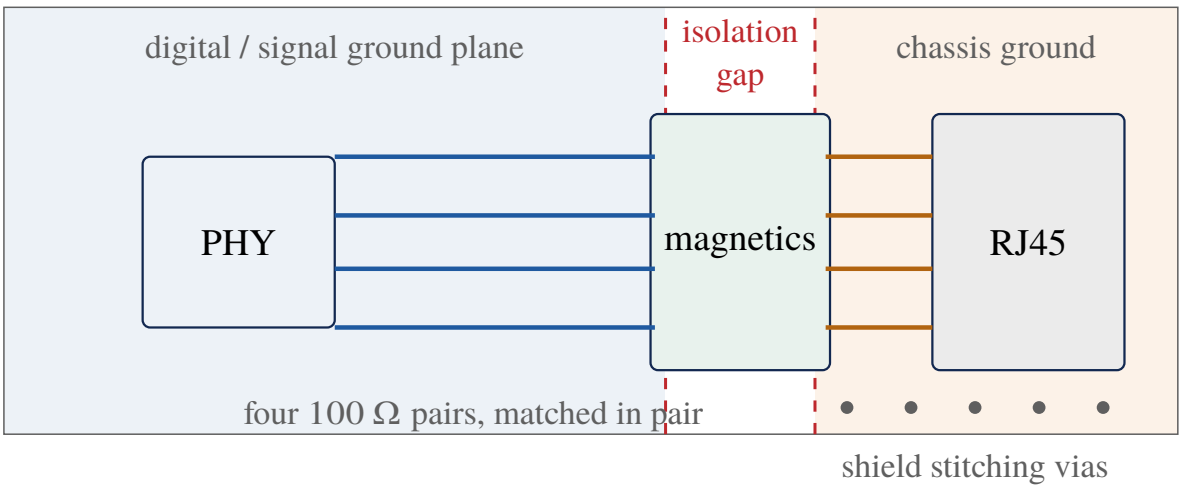


Figure 31.2. Layout zones of an Ethernet port: signal ground under the PHY, an isolation gap under the magnetics, and chassis ground under the connector.

Common Pitfall: Routing other signals through the isolation gap

Passing an LED line, a clock, or a power trace across the gap between PHY ground and chassis ground, or under the magnetics. The crossing defeats the isolation, couples noise into the cable's common mode, and often appears as an emissions failure at 125 MHz harmonics. LED lines of RJ45s with integrated LEDs should be routed out of the connector on the PHY side of the gap and filtered.

31.2 10GBASE-KR

10GBASE-KR runs 10.3125 Gb/s (64b/66b encoding) over a backplane, with a three-tap transmit FFE adapted by a training protocol, and with receiver equalization. The unit interval is 96.97 ps, and the Nyquist frequency is 5.156 GHz. IEEE 802.3 specifies the backplane channel by insertion-loss, return-loss, and crosstalk limits as functions of frequency rather than by a single number [74]; the same electrical framework is reused by many chip-to-chip 10 Gb/s SERDES links.

Worked Example 31.1: Backplane channel loss

Estimate the line loss at 5.156 GHz of a 20 in, 6 mil, 50 Ω-equivalent stripline in a mid-loss laminate ($D_f = 0.010$, $D_k = 3.8$) with 1 µm RMS roughness.

Solution. $\delta_s = 0.92$ µm and $K_H = 1.65$, so conductor loss is $36\sqrt{5.156}/(6 \times 50) \times 1.65 = 0.45$ dB/in. Dielectric loss is $2.31 \times 5.156 \times 0.010 \times \sqrt{3.8} = 0.23$ dB/in. Total 0.68 dB/in, or 13.6 dB for 20 in, before the two backplane connectors and the line cards' traces and vias.

31.3 MAC-to-PHY interfaces

31.3.1 RGMII

The Reduced Gigabit Media Independent Interface carries four data bits and a control signal in each direction, clocked on both edges of a 125 MHz clock for gigabit operation (250 Mb/s per data line). It is source synchronous (Chapter 17): the transmitter launches data aligned with its clock edges, and the receiver needs the clock delayed by roughly a quarter of the clock period so that it samples in the middle of the data eye

(Figure 31.3). Early versions of the RGMII specification placed this delay in the PCB routing; later versions allow it to be generated inside the transmitter or receiver ("RGMII-ID").[2]

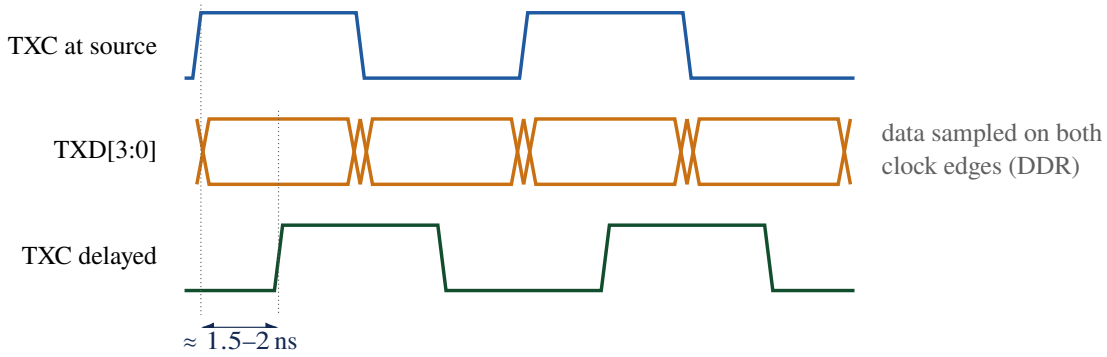


Figure 31.3. RGMII transmit timing. Data change at the source clock edges; delaying the clock by about 1.5–2 ns centers each edge in the 4 ns data eye.

Worked Example 31.2: Clock delay in routing

If a 2 ns RGMII clock delay were implemented in PCB routing with $t_{pd} = 150\,\text{ps/in}$, how much extra trace would be needed?

Solution. $2000/150 = 13.3$ in of extra clock trace—impractical and noisy. Enable the internal delay in exactly one of the two devices (MAC or PHY) for each direction, and match the data lines to their clock to a few tens of picoseconds.

Worked Example 31.3: A signed RGMII budget

A receiver needs (illustratively) 1.0 ns setup and 0.8 ns hold, the internal clock delay is 2.0 ns, and the data-to-clock skew at the receiver, $\Delta = t_{data} - t_{clk}$ from transmitter output skew plus board mismatch, ranges from −0.5–0.3 ns. Find the margins.

Solution. With a 4 ns data eye and sampling at the delayed clock edge (Eq. (17.3)), setup margin $= 2.0 - 1.0 - \Delta_{max} = 0.7$ ns and hold margin $= 4.0 - 2.0 - 0.8 + \Delta_{min} = 0.7$ ns. A symmetric ± 0.5 ns budget would have reported 0.5 ns of setup margin; the signed range shows that the skew is biased toward early data, so the board can tolerate 0.2 ns more late-data mismatch than a symmetric rule allows. Use the actual minimum and maximum values from the MAC and PHY data sheets.

Common Pitfall: Double or missing RGMII delay

Enabling the internal clock delay in both the MAC and the PHY, or in neither. The link may come up at 10 or 100 Mb/s, where the timing budget is enormous, and fail only at 1 Gb/s. Check the delay configuration in both devices' strapping and driver settings at bring-up.

31.3.2 SGMII and serial MAC interfaces

SGMII carries gigabit Ethernet over one differential pair in each direction at 1.25 Gb/s (8b/10b). Its signals are low-swing differential (LVDS-like), commonly AC coupled, and routed as 100 Ω pairs with modest intra-

[2]The required delay is usually quoted as about 1.5–2.0 ns; consult the RGMII version used by the MAC and PHY, and their data sheets.

pair matching. Its higher-rate relatives (2500BASE-X, USXGMII, 10GBASE-R interfaces such as XFI/SFI) follow the same SERDES rules as PCIe lanes of similar rate (Chapter 29).

Table 31.1. Ethernet interfaces on boards.

Interface	Rate	Board design notes
1000BASE-T MDI	4 × 125 MBd PAM5	100 Ω pairs; magnetics; isolation gap; chassis ground
10GBASE-KR	10.3125 Gb/s	Backplane SERDES; loss, RL, crosstalk masks
RGMII	4 × 250 Mb/s (DDR)	Source synchronous; one clock delay per direction
SGMII	1.25 Gb/s	100 Ω pair, usually AC coupled

Rule of Thumb: RGMII clock delay

The RGMII clock delay is about one quarter of the 8 ns clock period, i.e. about 2 ns, which places each clock edge near the center of the 4 ns data bit. *Limits:* the exact window depends on the MAC and PHY timing specifications and on the RGMII version; use the devices' data sheets.

Key Takeaways

- 1000BASE-T uses four bidirectional PAM5 pairs at 125 MBd; the PHY cancels echo and crosstalk.
- Magnetics provide isolation and common-mode rejection; keep the isolation gap free of copper and route the connector zone to chassis ground.
- 10GBASE-KR is a backplane SERDES at 10.3125 Gb/s; budget loss at 5.16 GHz.
- RGMII needs exactly one 1.5–2 ns clock delay per direction, preferably internal.

Further Reading

IEEE 802.3 [74] defines 1000BASE-T and 10GBASE-KR; PHY and magnetics vendors' application notes give layout recommendations for the media-side circuit.

Review Problems

P31.1 How many bits per symbol per pair does 1000BASE-T carry to reach 1 Gb/s at 125 MBd on four pairs?

P31.2 Find the UI and Nyquist frequency of SGMII.

P31.3 What is the bit rate per RGMII data line at gigabit speed?

P31.4 Find the UI of 10GBASE-KR.

P31.5 What trace length gives 1.8 ns of delay at 165 ps/in?

P31.6 Using Worked Example 31.1, estimate the line loss of a 24 in backplane route.

Chapter 32

MIPI CSI-2 and DSI: D-PHY and C-PHY

Chapter Overview

Cameras and displays in embedded systems connect to the application processor through MIPI CSI-2 (camera serial interface) and DSI (display serial interface). Both protocols run over one of two physical layers: D-PHY, a source-synchronous set of differential lanes with a forwarded clock, or C-PHY, an embedded-clock three-wire "trio" scheme. Both PHYs switch between a high-speed, low-swing, terminated mode and a low-power, large-swing, unterminated mode on the same wires. This chapter explains the two PHYs, their implications for routing on boards and flexible cables, and the common causes of failure.

Learning Objectives

1. Describe D-PHY high-speed and low-power modes and their voltage levels.
2. Explain C-PHY three-phase signaling and its throughput.
3. Derive lane-to-clock skew budgets for D-PHY.
4. Route trios and lanes through boards, connectors, and flex cables.
5. Identify failure modes particular to mode switching.

32.1 D-PHY

A D-PHY link has one clock lane and one to four data lanes, each a differential pair [75]. In *high-speed* (HS) mode, the transmitter drives a low-swing differential signal (about 200 mV differential on a common mode of about 200 mV) into a 100 Ω differential termination that the receiver switches on only during HS bursts. In *low-power* (LP) mode, each wire is driven single-ended between ground and 1.2 V, unterminated, at a few tens of megabits per second, to carry control sequences and escape-mode data. Each burst starts and ends with an LP sequence (Figure 32.1).

The data lanes are clocked by the forwarded HS clock at half the bit rate (DDR), so D-PHY is source synchronous: the receiver needs the clock-to-data skew within a fraction of a UI. Newer D-PHY versions add deskew calibration bursts that let the receiver compensate static skew at high rates.

Specification Note

D-PHY data rates have increased across versions (from 1.5 Gb/s per lane in early versions to several gigabits per second per lane later). HS and LP voltage ranges, termination tolerances, skew limits, and channel insertion-loss requirements are defined by the MIPI Alliance specifications [75], [76], which are available to members. Use the specification version supported by both the processor and the sensor or display.

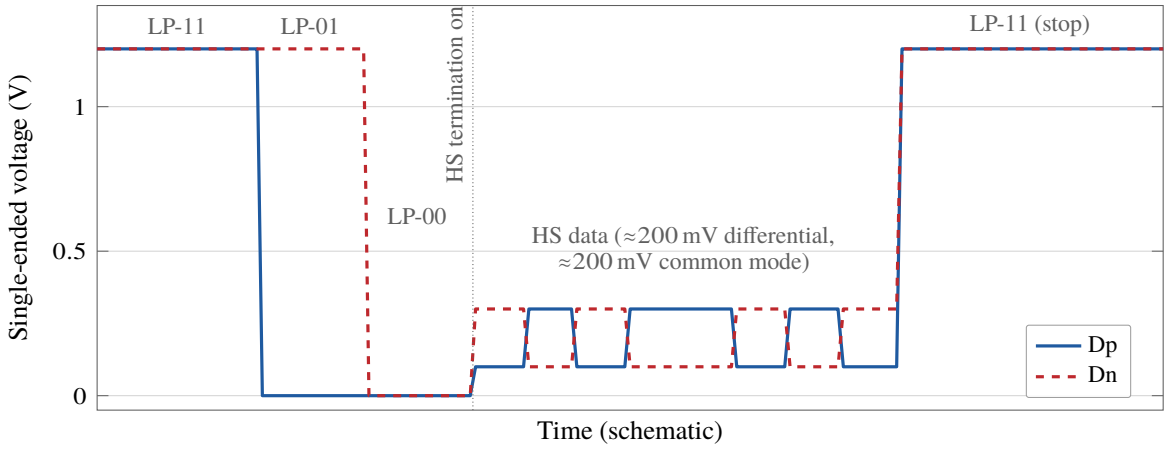


Figure 32.1. D-PHY lane voltages during a transition from the low-power stop state (LP-11) through LP-01 and LP-00 into high-speed data and back (schematic).

Worked Example 32.1: D-PHY lane-to-clock skew

A camera link runs at 2.5 Gb/s per lane. If 5 % of a UI is allocated to static lane-to-clock routing skew, what length mismatch is allowed at 165 ps/in?

Solution. UI = 400 ps; 5 % is 20 ps, or 121 mil. Within each pair, intra-pair skew should be held much tighter, because it distorts the HS differential crossing and the LP single-ended levels are received separately.

32.2 C-PHY

C-PHY replaces differential pairs with *trios* of three wires (Figure 32.2) [76]. In each symbol, one wire is driven high, one low, and one to the midpoint; three differential receivers measure A−B, B−C, and C−A. Every symbol differs from the previous one, so each symbol boundary carries a transition and the clock is embedded. The encoding maps 16 bits onto 7 symbols, or about 2.28 bits per symbol.

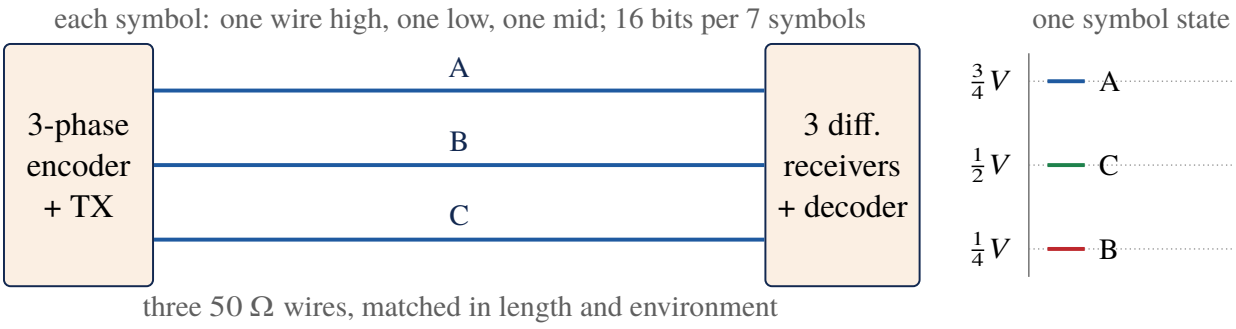


Figure 32.2. A C-PHY trio. Three wires each take one of three levels in every symbol; three differential receivers recover the symbol, and the guaranteed transitions carry the clock.

Worked Example 32.2: C-PHY versus D-PHY throughput

Compare a three-trio C-PHY link at 2.5 GS/s (symbols per second) with a four-lane D-PHY link at 2.5 Gb/s per lane.

Solution. C-PHY: $3 \times 2.5 \times 16/7 = 17.1$ Gb/s on nine wires. D-PHY: $4 \times 2.5 = 10$ Gb/s on ten wires (four data lanes plus the clock lane). C-PHY delivers about 70 % more throughput with fewer wires, at the cost of tighter matching among the three wires of each trio and a more complex PHY.

32.2.1 Routing trios

Because the receiver compares wires pairwise, the three wires of a trio must be matched in delay and in their electrical environment: each should have the same impedance to ground (nominally 50 Ω single-ended) and the same coupling to its two partners. Routing the three in a row makes the middle wire couple more strongly than the outer two. Whether that asymmetry is acceptable is a question for an extraction, not a rule of thumb: extract the 3×3 **L** and **C** matrices of the actual cross-section (Eq. (12.3)), compare the three wire self-terms and the three mutual terms, and simulate the trio with them. If the imbalance matters, symmetric arrangements (for example, rotating wire positions at intervals, or routing the trio in a triangle across two layers) can equalize it, but they add vias and should be justified by the same extraction rather than applied by default. Different trios are independent (each has its own embedded clock), so trio-to-trio skew is not critical.

32.3 Boards, connectors, and flex cables

Camera modules and displays usually connect through board-to-board connectors and flexible printed circuits (FPCs), whose stackups are thin and whose reference planes are often meshed (cross-hatched) for flexibility. A meshed plane raises impedance and loss and lets fields leak; its effect must be simulated, and the mesh pitch kept small compared with the trace width. Connector pin assignments should put ground pins between lanes and keep each pair (or trio) on adjacent pins.

Design Rule: MIPI routing

Route D-PHY lanes as 100 Ω differential pairs and C-PHY wires as 50 Ω single-ended lines with equal coupling within each trio. Match data lanes to the clock lane (D-PHY) or wires within each trio (C-PHY) in delay, including the flex and connector. Keep LP-mode aggressors in mind: the 1.2 V single-ended LP swing is six times the HS differential swing, so spacing between lanes must account for LP-to-HS crosstalk. No AC-coupling capacitors are used; both modes are DC coupled.

Common Pitfall: Treating the flex as "just a cable"

Designing the main board carefully and leaving the camera or display flex to the module vendor. The flex and its connectors are often the longest and least controlled part of the channel, with meshed references, impedance steps at each connector, and no stitching. Specify impedance, stackup, and ground-pin assignment for the flex, and include it in the channel simulation.

Table 32.1. D-PHY and C-PHY compared.

	D-PHY	C-PHY
Lane unit	Differential pair	Trio (three wires)
Clocking	Forwarded clock lane (source synchronous)	Embedded in symbol transitions
Bits per transfer	1 bit per UI per lane	16 bits per 7 symbols (2.28 per symbol)
Termination	100 Ω differential, HS only	Per-wire termination, HS only
Critical matching	Lane to clock; intra-pair	Within each trio
LP mode	1.2 V single-ended	1.2 V single-ended

Rule of Thumb: LP-mode crosstalk

The 1.2 V LP swing is about six times the 200 mV HS differential swing, so a coupling level that is harmless between HS lanes may disturb an HS lane when its neighbor is in LP mode. *Limits:* LP and HS phases on adjacent lanes overlap only during transitions in many systems; check whether the interface allows independent lane modes.

Key Takeaways

- MIPI CSI-2 and DSI run over D-PHY (pairs plus a clock lane) or C-PHY (embedded-clock trios).
- Both PHYs switch between terminated low-swing HS and unterminated 1.2 V LP modes on the same wires.
- D-PHY needs lane-to-clock matching; C-PHY needs matching within each trio.
- C-PHY carries 2.28 bits per symbol, giving higher throughput per wire.
- Flex cables and connectors are part of the channel and must be specified.

Further Reading

The MIPI Alliance specifications [75]–[78] define D-PHY, C-PHY, CSI-2, and DSI; they are available to MIPI members.

Review Problems

P32.1 Find the UI and Nyquist frequency of a 1.5 Gb/s D-PHY lane.

P32.2 A C-PHY trio runs at 3.5 GS/s. What is its bit rate?

P32.3 What is the aggregate throughput and wire count of a four-lane D-PHY link at 2.5 Gb/s per lane?

P32.4 What symbol rate must three C-PHY trios run to carry 10 Gb/s?

P32.5 Allocate 5 % of the UI at 4.5 Gb/s to lane-to-clock skew. What length mismatch is allowed at 165 ps/in?

P32.6 An LP-mode lane (1.2 V) couples 1.5 % into a neighbor. How much noise does the neighbor see?

Chapter 33

Clock Distribution

Chapter Overview

Every synchronous system depends on its clocks, and every serial link's jitter budget begins with its reference clock. Clocks are also the strongest sources of narrowband emissions on most boards. This chapter covers the components of a clock tree—crystals and oscillators, PLLs and jitter cleaners, and fanout buffers—and the board-level practices for routing and terminating clocks. It shows how to design a PLL loop filter for stability and how to convert an oscillator's phase-noise plot into RMS jitter.

Learning Objectives

1. Select between XO, TCXO, VCXO, and MEMS oscillators, and size crystal load capacitors.
2. Choose clock output standards (LVCMOS, LVDS, LVPECL, HCSL) and terminate them correctly.
3. Explain the reference- and VCO-noise transfer functions of a PLL and design a second-order loop filter.
4. Integrate phase noise to RMS jitter and combine jitter contributions.
5. Route clocks to minimize jitter pickup and emissions.

33.1 Clock sources

Crystal with on-chip oscillator
The cheapest source: a quartz crystal and two load capacitors connected to an inverter in the IC. Frequency accuracy depends on the load capacitance matching the crystal's specified C_L.

Crystal oscillator (XO)
A packaged crystal and oscillator; typical stabilities of ± 20–± 50 ppm and low phase noise.

TCXO, OCXO
Temperature-compensated and oven-controlled oscillators for sub-ppm stability.

VCXO
A voltage-controlled crystal oscillator, used inside jitter-cleaner PLLs.

MEMS oscillators
Silicon resonators with a PLL; robust and programmable, with phase noise that varies widely among products.

33.1.1 Crystal load capacitance

A crystal oscillates at its specified frequency only when loaded with its specified capacitance C_L. For the common Pierce circuit with capacitors C_1 and C_2 from each crystal terminal to ground,

$$C_L = \frac{C_1 C_2}{C_1 + C_2} + C_{\text{stray}}, \tag{33.1}$$

where C_{stray} includes the IC pins and board traces, typically 2–5 pF.

Worked Example 33.1: Crystal load capacitors

A crystal is specified for $C_L = 12\,\text{pF}$; stray capacitance is estimated at 3 pF. Choose equal load capacitors.

Solution. $C_1 C_2/(C_1 + C_2) = C/2 = 12 - 3 = 9\,\text{pF}$, so $C_1 = C_2 = 18\,\text{pF}$. Too much load capacitance pulls the frequency low; too little pulls it high, by a few ppm per picofarad.

Design Rule: Crystal layout

Place the crystal and its load capacitors next to the IC's oscillator pins, with short traces, a local ground under them connected to the IC's oscillator ground, and no other signals routed under or beside the crystal. Keep the loop of crystal, capacitors, and ground small: it is a sensitive high-impedance circuit.

33.2 Clock tree and output standards

A typical clock tree (Figure 33.1) starts from an oscillator, optionally passes through a PLL that multiplies the frequency and cleans the jitter, and fans out through buffers to the loads. Each load receives its own point-to-point output; daisy-chaining clocks to several loads is avoided because of reflections and loading.

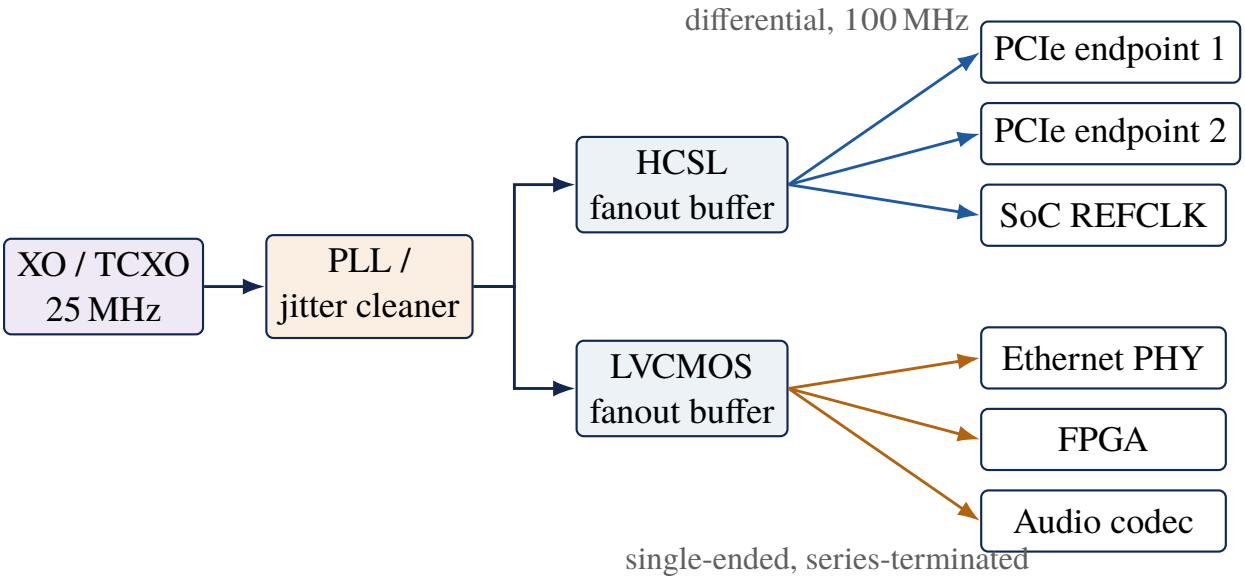


Figure 33.1. A clock tree: oscillator, PLL or jitter cleaner, and fanout buffers with point-to-point outputs to each load.

Table 33.1. Common clock output standards and their terminations.

Standard	Type	Swing (typ.)	Termination
LVCMOS	Single-ended	Rail-to-rail	Series resistor at source; one load per output
LVDS	Differential	350 mV	100 Ω across pair at receiver
LVPECL	Differential	800 mV	50 Ω to V_{CC} − 2 V (or Thevenin/Y equivalent)
HCSL	Differential, current mode	700 mV	50 Ω to ground on each leg (source or load end)
CML	Differential	400–800 mV	50 Ω to V_{CC}, internal

AC coupling between different standards is common for clocks, which are DC balanced; the receiver then needs its own bias. Each standard's data sheet defines its termination precisely; mixing standards without checking common-mode ranges is a frequent source of marginal clocks.

33.3 Jitter of clocks

Clock jitter is specified in several ways: *period jitter* (variation of individual periods), *cycle-to-cycle jitter* (difference between adjacent periods), and *phase jitter* or TIE (deviation of edges from an ideal clock), which is what matters for serial links. Oscillators specify their noise as single-sideband phase noise $\mathscr{L}(f)$, in dBc per hertz of offset from the carrier (Figure 33.2). The RMS phase jitter over an offset band $[f_1, f_2]$ is

$$\sigma_t = \frac{1}{2\pi f_0}\sqrt{2\int_{f_1}^{f_2} 10^{\mathscr{L}(f)/10}\,\mathrm{d}f} \quad \text{(seconds, RMS)}, \tag{33.2}$$

where f_0 is the carrier frequency and the factor of 2 accounts for both sidebands. The integration band must match the application: 12 kHz to 20 MHz is a conventional band for SONET and Ethernet reference clocks, whereas PCIe defines its own filtered measurement.

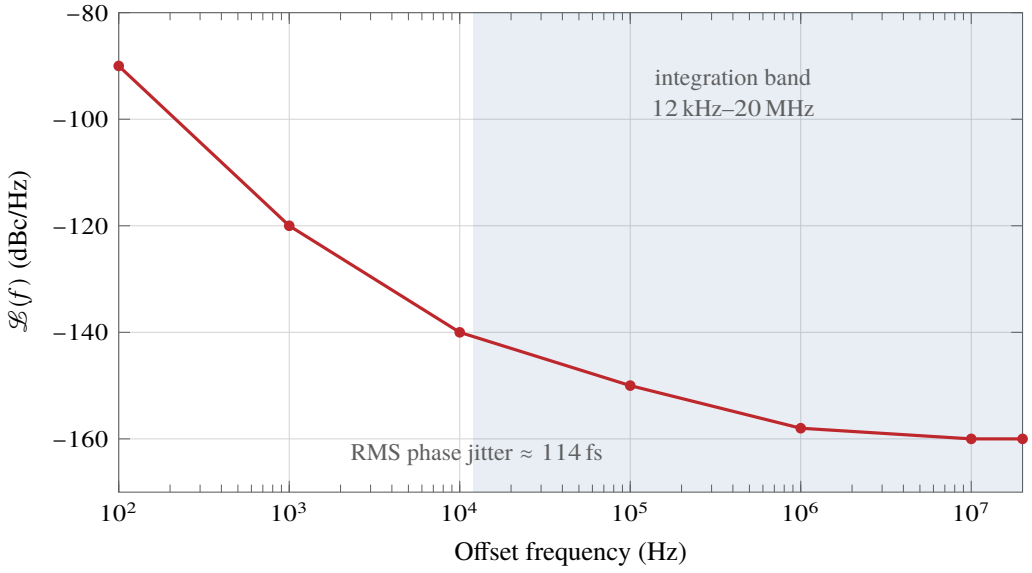


Figure 33.2. Representative phase noise of a 100 MHz oscillator. Integrating Eq. (33.2) over the shaded band gives about 114 fs RMS.

Worked Example 33.2: Phase noise to jitter

A 100 MHz clock has a flat phase-noise floor of −140 dBc/Hz from 12 kHz to 20 MHz. Find its RMS jitter in that band.

Solution. $10^{-14} \times (20 \times 10^{6} - 12 \times 10^{3}) = 1.999 \times 10^{-7}$; doubling and taking the square root gives 6.32×10^{-4} rad. Dividing by $2\pi \times 10^{8}$ gives $\sigma_t = 1.0$ ps. A floor 20 dB lower would give 100 fs.

Independent jitter sources add in root-sum-square fashion. A 300 fs reference through a fanout buffer with 100 fs of additive jitter gives $\sqrt{300^2 + 100^2} = 316$ fs.

33.4 PLLs and loop filters

A charge-pump PLL (Figure 33.3) compares the divided output with the reference in a phase-frequency detector, converts the phase error into current pulses, filters them into a control voltage, and steers a VCO. Its loop bandwidth determines how noise is shaped:

- reference and phase-detector noise are *low-pass* filtered by the loop and multiplied by N (raised by $20\log_{10} N$ dB) within the bandwidth;
- VCO noise is *high-pass* filtered: the loop suppresses it within the bandwidth and passes it outside.

In transfer-function form, with phase-detector gain K_{pd} (amperes per radian), loop-filter impedance $Z(s)$, VCO gain K_{vco} (radians per second per volt) and divider N, the open-loop gain is

$$G(s) = \frac{K_{\mathrm{pd}}\, Z(s)\, K_{\mathrm{vco}}}{s\, N}, \tag{33.3}$$

where the $1/s$ is the VCO itself: a VCO integrates frequency into phase, so it adds an integrator to the loop on top of the one in $Z(s)$. Phase noise then reaches the output through

$$\frac{\phi_{\mathrm{out}}}{\phi_{\mathrm{ref}}} = N\,\frac{G}{1+G}, \qquad \frac{\phi_{\mathrm{out}}}{\phi_{\mathrm{vco}}} = \frac{1}{1+G}, \tag{33.4}$$

low-pass and high-pass respectively, with the crossover near the loop bandwidth. The same low-pass response sets the *reference spurs*: charge-pump mismatch and leakage produce ripple at the phase-detector frequency f_{pd}, which modulates the VCO and appears as sidebands at $\pm f_{\mathrm{pd}}$; the loop filter's attenuation at f_{pd} (the pole ω_p and any extra filter pole) sets their level.

A jitter cleaner uses a narrow loop bandwidth (tens to hundreds of hertz) with a low-noise VCXO, so that the output follows the reference's frequency but inherits the VCXO's low phase noise. A frequency synthesizer uses a wider bandwidth to suppress the noise of an on-chip VCO.

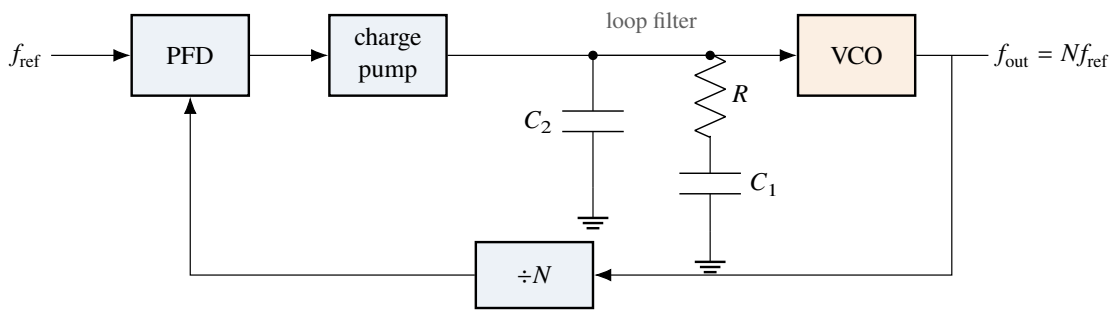


Figure 33.3. Charge-pump PLL with a second-order passive loop filter (R, C_1, C_2).

The second-order passive filter of Figure 33.3 has impedance

$$Z(s) = \frac{1 + sRC_1}{s(C_1 + C_2)\left(1 + sR\frac{C_1 C_2}{C_1 + C_2}\right)}, \tag{33.5}$$

with a zero at $\omega_z = 1/(RC_1)$ and a pole at $\omega_p = (C_1 + C_2)/(RC_1C_2)$. Stability requires the open-loop crossover ω_c to lie between them; the phase margin is maximized at $\omega_c = \sqrt{\omega_z \omega_p}$, where

$$\mathrm{PM}_{\max} = \arctan\sqrt{1 + \frac{C_1}{C_2}} - \arctan\frac{1}{\sqrt{1 + C_1/C_2}}. \tag{33.6}$$

Worked Example 33.3: Loop-filter phase margin

A loop filter uses $C_1 = 10C_2$. What is the maximum phase margin?

Solution. $\sqrt{11} = 3.317$; $\arctan 3.317 = 73.2°$ and $\arctan(1/3.317) = 16.8°$, so $\mathrm{PM}_{\max} = 56.4°$. Increasing the ratio to $C_1 = 16C_2$ raises it to 62.7°, at the cost of less attenuation of reference spurs above the crossover.

Rule of Thumb: PLL loop bandwidth

Set the loop bandwidth near the offset frequency at which the reference's phase noise, multiplied by N, crosses the VCO's phase noise; this minimizes integrated jitter. Keep the bandwidth below about one-tenth of the phase-detector frequency for the continuous-time analysis to hold. *Limits:* spur requirements, SSC tracking, and lock time may override the jitter optimum.

33.5 Clock routing

Design Rule: Clock routing

Route clocks point to point, on inner layers referenced to ground, with at least $3H$ to $5H$ spacing from other signals (and more from other clocks). Terminate every clock so that its edges are monotonic. Keep clock routes short and away from connectors and board edges. Give each clock buffer and PLL its own filtered supply (a ferrite or LDO plus local decoupling); supply noise in a PLL converts directly into jitter.

Common Pitfall: Supply noise in clock buffers

Sharing a noisy digital supply with a clock buffer or PLL. A few millivolts of supply ripple at a frequency inside the PLL bandwidth, or at the buffer's output, appears as periodic jitter on every clock in the tree—and often as spurs in the emission spectrum. Measure the clock's phase noise at bring-up with and without the noisy load active.

Key Takeaways

- Size crystal load capacitors from $C_L = C_1C_2/(C_1 + C_2) + C_{\text{stray}}$.
- Each clock output standard has a specific termination; use one load per output.
- RMS phase jitter is $\frac{1}{2\pi f_0}\sqrt{2\int \mathscr{L}\, \mathrm{d}f}$ over the application's band; independent contributions add in RSS.
- PLLs low-pass the reference and high-pass the VCO noise; the loop bandwidth trades between them.
- The maximum phase margin of a second-order filter depends only on C_1/C_2.

Further Reading

Li [54] relates phase noise to jitter and discusses PLL jitter transfer; Dally and Poulton [35] cover clock distribution and PLL design for digital systems.

Review Problems

P33.1 A crystal circuit uses $C_1 = C_2 = 22\,\text{pF}$ with $4\,\text{pF}$ stray. What load capacitance does the crystal see?

P33.2 A $156.25\,\text{MHz}$ clock has a flat $-150\,\text{dBc/Hz}$ floor from $12\,\text{kHz}$ to $20\,\text{MHz}$. Find its RMS jitter.

P33.3 Combine $250\,\text{fs}$, $150\,\text{fs}$, and $100\,\text{fs}$ of independent jitter.

P33.4 Find the maximum phase margin of a second-order loop filter with $C_1 = 16C_2$.

P33.5 Find the zero frequency of a loop filter with $R = 1\,\text{k}\Omega$ and $C_1 = 10\,\text{nF}$.

P33.6 A $25\,\text{MHz}$ oscillator is specified to $\pm 50\,\text{ppm}$. What is its frequency range?

Chapter 34

Interface Connectors, Contacts, and Generations

Chapter Overview

Part VI covered the electrical design of the main interfaces. This chapter collects the physical and naming facts that designers look up again and again, and that published diagrams most often get wrong: how many contacts a PCIe card edge has for each width and what the sideband signals do; how USB names its generations, connectors, and cables, and how Type-C detects orientation and negotiates power; how MIPI separates protocols from PHYs and how a flex cable flips the pin order; and how memory modules are keyed, how ECC widens them, and how DDR5 splits a module into two subchannels. Every fact that depends on a standard was checked against a public source (listed in the bibliography with an access date of 10 November 2026); the standards themselves are mostly members-only, and the governing document always wins.

Learning Objectives

1. State PCIe card-edge contact counts, keying, sideband signals, and the data rates of generations 1 through 7.
2. Distinguish USB generation names, connector contact counts, cable conductors, and Type-C configuration and power roles.
3. Separate MIPI protocols (CSI-2, DSI) from PHYs (D-PHY, C-PHY, M-PHY, A-PHY) and avoid flex-cable pin reversal.
4. Count the signals of a memory channel and explain DDR5 subchannels, ECC widths, SPD, and on-module PMICs.

34.1 PCI Express

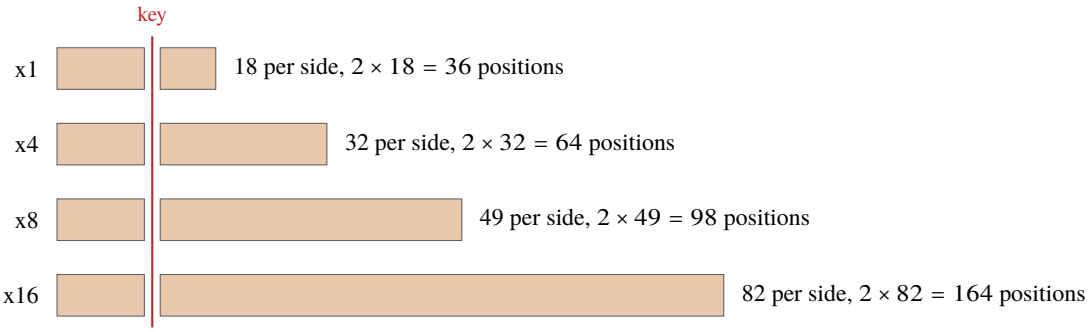
34.1.1 Card edge and receptacle

The PCIe Card Electromechanical (CEM) specification defines a card edge with contacts on both sides at 1.00 mm pitch and a matching receptacle on the system board. The number of positions depends on the width (Figure 34.1): **36 for x1, 64 for x4, 98 for x8, and 164 for x16**, that is 18, 32, 49, and 82 per side [8], [9]. Two words are worth keeping apart:

- the *card edge* (gold fingers) belongs to the add-in card; the *receptacle* (slot) belongs to the system board. A connector vendor sells receptacles; the card edge is a PCB feature with its own fabrication requirements (bevelled edge, hard gold, no solder mask);

- the *mechanical width* of a slot is the size of the receptacle; the *electrical width* is the number of lanes actually wired. An x16 slot wired as x4 accepts x16 cards, which then train to x4.

Vendors' web pages sometimes label the four sizes "x1, x2, x3, x4" or similar when they mean x1, x4, x8, x16; the front-matter audit records one such case. Read the drawing, not the catalogue title.



Each bar is one side of the card edge (1.00 mm pitch). Left of the key: positions 1–11 on each side (power, presence detect, SMBus, JTAG, wake, reset). Right of the key: the reference clock, the lanes with their grounds, and a presence-detect return near the end of each supported width.

Figure 34.1. PCIe CEM card-edge positions per side for each width, with the key after position 11. Schematic outline, not to scale. [Schematic]

Figure 34.2 shows the four receptacle sizes side by side; the short section at the left of each is the power and sideband section in front of the key.

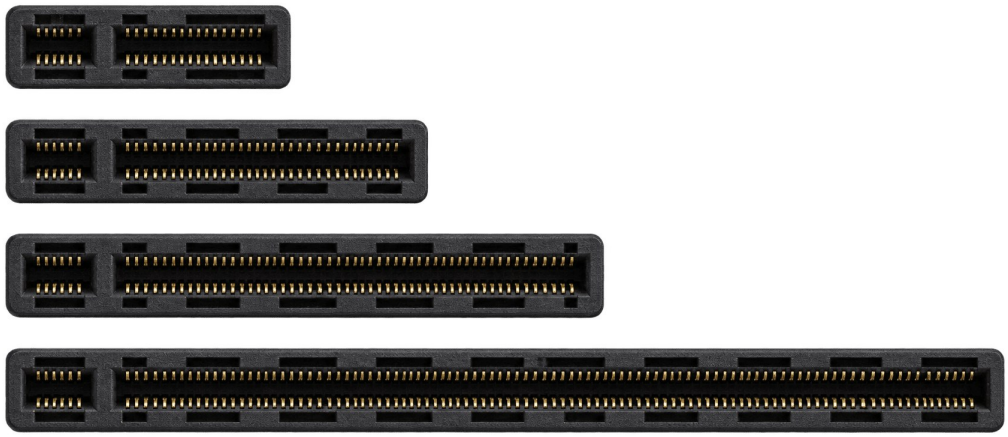


Figure 34.2. PCIe CEM receptacles for x1, x4, x8, and x16 cards (top to bottom). The short section left of the key carries power and sideband positions; the long section carries the reference clock and the lanes. Rendered illustration supplied with the manuscript, not a photograph of specific products; markings, contact counts, and proportions are not authoritative. [Illustration]

34.1.2 Sideband signals and power

Besides the lanes and the reference clock, the slot carries sideband signals that are low speed but essential:

- **PERST#**: fundamental reset from the system, released after power and the reference clock are stable (Chapter 25).

- **REFCLK±**: the 100 MHz differential reference clock (Chapter 29).
- **PRSNT1#, PRSNT2#**: presence detect; the card connects PRSNT1# to the PRSNT2# position at the end of its width, so the system can detect a card and its size.
- **CLKREQ#** (clock request for power management), **WAKE#** (wake from low-power states), **SMCLK/SMDAT** (SMBus), and **JTAG**.
- **Power**: 12 V, 3.3 V, and 3.3 V auxiliary. An x16 slot is commonly quoted as 75 W (5.5 A at 12 V plus 3 A at 3.3 V); higher-power cards use auxiliary power connectors. Check the current CEM revision for the exact limits.

Pin names in the CEM tables are given from a stated perspective (for example "transmitter" pairs); confirm whose transmitter is meant before mapping them to a device's TX and RX balls (Section 16.7).

34.1.3 Generations

Table 34.1. PCI Express generations. Raw bandwidth is the line rate times 16 lanes, both directions, before encoding and protocol overhead. Gen6 and Gen7 use PAM4, so their Nyquist frequency is a quarter of the bit rate.

Gen	Rate	Modulation / encoding	Nyquist	x16 raw, both directions	Year (spec)
1	2.5 GT/s	NRZ, 8b/10b	1.25 GHz	10 GB/s	2003
2	5 GT/s	NRZ, 8b/10b	2.5 GHz	20 GB/s	2007
3	8 GT/s	NRZ, 128b/130b	4 GHz	32 GB/s	2010
4	16 GT/s	NRZ, 128b/130b	8 GHz	64 GB/s	2017
5	32 GT/s	NRZ, 128b/130b	16 GHz	128 GB/s	2019
6	64 GT/s	PAM4, Flit (FEC + CRC)	16 GHz	256 GB/s	2022
7	128 GT/s	PAM4, Flit	32 GHz	512 GB/s	2025

Table 34.1 summarizes the generations. Gen6 doubled the rate by moving to PAM4 at 32 GBd, keeping the 16 GHz Nyquist frequency of Gen5 [10], [20]. Because PAM4's eyes are a third of the NRZ amplitude, the raw error rate rises; Gen6 therefore uses fixed 256-byte flits, each with 242 bytes of payload, 8 bytes of CRC, and 6 bytes of a lightweight FEC, and a first-bit-error-rate target of 10^{-6} before correction [10], [79]. PCI-SIG released the Gen7 specification to members in June 2025 at 128 GT/s, PAM4, and 512 GB/s for an x16 link in both directions [11]. For channel budgets see Chapter 29 and the Gen5 case study in Chapter 40.

34.2 USB

34.2.1 Names

USB names have changed several times, and the same silicon may be marketed under three names. Table 34.2 lists the current names with their predecessors [13].

USB4 Version 2.0 reaches 80 Gbit/s with two lanes of 40 Gbit/s, using PAM3 at 25.6 GBd; it works over existing 40 Gbit/s passive Type-C cables and new active cables, and it can be configured asymmetrically for 120 Gbit/s in one direction [13], [22].

Table 34.2. USB data-rate names. “Lanes” counts SuperSpeed or USB4 lanes per direction.

Rate	Specification name	Lanes	Earlier or alternative names
480 Mbit/s	USB 2.0 High Speed	—	Hi-Speed USB
5 Gbit/s	USB 3.2 Gen 1	1	USB 3.0, USB 3.1 Gen 1, SuperSpeed USB; logo “USB 5Gbps”
10 Gbit/s	USB 3.2 Gen 2	1	USB 3.1 Gen 2, SuperSpeed USB 10Gbps
20 Gbit/s	USB 3.2 Gen 2x2	2	SuperSpeed USB 20Gbps; Type-C only
40 Gbit/s	USB4 Gen 3x2	2	USB4 40Gbps
80 Gbit/s	USB4 Version 2.0 (Gen 4)	2	USB 80Gbps; PAM3 at 25.6 GBd; optional asymmetric 120/40

34.2.2 Contacts and conductors

A connector’s *contacts* are not the same as a cable’s *conductors*. Table 34.3 lists the contact counts; the cable may contain fewer wires (pins tied together inside the plug), more (drain wires), and a shield that is not counted as a contact.

Table 34.3. USB connector contact counts (shell not counted).

Connector	Contacts	Notes
Standard-A / B, USB 2.0	4	VBUS, D−, D+, GND
Mini-B, Micro-B (USB 2.0)	5	adds ID (used for OTG role detection on Micro-AB)
Standard-A, SuperSpeed	9	adds SSRX±, SSTX±, and GND_DRAIN (pin 7)
Standard-B, SuperSpeed	9	as above, in a B shell
Micro-B, SuperSpeed	10	a USB 2.0 Micro-B section plus a five-contact SuperSpeed section
Type-C receptacle	24	two rows of 12; symmetric for reversible insertion

Figure 34.3 shows four of the receptacles in Table 34.3.

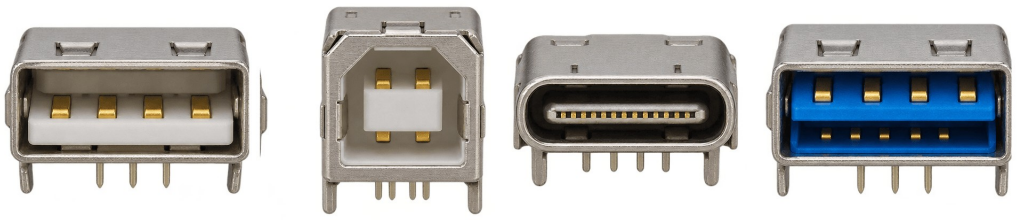


Figure 34.3. USB board-mount receptacles, left to right: USB 2.0 Standard-A (4 contacts), USB 2.0 Standard-B (4), Type-C, and SuperSpeed Standard-A (4 contacts at the front plus 5 SuperSpeed contacts behind them). Rendered illustration supplied with the manuscript, not a photograph of specific products; markings, contact counts, and proportions are not authoritative. [Illustration]

A USB 2.0 cable has four conductors (VBUS, GND, and the D± twisted pair) inside an overall shield. A USB 3.x cable for Standard-A to Micro-B adds two shielded differential pairs, each with a drain wire; the drains are joined to the GND_DRAIN contact [12]. The cable shield is bonded to the connector shell, not to a signal contact. In a Type-C cable, the D± positions A6/B6 and A7/B7 of the receptacle collapse to a single pair, because the plug carries D± only on one row.

34.2.3 Type-C

Figure 34.4 shows the receptacle contacts [14], [23]. The essential behaviour is in the configuration channel (CC):

- A **source** (downstream-facing port, DFP) pulls both CC contacts up with R_p (or a current source); a **sink** (upstream-facing port, UFP) pulls them down with $R_d = 5.1\,\text{k}\Omega$. The source finds which CC contact sees R_d, and so learns the plug orientation; it turns on VBUS only after it sees a sink. The value of R_p advertises the current the source offers (default USB, 1.5 A, or 3 A at 5 V).
- A **dual-role** port (DRP) toggles between the two roles until it finds a partner. Data roles (host or device) and power roles (source or sink) are separate and can be swapped.
- In the plug, the CC position not used for configuration becomes **VCONN**, which powers electronics in active or electronically marked cables.
- **USB Power Delivery** (PD) messages travel on CC to negotiate higher voltages; PD 3.1 Extended Power Range allows up to 240 W (48 V, 5 A) with suitable cables [22].
- The **SBU** contacts carry sideband functions of alternate modes (for example the DisplayPort AUX channel).

row A	A1 GND	A2 TX1+	A3 TX1−	A4 VBUS	A5 CC1	A6 D+	A7 D−	A8 SBU1	A9 VBUS	A10 RX2−	A11 RX2+	A12 GND
row B	B12 GND	B11 RX1+	B10 RX1−	B9 VBUS	B8 SBU2	B7 D−	B6 D+	B5 CC2	B4 VBUS	B3 TX2−	B2 TX2+	B1 GND

rotating the plug by 180° maps An onto Bn; the CC pin that sees the pull-down tells the source the orientation

Figure 34.4. USB Type-C receptacle contacts, viewed into the receptacle. The SuperSpeed pairs are named from the receptacle's own transmitter and receiver. [Schematic]

Common Pitfall: A Type-C receptacle with one CC resistor

A device that pulls only one CC contact down to ground with R_d, or ties both CC contacts together through one resistor, works in one orientation or with some cables only. Each CC contact needs its own 5.1 kΩ pull-down (or a CC controller). A missing R_d is the classic reason that a Type-C device charges from a Type-A-to-C cable but not from a C-to-C charger.

34.3 MIPI

34.3.1 Protocols and PHYs

MIPI specifications separate the *protocol* (what the bits mean) from the *physical layer* (how they travel). CSI-2 (camera) and DSI/DSI-2 (display) are protocols; D-PHY and C-PHY are PHYs that either protocol can use. M-PHY is a different PHY used with UniPro and JEDEC UFS storage, and A-PHY is a long-reach automotive SerDes. A camera described as "MIPI" therefore still needs two answers: CSI-2 over D-PHY or over C-PHY, and how many lanes.

Levels versus rails. D-PHY switches between two electrical modes: low-power signalling, a single-ended swing of roughly 1.2 V, and high-speed signalling, a differential swing of a few hundred millivolts around a common mode of about 200 mV. These are *signal levels* defined by the PHY. The SoC's MIPI supply rail may be 1.2 V, 1.8 V, or another value set by its process; do not infer one from the other.

Table 34.4. MIPI PHYs (public summaries of members-only specifications).

PHY	Wiring and clocking	Recent rates
D-PHY	one differential clock lane plus 1–4 differential data lanes; source-synchronous; low-power (LP) single-ended mode and high-speed (HS) differential mode	v3.0 (2021): 9 Gbit/s per lane standard channel, 11 Gbit/s short channel [17]
C-PHY	lanes of three wires (trios), three-phase symbols, clock embedded in the symbol transitions	v2.1: 6 Gsps (about 13.7 Gbit/s) per trio; v3.0 (2025): up to about 24.9 Gbit/s per trio on a short channel [24]
M-PHY	differential lanes, embedded clock, gears	v5.0: HS-G5 23.32 Gbit/s per lane (UFS 4.0) [25]; UFS 5.0 moves to M-PHY 6.0 with PAM4 [26]
A-PHY	asymmetric automotive SerDes over coax or shielded pairs, metres of reach	v1.1 added PAM4 and doubled the rate; v2.0 doubles it again

34.3.2 Connectors and flex cables

MIPI does not define a connector. Cameras and displays use board-to-board connectors or FPC/FFC connectors, commonly at 0.5 mm or 0.3 mm pitch, with top or bottom contacts. A flat flex cable can have its contacts on the same side at both ends or on opposite sides (Figure 34.5); for a given placement of the two connectors, one type keeps the pin order and the other reverses it, reversing every lane and swapping the polarity of every pair. Fix the cable type, connector contact side, and mounting side in one drawing, and check the mapping with the procedure of Section 16.7.

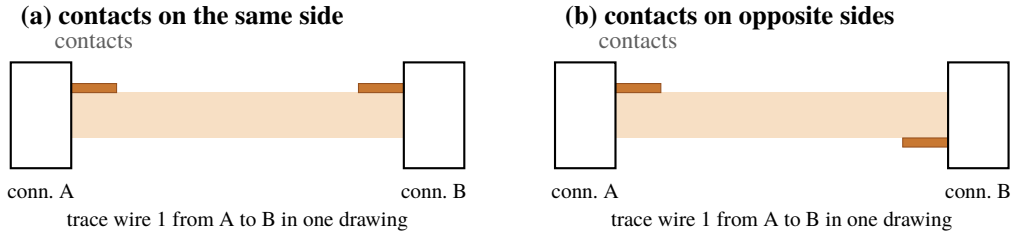


Figure 34.5. A flat flex cable with contacts on the same side (a) or opposite sides (b) at its two ends. Whether pin 1 meets pin 1 or pin N depends on the cable type and on how the two connectors are placed; draw both ends in one view and trace one wire. [Conceptual]

34.4 M.2

M.2 is a small card form factor with a 75-position edge connector; a key removes some positions so that cards fit only sockets of the matching key. The common keys are B and M (storage and WWAN cards: SATA or PCIe, with M carrying up to four PCIe lanes) and A and E (Wi-Fi and Bluetooth cards); card sizes are named by width and length in millimetres (for example 2230, 2242, 2280). Key positions that standard cards do not use are described as reserved for custom applications, which is relevant when a module with a non-standard pinout uses the M.2 mechanics (Section 41.5) [check the current M.2 specification]. Figures 34.6 and 34.7 show sockets and cards. The key is the only thing that stops a card from entering a socket with an incompatible pinout; a custom M.2 module must therefore use a key that no standard card uses.

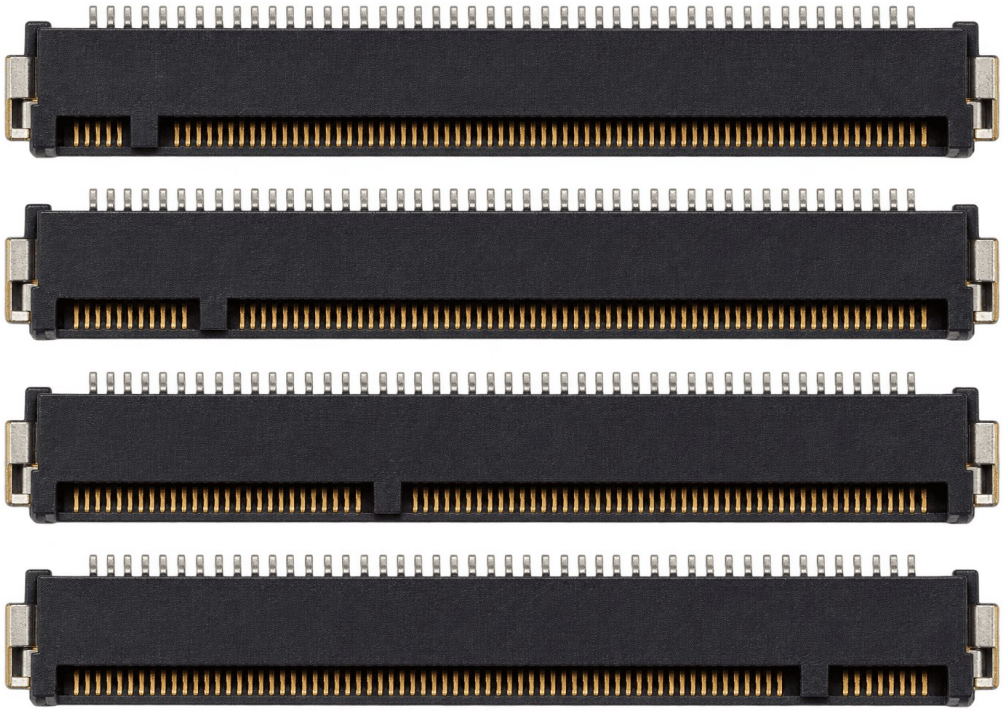


Figure 34.6. M.2 sockets with four different key positions. Identify a key from the connector drawing, not from the image. Rendered illustration supplied with the manuscript, not a photograph of specific products; markings, contact counts, and proportions are not authoritative. [Illustration]

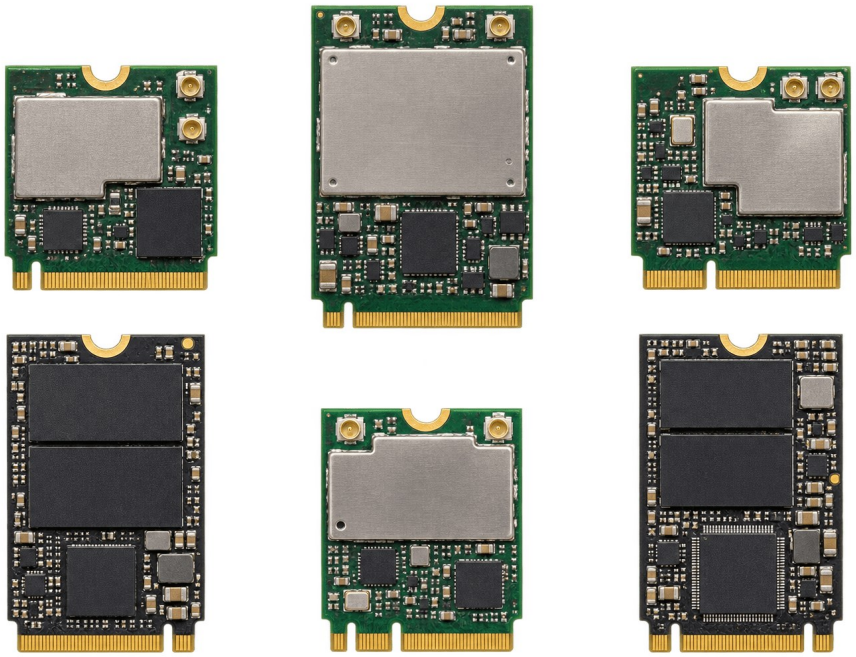


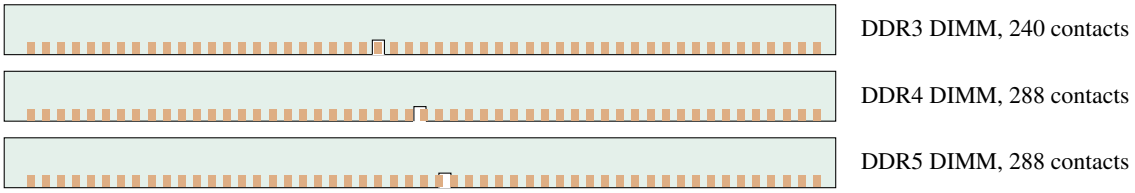
Figure 34.7. M.2 cards of several sizes: wireless cards with u.FL antenna connectors (green) and storage-style cards (black). Key notches in the rendering are not to scale. Rendered illustration supplied with the manuscript, not a photograph of specific products; markings, contact counts, and proportions are not authoritative. [Illustration]

34.5 Memory modules

Table 34.5. Module contact counts and core supply voltages (public module datasheets [15], [16], [27]).

Generation	DIMM	SO-DIMM	Supplies
DDR3	240	204	1.5 V (DDR3L 1.35 V)
DDR4	288	260	VDD = VDDQ = 1.2 V; VPP = 2.5 V
DDR5	288	262	VDD = VDDQ = 1.1 V, VPP = 1.8 V, generated by an on-module PMIC

Keying. Each generation moves the key notch, so a module cannot be inserted into the wrong socket (Figure 34.8). DDR4 and DDR5 DIMMs both have 288 contacts but are mechanically and electrically incompatible.



Each generation moves the key so that a module cannot be inserted into the wrong socket; DDR4 and DDR5 also share 288 contacts but are not interchangeable. Positions and the DDR4/DDR5 edge profile are not to scale.

Figure 34.8. Module outlines with their key notches. Schematic: notch positions and edge profiles are not to scale. [Schematic]

Figure 34.9 shows the parts a memory subsystem is built from.

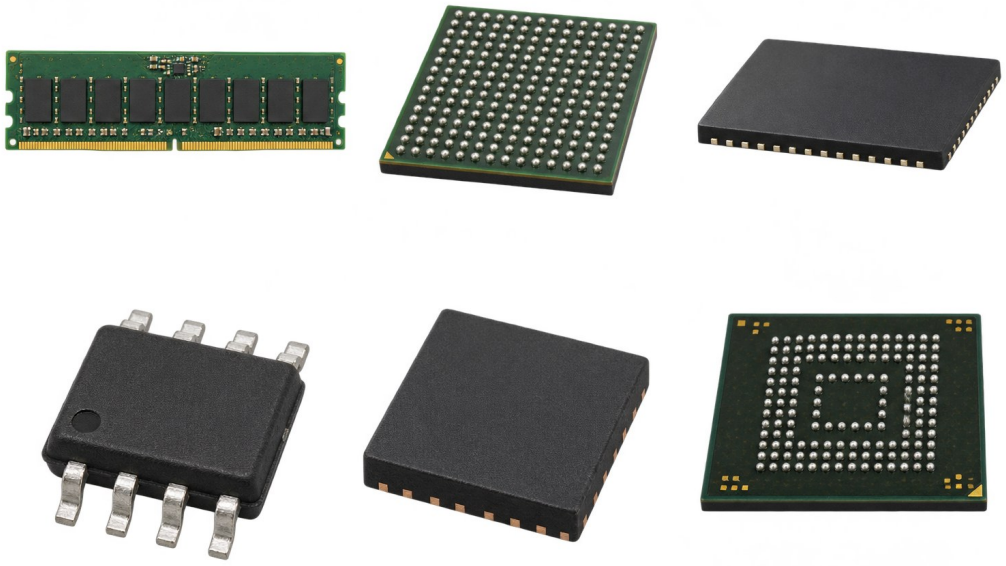


Figure 34.9. Memory silicon and module variants: a DIMM-style module with nine DRAMs (the count of a x72 ECC module built from x8 devices), a BGA DRAM, leadless packages, an SOIC-8 serial flash, and an eMMC BGA. Rendered illustration supplied with the manuscript, not a photograph of specific products; markings, contact counts, and proportions are not authoritative. [Illustration]

Width and ECC. A DDR4 channel is 64 data bits; an ECC module adds 8 check bits for 72. A DDR5 module carries *two independent 32-bit subchannels*; ECC adds 4 bits per subchannel on ECC UDIMMs and SO-DIMMs ($2 \times 36 = 72$) and 8 bits per subchannel on RDIMMs ($2 \times 40 = 80$) [16]. DDR5 DRAMs also contain on-die ECC, which corrects errors inside the DRAM and is invisible to the system; it does not replace module ECC.

Counting signals. For a byte lane built from x8 DRAMs, each byte needs 8 DQ, one DQS pair (2 wires), and one DM/DBI: 11 signals. A 64-bit DDR4 channel therefore has $8 \times 11 = 88$ data-group signals, and 99 with the ECC byte; the command/address, clock, and control signals come on top. A DDR5 subchannel of four bytes has $4 \times 11 = 44$ data-group signals per subchannel, plus its own command/address bus. With x4 DRAMs each nibble has its own strobe, so the strobe count doubles.

Management. DDR5 modules carry an SPD hub with a temperature sensor on an I^3C sideband (backward compatible with I^2C), and a PMIC that converts 5 V on client modules or 12 V on RDIMMs to the DRAM rails, moving part of the power design from the motherboard to the module [16]. Clocked DIMMs (CUDIMM, CSODIMM) add a clock driver on the module. LPDDR devices are soldered, not socketed; their board design is covered in Chapter 28.

Checklist: interface facts

- ☐ Connector part numbers checked against drawings for width, contact count, and keying.
- ☐ Mechanical versus electrical width recorded for every slot.
- ☐ Sideband signals (PERST#, CLKREQ#, WAKE#, PRSNT#, CC, SBU, SPD) designed, not left to defaults.
- ☐ Type-C CC network per contact; VCONN and PD behaviour defined.
- ☐ MIPI protocol, PHY, and lane count named; FPC cable type and orientation fixed in a drawing.
- ☐ Memory module type, ECC width, and sideband buses confirmed with the controller.

Key Takeaways

- PCIe card edges: 36, 64, 98, 164 positions for x1, x4, x8, x16; mechanical width is not electrical width.
- PCIe Gen6 and Gen7 use PAM4 and flits; Gen6 keeps Gen5's 16 GHz Nyquist frequency, Gen7 doubles it.
- USB contacts are not cable conductors; Type-C has 24 receptacle contacts and uses CC for orientation, roles, and power.
- CSI-2 and DSI are protocols; D-PHY and C-PHY are PHYs. FPC cable type can reverse every pin.
- DDR5 modules have two 32-bit subchannels, on-module PMICs, and x72 or x80 ECC widths.

Further Reading

The governing documents are the PCI-SIG CEM and base specifications, the USB 3.2, USB4, Type-C, and PD specifications, the MIPI PHY and protocol specifications, and the JEDEC memory and module standards. The public sources cited in this chapter summarize them and were current on the access date.

Review Problems

P34.1 How many card-edge positions does an x8 PCIe card have, and how many per side?

P34.2 A server motherboard has an x16 receptacle wired to eight lanes. What are its mechanical and electrical widths, and what happens when an x16 card is inserted?

P34.3 Compute the Nyquist frequency of PCIe Gen7 and explain why it is not 64 GHz.

P34.4 (Imperfect data) A cable listing says “USB 3.0 A to Micro-B, 19 pins”. Explain the number.

P34.5 Why does a Type-C sink need a pull-down on each CC contact?

P34.6 (Inverse design) A DDR5 RDIMM channel uses x4 DRAMs. How many DQS pairs does one 40-bit subchannel need?

P34.7 A camera is “MIPI, 4 lanes, 2.5 Gbit/s”. What information is still missing for board design?

Chapter 35

Related Interfaces: SATA, HDMI, DisplayPort, LVDS, CAN, and RS-485

Chapter Overview

Many boards carry interfaces that the main interface chapters did not cover in depth. This chapter gives each a compact board-design summary: signalling, impedance, coupling and termination, connector, and the mistakes most often seen in reviews. The values are the commonly published ones; each standard has revisions and options, and the governing document (or the transceiver datasheet that implements it) is the authority. Ethernet is covered in Chapter 31.

Learning Objectives

1. Apply the right impedance, coupling, and termination to each interface.
2. Recognize which interfaces are DC-coupled and which are AC-coupled.
3. Terminate multidrop buses correctly and bias them for fail-safe idle.

35.1 SATA

Figure 35.1 shows typical connectors for the interfaces of this chapter.

SATA links are point-to-point, AC-coupled differential pairs (one transmit, one receive per port) at 1.5, 3, or 6 Gbit/s, with 8b/10b encoding; the Nyquist frequency at 6 Gbit/s is 3 GHz. Board rules follow those of other SerDes lanes: 100 Ω differential, capacitors placed per the host or device guide, matched intra-pair length, and minimal stubs. The 7-contact data connector and the 15-contact power connector are separate; hot-plug support depends on the power-connector staging and the host. M.2 SATA devices use the M.2 connector, which can also carry PCIe; the host's key and configuration decide which.

35.2 HDMI and DisplayPort

HDMI. Transition-minimized differential signalling (TMDS) uses three data pairs and a clock pair. Unlike most SerDes, TMDS is *DC-coupled*: the sink terminates each leg with 50 Ω to its own 3.3 V supply, and the source sinks current from it. A source must therefore not back-drive an unpowered sink, and level shifters or redrivers must preserve the DC path (or be designed as AC-coupled re-timers with their own termination). HDMI 2.1 adds fixed-rate-link (FRL) signalling over four data lanes. The connector also carries the DDC (I^2C at 5 V), hot-plug detect (HPD), CEC, and a 5 V supply from the source; the DDC and HPD lines need ESD protection and correct pull-ups.

Table 35.1. Summary of related interfaces (commonly published values; confirm with the current standard and the transceiver datasheet).

Interface	Signalling	Impedance and coupling	Termination
SATA	point-to-point lanes, 1.5/3/6 Gb/s, 8b/10b; out-of-band (OOB) bursts for link start	100 Ω differential; AC-coupled	inside the PHYs
HDMI (TMDS)	3 data pairs + 1 clock pair; HDMI 2.1 FRL uses 4 data lanes at up to 12 Gb/s each	100 Ω differential; TMDS is DC-coupled	sink pulls each leg up to its 3.3 V supply through 50 Ω
DisplayPort	1, 2, or 4 main-link lanes; HBR3 8.1 Gb/s; DP 2.x UHBR up to 20 Gb/s per lane; AUX channel	100 Ω differential; AC-coupled	inside the PHYs
LVDS (TIA/EIA-644)	current-mode driver, about 3.5 mA, ≈350 mV across 100 Ω, common mode about 1.2 V	100 Ω differential; DC-coupled	100 Ω across the pair at the receiver (often internal)
CAN (ISO 11898-2)	two-wire bus, dominant/recessive states; CAN FD data phase at several Mb/s	twisted pair, about 120 Ω	120 Ω at each end of the trunk (split-termination option)
RS-485 (TIA-485-A)	multidrop differential bus, up to 32 unit loads, wide common-mode range (−7 to +12 V)	twisted pair, about 120 Ω	120 Ω at each end; fail-safe bias where receivers need it



Figure 35.1. Board-mount connectors for the interfaces of this chapter: SATA data, RJ45 with status LEDs, HDMI Type A, DisplayPort, a board-to-board mezzanine connector, and a pluggable terminal block of the kind used for CAN and RS-485 field wiring. Rendered illustration supplied with the manuscript, not a photograph of specific products; markings, contact counts, and proportions are not authoritative. [Illustration]

DisplayPort. The main link uses one, two, or four AC-coupled lanes; DP 1.4 reaches 8.1 Gbit/s per lane (HBR3), and DP 2.x adds UHBR rates of 10, 13.5, and 20 Gbit/s per lane with 128b/132b encoding. A half-duplex AUX channel carries link training and sideband data. DisplayPort over USB Type-C (alternate mode) maps the lanes onto the Type-C SuperSpeed pairs and AUX onto SBU (Section 34.2); a Type-C crosspoint or mux must support the rate. Figure 35.2 shows the receptacles that carry DisplayPort.



Figure 35.2. Receptacles that carry DisplayPort: standard DisplayPort, Mini DisplayPort, and USB Type-C (DisplayPort Alternate Mode). Rendered illustration supplied with the manuscript, not a photograph of specific products; markings, contact counts, and proportions are not authoritative. [Illustration]

35.3 LVDS

LVDS drivers steer a current of about 3.5 mA through a 100 Ω termination, giving roughly 350 mV differential around a common mode near 1.2 V [31]. The termination sits across the pair at the receiver and is often internal; a second external resistor halves the swing (Chapter 8). LVDS is DC-coupled; AC coupling is possible with a DC-balanced code and a receiver bias network. Display and camera interfaces such as FPD-Link and OpenLDI are built on LVDS-like signalling; bus LVDS (B-LVDS, M-LVDS) variants support multidrop with lower termination resistances.

35.4 CAN and RS-485

Both are multidrop differential buses over twisted pair, terminated at the two ends of the trunk with resistors matching the cable's characteristic impedance, about 120 Ω (Figure 35.3) [32], [33]. A CAN bus with both terminations measures about 60 Ω between CAN_H and CAN_L with power off, which is a quick check on a harness.

Transceivers for these buses come in small packages (Figure 35.4).

Board-level rules:

- Terminate only the ends of the trunk. A board that may sit at an end or in the middle needs a removable or switchable terminator, not a fixed one.
- Keep stubs short compared with the signal's rise time; slow-slew transceivers tolerate longer stubs.

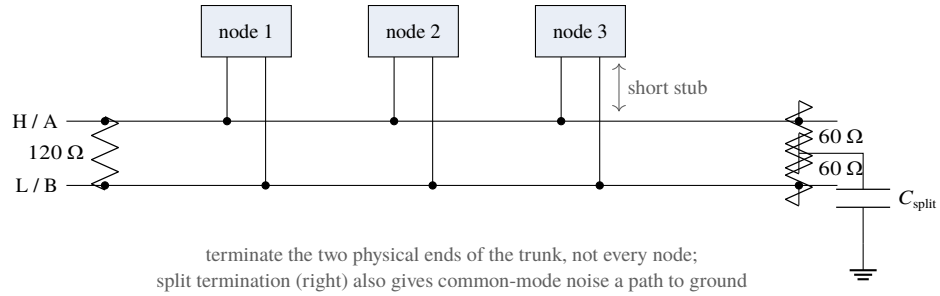


Figure 35.3. A multidrop differential bus terminated at its two ends. Nodes connect through short stubs. The right-hand termination is split, with a capacitor to ground for common-mode noise. [Schematic]

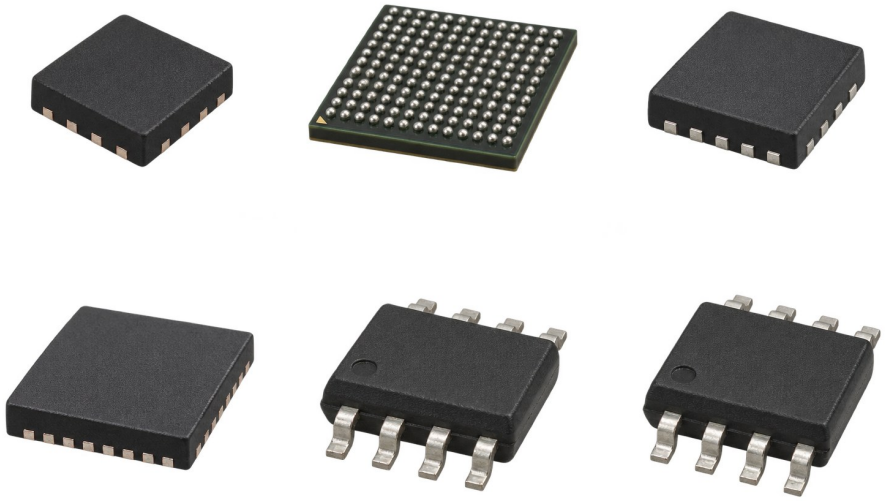


Figure 35.4. Typical packages of signal-conditioning parts and bus transceivers: leadless QFN/DFN packages (redrivers, retimers, level shifters), a fine-pitch BGA, and SOIC-8 packages (CAN and RS-485 transceivers). Rendered illustration supplied with the manuscript, not a photograph of specific products; markings, contact counts, and proportions are not authoritative. [Illustration]

- For RS-485, provide fail-safe biasing (pull-up on A, pull-down on B, sized with the terminations) unless the receivers have built-in fail-safe thresholds, so that an idle or open bus reads a defined state.
- Mind the common-mode range: RS-485 receivers tolerate about −7 to 12 V; ground differences between distant nodes beyond that need isolated transceivers.
- Protect the bus pins against ESD and surge at the connector; choose TVS diodes with low capacitance for CAN FD data rates.

Checklist: related interfaces

- ☐ Coupling (AC or DC) and termination location confirmed for each interface.
- ☐ TMDS DC path and back-drive protection handled; DDC/HPD pull-ups and ESD present.
- ☐ LVDS receivers checked for internal termination.
- ☐ CAN/RS-485 terminations only at trunk ends; fail-safe bias where needed; stubs short.

Key Takeaways

- SATA and DisplayPort main links are AC-coupled SerDes lanes; HDMI TMDS is DC-coupled with sink pull-ups.
- LVDS terminates across the pair at the receiver; avoid double termination.
- Multidrop buses are terminated at the two trunk ends, not at every node; RS-485 may need fail-safe bias.

Review Problems

P35.1 What resistance should an ohmmeter read between CAN_H and CAN_L on a correctly terminated, unpowered bus?

P35.2 An LVDS receiver has an internal 100 Ω termination and the board adds another. What differential swing results for a 3.5 mA driver?

P35.3 (Imperfect data) An HDMI redriver datasheet shows AC-coupling capacitors on its outputs. What must you check before using it with a TMDS sink?

P35.4 (Inverse design) An RS-485 bus has 120 Ω at each end. Size equal fail-safe bias resistors from 5 V so that the idle differential voltage is at least 200 mV.

Part VII

Simulation, Measurement, and Bring-Up

Chapter 36

Modeling and Simulation

Chapter Overview

Closed-form estimates choose the architecture; simulation confirms that it works. This chapter describes the models used in signal- and power-integrity simulation—IBIS and IBIS-AMI buffer models, SPICE models, field-solver extractions, and S-parameters—and the checks (passivity, causality, reciprocity) that keep those models physical. It defines mixed-mode S-parameters and shows how to read them, and lays out a pre-layout and post-layout simulation flow that catches problems while they are still cheap to fix.

Learning Objectives

1. Choose between IBIS, IBIS-AMI, and transistor-level models.
2. Select 2-D, 2.5-D, or 3-D field solvers for each part of a channel.
3. Define single-ended and mixed-mode S-parameters and convert between them.
4. Check S-parameter models for passivity, causality, and reciprocity.
5. Run a pre- and post-layout SI/PI flow with appropriate corners.

36.1 Buffer models

36.1.1 IBIS

The I/O Buffer Information Specification (IBIS) describes a buffer behaviorally, without revealing its circuit [80]. An IBIS model contains:

- current–voltage (V–I) tables for the pull-up and pull-down devices and the power and ground clamps;
- voltage–time (V–t) waveforms of the output switching into specified loads, which the simulator uses to reconstruct the transition;
- the die capacitance (C_comp) and package parasitics (lumped RLC or a package model);
- typical, minimum, and maximum corners for process, voltage, and temperature.

IBIS simulates quickly and is accurate for the reflections, crosstalk, and timing of single-ended and differential buffers. Its traditional form does not capture supply-noise dependence well; newer versions add power-aware features. The fast corner (strongest drive, fastest edges) is usually worst for ringing and crosstalk, the slow corner for timing.

36.1.2 IBIS-AMI

Multi-gigabit SERDES contain adaptive equalizers and CDRs that IBIS's analog tables cannot represent. IBIS-AMI (Algorithmic Modeling Interface) attaches executable models of the transmitter's FFE and the receiver's CTLE, DFE, and CDR to the analog IBIS front end. Two simulation modes are used:

Statistical
: The channel is characterized by its impulse response, assumed linear and time-invariant, and the eye and bathtub are computed by convolution of the impulse response with the data-pattern statistics. It is fast and gives BER contours down to very low probabilities.

Time-domain (bit-by-bit)
: Millions of bits of a pattern are passed through the channel and the equalizer models, including nonlinear and adaptive behavior. It is slower and limited in BER depth but captures adaptation.

A workable AMI flow is: (1) confirm that the TX and RX models match the silicon revision and the operating mode (rate, preset, equalization ranges); (2) build the channel from extracted and checked S-parameters (Section 36.3.4); (3) run a statistical analysis with adaptation enabled to find the optimal TX preset and the equalizer settings the RX converges to; (4) confirm in bit-by-bit mode, with a few million bits, that adaptation converges to similar settings and that the eye agrees with the statistical result at the BER it can reach; (5) sweep corners (silicon, channel tolerance, crosstalk aggressors) and report margins against the specification's reference receiver as well as the vendor's. When the two modes disagree, the difference is a finding, not noise to be averaged (Section 17.4.5).

36.1.3 Transistor-level models

Encrypted SPICE models from the silicon vendor are the most accurate but slowest option, and are typically used only to validate IBIS models or to analyze unusual conditions such as power-up or supply collapse.

36.2 Interconnect models and field solvers

Table 36.1. Field-solver classes and their uses.

Solver	Geometry	Typical use
2-D (cross-section)	Uniform lines	Z_0, RLGC matrices, coupling, loss of traces
2.5-D (planar MoM)	Layered planar structures	Planes, PDN impedance, large board regions
3-D full-wave (FEM, FDTD, FIT)	Arbitrary	Vias, connectors, BGA breakouts, packages
Quasi-static 3-D	Small structures	Lumped L, C of pins and pads

A complete channel model is usually assembled from pieces: 2-D models of the uniform trace segments, 3-D models of each via and connector transition, and vendor S-parameter models of packages and connectors, cascaded together. The segmentation boundaries must be placed where the fields are again quasi-TEM—a short distance away from each discontinuity—or the cascaded model will be inaccurate.

Common Pitfall: Using a laminate's nominal D_k and D_f

Building the 2-D models with the laminate family's data-sheet values. Each prepreg and core has its own resin content and therefore its own D_k and D_f, which also vary with frequency. Ask the fabricator for the values of the specific glass styles in the stackup, use a causal model (Section 11.3.2), and include roughness.

36.3 S-parameters

36.3.1 Definitions

An N-port network is described at each frequency by the relation between incident waves a_j and reflected waves b_i at its ports, normalized to a reference impedance Z_{ref} (usually 50 Ω):

$$b_i = \sum_{j=1}^{N} S_{ij}\, a_j, \qquad a_j = \frac{V_j + Z_{\text{ref}} I_j}{2\sqrt{Z_{\text{ref}}}}, \quad b_j = \frac{V_j - Z_{\text{ref}} I_j}{2\sqrt{Z_{\text{ref}}}}. \tag{36.1}$$

S_{ii} is the reflection coefficient at port i with all other ports matched; S_{ij} $(i \neq j)$ is the transmission from port j to port i. Insertion loss is $-20\log_{10}|S_{21}|$ and return loss $-20\log_{10}|S_{11}|$. S-parameters are exchanged in Touchstone files (.s2p, .s4p, …).

Worked Example 36.1: Return loss of an impedance error

A 55 Ω trace is measured with 50 Ω ports. Find the return loss of the impedance step, and the worst-case $|S_{11}|$ of a long section of it between the two ports.

Solution. $\Gamma = 5/105 = 0.0476$, a return loss of 26.4 dB. Reflections from the two ends add in phase when the section is an odd number of quarter-wavelengths long, giving $|S_{11}|_{\max} = 2\Gamma/(1+\Gamma^2) = 0.095$, or −20.4 dB. The return-loss curve therefore ripples between this value and zero, with period $1/(2T_{\text{D}})$.

36.3.2 Mixed-mode S-parameters

For differential channels the four single-ended ports are combined into two mixed-mode ports (Figure 36.1). With ports 1 and 2 forming the input pair (P and N) and ports 3 and 4 the output pair, the most used terms are [3], [4]

$$S_{\text{DD21}} = \tfrac{1}{2}\left(S_{31} - S_{32} - S_{41} + S_{42}\right), \qquad S_{\text{DD11}} = \tfrac{1}{2}\left(S_{11} - S_{12} - S_{21} + S_{22}\right), \tag{36.2}$$

$$S_{\text{CD21}} = \tfrac{1}{2}\left(S_{31} - S_{32} + S_{41} - S_{42}\right), \qquad S_{\text{CC21}} = \tfrac{1}{2}\left(S_{31} + S_{32} + S_{41} + S_{42}\right). \tag{36.3}$$

S_{DD21} is differential insertion loss; S_{DD11} differential return loss; S_{CD21} the conversion of an input differential signal to an output common-mode signal, which grows with asymmetry such as skew (Chapter 13). Figure 36.2 shows all three for a representative channel. Port numbering conventions differ between tools (some pair ports 1–3 and 2–4); always check the convention before applying these formulas.

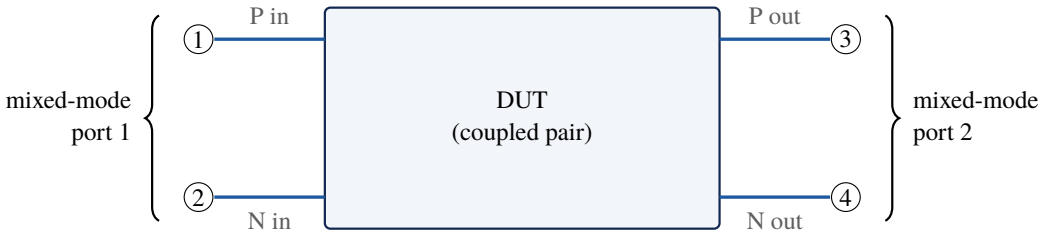


Figure 36.1. Port numbering for a four-port differential channel used in this book, and the corresponding mixed-mode ports.

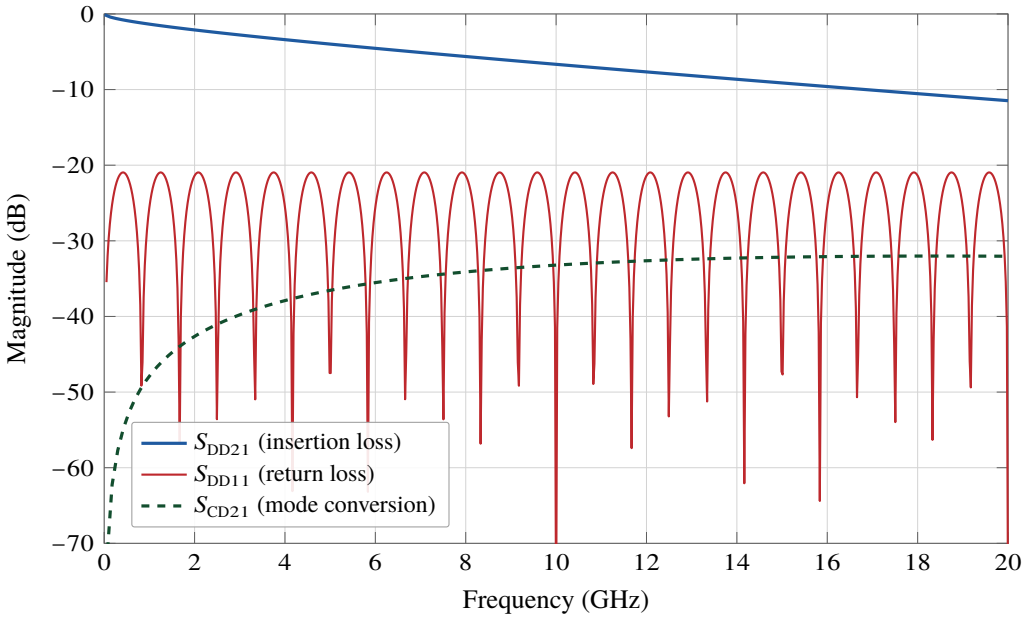


Figure 36.2. Mixed-mode S-parameters of a representative 10 in differential channel: insertion loss, return loss from a 93 Ω section ($T_D = 0.6$ ns) between 85 Ω ports, and mode conversion from 3 ps of intra-pair skew.

Worked Example 36.2: Mode conversion from skew

A pair has $S_{31} = 0.9$ and $S_{42} = 0.9\angle -10°$ at one frequency, with negligible cross terms. Find $|S_{DD21}|$ and $|S_{CD21}|$.

Solution. $S_{DD21} = \frac{1}{2}(0.9 + 0.9\angle -10°)$, magnitude $0.9\cos 5° = 0.897$ (−0.95 dB). $S_{CD21} = \frac{1}{2}(0.9 - 0.9\angle -10°)$, magnitude $0.9\sin 5° = 0.078$ (−22.1 dB). A 10° phase difference—about 2.8 ps at 10 GHz—already produces this conversion. Interpreted correctly (Section 13.4.2), 0.078 is a ratio of power-wave amplitudes: the converted *power* is $0.078^2 = 0.6\,\%$ of the incident differential power, and the common-mode *voltage* defined by Eq. (13.1) is half the ratio, 0.039 of the differential voltage (−28.1 dB). "Eight percent of the signal" would be ambiguous.

36.3.3 Physical consistency

Measured or extracted S-parameters must satisfy three physical properties before they are used in time-domain simulation:

Passivity

A passive network cannot create energy: all eigenvalues of $S^{H}S$ must be at most one at every frequency. A quick check is that every column satisfies $\sum_i |S_{ij}|^2 \le 1$; this is *necessary but not sufficient*—a matrix can pass every column test and still have an eigenvalue above one when the columns interfere, so the eigenvalue test is the one that decides. Small passivity violations from measurement noise cause time-domain simulations to grow without bound.

Causality

The response must not precede the stimulus. Non-causal models (from truncated bandwidth, constant-D_k material models, or interpolation errors) show precursors in the impulse response.

Reciprocity

For networks without magnetic materials or active devices, $S_{ij} = S_{ji}$.

S-parameter models must also extend to DC (or have a valid DC extrapolation) and cover a bandwidth of several times the Nyquist frequency, with frequency steps fine enough to resolve the longest delay in the model ($\Delta f \ll 1/T_{\mathrm{D}}$).

36.3.4 A Touchstone workflow

S-parameter data are exchanged as Touchstone files: .s2p, .s4p and so on for version 1, or the keyword-structured version 2 format. A reproducible workflow for each file is:

1. *Read the header.* Note the frequency unit, the data format (RI, MA or DB), the reference impedance of every port, and—for version 1 four-ports—the port order, which the file does not state (Figure 36.1).
2. *Inspect the frequency grid.* The lowest point should be close to DC, the highest at least $3f_{\mathrm{Nyq}}$, and the step fine enough for the longest delay.
3. *Check physics.* Run the passivity (eigenvalue), reciprocity and causality checks above, and record the worst violation and its frequency.
4. *Convert and plot.* Compute the mixed-mode terms with the port order confirmed, and plot S_{DD21}, S_{DD11}, S_{CD21} and the TDR (step response) of each through path.
5. *Sanity-check against expectations.* Compare delay with $\ell\, t_{\mathrm{pd}}$, loss at f_{Nyq} with the budget of Chapter 11, and TDR impedance with the stackup.
6. *Archive.* Keep the original file, any corrected version, and a note of every correction (passivity enforcement, DC extrapolation) applied.

Design Rule: S-parameter model hygiene

Before using any S-parameter file in a channel simulation, check passivity, causality, and reciprocity; confirm the port order and reference impedance; verify that it starts at or near DC and extends to at least three times the Nyquist frequency; and plot its TDR response to confirm that impedances and delays are plausible.

36.4 A pre- and post-layout flow

Figure 36.3 shows a practical SI/PI flow. *Pre-layout*, the designer explores topologies, stackups, terminations, and materials with idealized routing, sweeping lengths and corners to find where margins run out; the results become layout constraints. *Post-layout*, the routed board is extracted and every critical net class, the PDN, and the worst crosstalk victims are re-simulated with the real geometry. Corner analysis (fast/slow silicon, voltage, temperature, and fabrication tolerances such as $\pm 10\,\%$ impedance) or Monte Carlo analysis establishes the margin. Finally, simulations are correlated with measurements on the first boards (Chapter 37), which calibrates the models for the next design.

Rule of Thumb: Simulation bandwidth and resolution

Extract and simulate channel models to at least three times the Nyquist frequency, with a frequency step no larger than about one-tenth of $1/T_{\mathrm{D}}$ for the longest delay in the model. *Limits:* PAM4 and very low-BER analyses may need more bandwidth; very long channels need finer steps to avoid time-domain aliasing.

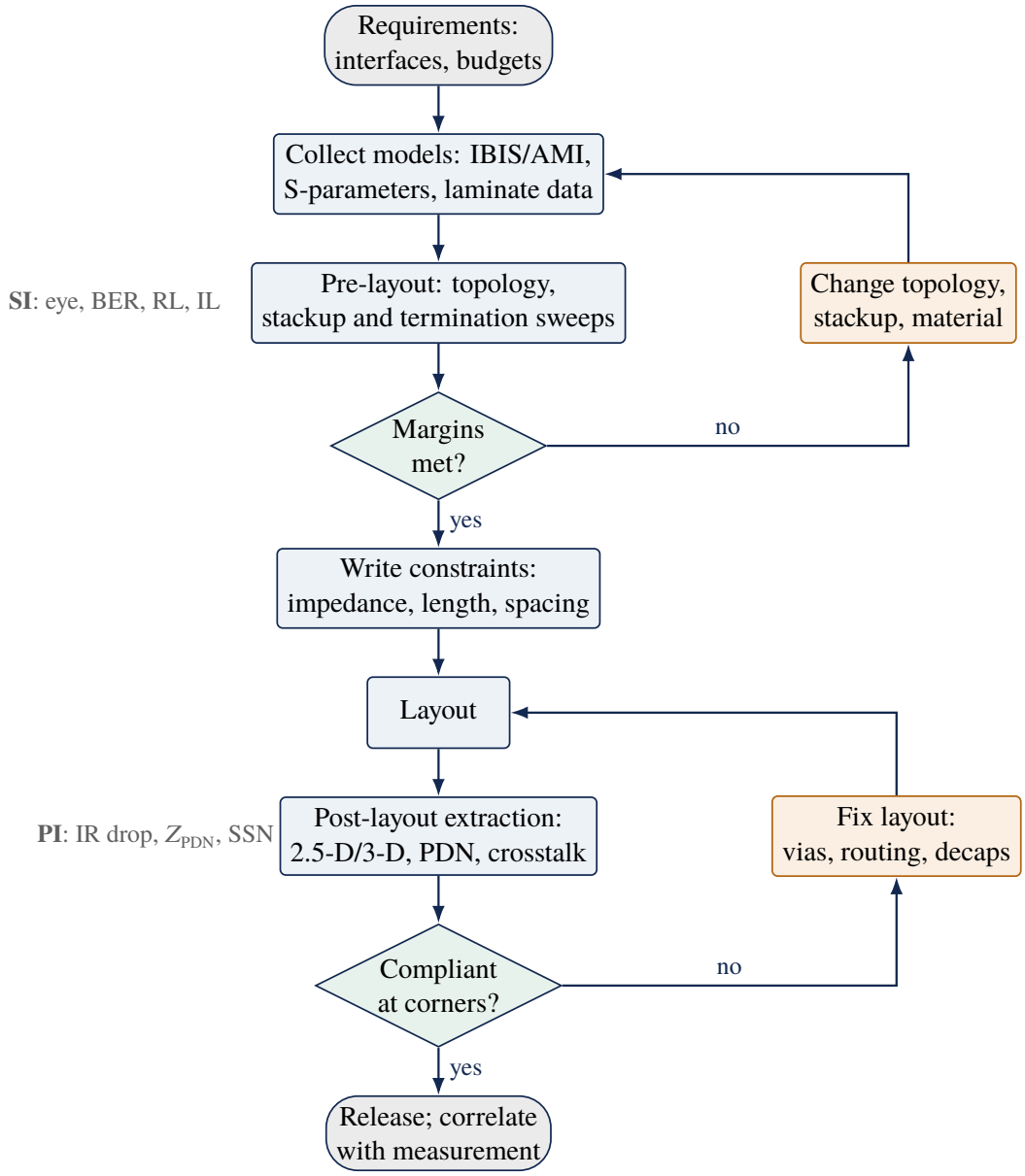


Figure 36.3. Pre- and post-layout signal- and power-integrity simulation flow.

Key Takeaways

- IBIS models buffers behaviorally from V–I and V–t tables; IBIS-AMI adds executable equalization and CDR models for SERDES.
- Use 2-D solvers for traces, 2.5-D for planes, and 3-D full-wave for vias and connectors; cascade at quasi-TEM boundaries.
- S-parameters describe incident and reflected waves; mixed-mode terms S_{DD21}, S_{DD11}, S_{CD21} characterize differential channels.
- Check every model for passivity, causality, reciprocity, port order, and bandwidth.
- Simulate pre-layout to set constraints and post-layout to verify them, across corners.

Further Reading

The IBIS specification [80] defines buffer models; Hall and Heck [4] cover S-parameters, passivity, and causality; IEEE 370 [29] addresses model and fixture quality.

Review Problems

P36.1 Find the return loss of a 60 Ω discontinuity measured with 50 Ω ports.

P36.2 What is the worst-case $|S_{11}|$ of a long 60 Ω line between 50 Ω ports?

P36.3 A reciprocal two-port has $|S_{11}| = 0.2$ and $|S_{21}| = 0.95$ at one frequency. Does it pass the column-sum passivity check?

P36.4 With 50 Ω single-ended ports, what are the differential- and common-mode reference impedances?

P36.5 Find S_{DD21} if $S_{31} = S_{42} = 0.8$ and $S_{32} = S_{41} = 0.1$ (all real).

P36.6 How many complex values per frequency point does a .s4p file contain?

Chapter 37

Measurement: TDR, VNA, Oscilloscopes, and Near-Field Probing

Chapter Overview

Simulation predicts; measurement decides. This chapter covers the instruments used to characterize interconnects and to verify signals on working boards: the time-domain reflectometer, the vector network analyzer with its calibration and de-embedding, the oscilloscope and its probes, and the near-field probes used to locate emission sources. For each, it explains what is actually measured, how the measurement setup itself corrupts the result, and how to avoid the most common errors.

Learning Objectives

1. Interpret TDR waveforms and extract impedance, excess capacitance, and excess inductance.
2. Estimate TDR spatial resolution from system rise time.
3. Explain VNA calibration and fixture de-embedding.
4. Choose oscilloscope bandwidth and probes, and quantify probe loading and ground-lead effects.
5. Use near-field probes to locate emission sources.

37.1 Time-domain reflectometry

A TDR launches a fast step into the device under test through a 50 Ω system and records the sum of the incident and reflected waves. At each instant, the reflection coefficient $\rho(t) = V_r(t)/V_i$ maps to the impedance seen at the corresponding round-trip position:

$$Z(t) = Z_{\text{ref}} \frac{1 + \rho(t)}{1 - \rho(t)}. \tag{37.1}$$

Time on the TDR display is round-trip time, so a feature at time t lies at distance $v_{\text{p}} t/2$ from the reference plane. Capacitive discontinuities appear as dips, inductive ones as bumps, and impedance steps as plateaus (Figure 37.1).

37.1.1 Excess capacitance and inductance

For a small discontinuity between uniform lines of impedance Z_0, integrating the reflected voltage over the dip or bump gives the excess reactance (Section 7.3):

$$C_{\text{exc}} = -\frac{2}{Z_0 V_i} \int v_r \, \mathrm{d}t, \qquad L_{\text{exc}} = \frac{2 Z_0}{V_i} \int v_r \, \mathrm{d}t. \tag{37.2}$$

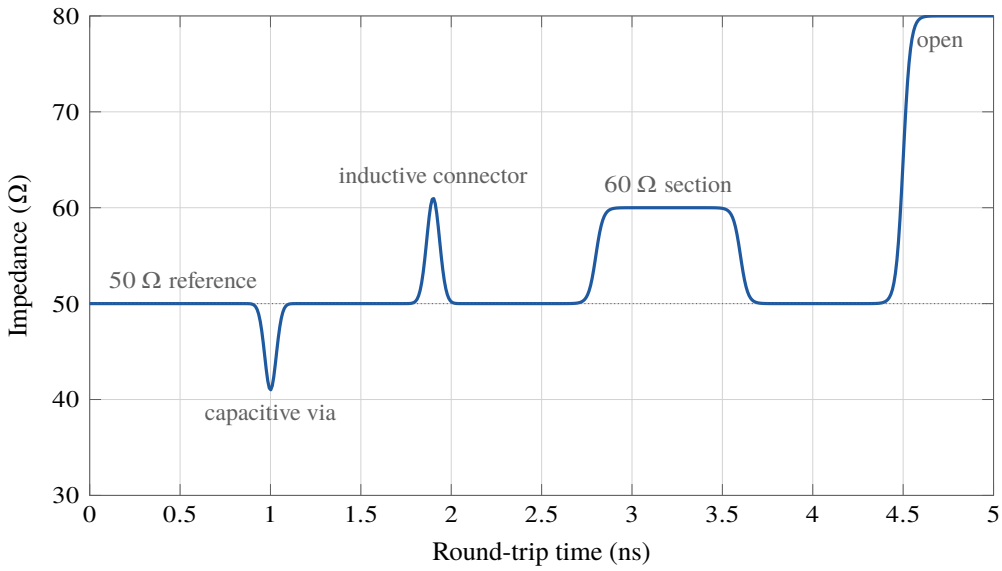


Figure 37.1. A TDR impedance profile: a capacitive via (dip), an inductive connector (bump), a 60 Ω trace section, and an open end.

Because the integral is insensitive to the instrument's rise time, this extraction is accurate even when the discontinuity is too small to resolve as a feature.

Worked Example 37.1: Reading a TDR

(a) A plateau shows $\rho = -0.10$. What is the impedance? (b) A via produces a triangular dip with peak $\rho = -0.15$ and base width 40 ps. Estimate its excess capacitance on a 50 Ω line.

Solution. (a) $Z = 50 \times 0.9/1.1 = 40.9\,\Omega$. (b) The dip's area is $\tfrac{1}{2} \times (-0.15) \times 40\,\text{ps} = -3\,\text{ps}$, so $C_{\text{exc}} = 2 \times 3\,\text{ps}/50\,\Omega = 0.12\,\text{pF}$.

37.1.2 Resolution and masking

A TDR displays round-trip time. Two discontinuities a distance ℓ apart produce reflections separated by $t = 2\ell/v_{\text{p}}$; they appear as separate features when that separation is at least about the system rise time t_{r}. The spatial resolution is therefore

$$\Delta x \approx \frac{t_{\text{r}}\, v_{\text{p}}}{2}, \tag{37.3}$$

where the factor $\tfrac{1}{2}$ comes from the round trip, not from "half the rise time." A 35 ps system edge and $v_{\text{p}} = 6\,\text{in/ns}$ give $\Delta x \approx 0.105\,\text{in}$. Resolution is not accuracy: a feature shorter than Δx is not invisible, but its apparent impedance is a blend of the feature and its surroundings (Figure 37.2). The usual conversion $Z = Z_{\text{ref}}(1+\rho)/(1-\rho)$ then gives an *apparent* impedance, not the local value; reconstructing local geometry requires a model at the same bandwidth.

Common Pitfall: A slow TDR that makes everything look good

Accepting a channel because its TDR is flat at 50 Ω when measured with a slow edge. Short discontinuities (pads, via barrels, connector pins) are averaged away by a slow edge but are fully visible to a fast serial link. Specify the TDR edge and the reference plane with every impedance acceptance window, and compare with a model filtered to the same bandwidth.

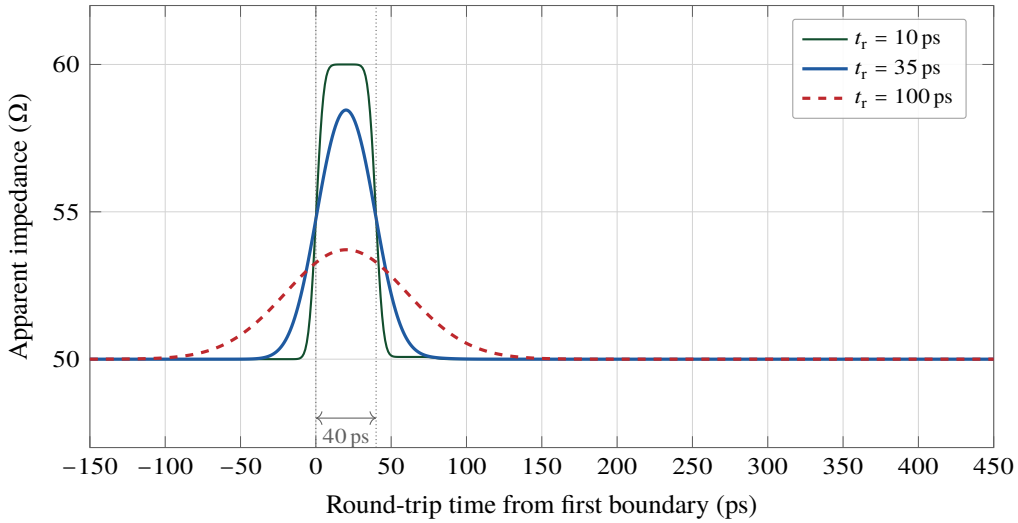


Figure 37.2. Simulated TDR of a 60 Ω segment with 20 ps one-way delay between matched 50 Ω lines, for Gaussian-equivalent system edges of 10, 35 and 100 ps. Multiple reflections in the segment are included; no loss or fixture. With the 100 ps edge, the 60 Ω plateau becomes a low bump. [Simulation (analytical lattice)]

Multiple reflections and the loss of the line mask features further down: a large discontinuity near the launch reduces the step that reaches later ones, and a lossy line slows the step so that distant features look smaller and smoother. Time-domain transmission (TDT), measured at the far end, gives delay, skew, and crosstalk; a differential TDR/TDT drives both lines of a pair to measure Z_{diff} and mode conversion.

37.2 Vector network analysis

A VNA measures S-parameters directly in the frequency domain, with far greater dynamic range than a TDR (often 100 dB or more), and its data can be transformed into the time domain. Two steps are essential (Figure 37.3):

Calibration

Measuring known standards (short, open, load, thru—SOLT—or thru, reflect, line—TRL) at the ends of the test cables moves the measurement reference plane to the cable connectors and removes the instrument's systematic errors.

De-embedding

The DUT on a board is reached through fixtures (launches, probe pads, traces). De-embedding removes their effect mathematically, using models or measurements of the fixtures. The "2×-thru" method measures a structure made of the two fixture halves joined directly, splits it into two halves, and removes them from the DUT measurement. IEEE 370 standardizes fixture design, de-embedding, and quality metrics for board interconnects up to 50 GHz [29].

Common Pitfall: De-embedding imperfect fixtures

Assuming that de-embedding removes any fixture. Fixtures with large discontinuities, poor repeatability between the 2×-thru and the DUT fixture, or resonances produce de-embedded results with non-physical ripple and passivity violations. Design fixtures for low reflection and good consistency, and check the de-embedded result with the quality metrics of IEEE 370.

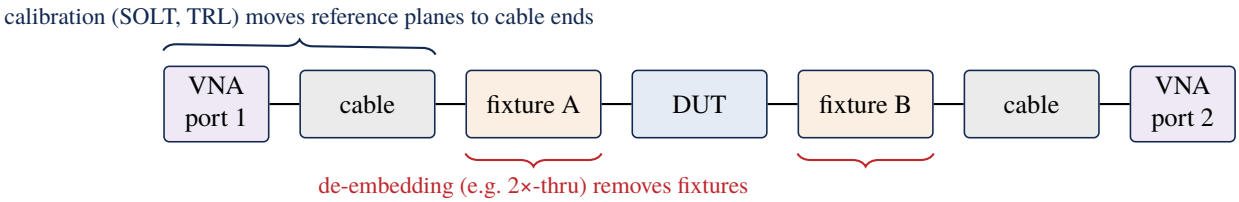


Figure 37.3. VNA measurement chain. Calibration places the reference planes at the cable ends; de-embedding removes the test fixtures to isolate the DUT.

37.2.1 Checking a de-embedding

A de-embedded result should pass three self-consistency checks before it is trusted: de-embedding the 2×-thru from *itself* should give a perfect thru (0 dB loss, zero phase, negligible reflection) to the accuracy claimed; the fixture halves extracted from the 2×-thru should be passive and causal; and a DUT of known behavior—a short line of the same geometry—should de-embed to the loss per inch measured from two line lengths. Repeating the fixture measurement on a second board or after reconnecting the cables shows how much of the residual is repeatability rather than fixture error, and that spread belongs in the reported uncertainty [29].

37.3 Measuring milliohm PDN impedance

A PDN of a few milliohms cannot be measured reliably with a one-port reflection measurement: at $Z \ll 50\,\Omega$ the reflection coefficient is close to -1, and the impedance is hidden in its small difference from -1, where calibration residuals dominate. The *two-port shunt-through* method connects port 1 and port 2 to the same power–ground location (Figure 37.4). For a shunt impedance Z between two matched 50 Ω ports,

$$S_{21} = \frac{2Z}{Z_0 + 2Z} \quad \Longleftrightarrow \quad Z = \frac{Z_0\, S_{21}}{2(1 - S_{21})} \approx \frac{Z_0}{2} S_{21} \quad (Z \ll Z_0), \tag{37.4}$$

using the complex S_{21} of a calibrated measurement [6]. The transmission is now proportional to Z instead of being a small difference.

Worked Example 37.2: How hard is a milliohm measurement?

A purely resistive 5 mΩ PDN is measured shunt-through with 50 Ω ports. Find $|S_{21}|$, and the error contributed by a residual fixture inductance of 10 pH at 100 MHz.

Solution. $S_{21} = 0.01/50.01 = 2.0 \times 10^{-4}$, or -74.0 dB: the VNA needs dynamic range, source power and IF bandwidth to measure that with low noise. The residual inductance has $|\mathrm{j}\omega L| = 2\pi \times 10^{8} \times 10^{-11} = 6.3\,\mathrm{m}\Omega$ at 100 MHz—larger than the DUT. Unless that inductance is calibrated out or the probe point is defined to include it, the high-frequency result is a measurement of the fixture.

Two further error sources are specific to this method. Above a few hundred hertz, the cable shields of the two ports connect the board ground to the instrument ground and form a loop in parallel with the DUT; at low frequency this ground loop can dominate a milliohm measurement, and it is broken with a common-mode choke (ferrite-loaded cable) or an isolation transformer on one port. And the result is the impedance *at the probe location*, not at the die; for correlation with a simulation, the model must be evaluated at the same port.

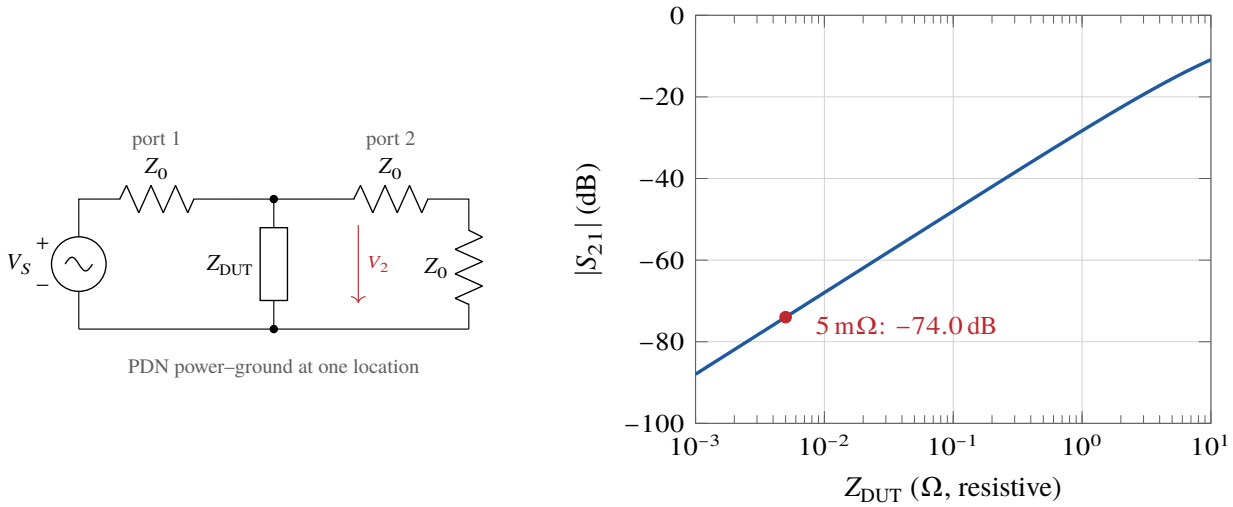


Figure 37.4. Left: ideal two-port shunt-through connection; cable shields, calibration standards and ground loops are omitted but matter in the laboratory. Right: $|S_{21}|$ versus a resistive DUT impedance from Eq. (37.4). [Analytical]

Design Rule: Plan PDN measurement points

For every critical rail, place a probe point (a pair of coaxial launch footprints or probe pads with adjacent ground) close to the load and record its location in the PDN model. Document the measurement floor (noise and residual inductance) with a shorted fixture before measuring the board.

37.4 Oscilloscopes and probing

37.4.1 Bandwidth

By Eq. (2.5), an oscilloscope adds its own rise time to the signal's. For a displayed rise time within 5 % of the true value, the scope's rise time must be less than about one-third of the signal's (Worked Example 2.2). For serial-link compliance, specifications usually state a required bandwidth tied to the data rate; at least two to three times the Nyquist frequency is a common minimum. Real-time oscilloscopes capture single events; equivalent-time (sampling) oscilloscopes offer higher bandwidth and lower noise for repetitive signals.

37.4.2 Probe loading

Every probe loads the circuit. A passive 10 × probe with 10 pF input capacitance has a reactance of only 32 Ω at 500 MHz; placed on a 50 Ω line, it forms a time constant of $Z_0 C/2 = 250$ ps. Active probes reduce input capacitance to a fraction of a picofarad; differential active probes measure pairs without referencing either side to ground.

37.4.3 Ground-lead inductance

The probe's ground lead forms a loop with the probe tip. Its inductance (about 1 nH/mm, so 100–200 nH for a 15 cm clip lead) resonates with the probe's input capacitance and the source impedance, producing ringing that is not on the signal (Figure 37.5).

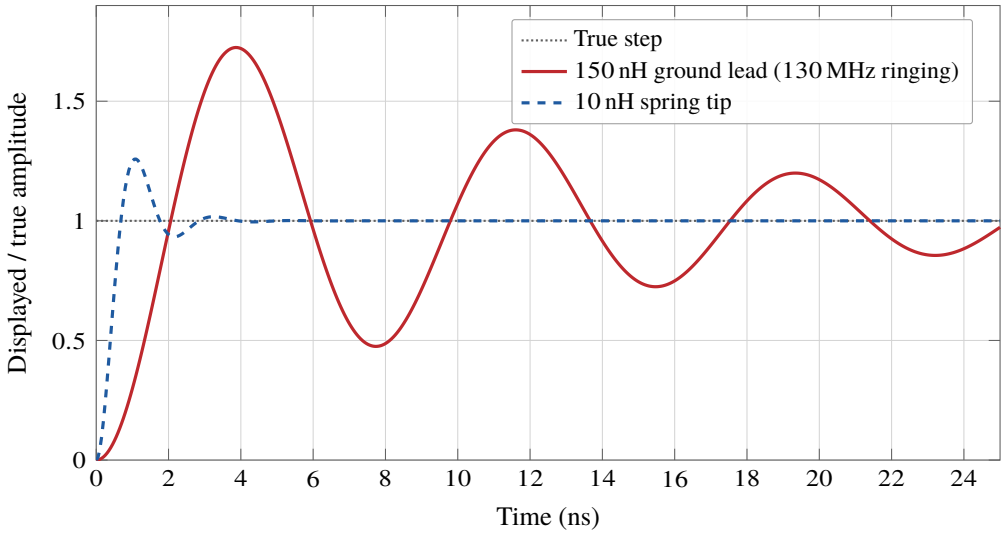


Figure 37.5. Step response through a passive probe with 10 pF tip capacitance and a 25 Ω source: a 150 nH ground lead rings at 130 MHz; a 10 nH spring tip resonates near 500 MHz and is well damped.

Worked Example 37.3: Ground-lead resonance

Find the resonant frequency of a 150 nH ground lead with a 10 pF probe, and with a 10 nH spring tip.

Solution. $f = 1/(2\pi\sqrt{LC})$: 130 MHz and 503 MHz. With a 25 Ω source the damping ratios are 0.10 and 0.40, so the long lead rings for many cycles and overshoots by about 70 %, while the spring tip overshoots by about 25 % and settles quickly.

Design Rule: Probing fast signals

Use active or solder-in probes with the shortest possible ground connection, probe at the receiver (where the signal matters), and provide probe points with an adjacent ground pad in the layout. For power-rail ripple, use a 50 Ω coaxial connection (1:1) with the scope input set to 50 Ω and a DC block if needed, rather than a 10 × probe with a clip lead.

37.5 Near-field probing

Small magnetic-loop (H-field) and electric-stub (E-field) probes connected to a spectrum analyzer locate emission sources on a board. An H-field probe responds to currents flowing in traces, planes, and cables beneath it and is the more useful for finding loop currents and return-path problems; an E-field probe responds to high-dV/dt nodes such as clock nets and switching regulators. Near-field measurements are relative—they do not predict the far-field result of a compliance test—but comparing amplitudes before and after a change, or across a board, quickly identifies the dominant sources and the cables that carry common-mode current.

Rule of Thumb: Probe ground leads

A probe ground lead has roughly 1 nH/mm of inductance. Keep its resonance with the probe's input capacitance at least three to five times above the signal bandwidth. *Limits:* the actual inductance depends on the loop shape; the safest choice is the shortest ground the probe allows.

Table 37.1. Measurement instruments and their uses.

Instrument	Measures	Limitations
TDR / TDT	Impedance profile, delay, skew, crosstalk	Resolution set by rise time; masking
VNA	S-parameters, wide dynamic range	Needs calibration and de-embedding
Real-time scope	Waveforms, eyes, jitter, single events	Bandwidth, noise, probe loading
Sampling scope	High-bandwidth repetitive waveforms	Needs trigger; repetitive signals only
Spectrum analyzer + near-field probe	Emission sources	Relative only
BERT	Bit-error ratio, bathtubs	Long test times at low BER

Key Takeaways

- TDR maps reflection coefficient to impedance, $Z = Z_{\text{ref}}(1 + \rho)/(1 - \rho)$; the area of a dip or bump gives excess C or L.
- Measure milliohm PDN impedance shunt-through, $Z \approx (Z_0/2)S_{21}$; residual fixture inductance and cable ground loops set the floor.
- TDR resolution is about $t_r v_p/2$ (the $\frac{1}{2}$ is the round trip); shorter features show an apparent, blended impedance, and distant ones are masked.
- VNA measurements require calibration to the cable ends and de-embedding of fixtures; IEEE 370 standardizes the process.
- Probe capacitance and ground-lead inductance corrupt fast measurements; use active probes with short grounds.
- Near-field probes locate emission sources by relative comparison.

Further Reading

Derickson and Müller [55] describe high-speed measurement instruments and methods; IEEE 370 [29] standardizes de-embedding; Novak and Miller [6] cover PDN measurement.

Review Problems

P37.1 A TDR plateau shows $\rho = +0.2$ in a 50 Ω system. What is the impedance?

P37.2 What reflection coefficient does a 42 Ω section produce in a 50 Ω system?

P37.3 An inductive bump has $\int v_r \, dt/V_i = 2\,\text{ps}$ on a 50 Ω line. Find the excess inductance.

P37.4 Find the resonance of a 150 nH ground lead with an 8 pF probe.

P37.5 Estimate the spatial resolution of a TDR with a 25 ps system rise time on a line with $v_p = 6\,\text{in/ns}$.

P37.6 What oscilloscope bandwidth displays a 50 ps edge within 5 %?

P37.7 A shunt-through measurement gives $S_{21} = 4.0 \times 10^{-4} \angle 0°$ at 1 MHz with 50 Ω ports. What is the PDN impedance?

P37.8 *(Imperfect data.)* A 60 Ω, 20 ps segment measured with a 100 ps TDR edge shows a peak of only about 54 Ω. A colleague concludes the segment is 54 Ω. What would you do?

Chapter 38

Board Bring-Up and Debug

Chapter Overview

The first powered board is where design assumptions meet reality. A disciplined bring-up procedure protects scarce prototypes from damage, isolates faults quickly, and produces the measurements that confirm—or correct—the simulation models. This chapter gives a step-by-step bring-up flow from bare-board inspection to high-speed link qualification, describes boundary scan and the built-in margining features of modern interfaces, and summarizes a root-cause method for the high-speed failures that remain.

Learning Objectives

1. Perform pre-power inspection and rail-impedance checks.
2. Power a board safely with current limits and verify sequencing and ripple.
3. Use JTAG and boundary scan to verify connectivity.
4. Interpret DDR training margins and serial-link margining results.
5. Apply a structured root-cause method to high-speed failures.

38.1 The bring-up flow

Figure 38.1 outlines the sequence. Its guiding principle is to verify each layer of the system before relying on it: power before clocks, clocks before logic, connectivity before firmware, and simple interfaces before fast ones.

38.2 Before power

1. **Visual and X-ray inspection.** Check component orientation, missing or tombstoned parts, solder bridges, and (by X-ray) BGA voiding and bridging, especially on the first article.
2. **Rail-to-ground resistance.** Measure each rail's resistance to ground and compare with a known-good board or with the expected load. Low-voltage, high-current core rails legitimately read low (a fraction of an ohm to a few ohms) because of the load's leakage, so comparison with a reference board is more meaningful than an absolute threshold.
3. **Rail-to-rail checks.** Look for shorts between adjacent rails, which are common under BGAs.

Worked Example 38.1: Interpreting a low rail resistance

A 0.8 V core rail measures 0.3 Ω to ground; a reference board measures 8 Ω. If powered at 0.8 V with a 2 A limit, what happens at the fault?

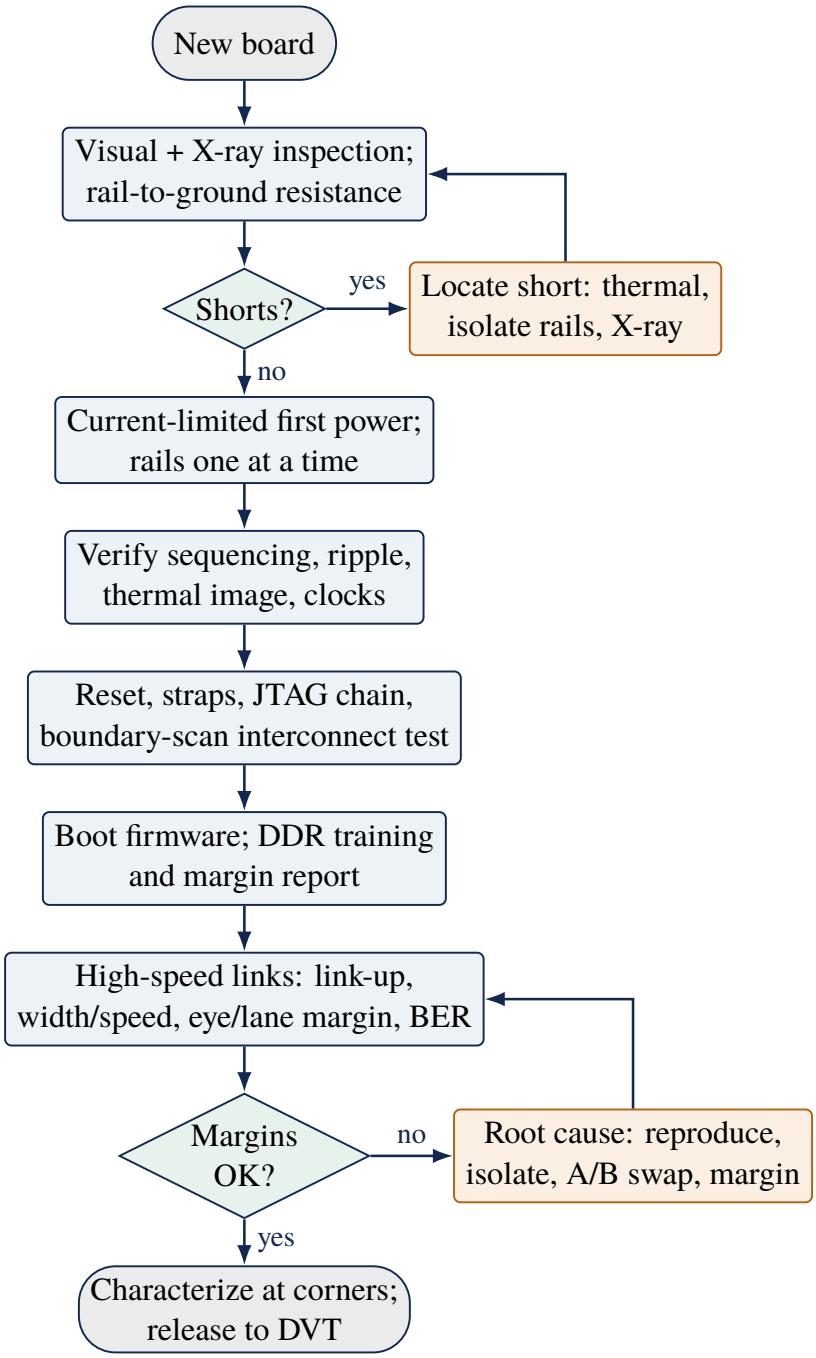


Figure 38.1. Board bring-up flow, from inspection to link qualification.

Solution. The rail is almost certainly shorted. At the current limit, the fault dissipates $I^2R = 2^2 \times 0.3 = 1.2\,\text{W}$, enough to make it visible on a thermal camera within seconds—which is how such shorts are located. That power is not guaranteed to be harmless: if the short is a small defect (a solder bridge under a BGA, a hairline inner-layer short or a damaged device), 1.2 W concentrated in it can overheat the joint or delaminate the laminate. Start with a lower limit—at 0.5 A the fault dissipates 75 mW—and raise it in steps while watching the thermal image, stopping as soon as the hot spot is located.

38.3 First power

Power the board from a bench supply with a current limit a little above the expected idle current, with each regulator enabled separately if the design allows it. For each rail:

- verify the DC voltage at the load (remote-sense point), not only at the regulator;
- verify sequencing and ramp times against the device requirements (Figure 38.2); many SoCs and memories specify the order, maximum ramp time, and maximum delay between rails, and the reset release relative to the last rail;
- measure ripple and transient response with a 50 Ω coaxial connection (Chapter 37), and compare with the PDN budget (Chapter 21);
- image the board thermally to catch hot spots.

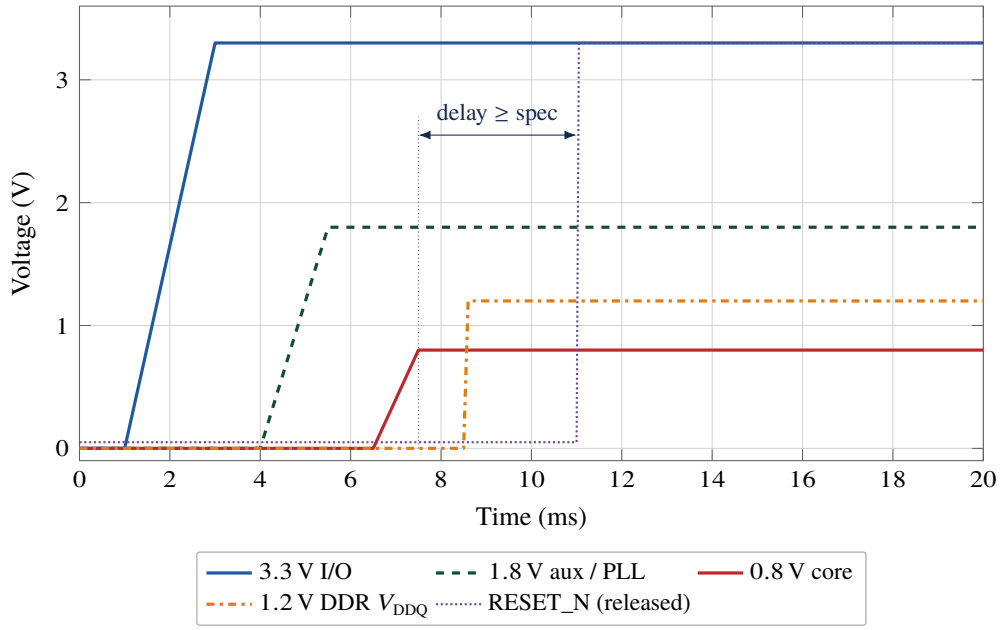


Figure 38.2. An example power-up sequence: I/O, auxiliary, core, and memory rails ramp in order, and reset is released after the last rail is stable. The required order and delays come from each device's data sheet.

Then verify clocks: frequency (against the ppm tolerance), amplitude and common mode at the receiver, and, for reference clocks of serial links, phase noise or jitter (Chapter 33).

38.4 JTAG and boundary scan

IEEE 1149.1 defines a test access port (TAP)—TCK, TMS, TDI, TDO, and optional TRST—and a boundary-scan register that lets a tester drive and sample every digital pin of a compliant device [81]. With all devices on a board chained TDI-to-TDO, boundary scan verifies:

- the chain itself (reading each device's IDCODE);
- interconnects between boundary-scan devices (opens and shorts);
- connections to non-scan devices that can be exercised indirectly, such as memories (by writing and reading through the controller's pins) and pull resistors.

Boundary scan tests DC connectivity only; it does not exercise signal integrity at speed. It is nevertheless invaluable for BGA-dense boards, where physical probing is impossible.

Design Rule: Design for JTAG

Bring the JTAG chain to a standard header, buffer TCK and TMS when they fan out to many devices, terminate TCK (it is a clock), provide a way to bypass or isolate devices that may be unpopulated, and keep the boundary-scan description (BSDL) files and the netlist under revision control for the test team.

38.5 Interface bring-up and margining

38.5.1 Memory

Memory controllers report training results: write-leveling delays per byte lane, read and write DQ-DQS delays per bit, and V_{REF} settings, often with a two-dimensional eye scan of each bit's timing and voltage margin. Byte lanes whose leveling delays are near the end of the range, or bits whose eyes are markedly smaller than their neighbors', point to layout problems (Section 28.5). Running memory stress tests at elevated temperature and at the supply voltage limits exposes marginal timing.

38.5.2 Serial links

Serial links first must train to the intended width and speed; a link that comes up at a lower width or generation indicates a failing lane. PCIe Gen4 and later define *lane margining at the receiver*, in which the receiver reports how far its sampling point can be moved in time and voltage before errors occur, without external equipment. Many SERDES also expose on-die eye monitors. PRBS generators and checkers in the PHY allow long BER tests (Worked Example 17.3).

38.5.3 Margins across conditions

A margin measured once, at room temperature and nominal voltage, on one board, says little about production. Plan a small matrix of process, voltage and temperature conditions (Table 38.1) and record the same quantities at each point, so that trends—a margin that shrinks with temperature, or one board that is an outlier—are visible. Keep the raw logs: the controller's training report, the PHY's eye-monitor output, the tool and firmware versions, the board serial number and the instrument settings.

38.6 Root-cause method

1. **Reproduce** the failure reliably, and record the board, temperature, voltage, data pattern, and firmware version.

Table 38.1. A minimal margining matrix and log template (illustrative).

Field	Content
Conditions	Boards: at least three, from different panels if possible; supply at nominal and both limits; temperature at cold, room and hot limits
Memory	Per-bit timing and V_{REF} eye widths; leveling delays per byte; stress-test result and duration
Serial links	Trained width and speed; TX preset chosen; lane-margining time and voltage per lane; PRBS error count and duration
Power	Rail voltages at the load under idle and stress; ripple and transient captured with stated bandwidth
Record	Board serial, firmware and tool versions, instrument and probe settings, date, operator

Table 38.2. Common high-speed symptoms and likely causes.

Symptom	Likely causes
PCIe link trains at reduced width	Swapped or open lane; missing AC cap; lane polarity or reversal not supported
PCIe link trains at lower speed	Excess loss; via stubs; reference-clock jitter; marginal equalization
DDR write leveling fails on outer lanes	CK-to-DQS skew beyond leveling range
DDR errors only when hot	Marginal timing; V_{REF} noise; insufficient decoupling
Intermittent lock loss on a SERDES	Supply noise in the PLL; reference-clock noise; SSC mismatch
Ethernet works at 100M, fails at 1G	RGMII delay misconfigured; MDI pair swap
Emissions at clock harmonics	Return-path discontinuity; unfiltered cable; missing SSC

2. **Isolate** it: which lane, which bit, which board, which direction. Patterns across boards distinguish design problems from assembly problems.
3. **Measure margin**, not pass/fail: vary voltage, temperature, frequency, or equalization settings and find how far the failure point is from nominal.
4. **Swap** parts or boards (A/B testing) to separate the device, the board, and the partner.
5. **Compare with simulation**: re-simulate the failing net with measured parameters (impedance from TDR, actual stackup from cross-section); a discrepancy between model and measurement is itself a finding.
6. **Confirm the fix** by restoring the margin, not merely by making the failure disappear once.

Worked Example 38.2: Two failure investigations

(a) One PCIe Gen4 lane trains only at Gen3, on two of five boards. Isolation: the same lane fails on both boards, and swapping the add-in card does not move it, so the board is suspected. Lane margining on the passing boards shows that lane with the smallest time margin. A VNA measurement of the lane (Chapter 37) shows a notch near 40 GHz, below the 48 GHz expected from the back-drill target (Section 15.5.1). A cross-section finds a residual stub of about 37 mil instead of the specified maximum: the back-drill depth on that via row is out of tolerance. *Finding:* a fabrication escape, not a design error; the fix is a drill-depth inspection requirement and coupon, and the design margin is confirmed by re-simulating with the measured stub.

(b) DDR read errors only above 70 °C. Isolation: errors are on reads, on all byte lanes, and disappear when the controller's V_{REF} is trained at temperature. A rail measurement with adequate bandwidth shows V_{DDQ} noise peaking at a frequency that matches a PDN anti-resonance predicted by the model but judged acceptable because its peak was "only slightly" above target; the capacitance of the bulk MLCCs falls with temperature and moves the peak. *Finding:* a design margin problem; the fix is damping the anti-resonance (Section 21.4) and re-running the PVT matrix of Table 38.1.

Common Pitfall: Debugging by changing several things at once

Swapping a component, changing a firmware setting, and moving a probe at the same time. When the failure disappears, the cause is unknown and the fix may not be real. Change one variable at a time, and record each experiment.

Worked Example 38.3: Checking a power-up sequence

A device requires its reset to be held for at least 2 ms after the last supply reaches 95 % of its final value. In the capture of Figure 38.2, the last rail (1.2 V V_{DDQ}) is stable at 8.6 ms and reset is released at 11.0 ms. Is the requirement met, and what is the margin?

Solution. The hold time is $11.0 - 8.6 = 2.4$ ms, meeting the 2 ms requirement with 0.4 ms of margin. Because regulator ramp times vary with load and temperature, verify the sequence also at the slowest expected ramp.

Rule of Thumb: First-power current limit

Set the bench supply's current limit at first power to about 1.2 to 1.5 times the expected idle current. *Limits:* bulk capacitance and regulator soft-start produce inrush currents above the idle value; if the supply folds back during start-up, raise the limit briefly or pre-charge through a resistor.

Key Takeaways

- Inspect and measure rail resistances against a reference before power.
- Power with current limits, one rail at a time; verify voltage at the load, sequencing, ripple, temperature, and clocks.
- Boundary scan verifies connectivity of BGA-dense boards without physical access.
- Use DDR training reports, PCIe lane margining, eye monitors, and PRBS BER tests to quantify margin.
- Root-cause by reproducing, isolating, margining, swapping, and comparing with simulation—one change at a time.

Further Reading

IEEE 1149.1 [81] defines boundary scan; Derickson and Müller [55] discuss BER testing and jitter measurement used in link qualification.

Review Problems

P38.1 A 0.8 V rail shows 30 mV peak-to-peak noise against a ±3 % tolerance. Is the dynamic noise within budget, assuming the full tolerance is available?

P38.2 A 0.3 Ω short carries a 2 A current limit. What power does it dissipate?

P38.3 Four devices in a JTAG chain have instruction registers of 4, 8, 6, and 5 bits. What is the total instruction-register length?

P38.4 A boundary-scan chain has 2000 cells and TCK runs at 10 MHz. How long does one full shift take?

P38.5 A PCIe ×16 link trains at ×8. What fraction of bandwidth is lost?

P38.6 How long must one 8 GT/s lane run error-free to show BER $\leq 10^{-12}$ at 95 % confidence?

Chapter 39

Design for Manufacturing and Release

Chapter Overview

A high-speed design is only as good as the board that the fabricator builds. The release package must tell the fabricator and the assembler exactly what to build, which tolerances matter, and how compliance will be verified—and the design must be manufacturable within those tolerances at reasonable yield. This chapter covers fabrication notes and drawings, stackup and impedance documentation, impedance coupons, tolerance analysis, design-for-manufacturing and design-for-test rules, the data formats of a release package, and change control.

Learning Objectives

1. Write fabrication notes and a stackup/impedance table that a fabricator can build from without ambiguity.
2. Specify impedance coupons and their test method.
3. Estimate the effect of fabrication tolerances and etch on impedance.
4. Apply DFM and DFT rules to a high-speed layout.
5. Assemble a complete release package and control subsequent changes.

39.1 Fabrication drawing and notes

The fabrication drawing (Figure 39.1) is the contract with the fabricator. It contains the board outline and dimensions, the stackup and impedance table, the drill table, and the fabrication notes. Typical notes specify:

- the performance class (IPC-6012 Class 2 for most commercial products, Class 3 for high-reliability ones) [82];
- the laminate (by IPC-4101 slash sheet or by named product) and its minimum T_g [50];
- finished thickness and tolerance, copper weights, surface finish, soldermask and legend;
- via treatment (tented, plugged, filled and capped) referenced to IPC-4761 types [83], and any via-in-pad requirements;
- back-drill locations, the layer that must not be cut, and the maximum residual stub (Chapter 15);
- finished-hole tolerances that make physical sense: a tolerance such as "−full" (the hole may plate closed) is a convention for unfilled vias only, and is wrong for filled or capped vias, whose fill needs an open barrel—the drill table of Figure 39.1 uses ± 0.05 mm;
- controlled-impedance requirements, permission (or not) for the fabricator to adjust widths, and the coupon and test method;
- bow and twist, electrical test (to an IPC-D-356 netlist [84]), and marking requirements.

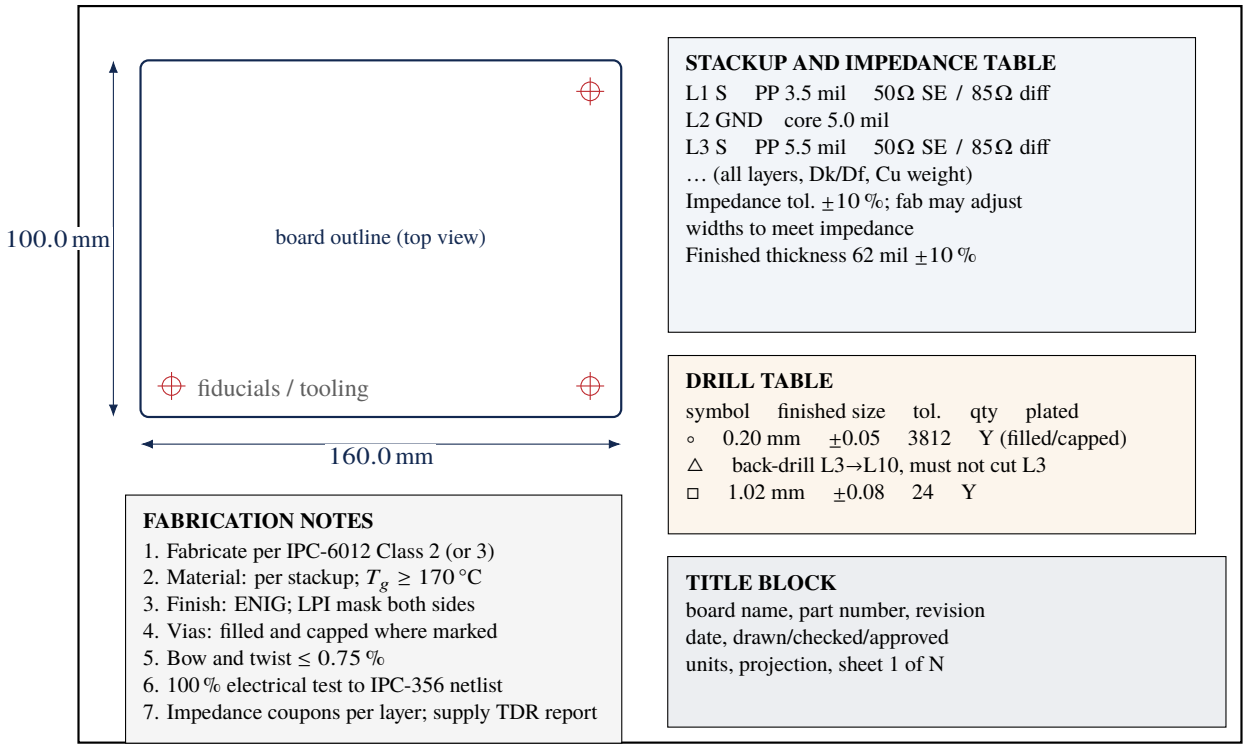


Figure 39.1. Organization of a fabrication drawing: outline and dimensions, stackup and impedance table, drill table, fabrication notes, and title block (schematic).

Design Rule: Let the fabricator tune impedance

Specify impedance targets and tolerances per layer and per structure, list the designed trace widths and spacings as nominal, and allow the fabricator to adjust widths (within stated limits) and dielectric thicknesses to meet the targets with its materials and process. Ask for the fabricator's stackup proposal before routing, and route to it.

39.2 Impedance coupons

Because impedance cannot be measured on the production traces without damaging them, fabricators build *coupons*—test structures on the panel border—with the same layers, widths, and references as the controlled-impedance traces (Figure 39.2). Each coupon is measured by TDR, and the impedance is read from the stable plateau of the trace, away from the launch and the far end. IPC-2141 and the IPC-TM-650 test methods describe coupon design and measurement [45], [85].

Worked Example 39.1: Coupon length

A TDR with 35 ps rise time measures a coupon in stripline at 170 ps/in. If the plateau should be at least 1 ns of one-way delay long after excluding launch effects, how long must the coupon be?

Solution. $1\,\mathrm{ns}/170\,\mathrm{ps/in} = 5.9\,\mathrm{in}$, plus the launch region. This is why coupons are typically 4–6 in or longer, and why they are placed on the panel rather than on the board.

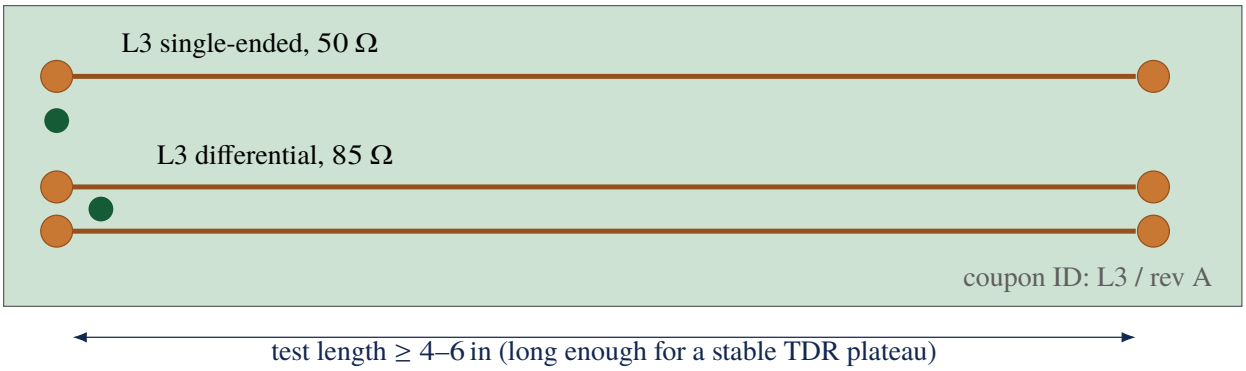


Figure 39.2. An impedance coupon with single-ended and differential test lines, ground launch vias (green) next to the probe pads, and layer identification.

39.2.1 Acceptance windows

A coupon result is meaningful only with its measurement conditions. The fabrication notes or the purchase specification should state: the TDR edge (rise time at the reference plane, or the filtering applied); the evaluation window along the trace, usually expressed as a fraction of the coupon's delay so that launch and far-end effects are excluded (for example, 30–70 % of the trace); whether the limit applies to the average over the window or to every point within it; whether a loss-induced upward slope is corrected; and the acceptance band itself. Two laboratories with different edges and windows can report different impedances for the same coupon (Section 37.1.2), so a disputed result is settled by repeating the measurement under the stated conditions, not by comparing numbers taken differently.

Design Rule: Specify how the coupon is judged

State the TDR rise time, the evaluation window, the statistic (average or all points) and the acceptance band for each controlled-impedance class on the fabrication drawing. Ask for the coupon reports with every lot and keep them with the board's traveler data.

39.3 Tolerances

Worked Example 9.4 showed that dielectric thickness, etched width, and D_k together produce a spread of about ± 10 % in impedance. Two manufacturing effects deserve particular attention:

Etch profile
: Etching undercuts the top of a trace, leaving a trapezoid narrower at the top than at the bottom. The fabricator compensates by adjusting the artwork width, but the effective width differs from the drawn width.

Pressed dielectric thickness
: Prepreg thickness after lamination depends on how much resin flows into the adjacent copper pattern. Sparse copper layers absorb more resin, thinning the dielectric. Balanced copper (with thieving where necessary) keeps thickness predictable.

Worked Example 39.2: Effect of etch on stripline impedance

The stripline of Worked Example 9.2 ($W = 4.2$ mil, $T = 0.6$ mil, $B = 10.6$ mil, $D_k = 3.8$) is etched so that its average width is 0.25 mil less than drawn. Estimate the impedance change.

Solution. From Eq. (9.5), $\partial Z_0/\partial W = -(60/\sqrt{3.8})\times 0.8/(0.8W+T) = -30.78\times 0.8/3.96 = -6.2\ \Omega/\text{mil}$. A 0.25 mil reduction raises Z_0 by about 1.6 Ω, or 3 %—a third of a ±10 % budget from etch alone.

Rule of Thumb: Bow and twist

For boards with surface-mount components, bow and twist is usually limited to 0.75 % of the diagonal (IPC-6012 gives the classes' limits). *Limits:* fine-pitch BGAs and large boards may need tighter values; check the assembler's requirement.

39.4 Package families and assembly

The package family sets the land pattern, the inspection method, and the rework options. Leaded packages (Figures 39.3 and 39.4) can be inspected optically and reworked with simple tools; J-leaded and leadless packages hide their joints and need X-ray inspection or careful process control; fine-pitch leads (TSSOP, TQFP) need accurate stencil design. Choose packages with the assembler's capabilities in mind, and record the inspection method for each in the fabrication and assembly notes.

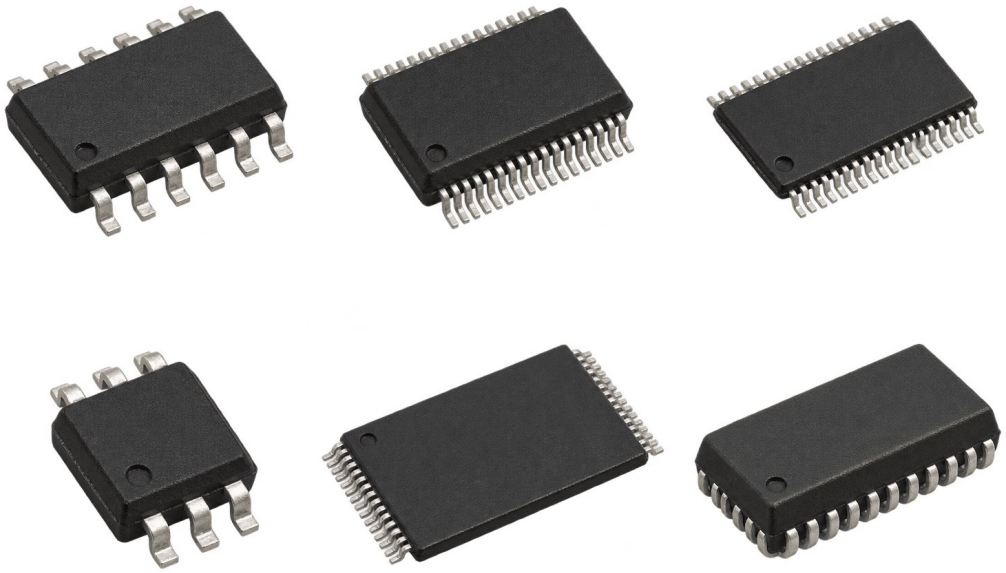


Figure 39.3. Leaded surface-mount packages with two rows of leads: SOIC, SSOP, TSSOP, a six-lead small-outline transistor package, TSOP, and SOJ (J-leads). Rendered illustration supplied with the manuscript, not a photograph of specific products; markings, contact counts, and proportions are not authoritative. [Illustration]

39.5 Design for manufacturing and test

Common Pitfall: Test points on multi-gigabit nets

Adding a through-hole test pad to every net, including PCIe and USB lanes. Each pad and its via is a stub (Chapter 15). On high-speed nets, omit test points, use small pads directly on the trace with no via, or rely on boundary scan and built-in self-test for connectivity.

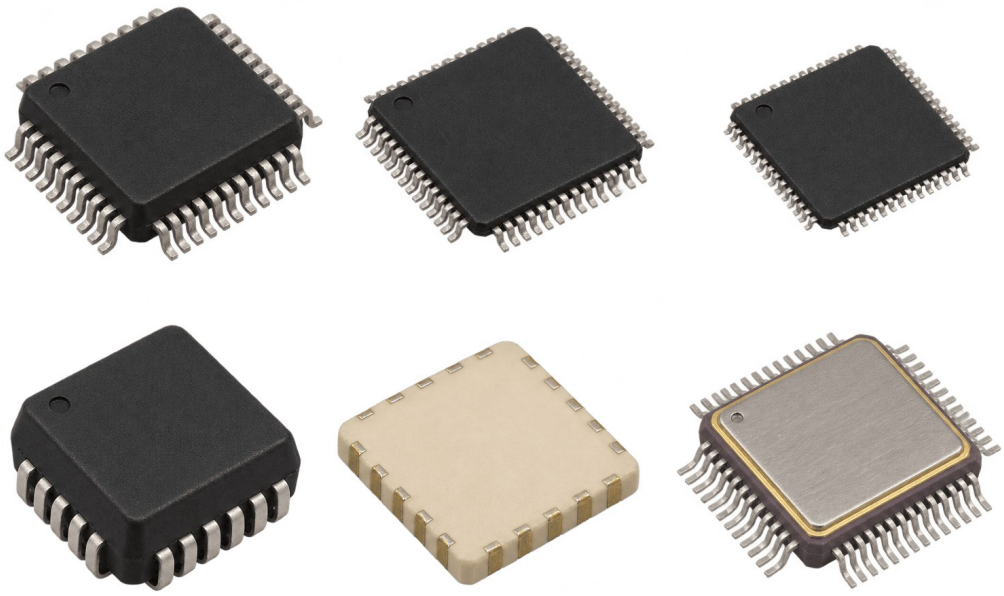


Figure 39.4. Four-sided surface-mount packages: QFP, LQFP, TQFP, PLCC (J-leads), a ceramic leadless chip carrier, and a ceramic QFP with a metal lid. Rendered illustration supplied with the manuscript, not a photograph of specific products; markings, contact counts, and proportions are not authoritative. [Illustration]

Table 39.1. Key DFM and DFT checks.

Check	Purpose
Minimum trace/space per layer	Within fabricator capability at the copper weight
Annular ring	Pad minus finished hole, minus drill wander, ≥ class minimum
Copper balance and thieving	Uniform plating and pressed thickness; reduces warp
Acid traps, slivers	Avoid acute copper angles and narrow mask/copper features
Soldermask dams and clearance	Prevent bridging between fine-pitch pads
Via-in-pad	Filled and planarized, or kept out of pads
Thermal relief	On plane connections of through-hole parts, for solderability
Fiducials and tooling	Global and local fiducials for placement accuracy
Component spacing	Rework access, pick-and-place and inspection clearances
Test points	One per net for ICT or flying probe, where signal integrity allows
JTAG access	Header and chain for boundary scan (Chapter 38)

39.6 The release package

Intelligent formats (ODB++ and IPC-2581) carry the stackup, netlist, drill spans, and component data in one database, reducing the risk of mismatched files; Gerber with X2 attributes adds much of the same information to the traditional format. Whatever format is used, the package should be generated from a single tagged design revision and checked by viewing the output files in an independent viewer before release.

39.6.1 Change control

After release, every change follows an engineering change order (ECO or ECN) that records what changed, why, which documents are affected, and which boards in the field or in production are impacted. Board

Table 39.2. Contents of a PCB release package.

Item	Content
Fabrication data	Gerber (with X2 attributes) [86], ODB++ [87], or IPC-2581 [88]
Drill data	NC drill files per drill span, including back-drill files and depths
Fabrication drawing	Outline, stackup, impedance table, drill table, notes
Netlist	IPC-D-356 for bare-board electrical test
Assembly data	Assembly drawings, pick-and-place (centroid) file, paste layers
Bill of materials	Manufacturer part numbers, approved alternates, reference designators
Stackup document	Materials by layer, thicknesses, D_k/D_f, copper weights, impedance targets
Readme	File list, units, layer order, revision, contacts

revisions are marked in copper on the board itself, and the BOM revision is tracked separately, since many changes are component-only.

Design Rule: Release checklist

Before release: run DRC with the fabricator's rules; confirm stackup and impedance with the fabricator; verify that back-drill and via-fill layers match the drill table; regenerate all outputs from one tagged revision; inspect every output in a viewer; check the BOM against the placement file; and archive the complete package with the simulation reports that justify it. Appendix C gives a full checklist.

Key Takeaways

- The fabrication drawing and notes are the contract: class, material, thickness, finish, vias, back-drill, impedance, and test.
- Specify impedance targets and allow the fabricator to tune widths; verify with TDR on coupons.
- Etch and pressed dielectric thickness dominate impedance tolerance; balance copper.
- Apply DFM and DFT rules, but keep test structures off multi-gigabit nets.
- Release a complete, consistent package from a single revision and control every change with an ECO.

Further Reading

IPC-6012 [82], IPC-4101 [50], IPC-4761 [83], and IPC-2221 [89] govern board performance, materials, via protection, and design; IPC-2581 [88], ODB++ [87], and the Gerber format [86] define the data exchange.

Review Problems

P39.1 What is the bow-and-twist limit at 0.75 % for a 6 in × 4 in board?

P39.2 A microstrip has $\partial Z_0/\partial W = -5.98\,\Omega/\text{mil}$. What impedance change results from 0.3 mil of etch reduction?

P39.3 What range corresponds to 85 Ω ±10 %?

P39.4 What coupon length gives 1 ns of one-way delay at 170 ps/in?

P39.5 A board has a 10 in × 8 in outline. What is its 0.75 % bow-and-twist limit?

P39.6 A via has an 18 mil pad and a 10 mil finished hole, and drill positional tolerance is ± 3 mil. What is the worst-case annular ring?

Part VIII

Case Studies

Chapter 40

Seven Case Studies

Chapter Overview

Each case below starts from a design question that comes up in real projects and works it through with the tools of the preceding chapters. The numbers are analytical estimates or simulations of synthetic models, labelled as such; none of them is a measurement of a product, and none should be quoted as one. What carries over is the reasoning: which quantity to compute first, which assumption dominates, and which measurement would confirm or refute the estimate.

Learning Objectives

1. Budget a PCIe Gen5 channel from its unit interval, Nyquist frequency, and loss allocation.
2. Quantify a length mismatch as skew and mode conversion, and decide where to fix it.
3. Lay out a USB SuperSpeed path through ESD protection, a mux, and a connector.
4. Plan a MIPI flex interconnect, a DDR fly-by bus, and a multi-rail board.
5. Diagnose a link that fails only when a regulator runs or the board heats.

40.1 Case 1: a PCIe Gen5 system-board channel

Question. How long can a PCIe Gen5 route on the system board be, from the root complex to a CEM slot?

Timing and frequency. At 32 GT/s NRZ, the unit interval is $1/32\,\text{GHz} = 31.25\,\text{ps}$ and the Nyquist frequency is 16 GHz (Table 5.2).

Budget. The PCIe Gen5 channel is specified bump to bump with a total insertion-loss budget of about 36 dB at 16 GHz. A published allocation [21] gives the root package 9.0 dB, the CEM connector 1.5 dB, the add-in card 9.5 dB, and the system board 16.0 dB (Figure 40.1). Even an 800 mV launch arrives as about 12.7 mV after 36 dB; the receiver's CTLE and DFE do the rest, which is why a Gen5 budget is not negotiable.

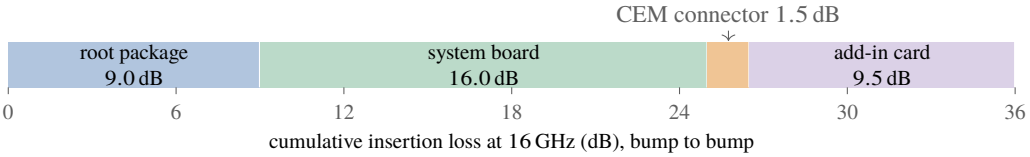


Figure 40.1. A PCIe Gen5 bump-to-bump insertion-loss allocation of 36 dB at 16 GHz, after a public vendor allocation. Other allocations are possible; the total is set by the specification. [Conceptual]

Loss per inch. For the L3 85 Ω pair of Chapter 10 ($W = 5.65$ mil, $Z_{odd} = 42.5\ \Omega$, $D_k = 3.7$), the closed-form estimates of Chapter 11 at 16 GHz give:

- low-loss laminate ($D_f = 0.004$, smooth foil, roughness factor 1.3): 0.78 dB/in conductor + 0.28 dB/in dielectric = 1.06 dB/in;
- mid-loss laminate ($D_f = 0.010$, roughness factor 1.5): 0.90 + 0.71 = 1.61 dB/in.

Reserving 2 dB of the board's 16 dB for the breakout, vias, and AC-coupling capacitors leaves 14 dB for trace: about 13 in on the low-loss laminate and 8.7 in on the mid-loss one. Longer routes need a retimer or a cabled interconnect.

What to verify. Loss per inch at 16 GHz on the fabricator's coupons; via stubs (back-drill, Chapter 15); a channel simulation with the specification's reference receiver; and, on hardware, link training results and lane margining.

40.2 Case 2: a 2.6 mm mismatch on a 5 Gb/s pair

Question. A review finds that one leg of a 5 Gbit/s pair on L1 is 2.6 mm longer than the other. Does it matter, and where should it be fixed?

Skew. At the odd-mode delay of the L1 pair (5.50 ps/mm), 2.6 mm is $\Delta t = 14.3$ ps, about 15 ps: 7 % of the 200 ps UI.

Mode conversion. At the 2.5 GHz Nyquist frequency, $|S_{cd21}| = |\sin(\pi f \Delta t)| = -19.0$ dB, or a common-mode voltage of 5.6 % of the differential voltage (Chapter 14). The differential signal loses the same energy, and the edge acquires a 15 ps step.

Where to fix it. If the mismatch is created at one end (say, at the connector breakout) and the pair runs coupled in microstrip, the modal model of Chapter 16 shows that correcting it at the far end leaves up to 9.5 % distortion of the differential edge, while correcting it next to the breakout removes the problem. Fix it locally, with small bumps on the short leg.

What this case does not establish. Common mode on a cable can increase radiated emissions, but whether *this* mismatch changes the product's EMC result depends on the cable, the connector, the enclosure, and the rest of the system. That is a question for measurement, not for this calculation.

40.3 Case 3: USB 3.2 Gen 2 through ESD, mux, and connector

Question. A Type-C port carries USB 3.2 Gen 2 (10 Gbit/s, Nyquist 5 GHz) through an ESD array and a 2:1 orientation mux. Each adds capacitance to a 90 Ω pair. What does that do, and what can the layout do about it?

Model. A synthetic odd-mode model (45 Ω per leg, 90 Ω differential reference): 20 mm of line, an ESD diode of 0.20 pF per leg, 5 mm of line, a mux modelled as 0.15 pF, 2 Ω, 0.15 pF, 27 mm of line, and a connector modelled as 6 mm of 82 Ω differential line. The "compensated" variant places short high-impedance sections (120 Ω differential: necked traces or a removed reference under the pads) around each capacitive load, sized so their inductance offsets the capacitance ($L \approx CZ_0^2$).

Result. Figure 40.2 shows the effect. At 5 GHz, S_{dd11} improves from −13.4 to −16.3 dB, and at 10 GHz from −10.0 to −13.7 dB; the impedance dip at the mux rises from about 73 to 80 Ω. The insertion loss at 5 GHz changes only from −1.56 to −1.43 dB: on a short path, return loss is the specification that the capacitive parts threaten.

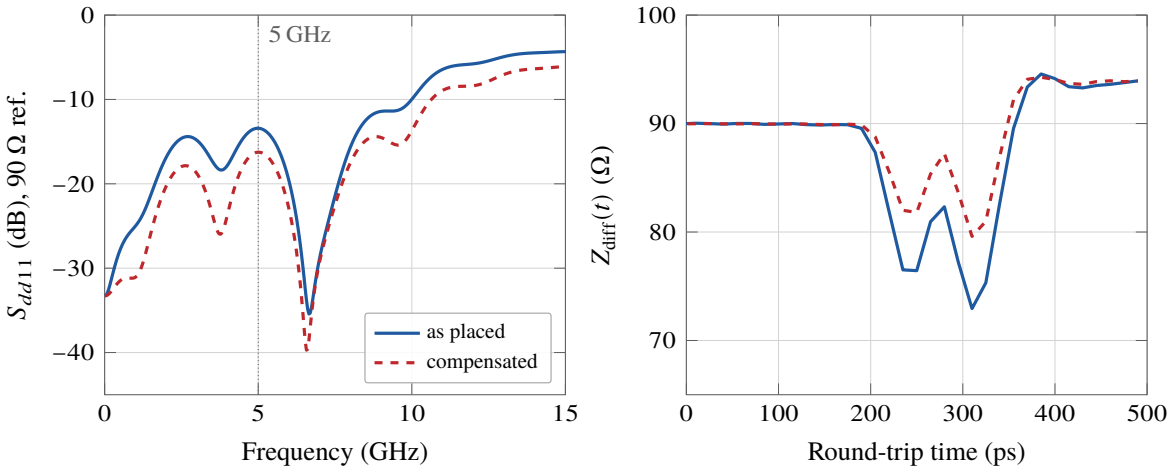


Figure 40.2. Synthetic model of a USB SuperSpeed path through an ESD array, a mux, and a connector, as placed and with compensation of each capacitive load. Left: S_{dd11}. Right: differential impedance profile. [Simulation]

Layout practice. Place the ESD array at the connector, with its pads on the signal path (no stubs); choose low-capacitance parts; remove the reference plane under large pads where the stackup allows (Chapter 15); place the mux close to the connector so that orientation switching happens before the long run; route the USB 2.0 pair separately; and check the result with the connector's S-parameter model.

40.4 Case 4: a MIPI D-PHY camera on a flex cable

Question. A camera module sends CSI-2 over four D-PHY lanes at 2.5 Gbit/s through a 150 mm flex cable. What are the risks?

Loss. At the 1.25 GHz Nyquist frequency, a 3.5 mil, 50 Ω odd-mode flex trace on polyimide ($D_k \approx 3.4$, $D_f \approx 0.005$, roughness factor 1.2) loses about 0.30 dB/in, mostly in the conductor: 1.8 dB over 6 in, about 2.8 dB with two connectors. Loss is not the problem at this rate.

Impedance. Flex stackups are thin; a solid reference may be replaced by a hatched one for flexibility, which raises the impedance and the loss and must be modelled with the fabricator. Keep the clock lane in the middle of the data lanes and ground conductors between lanes.

Orientation. The dominant risk is mechanical: the cable type (contacts on the same or opposite sides), the connectors' contact sides, and the mounting sides together decide whether lane order and polarity survive (Figure 34.5). D-PHY cannot correct a swapped clock-lane polarity. Fix all three in one drawing and check it with the pin-map procedure (Section 16.7).

What to verify. Impedance of the flex on coupons; the cable's actual contact layout; eye or error-rate tests at the highest lane rate the sensor uses.

40.5 Case 5: a DDR4 fly-by command bus

Question. Four DDR4-3200 DRAMs on a fly-by command/address bus are 12 mm apart on L3. What impedance and termination should the bus have, and what timing spread must write levelling absorb?

Loaded line. A 50 Ω L3 line has $C' = t_{\mathrm{pd}}/Z_0 = 0.128\,\mathrm{pF/mm}$, so a 12 mm segment holds 1.54 pF. With about 1.0 pF of device, pad, and via load per DRAM, the loaded segment's delay and impedance are

$$t'_{\mathrm{pd}} = t_{\mathrm{pd}}\sqrt{1 + \frac{C_d}{C_{\mathrm{seg}}}} = 8.24\,\mathrm{ps/mm}, \qquad Z'_0 = \frac{Z_0}{\sqrt{1 + C_d/C_{\mathrm{seg}}}} = 38.9\,\Omega. \tag{40.1}$$

The loaded bus therefore looks like a 39 Ω line; terminating it with $R_{TT} \approx 39\,\Omega$ to V_{TT} at the end matches it better than 50 Ω, and many guides specify a lower unloaded impedance for the fly-by section for the same reason.

Timing. Each segment adds about 99 ps; the clock reaches the last DRAM about 297 ps after the first, almost one 312.5 ps UI. Write levelling measures and compensates that spread for the data strobes. Within a byte, DQ-to-DQS matching comes from the budget: at 5 % of the UI it is 15.6 ps, or 2.4 mm on L3.

V_{TT}. With $R_{\mathrm{on}} = 34\,\Omega$ and $R_{TT} = 40\,\Omega$ each line carries ±8.1 mA (Section 8.3); the regulator must source and sink the worst-case sum.

40.6 Case 6: the multi-rail board

Question. Is the SBC power design of Chapters 24 and 25 ready for layout?

Review. The rail budget shows 30.3 W from 12 V and a 5 V buck that carries 5.7 A, not the 2.1 A of its direct load. The core buck runs from 5 V at 16 % duty instead of 6.7 % from 12 V. The 1.8 V LDO was replaced by a buck (1.2 W versus 0.2 W of loss). The core buck's loop crosses over at about 186 kHz with about 60° of phase margin, and a 3 A step is estimated at about 12 mV. The sequence needs a load switch on the SoC's 3.3 V I/O branch because the 1.8 V rail is derived from the 3.3 V bus.

Open items before layout. Confirm every load figure against the final datasheets; compute each regulator's dissipation and its thermal path (Chapter 26); design the PDN of each high-current rail with the method of Chapter 22; write the bring-up acceptance criteria (Section 25.6); and reserve test points for every rail, enable, power-good, and reset.

40.7 Case 7: a link that fails when a regulator runs or the board heats

Symptom. A 16 GT/s link passes at the bench but logs errors when a nearby high-current regulator is enabled, and again after the enclosure warms up.

Supply-coupled jitter. A switching regulator's ripple and spikes at its switching frequency and harmonics can reach a reference-clock buffer, PLL, or SerDes supply. If a clock device's datasheet gives a supply sensitivity, the periodic jitter follows from the ripple; for an illustrative 0.2 ps/mV and 10 mV of ripple, about 2 ps of periodic jitter appears at the switching frequency. The signature is a jitter spur, or an error rate that changes when the regulator's frequency or load changes.

Temperature. Copper resistivity rises by about 0.39 % per °C, so from 25 to 85 °C the resistivity rises 24 % and the skin-effect conductor loss about 11 %. Many epoxy laminates also lose more at higher temperature and humidity. If a 20 dB channel's loss is 55 % conductor and 45 % dielectric, and the dissipation factor rises 20 %, the loss grows to about 23 dB: 3 dB of margin gone with no change in the design.

Diagnosis. Figure 40.3 gives the order of experiments. The first question separates the two mechanisms; each branch then localizes the cause and tests a fix. Measure the channel's loss over temperature on a coupon of the same stackup, re-run link training at temperature, and compare the receiver's margining results before and after.

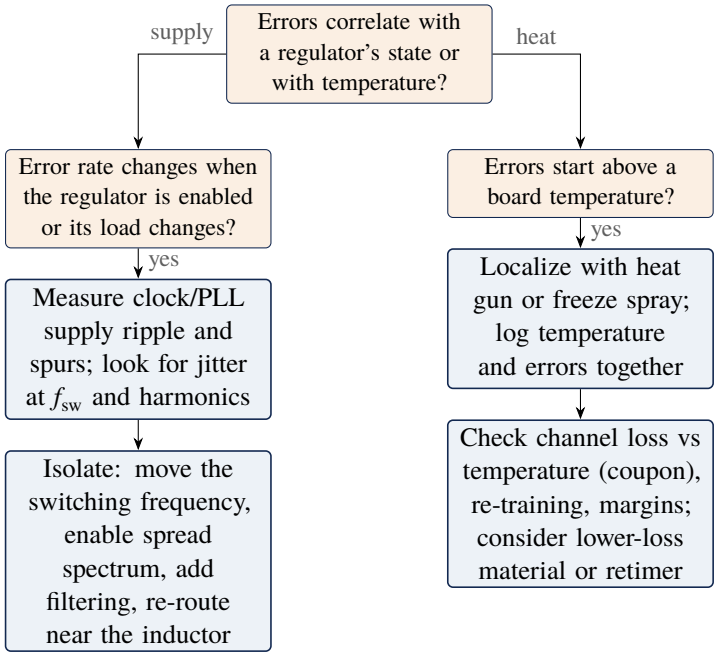


Figure 40.3. Troubleshooting tree for a link that fails when a regulator runs or the board heats. [Conceptual]

Key Takeaways

- Start every channel question from the UI, the Nyquist frequency, and the loss budget.
- Quantify skew as both time and mode conversion; fix it where it is created.
- Capacitive parts on a short path threaten return loss more than insertion loss; compensate them in the layout.
- On flex and connectors, orientation errors are more likely than loss problems.
- Fly-by buses are loaded lines; terminate to the loaded impedance and let training absorb the spread.

- Failures that depend on a regulator or temperature have physical signatures: jitter spurs, or loss that grows with temperature.

Review Problems

P40.1 Repeat Case 1 for a board budget of 13.5 dB (Gen4 allocation) at 8 GHz on the mid-loss laminate, reserving 1.5 dB.

P40.2 In Case 2, what mismatch would keep $|S_{cd21}|$ below -30 dB at 2.5 GHz?

P40.3 (Inverse design) In Case 3, what series inductance offsets a 0.2 pF shunt on a 45 Ω line, and how long is a 60 Ω section that provides half of it (5.78 ps/mm)?

P40.4 In Case 5, how does the loaded impedance change if the DRAMs are 8 mm apart?

P40.5 (Approximation failure) In Case 7 a technician sprays freeze spray on the regulator and the errors stop. Does that prove a thermal cause?

Chapter 41

Case Study: A Modular ARM and FPGA Embedded Platform

Chapter Overview

This chapter follows one design from requirements to bring-up: a heterogeneous embedded platform built from three boards. An ARM system-on-module (SOM) based on the Rockchip RK3588 plugs into a SO-DIMM socket; an FPGA SOM based on the Xilinx Zynq-7020 plugs into an M.2 socket; and a carrier board holds both sockets, the external ports, the power input, and the links between the two modules. The chapter applies the methods of the preceding parts (rail budgets, sequencing, impedance control, connector and pin-map review, thermal design) to this system. All three boards were designed in Altium Designer; the design has been rendered but not implemented, so the board figures are renderings of the design's 3D models, not photographs of manufactured hardware, and every test remains to be done once the boards are built (Section 41.1). Anything the design description does not fix is either stated as an **assumption** or marked **[TO BE CONFIRMED]**; all such items are listed in Section 41.11. No measurements, test results, performance figures, or costs are reported, because none exist for this design.

Learning Objectives

1. Explain why an application processor and an FPGA are combined, and why the design is split into modules on a carrier.
2. Derive an architecture and an inter-module interface plan from requirements.
3. Identify the design decisions each board needs and the risks of using standard card-edge connectors with custom pinouts.
4. Plan the power tree, sequencing, stack-up, layout, review, and bring-up of a multi-board system.
5. Carry loss, timing, and power-distribution budgets through the design with a requirements ledger, and see how card-edge mechanics constrain the module stack-ups.

41.1 Introduction

Why combine a processor and an FPGA? An application processor such as the RK3588 runs a full operating system, network stacks, graphics, video codecs, and neural-network inference efficiently, but its I/O is fixed by its silicon and its timing is subject to the operating system's scheduling. An FPGA offers the opposite: logic and I/O that can be defined for a specific sensor or protocol, with cycle-accurate, deterministic timing and parallel processing, but with far less capability for general software. Pairing the two lets each do what it does well: the processor handles applications, user interfaces, storage, and networking; the FPGA handles custom interfaces, hard real-time control, and pre-processing of data before it reaches software.

Why modules on a carrier? Both processors need dense, fine-pitch packages, high-speed memory, many supply rails, and layout rules (for example LPDDR4X routing) that call for a high layer count. Placing each on a small module confines that complexity to a small, expensive board, and lets the large carrier use a simpler, cheaper stack-up. Modules can also be replaced independently: a different ARM module or a larger FPGA can be fitted to the same carrier if the module pinouts are kept, and one validated module can serve several carriers. The price is two connectors in every inter-module path, a mechanical and electrical interface that must be specified precisely, and power and sequencing that cross board boundaries.

Design status. All three boards, the ARM SOM, the FPGA SOM, and the carrier, were designed in Altium Designer. The design exists as schematics, board layouts, and 3D board models; it has been *rendered*, not *implemented*. No board has been fabricated or assembled, and so no board has been powered, measured, or tested.

How the figures were made. The 3D models of the three boards, with their component bodies placed as in the layout, were exported from Altium Designer and rendered in KeyShot, a 3D rendering application, to produce Figures 41.2 to 41.5. A rendering shows the board outline, the placement, and the mechanical arrangement of the design; it does not show that the board can be built, that it works, or that it meets its requirements. Small details such as part markings come from the component models and textures and are not authoritative; where they disagree with the design description, this chapter says so and lists the item in Table 41.7.

What remains. Every verification step, from fabrication checks to interface compliance, thermal, and EMC tests, remains to be carried out after implementation. Section 41.8.11 lists these tests; until they are done, no performance claim can be made for the platform.

41.2 Requirements and design goals

The design description fixes the following requirements; the goals that follow from them are this chapter's interpretation.

- **Processing:** an RK3588 application processor with LPDDR4X memory, eMMC storage, a PMIC, and a Wi-Fi/Bluetooth module with two u.FL antenna connectors; a Zynq-7020 with DDR3L memory, QSPI flash, a clock oscillator, switching regulators, and status LEDs.
- **Modularity:** the ARM SOM in SO-DIMM form factor (about 70×50 mm, lying flat); the FPGA SOM in M.2 form factor (about 30×60 mm, held by one screw).
- **External I/O:** two Gigabit Ethernet ports (RJ45), HDMI output, HDMI input, two USB 3.0 Type-A ports, one USB-C port, a 12 V DC input, a microSD slot, and two SMA antenna connectors.
- **On-board I/O:** a 40-pin GPIO header, two Pmod headers, a JTAG header, two flat-flex camera/display connectors, a fan header, a coin-cell holder, power and reset buttons, and status LEDs.
- **Inter-module links** (assumed): Gigabit Ethernet, several LVDS pairs, SPI, UART, I^2C, and GPIO/interrupt lines. The Zynq-7020 has no multi-gigabit transceivers, so there is no PCIe link to it.

Design goals (interpretation). A single 12 V input powering all three boards; modules that cannot be damaged by insertion into the wrong socket or by power applied in the wrong order; inter-module links that meet their timing through two connectors; controlled impedance for every high-speed external port; a thermal

design that keeps both processors within their ratings in the intended enclosure [TO BE CONFIRMED]; and test access for bring-up.

41.3 System architecture

Figure 41.1 shows the architecture. The two SOMs sit above the carrier and connect to it through their card edges; every link between them passes through both connectors and across the carrier. The carrier also owns all external ports, the power input, and the conversion from 12 V to the rails delivered to the modules.

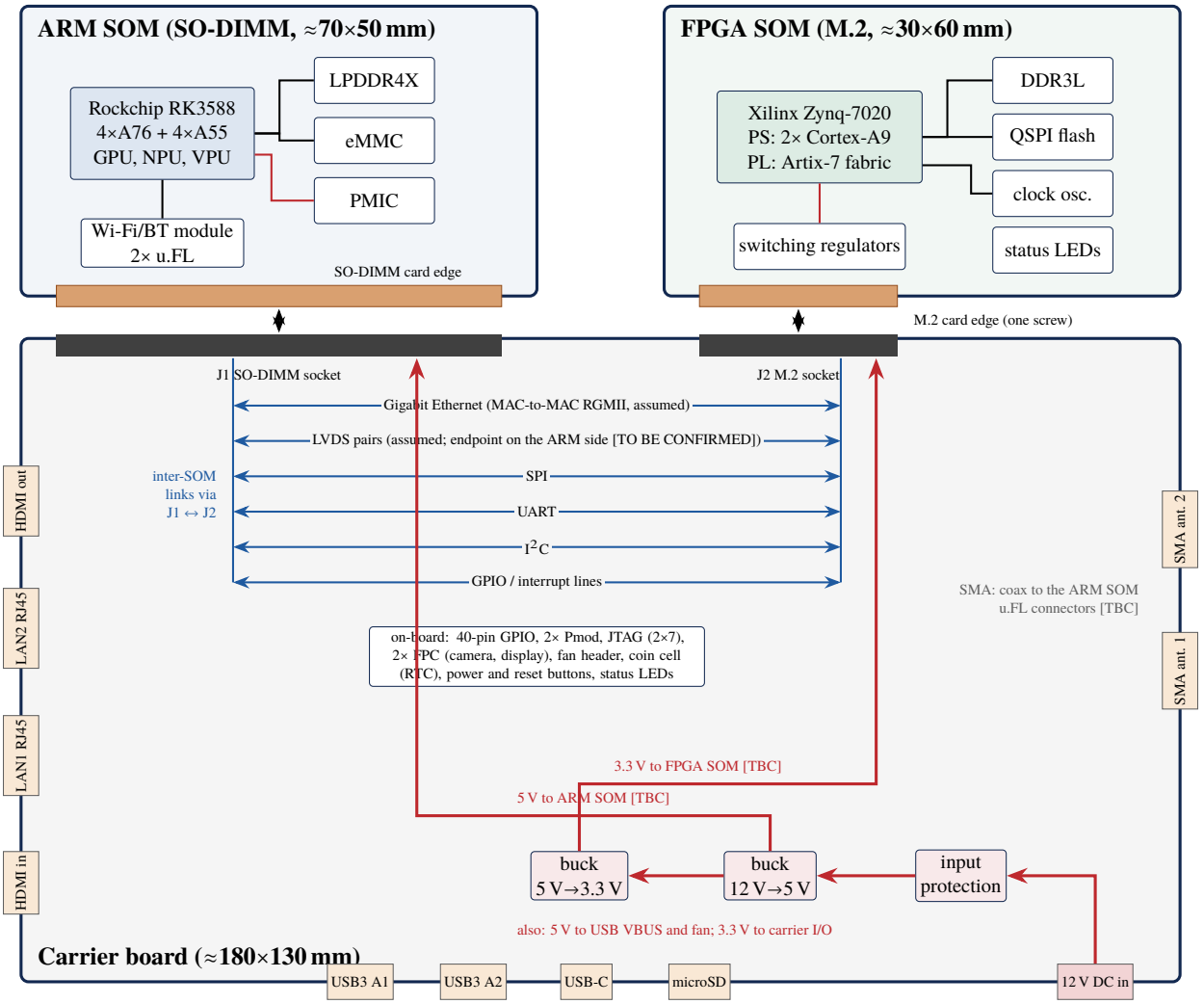


Figure 41.1. System block diagram. The ARM SOM and the FPGA SOM connect to the carrier through the SO-DIMM (J1) and M.2 (J2) sockets; the inter-SOM links (blue) pass through both sockets; the external ports sit on the carrier's edge; the power path (red) runs from the 12 V input through protection and converters to both modules. Inter-SOM links and module supply voltages are assumptions. [Conceptual]

Which device serves which port. The design description does not assign the external ports to a processor. Based on the public characteristics of the two devices, a natural assignment is (all **assumed**):

- RK3588: HDMI output and HDMI input (the RK3588 has HDMI transmitters and an HDMI receiver [90], [91]), the USB 3.0 Type-A ports and the USB-C port, the microSD slot, the camera and

display connectors (MIPI CSI and DSI), the 40-pin GPIO header, and one RJ45 port through an Ethernet PHY on the carrier.

- Zynq-7020: the two Pmod headers, the JTAG header, and the second RJ45 port through a second carrier PHY.

The second assumption matters because the RK3588 provides two RGMII Ethernet MACs and the Zynq-7020 processing system provides two Gigabit Ethernet MACs [90], [92]. If both RJ45 ports belonged to the RK3588, it would have no MAC left for the inter-module Ethernet link. Using one MAC on each side for the external ports and one on each side for a direct MAC-to-MAC link resolves this [TO BE CONFIRMED].

Table 41.1. Key specifications of the three boards. Values not fixed by the design description are placeholders.

Item	ARM SOM	FPGA SOM	Carrier
Main device	Rockchip RK3588: 4× Cortex-A76 and 4× Cortex-A55, Mali-G610 MP4 GPU, 6-TOPS NPU [90], [91]	Xilinx Zynq-7020: dual Cortex-A9 PS, Artix-7-class PL with 85K logic cells, 53,200 LUTs, 220 DSP slices, 4.9 Mb block RAM [92]	—
Memory	LPDDR4X, capacity [TO BE CONFIRMED]	DDR3L, width and capacity [TO BE CONFIRMED]	—
Storage	eMMC, capacity [TBC]	QSPI flash, capacity [TBC]	microSD slot
Power	on-module PMIC; input [TBC]	on-module switching regulators; input [TBC]	12 V DC input; carrier converters
Wireless	Wi-Fi/Bluetooth module, 2× u.FL	—	2× SMA (cabled to u.FL, [TBC])
Form factor	SO-DIMM, ≈70×50 mm, lies flat	M.2, ≈30×60 mm, one screw	≈180×130 mm
Connector	SO-DIMM card edge; type [TBC]	M.2 card edge; key [TBC]	J1 SO-DIMM socket, J2 M.2 socket
Indicators	—	status LEDs	status LEDs; power and reset buttons
External ports	—	—	2× GbE RJ45, HDMI out, HDMI in, 2× USB 3.0 A, USB-C, 12 V DC, microSD, 2× SMA
On-board I/O	—	—	40-pin GPIO, 2× Pmod, JTAG, 2× FPC, fan header, coin cell

41.4 ARM SOM design

Figure 41.2 shows the ARM SOM: the RK3588 at the centre, two LPDDR4X packages to its right, the eMMC at the top left, the PMIC with its inductors and two smaller step-down converters on the left and lower edge, a crystal, and the Wi-Fi/Bluetooth module with its two u.FL connectors in the top right corner. Two mounting holes at the top corners take the screws that hold the module flat, and the notched card edge along the bottom carries every signal and supply to the carrier.

Processor and memory. The RK3588 combines four Cortex-A76 and four Cortex-A55 cores with a Mali-G610 GPU and a 6-TOPS NPU, and supports LPDDR4, LPDDR4X, and LPDDR5 memory [90], [91]. The

Table 41.2. Inter-SOM interfaces. Every row is an **assumption** of the design description; electrical details are proposals to be confirmed.

Link	ARM side (RK3588)	FPGA side (Zynq-7020)	Signalling	Purpose
Gigabit Ethernet	RGMII MAC	PS Gigabit Ethernet MAC (MIO, or EMIO through PL logic)	RGMII, MAC-to-MAC without PHYs; one side adds the clock delay [TBC]	bulk data, network-style software interface
LVDS pairs	endpoint [TO BE CONFIRMED]; RK3588 general-purpose I/O is single-ended	PL HR banks, LVDS_25	100 Ω differential; count and direction [TBC]	high-rate streaming
SPI	SPI controller	PL logic or PS SPI	single-ended, clock rate [TBC]	register access, control
UART	UART	PS UART or PL	single-ended	console, debug, simple messages
I^2C	I^2C controller	PS I^2C or PL	open drain with pull-ups	management, module identification [TBC]
GPIO / interrupts	GPIO	PS MIO or PL	single-ended	events, resets, handshakes

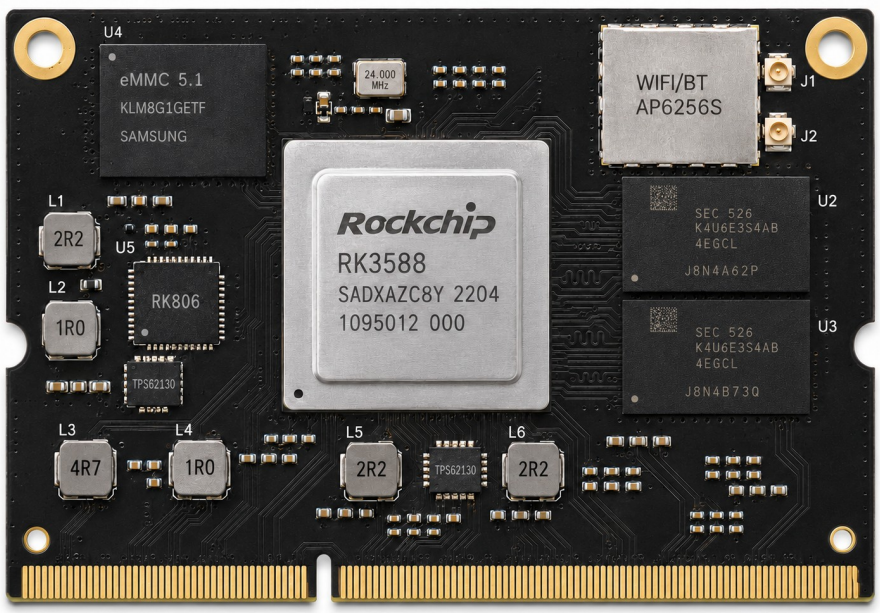


Figure 41.2. ARM SOM in SO-DIMM form factor, top view: RK3588 (U1), two LPDDR4X packages (U2, U3), eMMC (U4), PMIC (U5) with its inductors, two auxiliary step-down converters, crystal, Wi-Fi/Bluetooth module with two u.FL connectors (J1, J2), and the SO-DIMM card edge. Rendering of the board's Altium Designer 3D model, made in KeyShot; the board has not been manufactured, and this is not a photograph of hardware; part markings are illustrative and do not identify selected parts. [Illustration]

design uses LPDDR4X; the capacity, channel configuration, and memory parts are [TO BE CONFIRMED]. LPDDR4X routing (point-to-point, high data rate, tight length and impedance control) is the reason for the module's high layer count and fine-pitch via technology; follow the SoC vendor's hardware design guide for the memory layout, as Chapter 28 describes.

Power. The on-module PMIC generates the SoC and memory rails from the module input. The module's input voltage range and maximum current are [TO BE CONFIRMED]; this chapter assumes a single 5 V input from the carrier (Section 41.8.5). The PMIC also sequences the SoC rails, so the carrier only needs to supply the input and observe a module power-good signal [TBC].

Wireless. The Wi-Fi/Bluetooth module connects to the carrier's SMA connectors through coaxial cables to its u.FL connectors (assumed). The module's regulatory certification and the antenna types determine whether the product can use the module's existing approvals [TO BE CONFIRMED].

The SO-DIMM interface. SO-DIMM sockets are inexpensive, latch the module flat, and offer a few hundred contacts at fine pitch. The SOM uses its own pinout, not a memory pinout. A memory module inserted into the carrier's J1, or the SOM inserted into a computer's memory socket, would connect power to signal pins. The socket type (contact count and key position) and protective measures (a key position not used by standard memory, labelling, or a power-up check of a module identification line) are [TO BE CONFIRMED].

41.5 FPGA SOM design

Figure 41.3 shows the FPGA SOM: the Zynq-7020 (U1) in the centre, a DDR3L package (U2), a QSPI flash in an SOIC-8 package (U3), a clock oscillator (Y1), a switching regulator (U4) with its two inductors (L1, L2), the PWR and DONE status LEDs, the keyed M.2 card edge on the left, and the half-moon screw notch on the right.



Figure 41.3. FPGA SOM in M.2 form factor, top view: Zynq-7020 (U1), DDR3L (U2), QSPI flash (U3), clock oscillator (Y1), switching regulator (U4) with inductors L1 and L2, PWR and DONE LEDs, keyed card edge (left), and screw notch (right). Rendering of the board's Altium Designer 3D model, made in KeyShot; the board has not been manufactured, and this is not a photograph of hardware; part markings, including the package code and the oscillator frequency, are illustrative and do not identify selected parts. [Illustration]

Device. The Zynq-7020 combines a dual Cortex-A9 processing system (PS) with programmable logic (PL) of 85K logic cells, 53,200 LUTs, 106,400 flip-flops, 140 block RAMs (4.9 Mb), and 220 DSP slices; the PS supports DDR3, DDR3L, DDR2, and LPDDR2 memory [92]. Unlike the larger family members, it has no

multi-gigabit transceivers and no PCI Express block [92], which is why the inter-module links are parallel, LVDS, and serial low-speed interfaces. The package and speed grade are [TO BE CONFIRMED].

Memory and configuration. DDR3L connects to the PS memory controller; its width (16 or 32 bits) and capacity are [TO BE CONFIRMED]. The QSPI flash holds the boot image and the PL bitstream. The DONE LED indicates that the PL is configured.

Clock. The PS needs a reference clock. Published Zynq-7000 module designs commonly use 33.333 MHz for the PS clock and a separate 25 MHz oscillator for an Ethernet PHY [93]. The rendering marks its oscillator 25.000 MHz; since this SOM carries no Ethernet PHY, the oscillator's frequency must be checked against the PS clock requirements in the technical reference manual [94] [TO BE CONFIRMED].

The M.2 interface. M.2 gives a thin, screw-retained module with 67 usable contacts (75 positions minus a key). The standard M.2 pinouts carry PCIe, SATA, or USB, none of which this module provides, so the FPGA SOM uses a custom pinout. Two consequences follow:

- the key must stop standard cards from entering J2 and the FPGA SOM from entering a standard socket. The carrier rendering labels J2 "M.2 KEY M", which is the key of NVMe SSDs; a key position intended for non-standard use (key G is commonly described as reserved for custom applications; see Figure 34.6) would be safer [TO BE CONFIRMED];
- the module outline (≈30×60 mm) should match one of the standard M.2 sizes so that standard sockets and standoffs fit, or the standoff position must be specified on the carrier [TO BE CONFIRMED].

M.2 contacts carry limited current, so the module input should be at the highest convenient voltage for its regulators (assumed 3.3 V), with several contacts in parallel for power and ground [TO BE CONFIRMED].

41.6 Carrier board design

Figure 41.4 shows the empty carrier. The external connectors line the top edge: the Ethernet jack position labelled LAN1/LAN2, HDMI out and in, two USB 3.0 Type-A ports, USB-C, and the 12 V DC jack. The microSD slot is on the left edge. The SO-DIMM socket J1, with its latches and two threaded standoffs, and the M.2 socket J2, with the outline of the FPGA SOM and its standoff, occupy the centre. The two flat-flex connectors (camera and display) are on the right edge; the 40-pin GPIO, two Pmod, and JTAG headers run along the bottom edge; the power and reset buttons, the six status LEDs (STAT, PWR, ACT, LAN1, LAN2, FPGA), the fan header, and the coin cell sit between them. The design calls for two Gigabit Ethernet ports; whether they are provided by a dual-port jack at the LAN1/LAN2 position is [TO BE CONFIRMED].

Sockets. J1 holds the ARM SOM flat under two latches, with two screws into the standoffs at its far corners; J2 holds the FPGA SOM with a standoff and screw at its far end. The rendering marks J1 with "260", the contact count of a DDR4 SO-DIMM socket, which makes the keying question of Section 41.4 concrete. The two sockets are close together so that the inter-module links are short; this also concentrates the heat of both modules in one area (Section 41.8.12).

Ports. Each external port has its own requirements from Part VI and Chapters 34 and 35:

- **Ethernet:** each RJ45 needs a PHY with its magnetics (inside the jack or separate), 100 Ω MDI pairs, and an RGMII connection to its MAC on one of the modules.

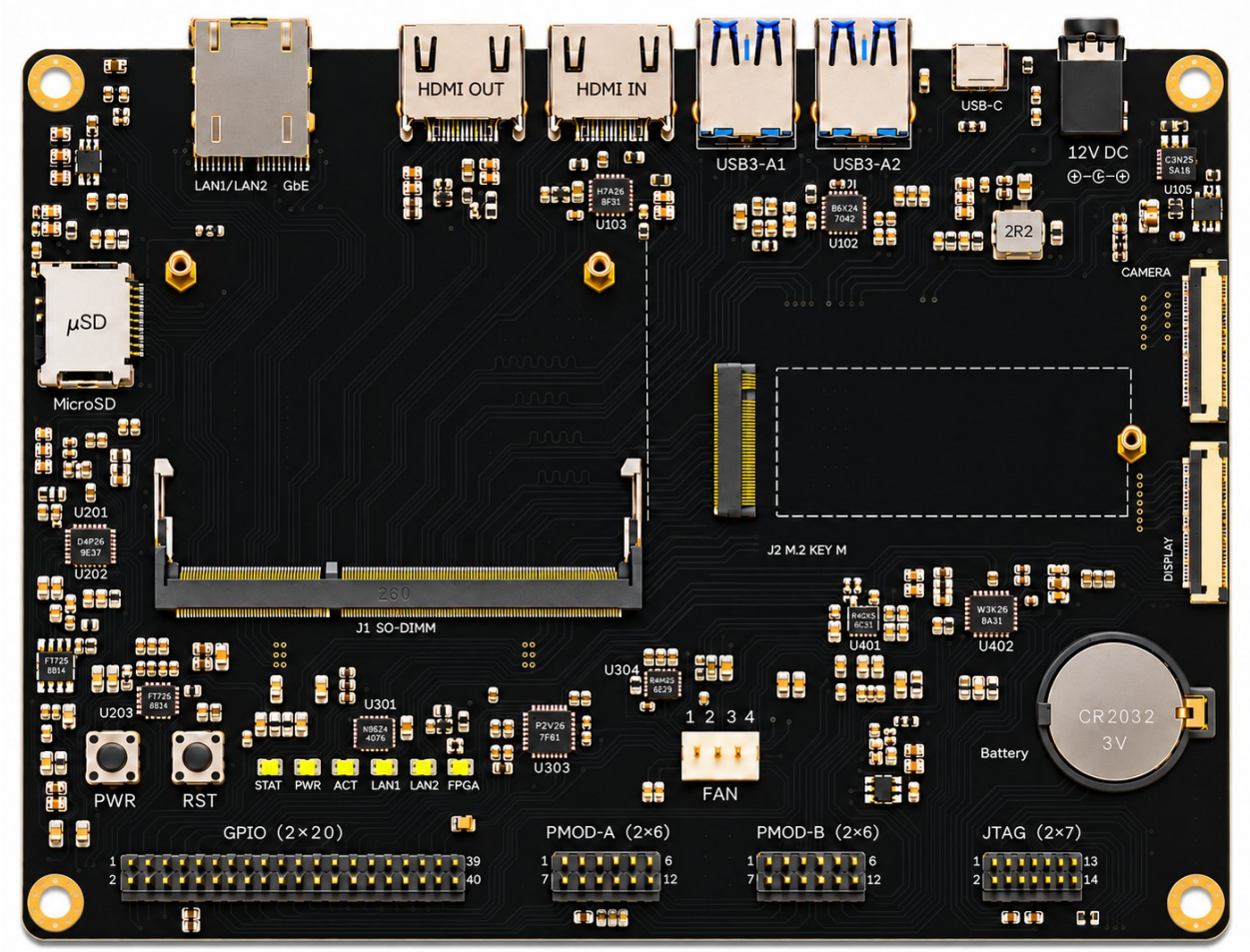


Figure 41.4. Carrier board, top view, without modules: external ports on the top and left edges, J1 SO-DIMM socket with standoffs, J2 M.2 socket with the FPGA SOM outline, camera and display FPC connectors, GPIO, Pmod and JTAG headers, fan header, coin cell, buttons, and status LEDs. Rendering of the board's Altium Designer 3D model, made in KeyShot; the board has not been manufactured, and this is not a photograph of hardware; part markings and small labels are illustrative. The rendering does not show the two SMA antenna connectors, whose position is [TO BE CONFIRMED]. [Illustration]

- **HDMI out and in:** TMDS pairs at 100 Ω differential, ESD protection, the DDC and HPD lines, and the 5 V supply pin (sourced on the output, sensed on the input); TMDS is DC-coupled (Section 35.2).
- **USB 3.0 Type-A:** SuperSpeed pairs at 90 Ω, AC-coupling capacitors on the transmit pairs, low-capacitance ESD protection, and a current-limited VBUS switch per port (Section 40.3).
- **USB-C:** a CC network on each CC contact and a defined role (Section 34.2); whether it is a data, recovery, or power port is [TO BE CONFIRMED].
- **microSD:** 50 Ω single-ended routing, card-detect, and the card's I/O voltage switching if UHS modes are used [TBC].
- **Headers:** the 40-pin GPIO, Pmod, and JTAG headers expose module I/O directly; each needs series resistors or buffers where its signals could be back-driven by external hardware while a module is unpowered (Section 24.5). A 2×7 JTAG header matches the common Xilinx cable pinout [TO BE CONFIRMED].

Inter-SOM links. The carrier routes each link of Table 41.2 between J1 and J2:

- **RGMII.** Two groups (transmit and receive) of a clock, four data lines, and a control line, at 125 MHz double data rate. Without PHYs, one side must provide the 2 ns-class clock-to-data delay that RGMII expects, either in the MAC or the PL logic or by PCB delay; the choice is [TO BE CONFIRMED]. Route each group as 50 Ω single-ended lines matched within a budget derived from the RGMII timing (Section 16.5). Both ends must use the same I/O voltage; a level translator in an RGMII path adds delay and should be avoided.
- **LVDS.** 100 Ω differential pairs with 100 Ω termination at each receiver (often the FPGA's internal termination). On the Zynq side, LVDS uses the HR I/O banks, whose supply voltage must suit the LVDS standard chosen; every other signal in such a bank must use the same bank voltage [TO BE CONFIRMED]. On the RK3588 side the endpoint is open: the RK3588 does not, as far as its public summaries show, offer general-purpose LVDS I/O, so these pairs would either end at a MIPI CSI receiver driven by a D-PHY-compatible circuit in the FPGA, or serve FPGA-to-carrier connections (for example the FPC connectors) instead [TO BE CONFIRMED].
- **SPI, UART, I^2C, GPIO.** Short single-ended nets at a common I/O voltage, with series resistors at the drivers where edges are fast relative to the length (Chapter 8); I^2C pull-ups sized for the bus capacitance.

Power distribution. See Section 41.8.5.

41.7 Design budgets and worked analyses

This section carries the budget methods of the book through the platform: the rates and impedances of its interfaces, a requirements ledger, the floorplans, the stack-ups, and the loss, timing, power-distribution, and EMC budgets. Data rates, rail voltages and currents, and route lengths that the design description does not fix are **illustrative** values, marked as such; on the real design they come from the RK3588 and Zynq-7000 hardware design guides and from the Altium layout, which take precedence over every estimate here. The calculations show the method, not results for the platform, and every verification named in this section is planned, not done (Section 41.1).

41.7.1 Interface rates and impedances

Table 41.3 collects the interfaces. Without PCI Express in the base design, the highest Nyquist frequencies are those of USB 3.0 (2.5 GHz) and of the LPDDR4X data lines (2.13 GHz at the illustrative rate), and HDMI's depends on the version supported [TBC]. Channel models for these paths, including the SO-DIMM contacts, must be valid to at least three times the highest Nyquist frequency, that is 7.5 GHz for USB 3.0 (Chapter 2). Every external high-speed path except the memories crosses the SO-DIMM socket J1, so the socket's insertion and return loss at these frequencies must be obtained from its manufacturer [TO BE CONFIRMED].

41.7.2 Requirements ledger

Each requirement is carried through the design in a *ledger* (Table 41.4) that records where it came from, the budget derived from it, how it will be verified, and its status. A row is closed only when its verification has been done on real hardware, not when a simulation passes. Because the platform has been designed and rendered but not built, every row is open; the verifications are the tests of Table 41.6.

Table 41.3. Rates, unit intervals, Nyquist frequencies, and impedance targets of the platform's interfaces. Rates marked (illus.) are illustrative; the others follow from the interface standards.

Interface	Rate	UI	f_{Nyq}	Impedance	Where
LPDDR4X	4266 MT/s (illus.)	234 ps	2.13 GHz	40 Ω SE / 80 Ω diff	ARM SOM only, point-to-point
DDR3L	per speed grade [92] [TBC]	—	—	per UG933 [95]	FPGA SOM only
USB 3.0 (5 Gb/s)	5 Gb/s	200 ps	2.5 GHz	90 Ω diff	ARM SOM, J1, carrier, Type-A and USB-C
USB 2.0	480 Mb/s	2.08 ns	240 MHz	90 Ω diff	Same path; short orientation stubs at USB-C
MIPI CSI-2 (D-PHY)	2.5 Gb/s (illus.)	400 ps	1.25 GHz	100 Ω diff	ARM SOM, J1, carrier, camera FPC
HDMI TMDS	per HDMI version [TBC]	—	—	100 Ω diff	ARM SOM, J1, carrier, HDMI out and in
RGMII	250 Mb/s per line	4 ns	125 MHz	50 Ω SE	Inter-SOM link and carrier PHYs
1000BASE-T	125 MBd	8 ns	62.5 MHz	100 Ω diff	Carrier PHYs to RJ45
LVDS	[TBC]	—	—	100 Ω diff	Inter-SOM, through J1 and J2

Table 41.4. Requirements ledger for the platform (excerpt; budget values illustrative). No row is closed, because no board has been built.

Requirement	Source	Budget or constraint	Verification	Status
LPDDR4X DQ–DQS skew (ARM SOM)	RK3588 design guide [TBC]	3 % UI static (Worked Example 41.2)	Training report; 2-D eye scan per bit	Open
SoC core rail ±3 % (ARM SOM)	RK3588 data sheet [TBC]	3.2 mΩ dynamic target, illustrative (Worked Example 41.3)	PDN sweep and load step	Open
USB 3.0 channel loss	USB 3.x spec. and socket data	Board share at 2.5 GHz (Worked Example 41.1)	Post-layout extraction; TDR/VNA of one carrier	Open
Inter-SOM RGMII timing	RGMII spec. and MAC data	Clock-to-data delay and matching (Section 41.7.6)	Timing analysis; link test	Open
Controlled impedance	Stack-up plan	50 Ω SE; 90 Ω and 100 Ω diff, ±10 %	Coupons with stated edge and window (Section 39.2.1)	Open
Radiated emissions	Regulatory target [TBC]	Margin target [TBC]	Pre-scan with cables attached	Open

41.7.3 Floorplans

The renderings show floorplans that follow the principles of the earlier chapters:

- **ARM SOM** (Figure 41.2): the two LPDDR4X packages sit immediately beside the RK3588, so that the byte lanes are short and can be matched without serpentines (Chapter 28); the PMIC and its inductors are as close to the SoC as the module area allows, to keep the IR drop and the inductance of the high-current paths low (Chapter 21); and the radio module with its u.FL connectors is in the corner farthest from the card edge.
- **FPGA SOM** (Figure 41.3): the DDR3L package is immediately beside the Zynq-7020; the QSPI flash and the oscillator sit near the card edge, and the regulator with its inductors at the far end, near the screw notch.
- **Carrier** (Figure 41.4): the fast external connectors share the top edge, so their ESD parts and AC-coupling capacitors can sit between connector and socket with short routes (Chapter 30); the Ethernet jack occupies the top-left corner, where a chassis zone with an isolation gap can be formed (Chapter 31); and the two sockets are close together so that the inter-module links are short.

41.7.4 Stack-ups and impedance plan

Module thickness is fixed by the card edge. An M.2 card is 0.8 mm thick (±10 %) [96], and SO-DIMM modules are about 1.2 mm thick at the connector [97]; the exact values for the chosen sockets are [TO BE CONFIRMED]. Neither module can therefore use the 62.6 mil (1.59 mm) ten-layer stack-up of Figure 9.3. Fitting the layers the RK3588 and its LPDDR4X need into about 1.2 mm calls for thin cores and prepregs and, typically, microvias. Thinner dielectrics lower the impedance of a given trace (a smaller H in Section 10.5), so the module traces for each impedance class are narrower than on the carrier, and the fabricator's achievable minimum width and its tolerance must be checked before the layer count is fixed [TO BE CONFIRMED].

Carrier. The carrier has no memory interface and fewer dense parts, so its layer count is set by the number of high-speed routing layers and the need for every one to have a solid reference. One candidate is the ten-layer, 62.6 mil stack-up of Figure 9.3, used as follows (an **assumption**; a six- or eight-layer build may suffice, and the choice is [TO BE CONFIRMED]):

- L1 and L10 (microstrip): connector and socket breakouts, short escapes, and components;
- L3 and L8 (stripline, ground on both sides): USB SuperSpeed, HDMI, MIPI, LVDS, and RGMII;
- L5 and L6: the 5 V and 3.3 V distribution, each coupled to ground through a 3 mil core;
- L2, L4, L7, L9: ground.

A mid-loss laminate ($D_k \approx 3.7$, $D_f \approx 0.010$ at a few gigahertz) with HVLP foil on L3 and L8 is **assumed** in the examples below. The fabricator is asked to confirm widths for 50 Ω single-ended and 90 Ω and 100 Ω differential on each signal layer (and 85 Ω if the PCI Express extension of Section 41.10 is adopted), with ±10 % tolerance and permission to adjust widths. Starting values from Worked Examples 9.1 and 9.2 are about 4.2 mil for 50 Ω on L3/L8; differential geometries are taken directly from the fabricator's solver, and their sensitivity to W, H, t, ε_r, and S is that of Table 10.4.

41.7.5 Loss budgets

Worked Example 41.1: USB 3.0 route from the RK3588 to a Type-A port

The SuperSpeed lanes run from the RK3588 across the ARM SOM, through J1, and across the carrier to a Type-A connector; assume 5 in of stripline in total (**illustrative**) with the laminate and foil assumed above. Estimate the trace loss at 2.5 GHz.

Solution. With $W \approx 5$ mil and HVLP foil ($\Delta = 1\,\mu$m): at 2.5 GHz, $\delta_s = 1.32\,\mu$m and $K_H = 1.43$. The smooth-copper conductor loss scales as $\sqrt{f}$ from 0.288 dB/in at 4 GHz to 0.228 dB/in, or $0.228 \times 1.43 = 0.33$ dB/in with roughness. Dielectric loss is $2.3 \times 2.5 \times 0.010 \times \sqrt{3.7} = 0.11$ dB/in. The total, 0.44 dB/in, gives 2.2 dB for 5 in; vias and AC-coupling capacitors add about 0.5 dB. This is small compared with what the receiver's equalization can recover, so the dominant risks are the discontinuities: the SO-DIMM contacts, the ESD device, and the Type-A connector, whose return loss must come from their manufacturers' models [TO BE CONFIRMED].

The MIPI lanes to the camera connector are short on the carrier; their channel is dominated by the camera FPC and its two connectors, plus the SO-DIMM contacts, which must be specified with the camera-module vendor (Chapter 32). The HDMI TMDS pairs follow the USB analysis at the frequency set by the HDMI version [TBC], with the difference that they are DC-coupled (Section 35.2).

41.7.6 Timing and matching

Worked Example 41.2: LPDDR4X byte-lane matching on the ARM SOM

Allocate 3 % of the UI at the illustrative LPDDR4X rate of 4266 MT/s to static DQ-to-DQS mismatch. What length tolerance results in stripline at 165 ps/in?

Solution. UI = 234 ps; 3 % is 7.0 ps, or 43 mil. The RK3588 design guide's own rule (for example 10–20 mil within a byte) is used for layout; the computed value shows how much of the timing budget that rule represents.

For the MIPI CSI-2 lanes at the illustrative 2.5 Gb/s (UI 400 ps), 5 % of a UI corresponds to about 120 mil of lane-to-clock mismatch (Worked Example 32.1), counted over the module, J1, the carrier, and the FPC; intra-pair matching is held to a few mils. For the carrier's Ethernet PHYs, the RGMII clock delay is taken from the PHY's internal delay, which avoids about 13 in of delay routing (Worked Example 31.2), with data matched to clock within about 100 mil. On the MAC-to-MAC inter-SOM link there is no PHY, so one of the two MACs, the PL logic, or the PCB must provide that delay [TO BE CONFIRMED], and the matching budget must include the lengths on both modules and the two socket contacts. The LVDS pairs need tight intra-pair matching; their pair-to-pair matching depends on whether the receiver deskews [TBC].

41.7.7 Power distribution

Worked Example 41.3: Core-rail decoupling plan on the ARM SOM

Take an **illustrative** SoC core rail of 0.8 V with a ± 3 % allowance for dynamic noise and a 7.5 A transient; the RK3588's actual rails and currents are [TO BE CONFIRMED]. Find the target impedance and the number of 100 nF capacitors (ESR 10 mΩ, mounted ESL 0.4 nH) needed at 20 MHz.

Solution. $Z_{\text{target}} = 0.024/7.5 = 3.2\,\text{m}\Omega$. The capacitor's SRF is 25.2 MHz, so at 20 MHz it is still capacitive: $X_L = 50.3\,\text{m}\Omega$, $X_C = 79.6\,\text{m}\Omega$, and $|Z_C| = 31.0\,\text{m}\Omega$. By Eq. (21.4), $N \geq \lceil 31.0/3.2 \rceil = 10$ (the inductive-only shortcut would give 16). The same bank needs 8 at 30 MHz but 42 at 10 MHz and

30 at 50 MHz: one value cannot cover the band, so 1 µF and 10 µF banks take the lower frequencies and the planes and package the higher ones. The plan places at least 10 of the 100 nF parts with side-via mounting on the module's bottom side directly below the core balls, connected through short vias to the core-rail plane and its adjacent ground, and the count is confirmed by a sweep of the whole network and a load-step simulation (Section 21.6). Bulk capacitors near the PMIC cover the range below 1 MHz; the die–package anti-resonance above that is left to the SoC's package design, and the module's contribution is checked by PDN simulation (Figure 21.2).

The DC drop from the regulator to the SoC across the core plane is estimated with Eq. (21.11): with the regulator about two squares away, an illustrative 15 A maximum, and 1 oz copper, the drop is about 15 mV. Module copper may be thinner [TBC], which raises the drop in proportion. If the PMIC's core regulator offers remote sense, it is connected at the SoC's core balls to remove this static error [TO BE CONFIRMED]. On the carrier, the module input currents pass through the socket contacts, whose current rating sets the number of power contacts (Section 41.8.5).

41.7.8 EMC and protection

- Fast nets run on inner stripline layers between ground planes where possible, on the modules and on the carrier; microstrip segments are limited to breakouts.
- The USB-C and Type-A connectors have low-capacitance ESD protection on SuperSpeed and USB 2.0 lines, a common-mode choke option on USB 2.0, and VBUS protection; the HDMI connectors have ESD protection on TMDS, DDC, and HPD lines.
- The Ethernet corner has an isolation gap under the magnetics (or a jack with integrated magnetics), a Bob Smith network, and a chassis-ground connection to the RJ45 shield.
- Memory clocks run with spread-spectrum clocking where the SoC supports it [TBC]; clock sources have filtered supplies.
- A via fence along each board edge stitches the ground planes at about 100 mil spacing, tighter near the connectors.
- The coaxial cables from the u.FL connectors to the SMA connectors are routed away from the fast connectors and the switching regulators [TBC].

41.7.9 Summary of design decisions

Table 41.5 collects the decisions of this section with their basis. Each is a design intent; none has been verified on hardware.

Table 41.5. Design decisions carried through this section and their basis. None has been verified on hardware.

Decision	Basis
Module stack-ups within the card-edge thickness (0.8 mm M.2, about 1.2 mm SO-DIMM)	Mechanical standard; thinner dielectrics need narrower traces (Section 10.5)
Carrier: every signal layer referenced to ground; ten layers as one candidate	Chapter 9; layer count [TBC]
Mid-loss laminate, HVLP on L3/L8 (assumed)	USB 3.0 loss budget (Worked Example 41.1) and the PCIe extension
DQ–DQS matching per the RK3588 guide	≈ 3 % of UI equals 43 mil (Worked Example 41.2)
≥10 100 nF plus 1 µF/10 µF banks (illustrative rail)	Z_{target} = 3.2 mΩ; R–L–C sizing and sweep (Worked Example 41.3)
Remote sense at the SoC, if available	15 mV static plane drop at 1 oz (Chapter 21)
RGMII delay in the carrier PHYs; inter-SOM delay location [TBC]	Avoids 13 in of delay routing (Worked Example 31.2)

41.8 Design process, step by step

41.8.1 Requirements capture

Write the requirements of Section 41.2 as a numbered list, each with a verification method (inspection, analysis, or test). Add the missing ones explicitly: operating temperature range, enclosure and airflow, power budget, regulatory targets (EMC, radio), and the software that will run on each processor. Each [TO BE CONFIRMED] item in this chapter is a requirement that has not yet been captured.

41.8.2 Processor and FPGA selection

The RK3588 was chosen for its CPU, GPU, NPU, and video capability and its HDMI input and outputs [90], [91]; the Zynq-7020 for an FPGA fabric with an integrated ARM processing system and a mature toolchain, at the cost of having no transceivers [92]. The selection should be recorded with the requirements it satisfies and the alternatives rejected (for example a Zynq device with transceivers, which would allow a PCI Express link; see Section 41.10).

41.8.3 Choice of module form factors

- **SO-DIMM for the ARM SOM:** enough contacts for the RK3588's many interfaces, a flat, latched mounting with a low profile, a module area large enough for the SoC, memory, PMIC, and radio, and a widely available, low-cost socket.
- **M.2 for the FPGA SOM:** a smaller module that needs fewer contacts, a thin profile, and screw retention; the M.2 socket is small and inexpensive.

Both form factors are borrowed mechanical standards with custom pinouts, which brings the keying risks discussed in Sections 41.4 and 41.5.

41.8.4 Interface and pin allocation

Build one table per connector listing every contact, its signal, its voltage domain, its direction, and its source document, and review both ends together with the procedure of Section 16.7. Allocate power and ground

contacts first (enough in parallel for the module current, with grounds interleaved with fast signals), then the high-speed pairs and RGMII groups (short, grouped, with adjacent grounds), then the low-speed signals. On the Zynq side, allocate by I/O bank: all signals in a bank share its supply voltage, and LVDS pairs need the bank voltage their standard requires [TO BE CONFIRMED].

41.8.5 Power tree and sequencing

The carrier converts the 12 V input to the module input rails and its own rails (Figure 41.1). An **assumed** tree, to be sized with the rail-budget method of Chapter 24:

- input protection (reverse polarity, overvoltage and surge, inrush limiting) at the 12 V jack;
- a 12 V-to-5 V buck converter feeding the ARM SOM input, the USB VBUS switches, and the fan;
- a 5 V-to-3.3 V buck converter feeding the FPGA SOM input and the carrier's PHYs and peripherals;
- local regulators for any carrier rail with tighter noise limits (for example PHY analog supplies) [TBC].

Every current in this tree is [TO BE CONFIRMED] until the module datasheets give their maximum input currents. For sequencing (Chapter 25), two rules dominate:

- **No back-powering across the modules.** If one module is powered and the other is not, the powered one must not drive the other's inputs; either both modules come up together under one carrier sequencer, or the inter-module signals are isolated (buffers with power-off isolation, or drivers held in a high-impedance state) until both modules report power-good (Section 24.5).
- **Resets after supplies.** The carrier releases each module's reset only after that module's power-good signal, and the RK3588 software should not access the FPGA until the Zynq's PL is configured (its DONE signal, made available to the ARM side through a GPIO [TBC]).

41.8.6 Schematic capture

Capture each board with hierarchical sheets that mirror the block diagram: power, each processor, memory, each interface, and the connector. Annotate every net with its voltage domain and, for high-speed nets, its impedance class. Use the same net names for inter-module signals on all three schematics so that the connector tables can be checked by script.

41.8.7 PCB stack-up and layout

- **Stack-up.** The modules need high layer counts and fine-pitch vias for their BGAs and memory [TO BE CONFIRMED]; the carrier can use a conventional stack-up with solid reference planes under every high-speed layer, for example a six- or eight-layer build like that of Chapter 10 [assumption].
- **Impedance control.** 100 Ω differential for HDMI, Ethernet MDI, and LVDS; 90 Ω for USB; 50 Ω single-ended for RGMII and SD; computed for the actual stack-up and specified to the fabricator with coupons (Section 10.8).
- **High-speed routing.** Short, direct routes from J1 and J2 to the external connectors and between the sockets; reference-plane continuity (no splits under fast traces); stitching vias next to every layer change; ESD parts at the connectors; AC-coupling capacitors placed per the interface guides.
- **DDR length matching.** The DDR routing lives on the modules: LPDDR4X on the ARM SOM and DDR3L on the FPGA SOM, each following its vendor's guide (for the Zynq-7000, the PCB design guide UG933 [95]). On the carrier, length matching applies to the RGMII groups, the LVDS pairs (intra-pair), and the external pairs, with budgets derived from timing (Chapter 16).
- **Thermal design.** See Section 41.8.12.

41.8.8 Design review

Review each board against its checklist (Appendix C) and the system as a whole: the connector tables at both ends, the power tree and sequence across boards, the I/O voltage of every inter-module net, the key and mechanical stack-up of each module in its socket, and the thermal path. Record each review finding and its resolution.

41.8.9 Prototyping

This step and the ones after it have not been started; they follow fabrication of the boards designed in Altium Designer. Build the carrier first with test points on every rail, enable, power-good, and reset, and with zero-ohm links that let each module's supply be enabled separately. A dummy module (a bare PCB with the card edge, loads, and test points) allows the carrier's power tree and sequencing to be checked before real modules are inserted [assumption].

41.8.10 Board bring-up and testing

1. Inspect: shorts between rails, correct parts on the power path, socket orientation. Record the rail-to-ground resistances of the first boards; they become the reference for later builds (Figure 38.1).
2. Power the carrier without modules from a current-limited supply; check every rail, ramp, and sequence against written acceptance criteria (Section 25.6).
3. Insert the dummy module, then the real modules one at a time; check input currents and module power-good signals.
4. Bring up the Zynq over JTAG: confirm the device ID, configure the PL, observe DONE.
5. Bring up the RK3588 through its serial console: boot loader, memory training, then the operating system. Record the LPDDR4X and DDR3L training reports and the two-dimensional eye margins of every byte lane on the first boards.
6. Test each interface: Ethernet links, USB enumeration (USB-C in both plug orientations), HDMI, microSD, the camera link with PRBS patterns if the sensor supports them, the inter-module links (loopback and data integrity), and the headers.
7. Compare TDR measurements of the impedance coupons and of one carrier's USB and HDMI routes with the fabricator's report and with the post-layout models, closing the loop for the next revision.
8. Characterize: temperatures under the intended workload (Section 26.6), supply ripple, and link error counts over time and temperature.

No results are reported here; record them in a bring-up report (Chapter 38).

41.8.11 Tests remaining after implementation

Because the platform has been designed and rendered but not built, all of the tests in Table 41.6 remain open. They are listed in the order in which they would normally be run, with the chapter that describes each method. Pass criteria are written before the test, from the requirements and the vendor specifications; they are [TO BE CONFIRMED] for this design.

Table 41.6. Tests that remain after the boards are fabricated and assembled. None has been performed.

Test	Boards	What it establishes	Method
Bare-board electrical test and impedance coupons	All three	Opens and shorts; controlled impedance as built	Section 10.8
Assembly inspection (optical, X-ray of BGAs)	All three	Solder joints under the RK3588, Zynq-7020, memory, and PMIC	Chapter 39
Power-rail and sequencing check	Carrier, then each SOM	Rail values, ramps, order, power-good, and reset behaviour	Chapter 25
Rail ripple and transient response	All three	PDN behaviour under load steps	Chapters 19 and 37
Processor bring-up	ARM SOM, FPGA SOM	Boot of the RK3588; JTAG access and PL configuration of the Zynq	Chapter 38
Memory test	ARM SOM, FPGA SOM	LPDDR4X and DDR3L training and data integrity over temperature	Chapter 28
Inter-module links	All three	RGMII link, LVDS bit-error rate, SPI, UART, I^2C, and GPIO through both sockets	Chapters 31 and 35
External interfaces	Carrier	Ethernet, USB 3.0 and USB-C, HDMI out and in, microSD	Chapters 30, 31 and 35
Signal-integrity measurements	Carrier	TDR of critical nets; eye diagrams of the fastest links	Chapter 37
Thermal test	Assembled system	Junction and board temperatures under the intended workload, with the heat sink and fan	Chapter 26
EMC and ESD pre-compliance	Assembled system	Emissions and immunity before formal testing	Chapter 27
Wireless and regulatory	ARM SOM, assembled system	Antenna performance; radio and product certification	[TBC]

41.8.12 Thermal design

The RK3588 is the main heat source. Because the ARM SOM lies flat with its components facing up, a heat sink or heat spreader on the SoC (with a fan if needed) is the main heat path; the module's own board and the SO-DIMM contacts conduct little heat to the carrier. The FPGA SOM's power depends on its logic design; its top surface is accessible for a spreader if needed. The fan header, the module spacing, and the enclosure's airflow must be designed together, and the power and thermal values are [TO BE CONFIRMED] for both modules. Estimate junction temperatures with the methods of Chapter 26, not with $R_{\theta JA}$ alone.

41.9 The assembled system

Figure 41.5 shows both modules installed: the ARM SOM lies flat in J1 under its latches and is screwed to the two standoffs, and the FPGA SOM sits in J2 with its screw at the far end. The camera and display FPC connectors on the right edge and the headers along the bottom edge remain accessible.

How the work is divided. A typical division (**assumed**):

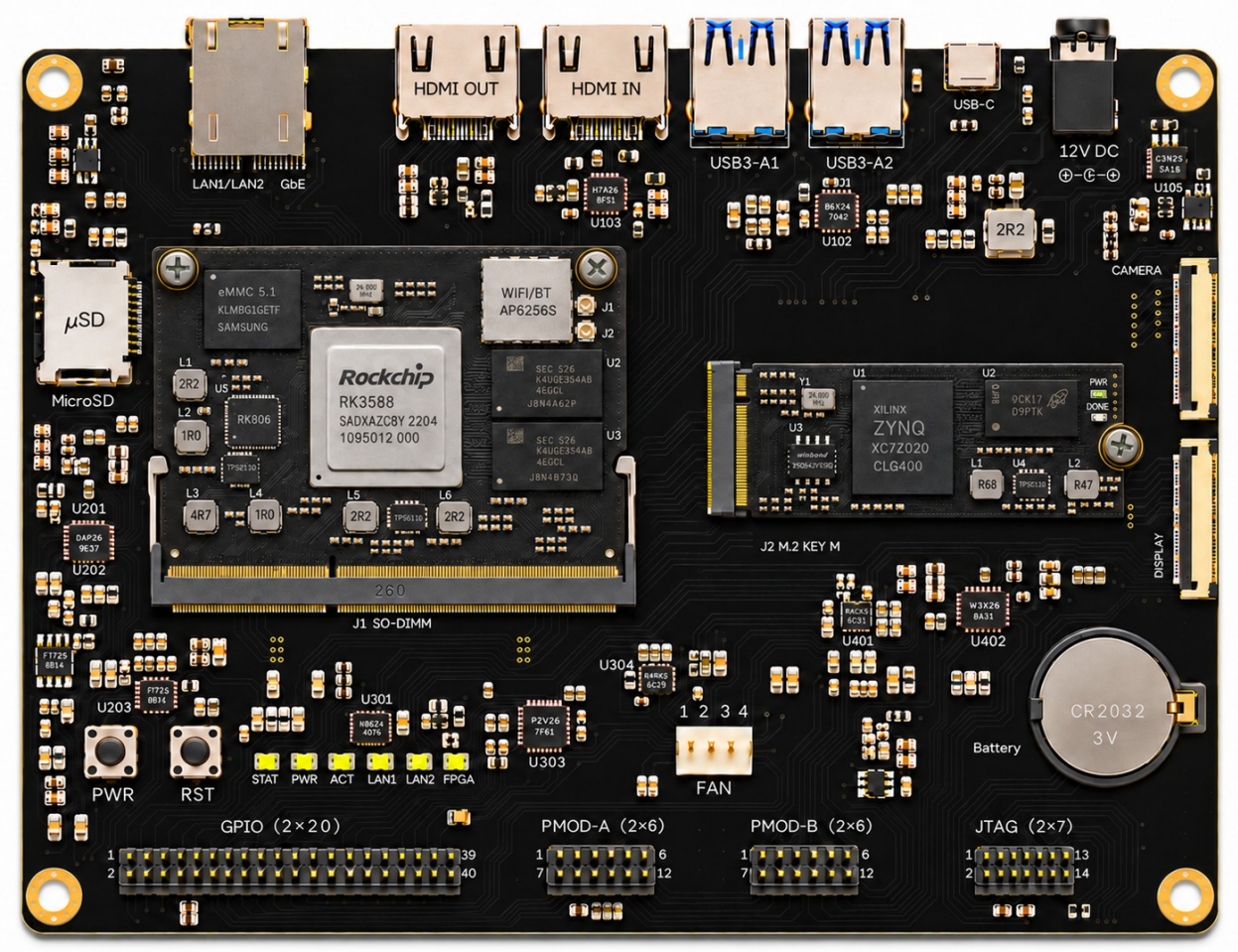


Figure 41.5. Assembled system, top view: the ARM SOM installed flat in the SO-DIMM socket and secured by two screws, and the FPGA SOM installed in the M.2 socket and secured by one screw. Rendering of the Altium Designer 3D models of the three boards, made in KeyShot; the system has not been manufactured or assembled, and this is not a photograph of hardware. Small part markings differ from those in the separate module renderings and are illustrative. Antenna cables, heat sink, and fan are not shown. [Illustration]

- **ARM side:** the operating system and applications, user interface and display, HDMI capture and output, video encoding and decoding, neural-network inference on the NPU, storage, external networking, and USB devices.
- **FPGA side:** custom and timing-critical I/O (Pmod peripherals, sensors, actuators), deterministic control loops, data acquisition with precise timestamps, and pre-processing (filtering, decimation, framing) that reduces the data rate before it reaches the ARM side.
- **Between them:** Gigabit Ethernet for bulk data with a standard software interface; SPI for register access to FPGA functions; UART for consoles and simple messages; I^2C for management; GPIO lines for interrupts and handshakes; LVDS pairs for high-rate streams where an endpoint exists on the ARM side [TO BE CONFIRMED].

The aggregate inter-module bandwidth is bounded by these links: Gigabit Ethernet carries at most its 1 Gbit/s line rate, and the other links are far slower except for the LVDS pairs, whose rate and use are open. Applications that need more must either reduce data in the FPGA or use the extensions of Section 41.10.

41.10 Design trade-offs, limitations, and possible extensions

Trade-offs.

- **Modularity versus signal integrity:** every inter-module signal crosses two connectors and the carrier; connector discontinuities and stubs limit the usable rates, and the connector models must be included in any channel analysis.
- **Standard mechanics, custom pinouts:** low-cost sockets, but a risk of insertion into the wrong socket that the keying and documentation must address.
- **Cost and complexity split:** high-layer-count modules and a simpler carrier, at the price of three designs to maintain and a system-level specification of the interfaces between them.
- **Concentrated heat:** short inter-module links put both heat sources close together.

Limitations.

- No PCI Express between the processors: the Zynq-7020 has no transceivers or PCIe block [92].
- Inter-module bandwidth limited to Gigabit Ethernet plus the LVDS pairs, whose ARM-side endpoint is unresolved.
- Many details remain open (Section 41.11).

Possible extensions.

- Replace the FPGA SOM with a Zynq-7000 device that has transceivers and a PCI Express block (the Z-7015 and larger devices [92]) and connect it to one of the RK3588's PCIe controllers, which the RK3588 provides in PCIe 3.0 and PCIe 2.0 forms [90]; this needs a new module pinout and an 85 Ω channel through both sockets.
- Add an M.2 socket for NVMe storage on the RK3588's PCIe 3.0 lanes, keyed as a standard socket and separated from J2 (see the example below). A second, standard key M socket on the same carrier makes the keying of J2 (Section 41.5) even more important.
- Add a time-synchronization path (a pulse-per-second line or IEEE 1588 over the inter-module Ethernet) for systems where the FPGA timestamps data.
- Add power over Ethernet, an isolated fieldbus, or additional camera inputs as carrier variants that reuse both modules.

Worked Example 41.4: PCIe Gen3 route to an added M.2 NVMe socket

In a carrier variant, PCIe Gen3 lanes run from the RK3588 across the ARM SOM, through J1, and on L3 of the carrier to an M.2 socket, 4 in of stripline in total (**illustrative**); the lane count is set by the RK3588's PCIe 3.0 controller [TBC]. Estimate the board's insertion loss at 4 GHz and compare it with a Gen3 budget of about 22 dB end to end.

Solution. With $W \approx 5$ mil, 50 Ω-equivalent, HVLP ($\Delta = 1$ µm): $\delta_s = 1.045$ µm, $K_H = 1.58$, conductor loss $0.288 \times 1.58 = 0.45$ dB/in. Dielectric loss $2.3 \times 4 \times 0.010 \times \sqrt{3.7} = 0.18$ dB/in. Total 0.63 dB/in, or 2.5 dB for 4 in. Adding about 0.5 dB for vias and AC caps and 1 dB for the M.2 connector, the board's share is about 4 dB plus the loss of the SO-DIMM contacts, which must come from the socket vendor [TBC]. Most of the budget remains for the SoC package, the SSD's package and traces, and margin. Via stubs from L3 run to L10 (about 50 mil), resonating near 30 GHz (Worked Example 15.4), well above

$3f_{Nyq} = 12$ GHz, so back-drilling is not needed at Gen3. The 85 Ω impedance class is then added to the carrier's stack-up plan (Section 41.7.4).

41.11 Chapter summary

Key Takeaways

- Combining an RK3588 application processor with a Zynq-7020 lets software and deterministic, custom I/O each run where they fit best.
- Splitting the design into a SO-DIMM ARM SOM, an M.2 FPGA SOM, and a carrier confines dense, high-layer-count design to small modules and keeps the carrier simpler, at the cost of two connectors in every inter-module path.
- Standard sockets with custom pinouts need deliberate keying and documentation; the M.2 key shown in the rendering (key M) would admit standard SSDs.
- With no transceivers on the Zynq-7020, the modules communicate over RGMII Gigabit Ethernet, LVDS, SPI, UART, I^2C, and GPIO; the ARM-side endpoint of the LVDS pairs must be resolved.
- Power, sequencing, and back-powering must be designed across board boundaries; the carrier owns the sequence.
- The card edges fix the module thickness (0.8 mm for M.2, about 1.2 mm for SO-DIMM), so the modules need thin, dense stack-ups; loss, timing, and PDN budgets are carried in a requirements ledger whose rows close only on hardware.
- The three boards were designed in Altium Designer and rendered in KeyShot; the design has not been implemented, and all tests in Table 41.6 remain to be done after fabrication and assembly.
- Every unresolved item is listed below, and no performance or test result is claimed for the design.

Placeholders and assumptions to resolve

Table 41.7. Items marked [TO BE CONFIRMED] or stated as assumptions in this chapter.

No.	Area	Item
1	ARM SOM	LPDDR4X capacity, channel configuration, and parts
2	ARM SOM	eMMC capacity
3	ARM SOM	Module input voltage range and maximum current; PMIC selection; power-good output
4	ARM SOM	Wi-Fi/Bluetooth module selection, certification, and antenna types
5	ARM SOM	SO-DIMM socket type (contacts, key, height) and the SOM pinout specification
6	ARM SOM	Layer count and via technology
7	FPGA SOM	Zynq-7020 package and speed grade
8	FPGA SOM	DDR3L width and capacity; QSPI flash capacity
9	FPGA SOM	PS reference-clock frequency (the module rendering shows 25.000 MHz, the assembled rendering 24.000 MHz)
10	FPGA SOM	Module input voltage (assumed 3.3 V), current, and number of power contacts
11	FPGA SOM	M.2 key (the rendering shows key M), outline size, and standoff position
12	Carrier	Input protection, converter ratings, and the full rail budget

No.	Area	Item
13	Carrier	Ethernet jack type (dual-port jack at the LAN1/LAN2 position or two jacks); assignment of each port to a MAC and PHY; MAC-to-MAC RGMII implementation and clock-delay location
14	Carrier	Number, direction, rate, and ARM-side endpoint of the LVDS pairs; Zynq bank voltages
15	Carrier	SPI roles and clock rate; UART, I^2C addresses and pull-ups; GPIO count and functions
16	Carrier	USB-C role (data, recovery, power, display) and CC configuration; source of the USB 3.0 ports
17	Carrier	Assignment of HDMI in/out, microSD, camera and display connectors, and the 40-pin header to a module
18	Carrier	Pmod and JTAG connections and JTAG chain; header I/O voltage and protection
19	Carrier	Real-time clock device powered by the coin cell; fan control
20	Carrier	Stack-up and layer count; SMA connector positions (not shown in the rendering)
21	System	Power and thermal values of both modules; heat-sink, fan, and enclosure design
22	System	Power-up order, back-powering protection, and reset release across modules
23	System	Boot sources and boot-mode configuration of both processors
24	ARM SOM, FPGA SOM	Module thickness for the chosen sockets; module stack-ups, minimum trace widths, and via technology
25	System	Data rates (LPDDR4X, DDR3L, MIPI, HDMI, LVDS), route lengths, and rail voltages and currents used in the worked examples of Section 41.7
26	System	SO-DIMM and connector channel models (insertion and return loss) and contact current ratings; regulatory targets for emissions

Further Reading

The Zynq-7000 overview data sheet [92], technical reference manual [94], and PCB design guide [95] govern the FPGA SOM; Rockchip's RK3588 datasheet and hardware design guide (summarized publicly in [90], [91], [98]) govern the ARM SOM. Commercial SOM vendors' carrier-design guides are useful examples of how module interfaces are specified.

Review Problems

P41.1 Why can the Zynq-7020 SOM not be connected to the RK3588 over PCI Express, and what change would allow it?

P41.2 The carrier rendering labels J2 "M.2 KEY M". Describe the risk and a mitigation.

P41.3 (Inverse design) Both RJ45 ports and an inter-module Ethernet link are required, but each processor has two Ethernet MACs. Propose an assignment.

P41.4 (Imperfect data) The FPGA SOM rendering shows a 25.000 MHz oscillator. What should you check before accepting it as the PS clock?

P41.5 Why must the inter-module signals be isolated or held inactive when only one module is powered?

P41.6 List the impedance classes the carrier needs and one interface for each.

Appendices

Appendix A

Formula Reference Sheet

This appendix collects the principal equations of the book. Units are SI unless a practical unit is shown; "(mil)" and "(GHz)" indicate the unit of the variable in that formula. Each entry refers to the equation in the text, where its derivation and limits are given.

Edges, bandwidth, and delay

Quantity	Formula	Ref.
RC rise time	$t_r = \tau \ln 9 \approx 2.2\tau$	Eq. (2.2)
Bandwidth (single pole)	$f_{3\text{dB}} \approx 0.35/t_r$	Eq. (2.3)
Knee frequency	$f_{\text{knee}} \approx 0.5/t_r$	Eq. (2.4)
Cascaded rise times	$t_r = \sqrt{\sum t_{r,i}^2}$	Eq. (2.5)
Trapezoid corners	$f_1 = 1/(\pi\tau)$, $f_2 = 1/(\pi T_{\text{ramp}})$, $T_{\text{ramp}} = t_r/0.8$ (linear)	Eq. (2.8)
Propagation delay	$t_{\text{pd}} = \sqrt{\varepsilon_{\text{eff}}}/c \approx 84.7\,\text{ps/in}\sqrt{\varepsilon_{\text{eff}}}$	Eq. (2.10)
Critical length	$\ell_{\text{crit}} = t_r/(k\,t_{\text{pd}})$, $k \approx 6$	Eq. (2.11)
Nyquist frequency	$f_{\text{Nyq}} = 1/(2\,\text{UI})$	Eq. (2.12)

Electromagnetics

Quantity	Formula	Ref.
Plate capacitance	$C = \varepsilon_0\varepsilon_r A/h$; $\varepsilon_0 = 0.2249\,\text{pF/in}$	Eq. (3.1)
Loop inductance	$L_{\text{loop}} = L_{11} + L_{22} - 2L_{12}$	Eq. (3.5)
Two-wire loop	$L_{\text{loop}} \approx (\mu_0\ell/\pi)\ln(s/r)$	Eq. (3.8)
Sheet inductance	$L_\square = \mu_0 h \approx 32\,\text{pH per square per mil}$	Eq. (3.9)
Return current	$J(x) = \dfrac{I}{\pi H}\dfrac{1}{1+(x/H)^2}$	Eq. (3.10)
Skin depth (Cu)	$\delta_s \approx 66.1\,\mu\text{m}/\sqrt{f/\text{MHz}}$	Eq. (3.13)

Transmission lines and reflections

Quantity	Formula	Ref.
Characteristic impedance	$Z_0 = \sqrt{(R' + \mathrm{j}\omega L')/(G' + \mathrm{j}\omega C')} \to \sqrt{L'/C'}$	Eqs. (4.6) and (4.7)
Per-unit-length L, C	$C' = t_{\mathrm{pd}}/Z_0$, $L' = Z_0 t_{\mathrm{pd}}$	Eq. (4.9)
Attenuation	$\alpha \approx R'/(2Z_0) + G'Z_0/2$	Eq. (4.12)
Input impedance	$Z_{\mathrm{in}} = Z_0 \dfrac{Z_{\mathrm{L}} + \mathrm{j}Z_0 \tan\theta}{Z_0 + \mathrm{j}Z_{\mathrm{L}} \tan\theta}$	Eq. (4.16)
Reflection coefficient	$\Gamma = (Z_{\mathrm{L}} - Z_0)/(Z_{\mathrm{L}} + Z_0)$, $T = 1 + \Gamma$	Eq. (7.2)
Launch voltage	$V_S Z_0/(Z_0 + R_S)$	Eq. (7.3)
Capacitive load (end, middle)	$\tau = Z_0 C$; $\tau = Z_0 C/2$	Eqs. (7.6) and (7.7)
Shunt-C reflection	$\lvert\Gamma\rvert = x/\sqrt{4 + x^2}$, $x = \omega C Z_0$	Eq. (30.1)

Impedance approximations (see limits in Chapter 9)

Geometry	Formula	Ref.
Microstrip (IPC-2141)	$\dfrac{87}{\sqrt{\varepsilon_{\mathrm{r}} + 1.41}} \ln \dfrac{5.98H}{0.8W + T}$	Eq. (9.1)
Stripline (IPC-2141)	$\dfrac{60}{\sqrt{\varepsilon_{\mathrm{r}}}} \ln \dfrac{1.9B}{0.8W + T}$	Eq. (9.5)
Diff. microstrip	$2Z_0[1 - 0.48\mathrm{e}^{-0.96S/H}]$	Eq. (9.7)
Diff. stripline	$2Z_0[1 - 0.347\mathrm{e}^{-2.9S/B}]$	Eq. (9.8)
Odd/even modes	$Z_{\mathrm{odd}} = \sqrt{\frac{L_{11} - L_{12}}{C_g + 2C_m}}$, $Z_{\mathrm{even}} = \sqrt{\frac{L_{11} + L_{12}}{C_g}}$	Eq. (13.2)
Differential, common	$Z_{\mathrm{diff}} = 2Z_{\mathrm{odd}}$, $Z_{\mathrm{cm}} = Z_{\mathrm{even}}/2$	Eq. (13.3)

Loss, crosstalk, skew, vias

Quantity	Formula	Ref.
Conductor loss (stripline)	$36\sqrt{f_{\mathrm{GHz}}}/(W_{\mathrm{mil}} Z_0)$ dB/in	Eq. (11.3)
Dielectric loss	$2.31 f_{\mathrm{GHz}} D_{\mathrm{f}} \sqrt{D_{\mathrm{k}}}$ dB/in	Eq. (11.6)
Hammerstad roughness	$1 + \frac{2}{\pi} \arctan[1.4(\Delta/\delta_{\mathrm{s}})^2]$	Eq. (11.4)
NEXT coefficient	$K_b = \frac{1}{4}(C_m/C + L_m/L)$	Eq. (12.1)
FEXT coefficient	$K_f = \frac{1}{2v_{\mathrm{p}}}(C_m/C - L_m/L)$ ($\lvert\Delta T\rvert \ll t_{\mathrm{r}}$)	Eq. (12.2)
FEXT, modal	$V = \frac{1}{2}[a(t - T_e) - a(t - T_o)]$, $K_f \ell = -\Delta T/2$	Eq. (12.7)
Saturation length	$\ell_{\mathrm{sat}} = t_{\mathrm{r}} v_{\mathrm{p}}/2$	Eq. (12.5)
Skew mode conversion	$\lvert V_c/V_d\rvert = \frac{1}{2}\lvert\sin \pi f \Delta t\rvert$	Eq. (13.4)
Via inductance	$5.08h[\ln(4h/d) + 1]$ nH (h, d in inches)	Eq. (15.1)
Via pad capacitance	$1.41\varepsilon_{\mathrm{r}} T D_1/(D_2 - D_1)$ pF	Eq. (15.2)
Stub resonance	$f = c/(4\ell\sqrt{\varepsilon_{\mathrm{eff}}})$	Eq. (15.7)

Timing, jitter, and clocks

Quantity	Formula	Ref.
Setup margin	$T_{\text{clk}} - (t_{\text{co,max}} + t_{\text{flight}} + t_{\text{su}} + t_{\text{skew}} + t_{\text{jit}})$	Eq. (17.1)
Total jitter	$\text{TJ} = \text{DJ}_{\delta\delta} + 2Q_{\text{BER}}\sigma;\ 2Q = 14.07$ at 10^{-12}	Eq. (17.5)
Two-tap FFE	$\lvert H\rvert = \sqrt{c_0^2 + c_1^2 + 2c_0c_1\cos(2\pi f\ \text{UI})}$	Eq. (17.7)
Phase noise to jitter	$\sigma_t = \frac{1}{2\pi f_0}\sqrt{2\int 10^{\mathscr{L}/10}\,\mathrm{d}f}$	Eq. (33.2)
Crystal load	$C_L = C_1C_2/(C_1 + C_2) + C_{\text{stray}}$	Eq. (33.1)

Power integrity and EMC

Quantity	Formula	Ref.
Ground bounce	$V = L_{\text{gnd}}N\ \mathrm{d}i/\mathrm{d}t$	Eq. (20.1)
Parallel pins	$L_{\text{eff}} = [L + (N_g - 1)M]/N_g$	Eq. (20.3)
Target impedance	$Z_{\text{target}} = \Delta V/\Delta I$	Eq. (21.1)
Capacitor SRF	$f_{\text{SRF}} = 1/(2\pi\sqrt{\text{ESL}\,C})$	Eq. (21.3)
Capacitor count	$N(f) \geq \lceil \lvert Z_C(f)\rvert / Z_{\text{target}} \rceil$ (full R–L–C)	Eq. (21.4)
Anti-resonance	$f = \frac{1}{2\pi}\sqrt{\frac{C_1+C_2}{(L_1+L_2)C_1C_2}}$, $\lvert Z\rvert \approx X^2/(R_1 + R_2)$	Eqs. (21.6) and (21.7)
Bounce, capacitive load	$\mathrm{d}i/\mathrm{d}t\rvert_{\max} = \frac{\pi^2}{2}C_L\Delta V/T_e^2$ (raised cosine)	Eq. (20.2)
Plane resistance	$R_\square = \rho/t$ (0.49 mΩ for 1 oz)	Eq. (21.11)
Cavity resonance	$f_{mn} = \frac{c}{2\sqrt{\varepsilon_r}}\sqrt{(m/a)^2 + (n/b)^2}$	Eq. (23.1)
DM emission	$E = 131.6 \times 10^{-16} f^2 AI/r$	Eq. (27.1)
CM emission	$E = 1.257 \times 10^{-6} fLI/r$	Eq. (27.2)

Measurement

Quantity	Formula	Ref.
TDR resolution	$\Delta x \approx t_r v_p/2$ (round trip)	Eq. (37.3)
Shunt-through PDN	$Z = Z_0S_{21}/[2(1 - S_{21})] \approx (Z_0/2)S_{21}$	Eq. (37.4)
TDR impedance	$Z = Z_{\text{ref}}(1 + \rho)/(1 - \rho)$	Eq. (37.1)
Excess C, L	$C = -\frac{2}{Z_0V_i}\int v_r\,\mathrm{d}t;\ L = \frac{2Z_0}{V_i}\int v_r\,\mathrm{d}t$	Eq. (37.2)
Mixed-mode S_{DD21}	$\frac{1}{2}(S_{31} - S_{32} - S_{41} + S_{42})$	Eq. (36.2)

Appendix B

Material Property Tables

The values in this appendix are representative ranges compiled for orientation. Laminate D_k and D_f depend on resin content, glass style, frequency, temperature, humidity, and the test method used to measure them. Always design with the manufacturer's current data for the specific construction in the stackup, and with the fabricator's recommendation.

B.1 Laminate classes

Table B.1. Representative laminate properties at about 1–10 GHz. Ranges, not guaranteed values; consult product data sheets.

Laminate class (examples)	D_k	D_f	T_g (°C)	Typical use
Standard FR-4, high T_g	4.0–4.6	0.015–0.022	150–180	General digital
Mid-loss epoxy/modified epoxy	3.6–4.1	0.008–0.012	170–200	DDR4/5, PCIe Gen3/4
Low-loss (modified PPE, etc.)	3.3–3.8	0.004–0.007	180–210	PCIe Gen4/5, USB4
Very/ultra-low loss	3.0–3.6	0.0015–0.003	200+	PCIe Gen5/6, 56G+
Ceramic-filled hydrocarbon (RF)	3.0–3.7	0.002–0.004	>280	RF, mixed builds
PTFE (RF)	2.1–3.0	0.0009–0.002	—	Microwave, antennas

B.2 Frequency dependence of a causal model

Table B.2 evaluates the Djordjevic–Sarkar model (Eq. (11.8)) with relaxation limits at 1 kHz and 1 THz, fitted to three laminates at 1 GHz. It illustrates how D_k falls slowly with frequency while D_f stays nearly constant; real materials depart from this ideal, so measured data should be used where available.

Table B.2. D_k / D_f versus frequency from a Djordjevic–Sarkar fit at 1 GHz.

Fit at 1 GHz	0.1 GHz	1 GHz	5 GHz	10 GHz	20 GHz	50 GHz
4.0 / 0.020	4.12 / 0.019	4.00 / 0.020	3.92 / 0.020	3.88 / 0.021	3.85 / 0.021	3.80 / 0.020
3.7 / 0.010	3.75 / 0.010	3.70 / 0.010	3.66 / 0.010	3.65 / 0.010	3.63 / 0.010	3.61 / 0.010
3.5 / 0.004	3.52 / 0.004	3.50 / 0.004	3.49 / 0.004	3.48 / 0.004	3.47 / 0.004	3.47 / 0.004

B.3 Glass styles

Table B.3. Common E-glass styles in prepreg (approximate; pressed thickness depends on resin content and adjacent copper).

Style	Approx. pressed thickness (mil)	Notes
106	1.5–2.5	Very thin, open weave, resin rich (lower D_k)
1080	2.5–3.5	Thin, open weave; common for thin prepregs
1078 / 1067 / 1035	2–3	Spread (flat) glass; reduced weave skew
2116	4–5.5	Medium, tighter weave
1652	4.5–6	Medium-heavy
7628	6.5–8	Heavy, glass rich (higher D_k); not for high-speed layers

B.4 Copper

Table B.4. Copper foil weights and properties.

Foil weight	Thickness (µm)	Thickness (mil)	$R_\square$ at 20 °C (mΩ)
0.5 oz (0.5 oz/ft^2)	17.4	0.69	0.99
1 oz	34.8	1.37	0.49
2 oz	69.6	2.74	0.25

Copper resistivity is $1.724 \times 10^{-8}\ \Omega \cdot \mathrm{m}$ at 20 °C (annealed) with a temperature coefficient of about 0.39 %/°C. Electrodeposited foils are offered in decreasing roughness grades (standard, reverse-treated, VLP, HVLP/ULP); the fabricator or foil supplier provides roughness parameters for each, preferably as fitted Hammerstad or Huray model parameters.

B.5 Surface finishes

Table B.5. PCB surface finishes and their high-speed relevance.

Finish	Notes
ENIG	Flat, good shelf life; ferromagnetic nickel adds loss on outer-layer traces at multi-GHz
ENEPIG	As ENIG, with palladium; wire-bondable
Immersion silver	Flat, low loss; tarnish sensitive
OSP	Lowest loss, flat; limited shelf life and reflow cycles
HASL (lead-free)	Inexpensive; not flat enough for fine-pitch BGAs

Appendix C

Design-Rule Checklists

These checklists summarize the design rules of the book in review order. They are starting points; adapt them to the interfaces, the silicon vendors' guides, and the fabricator's capabilities for each project.

C.1 Schematic review

- ☐ Every high-speed net class identified, with driver edge rate and ℓ_{crit} computed (Chapter 2).
- ☐ Terminations present and correct (series values, V_{TT}, ODT settings, differential terminations) (Chapter 7).
- ☐ AC-coupling capacitors on the correct side of each serial link, with values in the specified range (Chapters 29 and 30).
- ☐ Reference-clock architecture, SSC, and termination standard consistent at both ends (Chapter 33).
- ☐ Power sequencing, reset, and power-good logic match device requirements (Chapter 38).
- ☐ Decoupling quantities and values per rail from the PDN analysis (Chapter 21); filtered supplies for PLLs and clock buffers.
- ☐ ESD protection at every connector, with capacitance suitable for the data rate (Chapter 30).
- ☐ JTAG chain complete; test access for rails and critical signals.
- ☐ RGMII delay, strapping options, and PHY addresses verified.

C.2 Stackup

- ☐ Each signal layer adjacent to a solid ground plane (Chapter 9).
- ☐ Each power plane within 2–4 mil of a ground plane.
- ☐ Symmetric construction; copper balance reviewed.
- ☐ Laminate chosen from the loss budget at Nyquist; foil roughness specified (Chapter 11).
- ☐ Glass styles for high-speed layers reviewed for weave skew.
- ☐ Fabricator stackup and impedance calculation received and accepted.
- ☐ Via spans, back-drill needs, and HDI structures defined (Chapter 15).

C.3 Routing

- ☐ Controlled-impedance widths per layer from the fabricator stackup.
- ☐ No high-speed trace over a plane split or near a plane edge (Chapter 23).
- ☐ Ground return via next to every high-speed layer transition.
- ☐ Spacing to aggressors $\geq 2H$–$3H$ (more for long runs); TX and RX groups separated (Chapters 12 and 29).
- ☐ Intra-pair skew compensated near its source; delay matching in time units (Chapter 13).

- ☐ DDR byte lanes on one layer, matched per controller guide, fly-by order correct, write-leveling range respected (Chapter 28).
- ☐ Via stubs within limits (with drill tolerance) or back-drilled; AC-cap pad treatment checked with a 3-D model.
- ☐ LPDDR5/5X: separate constraints for the CK/CA, WCK/write-data and RDQS/read-data groups (Section 28.1.2).
- ☐ Serpentines meet spacing rules; no test points on multi-gigabit nets.
- ☐ Clocks routed point-to-point, terminated, and kept away from connectors.

C.4 Power distribution

- ☐ Target impedance computed for each rail (Chapter 21).
- ☐ Rail tolerance allocated among set point, DC drop and dynamic noise (Table 21.1); observation port defined.
- ☐ Capacitor counts from full R–L–C impedance with biased capacitance, not from ωESL alone (Eq. (21.4)).
- ☐ PDN impedance simulated to the package's handoff frequency; anti-resonances reviewed; load-step transient simulated with the specified current profile.
- ☐ PDN measurement points placed and recorded in the model (Section 37.3).
- ☐ Decoupling capacitors mounted with side vias, close to the plane pair.
- ☐ DC IR drop simulated; remote sense connected at the load.
- ☐ Current density in vias and necks within IPC-2152 limits.
- ☐ Each power and ground ball of high-current devices has its own via.

C.5 EMI/EMC

- ☐ Slowest acceptable edge rates selected; SSC enabled where supported (Chapter 27).
- ☐ Connector region with solid ground, filtering, and shield bonding.
- ☐ No slots in planes under signals; via fences on board edges.
- ☐ Common-mode chokes and ferrites placed at cables where required.
- ☐ Ethernet isolation gap clean; chassis ground connections defined (Chapter 31).

C.6 Release

- ☐ DRC with fabricator rules clean; all waivers documented (Chapter 39).
- ☐ Post-layout SI/PI simulation reports archived (Chapter 36).
- ☐ Fabrication drawing: class, materials, stackup, impedance table, drill and back-drill tables, via treatment, finish, test requirements.
- ☐ Coupons specified per controlled-impedance layer.
- ☐ Outputs (Gerber/ODB++/IPC-2581, drill, netlist, assembly, centroid, BOM) regenerated from one tagged revision and inspected in a viewer.
- ☐ Revision marked in copper; ECO process in place for changes.

C.7 Simulation

- ☐ Buffer models (IBIS/IBIS-AMI) at fast, typical, and slow corners from the silicon vendor (Chapter 36).

- ☐ Laminate D_k/D_f per glass style, causal model, roughness parameters.
- ☐ S-parameter models checked for passivity, causality, reciprocity, port order, DC point, and bandwidth ($\geq 3f_{Nyq}$).
- ☐ Channel segmented at quasi-TEM boundaries; 3-D models for vias, connectors, and BGA breakouts.
- ☐ Crosstalk simulated with the two nearest aggressors on each side.
- ☐ PDN impedance and DC IR drop simulated with the routed planes.
- ☐ Margins reported at corners, including $\pm 10\,\%$ impedance.

C.8 Bring-up

- ☐ Visual and X-ray inspection; rail-to-ground resistance compared with a reference (Chapter 38).
- ☐ Current-limited first power; voltage at the load; sequencing captured.
- ☐ Ripple measured with a $50\,\Omega$ coaxial connection; thermal image.
- ☐ Clocks: frequency, amplitude, common mode, jitter.
- ☐ JTAG chain IDCODEs; boundary-scan interconnect test.
- ☐ DDR training and 2-D eye margins recorded per bit.
- ☐ Serial links: width, speed, lane margining, PRBS BER.
- ☐ TDR of coupons and critical routes compared with fabricator data and models.
- ☐ Near-field scan for emission sources before the first compliance test.

Appendix D

Unit Conversions and Rules of Thumb

D.1 Physical constants

Table D.1. Physical constants.

Constant	Value	Board units
Speed of light c	2.998×10^8 m/s	11.80 in/ns; 84.7 ps/in
Permittivity ε_0	8.854×10^{-12} F/m	0.2249 pF/in
Permeability μ_0	$4\pi \times 10^{-7}$ H/m	31.9 nH/in; $\mu_0/2\pi$ = 5.08 nH/in
Free-space impedance η_0	376.7 Ω	—
Copper resistivity ρ	1.724×10^{-8} Ω · m (20 °C)	0.49 mΩ per square, 1 oz

D.2 Length and copper

Table D.2. Length conversions.

Unit	Equivalent
1 mil	0.001 in = 25.4 µm
1 mm	39.37 mil
1 in	25.4 mm = 1000 mil
1 µm	0.039 37 mil
1 oz copper	34.8 µm = 1.37 mil

D.3 Decibels

D.4 Rules of thumb and their limits

Table D.3. Decibel conversions.

Quantity	Relation
Voltage ratio	$\mathrm{dB} = 20\log_{10}(V_2/V_1)$; 6.02 dB = 2×, 20 dB = 10×
Power ratio	$\mathrm{dB} = 10\log_{10}(P_2/P_1)$
Nepers	1 Np = 8.686 dB
dBm	$10\log_{10}(P/1\,\mathrm{mW})$; 0 dBm = 224 mV rms in 50 Ω
dBµV	$20\log_{10}(V/1\,\mu\mathrm{V})$; 0 dBm = 107 dBµV in 50 Ω
Field	1 µV/m = 0 dBµV/m; 100 µV/m = 40 dBµV/m

Table D.4. Rules of thumb used in this book, with their limits.

Rule	Statement	Limits
Bandwidth	$f_{3\mathrm{dB}} \approx 0.35/t_\mathrm{r}$	Exact for single pole; 0.34–0.45 for others
Knee frequency	$f_\mathrm{knee} \approx 0.5/t_\mathrm{r}$	Heuristic; content above it still matters for EMI
Propagation	≈ 6 in/ns, 170 ps/in in FR-4 stripline	D_k varies with resin, frequency
Critical length	$\ell_\mathrm{crit} = t_\mathrm{r}/(6t_\mathrm{pd})$	k = 4–10 depending on tolerance to ringing
50 Ω line	3.4 pF/in, 8.5 nH/in	FR-4 stripline; scale with Z_0 and t_pd
Wire inductance	≈ 1 nH/mm	Logarithmic in ℓ/r; order of magnitude only
Sheet inductance	32 pH per square per mil	Plane pairs, uniform current
Return current	80 % within $\pm 3H$	Thin trace approximation
Small reflection	$\Gamma \approx \Delta Z/(2Z_0)$	$\lvert\Delta Z\rvert \lesssim 0.2Z_0$
Dielectric loss	$2.31 f_\mathrm{GHz} D_\mathrm{f}\sqrt{D_\mathrm{k}}$ dB/in	Stripline; less in microstrip
Conductor loss	$36\sqrt{f_\mathrm{GHz}}/(WZ_0)$ dB/in	Smooth Cu, stripline; times K_rough
3W rule	$D \geq 3W$ for low coupling	Only when $W \approx 2H$; specify in H
Stub length	$f_{\lambda/4} \geq 3f_\mathrm{Nyq}$	Heuristic; include drill and ε_eff tolerance; verify by simulation
Lumped stub	Within 5 % below $\lambda/16$	$\lambda/10$ already gives 14 % error
Edge duration	$T_\mathrm{ramp} = t_\mathrm{r}/0.8$	Linear ramps only
Stitching vias	$\leq \lambda/20$ spacing	Heuristic; denser near connectors
Ground bounce	$\propto N/t_\mathrm{r}$ (line-driven); $\propto N/t_\mathrm{r}^2$ (lumped C)	Doubling t_r gives 6 dB or 12 dB
Guard-via spacing	$\leq t_\mathrm{r} v_\mathrm{p}/10$	$\lambda/10$ at f_knee is twice as wide
TDR resolution	$t_\mathrm{r} v_\mathrm{p}/2$	Resolution, not accuracy

Appendix E

Python Scripts

The three scripts below implement calculations from the text. They use only the Python standard library (Python 3.8 or later) and can be run directly, e.g. python3 impedance.py. Each script's __main__ block reproduces a worked example; the printed output is shown after each listing. The source files are in appendices/code/. A separate script, tools/gen_figdata.py, regenerates the data-driven figures of the book.

E.1 Microstrip and stripline impedance

Listing E.1. impedance.py — closed-form impedance estimates (Chapter 9).

```
#!/usr/bin/env python3
"""impedance.py - closed-form PCB impedance estimates (Chapter 7).

All dimensions in mils. Results are approximations; see the accuracy
limits stated in Chapter 7 and confirm with a field solver.
"""
import math


def microstrip_ipc(w, h, t, er):
    """IPC-2141 microstrip. Valid for 0.1 < w/h < 2.0 and 1 < er < 15."""
    if not 0.1 < w / h < 2.0:
        print("  warning: w/h outside IPC-2141 range")
    return 87.0 / math.sqrt(er + 1.41) * math.log(5.98 * h / (0.8 * w + t))


def microstrip_hammerstad(w, h, er, t=0.0):
    """Simplified Hammerstad-type closed forms of Chapter 7 with optional
    Wheeler thickness correction. Not the full Hammerstad-Jensen model.

    Returns (Z0, e_eff).
    """
    if t > 0:
        w = w + (t / math.pi) * (1.0 + math.log(2.0 * h / t))
    u = w / h
    a = (er + 1.0) / 2.0
    b = (er - 1.0) / 2.0
    if u <= 1.0:
        eeff = a + b * ((1.0 + 12.0 / u) ** -0.5 + 0.04 * (1.0 - u) ** 2)
        z0 = 60.0 / math.sqrt(eeff) * math.log(8.0 / u + u / 4.0)
    else:
        eeff = a + b * (1.0 + 12.0 / u) ** -0.5
```

```
        z0 = 120.0 * math.pi / (math.sqrt(eeff) * (u + 1.393 + 0.667 * math.log(u + 1.444)))
    return z0, eeff


def microstrip_hj_full(w, h, er, t=0.0):
    """Full Hammerstad-Jensen (1980) quasi-static microstrip, with the
    Hammerstad-Jensen thickness correction. Returns (Z0, e_eff).
    Idealized geometry: rectangular strip, no soldermask, infinite ground."""
    eta0 = 376.73
    u = w / h

    def z01(x):
        f = 6.0 + (2.0 * math.pi - 6.0) * math.exp(-(30.666 / x) ** 0.7528)
        return eta0 / (2.0 * math.pi) * math.log(f / x + math.sqrt(1.0 + 4.0 / x ** 2))

    def ee(x):
        a = (1.0 + math.log((x ** 4 + (x / 52.0) ** 2) / (x ** 4 + 0.432)) / 49.0
             + math.log(1.0 + (x / 18.1) ** 3) / 18.7)
        b = 0.564 * ((er - 0.9) / (er + 3.0)) ** 0.053
        return (er + 1.0) / 2.0 + (er - 1.0) / 2.0 * (1.0 + 10.0 / x) ** (-a * b)

    if t > 0:
        th = t / h
        du1 = th / math.pi * math.log(1.0 + 4.0 * math.e /
                                      (th / math.tanh(math.sqrt(6.517 * u)) ** 2))
        dur = 0.5 * (1.0 + 1.0 / math.cosh(math.sqrt(er - 1.0))) * du1
        u1, ur = u + du1, u + dur
    else:
        u1 = ur = u
    eeff = ee(ur) * (z01(u1) / z01(ur)) ** 2
    return z01(ur) / math.sqrt(ee(ur)), eeff


def stripline_ipc(w, b, t, er, warn=True):
    """IPC-2141 symmetric stripline; b = plane-to-plane spacing."""
    if warn and (w / (b - t) >= 0.35 or t / b >= 0.25):
        print("  warning: outside IPC-2141 stripline range")
    return 60.0 / math.sqrt(er) * math.log(1.9 * b / (0.8 * w + t))


def stripline_wheeler(w, b, t, er):
    """Wheeler's centered stripline with finite thickness (Chapter 7)."""
    if t > 0:
        m = 2.0 / (1.0 + 2.0 / 3.0 * t / (b - t))
        dw = t / math.pi * (1.0 - 0.5 * math.log((t / (2 * b - t)) ** 2
                                                 + (0.0796 * t / (w + 1.1 * t)) ** m))
    else:
        dw = 0.0
    x = (b - t) / (w + dw)
    return 30.0 / math.sqrt(er) * math.log(
        1.0 + 4.0 / math.pi * x * (8.0 / math.pi * x
                                   + math.sqrt((8.0 / math.pi * x) ** 2 + 6.27)))

```

```

def stripline_exact_t0(w, b, er):
    """Exact zero-thickness stripline (conformal mapping), via the AGM."""
    def ellipk(k):
        a, g = 1.0, math.sqrt(1.0 - k * k)
        for _ in range(30):
            a, g = 0.5 * (a + g), math.sqrt(a * g)
        return math.pi / (2.0 * a)
    k = 1.0 / math.cosh(math.pi * w / (2.0 * b))
    kp = math.sqrt(1.0 - k * k)
    return 30.0 * math.pi / math.sqrt(er) * ellipk(k) / ellipk(kp)


def zdiff_microstrip(z0, s, h):
    """IPC-2141 edge-coupled microstrip differential impedance."""
    return 2.0 * z0 * (1.0 - 0.48 * math.exp(-0.96 * s / h))


def zdiff_stripline(z0, s, b):
    """IPC-2141 edge-coupled stripline differential impedance."""
    return 2.0 * z0 * (1.0 - 0.347 * math.exp(-2.9 * s / b))


def tpd_ps_per_inch(eeff):
    """Propagation delay per inch for an effective permittivity."""
    return 84.73 * math.sqrt(eeff)


def solve_width(func, target, lo=0.5, hi=40.0):
    """Bisection for the width that gives a target impedance (Z falls with W)."""
    for _ in range(100):
        mid = 0.5 * (lo + hi)
        if func(mid) > target:
            lo = mid
        else:
            hi = mid
    return 0.5 * (lo + hi)


if __name__ == "__main__":
    # Worked Example 5.1: 4-mil trace on 3.5-mil prepreg, Dk = 3.8, T = 1.9 mil
    w, h, t, er = 4.0, 3.5, 1.9, 3.8
    z_ipc = microstrip_ipc(w, h, t, er)
    z_s0, ee0 = microstrip_hammerstad(w, h, er)
    z_s, ee = microstrip_hammerstad(w, h, er, t)
    z_f, eef = microstrip_hj_full(w, h, er, t)
    print("Microstrip W=%.1f H=%.1f T=%.1f Dk=%.2f" % (w, h, t, er))
    print("  IPC-2141                    Z0 = %.1f ohm" % z_ipc)
    print("  simplified set, T=0         Z0 = %.1f ohm, e_eff = %.2f" % (z_s0, ee0))
    print("  simplified + Wheeler dW     Z0 = %.1f ohm, e_eff = %.2f" % (z_s, ee))
    print("  full Hammerstad-Jensen, T   Z0 = %.1f ohm, e_eff = %.2f, tpd = %.0f
    ps/in"
          % (z_f, eef, tpd_ps_per_inch(eef)))

    # Worked Example 5.2: stripline, B = 10.6 mil, T = 0.6 mil
    b, t2 = 10.6, 0.6
```

```
    w50 = solve_width(lambda x: stripline_ipc(x, b, t2, er, warn=False), 50.0)
    print("Stripline B=%.1f T=%.1f Dk=%.2f" % (b, t2, er))
    print("  IPC width for 50 ohm  W = %.2f mil  (W/(B-T) = %.3f)" % (w50, w50 / (b
    - t2)))
    stripline_ipc(w50, b, t2, er)          # prints the range warning
    w50w = solve_width(lambda x: stripline_wheeler(x, b, t2, er), 50.0)
    print("  Wheeler width for 50 ohm W = %.2f mil" % w50w)
    print("  check T=0, W=4.2: Wheeler %.2f ohm, exact %.2f ohm"
          % (stripline_wheeler(4.2, b, 0.0, er), stripline_exact_t0(4.2, b, er)))
    print("  Zdiff (S = 6 mil) = %.1f ohm" % zdiff_stripline(50.0, 6.0, b))
```

Output:

```
Microstrip W=4.0 H=3.5 T=1.9 Dk=3.80
  IPC-2141                    Z0 = 53.8 ohm
  simplified set, T=0         Z0 = 70.9 ohm, e_eff = 2.81
  simplified + Wheeler dW     Z0 = 60.7 ohm, e_eff = 2.87
  full Hammerstad-Jensen, T   Z0 = 61.4 ohm, e_eff = 2.57, tpd = 136 ps/in
Stripline B=10.6 T=0.6 Dk=3.80
  IPC width for 50 ohm  W = 4.21 mil  (W/(B-T) = 0.421)
  warning: outside IPC-2141 stripline range
  Wheeler width for 50 ohm W = 4.42 mil
  check T=0, W=4.2: Wheeler 58.17 ohm, exact 58.20 ohm
  Zdiff (S = 6 mil) = 93.3 ohm
```

E.2 PDN target impedance and capacitor sizing

Listing E.2. pdn_target.py — target impedance, capacitor impedance, anti-resonance, and IR drop (Chapter 21).

```
#!/usr/bin/env python3
"""pdn_target.py - PDN target impedance and capacitor bank sizing (Chapter 16)."""
import math


def target_impedance(vdd, ripple_fraction, i_transient):
    """Z_target = Vdd * ripple / I_transient (ohms)."""
    return vdd * ripple_fraction / i_transient


def cap_impedance(f, c, esr, esl):
    """|Z| of a capacitor with series ESR and ESL at frequency f (Hz)."""
    w = 2.0 * math.pi * f
    return math.hypot(esr, w * esl - 1.0 / (w * c))


def srf(c, esl):
    """Self-resonant frequency (Hz)."""
    return 1.0 / (2.0 * math.pi * math.sqrt(esl * c))


def caps_needed(z_target, f, c, esr, esl):
    """Pointwise lower bound on the number of identical, independent capacitors
    whose parallel bank meets z_target at frequency f: ceil(|Z_branch|/Z_target).
    Uses the full series R-L-C branch; ignores shared mounting inductance."""
```

```
    return math.ceil(cap_impedance(f, c, esr, esl) / z_target)


def caps_needed_inductive_only(z_target, f, esl):
    """The common shortcut ceil(w*ESL/Z_target); valid only well above the SRF."""
    return math.ceil(2.0 * math.pi * f * esl / z_target)


def branch(f, c, esr, esl):
    """Complex impedance of one series R-L-C branch."""
    w = 2.0 * math.pi * f
    return complex(esr, w * esl - 1.0 / (w * c))


def parallel_peak(branches, f_lo, f_hi, n=200000):
    """Frequency (Hz) and magnitude (ohm) of the largest |Z| of parallel
    R-L-C branches between f_lo and f_hi (log-spaced brute-force sweep)."""
    best = (0.0, 0.0)
    for i in range(n + 1):
        f = f_lo * (f_hi / f_lo) ** (i / n)
        y = sum(1.0 / branch(f, *b) for b in branches)
        z = abs(1.0 / y)
        if z > best[1]:
            best = (f, z)
    return best


def antiresonance_lossless(c1, l1, c2, l2):
    """Admittance zero of two lossless series L-C branches in parallel (Hz)."""
    return math.sqrt((c1 + c2) / ((l1 + l2) * c1 * c2)) / (2.0 * math.pi)


def ir_drop(current, squares, copper_um=34.8, rho=1.724e-8):
    """DC drop across a plane path of a given number of squares."""
    return current * squares * rho / (copper_um * 1e-6)


if __name__ == "__main__":
    zt = target_impedance(0.8, 0.03, 10.0)
    print("Target impedance (0.8 V, 3 %%, 10 A): %.2f mOhm" % (zt * 1e3))
    print("SRF of 100 nF with 0.5 nH: %.1f MHz" % (srf(100e-9, 0.5e-9) / 1e6))
    print("|Z| of that capacitor at 100 MHz: %.3f ohm"
          % cap_impedance(100e6, 100e-9, 0.01, 0.5e-9))
    zb = cap_impedance(20e6, 100e-9, 0.010, 0.5e-9)
    print("100 nF / 10 mOhm / 0.5 nH at 20 MHz: |Z| = %.2f mOhm" % (zb * 1e3))
    print("  capacitors for Z_target (full R-L-C):   %d"
          % caps_needed(zt, 20e6, 100e-9, 0.010, 0.5e-9))
    print("  capacitors (inductive-only shortcut):   %d   <- invalid below SRF"
          % caps_needed_inductive_only(zt, 20e6, 0.5e-9))
    fp, zp = parallel_peak([(100e-9, 0.020, 0.5e-9), (10e-9, 0.030, 0.4e-9)], 10e6,
    200e6)
    print("Anti-resonance 100 nF/20 mOhm/0.5 nH || 10 nF/30 mOhm/0.4 nH:")
    print("  full model peak %.2f MHz, %.3f ohm" % (fp / 1e6, zp))
    print("  lossless admittance zero %.2f MHz"
          % (antiresonance_lossless(100e-9, 0.5e-9, 10e-9, 0.4e-9) / 1e6))
    print("IR drop, 15 A over 3 squares of 1 oz: %.1f mV" % (ir_drop(15, 3) * 1e3))
```

```
    print()
    print("  f (MHz)   |Z| 10uF   |Z| 1uF   |Z| 100nF  (ohm)")
    for fmhz in (0.1, 1, 3, 10, 30, 100, 300):
        f = fmhz * 1e6
        print("  %7.1f   %8.4f   %8.4f   %8.4f" % (
            fmhz,
            cap_impedance(f, 10e-6, 0.003, 0.8e-9),
            cap_impedance(f, 1e-6, 0.005, 0.6e-9),
            cap_impedance(f, 100e-9, 0.010, 0.5e-9)))
```

Output:

```
Target impedance (0.8 V, 3 %, 10 A): 2.40 mOhm
SRF of 100 nF with 0.5 nH: 22.5 MHz
|Z| of that capacitor at 100 MHz: 0.298 ohm
100 nF / 10 mOhm / 0.5 nH at 20 MHz: |Z| = 19.50 mOhm
  capacitors for Z_target (full R-L-C):   9
  capacitors (inductive-only shortcut):   27   <- invalid below SRF
Anti-resonance 100 nF/20 mOhm/0.5 nH || 10 nF/30 mOhm/0.4 nH:
  full model peak 55.16 MHz, 0.443 ohm
  lossless admittance zero 55.64 MHz
IR drop, 15 A over 3 squares of 1 oz: 22.3 mV

  f (MHz)   |Z| 10uF   |Z| 1uF   |Z| 100nF  (ohm)
      0.1     0.1587     1.5912    15.9152
      1.0     0.0113     0.1555     1.5884
      3.0     0.0102     0.0420     0.5212
     10.0     0.0488     0.0224     0.1281
     30.0     0.1503     0.1079     0.0424
    100.0     0.5025     0.3754     0.2984
    300.0     1.5079     1.1305     0.9372
```

The capacitor count (27) is the rounded-up value of the 26.2 estimated in Worked Example 21.2.

E.3 Reflection and bounce-diagram calculation

Listing E.3. bounce.py — lattice-diagram waves and end voltages (Chapter 7).

```
#!/usr/bin/env python3
"""bounce.py - lattice (bounce) diagram for a resistive source and load (Chapter 6).

Prints each traveling wave and the cumulative voltage at the source and load
ends, for a step source of amplitude vs, source resistance rs, line
impedance z0 and one-way delay td, and a load resistance rl (use
float('inf') for an open circuit).
"""


def gamma(z, z0):
    """Reflection coefficient of impedance z on a line of impedance z0."""
    if z == float("inf"):
        return 1.0
    return (z - z0) / (z + z0)


def lattice(vs, rs, z0, rl, td, n_trips=6):
```

```
    gs, gl = gamma(rs, z0), gamma(rl, z0)
    wave = vs * z0 / (z0 + rs)              # launched wave
    v_src, v_load = wave, 0.0
    events = [(0.0, "source", "launch", wave, v_src)]
    t = 0.0
    for _ in range(n_trips):
        t += td                             # wave reaches the load
        refl = wave * gl
        v_load += wave + refl
        events.append((t, "load", "%+.4f in, %+.4f out" % (wave, refl), refl, v_load))
        wave = refl
        t += td                             # wave reaches the source
        refl = wave * gs
        v_src += wave + refl
        events.append((t, "source", "%+.4f in, %+.4f out" % (wave, refl), refl,
    v_src))
        wave = refl
    v_final = vs if rl == float("inf") else vs * rl / (rs + rl)
    return gs, gl, events, v_final


if __name__ == "__main__":
    # Worked Example 4.1: 3.3 V, Rs = 10 ohm, Z0 = 50 ohm, open load, TD = 1 ns
    gs, gl, events, vf = lattice(3.3, 10.0, 50.0, float("inf"), 1.0, n_trips=4)
    print("Gamma_S = %+.4f, Gamma_L = %+.4f, final value = %.3f V" % (gs, gl, vf))
    print("  t (ns)   end      waves                          V (V)")
    for t, end, desc, _, v in events:
        print("  %5.1f    %-7s  %-26s  %7.3f" % (t, end, desc, v))
```

Output:

```
Gamma_S = -0.6667, Gamma_L = +1.0000, final value = 3.300 V
  t (ns)   end      waves                          V (V)
     0.0   source   launch                           2.750
     1.0   load     +2.7500 in, +2.7500 out          5.500
     2.0   source   +2.7500 in, -1.8333 out          3.667
     3.0   load     -1.8333 in, -1.8333 out          1.833
     4.0   source   -1.8333 in, +1.2222 out          3.056
     5.0   load     +1.2222 in, +1.2222 out          4.278
     6.0   source   +1.2222 in, -0.8148 out          3.463
     7.0   load     -0.8148 in, -0.8148 out          2.648
     8.0   source   -0.8148 in, +0.5432 out          3.191
```

These values match Figure 7.1 and Worked Example 7.1.

E.4 S-parameter file check

Listing E.4. sp_workflow.py — checks a Touchstone file for loss, passivity, reciprocity, and its impedance profile (Chapter 14).

```
"""sp_workflow.py -- a minimal S-parameter sanity-check workflow (scikit-rf 2.2).

Reads a two-port Touchstone file, reports insertion and return loss at chosen
frequencies, checks passivity and reciprocity, and estimates the impedance
profile from S11 after DC extrapolation.  The default input is the SYNTHETIC
network figures/data/synthetic_dut.s2p made by tools/gen_r3_sp.py.
```

```
"""
import sys
import numpy as np
import skrf as rf

path = sys.argv[1] if len(sys.argv) > 1 else "figures/data/synthetic_dut.s2p"
net = rf.Network(path)
f = net.f / 1e9
for fq in (1.0, 5.0, 10.0):
    i = np.argmin(abs(f - fq))
    print(f"{f[i]:5.2f} GHz  S11 {net.s_db[i,0,0]:6.1f} dB  S21
    {net.s_db[i,1,0]:6.2f} dB")
sv = np.linalg.svd(net.s, compute_uv=False).max()
print(f"largest singular value {sv:.4f}  (passive if <= 1)")
print(f"reciprocity |S12-S21| max {abs(net.s[:,0,1]-net.s[:,1,0]).max():.1e}")
dc = net.extrapolate_to_dc(kind="linear")
t, step = dc.s11.step_response(window="hamming", pad=8000)
z = 50 * (1 + step) / (1 - step)
sel = (t > 0) & (t < 0.9e-9)
print(f"impedance profile: min {z[sel].min():.1f} ohm at
    {t[sel][z[sel].argmin()]*1e12:.0f} ps")
```

Output for the synthetic file *figures/data/synthetic_dut.s2p*:

```
 1.00 GHz  S11  -35.1 dB  S21  -0.41 dB
 5.00 GHz  S11  -22.7 dB  S21  -1.27 dB
10.00 GHz  S11   -6.8 dB  S21  -3.55 dB
largest singular value 0.9963  (passive if <= 1)
reciprocity |S12-S21| max 5.8e-14
impedance profile: min 39.6 ohm at 645 ps
```

Appendix F

Interface Quick-Reference Sheets

Each sheet condenses the board-level rules for one interface into a single page for use during schematic and layout review. The values are typical starting points drawn from the chapters of Part V; the controlling documents are the interface specification and the silicon vendor’s design guide, which must be checked for every project.

F.1 DDR4 / DDR5 / LPDDR4 / LPDDR5

Signals	DQ, DM/DBI, DQS (byte lanes); CA, control; CK. LPDDR5/5X: WCK per byte and RDQS where used
Clocking	DDR4/5, LPDDR4/4X: DQ source synchronous to DQS; CA to CK. LPDDR5/5X: writes to WCK, reads to RDQS, CA to CK (Table 28.2)
Topology	Fly-by for CA/CK on multi-device ranks; point-to-point for LPDDR
Impedance	36–50 Ω single-ended (often 40 Ω); 80–100 Ω diff. CK/DQS
Termination	ODT on DQ; V_{TT} resistors at fly-by end (DDR4 CA); on-die CA termination (DDR5)
Matching	Per timing group, per controller guide; CK–DQS (or WCK–CK) within the leveling range
Layers	Byte lane on one layer, same via count, ground referenced
Spacing	$\geq 3H$ between byte lanes; more for long parallel runs
VREF	Internal and trained for DQ; external CA VREF (DDR4) decoupled and isolated
PDN	Low-impedance V_{DDQ}; V_{TT} that sources and sinks
Bring-up	Capture training results and 2-D eye margins per bit
Common failures	Leveling range exceeded; byte-lane crosstalk; V_{TT}/V_{DDQ} noise
References	Chapter 28; JEDEC [44], [57], [65], [66]

F.2 PCI Express

Signals	TX and RX differential pairs per lane; REFCLK (HCSL); PERST#, CLKREQ#, WAKE#
Rates	Gen3 8, Gen4 16, Gen5 32 GT/s NRZ; Gen6 64 GT/s PAM4
Impedance	85 Ω differential nominal; ±10 %
Coupling	AC caps on TX, symmetric; void reference under pads only where a 3-D model shows it helps; $f_{\mathrm{HP}} = 1/[2\pi(R_S + R_L)C]$
Skew	Intra-pair tight and compensated near bends; lane-to-lane loose
Loss	Budget at Nyquist across packages, board, connector, card
Vias	Back-drill or deep-layer routing at Gen4+; return via at each transition
Crosstalk	Separate TX and RX groups; ≥ $3H$–$5H$ pair spacing
REFCLK	100 MHz, CC/SRNS/SRIS architecture chosen early; SSC compatibility
Laminate	Mid-loss to Gen4; low or very-low loss at Gen5/Gen6
Bring-up	Link width and speed; lane margining; eye monitors
References	Chapter 29; PCI-SIG [67]–[69]

F.3 USB and Type-C

Signals	USB 2.0 D+/D−; SuperSpeed/USB4 TX/RX pairs; CC1/CC2; SBU; VBUS
Impedance	90 Ω differential nominal (check per generation)
USB 2.0	Tie A6/B6 and A7/B7 at the receptacle with short stubs
SuperSpeed	TX AC caps; orientation mux or two-lane PHY; redriver/retimer if needed
ESD	At connector; sub-picofarad on high-speed lanes; short ground
CC	R_p/R_d or PD controller; $R_d = 5.1\,\text{k}\Omega$ in sinks
VBUS	Wide copper; inrush limits; overvoltage protection
EMI	CM choke option on USB 2.0; shield bonded to chassis at connector
Bring-up	Test both plug orientations; all supported speeds
References	Chapter 30; USB-IF [70]–[73]

F.4 Ethernet

Media side	Four 100 Ω MDI pairs; magnetics; RJ45
Isolation	Gap under magnetics free of copper on all layers; creepage per safety standard
Termination	Center taps per PHY type; Bob Smith network to chassis
Layout	Magnetics close to RJ45; pairs matched within pair; minimal vias
RGMII	One 1.5–2 ns clock delay per direction (internal preferred)
SGMII	1.25 Gb/s, 100 Ω, usually AC coupled
10GBASE-KR	10.3125 Gb/s; loss/RL/crosstalk per IEEE 802.3
EMI	LED and other lines routed on PHY side of the gap
Bring-up	Link at all speeds; RGMII delay settings in both devices
References	Chapter 31; IEEE 802.3 [74]

F.5 MIPI CSI-2 / DSI

PHYs	D-PHY (pairs + clock lane) or C-PHY (trios, embedded clock)
Impedance	D-PHY 100 Ω differential; C-PHY 50 Ω per wire
Modes	HS (terminated, low swing) and LP (1.2 V, unterminated) on same wires
Matching	D-PHY lane-to-clock and intra-pair; C-PHY within each trio
Coupling	DC coupled; no AC caps
Crosstalk	Spacing for LP-to-HS coupling (LP swing ≈ 6× HS)
Flex	Specify FPC stackup, impedance, ground pins, and hatch pitch
Bring-up	PRBS or test patterns where supported; check both modes
References	Chapter 32; MIPI [75]–[78]

F.6 Clocks

Sources	XO, TCXO, VCXO, MEMS; crystals with C_L-matched load caps
Standards	LVCMOS (series term.), LVDS, LVPECL, HCSL, CML—terminate per data sheet
Topology	Point to point; one load per output; fanout buffers
Jitter	Integrate phase noise over the application's band; RSS independent terms
PLL	Loop bandwidth at crossover of scaled reference and VCO noise; phase margin from C_1/C_2
Supplies	Filtered supply (ferrite or LDO) for each PLL and buffer
Routing	Inner layers, ground referenced, $\geq 3H$–$5H$ from other nets
EMI	SSC where allowed; slowest acceptable edges; keep away from connectors
References	Chapter 33

Appendix G

Glossary

AC coupling Series capacitors that block DC between transmitter and receiver while passing the signal (Chapter 29).

Aggressor The net whose switching induces crosstalk in another (the victim) (Chapter 12).

Anti-pad Clearance hole in a plane around a via that is not connected to it (Chapter 15).

Anti-resonance Impedance peak formed by an inductive branch in parallel with a capacitive branch of a PDN (Chapter 21).

Back-drilling Removing the unused portion of a plated via barrel with an oversized drill to shorten its stub (Chapter 15).

Baseline wander Drift of the received level of an AC-coupled link during long runs of identical symbols (Section 29.3).

Bathtub curve Plot of BER versus sampling phase across a unit interval (Chapter 17).

BER Bit-error ratio: the fraction of received bits in error.

Bob Smith termination Resistor network from Ethernet cable-side center taps to chassis through a high-voltage capacitor (Chapter 31).

Boundary scan IEEE 1149.1 test method that drives and samples device pins through a serial chain (Chapter 38).

Causality Property of a physical system that its response does not precede its stimulus; required of dielectric and S-parameter models (Chapters 11 and 36).

CDR Clock and data recovery: a receiver circuit that extracts timing from the data transitions (Chapter 17).

Common mode The average of the voltages on the two conductors of a pair (Chapter 13).

Critical length Length beyond which an interconnect must be treated as a transmission line, $\ell_{\text{crit}} = t_r/(k t_{\text{pd}})$ (Chapter 2).

CTLE Continuous-time linear equalizer: an analog peaking filter in a receiver (Chapter 17).

Cursor One sample per UI of the channel's pulse response; the main cursor is the largest, with precursors before and postcursors after it (Section 17.4.3).

DBI Data-bus inversion: optional inversion of a byte to reduce simultaneous switching (Chapter 20).

De-embedding Mathematical removal of test fixtures from a measurement (Chapter 37).

DFE Decision-feedback equalizer: cancels post-cursor ISI using previous decisions (Chapter 17).

Dk, Df Dielectric constant (relative permittivity) and dissipation factor (loss tangent) of a laminate (Chapters 9 and 11).

Dual-Dirac model Jitter model representing DJ by two delta functions convolved with Gaussian RJ (Chapter 17).

ESL, ESR Equivalent series inductance and resistance of a capacitor (Chapter 21).

Evidence label The bracketed tag at the end of each figure caption stating whether the figure is analytical, simulated, extracted, measured or conceptual (Section 1.5).

Eye diagram Overlay of many unit intervals of a waveform showing amplitude and timing margin (Chapter 17).

FEXT Far-end crosstalk: forward-traveling coupled noise arriving at the victim's far end (Chapter 12).

FFE Feed-forward equalizer: a finite-impulse-response filter, usually in the transmitter (Chapter 17).

Flit Fixed-size flow-control unit (256 bytes in PCIe Gen6) that carries payload, CRC, and FEC together (Section 34.1).

Fly-by topology Daisy-chained routing of DDR command, address, and clock past each memory device (Chapter 28).

Glass-weave skew Skew caused by local D_k differences between glass and resin in a laminate (Chapter 9).

Ground bounce Shift of a die's ground relative to the board due to $L\,\mathrm{d}i/\mathrm{d}t$ in shared return paths (Chapter 20).

Group delay The negative slope of the unwrapped transmission phase with respect to angular frequency (Section 14.6).

IBIS I/O Buffer Information Specification: behavioral buffer models (Chapter 36).

Insertion loss $-20\log_{10}|S_{21}|$: attenuation of a signal through a channel (Chapter 11).

ISI Inter-symbol interference: residual energy from one bit corrupting others, mainly due to loss and reflections (Chapters 11 and 17).

Knee frequency $f_{\text{knee}} \approx 0.5/t_r$, the upper limit of a digital edge's significant spectrum (Chapter 2).

Lattice diagram Distance–time diagram of reflections on a line (Chapter 7).

Loop inductance Inductance of a closed current path, including its return (Chapter 3).

Microvia Laser-drilled blind via spanning one thin dielectric layer (Chapter 15).

Mixed-mode S-parameters Differential and common-mode scattering parameters formed from single-ended ports, including mode-conversion terms S_{cd} and S_{dc} (Section 14.8).

Mode conversion Transfer of energy between differential and common modes due to asymmetry (Chapter 13).

NEXT Near-end crosstalk: backward-traveling coupled noise at the victim's near end (Chapter 12).

ODT On-die termination (Chapters 7 and 28).

PAM3, PAM4 Pulse-amplitude modulation with three or four levels; about 1.58 or 2 bits per symbol (Section 5.4).

PAM4 Four-level pulse amplitude modulation, two bits per symbol (Chapter 29).

Passivity Property that a network does not generate energy; required of S-parameter models (Chapter 36).

PDN Power distribution network from regulator to die (Chapter 21).

Phase noise Single-sideband noise power density of an oscillator relative to its carrier, in dBc/Hz (Chapter 33).

Power-good (PG) Regulator output that asserts when its output is within a window of its setpoint; used to chain sequencing (Section 25.3).

Ψ_{JT} Junction-to-top characterization parameter used to estimate junction temperature from a measured package-top temperature (Section 26.2).

PSIJ Power-supply-induced jitter: edge-timing variation caused by supply noise acting on driver delay (Chapter 20).

Pulse response A channel's received response to a single one-UI pulse; sampled once per UI it gives the cursors used for equalization analysis (Section 17.4.3).

RDQS LPDDR5/5X differential read data strobe, returned by the DRAM with read data in configurations that use it (Section 28.1.2).

Renormalization Converting S-parameters from one reference impedance to another (Section 14.10).

Retimer A protocol-aware device that recovers and re-transmits a serial signal, splitting a channel into two electrically independent segments (Chapter 29).

Return loss $-20\log_{10}|S_{11}|$: a measure of reflection (Chapter 36).

Rise time Time for an edge to go from 10 % to 90 % of its swing (Chapter 2).

S-parameters Scattering parameters relating incident and reflected waves of a network (Chapter 36).

Saturation length Coupled length beyond which NEXT stops growing, $t_r v_p/2$ (Chapter 12).

Shunt-through measurement Two-port VNA method for milliohm impedances, with both ports connected to the same node; $Z \approx (Z_0/2)S_{21}$ (Section 37.3).

Skin depth Depth at which current density falls by $1/e$ in a conductor at high frequency (Chapter 3).

SSC Spread-spectrum clocking (Chapter 27).

SSN Simultaneous switching noise (Chapter 20).

Stitching capacitor Capacitor that provides a high-frequency return path across a plane split or between planes (Chapter 23).

Stitching via Via connecting ground planes on different layers (Chapter 23).

Stub Unused branch of a signal path, such as the barrel of a via below its last connection (Chapter 15).

Supervisor Voltage monitor that holds a reset output asserted until a rail is within tolerance for a set time (Section 25.3).

T network Differential termination that matches both modes: Z_{odd} from each line to a center node and $(Z_{\text{even}} - Z_{\text{odd}})/2$ to ground (Chapters 7 and 13).

Target impedance Maximum PDN impedance that keeps supply noise within tolerance, $\Delta V/\Delta I$ (Chapter 21).

TDR Time-domain reflectometry (Chapter 37).

Touchstone Text file format for network parameters (.s*n*p) (Section 14.9).

Unit interval (UI) Duration of one symbol (Chapter 2).

VCONN Power supplied on the unused CC position of a Type-C plug to electronics in the cable (Section 34.2).

Victim The net receiving crosstalk (Chapter 12).

WCK LPDDR5/5X differential data (write) clock, separate from the command clock CK (Section 28.1.2).

Write leveling DDR training that aligns each byte lane's DQS to CK at its memory device (Chapter 28).

Appendix H

Study Paths and Integrated Exercises

The chapters are written so that each can be read on its own, but design problems cross chapter boundaries. This appendix gives two things: study paths through the book for readers with different goals, and integrated exercises that carry one problem from the interconnect to a decision. The exercises use illustrative requirements and make no compliance claims; each states what its model still lacks.

H.1 Study paths

Table H.1 groups the material into a core path and six advanced modules. The core path is enough to design and review a typical ten-layer board; each module adds the analysis needed for one class of hard problem. Readers already fluent in the foundations can take the diagnostic problems marked *(Approximation failure)* in Chapters 2, 4 and 7 as a self-test before skipping ahead.

Table H.1. Core path and advanced modules.

Path	Sections	Integrated exercise
Core	Chapters 1 and 2; Chapter 4; Chapters 6 and 7; Chapter 18; Chapter 9; Section 21.1.1; Section 41.7	Case-study ledger (Table 41.4)
A: Channel modeling	Sections 4.6.1, 11.1, 36.3.4 and 37.2.1	—
B: SERDES and PAM4	Sections 17.4.3, 17.4.5 and 29.6	Appendix H.2
C: Source-synchronous timing	Sections 17.1, 28.1.1 and 28.1.2	—
D: PDN co-design	Chapter 19; Sections 21.1.1, 21.4, 21.6 and 37.3	Appendix H.3
E: Tolerance and validation	Sections 9.7, 15.5.1, 37.1.2 and 39.2.1	Appendix H.4
F: EMC diagnosis	Sections 2.3.1 and 27.2; Chapter 23	—

Three kinds of review problem recur in the chapters and are labeled in the text: *approximation failure* problems ask where a quick formula breaks; *inverse design* problems start from a requirement and solve for a dimension or a tolerance; and *imperfect data* problems give a measurement or a coupon set that does not, by itself, support the obvious conclusion.

H.2 Exercise A: from pulse response to receiver margin

Purpose: connect the interconnect chapters, the equalization of Chapter 17 and the simulation workflow of Chapter 36. Synthetic sampled channel; no compliance claim.

Table H.2. Design inputs for Exercise A.

Input	Value or condition
Symbol rate and signaling	16 GBd NRZ; UI = 62.5 ps; levels ± 1
Sampled channel	$h_0 = 1$; $h_{-1} = -0.08$; $h_1, h_2, h_3 = 0.25, 0.12, 0.05$
Noise at the slicer	Independent Gaussian, $\sigma = 0.03$ (cursor units)
Initial receiver	Threshold 0; fixed sampling phase; no jitter or CDR error
Candidate receiver	Three-tap DFE with exact taps and correct prior decisions

Step 1: amplitude budget. Without feedback the worst-case ISI is $0.08 + 0.25 + 0.12 + 0.05 = 0.50$, so the minimum distance to threshold is 0.50 (16.7σ). With the ideal DFE only the precursor remains: 0.92 (30.7σ). These are static amplitude distances, not BER values (Worked Example 17.5).

Step 2: a plausible failure. Add a reflection postcursor of 0.15 at $k = 8$ (500 ps). It lies outside the tap span; the half-height falls to 0.77. Find the via, connector or package discontinuity whose round-trip delay is 500 ps ($\approx$ 1.5 in of stripline one way), and compare two remedies: a longer-span or floating-tap DFE, and a board change that removes the echo.

Step 3: what the model lacks. Sampling-phase error and CDR response, noise correlation, tap quantization and adaptation limits, transmitter swing limits, and decision errors (one wrong decision on the first tap costs 0.50). The correct-decision subtraction is an upper bound; an adaptive bit-by-bit simulation should report convergence time and error propagation, and a statistical analysis should state its assumptions (Section 17.4.5).

Required submission. Cursor table, tap values, residual-ISI budget, before-and-after eyes at the same reference plane, and a short sensitivity sweep. The decision must name the impairment that controls the margin and argue whether the board or the receiver is the more robust place to fix it.

Extension. Repeat with PAM4 levels ± 1, $\pm \frac{1}{3}$ and three thresholds; track the three eye margins and level-dependent distortion, and explain why FEC changes the reliability interpretation without restoring analog eye height (Section 29.6).

H.3 Exercise B: from a rail tolerance to a validated PDN

Purpose: turn the target-impedance calculation of Chapter 21 into a design process. Allocations are illustrative requirements for the exercise.

Step 1: define the observation port. Board load pads or die bumps? State whether the load current is board or die current, include the package model and on-die capacitance when predicting die voltage, and reserve measurement uncertainty inside the allocation before using it as an acceptance threshold.

Step 2: full model and meaningful corners. Use each capacitor's biased capacitance, ESR and mounted ESL (Eq. (21.4)); include shared vias, spreading inductance, regulator output impedance and the package. Sweep frequency and port location, and check the corners and the correlations that the mounting geometry creates (Figures 21.4 and 21.6).

Table H.3. Rail budget for Exercise B.

Item	Allocation
Nominal and allowed deviation	0.8 V ±3 % = ±24 mV
Set point and regulator error	±4 mV
Uncompensated DC path drop	6 mV maximum
Dynamic allocation	14 mV for the specified worst-direction transient
Load change	10 A increase, defined ramp and dwell
Initial dynamic target	14 mV / 10 A = 1.4 mΩ at the chosen port

Step 3: excite with the real current. Apply the specified rise time, amplitude, repetition and dwell; a periodic load can excite a narrow resonance that an isolated step does not (Section 21.6). Compare peak droop, overshoot and settling with the asymmetric budget.

Step 4: plan correlation before fabrication. Place a shunt-through measurement point (Section 37.3), document the measurement floor, and compare simulated and measured impedance at the same port. Record temperature, probe setup and bias. Explain residual differences rather than forcing a fit by changing several model parameters at once.

Required submission. Allocation table, network diagram, biased capacitor inventory, frequency and transient plots, corner table, measurement plan, and one paragraph defending the remaining margin.

H.4 Exercise C: from a coupon report to a production decision

Purpose: practice reasoning from imperfect data (Chapters 9, 37 and 39).

A first lot of boards arrives with coupon reports for the 85 Ω differential class: ten coupons from two panels, mean 88.9 Ω, all within ±10 %. The reports state a 35 ps TDR edge but no evaluation window. A PCIe Gen5 link on the board shows less margin than simulated.

1. List what the coupon data can and cannot establish: within-lot versus lot-to-lot variation; coupon versus board geometry; average versus local impedance; the effect of the unstated window (Section 39.2.1).
2. Estimate how much of the margin loss a 4.6 % impedance offset could explain, using the return-loss rule of thumb of Chapter 7, and what else could explain it (stub length after back-drill, Section 15.5.1; de-embedding error, Section 37.2.1).
3. Specify the additional measurements—TDR of a production trace with a stated edge and window, a back-drill cross-section, a fixture self-check—and the decision each could change.

What a good answer avoids: concluding from the coupon mean alone that the fabrication is either the cause or not the cause.

Appendix I

Templates, Trees, Interface Table, Credits, and Build

This appendix collects the working material of the revision-3 chapters: blank templates for rail, sequencing, and thermal budgets and for a power bring-up record; troubleshooting trees; a one-page interface comparison; a formula supplement; the image-credits manifest; and the list of data files, scripts, and build steps.

I.1 Templates

Rail budget (Section 24.1). One row per rail; compute from the leaves toward the input.

Rail	V, tol.	$I_{\max}$	Step, $\mathrm{d}I/\mathrm{d}t$	Noise limit	Source	Regulator, η	Loads and datasheet references

Sequencing constraints (Section 25.1).

Rail/signal A	Relation	Rail/signal B	Value (min/max)	Source (document, revision, page)
	before / tracks / $\leq$			

Thermal budget (Section 26.1).

Part	P (W)	Path (top/board)	Metric used	T_J estimate	Assumptions (airflow, T_A, copper)

Power bring-up record (Section 25.6).

Field	Entry
Board, revision, serial; date; operator	
Instruments, probes, bandwidth limits, trigger	
Input voltage(s) and temperature(s) tested	
Order and gaps (measured vs required)	
Ramp times, monotonicity, overshoot	
Relative-voltage limits during ramps	
Reset timing (assert and release)	
Power-down order and residual voltages	
Cycles run, fast-cycle test, violations	
Screenshots and file names	

I.2 Troubleshooting trees

Figure 40.3 in Case 7 gives the tree for failures that depend on a regulator or on temperature. Figure I.1 is a general tree for a high-speed link that does not train or shows errors.

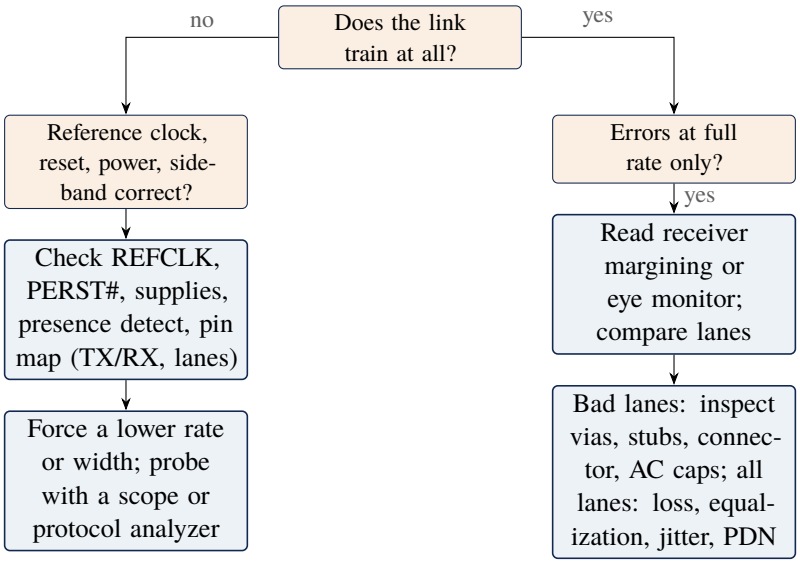


Figure I.1. A general troubleshooting tree for a high-speed serial link. [Conceptual]

I.3 Interface comparison

Table I.1. Interface comparison (one lane; commonly published values, access date 10 November 2026). See Chapters 34 and 35 and Part VI for sources and details.

Interface	Lane rate	Modulation	Nyquist	Z_{diff}	Coupling / termination
PCIe Gen5	32 GT/s	NRZ	16 GHz	85 Ω	AC; internal
PCIe Gen6	64 GT/s	PAM4	16 GHz	85 Ω	AC; internal
USB 3.2 Gen 2	10 Gb/s	NRZ	5 GHz	90 Ω	AC; internal
USB4 v2	40 Gb/s	PAM3	12.8 GHz	85 Ω	AC; internal
SATA 6 Gb/s	6 Gb/s	NRZ	3 GHz	100 Ω	AC; internal
DisplayPort UHBR20	20 Gb/s	NRZ	10 GHz	100 Ω	AC; internal
HDMI (TMDS)	up to 6 Gb/s	NRZ	3 GHz	100 Ω	DC; sink pull-up 50 Ω to 3.3 V
MIPI D-PHY v3.0	up to 9 Gb/s	NRZ	4.5 GHz	100 Ω	DC; receiver termination in HS
DDR5-6400	6.4 GT/s	NRZ	3.2 GHz	per guide	on-die (POD to VDDQ)
10GBASE-KR	10.3125 Gb/s	NRZ	5.16 GHz	100 Ω	AC; internal
LVDS	up to ~1 Gb/s	NRZ	—	100 Ω	DC; 100 Ω at receiver

I.4 Formula supplement

Quantity	Formula (chapter)
Return loss, VSWR, input impedance	$\mathrm{RL} = -20\log_{10}\lvert S_{11}\rvert$; $\mathrm{VSWR} = \frac{1+\lvert S_{11}\rvert}{1-\lvert S_{11}\rvert}$; $Z_{\text{in}} = Z_{\text{ref}}\frac{1+S_{11}}{1-S_{11}}$ (Chapter 14)
Group delay; unwrapping step	$\tau_g = -\frac{1}{2\pi}\frac{\mathrm{d}\phi}{\mathrm{d}f}$; $\Delta f < 1/(2\tau)$
Skew mode conversion	$\lvert S_{cd21}\rvert = \lvert\sin \pi f \Delta t\rvert$; $\lvert V_c/V_d\rvert = \frac{1}{2}\lvert\sin \pi f \Delta t\rvert$
Impedance from capacitances	$Z_0 = 1/(c_0\sqrt{CC_{\text{air}}})$; $\varepsilon_{\text{eff}} = C/C_{\text{air}}$; $Z_{\text{diff}} = 2Z_{\text{odd}}$ (Chapter 10)
Turn skew	$\Delta\ell = 2pn\tan(\theta/2n)$; arc: $p\theta$ (Chapter 16)
V_{TT} current	$I = \pm\frac{V_{DDQ}/2}{R_{\text{on}}+R_{TT}}$ (Chapter 8)
T termination	$R_1 = Z_{\text{odd}}$, $R_2 = \frac{1}{2}(Z_{\text{even}} - Z_{\text{odd}})$
Self-resonance; bead resonance	$f = 1/(2\pi\sqrt{LC})$; $Z_c = \sqrt{L/C}$ (Chapter 22)
Buck ripple	$\Delta I_L = \frac{(V_{\text{in}}-V_{\text{out}})D}{f_{\text{sw}}L}$; $\Delta V \approx \frac{\Delta I_L}{8f_{\text{sw}}C} + \Delta I_L\mathrm{ESR}$ (Chapter 24)
Input-capacitor RMS	$I_{\text{out}}\sqrt{D(1-D)}$
RC delay	$t_d = RC\ln\frac{V_S}{V_S-V_{\text{th}}}$ (Chapter 25)
Via conduction	$R = L/(k\pi(r_o^2 - r_i^2))$ (Chapter 26)
Junction from case top	$T_J = T_{\text{top}} + P\Psi_{JT}$
Loaded line	$t'_{\text{pd}} = t_{\text{pd}}\sqrt{1 + C_d/C_{\text{seg}}}$; $Z'_0 = Z_0/\sqrt{1 + C_d/C_{\text{seg}}}$ (Chapter 40)

I.5 Image-credits manifest

No photographs, scanned drawings, or third-party images are used in this book. Diagrams, schematics, and plots were drawn for this book in TikZ, CircuiTikZ, and PGFPlots, or were plotted by PGFPlots from data generated by the scripts listed below. Pin maps and contact counts shown in diagrams were redrawn from the cited public sources; no vendor artwork was copied.

The figures labelled *Illustration* are rendered illustrations supplied with the manuscript (directory *figures/images*). The three platform boards and the assembled system of Chapter 41 were designed in Altium Designer, and their 3D models were rendered in KeyShot; the boards have not been manufactured. The component and connector illustrations in Chapters 8, 22, 24 to 26, 34 and 35 and Chapter 39. They are not photographs of specific products. Part markings and reference designators visible in them are illustrative.

The USB and DisplayPort illustrations were cropped to remove renderings whose contacts were not accurate. Product and organization names are used only to identify the standards and sources cited.

I.6 Data files, scripts, and build instructions

Scripts (directory *tools*; Python 3.11 or later):

- *fd2d.py*: non-uniform finite-difference Laplace solver for cross-sections (validated against Hammerstad–Jensen to about 1 %).
- *gen_r3_imp.py*: impedance designs, sensitivities, Monte Carlo (*imp_*.csv*, *imp_designs.json*); a few minutes.
- *gen_r3_sp.py*: synthetic S-parameter networks and their views (*sp_*.csv*, *synthetic_dut.s2p*, *synthetic_pair.s4p*).
- *gen_r3_route.py*: bend reflection, turn skew, modal skew-correction model (*route_**); needs *imp_designs.json*.
- *gen_r3_term.py*: Bergeron waveforms of terminations (*term_**).
- *gen_r3_power.py*: capacitor, bead, PDN, buck loop, RC delay, and thermal data (*cap_**, *pdn_**, *buck_loop*, *seq_rc*, *therm_**).
- *gen_r3_rails.py*: the SBC rail budget (*rails_**).
- *gen_r3_cases.py*: case-study numbers and the USB path model (*case_**).
- *make_figplan.py*: regenerates the figure-plan table of the front matter.
- *appendices/code/sp_workflow.py*: the S-parameter check of Chapter 14.

The JSON files hold the numbers quoted in the text; each CSV has a header row naming its columns, and every Touchstone file written by the scripts begins with a comment stating that it is synthetic.

Python requirements (file *requirements.txt*): numpy, scipy, scikit-rf 2.2, pandas, and matplotlib (optional, for quick checks).

Build.

1. Install a TeX distribution with XeLaTeX, biber, and makeindex, and the TeX Gyre Termes fonts (used when Times New Roman is not installed).
2. Optionally regenerate the data: run *gen_r3_imp.py* first, then the other generators in any order, then *make_figplan.py*.
3. Run *latexmk* in the project root; the supplied *latexmkrc* selects XeLaTeX, biber, and makeindex. A full build takes several minutes because of the plots.

Answers to Review Problems

Numerical answers are given to two or three significant figures, which is the accuracy justified by the approximations used in most problems. Where a problem asks for a design choice, a brief justification is given instead.

Chapter 1: The PCB as an Electrical System

P1.1 Pressed dielectric thickness (impedance rises with thickness); etched width and trapezoid (narrower traces raise impedance); dielectric permittivity of the chosen glass and resin (higher D_k lowers impedance). Plated thickness and roughness are also acceptable answers.

P1.2 $T_D = 290\,\text{ps}$. With 1 ns: $T_D/t_r = 0.29$, between lumped and distributed, marginal. With 80 ps: $T_D/t_r = 3.6$, a transmission line.

P1.3 $R = 1.724 \times 10^{-8} \times 0.254/(203 \times 10^{-6} \times 34.8 \times 10^{-6}) \approx 0.62\,\Omega$.

P1.4 $C_{tot} = T_D/Z_0 = 10\,\text{pF}$; $L_{tot} = Z_0 T_D = 25\,\text{nH}$.

P1.5 The display is round-trip: $T_D = 0.6\,\text{ns}$, so the distance is $0.6/0.170 \approx 3.5\,\text{in}$.

P1.6 Slot: discontinuity and crosstalk from shared detour; noise coupled into the plane cavity; loop and slot radiation. Unstitched reference change: return-path discontinuity; noise injected between the planes; excitation of plane-pair resonances and radiation from the board edge.

Chapter 2: What Makes a Design "High-Speed"

P2.1 $f_{3dB} \approx 10\,\text{GHz}$; $f_{knee} \approx 14.3\,\text{GHz}$.

P2.2 $T_D = 580\,\text{ps}$; $\ell_{crit} = 2.30\,\text{in}$; the 4 in line is distributed ($T_D/t_r = 0.29 > 1/6$).

P2.3 Scope $t_r = 43.75\,\text{ps}$; displayed $\approx 74.3\,\text{ps}$.

P2.4 $t_{pd} \approx 156\,\text{ps/in}$; $T_D \approx 1.09\,\text{ns}$.

P2.5 $f_1 = 15.9\,\text{MHz}$; $T_{ramp} = 1.25\,\text{ns}$, so $f_2 = 255\,\text{MHz}$; $f_{knee} = 500\,\text{MHz}$.

P2.6 UI = 62.5 ps; $f_{Nyq} = 8\,\text{GHz}$; $t_r = 25\,\text{ps}$, so $f_{knee} = 20\,\text{GHz}$.

Chapter 3: Electromagnetic Fundamentals

P3.1 6.61 µm and 0.661 µm.

P3.2 $C \approx 10.8\,\text{nF}$.

P3.3 70.5 %; 87.4 %.

P3.4 $L_\square = \mu_0 h \approx 128\,\text{pH}$ per square.

P3.5 $L_{loop} \approx 0.94\,\text{nH}$.

P3.6 $v = L\,\mathrm{d}i/\mathrm{d}t = 0.4\,\text{V}$.

P3.7 $R \approx 2.48\,\Omega$.

Chapter 4: Transmission Line Theory

P4.1 $Z_0 = 50\,\Omega$; $t_{pd} = 180\,\text{ps/in}$; $\varepsilon_r \approx 4.51$.

P4.2 $t_{pd} = 158.5\,\text{ps/in}$; $C' = 3.52\,\text{pF/in}$; $L' = 7.13\,\text{nH/in}$.

P4.3 $\alpha_d \approx 0.140\,\text{dB/in}$.

P4.4 $\sqrt{50 \times 100} = 70.7\,\Omega$.

P4.5 $\theta = 1.068\,\text{rad}$; $Z_{in} = \text{j}90.95\,\Omega$, i.e. 14.47 nH, versus $L'\ell = 8.50\,\text{nH}$: a ratio of $\tan\theta/\theta = 1.70$ (Eq. (4.17)).

P4.6 $f = R'/(2\pi L') \approx 4.0\,\text{MHz}$.

P4.7 $\theta = 0.387$, so $T_D \le 0.387/(2\pi \times 8 \times 10^9) = 7.7\,\text{ps}$ (about 45 mil of stripline). Exact: $1 - 0.387/\tan 0.387 = 5.0\,\%$.

P4.8 $T_D = Z_0 C = 22.5\,\text{ps}$, i.e. 0.13 in (132 mil).

Chapter 5: Signals, Symbols, and Skew

P5.1 $W = 127\,\mu\text{m}$, $t = 30.5\,\mu\text{m}$: $R = 1.72 \times 10^{-8} \times 0.2/(127 \times 10^{-6} \times 30.5 \times 10^{-6}) = 0.89\,\Omega$.

P5.2 56 GBd; UI 17.9 ps; Nyquist 28 GHz.

P5.3 $16 \times 4 \times 128/130/8 = 7.88\,\text{GB/s}$.

P5.4 Gen6 is PAM4 at 32 GBd; Nyquist is 16 GHz.

P5.5 $20 + 14.07 \times 1.5 = 41.1\,\text{ps}$.

P5.6 $\frac{2}{\pi}\arctan 1 = 50\,\%$.

Chapter 6: What Is Signal Integrity?

P6.1 The received signal is on the correct side of the decision threshold by more than the receiver's sensitivity, within its timing window, with margin for noise and variation, so that the bit-error ratio meets the requirement.

P6.2 $\Gamma = (40 - 50)/90 = -0.111$; $-111\,\text{mV}$; 19.1 dB.

P6.3 (a) ISI from loss; (b) crosstalk; (c) overshoot from reflection/ringing.

P6.4 $10^{-12/20} = 0.25$; yes, by the rule of thumb of this chapter an unequalized NRZ eye would be small or closed.

P6.5 A TDR for location (impedance versus round-trip time); a VNA for S_{11} at 8 GHz.

P6.6 $16 - 1 - 1.5 - 3 = 10.5\,\text{dB}$, so $10.5/0.9 \approx 11.7\,\text{in}$.

P6.7 Supply noise modulates driver delay and PLL phase, producing jitter; simultaneous switching noise shifts the ground reference and closes the eye vertically.

Chapter 7: Reflections and Termination

P7.1 75 Ω: $\Gamma = 0.2$, $T = 1.2$. 25 Ω: $\Gamma = -0.333$, $T = 0.667$.

P7.2 $V_{launch} = 0.8\,\text{V}$; $\Gamma_S = -1/3$.

P7.3 1.333, 0.889 and 1.037 V.

P7.4 $R_S = 38\,\Omega$ ideal; use $39\,\Omega$.

P7.5 $\tau = Z_0 C/2 = 100\,\text{ps}$; added rise time $\approx 220\,\text{ps}$.

P7.6 $C \ge 96\,\text{pF}$; choose 100 pF.

P7.7 $R_1 = 45\,\Omega$ to the center node and $R_2 = 7.5\,\Omega$ from it to ground. The split termination gives $\Gamma_e = (45 - 60)/(45 + 60) = -0.143$.

P7.8 Only the launch and first arrival are reasonably approximated; later Γ_S values depend on the instantaneous operating point. Use the IBIS V–I and V–t tables in a nonlinear simulator (Chapter 36).

Chapter 8: Termination Networks in Detail

P8.1 $R_S = 38\ \Omega$ (use $39\ \Omega$); plateau ≈ 0.9 V for one round trip.

P8.2 $R_2/(R_1 + R_2) = 1.2/3.3$ and $R_1R_2/(R_1 + R_2) = 60$: $R_1 = 165\ \Omega$, $R_2 = 94.3\ \Omega$.

P8.3 $I = 0.675/73 = 9.25$ mA per line; $1.5/0.00925 = 162$ lines.

P8.4 $R_1 = 45\ \Omega$, $R_2 = 6.5\ \Omega$.

P8.5 $RC = 0.5$ ns is shorter than the 3 ns round trip; the capacitor is charged before the reflection returns, so the far end acts almost as an open and the line rings.

P8.6 The intent is AC coupling: the capacitors belong in series with each leg (TX side), not as shunts to ground.

P8.7 The bus is bidirectional and multi-rank; the termination must be enabled in the receiving device and disabled or parked elsewhere, which only switched on-die termination can do without stubs.

Chapter 9: PCB Stackup and Impedance Control

P9.1 $W \approx 4.7$ mil.

P9.2 $W \approx 7.5$ mil; the zero-thickness formula ignores the large plated thickness, which widens the effective strip, and the two approximations differ intrinsically.

P9.3 $Z_{\text{diff}} \approx 88\ \Omega$.

P9.4 IPC-2141 gives $Z_0 \approx 51.3\ \Omega$, but $W/(B - T) = 0.40 > 0.35$, so this is an extrapolation; Wheeler's formula gives $52.4\ \Omega$.

P9.5 $\varepsilon_r' \approx 3.63$; $Z_0 \approx 44.5\ \Omega$.

P9.6 ≈ 17.6 ps (2.2 ps/in).

P9.7 ΔH gives $1.9\ \Omega$, ΔD_k about $0.6\ \Omega$; the remaining $2.9\ \Omega$ allows $\Delta W = 2.9/5.98 \approx \pm 0.48$ mil.

P9.8 No. They show within-panel variation only (mean $52.5\ \Omega$, $s \approx 0.4\ \Omega$); lot-to-lot dielectric thickness and D_k variation, which dominate, are absent. Collect coupons across panels and lots.

Chapter 10: Impedance Geometry, Sensitivity, and Fabrication

P10.1 $W = 4$, edge gap $S = 6$: pitch 10 mil. Or $W = 4$, centre pitch 6: gap 2 mil (unlikely but possible). Always state which.

P10.2 $Z_0 = 1/(3 \times 10^8 \sqrt{122 \times 10^{-12} \cdot 41 \times 10^{-12}}) = 47.1\ \Omega$; $\varepsilon_{\text{eff}} = 2.98$.

P10.3 $Z_{\text{diff}} = 88\ \Omega$, $Z_{\text{cm}} = 26\ \Omega$; fairly loose ($Z_{\text{even}}/Z_{\text{odd}} = 1.18$).

P10.4 ε_r +5.4 % lowers Z_{diff} by about $0.54 \times 4.7 = 2.5\ \Omega$; W must shrink by about $2.5/0.42 \approx 6$ %, i.e. to about 3.73 mil.

P10.5 About $47\ \Omega$: the mask lowers the impedance by about $3\ \Omega$ (Table 10.2).

P10.6 Over-etch reduces W (raising Z) and increases S (also raising Z), so the two effects add instead of averaging.

P10.7 Not determinable: the reading must be taken in the stated window in the middle of the trace; the launch region is affected by the probe. Ask for the windowed value.

Chapter 11: Losses

P11.1 ≈ 0.57 dB/in.

P11.2 $\approx 0.127\,\text{dB/in}$.

P11.3 $f_\times \approx 2.3\,\text{GHz}$.

P11.4 $\delta_s = 0.935\,\mu\text{m}$; $K_H \approx 1.90$.

P11.5 $K \to 1 + 1.5 \times 1.2 = 2.8$.

P11.6 Conductor $\approx 1.08\,\text{dB/in}$, dielectric $\approx 0.28\,\text{dB/in}$; total $\approx 8.1\,\text{dB}$.

Chapter 12: Crosstalk

P12.1 $K_b = 0.0325$.

P12.2 $\ell_{\text{sat}} = 0.15\,\text{in}$.

P12.3 $2T_D = 33.3\,\text{ps}$; NEXT $\approx 0.0325 \times 33.3/50 = 21.7\,\text{mV}$.

P12.4 $\approx -115\,\text{mV}$.

P12.5 $K_b = 0.025$, so NEXT = 25 mV (saturated); FEXT = 0.

P12.6 From 0.100 to 0.027, a factor of 3.7 (11.4 dB).

Chapter 13: Differential Signaling

P13.1 $Z_{\text{diff}} = 96\,\Omega$; $Z_{\text{cm}} = 30\,\Omega$.

P13.2 $R_1 = 48\,\Omega$ on each trace; $R_2 = 6\,\Omega$ to ground.

P13.3 $\Delta t = 3.4\,\text{ps}$.

P13.4 0.050, i.e. −26 dB.

P13.5 $\Delta\ell = 75.4\,\text{mil}$; $\Delta t = 12.8\,\text{ps}$.

P13.6 $\Delta t = 3.75\,\text{ps}$; $\Delta\ell \approx 22\,\text{mil}$.

Chapter 14: S-Parameters for Board Designers

P14.1 $|\Gamma| = 0.200$; 4.0 %; RL = 14 dB; VSWR = 1.50.

P14.2 No; $|\Gamma| \le 10^{-12/20} = 0.251$, and −11 dB is 0.282.

P14.3 $100(1.1/0.9) = 122\,\Omega$ or $100(0.9/1.1) = 81.8\,\Omega$.

P14.4 $1 - 0.0316 - 0.631 = 0.337$, i.e. 34 %.

P14.5 $\tau = 0.8\sqrt{3.6}/c = 5.06\,\text{ns}$; $1/(2\tau) = 98.8\,\text{MHz}$. The step is just too coarse; unwrapping may skip cycles. Use a step below about 50 MHz.

P14.6 $\left|\sin(\pi \cdot 8 \times 10^9 \cdot 4 \times 10^{-12})\right| = 0.100$, i.e. −20 dB; voltage ratio 0.050.

P14.7 The file uses the side-by-side numbering (1, 2 in; 3, 4 out): S_{31} is the through path and S_{21} is near-end crosstalk.

P14.8 Investigate calibration and connection, then apply passivity enforcement or truncate the band; do not conclude the board has gain.

Chapter 15: Vias and Discontinuities

P15.1 $L \approx 2.1\,\text{nH}$.

P15.2 $C \approx 0.50\,\text{pF}$.

P15.3 $f \approx 22.2\,\text{GHz}$.

P15.4 $3f_{\text{Nyq}} = 24\,\text{GHz}$; $\ell \le 63\,\text{mil}$.

P15.5 $Z_{\text{via}} \approx 41.6\,\Omega$.

P15.6 $3/0.8 = 3.75$ mil; specify 4 mil.

P15.7 Clearance $(0.80 - 0.50)/2 = 0.15$ mm; $Z_{\text{via}} = (60/\sqrt{3.6})\ln(0.80/0.25) \approx 36.8\ \Omega$.

P15.8 $0.25\, e^{50\sqrt{3.6}/60} \approx 1.22$ mm; no—the maximum on a 1.0 mm grid is 0.85 mm.

Chapter 16: Routing Geometry, Length Matching, and Connectors

P16.1 $1.657 \times 12 = 19.9$ mil; arc $1.571 \times 12 = 18.8$ mil; saving 1.0 mil.

P16.2 Three net turns: $3 \times 15.7 = 47.1$ mil; compensate near the turns that create it.

P16.3 The bend capacitance scales with width (and W/H is large); the trace width is no longer small compared with the wavelength, and the closed form may be outside its validity range. Use a mitred bend and a field solver.

P16.4 $2.5/5.50 = 0.45$ mm (18 mil).

P16.5 In homogeneous stripline the odd and even modes travel at the same speed, so the converted common mode stays aligned and re-combines; in microstrip it falls behind.

P16.6 Overlay the footprint on the drawing's PCB-side view (3-D model or printout) and check a few asymmetric pins (pin 1, keys, power) against both ends' tables.

Chapter 17: Timing, Jitter, and Eye Diagrams

P17.1 Setup margin $= -0.05$ ns (fails); hold margin $= 0.6$ ns.

P17.2 TJ ≈ 23.3 ps.

P17.3 3.0×10^{12} bits; 187 s.

P17.4 $20\log_{10}(1.0/0.6) = 4.4$ dB.

P17.5 $20\log_{10}(8/2) = 12$ dB.

P17.6 $2b_1 = 0.50$ of the cursor. The half-height drops from 0.92 to 0.42: still positive without noise, but at 14σ rather than 30.7σ, and a further error on tap 2 (0.24) would leave 0.18.

P17.7 $1 - 0.1 - |h_3| \geq 0.6$, so $|h_3| \leq 0.30$.

P17.8 At 40 MHz: $|\mathrm{j}10/(1 + \mathrm{j}10)| = 0.995$ (practically not attenuated). At 400 kHz: 0.0995, about -20 dB.

Chapter 18: What Is Power Integrity?

P18.1 The ability of the PDN to keep each supply within its tolerance, under all operating conditions, at the point the device specifies (package balls or die).

P18.2 $L\ \mathrm{d}I/\mathrm{d}t = 50$ mV during the ramp; $IR = 10$ mV afterward.

P18.3 $\Delta V = 18$ mV; $Z_{\text{target}} = 3\ \mathrm{m}\Omega$.

P18.4 $C = L/Z_c^2 = 10^{-9}/(25 \times 10^{-6}) = 40\ \mu$F; droop about $10 \times 5 = 50$ mV plus IR drop.

P18.5 It forms a resonance at about 5 MHz with $\sqrt{L/C} = 32\ \mathrm{m}\Omega$; the ramp, about half a resonant period long, excites it and the droop grows (Figure 18.3).

P18.6 (a) Oscilloscope with a rail probe; (b) VNA shunt-through impedance; (c) oscilloscope at the load during the workload, or an injected load step.

P18.7 The noise modulates the VCO frequency and phase; the resulting clock jitter transfers to the transmitted data or the receiver's sampling clock, shrinking the eye width and raising the BER.

Chapter 19: Transient Current, Inductance, and the PDN Response

P19.1 $\mathrm{d}I/\mathrm{d}t = 2 \times 10^8$ A/s; 100 mV and 400 mV.

P19.2 $6 \times 2 \times 10^{-6}/470 \times 10^{-6} \approx 25.5\,\text{mV}$.

P19.3 $f_0 \approx 240\,\text{kHz}$; $Z_c \approx 3.0\,\text{m}\Omega$; $\zeta = 0.0015\sqrt{1.1 \times 10^5} \approx 0.50$.

P19.4 $+0.2\,\text{V}$ for 20 ns, zero for 50 ns, then $-0.4\,\text{V}$ for 10 ns.

P19.5 The capacitor supplies the fast current, so the path current does not change at $10^8\,\text{A/s}$; the simulated droop is 75 mV.

P19.6 $L = Z_c^2 C = 4 \times 10^{-6} \times 10^{-4} = 0.4\,\text{nH}$.

P19.7 The third harmonic (6.9 MHz); a square wave has strong odd harmonics, so this component, about one third of the fundamental, is multiplied by $35\,\text{m}\Omega$ instead of a few milliohms.

P19.8 The first droop (die–package). Board capacitors cannot reach it through the package inductance; it is set by on-die and package capacitance. Also confirm the probe bandwidth and grounding before trusting a nanosecond feature.

Chapter 20: Ground Bounce and SSN

P20.1 $V_{\text{gb}} = 0.32\,\text{V}$.

P20.2 0.30 nH.

P20.3 $(1.2 + 3 \times 0.2)/4 = 0.45\,\text{nH}$.

P20.4 $N \leq 15$.

P20.5 $f \approx 113\,\text{MHz}$.

P20.6 Line-driven: by a factor of two, 6 dB. Lumped capacitive loads (Eq. (20.2)): by a factor of four, 12 dB.

Chapter 21: Power Distribution Network Design

P21.1 $Z_{\text{target}} = 10\,\text{m}\Omega$.

P21.2 $f_{\text{SRF}} \approx 6.5\,\text{MHz}$.

P21.3 $N \geq 2\pi f\,\text{ESL}/Z_{\text{target}} = 15.1$; use 16. The inductive-only formula is acceptable only because the problem states that $f \gg f_{\text{SRF}}$.

P21.4 $R_\square = 0.99\,\text{m}\Omega$; $\Delta V \approx 40\,\text{mV}$.

P21.5 $f_{\text{ar}} \approx 178\,\text{MHz}$.

P21.6 $f_{\text{SRF}} = 6.5\,\text{MHz}$; $|Z_C| = 14.3\,\text{m}\Omega$, so $N = 6$. The shortcut gives $\lceil 7.85 \rceil = 8$ but ignores the capacitive reactance below the SRF, and would give an even larger error further below it.

P21.7 With $0.5\,\mu\text{F}$, $|Z_C| = 45.2\,\text{m}\Omega$ and $N = 19$: the DC-bias derating, not the ESL, now dominates. Always use biased capacitance.

P21.8 $R = \sqrt{L/C}/Q = 7.07/0.7 \approx 10\,\text{m}\Omega$; the DC drop is then 10 mV per ampere.

Chapter 22: Capacitor Technologies and Decoupling Practice

P22.1 $1/(2\pi\sqrt{1.2 \times 10^{-9} \cdot 4.7 \times 10^{-6}}) = 2.12\,\text{MHz}$.

P22.2 Effective $8.8\,\mu\text{F}$ each; $100/8.8 = 11.4$, so 12 parts (before temperature and ageing margins).

P22.3 The ESR zero of the regulator's loop moved to a much higher frequency, removing phase boost the compensation relied on; the loop lost phase margin. Recompensate or keep some ESR.

P22.4 $Z_c = \sqrt{0.6 \times 10^{-6}/22 \times 10^{-6}} = 0.165\,\Omega$; use about 0.15–0.2 Ω in series with a capacitor several times $22\,\mu\text{F}$.

P22.5 The peak is an anti-resonance between the 100 nF group's inductance and the 10 µF group; more of the same parts shift and lower it somewhat but the gap remains. Fill the band with intermediate values or lower-inductance parts and add damping.

P22.6 Capacitance versus DC bias (and temperature), and impedance/ESR versus frequency; the part may retain only a small fraction of 10 µF at 5 V, and 6.3 V rating leaves little margin.

Chapter 23: Planes and Return Paths

P23.1 492, 738 and 887 MHz.

P23.2 $\lambda = 0.622$ in; spacing ≈ 31 mil.

P23.3 $\approx 5.0\,\Omega$.

P23.4 $\Gamma = 0.23$.

P23.5 80 mil.

P23.6 $1/(\omega C) \approx 80\,\mathrm{m}\Omega$.

Chapter 24: Power-Rail Architecture and Regulator Design

P24.1 1.5 A: an LDO's input current equals its output current (plus ground current).

P24.2 $D = 4.2\,\%$; $t_{\mathrm{on}} = 52$ ns: close to typical minimum on-times; consider a lower frequency or an intermediate rail.

P24.3 $\Delta I = 0.9$ A; $D = 0.275$; $L = (12 - 3.3) \times 0.275/(5 \times 10^5 \times 0.9) = 5.3\,\mu\mathrm{H}$ (use 5.6 µH).

P24.4 0.68 W; 66 %.

P24.5 The loop does not respond instantly: until about $1/(2\pi f_c)$ the capacitor supplies the step, giving $\Delta I/(2\pi f_c C)$, here about 12 mV, plus the ESR term.

P24.6 $I = 470 \times 10^{-6} \times 48/0.02 = 1.13$ A; $E = \frac{1}{2}CV^2 = 0.54$ J.

P24.7 Input droop or ringing from the USB device's inrush on the 5 V rail; ground bounce or ESD injection at the connector; insufficient bulk on 5V0; reset supervisor threshold too close to the rail minimum.

Chapter 25: Power Sequencing and Reset

P25.1 $t_d = 0.0221 \ln(5/3.5) = 7.9$ ms.

P25.2 No: the ramp is longer than RC, so the capacitor follows the ramp and the crossing occurs roughly when the rail itself passes about 1.5 V plus a lag of order RC; the delay is measured from an ill-defined start.

P25.3 Feed the 1.0 V buck from another rail (for example 5 V) and enable 1.8 V from its PG; or keep the tree and add a load switch on the device's 1.8 V I/O branch enabled by the 1.0 V PG.

P25.4 The lowest threshold must be at or above 3.0 V: $V_{\mathrm{nom}} \geq 3.0/0.985 = 3.05$ V; check that the highest threshold, $3.05 \times 1.015 = 3.10$ V, stays below the rail's minimum regulated value.

P25.5 A downstream load or regulator turning on (inrush) during the ramp, or a current-limited source; it violates monotonicity.

Chapter 26: Thermal Design of the Board

P26.1 $71 + 1.6 = 72.6\,^{\circ}\mathrm{C}$.

P26.2 $R_{\theta JA}$ is for a JEDEC board; the real board's copper, neighbours, and airflow can change it greatly. Model the board or measure with Ψ_{JT}.

P26.3 $A = \pi(0.125^2 - 0.105^2)\,\text{mm}^2 = 0.0145\,\text{mm}^2$; $R = 1.2 \times 10^{-3}/(390 \times 1.45 \times 10^{-8}) = 212\,\text{K/W}$ each; 23.7 K/W for nine (barrels only).

P26.4 $R \leq 5\,\text{K/W}$; $190/5 = 38$ vias, which may not fit: consider filled vias (58 K/W each, 12 needed) or a thinner path to the plane.

P26.5 Unknown: bare copper has very low emissivity, so the camera reads mostly reflected ambient. Apply high-emissivity tape or use a thermocouple.

P26.6 Nearby planes and copper conduct heat away efficiently; the legacy data did not include them, while IPC-2152 does with modifiers.

Chapter 27: Emission Mechanisms and EMC Design

P27.1 $E \approx 351\,\mu\text{V/m}$ (50.9 dBµV/m).

P27.2 $\lambda = 1.5\,\text{m}$, so 0.15 m is $\lambda/10$ (just valid); $I_{CM} \approx 12\,\mu\text{A}$. A 1.5 m cable is a full wavelength long, four times $\lambda/4 = 0.375\,\text{m}$: the short-monopole formula does not apply and the cable must be modeled or measured.

P27.3 $\approx 5.9\,\text{in}$ (150 mm).

P27.4 $10\log_{10}(1.5/0.12) \approx 11\,\text{dB}$.

P27.5 By 6 dB.

P27.6 $\delta_s = 2.09\,\mu\text{m}$; $A \approx 145\,\text{dB}$.

Chapter 28: DDR4/DDR5 and LPDDR4/LPDDR5

P28.1 UI = 156.25 ps; $f_{Nyq} = 3.2\,\text{GHz}$.

P28.2 4.2 ps; about 25 mil.

P28.3 $Z_{loaded} \approx 37\,\Omega$.

P28.4 $\pm 192\,\text{mA}$.

P28.5 $4 \times 0.55 \times 210 = 462\,\text{ps}$.

P28.6 $\pm 12\,\text{mV}$.

P28.7 LPDDR5 writes are referenced to WCK and reads to RDQS (where used), not a single bidirectional DQS. You need the controller's list of timing groups, clock ratio, read mode, training ranges and the package delays to derive separate write, read and CK/CA budgets.

Chapter 29: PCI Express

P29.1 UI = 31.25 ps; $f_{Nyq} = 16\,\text{GHz}$.

P29.2 15.75 Gb/s.

P29.3 32 GBd; UI = 31.25 ps; one-third, −9.5 dB.

P29.4 $f_{HP} = 1/[2\pi(50 + 50) \times 100 \times 10^{-9}] \approx 15.9\,\text{kHz}$.

P29.5 12.7 in.

P29.6 99.5 MHz.

Chapter 30: USB and Type-C

P30.1 UI = 100 ps; $f_{Nyq} = 5\,\text{GHz}$.

P30.2 UI = 50 ps; $f_{Nyq} = 10\,\text{GHz}$.

P30.3 4 Gb/s and 9.70 Gb/s.

P30.4 $|\Gamma| = 0.333$ (9.5 dB return loss).

P30.5 $R_d = 5.1\,\text{k}\Omega$.

P30.6 ≈ 0.56 in.

Chapter 31: Ethernet

P31.1 Two bits per symbol per pair ($4 \times 2 \times 125 = 1000$).

P31.2 UI = 800 ps; $f_{\text{Nyq}} = 625$ MHz.

P31.3 250 Mb/s (DDR at 125 MHz).

P31.4 96.97 ps.

P31.5 ≈ 10.9 in.

P31.6 ≈ 16.4 dB.

Chapter 32: MIPI D-PHY and C-PHY

P32.1 UI = 667 ps; $f_{\text{Nyq}} = 750$ MHz.

P32.2 $3.5 \times 16/7 = 8.0$ Gb/s.

P32.3 10 Gb/s on ten wires.

P32.4 ≈ 1.46 GS/s.

P32.5 11.1 ps; ≈ 67 mil.

P32.6 18 mV, almost 10 % of a 200 mV HS swing.

Chapter 33: Clock Distribution

P33.1 $C_L = 15$ pF.

P33.2 ≈ 204 fs.

P33.3 ≈ 308 fs.

P33.4 $\approx 62.7°$.

P33.5 $f_z = 15.9$ kHz.

P33.6 25 MHz $\pm$ 1.25 kHz.

Chapter 34: Interface Connectors, Contacts, and Generations

P34.1 98 in total, 49 per side.

P34.2 Mechanical x16, electrical x8; the card trains to x8.

P34.3 PAM4 carries 2 bits per symbol: 128 GT/s is 64 GBd; Nyquist is half the symbol rate, 32 GHz.

P34.4 9 contacts on the SuperSpeed Type-A plus 10 on the SuperSpeed Micro-B plug.

P34.5 The plug connects only one CC contact; with a pull-down on each, the source detects the sink in either orientation.

P34.6 One per nibble: $40/4 = 10$ pairs.

P34.7 Protocol (CSI-2 version), PHY (D-PHY or C-PHY; "lanes" means pairs or trios), connector and cable type and orientation, and supply/IO levels.

Chapter 35: Related Interfaces

P35.1 About 60 Ω (two 120 Ω terminations in parallel).

P35.2 3.5 mA × 50 Ω = 175 mV, half the intended swing.

P35.3 Whether the redriver re-establishes the TMDS DC termination behaviour (it must present the DC-coupled interface the sink expects) and whether it supports the needed mode and rates.

P35.4 Bus load 60 Ω: $5 \times 60/(2R + 60) \geq 0.2$ gives $R \leq 720\,\Omega$; use 680 Ω pull-up on A and pull-down on B (check transceiver loading and power).

Chapter 36: Modeling and Simulation

P36.1 $\Gamma = 0.091$; return loss 20.8 dB.

P36.2 $2\Gamma/(1 + \Gamma^2) = 0.180$ (−14.9 dB).

P36.3 Yes: $0.04 + 0.9025 = 0.9425 \leq 1$.

P36.4 100 Ω and 25 Ω.

P36.5 0.7 (−3.1 dB).

P36.6 16.

Chapter 37: Measurement

P37.1 75 Ω.

P37.2 $\rho = -0.087$.

P37.3 $L = 2 \times 50 \times 2\,\text{ps} = 200\,\text{pH}$.

P37.4 ≈ 145 MHz.

P37.5 ≈ 75 mil.

P37.6 Scope rise time ≤ 16 ps; bandwidth ≳ 22 GHz.

P37.7 $Z \approx 25 \times 4.0 \times 10^{-4} = 10\,\text{m}\Omega$ (exact expression: 10.004 mΩ).

P37.8 The feature is shorter than Δx for that edge, so 54 Ω is an apparent, blended value (Figure 37.2). Re-measure with a faster edge, or compare with a model of the expected geometry filtered to the same 100 ps edge.

Chapter 38: Board Bring-Up and Debug

P38.1 ±3 % is ±24 mV (48 mV peak-to-peak); 30 mV fits, but leaves little for DC error.

P38.2 1.2 W.

P38.3 23 bits.

P38.4 About 200 µs.

P38.5 Half.

P38.6 3.0×10^{12} bits; about 374 s.

Chapter 39: Design for Manufacturing and Release

P39.1 Diagonal 7.21 in; limit ≈ 0.054 in.

P39.2 ≈ +1.8 Ω.

P39.3 76.5–93.5 Ω.

P39.4 ≈ 5.9 in.

P39.5 ≈ 0.096 in.

P39.6 Nominal 4 mil; worst case 1 mil.

Chapter 40: Seven Case Studies

P40.1 At 8 GHz: conductor $0.90/\sqrt{2} = 0.64$, dielectric $0.71/2 = 0.35$ dB/in; total 0.99 dB/in; $12/0.99 \approx 12$ in.

P40.2 $|\sin(\pi f\,\Delta t)| \le 0.0316$: $\Delta t \le 4.0$ ps, i.e. 0.73 mm at 5.50 ps/mm.

P40.3 $L = CZ_0^2 = 0.405$ nH; half is 0.20 nH; $\tau = L/Z_h = 3.4$ ps, about 0.58 mm.

P40.4 $C_{\text{seg}} = 1.03$ pF; factor $\sqrt{1 + 1/1.03} = 1.40$; $Z_0' = 35.6\,\Omega$.

P40.5 No: cooling changes the regulator's behaviour (frequency, ripple, current limit) as well as the channel; the result is consistent with both branches. Separate them by changing the regulator's frequency or load at constant temperature.

Chapter 41: Case Study: A Modular ARM and FPGA Embedded Platform

P41.1 The Zynq-7020 has no transceivers or PCIe block; a Zynq-7000 device with transceivers and PCIe (Z-7015 or larger) on a redesigned module would allow it.

P41.2 A standard M-key NVMe SSD fits J2 and would meet a custom pinout, risking damage; use a key position not used by standard cards, label the socket, and add a module-ID check before enabling power.

P41.3 One external port on each processor, and the remaining MAC on each side connected MAC-to-MAC over RGMII between the modules.

P41.4 The allowed PS reference-clock range and the boot and PLL configuration in the TRM; published designs commonly use 33.333 MHz, and 25 MHz is typical of an Ethernet PHY clock.

P41.5 Driven inputs back-power the unpowered module through its clamp diodes, leaving it in an undefined state and risking latch-up (Section 24.5).

P41.6 100 Ω differential (HDMI, Ethernet MDI, LVDS); 90 Ω differential (USB); 50 Ω single-ended (RGMII, SD).

Bibliography

[1] H. W. Johnson and M. Graham, *High-Speed Digital Design: A Handbook of Black Magic*. Upper Saddle River, NJ: Prentice Hall, 1993, ISBN: 0-13-395724-1.

[2] H. W. Johnson and M. Graham, *High-Speed Signal Propagation: Advanced Black Magic*. Upper Saddle River, NJ: Prentice Hall, 2003, ISBN: 0-13-084408-X.

[3] E. Bogatin, *Signal and Power Integrity—Simplified*, 3rd ed. Boston, MA: Prentice Hall, 2018, ISBN: 978-0-13-451341-6.

[4] S. H. Hall and H. L. Heck, *Advanced Signal Integrity for High-Speed Digital Designs*. Hoboken, NJ: Wiley-IEEE Press, 2009, ISBN: 978-0-470-19235-1.

[5] H. W. Ott, *Electromagnetic Compatibility Engineering*. Hoboken, NJ: Wiley, 2009, ISBN: 978-0-470-18930-6.

[6] I. Novak and J. R. Miller, *Frequency-Domain Characterization of Power Distribution Networks*. Norwood, MA: Artech House, 2007.

[7] L. D. Smith and E. Bogatin, *Principles of Power Integrity for PDN Design—Simplified: Robust and Cost Effective Design for High Speed Digital Products*. Boston, MA: Prentice Hall, 2017.

[8] TE Connectivity. "PCIe gen 5 CEM connectors." (), [Online]. Available: https://www.te.com/en/products/connectors/pcb-connectors/card-edge-connectors/pcie-gen-5cem-connectors.html (visited on 11/10/2026).

[9] *PCIE-G4/PCIE-G5 series card-edge connector datasheet*, F-224, Samtec, Jun. 3, 2025. (visited on 11/10/2026).

[10] Teledyne LeCroy. "PCI Express 6.0 physical-layer overview." (), [Online]. Available: https://teledynelecroy.com/doc/docview.aspx?id=14227 (visited on 11/10/2026).

[11] Signal Integrity Journal. "PCI-SIG releases PCIe 7.0 specification." (Jun. 11, 2025), [Online]. Available: https://www.signalintegrityjournal.com/articles/3984 (visited on 11/10/2026).

[12] *USB3SAUB3 SuperSpeed USB 3.0 cable A to Micro B datasheet*, StarTech.com. [Online]. Available: https://www.mouser.lt/datasheet/3/6177/1/usb3saub3_datasheet.pdf (visited on 11/10/2026).

[13] USB Implementers Forum. "USB-IF announces publication of new USB4 specification to enable USB 80Gbps performance." (Oct. 2022), [Online]. Available: https://www.usb.org/sites/default/files/2022-10/USB-IF%20USB%2080Gbps%20Announcement_FINAL.pdf (visited on 11/10/2026).

[14] Cambrionix. "USB Type-C pinout." (), [Online]. Available: https://knowledge.cambrionix.com/Content/Articles/USB/USB-Type-C-Pinout.htm (visited on 11/10/2026).

[15] Kingston Technology. "DDR4 memory standard." (), [Online]. Available: https://kingston.com/en/memory/ddr4 (visited on 11/10/2026).

[16] *DDR5 UDIMM/RDIMM module datasheet*, SMART Modular Technologies. [Online]. Available: https://www.smartm.com/api/download/fetch/81 (visited on 11/10/2026).

[17] MIPI Alliance. "MIPI D-PHY v3.0 doubles data rate of physical layer interface while extending power efficiency." (Sep. 2, 2021), [Online]. Available: https://www.mipi.org/press-releases (visited on 11/10/2026).

[18] D. Edwards and H. Nguyen, *Semiconductor and IC package thermal metrics*, SPRA953D, Texas Instruments, Mar. 2024. [Online]. Available: https://www.ti.com/lit/SPRA953 (visited on 11/10/2026).

[19] *Standard for determining current carrying capacity in printed board design*, IPC-2152, IPC, Bannockburn, IL, 2009.

[20] PCI-SIG. "PCI-SIG releases PCIe 6.0 specification delivering record performance to power big data applications." Press release text as republished. (Jan. 11, 2022), [Online]. Available: https://www.design-reuse.com/news/news51221.html (visited on 11/10/2026).

[21] Astera Labs. "PCI Express 5.0 architecture channel insertion loss budget." (), [Online]. Available: https://www.asteralabs.com/insights/pci-express-5-0-architecture-channel-insertion-loss-budget/ (visited on 11/10/2026).

[22] Keysight Technologies. "USB 80Gbps: What's new in USB4 version 2?" (Nov. 8, 2022), [Online]. Available: https://www.keysight.com/blogs/tech/bench/2022/11/08/usb-80gbps-whats-new-in-usb4-version-2 (visited on 11/10/2026).

[23] *Universal serial bus type-c cable and connector specification*, Release 2.x; obtain the current release from usb.org, USB Implementers Forum. [Online]. Available: https://www.usb.org/document-library/usb-type-cr-cable-and-connector-specification (visited on 11/10/2026).

[24] CNX Software. "MIPI C-PHY v3.0 brings up to 75 Gbps over three lanes." (May 9, 2025), [Online]. Available: https://www.cnx-software.com/2025/05/09/ (visited on 11/10/2026).

[25] MIPI Alliance. "MIPI M-PHY." (), [Online]. Available: https://mipi.org/specifications/m-phy (visited on 11/10/2026).

[26] Synopsys. "UFS 5.0 verification solutions for mobile, automotive, and AI storage." (), [Online]. Available: https://www.synopsys.com/blogs/chip-design/ufs-5-0-verification-mobile-automotive-ai-storage.html (visited on 11/10/2026).

[27] *DDR4 288-pin ECC DIMM datasheet*, Apacer Technology. [Online]. Available: https://www.mouser.com/datasheet/3/846/1/D31.23183S.001.pdf (visited on 11/10/2026).

[28] *Integrated circuits thermal measurement method – electrical test method (JESD51 series, incl. JESD51-2A, JESD51-7, JESD51-14)*, See the current revisions at jedec.org, JEDEC.

[29] *IEEE standard for electrical characterization of printed circuit board and related interconnects at frequencies up to 50 GHz*, IEEE Std 370-2020, IEEE, New York, 2021.

[30] *Touchstone file format specification, version 2.1*, IBIS Open Forum. [Online]. Available: https://ibis.org/touchstone_ver2.1/ (visited on 11/10/2026).

[31] *Electrical characteristics of low voltage differential signaling (LVDS) interface circuits*, TIA/EIA-644-A, See the current revision, TIA.

[32] *Electrical characteristics of generators and receivers for use in balanced digital multipoint systems*, TIA-485-A, See the current revision, TIA.

[33] *Road vehicles – controller area network (CAN) – part 2: High-speed medium access unit*, ISO 11898-2, See the current edition, ISO.

[34] L. W. Ritchey, *Right the First Time: A Practical Handbook on High Speed PCB and System Design*. Glen Ellen, CA: Speeding Edge, 2003, vol. 1.

[35] W. J. Dally and J. W. Poulton, *Digital Systems Engineering*. Cambridge, UK: Cambridge University Press, 1998.

[36] C. R. Paul, *Inductance: Loop and Partial*. Hoboken, NJ: Wiley-IEEE Press, 2010.

[37] S. Ramo, J. R. Whinnery, and T. Van Duzer, *Fields and Waves in Communication Electronics*, 3rd ed. New York: Wiley, 1994.

[38] D. M. Pozar, *Microwave Engineering*, 4th ed. Hoboken, NJ: Wiley, 2012.

[39] A. R. Djordjević, R. M. Biljic, V. D. Likar-Smiljanic, and T. K. Sarkar, "Wideband frequency-domain characterization of FR-4 and time-domain causality," *IEEE Transactions on Electromagnetic Compatibility*, vol. 43, no. 4, pp. 662–667, Nov. 2001.

[40] C. R. Paul, *Analysis of Multiconductor Transmission Lines*, 2nd ed. Hoboken, NJ: Wiley-IEEE Press, 2008.

[41] S. H. Hall, G. W. Hall, and J. A. McCall, *High-Speed Digital System Design: A Handbook of Interconnect Theory and Design Practices*. New York: Wiley-Interscience, 2000.

[42] D. Brooks, *Signal Integrity Issues and Printed Circuit Board Design*. Upper Saddle River, NJ: Prentice Hall, 2003.

[43] *Versal adaptive SoC PCB design user guide (UG863): Signals and connections for LPDDR5/5x interfaces*, AMD. [Online]. Available: https://docs.amd.com/r/en-US/ug863-versal-pcb-design/Signals-and-Connections-for-LPDDR5/5x-Interfaces (visited on 11/07/2026).

[44] *Low power double data rate 5 (LPDDR5)*, JESD209-5, See the current revision, JEDEC Solid State Technology Association, Arlington, VA.

[45] *Design guide for high-speed controlled impedance circuit boards*, IPC-2141A, IPC, Bannockburn, IL, 2004.

[46] B. C. Wadell, *Transmission Line Design Handbook*. Norwood, MA: Artech House, 1991.

[47] H. A. Wheeler, "Transmission-line properties of a strip on a dielectric sheet on a plane," *IEEE Transactions on Microwave Theory and Techniques*, vol. 25, no. 8, pp. 631–647, Aug. 1977.

[48] E. Hammerstad and Ø. Jensen, "Accurate models for microstrip computer-aided design," in *IEEE MTT-S International Microwave Symposium Digest*, 1980, pp. 407–409.

[49] Qucs project. "Qucs technical documentation: Single microstrip line." (), [Online]. Available: https://qucs.sourceforge.net/tech/node75.html (visited on 11/07/2026).

[50] *Specification for base materials for rigid and multilayer printed boards*, IPC-4101, See the current revision, IPC.

[51] P. G. Huray, O. Oluwafemi, J. Loyer, E. Bogatin, and X. Ye, "Impact of copper surface texture on loss: A model that works," in *DesignCon*, Santa Clara, CA, 2010.

[52] P. G. Huray, *The Foundations of Signal Integrity*. Hoboken, NJ: Wiley, 2009.

[53] A. Arsenovic *et al.*, *Scikit-rf: Open source RF engineering*, version 2.2.0. [Online]. Available: https://scikit-rf.org (visited on 11/10/2026).

[54] M. P. Li, *Jitter, Noise, and Signal Integrity at High-Speed*. Upper Saddle River, NJ: Prentice Hall, 2007.

[55] D. Derickson and M. Müller, Eds., *Digital Communications Test and Measurement: High-Speed Physical Layer Characterization*. Upper Saddle River, NJ: Prentice Hall, 2008.

[56] M. Swaminathan and A. E. Engin, *Power Integrity Modeling and Design for Semiconductors and Systems*. Upper Saddle River, NJ: Prentice Hall, 2007.

[57] *DDR4 SDRAM standard*, JESD79-4, See the current revision, JEDEC Solid State Technology Association, Arlington, VA.

[58] L. D. Smith, R. E. Anderson, D. W. Forehand, T. J. Pelc, and T. Roy, "Power distribution system design methodology and capacitor selection for modern CMOS technology," *IEEE Transactions on Advanced Packaging*, vol. 22, no. 3, pp. 284–291, Aug. 1999.

[59] I. Novak, Ed., *Power Distribution Network Design Methodologies*. Chicago, IL: International Engineering Consortium, 2008.

[60] B. R. Archambeault, *PCB Design for Real-World EMI Control*. Boston, MA: Kluwer Academic Publishers, 2002.

[61] M. I. Montrose, *Printed Circuit Board Design Techniques for EMC Compliance: A Handbook for Designers*, 2nd ed. New York: Wiley-IEEE Press, 2000.

[62] C. R. Paul, *Introduction to Electromagnetic Compatibility*, 2nd ed. Hoboken, NJ: Wiley-Interscience, 2006.

[63] U.S. Federal Communications Commission, *Title 47 CFR part 15—radio frequency devices*, Code of Federal Regulations, See the current version. [Online]. Available: https://www.ecfr.gov/current/title-47/chapter-I/subchapter-A/part-15.

[64] *CISPR 32: Electromagnetic compatibility of multimedia equipment—emission requirements*, See the current edition, International Electrotechnical Commission, Geneva.

[65] *DDR5 SDRAM standard*, JESD79-5, See the current revision, JEDEC Solid State Technology Association, Arlington, VA.

[66] *Low power double data rate 4 (LPDDR4)*, JESD209-4, See the current revision, JEDEC Solid State Technology Association, Arlington, VA.

[67] *PCI Express base specification, revision 5.0*, PCI-SIG, Beaverton, OR, 2019.

[68] *PCI Express base specification, revision 6.0*, PCI-SIG, Beaverton, OR, 2022.

[69] *PCI Express card electromechanical specification*, See the revision matching the target PCIe generation, PCI-SIG, Beaverton, OR.

[70] *Universal serial bus specification, revision 2.0*, USB Implementers Forum, 2000.

[71] *Universal serial bus 3.2 specification*, See the current revision, USB Implementers Forum.

[72] *USB4 specification*, See the current version, USB Implementers Forum.

[73] *Universal serial bus Type-C cable and connector specification*, See the current release, USB Implementers Forum.

[74] *IEEE standard for ethernet*, IEEE Std 802.3-2022, IEEE, New York, 2022.

[75] *MIPI alliance specification for D-PHY*, See the current version, MIPI Alliance.

[76] *MIPI alliance specification for C-PHY*, See the current version, MIPI Alliance.

[77] *MIPI alliance specification for camera serial interface 2 (CSI-2)*, See the current version, MIPI Alliance.

[78] *MIPI alliance specification for display serial interface (DSI)*, See the current version, MIPI Alliance.

[79] G. Ruggles. "Successful PCI Express 6.0 designs at 64 GT/s with IP," Synopsys. (Apr. 19, 2021), [Online]. Available: https://www.synopsys.com/articles/pcie-6-designs.html (visited on 11/10/2026).

[80] *I/O buffer information specification (IBIS)*, See the current version; earlier versions also published as ANSI/EIA-656, IBIS Open Forum. [Online]. Available: https://ibis.org.

[81] *IEEE standard for test access port and boundary-scan architecture*, IEEE Std 1149.1, See the current revision, IEEE, New York.

[82] *Qualification and performance specification for rigid printed boards*, IPC-6012, See the current revision, IPC.

[83] *Design guide for protection of printed board via structures*, IPC-4761, See the current revision, IPC.

[84] *Bare substrate electrical test data format*, IPC-D-356, See the current revision, IPC.

[85] *IPC-TM-650 test methods manual*, See the current revision of the individual test method, IPC.

[86] *The gerber layer format specification*, See the current revision, Ucamco. [Online]. Available: https://www.ucamco.com/en/gerber.

[87] Siemens Digital Industries Software, *ODB++ design format specification*, See the current release.

[88] *Generic requirements for printed board assembly products manufacturing description data and transfer methodology*, IPC-2581, See the current revision, IPC.

[89] *Generic standard on printed board design*, IPC-2221, See the current revision, IPC.

[90] Rutronik. "Rockchip application microprocessor highlights (RK3588 key specifications)." (), [Online]. Available: https://www.Rutronik.com/fileadmin/rutronik/Downloads/printmedia/products/01_semiconductors/Rockchip_Flyer_Microprocessor_Highlights_Online.pdf (visited on 11/11/2026).

[91] CNX Software. "Rockchip RK3588 datasheet available, SBCs coming soon." (Dec. 16, 2021), [Online]. Available: https://www.cnx-software.com/2021/12/16/rockchip-rk3588-datasheet-sbc-coming-soon/ (visited on 11/11/2026).

[92] *Zynq-7000 SoC data sheet: Overview*, DS190, version 1.11.1, Xilinx (AMD), Jul. 2, 2018. [Online]. Available: https://www.xilinx.com/support/documentation/data_sheets/ds190-Zynq-7000-Overview.pdf (visited on 11/11/2026).

[93] Trenz Electronic. "TE0720 zynq-7000 module technical reference manual (clocking)." (), [Online]. Available: https://wiki.trenz-electronic.de/x/ExgpAg (visited on 11/11/2026).

[94] *Zynq-7000 SoC technical reference manual*, UG585, Consult the current revision; not read in full for this edition, Xilinx (AMD).

[95] *Zynq-7000 SoC PCB design guide*, UG933, Consult the current revision; not read in full for this edition, Xilinx (AMD).

[96] Wikipedia contributors. "M.2 (form factors and keying)." (), [Online]. Available: https://en.wikipedia.org/wiki/M.2 (visited on 11/12/2026).

[97] Silicon Power. "Industrial DDR5 SODIMM product specification (physical dimensions)." (), [Online]. Available: https://docs.rs-online.com/00fa/A700000015828382.pdf (visited on 11/12/2026).

[98] Forlinx Embedded. “OK3588-C user’s hardware manual.” (), [Online]. Available: https://docs.forlinx.net/rockchip/ok3588-c/OK3588-C_User_Hardware_Manual.html (visited on 11/11/2026).

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